



Intelligent Security API

Developer Guide

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1 Reading Guide

Chapter	Description
Overview	Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history.
ISAPI Framework	Read the chapter to take a quick look at the ISAPI framework and basic functions.
Quick Start Guide	Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading.
API Reference	Start programming according to API definitions.
How-To Video Guidance	How-to videos demonstrate detailed steps of different integration tasks.

2 Overview

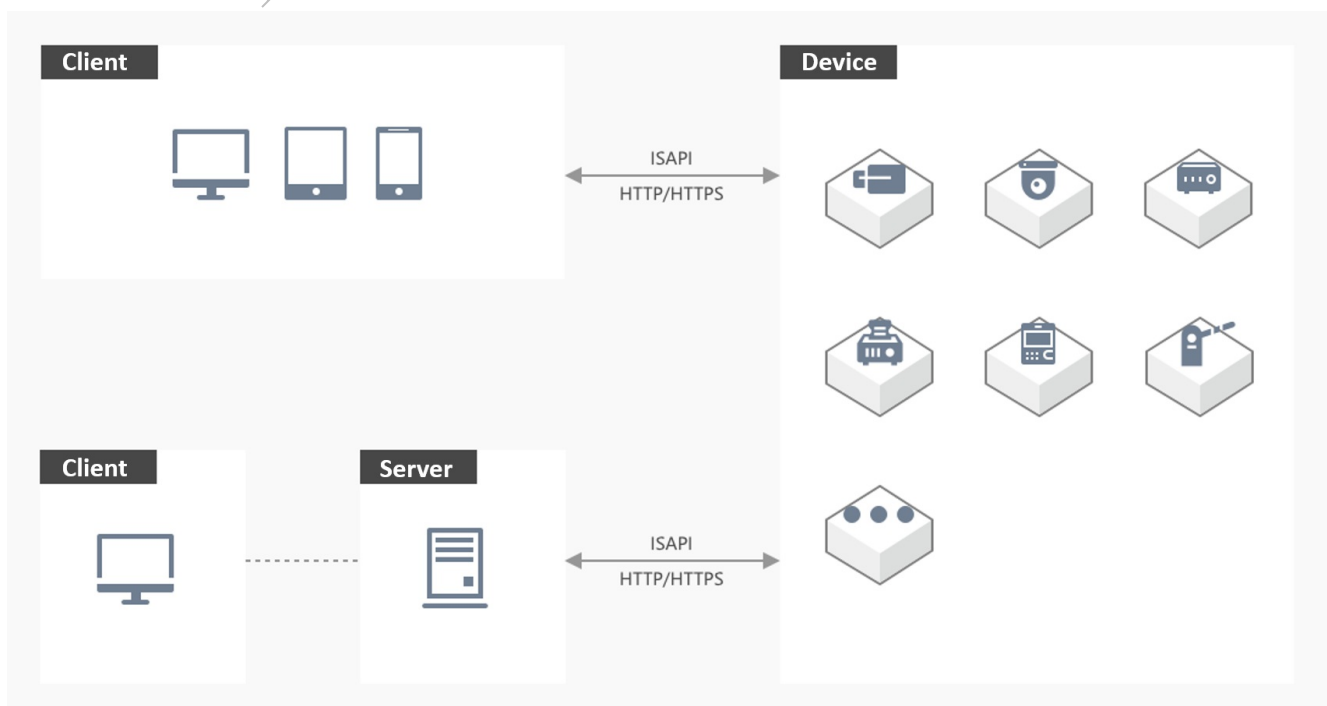
2.1 Introduction

Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on [HTTP \(Hypertext Transfer Protocol\)](#) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.



2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP.

Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, [RTSP \(Real-Time Streaming Protocol\)](#) based on TCP/UDP for live view and video playback of devices, etc.

Application Layer	Device Discovery SADP	Signaling Interaction ISAPI (HTTP)	Media Stream RTSP/RTP
Transport Layer	MCAST	TCP	TCP/UDP
Network Layer	IP		
Network Interface	Hardware Drive Interface		

2.2 Product Scope

- Access Control Accessories

- Enrollers

DS-K1F510-P, DS-K1F600-D6E, DS-K1F600-D6E-F, DS-K1F600U-D6E, DS-K1F600U-D6E-F

2.3 Terms And Definitions

2.3.2 Device Operation Log

Logs generated during the operation of the device's firmware. These logs are recorded in log files or log systems in text format, and are primarily used by device developers and maintenance personnel to identify device issues. #

2.3.5 Event

It refers to the information uploaded by the device. The event is uploaded by the device in real time for the immediate response from the client or the platform. If the device is offline, the event will be stored in the cache first and then it will be uploaded again when the connection is restored.

2.3.4 Arming

Arming means that the client establishes connection with the device so that events can be uploaded to the client via the connection. The client can subscribe to some event types, and the device will upload the specified events only, otherwise the device will upload all types of events to the client.

2.3.6 Listen

After the platform starts listening service, when an event occurs, the event information will be sent to the listening port of the platform based on the IP address and port No., then the connection will be closed.

2.3.7 Listening Host

The listening host service can be enabled to receive the event information from devices.

2.3.8 Center

A group of independent services, including at least one device access service and one or multiple application platforms.

2.3.9 Push Mode

In push mode, the device actively sends related information to the center.

2.3.10 Pull Mode

In pull mode, the center actively accesses the device to obtained related information.

2.3.11 Alarm Receiving Center

The ARC receives alarm information from devices and provides alarm services.

2.3.12 Event Code

An event code is used to identify a specific event and corresponds to a CID code.

2.4 Symbols And Acronyms

admin: administrator

HC: Connect Mobile Client

HPP: Partner Pro

HTTP: Hypertext Transfer Protocol

ISAPI: Intelligent Security API.

ISUP: Intelligent Security Uplink Protocol.

CMS: Central Management Server.

AMS: Alarm Management Server.

RTSP: Real-Time Streaming Protocol.

SDP: Session Description Protocol.

ARC: Alarm Receiving Center

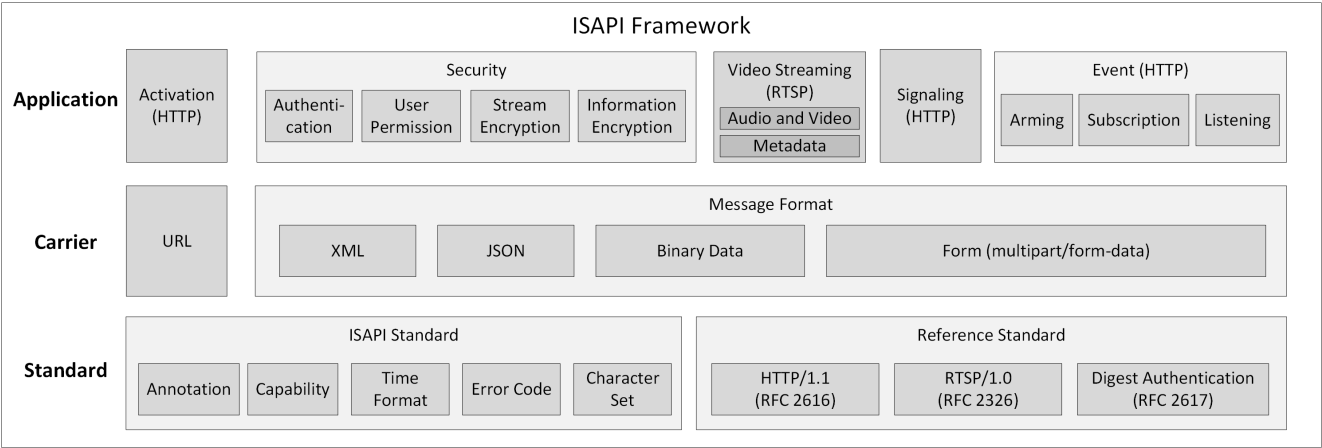
CID: Event Code (Contac ID)

2.5 Update History

No update record

3 ISAPI Framework

3.1 Overview



Notes:

In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

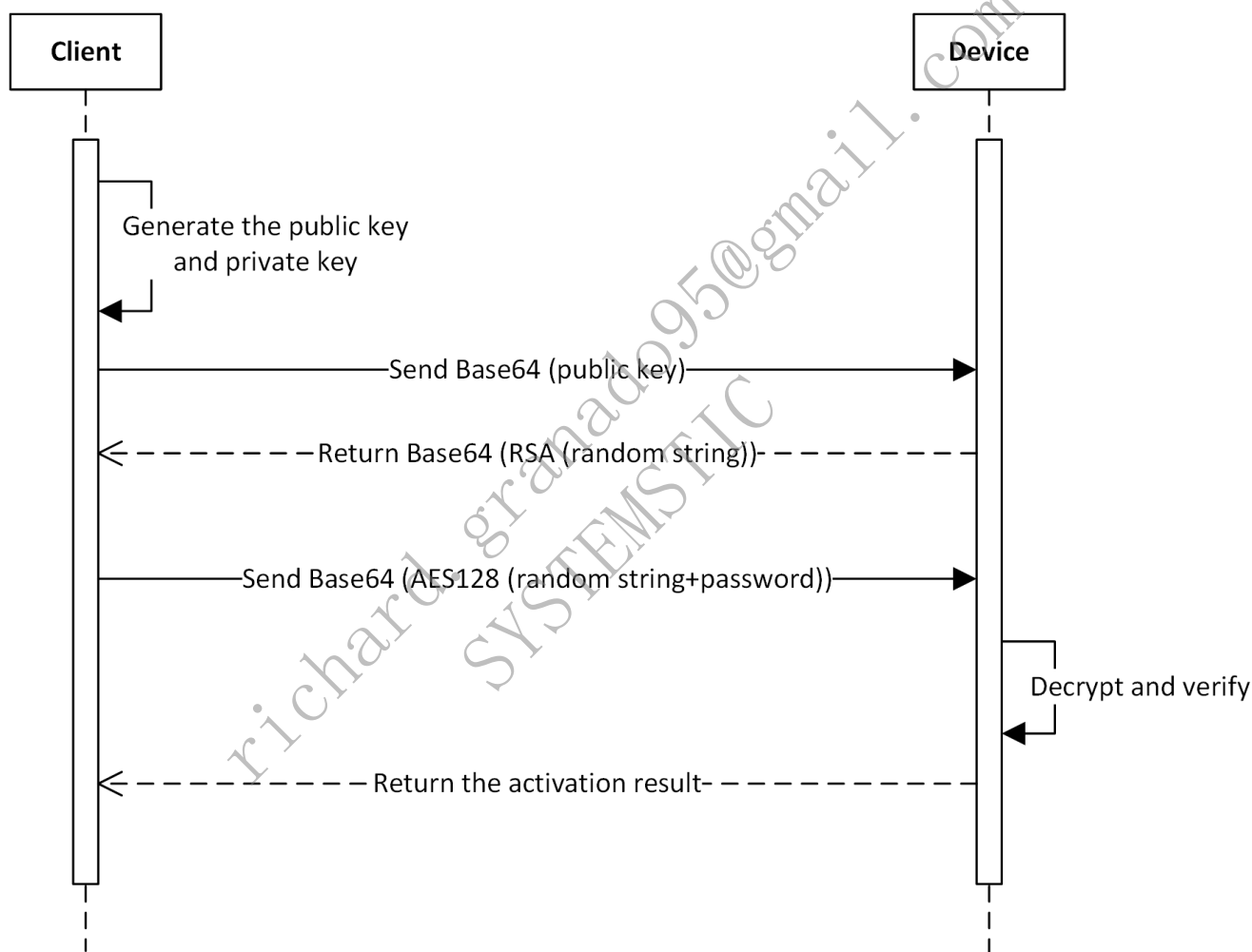
3.2 Activation

The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.



Firstly, two operations are defined:

- `bytesToHexString`: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23` -> `7f0a17`
 - `hexStringToBytes`: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17` -> `127,10,23`
1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
 2. The client converts the public key modulus to a 256-byte public key string via `bytesToHexString` and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.
 3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via `hexStringToBytes`. The complete public key is the combination of obtained public key

modulus and public exponent (the default value is '010001').

4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via `bytesToHexString`, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via `hexStringToBytes` to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via `hexStringToBytes` to get a 16-byte AES key. Then the client uses the AES key to encrypt the "string consisting of the first 16 characters of the random string and the real password" by AES128 ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via `bytesToHexString`, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`).
Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.
6. The device decodes the string by Base64, converts it via `hexStringToBytes` to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.
7. The device verifies the password and returns the activation result.

Notes:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

3.3 Security Mechanism

3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in [RFC 7616](#)) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in [Authentication](#) of Quick Start Guide.

3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

Administrator: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

Operator: Has the permission to access general resources and a part of advanced resources.

Normal User: Only has the permission to access general resources.

3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

3.4 Video Streaming

3.4.1 Audio and Video Stream

ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in [RFC 7826](#)). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

4 Quick Start Guide

4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in [RFC 7616](#)) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

4.1.1 C/C++ (libcurl)

```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string*)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```


4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

4.1.3 Java (HttpClient)

```
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsByteArray().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

4.1.4 Python (requests)

```
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

4.2 Message Parsing

4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

4.2.1.1 XML

Generally, the Content-Type in the headers of the HTTP request is application/xml; charset="UTF-8".

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace `http://www.isapi.org/ver20/XMLSchema` and ISAPI version number `2.0` of XML messages are configured by default, see the example below.

```
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

4.2.1.2 JSON

The `Content-Type` in the headers of the HTTP request is often `application/json`.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter `format=json` to all request URLs with JSON messages, e.g.,

`http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json`. Messages of request URLs without the query parameter `format=json` are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

4.2.1.3 Binary Data

For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data`, `boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length. For the detailed format description, refer to [RFC 1867 \(Form-Based File Upload in HTML\)](#). Pay attention to two hyphens (`--`) before and after the boundary.

Notes

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

```

POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--

```

The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (pid, contentid, and filename) are used for linking form units. The corresponding relations are as follows:

Node Name	Form Field	Description
pid	name	pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers.
contentid	Content-ID	contentid in XML/JSON messages corresponds to Content-ID in form headers.
filename	filename	filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers.

```

HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
  "ipAddress": "172.6.64.7",
  "macAddress": "01:17:24:45:D9:F4",
  "channelID": 1,
  "dateTime": "2009-11-14T15:27+08:00",
  "eventType": "mixedTargetDetection",
  "eventDescription": "Mixed target detection",
  "deviceID": "123456789",
  "CaptureResult": [{
    "targetID": 1,
    "Human": {
      "Rect": {
        "height": 1.0,
        "width": 1.0,
        "x": 0.0,
        "y": 0.0
      },
      "contentID1": "humanImage", /*human body thumbnail*/
      "contentID2": "humanBackgroundImage", /*human body background picture*/
      "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
      "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
    }
  ]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--

```

4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```

<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#level 1,level2#level 2,level3#level 3]-->level1
    </level>
  </Node>
</NodeList>

```

```

{
  "name": "test",
  /*ro, req, string, name, range:[1,32]*/
  "type": "type1",
  /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
  "enabled": true,
  /*ro, opt, bool, enable or not, desc:xxxxxx*/
  "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
    "scene": 1,
    /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
    "ID": 1
  }
  /*req, int, No., range:[1,8], step:, unit:, unitType:*/
}

```

Key annotations are shown in the table below.

Annotation	Description	Remark
ro	Attribute: Read-Only	This field can only be obtained and cannot be edited.
wo	Attribute: Write-Only	This field can only be edited and cannot be obtained.
req	Attribute: Required	This field is required for request messages sent to the device and response messages returned from the device.
opt	Attribute: Optional	This field is optional for request messages sent to the device and response messages returned from the device.
dep	Attribute: Dependent	This field is valid and required when specific conditions are satisfied.
object	Field Type: Object	The field of type object contains multiple sub-fields.
list	Field Type: List	The subType following it refers to the data type of sub-items in the list.
subType	Field Type: String	The range following it refers to the maximum and the minimum string size of the field.
int	Field Type: Int	The range following it refers to the maximum and the minimum value of the field.
float	Field Type: Float	The range following it refers to the maximum and the minimum value of the field.
bool	Field Type: Boolean	The value can be true or false.
enum	Field Type: Enumeration	The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators.
subType	Sub-Type of Field	When the type of field is list or enum, the value of subType is the data type of each sub-object.
desc	Field Description	The detailed description of the field.

4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

Whether the device supports a function: it is often in the format `isSupporttXXXXXXX`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```

{
  "isSupportMap": true,
  /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
  "isSupportAlgTrainResourceInfo": true,
  /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
  desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
  "isSupportAlgTrainAuthInfo": true,
  /*ro, opt, bool, whether it supports only getting the authorization information of the algorithm training platform,
  desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
  "isSupportAlgTrainNodeList": true,
  /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
  format=json*/
  "isSupportNAS": true
  /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}

```

The capability message example in XML format is shown below:

```

<isSupportNetworkStatus>
  <!--ro, opt, bool, whether it supports searching the network status, desc: reLated API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>

```

The value range of the field: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```

{
  "boolType": {
    /*req, object, example of the capability of type bool*/
    "@opt": [true, false]
    /*req, array, options, subType: bool*/
  },
  "integerType": {
    /*req, object, example of the capability of type integer*/
    "@min": 0,
    /*ro, req, int, the minimum value*/
    "@max": 100
    /*ro, req, int, the maximum value*/
  },
  "stringType": {
    /*req, object, example of the capability of type string*/
    "@min": 0,
    /*ro, req, int, the minimum string size*/
    "@max": 32
    /*ro, req, int, the maximum string size*/
  },
  "enumType": {
    /*req, object, capability example of type enum*/
    "@opt": ["enum1", "enum2", "enum3"]
    /*req, array, options, subType: string*/
  }
}

```

The example of XML format is shown below:

```

<boolType opt="true,false" def="true">
  <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
  <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
  <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
  <!--ro, opt, string, example of the capability of type enum-->white
</enumType>

```

Note: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

4.2.4 Time Format

ISAPI adopts [ISO 8601 Standard Time Format](#), which is the same as [W3C Standard Date and Time Formats](#).

Format: YYYY-MM-DDThh:mm:ss.sTZD

YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)

Example: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

Example: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

Notes:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
  "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the device's local time and UTC*/
  "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert the dateTime into the local time for use*/
}
```

4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to [RFC 2044](#)).

No.	Name	Special Character	No.	Name	Special Character
1	Open Parenthesis	(	18	Dollar Sign	\$
2	Close Parenthesis	)	19	Percent Sign	%
3	Plus Sign	+	20	Ampersand	&
4	Comma	,	21	Close Single Quotation Mark	'
5	Minus Sign	-	22	Asterisk	*
6	Period	.	23	Slash	/
7	Semicolon	;	24	Smaller Than	<
8	Equal Sign	=	25	Greater Than	>
9	At Sign	@	26	Question Mark	?
10	Open Square Bracket	[	27	Caret	^
11	Close Square Bracket	]	28	Open Single Quotation Mark	'
12	Underscore	_	29	Vertical Bar	
13	Open Brace	{	30	Tilde	~
14	Close Brace	}	31	Double Quotation Marks	"
15	Space		32	Colon	:
16	Exclamation Mark	!	33	Backslash	\
17	Octothorpe	#			

The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (a-z), uppercase letters (A-Z), digits (0-9), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in [RFC 2616](#). For ISAPI error codes, refer to Error Code Dictionary.

Message Example:


```

HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
  "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
  "statusCode": 4,
  "statusString": "Invalid Operation",
  "subStatusCode": "notSupport",
  "errorCode": 1073741825,
  "errorMsg": "notSupport"
}

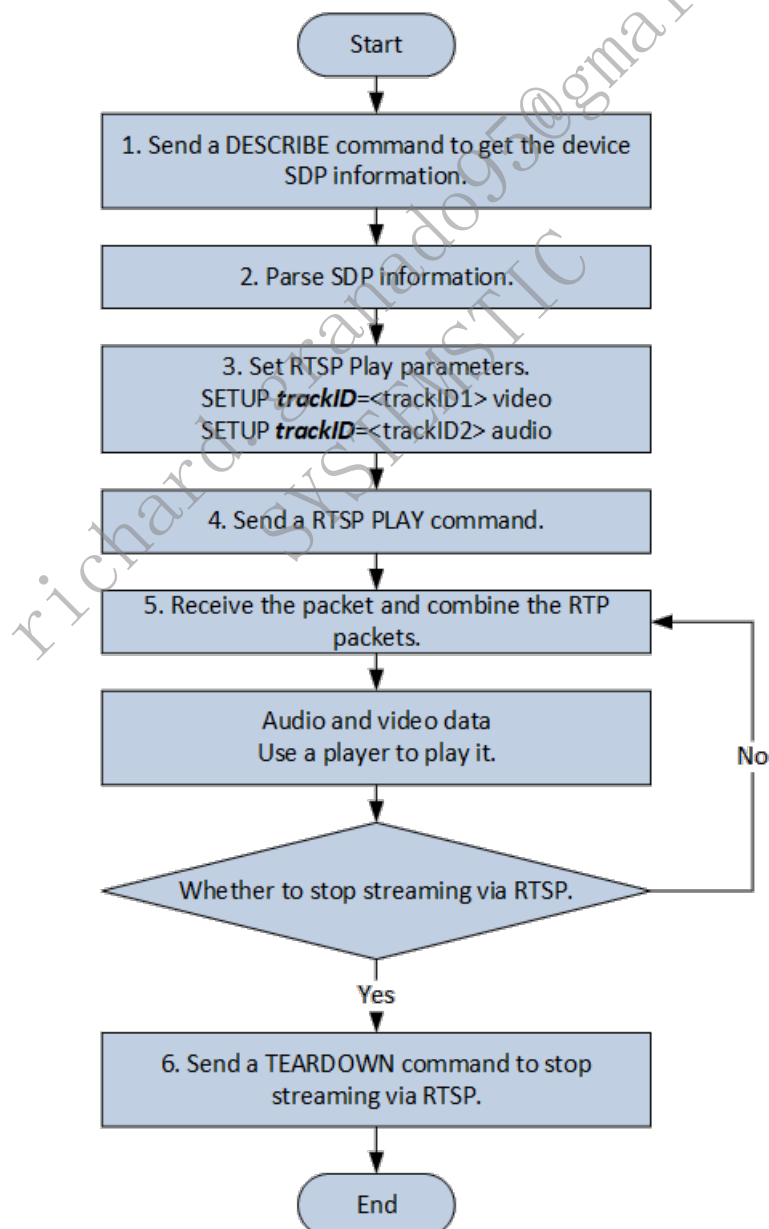
```

4.3 Real-Time Live View

4.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type. Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in [RFC 7826](#)).

4.3.2 API Calling Flow



1. A client sends RTSP DESCRIBE commands such as DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0. Digest authentication with devices is required before this step.
2. The client parses the media SDP information returned by the device.

3. Set RTSP play parameters, that is to set the track ID parsed from SDP information via SETUP commands. For example, trackID=1 indicates videos while trackID=2 indicates audios.
4. The client sends an RTSP PLAY command, and the device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`.
5. The client receives the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed.
6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address; `[:port]` is optional, and 554 by default; `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, the IP address of the target device is `172.7.203.11`, and the streaming address of main stream for channel 17 will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2=-value2...)`, such as `/Streaming/Channels/101?transportmode=unicast`.

4.3.3 Example

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

```
NFO=494D4B4802010000020000111710110401F00000FA0000000000000000000000000000;

ds RTSP SETUP requests, and the server responds to them.

.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0

username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/St
deb028dd9b4932cc44201"
;unicast;interleaved=0-1:ssrc=0
.00.00.081112

imeout=60
P;unicast;interleaved=0-1:ssrc=433122aa

Seeking, Time-Progressing, Time-Duration=0
00:09:45 GMT
```

```
.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0

username="admin", realm="3521781c29acb312330dd668", nonce="026019833", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1"
deb028dd9b4932cc44201"
;unicast;interleaved=0-1;ssrc=0
.00.00.081112

timeout=60
P;unicast;interleaved=0-1;ssrc=433122aa

Seeking, Time-Progressing, Time-Duration=0
0 02:09:45 GMT
```

```
00.00.081112
timeout=60
P;unicast;interleaved=0-1;ssrc=433122aa
Seeking, Time-Progressing, Time-Duration=0
0 02:09:45 GMT
```

```
00.00.081112
timeout=60
P;unicast;interleaved=0-1;ssrc=433122aa
Seeking, Time-Progressing, Time-Duration=0
0 02:09:45 GMT
```

```
147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
user: name="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri=
deb028dd9b4932cc44201"
timeout=60
unicast;interleaved=2-3;ssrc=0
.00.00.081112

timeout=60
P;unicast;interleaved=2-3;ssrc=433122ab
```

```

Seeking, Time-Progressing, Time-Duration=0
0 02:09:45 GMT

ds an RTSP PLAY request.

147:554/ISAPI/Streaming/channels/101 RTSP/1.0

username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri=
ef767f7c49d1c847200bd"

meout=60
000000

```

```
147:554/ISAPI/Streaming/channels/101 RTSP/1.0

  username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101"
  meout=60
000000
```

ends audio and video stream data.

ends audio and video stream data.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. ....d1.w....C....".T....g....).i.....a....7.S..~J.....X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

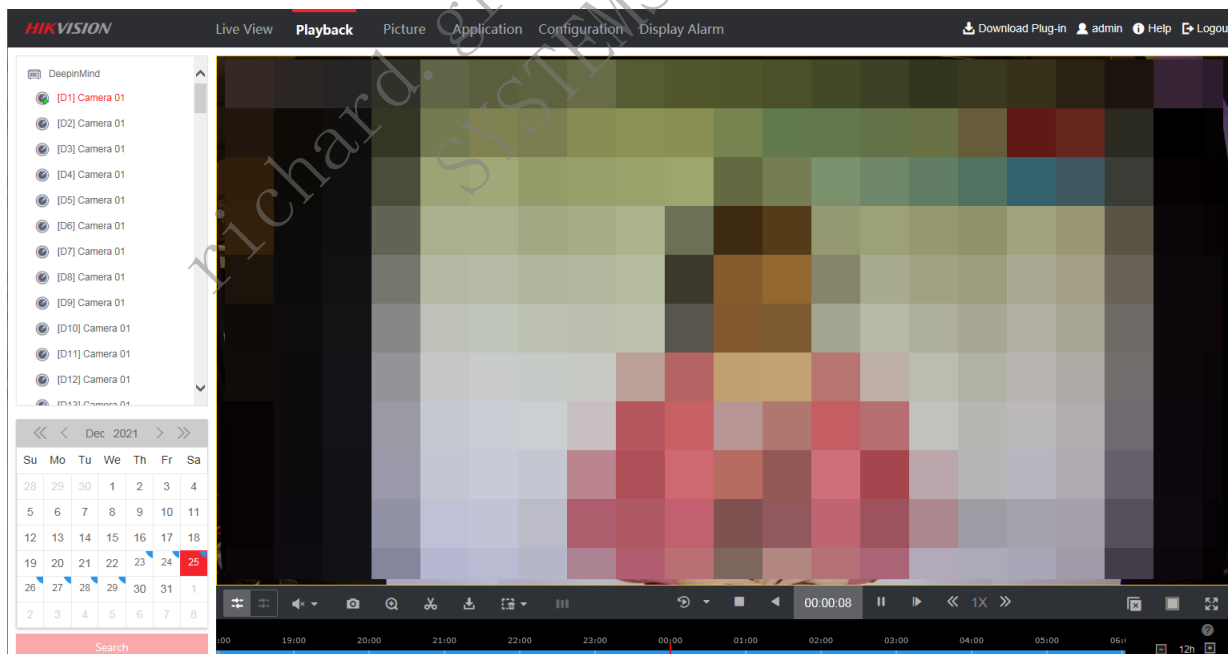
```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

4.4 Playback

4.4.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in [RFC 7826](#)).

Below is an example of playback on a web client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos are shown on the calendar and you can click a date to play back searched videos of the corresponding day.



4.4.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.

3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. Refer to streaming via RTSP in real-time streaming. See details in [RFC 7826](#) (Real Time Streaming Protocol).

Notes:

- Playback via RTSP needs digest authentication. The method is the same as the one of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in [RFC 7826](#) (Real Time Streaming Protocol).

4.4.3 Example

4.4.3.1 Search for Videos According to Calendar

Request from Client

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
  <year>2021</year>
  <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

Response from Device

```

HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <dayList>
    <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
    <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
    <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
    <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
    <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
    <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
    <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
    <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
    <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
    <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
    <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
    <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
    <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
    <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
    <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
    <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
    <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
    <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
  </dayList>
</trackDailyDistribution>

```

4.4.3.2 Search for Videos

Request from Client

```

POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

```

```

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
  <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
  <trackList>
    <trackID>101</trackID>
  </trackList>
  <timeSpanList>
    <timeSpan>
      <startTime>2021-08-16T00:00:00Z</startTime>
      <endTime>2021-08-18T23:59:59Z</endTime>
    </timeSpan>
  </timeSpanList>
  <maxResults>100</maxResults>
  <searchResultPostion>0</searchResultPostion>
  <metadataList>
    <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
  </metadataList>
</CMSearchDescription>

```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
  <responseStatus>true</responseStatus>
  <responseStatusStrg>OK</responseStatusStrg>
  <numOfMatches>1</numOfMatches>
  <matchList>
    <searchMatchItem>
      <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
      <trackID>101</trackID>
      <timeSpan>
        <startTime>2021-08-18T15:18:15Z</startTime>
        <endTime>2021-08-18T15:19:08Z</endTime>
      </timeSpan>
      <mediaSegmentDescriptor>
        <contentType>video</contentType>
        <codecType>H.264-BP</codecType>
        <playbackURI>rtsp://10.14.97.40/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788</playbackURI>
        <lockStatus>unlock</lockStatus>
        <name>00000004667000100</name>
      </mediaSegmentDescriptor>
      <metadataMatches>
        <metadataDescriptor>recordType.meta..com/timing</metadataDescriptor>
      </metadataMatches>
    </searchMatchItem>
  </matchList>
</CMSearchResult>
```

4.4.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

```
O=49404B4802010000040005001710110401F00000FA000000000000000000000000000000;

s RTSP SETUP commands two times, and the server responds to them.

0:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464c3 (LIVE555 Streaming Media v2016.11.28)
ast;client_port=61048-61049

out=60
ast;client_port=61048-61049;server_port=62000-62001;ssrc=46d3ed5

m-Access=1.0s, Unlimited, Imutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
07:29:25 GMT
```

```

ast;client_port=61048-61049

out=60
ast;client_port=61048-61049;server_port=62000-62001;src=dc63ed5

m-Account=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4
07:29:25 GMT

```

```

ast;client_port=61048-61049

out=60
ast;client_port=61048-61049;server_port=62000-62001;src=dc63ed5

m-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4
07:29:25 GMT

```

```

ast;client_port=61048-61049

out=60
ast;client_port=61048-61049;server_port=62000-62001;src=dc63ed5

m-Account=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4
07:29:25 GMT

```

```

ast;client_port=61048-61049

out=60
ast;client_port=61048-61049;server_port=62000-62001;src=dc63ed5

m-Account=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4
07:29:25 GMT

```

```

ast;client_port=61048-61049

out=60
ast;client_port=61048-61049;server_port=62000-62001;src=dc63ed5

m-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4
07:29:25 GMT

```

s an RTSP PLAY request, and the server starts to push streams for playback.

s an RTSP PLAY request, and the server starts to push streams for playback.


```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

4.5 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

4.5.1 Arming

The client establishes a HTTP persistent connection with the device to receive event messages from the device.

There are two methods (arming with subscription and arming without subscription) to receive events from the device. The arming without subscription is to get all event messages from the device via HTTP GET method, while the arming with subscription is to get messages of subscribed events via HTTP POST method.

Notes

- ISAPI arming (with or without subscription) uses the HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send event messages continuously, while it's not supported for clients to send any message to the device via the connection.
- When the heartbeat timed out and no message is received from the device, you should terminate the arming connection and try establishing a new one.

4.5.1.1 Arming without Subscription

- Establish the connection of arming without subscription: `GET /ISAPI/Event/notification/alertStream` and keep the connection alive via configuring `Connection: keep-alive` in HTTP headers on the client.
- Receive events sent by the device. The event message will be separated and parsed by boundary. For parsing details, see **Event Message Parsing** below.
- Terminate the arming connection when no event message needs to be received.

Event Message Parsing:

```

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP
Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8" <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message format according to Content-Type when parsing event messages-->
Content-Length: text_length

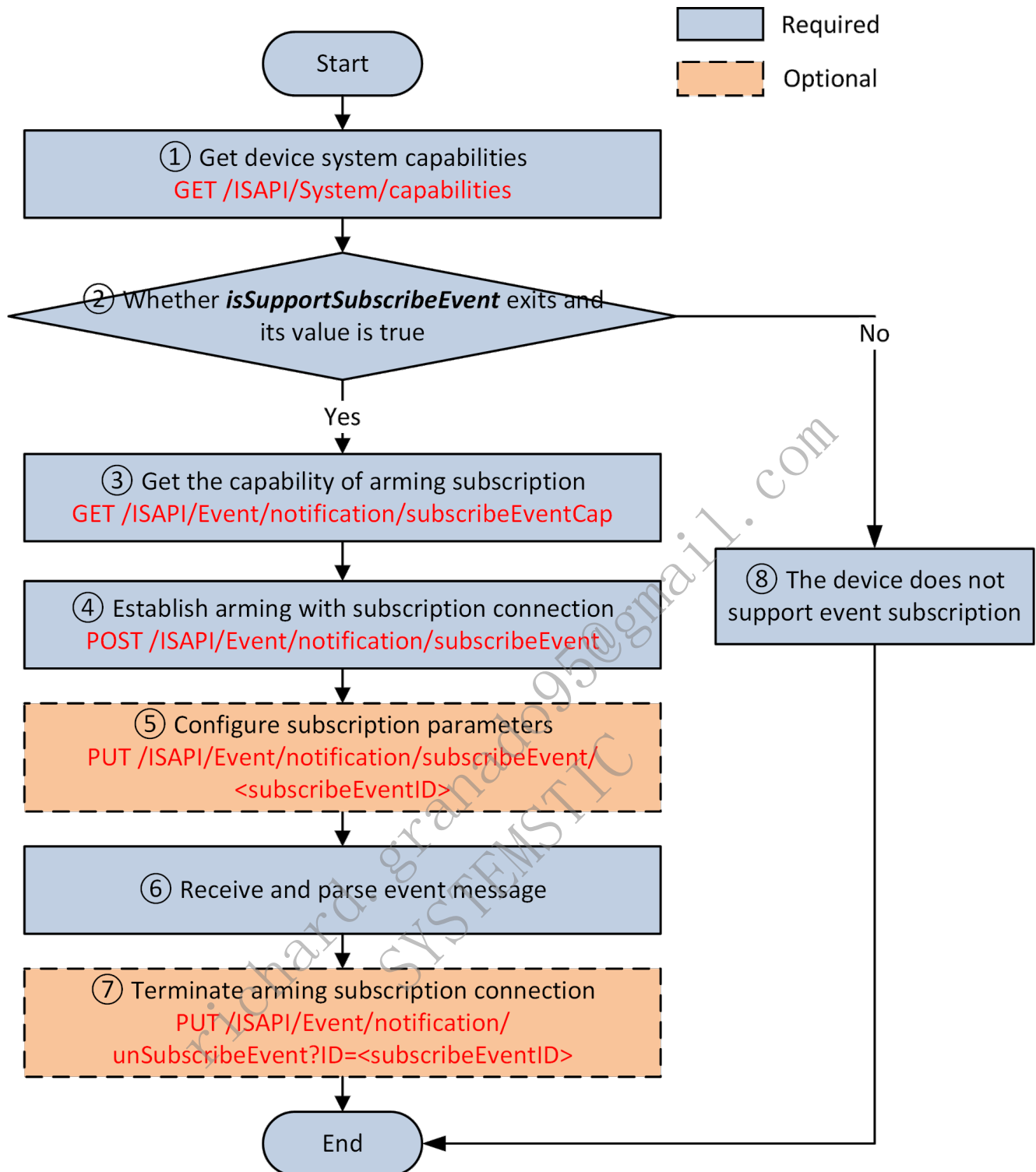
<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

[Picture Data]
--<frontier>

```

Note: <data_gateway_ip> and <frontier> are variables, [Picture Data] indicates the raw data of a picture.

4.5.1.2 Arming with Subscription



1. Get device system capabilities: GET /ISAPI/System/capabilities.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: GET /ISAPI/Event/notification/subscribeEventCap.
4. Establish a connection of arming with subscription: POST /ISAPI/Event/notification/subscribeEvent. You need to set `Connection: keep-alive` in HTTP headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>. Then, edit the parameters based on the existing subscription configurations: PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>.
6. Receive events sent by the device. The event messages will be separated and parsed by the boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: <SubscribeEventResponse/>, <EventNotificationAlert/>, and picture data. <SubscribeEventResponse/> is the data of first form sent by the device after arming established, see the response parameters of URL (POST /ISAPI/Event/notification/subscribeEvent) for details; and <EventNotificationAlert/> is the event content or heartbeat, you can identify the event type via field eventType, e.g., for heartbeat, the value of eventType is heartBeat.

Event Messages Parsing:

Client Creates Arming with Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin", realm="IP
Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

Server Responds to Request and Push Event Message

When the client sends a request to the device for establishing an arming connection, the device sends data in HTTP form format (multipart/form-data). In HTTP request headers, the Content-Type is multipart/form-data, boundary=AaB03x, of which the boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens -- before and after boundary. In normal conditions, devices will not close the arming link actively, and the end mark --AaB03x-- will not be sent on the arming link.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

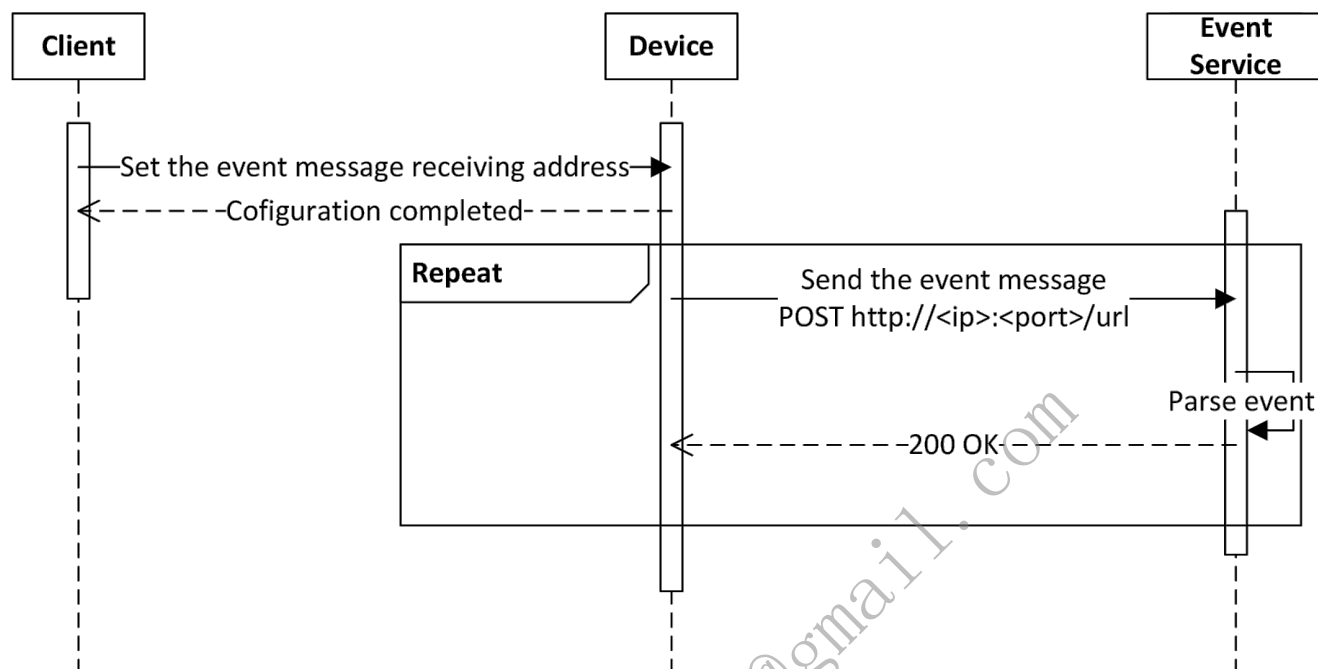
Picture Data
--AaB03x--
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. A form message which starts with --boundary and ends with --boundary--.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description. form-data is a piece of form data.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next \r\n to the next --boundary.

4.5.2 Listening

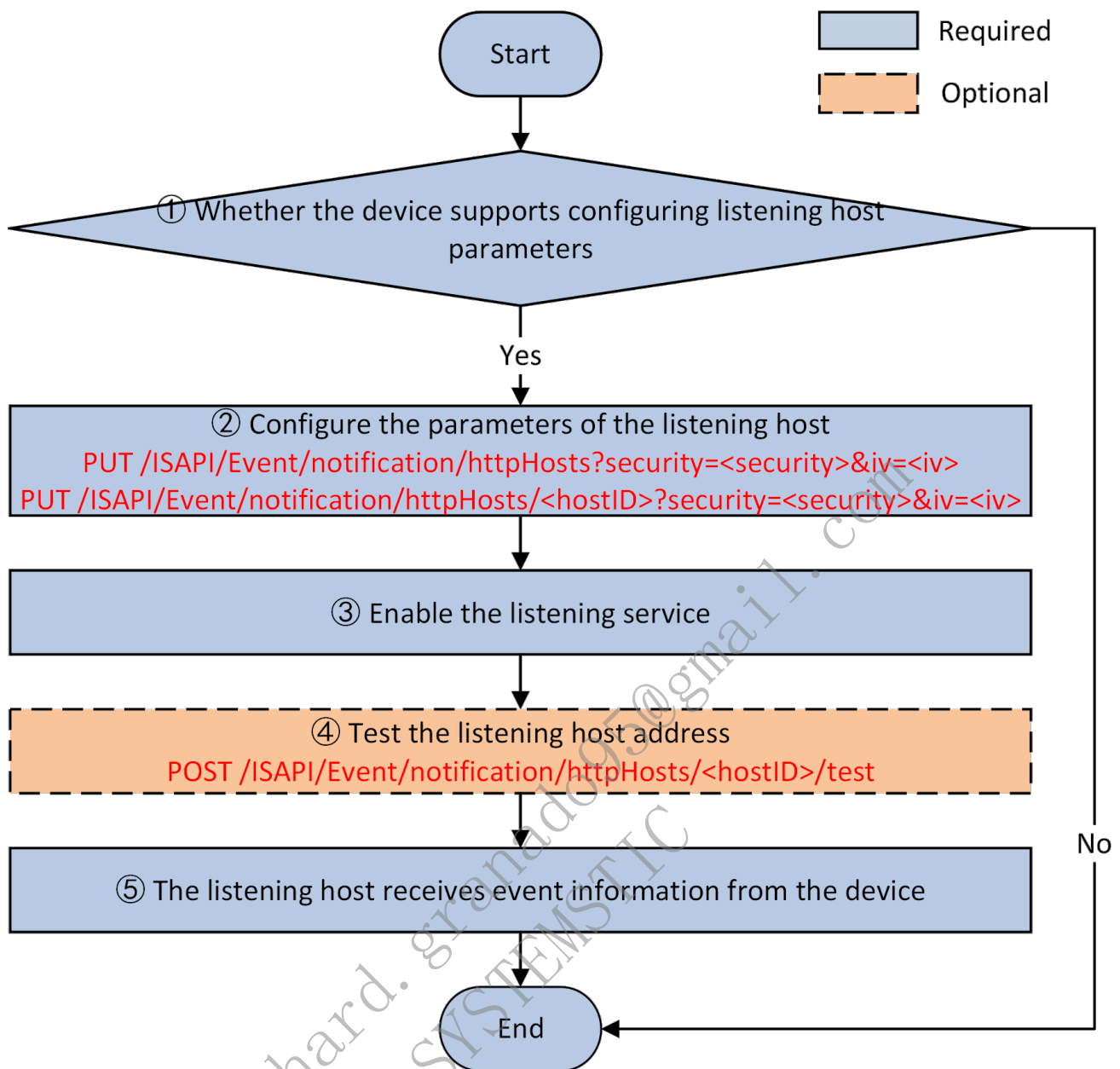
After a client enables the listening service, when an event occurs, the device will send the event information actively to the configured event receiving address. The event receiving address should be valid and configured on the device.



Notes:

- The client and event service can be the same program.
- In listening mode, no heartbeat information is generated on devices.

4.5.2.1 API Calling Flow



1. Check whether the device supports configuring listening host parameters.

Get the configuration capability of the listening host: `GET /ISAPI/Event/notification/httpHosts/capabilities`.
 If the node `<HttpHostNotificationCap>` is returned and its value is true, it indicates that the device supports configuring listening host parameters.

2. Configure parameters of the listening host.

Configure parameters of all listening hosts: `PUT /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

Get parameters of all listening hosts: `GET /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

Configure parameters of a listening host: `PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`;

Get parameters of a listening host: `GET /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`.

3. Enable the listening service.

You need to enable the listening service of the listening host.

4. (Optional) Test the listening service.

The platform applies the command to the device to test whether the listening host is available for the device: `POST /ISAPI/Event/notification/httpHosts/<hostID>/test`.

5. The listening host receives event information from the device.

When an event occurs, the device creates an connection with the client and uploads alarm information actively. Meanwhile, the listening host receives data from the device. See details in **Event Messages**.

Note: You can also configure the listening parameters such as the time out via URL

`/ISAPI/Event/notification/httpHosts/<hostID>/uploadCtrl`.

4.5.2.2 Event Messages

When an event occurs or an alarm is triggered in listening mode, the event/alarm information can be uploaded with binary data (such as pictures) and without binary data.

1. Without Binary Data:

The `Content-Type` in headers of the HTTP request sent by the device is usually `application/xml` or `application/json` as follows:

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: data_gateway_ip:port <!--HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: en-us
Date: YourDate
Content-Type: application/xml; <!--Content Type, which is used for the upper Layer to distinguish different formats when parsing the message-->
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

<EventNotificationAlert/>
```

Response by the Listening Host

```
HTTP/1.1 200 OK
Date: YourDate
Connection: close
```

2. With Binary Data:

The format of the data sent by the device is HTTP form (multipart/form-data). The `Content-Type` in headers of the HTTP request is usually `multipart/form-data; boundary=<frontier>`, of which boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. See details in [RFC 1867 \(Form-based File Upload in HTML\)](#). An example is shown below. Please note two hyphens `--` before and after the boundary.

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, , related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: device_ip:port <!--HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: en-us
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml <!--some event messages are uploaded in JSON format, and the upper Layer needs to distinguish the message format according to Content-Type when parsing event messages-->

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture Data]
--<frontier>--
```

Response by the Listening Host

HTTP/1.1 200 OK
Date: YourDate
Connection: close

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=frontier	Content type, multipart/form-data refers to data in form format.
boundary	frontier	Delimiter of the form message. A form message which starts with --boundary and ends with --boundary--.
Content-Disposition	form-data; name="Picture_Name";	Content description. form-data is a piece of form data.
filename	"Picture_Name"	File name. The file refers to the form message.
Content-Length	10	Content length, starting from the next \r\n to the next --boundary.

4.5.2.3 Exception Handling

Error Codes

statusCode	statusString	subStatusCode	errorCode	Description
6	Invalid Content	eventNotSupport	0x60001024	

5 Device Management (General)

5.1 Arming and Subscription

5.1.1 Introduction to the Function

With arming and subscription, the client can establish HTTP persistent connection with the device, and continuously receive the event messages from the device.

For ISAPI event arming, the client can receive all types of events by GET method, or receive the subscribed events only by POST method.

5.1.2 API Calling Flow

5.1.2.1 Without Subscription

1. Establish a connection for arming: GET /ISAPI/Event/notification/alertStream. You need to set Connection: keep-alive in HTTP Headers.
2. When receiving events sent by the device, the event messages can be separated and parsed by boundary. See "Parsing Event Messages" below for details.
3. Disable the arming connection when you do not need to receive event messages.

5.1.2.1.2 Syntax


```

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/form-data; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8" <!-- Some alarms are in JSON format, so the upper layer should parse based on the Content-Type field -->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

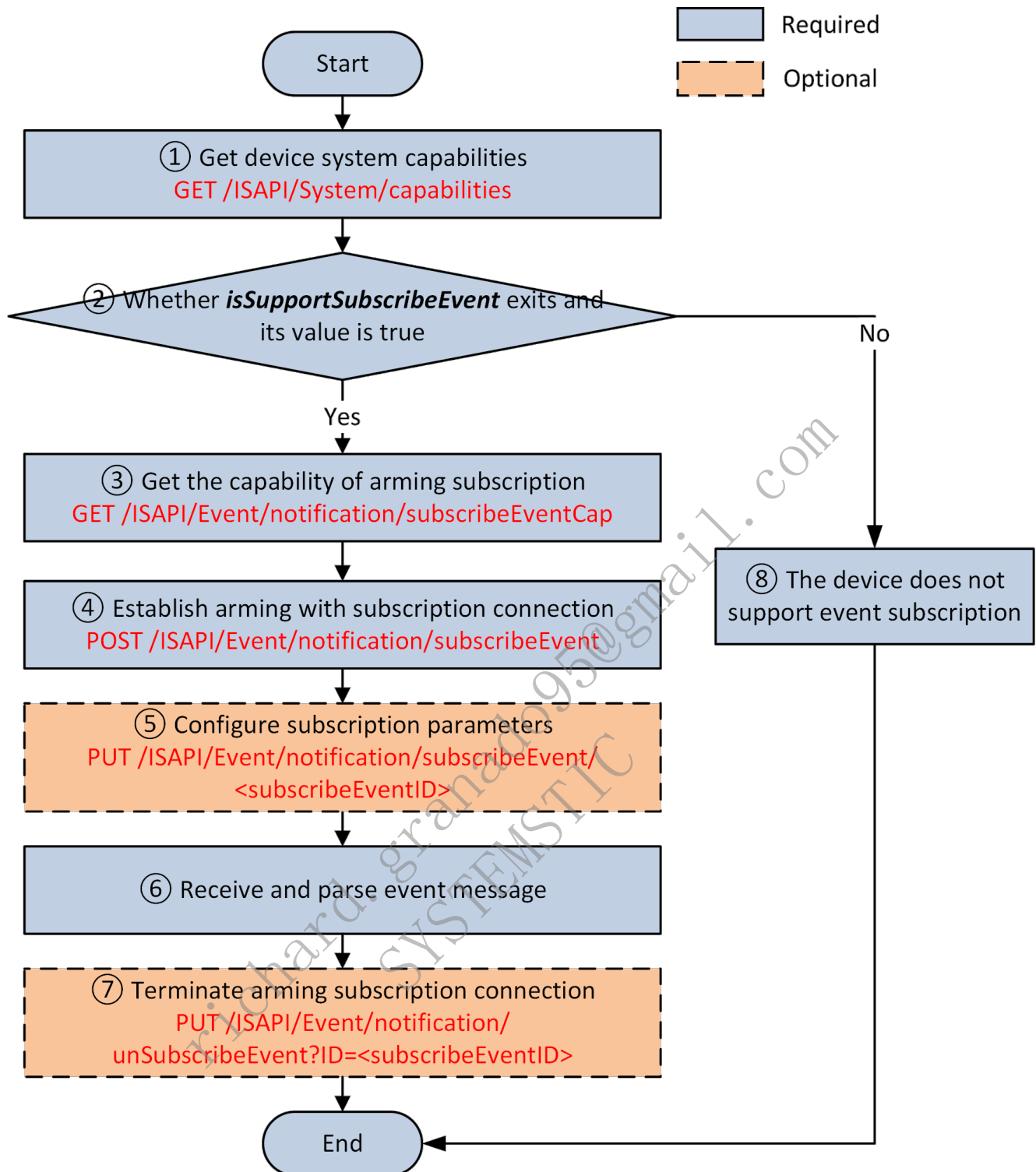
[Image Data]
--<frontier>

```

Note: <data_gateway_ip> and <frontier> are variables, and [Image Data] is an abbreviated representation indicating the raw data of an image at this location.

5.1.2.2 Subscription

5.1.2.2.1 API Calling Flow



1. Get device system capabilities: GET /ISAPI/System/capabilities.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: GET /ISAPI/Event/notification/subscribeEventCap.
4. Establish a connection of arming with subscription: POST /ISAPI/Event/notification/subscribeEvent. You need to set `Connection: keep-alive` in HTTP Headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>. Then, edit the parameters based on the existing subscription configurations: PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>.
6. Receive events sent by the device. The event messages will be separated and parsed by boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: PUT
/ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: <SubscribeEventResponse/>, <EventNotificationAlert/>, and picture data. <SubscribeEventResponse/> is the data of first form sent by the device after arming established, see the response parameters of URL (POST /ISAPI/Event/notification/subscribeEvent) for details; and <EventNotificationAlert/> is the event content or heartbeat, you can identify the event type via field eventType, e.g., for heartbeat, the value of eventType is heartBeat.

5.1.2.2.2 Example

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Host: device_ip
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml;
Content-Length: text_length
Connection: Keep-Alive

<SubscribeEvent/>
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin", realm="IP
Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", uri="/ISAPI/Event/notification/alertStream", ncnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8" <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message format according to Content-Type when parsing event messages-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8" <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message format according to Content-Type when parsing event messages-->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

5.1.2.3 Event Messages Parsing

After the arming connection with the device is established, the data sent by the device is in HTTP form format (multipart/form-data). In an HTTP request, Content-Type in Headers is usually multipart/form-data, boundary=AaB03x, and boundary is a variable used to divide HTTP Body into multiple units, each being a set of data and has its own Headers and Body. For detailed format description, see [RFC 1867 \(Form-based File Upload in HTML\)](#). An example is shown below. Note the dash -- before and after boundary. Under normal circumstances, the device will not actively close the arming connection, so the device will not send the form format end symbol --AaB03x-- on the arming connection.

```

HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Image Data
--AaB03x--

```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. --boundary is the start of a form. --boundary-- is the end of the whole HTTP form message.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next \r\n to the next --boundary.

5.1.3 Restriction Description

Note that ISAPI arming (with or without subscription) uses HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send out event messages continuously, while you cannot send any message to the device via the connection.

After the heartbeat time, if you do not receive any message from the device, you should disable the arming connection and try establishing a new one.

5.1.4 Sample Messages

5.1.4.1 Establish Arming Subscription

```

POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin", realm="IP
Camera(C2183)", nonce="4e5468694e7a4d364f4449354d7a6b354d54513d", uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>

```

5.1.4.2 The Device Responses and Uploads an Event Message

```

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8" <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8" <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
  <ipAddress>10.17.133.46</ipAddress>
  <portNo>80</portNo>
  <protocol>HTTP</protocol>
  <macAddress>44:19:b6:6d:24:85</macAddress>
  <channelID>1</channelID>
  <dateTime>2017-05-04T11:20:02+08:00</dateTime>
  <activePostCount>0</activePostCount>
  <eventType>heartBeat</eventType>
  <eventState>active</eventState>
  <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length
Content-ID: image_ID

[Picture Data]
--<frontier>

```

5.2 Calling Flow of Device Packet Capture

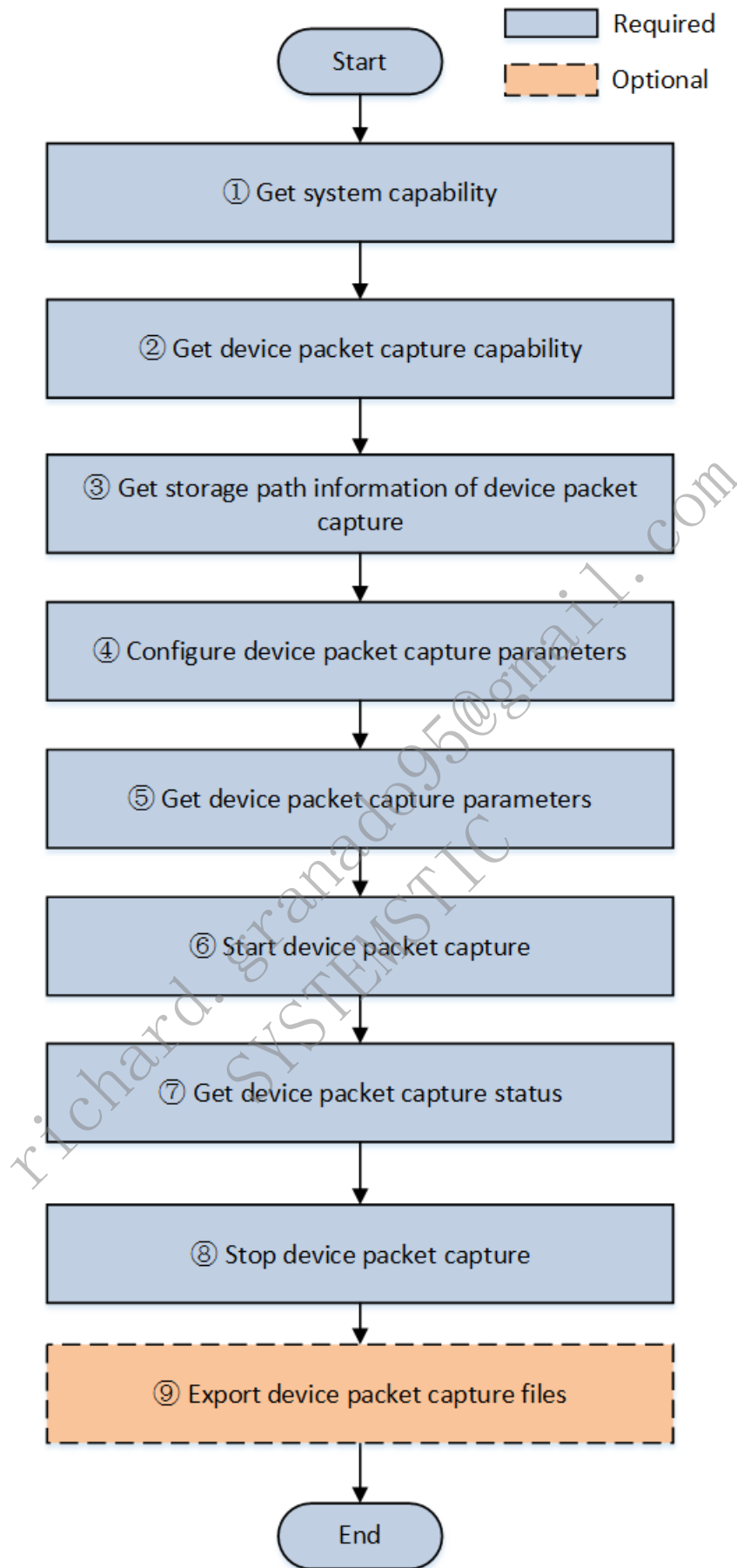
5.2.1 Function Introduction

When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.
2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

5.2.2 API Calling Flow

5.2.2.1 Device Packet Capture



1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: GET /ISAPI/System/networkCapture/StoragePathInfo?format=json.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: PUT /ISAPI/System/networkCapture/captureParams?format=json.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: GET /ISAPI/System/networkCapture/captureParams?format=json.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>.

Note:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: GET /ISAPI/System/networkCapture/manualStatus?format=json.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

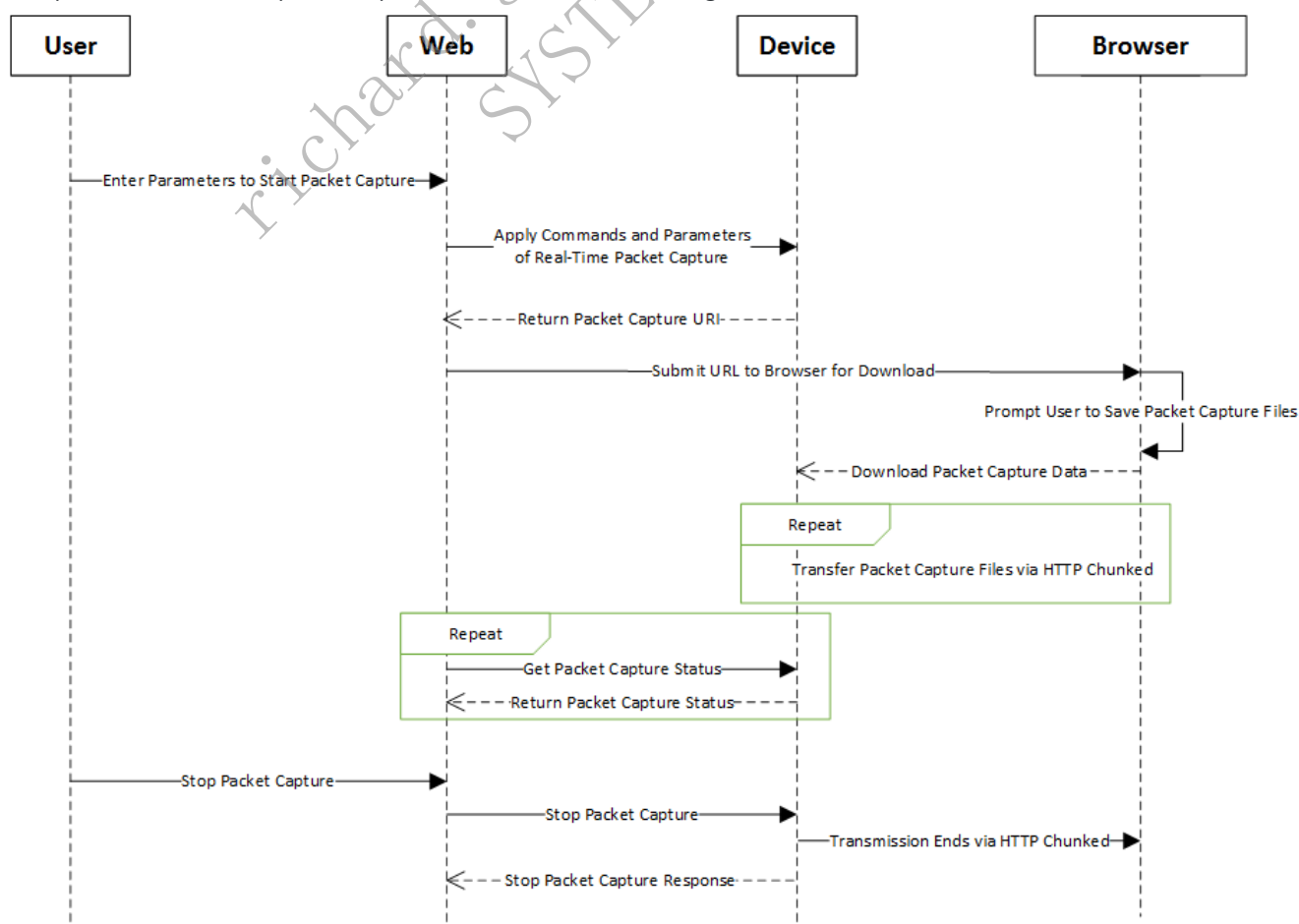
Stop device packet capture: PUT /ISAPI/System/networkCapture/manualStop?format=json.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

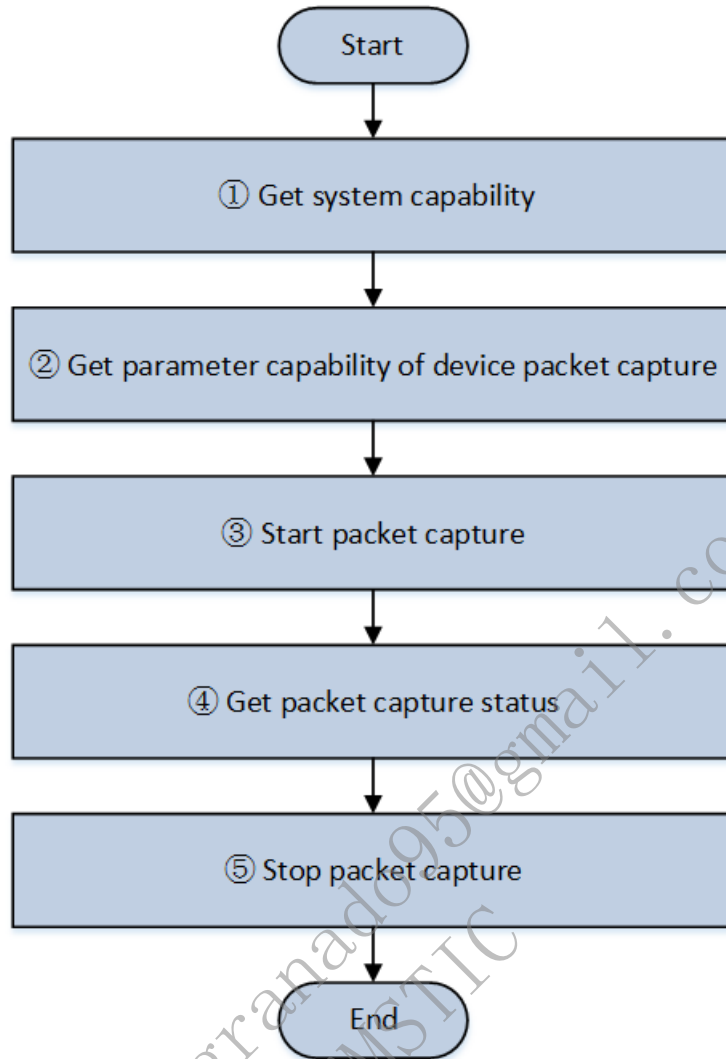
Export device packet capture files: GET /ISAPI/System/networkCapture/exportFile?format=json.

5.2.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:



See the following figure for the calling flow:



1. Get device system capabilities: `GET /ISAPI/System/capabilities`.

`<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture>` indicates the device supports starting packet capture. `<isSupportStopNetworkCapture>true</isSupportStopNetworkCapture>` indicates the device supports stopping packet capture. `<isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus>` indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: `GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json`. The `realTimeEnabled` field indicates whether the device supports real-time packet capture.

3. Set the field `realTimeEnabled` as `true` in the parameters applied to the device to start device packet capture. The device returns an URI, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: `POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>`.

Note:

- If `realTimeEnabled=true` is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: `GET /ISAPI/System/GetNetworkCaptureStatus?format=json`.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: `POST /ISAPI/System/StopNetworkCapture?format=json`.

5.3 Device Hardware Asset Management

5.3.1 Introduction to the Function

Hardware assets: the host assets (CPU, memory, and HDD) and peripheral assets (hardware connected to the host including camera, sensor, and USB flash drive).

Typical application: financial industry, where the head office needs to regularly count the security device assets of each branch to collect the number, operation status, and other basic information of deployed host/storage/peripheral cameras.

Extended application: management of industry-related service applications for assets customized by the industry platform (e.g., information and software assets, person assets, and service assets) based on the hardware asset integration.

5.3.2 API Calling Flow

1. Get the search capability of the device hardware asset data via `GET /ISAPI/System/deviceInfo/capabilities`: when `isSupportSearchHardwareAssets` is true, it indicates the device supports searching for asset data.
2. Search for device hardware asset data via `GET /ISAPI/System/deviceInfo/ExportDeviceAssets?format=json` to get the information of hardware assets on device including host assets, connected sub-device assets, HDD assets, etc.
3. Export the device hardware asset information in binary data in Excel format.

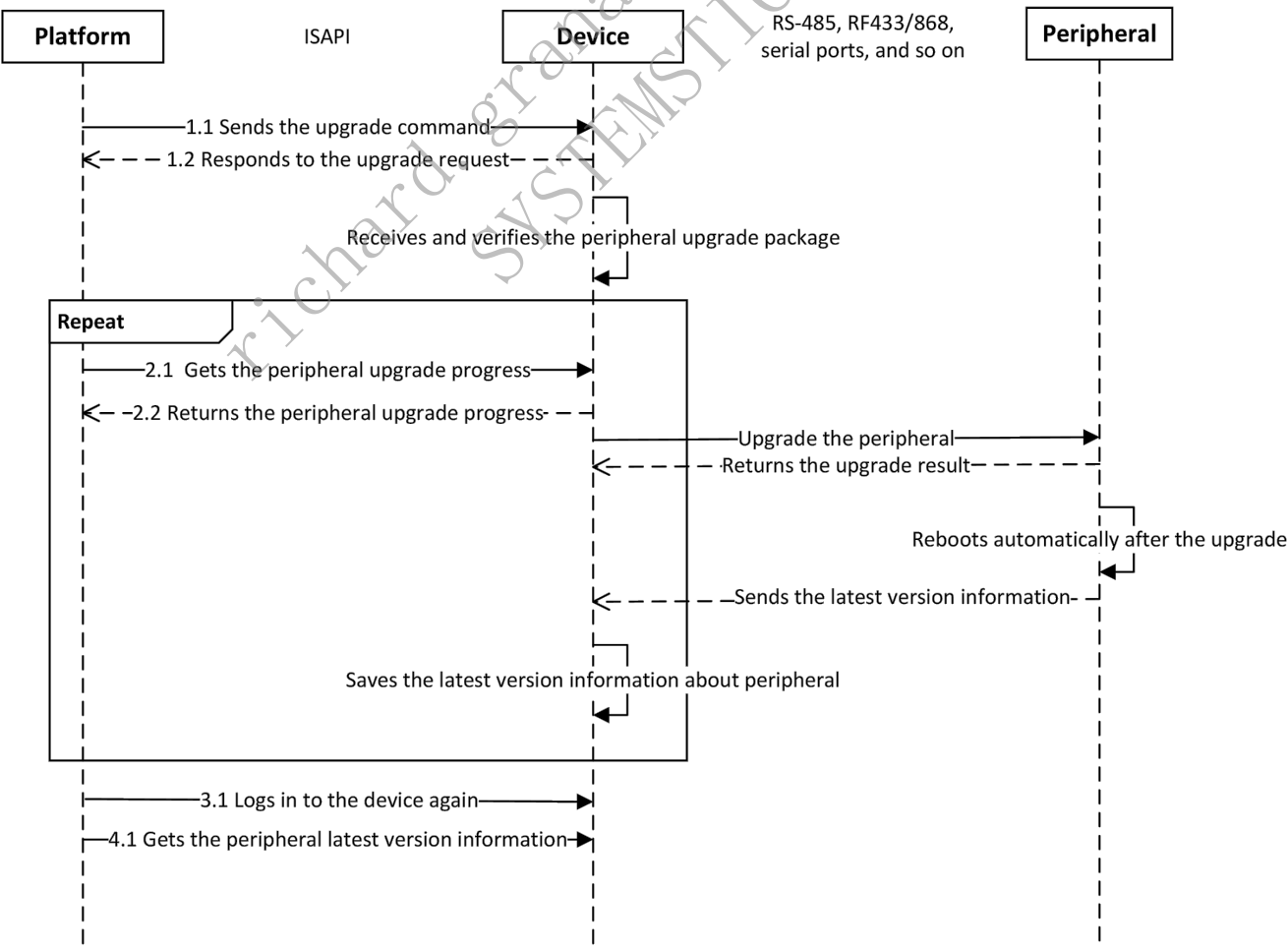
5.4 Device Peripherals Upgrade

5.4.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

5.4.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.



1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.

2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

5.5 Device Time Sync

5.5.1 Introduction to the Function

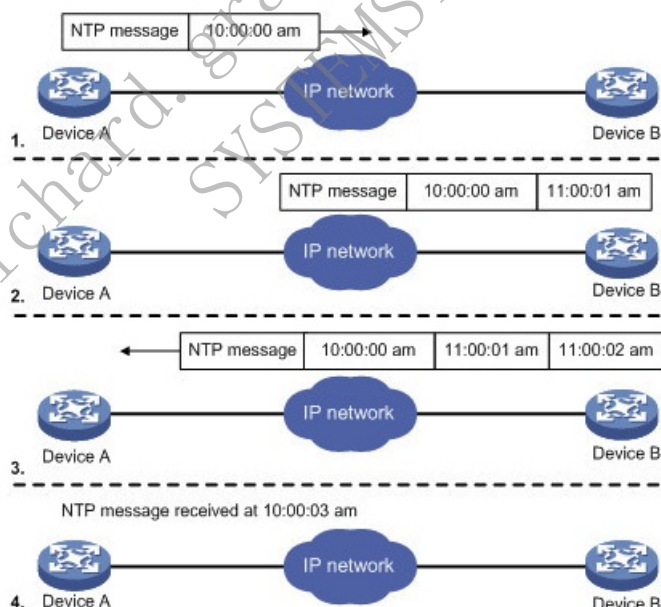
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

5.5.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.



The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B. Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: $\text{Delay} = (T4 - T1) - (T3 - T2) = 2 \text{ seconds}$.

Time difference between Device A and Device B: $\text{offset} = ((T2 - T1) + (T3 - T4)) / 2 = 1 \text{ h}$.

Device A can sync its own time with that of Device B according to calculation results.

5.5.2 API Calling Flow

5.5.2.1 Time Sync Configuration

1. Get the Capability of Device Time Synchronization Management

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: GET /ISAPI/System/time/capabilities.

2. Set device time synchronization management parameters

You can configure the time synchronization mode as follows:

Get device time synchronization management parameters: GET /ISAPI/System/time;

Set device time synchronization management parameters: PUT /ISAPI/System/time;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime`, `timeZone`.

Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

Note: For manual time synchronization (time offset including time zone offset): Set manual time synchronization: `localTime` refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); `timeZone` refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: `localTime` refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); `timeZone` refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

5.5.2.2 Time Zone Configuration

1. Get device time zone configuration capability

Call GET /ISAPI/System/capabilities to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

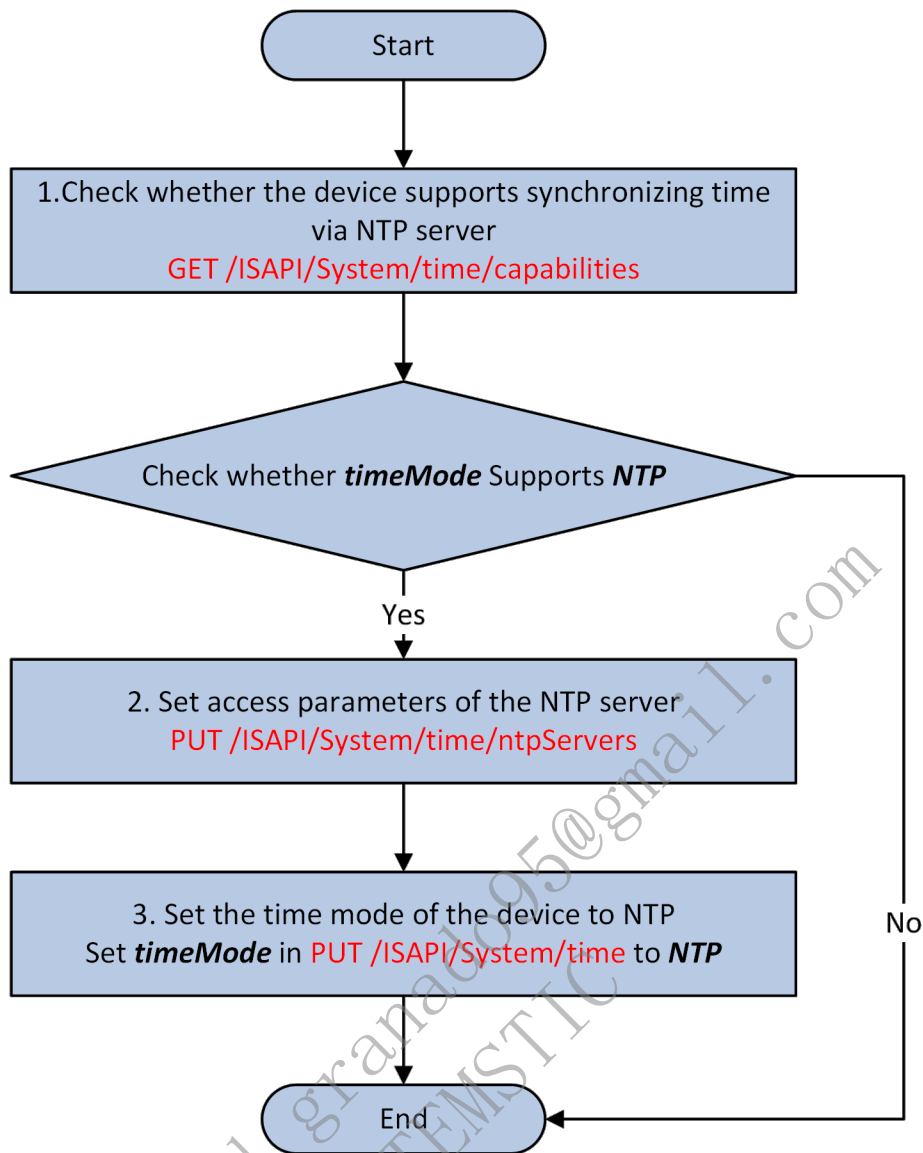
2. Configure time zone parameters

Get the device's time zone parameters: GET /ISAPI/System/time/timeZone. Set the device's time zone parameters: PUT /ISAPI/System/time/timeZone.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

5.5.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):



1. Check whether the device supports synchronizing time via NTP server Get the capability of the device: GET /ISAPI/System/time/capabilities; and check whether timeMode supports NTP.

2. Set access parameters of the NTP server

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: GET /ISAPI/System/time/ntpServers/capabilities

Set access parameters of the NTP server: PUT /ISAPI/System/time/ntpServers

Get access parameters of the NTP server: GET /ISAPI/System/time/ntpServers

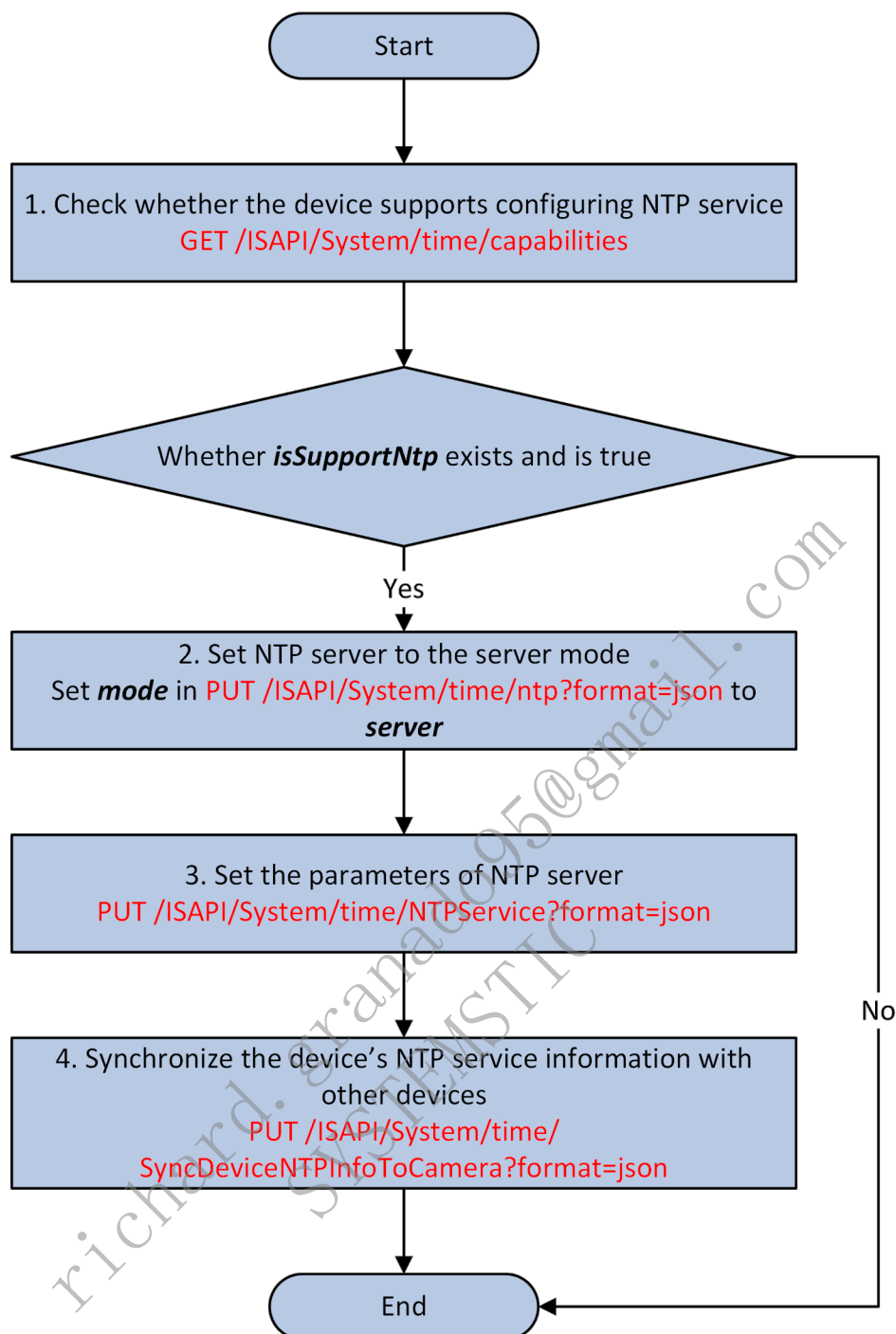
3. Set the time mode of the device to NTP

Supports setting the value of timeMode to NTP.

Get device time synchronization management parameters: GET /ISAPI/System/time Set device time synchronization management parameters: PUT /ISAPI/System/time

5.5.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):



1. Check whether the device supports configuring NTP service Get the capability of device time synchronization management: GET /ISAPI/System/time/capabilities; If isSupportNtp is returned, it indicates that the device supports time synchronization management.

2. Set NTP server to the server mode

Supports setting the value of mode to server.

Get the capability of server mode: GET /ISAPI/System/time/ntp/capabilities?format=json

Set NTP to server mode: PUT /ISAPI/System/time/ntp?format=json

Get parameters of NTP server mode: GET /ISAPI/System/time/ntp?format=json

3. Set the parameters of NTP server

Supports setting the IP address of the NTP server.

Get the capability of NTP server: GET /ISAPI/System/time/NTPService/capabilities?format=json

Set the NTP server parameters: PUT /ISAPI/System/time/NTPService?format=json

Get the parameters of the NTP server: GET /ISAPI/System/time/NTPService?format=json

4. Synchronize the device's NTP service information with other devices

Supports synchronizing the time information to the camera.

Get the capability set of synchronizing device's NTP service information with the camera: GET
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json

Synchronize device's NTP service information with the camera: PUT
/ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json

Get the progress of synchronizing device's NTP service information with the camera: GET
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json

Search for the results of synchronizing device's NTP service information with the camera: POST
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json

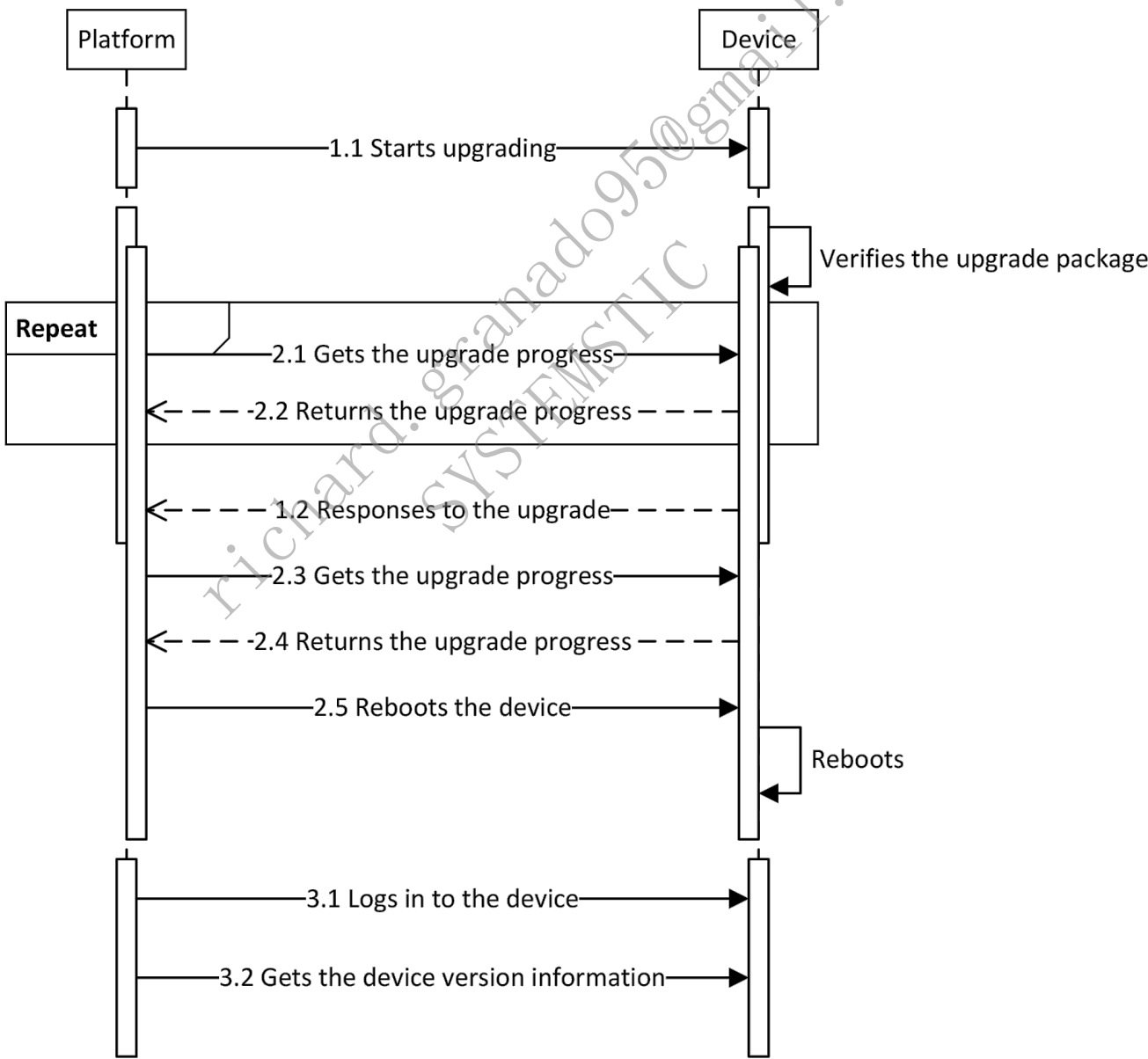
5.6 Device Upgrade

5.6.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

5.6.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.



1. Upgrade devices.

Upgrade the device firmware: POST /ISAPI/System/updateFirmware.

2. Get the device upgrade progress.

Get the device upgrade progress: GET /ISAPI/System/upgradeStatus.

3. Reboot devices.

Reboot devices: PUT /ISAPI/System/reboot.

5.7 Mutually Exclusive Functions

5.7.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e, only one of them is allowed at one time).

5.7.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: GET /ISAPI/System/mutexFunction/capabilities?format=json. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exclusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).
2. Search for the functions that are mutually exclusive with a specified function: POST /ISAPI/System/mutexFunction?format=json. Based on the list of mutually exclusive functions returned by GET /ISAPI/System/mutexFunction/capabilities?format=json, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disable the mutually exclusive function.
3. Get the mutual exclusion information when device function exception occurs: GET /ISAPI/System/mutexFunctionErrorMsg. After getting the error code, you can call this API to get the current mutually exclusive functions.

5.8 Query Device Operation Log

5.8.1 Introduction to the Function

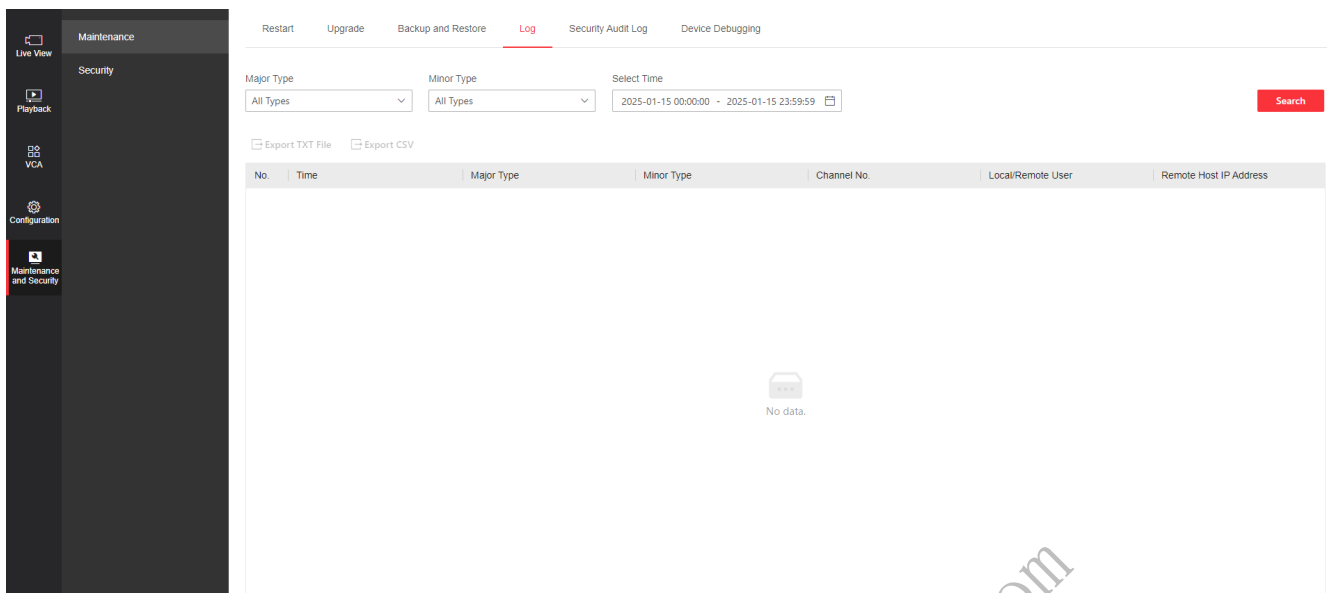
Device operation logs primarily include:

- Operation records for the device, such as startup, restart, PTZ control, etc.
- Events from the device, such as the start and end of motion detection, face capture, etc.
- Device status exceptions, such as network disconnection, network recovery, IP address conflict, etc.
- Device trigger information, such as starting recording, stopping recording, and periodically recording hard drive status.

Log query functions include log query and security audit log query, which overlap to some extent. The differences are as follows:

- Log query is mainly designed to query all operational status and operation records of the device.
- Security audit log query is mainly designed to query security-related status and operation records of the device, such as user login/logout records, device SSH service start/stop records, etc.

Operation logs can be queried and displayed item by item through the device web page, remote client, or platform. The device web page log query is shown in the following figure.



5.8.2 API Calling Flow

1. Call `POST /ISAPI/ContentMgmt/logSearch` to query logs except security audit logs.

Note 1:

- When you query logs of a specified type, first set the `metaId` to `log.std-cgi.com/Information`. If the total number returned is not 0, it indicates that the log query is standardized, and subsequent queries should follow the standard. If the total number returned is 0, use `log.std-cgi.com/Infomation` to query again. If there is a return value, subsequent queries should use `log.std-cgi.com/Infomation`.
- When you set the `metaId` to `all`, the response results should be parsed for both `log.std-cgi.com/Infomation` and `log.std-cgi.com/Information`.
- When you query logs of a specified type, the recommended format for the `searchID` field is the GUID (8-4-4-4-12) format, such as `812F04E0-4089-11A3-9A0C-0305E82C2906`.

2. Call `POST /ISAPI/ContentMgmt/security/logSearch` to query security audit logs.

**Note **:

- Standard definition: Query log `POST /ISAPI/ContentMgmt/logSearch: log.std-cgi.com/Information` indicates querying information logs.

Device Type	Implementation Differences
Encoding Devices (General Cameras/Storage Devices/Traffic Cameras/Thermal Cameras/Security Inspection)	Input and output log information are <code>log.std-cgi.com/Infomation</code>
Non-Encoding Devices (Access Control Devices/Transmission Devices/Display & Control Devices)	Input and output log information are <code>log.std-cgi.com/Information</code>

5.8.Export Device Operation Log

5.8.1 Introduction to the Function

Operation logs can be exported to a client over the network or to an external storage medium (USB drive, external hard drive, etc.) via a USB interface.

5.8.2 API Calling Flow

5.8.2.1 Export to Client

1. Call `POST /ISAPI/ContentMgmt/logSearch/dataPackage` to get the URL to download the exported device log.

2. The client downloads the log files via the URL returned by the device.

5.8.2.2 Export to USB Storage Medium

1. Call `GET /ISAPI/System/exportLogToUSB/capabilities?format=json` to determine if the device supports log output to a USB drive.
2. Call `PUT /ISAPI/System/exportLogToUSB/mode?format=json` to set the parameters of exporting logs to a USB drive.
3. Call `GET /ISAPI/System/exportLogToUSB/status?format=json` to get the status of log output to a USB drive.

5.8.Device Operation Log

5.8.1 Introduction to the Function

Device operation logs are used in the following two scenarios:

Scenario 1: View the detailed operation log files to locate the issue when a device fails.

Scenario 2: Before a device fails, basic operation status information is recorded in the logs. Maintenance personnel can proactively query the device operation status to promptly identify potential issues and faults in the device.

- The operation status logs generally include host information (network bandwidth, online users, video output port/USB port status), linked device information (online status, video recording schedule/status), hard disk information (capacity, status, runtime, temperature), etc.
- For example, to ensure the normal operation of bank equipment and facilities, and to promptly identify hidden dangers and faults in equipment operations, maintenance personnel of bank outlets are required to check the equipment operation status at regular intervals each day and manually record data for subsequent data traceability. This management approach consumes manpower and may result in data omissions. Therefore, you can configure the equipment to automatically record operational status by schedule for querying and exporting history data, thus saving labor and ensuring data integrity.

5.8.2 API Calling Flow

5.8.2.1 Upload to Specified FTP/HTTP(S) Server

1. Call `PUT /ISAPI/System/operationData/server?format=json` to allow devices to upload operation log files to a specified FTP/HTTP(S) Server.
2. Check the device logs on the FTP/HTTP(S) server.

Note:

- See the "Device Operation Status Diagnosis" module for FTP/HTTP(S) Server API details.
- We recommend that use this method to remotely obtain device operation logs for device operation and maintenance platforms.

5.8.2.2 Upload to Syslog Server

Syslog is a standard log transmission protocol widely used in system logs. It is defined in [RFC 5424](#) (The Syslog Protocol). After you configure the Syslog service address and enable the Syslog function on the device, the device will send its operation logs to the Syslog server using the Syslog protocol. The Syslog server can manage the operation logs of multiple devices.

1. Get the Syslog management capabilities to determine if the device supports the Syslog function and the parameter value range.
2. Call `PUT /ISAPI/System/logServer` to configure Syslog server settings such as the IP address, port No., and certificate.
3. When the device upload logs to the Syslog server using the Syslog protocol, the logs can be viewed on the Syslog server.

5.8.2.3 Query Device Operation Status Logs

1. Call `/ISAPI/ContentMgmt/RuningLogPlan?format=json` to configure the device operation log recording schedule.

Supports configuring up to 8 time points per day. The device will automatically store and back up the operation status data at the corresponding time points for subsequent historical log viewing or export. The operation status data includes host information (network bandwidth, online users, video output port/USB port status), external device information (online status, video device recording plan/status), disk information (capacity, status, runtime, temperature).

2. Query historical device operation logs: `/ISAPI/ContentMgmt/SearchRuningLogData?format=json`.
3. Export historical device operation logs: `/ISAPI/ContentMgmt/ExportRuningLogData?format=json`. Export based on the query conditions set in step 2.

5.9 Serial Port Accessed External Device Management

5.9.1 Introduction to the Function

Information management of device accessed the serial are as follows: 1. Configure manufacturer, type, and model information of the specific serial port access device. 2. Search for the device type or model supported by the specific serial port.

5.9.2 API Calling Flow

1. Check whether the device supports information management of serial port. Get the capability of the device serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportDeviceInfo>` is returned, the device supports information configuration of devices access the serial port.

2. Set the information of serial port:

- Get the capability of device information parameters of a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Get device information parameters access single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Set device information parameter of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.

3. Check whether the device supports linking information of devices access the serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportSearchDeviceInfoRelations>` is returned, it indicates that the device supports searching for linked information of devices access the serial port.

4. Search for linked information of devices access the serial port.

- Get the capability of searching for linked parameters of information of devices access a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations/capabilities?format=json`.
- Search for linked parameters of information of devices access a single serial port: `POST /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations?format=json`.

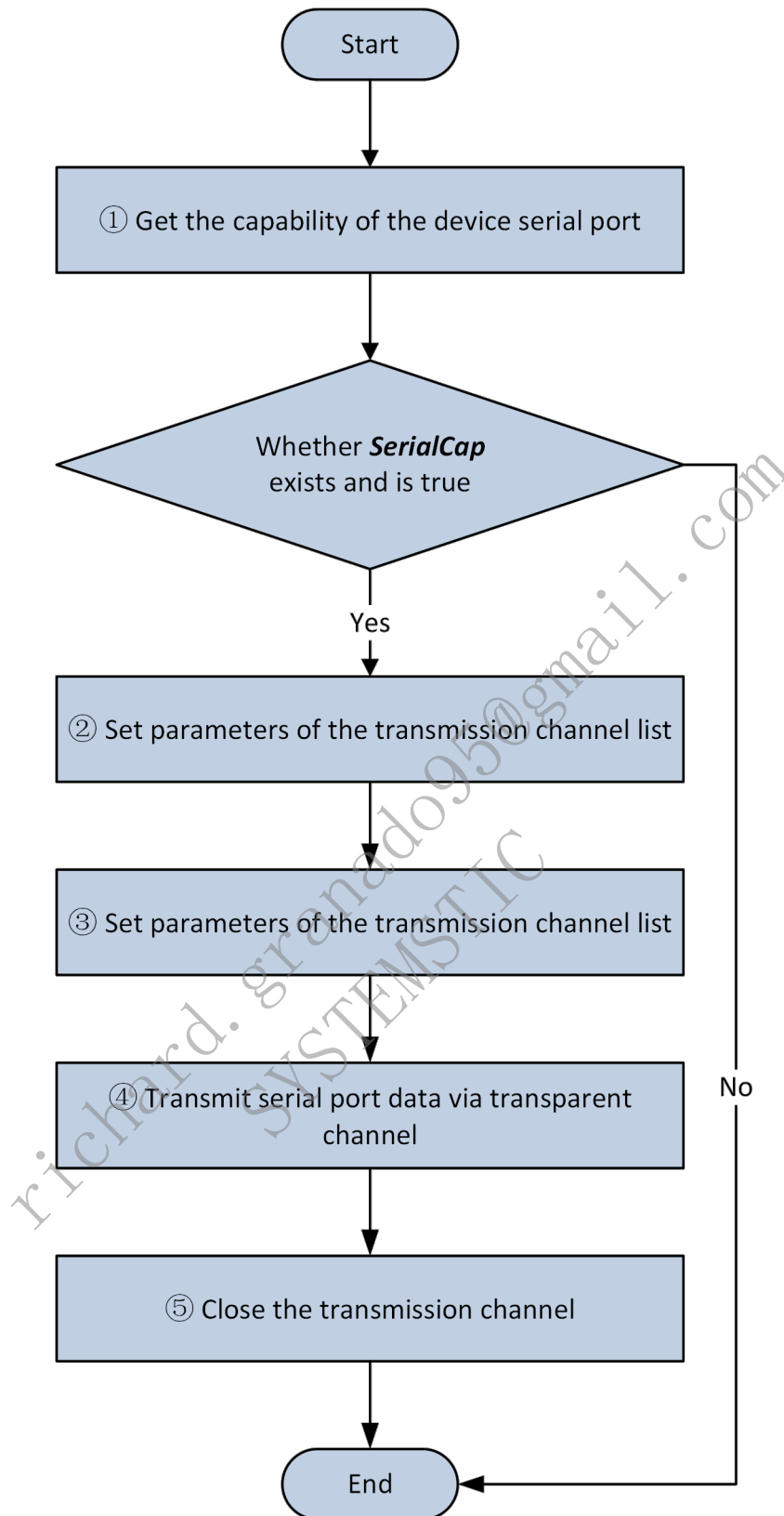
5.10 Serial Port Data Transparent Transmission

5.10.1 Introduction to the Function

RS485, RS422 and RS232 serial ports external to the device are used as transparent channels to transmit serial port data. Supports the client sending serial data to the device, which then forwards the data to the serial port; conversely, when the serial port sends data to the device, which transparently transmits the data to the client. Note that the device's transparent transmission of serial data is half-duplex communication, meaning that bidirectional communication can be implemented between the device and the client, but the communication data is not necessarily in a question-and-answer format, and the client must match the request and response relationship on its own.

Since most standard HTTP request libraries do not support the persistent connection for receiving and sending data in real time, it is recommended that the client uses two TCP clients to implement the sending and receiving of transparently transmitted serial data.

5.10.2 API Calling Flow



1. Check whether the device supports serial port data transmission.

Get the capability of the device serial port: `GET /ISAPI/System/capabilities`. If `SerialCap` is returned and the value is true, it indicates that the device supports the functions of the serial port.

2. Set parameters of the transmission channel list.

- Get parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.
- Configure parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.

3. Open the transmission channel: PUT

/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open;

4. Transmit serial port data via transparent channel.

- Receive data uploaded by device serial port through transmission channel: GET
/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData.
- Send data to device serial port through transmission channel: PUT
/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData.

5. Close the transmission channel: PUT

/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close.

5.11 Serial Port Parameter Configuration

5.11.1 Introduction to the Function

Serial port parameter configuration.

5.11.2 API Calling Flow

1. Check whether the device supports configuring serial port parameters.

Get the capability of device serial port: GET /ISAPI/System/capabilities. If is returned and its value is true, the device supports functions of serial port.

3. Get parameters of all serial ports.

- Get the capability of all serial ports: GET /ISAPI/System/Serial/capabilities.
- Get control parameters of all serial ports: GET /ISAPI/System/Serial/ports?permissionController=<indexID>.

4. Set control parameters of a single serial port.

- Get control parameters of single serial port: GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>.
- Configure control parameters of single serial port: PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>.

5. Get the status of single serial port: GET /ISAPI/System/Serial/ports/<portID>/status.

5.12 Sub-device Batch Upgrade

5.12.1 Introduction to the Function

Sub-device Batch Upgrade (Single Task): applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

Management of Sub-device Batch Upgrade Tasks: There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

5.12.2 API Calling Flow

Sub-device Batch Upgrade (Single Task)

Get the device capability: /ISAPI/System/capabilities.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgrading sub-devices.

Sub-device batch upgrade: POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json.

Batch sub-device upgrade progress search: POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json.

Sub-device Batch Upgrade Task Management

Get device system capabilities /ISAPI/System/capabilities, If the node isSupportBulkUpgradeChildDevice exists and is true, it indicates that batch upgrade of sub devices is supported,

If the node isSupportSearchBulkUpgradeChildDeviceProgress exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node isSupportSearchBulkUpgradeChildDeviceTask exists and is true, it indicates that batch upgrading devices can be searched,

If the node isSupportModifyBulkUpgradeChildDeviceTask exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node isSupportDeleteBulkUpgradeChildDeviceTask exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json.

Batch upgrade sub devices: POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json.****

Search for tasks of batch sub-device upgrade: POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json.

Edit tasks of batch sub-device upgrade: PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json.****

Delete tasks of batch sub-device upgrade: PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json.

Note: 1. The two methods for batch upgrading child devices are the same API: POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json. If the BulkUpgradeChildDeviceListCap.taskID is returned in GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json. Sub devices with the tag isUpgradable and value true indicate that the sub devices support upgrades.

5.13 User Management

5.13.1 Introduction to the Function

When the device is activated, you can log in to it via the admin account and corresponding password, and manage users as needed, including:

1. Change the password of the admin account. The user name cannot be edited.
2. Add, edit, and delete other users, including the user type, password, user name, and so on. A Non-admin user can log in to and operate the device after being created.

Remarks

1. Common user types:
 - o Administrator (admin): has the permission of accessing all resources supported by the device, and can operate all functions of the device. The admin account cannot be deleted.
 - o Operator (operator): has the permission to view. Their operation permissions are assigned by admin. Operator accounts are created by the administrator only.
 - o User (viewer): has the permission to view only. They have no operation permission. User accounts are created by the administrator only.
2. User types related to the installer:
 - o Local Administrator (localAdmin) : Activated by the device owner on the local software side (e.g., WEB) that accompanies the device. By default, a localAdmin has the permission to access the device and perform all functions supported by the device.
 - o Local Installer (localInstaller) : Activated by localAdmin. The identity can be given to the installer for device installation and debugging on site. By default, a localInstaller has the permission to access and perform all

functions supported by the device.

- Local Operator (`localOperator`) : Created by `localAdmin` or `cloudAdmin`. A password is required to be set by `localAdmin`, `cloudAdmin`, `localInstaller`, `installerAdmin`, or `installerEmployee` before the local operator can log in via the local keypad. A local operator's business functions are limited, and cannot configure device parameters. By default, only arming, disarming (including alarm clearing), and relay control operations are allowed. According to the actual business needs, there can be two types of local operator: one-time local operator (can only log in once and the identity will expire after logging out), and temporary local operator (can log in within the validity period), both created by `localAdmin` or `cloudAdmin`.
- Cloud Administrator (`cloudAdmin`) : Created by the device owner in the mobile software that accompanies the device (e.g. HC over the cloud) and then synchronized to the device. By default, a `CloudAdmin` has the permission to access and perform all functions supported by the device.
- Cloud Operator (`cloudOperator`) : Created by `cloudAdmin` in the mobile software that accompanies the device (e.g. HC over the cloud) and then synchronized to the device for ordinary mobile users. A cloud operator's business functions are limited, and cannot configure device parameters, access and view the device, arm/disarm (including alarm clearing), or operate relays.
- Installer Administrator (`installerAdmin`) : User on the installer business application, created on the business platform (e.g. HPC deployed in the cloud) and then synchronized to the device for remote device management by the business platform's admin. By default, an `installerAdmin` has the permission to access the device and perform all functions supported by the device. To facilitate installation management, when the business platform sets `installerAdmin` as the highest authority and synchronizes to the device, `installerAdmin` is allowed to modify user permissions of `localAdmin` and `cloudAdmin`.
- Installer Employee (`installerEmployee`) : User on the installer business application, created on the business platform (e.g. HPC deployed in the cloud) and then synchronized to the device for the employees on the business platform to remotely install and debug the device. By default, an `installerEmployee` has the permission to access the device and perform all functions supported by the device.

3. User password:

- To ensure the security of account information, it is recommended to create a password using eight to sixteen characters, including at least two kinds of the following categories: digits, lower case letters, upper case letters, and special characters, and the user name is not allowed in the password.
- Risky passwords include the following categories: less than 8 characters, containing only one type of characters, same as the user name or reversed user name. To protect user data privacy and improve security, it is recommended to use a strong password.
- The password strength rule is as follows:
 - Strong password: including at least three kinds of the categories (digits, lower case letters, upper case letters, and special characters).
 - Medium password: a combination of digits and special characters, lower case letters and special characters, upper case letters and special characters, or lower case letters and upper case letters.
 - Weak password: a combination of digits and lower case letters or digits and upper case letters.

5.13.2 API Calling Flow

- Get the user management capability of devices on the client: `GET /ISAPI/Security/users/<indexID>/capabilities`.
- Add device users on the client: `POST /ISAPI/Security/users?security=<security>&iv=<iv>`.

Remarks:

- Only `admin` can create other types of users, and creating users requires login password verification (`<loginPassword>`) of `admin`.
- If the user account is inactivated, it's required to log in to the account and change the user password (`PUT /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>`). The account is activated when the password is changed.
- When the account is inactivated, it's not allowed to perform any operations except changing the user

password. Otherwise, an error (0x0020000f) will be returned.

3. Edit the user information on the client: PUT /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>.

Remarks:

- It requires password verification of admin when admin changes the user password. The account turns inactivated when the user password is changed by admin. Once logging out, the user needs to change the password first before the next login.
- When non-admin users changed their passwords, the account remains activated.

4. Delete users on the client: DELETE /ISAPI/Security/users?loginPassword=<loginPassword>&security=<security>&iv=<iv>.

Remarks:

Only admin can delete users, and deleting users requires login password verification of admin.

5. Get the user information, including user name, activation status (<userActivationStatus>), and so on.

Get a single user information: GET /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>.

Get the information of all users: GET /ISAPI/Security/users?security=<security>&iv=<iv>.

Get the information of online users: GET /ISAPI/Security/onlineUser. Online users refer to users who have logged in to the device. The information such as user name, user type, and IP address can be obtained.

Remarks:

- If multiple attempts of admin login password verification failed in the process of adding, editing, or deleting users, the admin will be locked. The remaining attempts are defined by the field retryTimes in the response message.
- The new password cannot be the same as the last password. Otherwise, an error (0x400010E8) will be returned.

5.13.3 Exception Handling

Error Code

statusCode	statusString	subStatusCode	errorCode	errorMsg	Descript
4	Invalid Operation	theAccountIsNotActivated	0x0020000f	The account is not activated.	
4	Invalid Operation	loginPasswordError	0x4000000C	Incorrect login password.	
4	Invalid Operation	theAnswerToTheUserSecurityQuestionIsDuplicate	0x4000A0B6	The answer to the user security question is duplicate.	Please se different answer.
4	Invalid Operation	theAnswerToTheUserSecurityQuestionIsTooShort	0x4000A0B7	The answer to the user security question is too short.	Please se longer answer.
4	Invalid Operation	cannotSameAsOldPassword	0x400010E8	New password cannot be the same as the old one.	Please se different password
6	Invalid Content	administratorPasswordError	0x60000042	Incorrect administrator password.	Please en the corre password you forgot the password you can reset the password

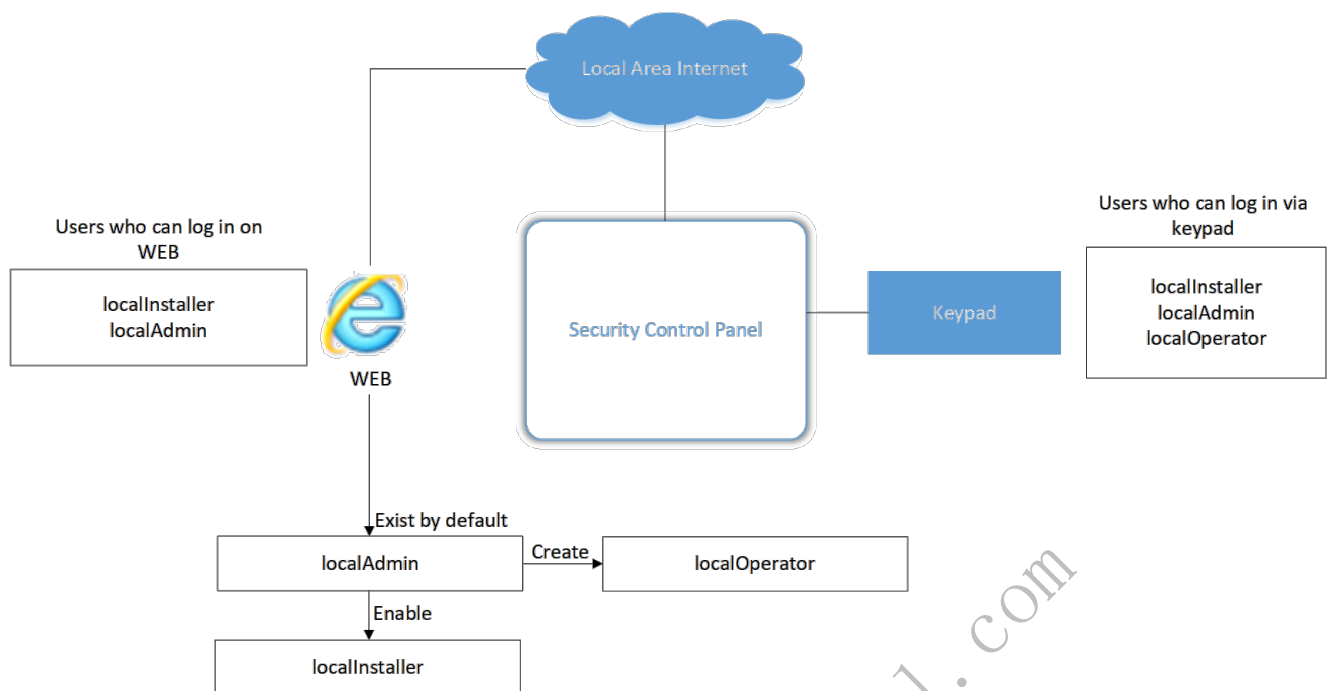
5.14 User Types Related to the Installer (supported by the security control panel)

For different application scenarios, e.g., local environment, cloud environment, and HPP installer environment, users related to the installer can be classified into seven types: `localAdmin`, `localInstaller`, `localOperator`, `cloudAdmin`, `installerAdmin`, `installEmployee`, and `cloudOperator`.

5.14.1 Create the User

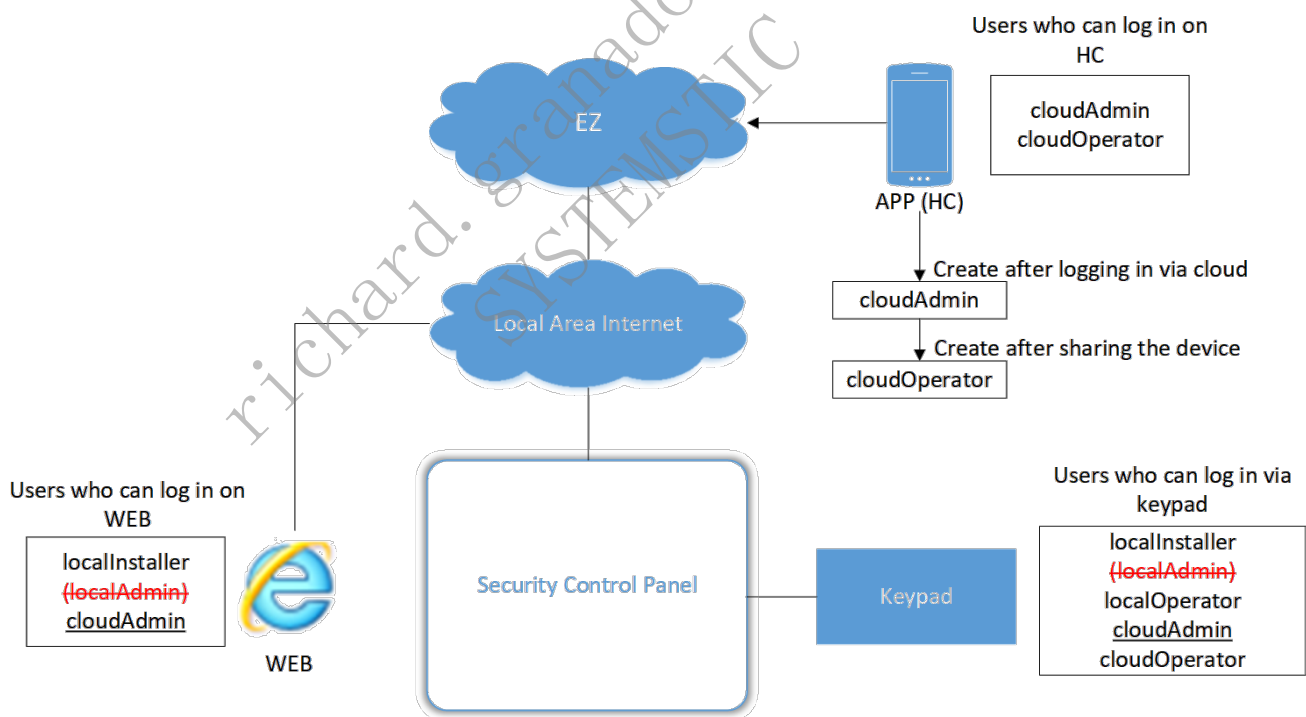
5.14.1.1 Local Environment

In a local environment (LAN), three types of users are involved: `localAdmin`, `localInstaller`, and `localOperator`. By default, two types of users exist on the device: `localAdmin` and `localInstaller`, but `localInstaller` is not enabled, which means that although a user ID is created, but the user cannot log in as `localInstaller` before the user type is enabled by `localAdmin`. `localOperator` is created by `localAdmin`, and can only log in via local keypad. Users who have been set with the keypad password can log in via keypad.



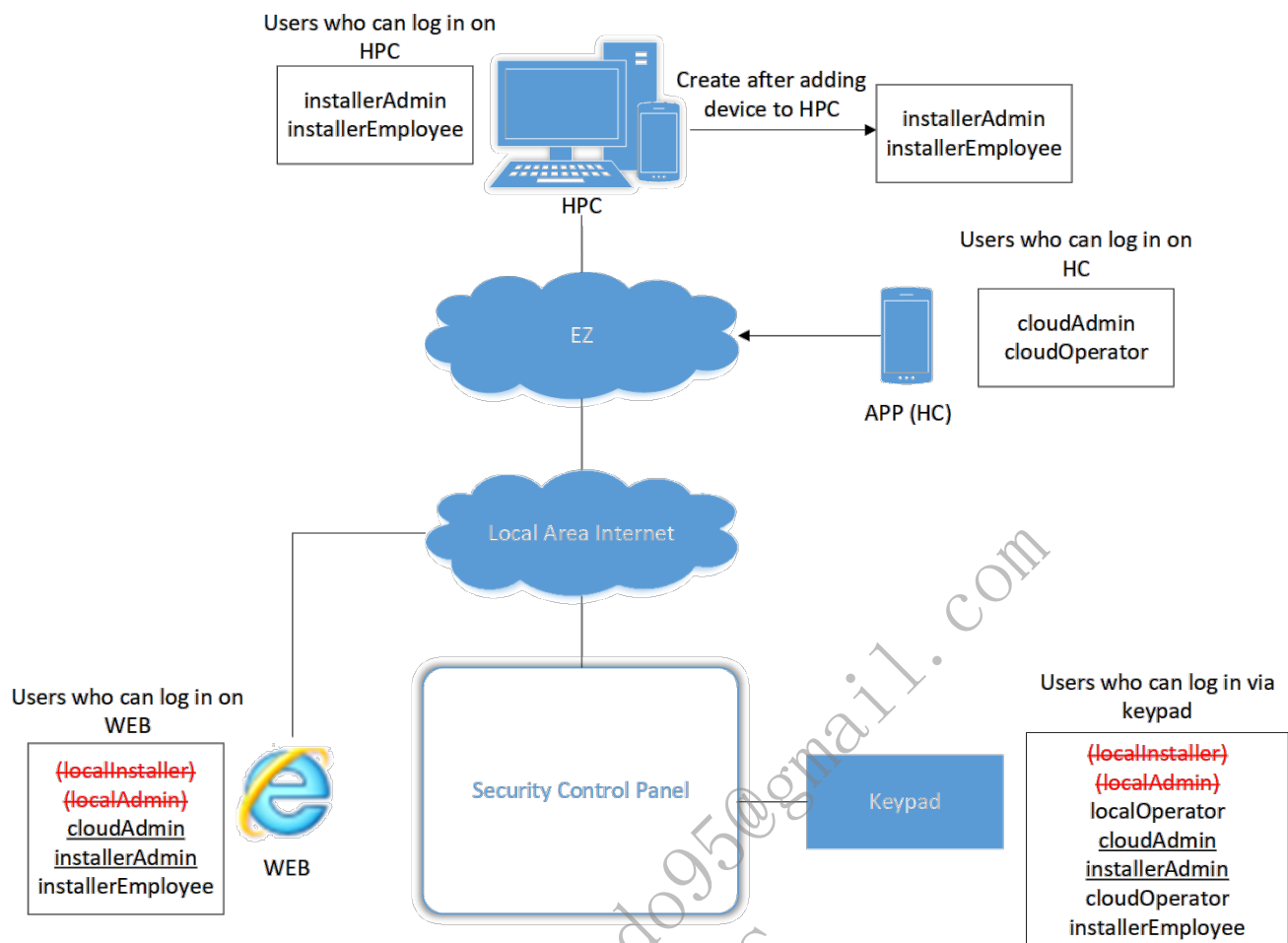
5.14.1.2 Cloud Environment

In a cloud environment, four types of users are involved: `cloudAdmin`, `cloudOperator`, `localInstaller`, and `localOperator`. After logging in to the device on HC application via cloud, a `cloudAdmin` can be created. Then, the `cloudAdmin` can share the device to create a `cloudOperator`. Note that after a `cloudAdmin` is created, the existing `localAdmin` will expire. Users who have been set with the keypad password can log via keypad.



5.14.1.3 HPP Installer Environment

In an HPP installer environment, five types of users are involved: `cloudAdmin`, `cloudOperator`, `localOperator`, `installerAdmin`, and `installerEmployee`. After a device is added to HPC, the user adding protocol will be applied, and `installerAdmin` and `installerEmployee` can be created (batch creating is supported). Note that after an `installerAdmin` is created, the existing `localInstaller` will expire. Users who have been set with the keypad password can log via keypad.



5.14.2 User Permissions

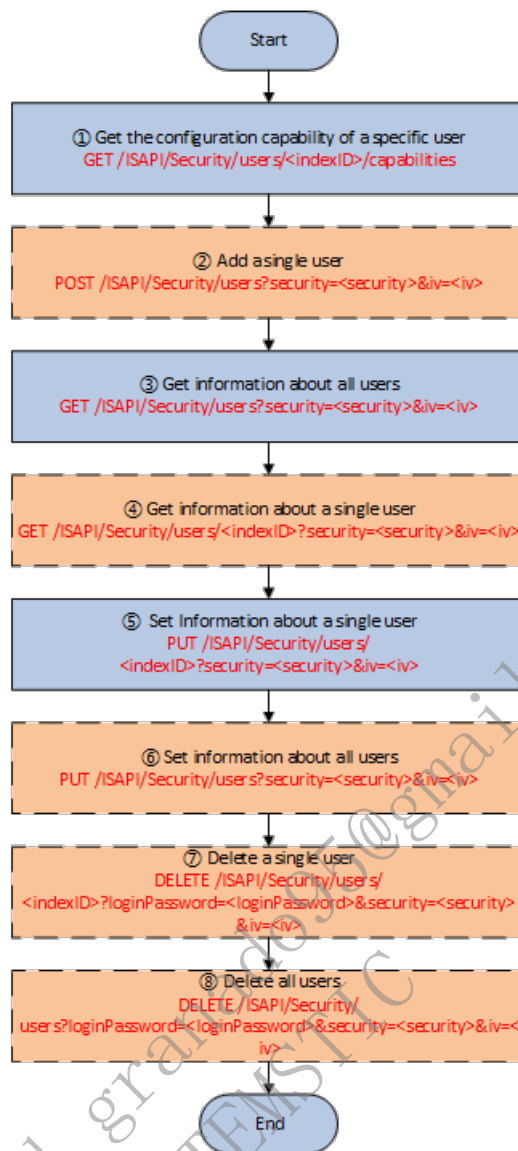
User Type / Permission	Local Admin	Local Installer	Local Operator	Cloud Admin	Cloud Operator	Installer Admin	Installer Employee
Arming	√	√	√	√	√	√	√
Disarming (Alarm Clearing)	√	√	√	√	√	√	√
Bypass	√	√	×	√	√	√	√
View logs and status	√	√	×	√	√	√	√
Configure parameters	√	√	×	√	×	√	√
Manage partitions	√	√	×	√	×	√	√
Operate relays	√	√	one-time local operator: × temporary local operator: √	√	√	√	√
Edit localAdmin's keypad password	√	√	×	×	×	√	√
Edit cloudAdmin's keypad password	×	√	×	√	×	√	√
Edit localInstaller's keypad password	×	√	×	×	×	×	×
Edit							

installerAdmin's keypad password	x	x	x	x	x	√	x
Edit installerEmployee's keypad password	x	x	x	x	x	√	√ (only the employee's own password)
Edit localOperator's keypad password	√	√	x	√	x	√	√
Edit cloudOperator's keypad password	x	√	x	√	√ (only the operator's own password)	√	√
Edit localAdmin's permission	x	x	x	x	x	x	x
Edit cloudAdmin's permission	x	x	x	x	x	x	x
Edit localInstaller's permission	√	x	x	√	x	x	x
Edit installerAdmin's permission	√	x	x	√	x	x	x
Edit installerEmployee's permission	√	x	x	√	x	x	x
Edit localOperator's permission	√	√	x	√	x	√	√
Edit cloudOperator's permission	x	√	x	√	x	√	√

Note: "Configure parameters" includes parameters of zones, sounders, keypads, card readers, keyfobs, cards, relays, repeaters, transmitters, network cameras, partitions, and so on.

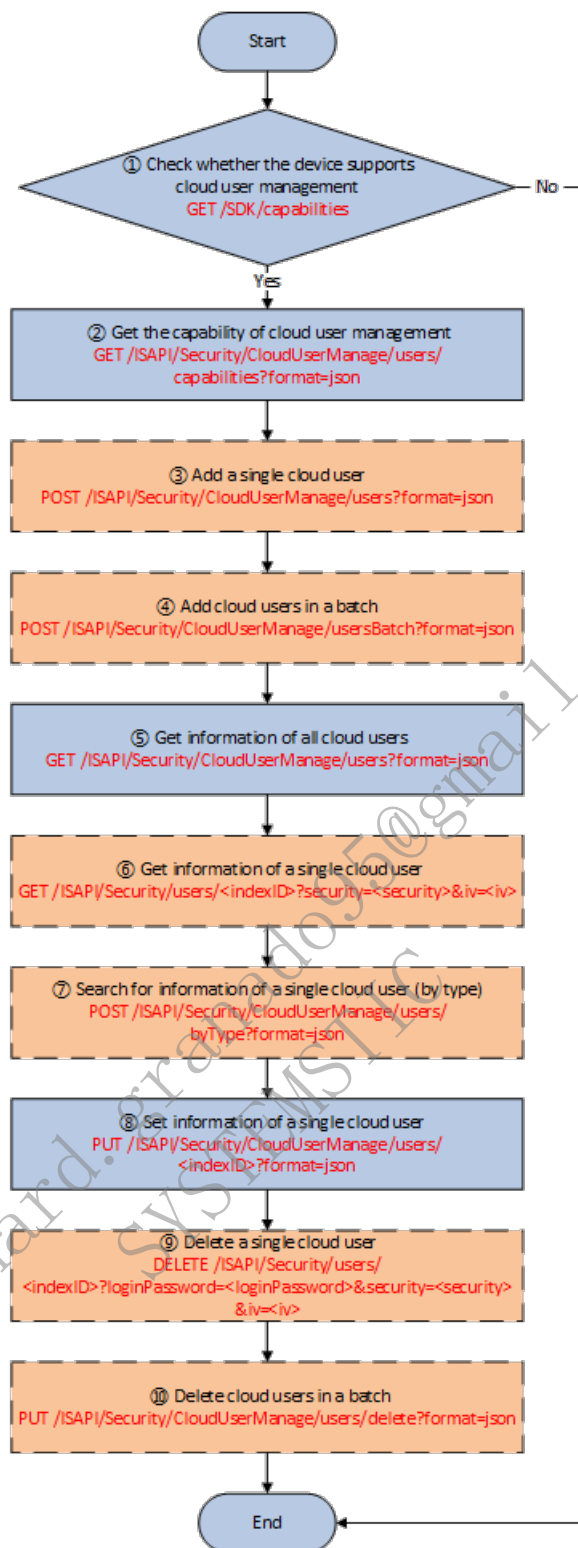
5.14.3 Manage User Information

5.14.3.1 Local User



1. Get the configuration capability of a specific user: GET /ISAPI/Security/users//capabilities
2. (Optional) Add a user: POST /ISAPI/Security/users?security=&iv=. The nodes userName, password, keypadPassword, and loginPassword in the message will be encrypted.
3. Get information of all users: GET /ISAPI/Security/users?security=&iv=. The nodes phoneNum, emailAddress, password, duressPassword, keypadPassword, and loginPassword in the message will be encrypted.
4. Get information of a single user: GET /ISAPI/Security/users/?security=&iv=. The index in the URL is the user ID.
5. Set permissions for a single user: PUT /ISAPI/Security/users/?security=&iv=. The index in the URL is the user ID.
6. (Optional) Set information of all users: PUT /ISAPI/Security/users?security=&iv=. This API can be called for batch configuring user information.
7. (Optional) Delete a single user: DELETE /ISAPI/Security/users/?loginPassword=&security=&iv=. Delete the user by indexID, and loginPassword is required for deleting the user. The loginPassword should be encrypted.
8. (Optional) Delete all users: DELETE /ISAPI/Security/users?loginPassword=&security=&iv=. loginPassword should be encrypted.

5.14.3.2 Cloud User



1. Check whether the device supports cloud user management: GET /SDK/capabilities. If the node value of isSupportCloudUserManage is true, cloud user management is supported in these URLs:

/ISAPI/Security/CloudUserManage/users/capabilities?format=json

/ISAPI/Security/CloudUserManage/users/?format=json

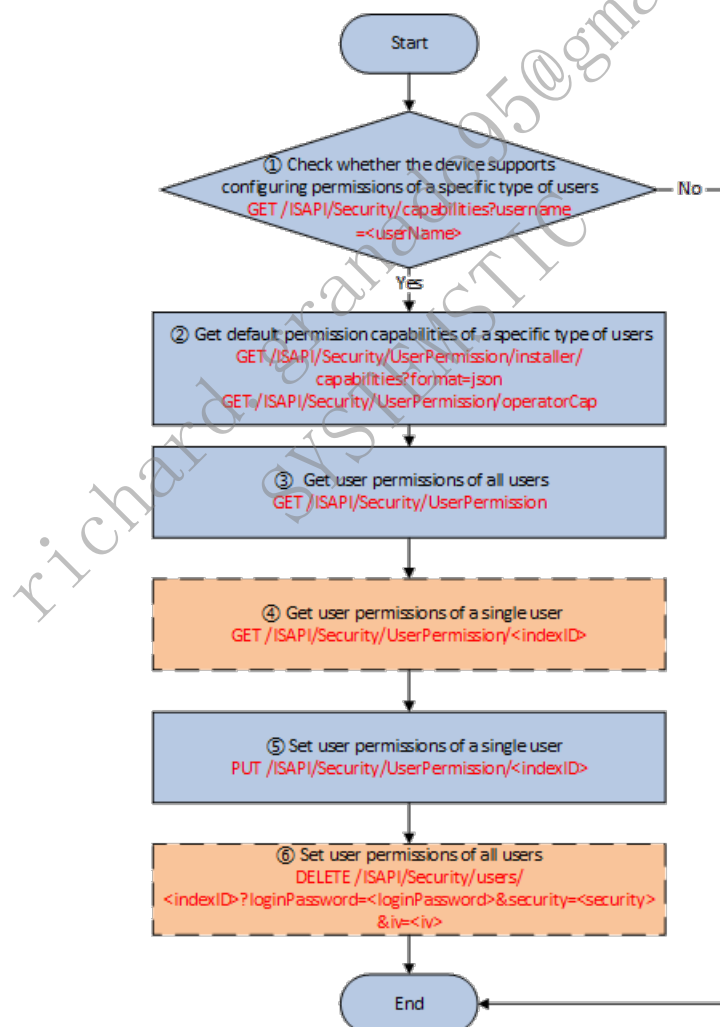
/ISAPI/Security/CloudUserManage/users?format=json

2. Get the capability of cloud user management: GET /ISAPI/Security/CloudUserManage/users/capabilities?format=json. If the node value of isSupportAddCloudUserList is true, it supports batch adding cloud users.
3. Add a single cloud user: POST /ISAPI/Security/CloudUserManage/users?format=json. The following cloud users need to create the user accounts on their own: coludAdmin, installerAdmin, installerEmployee, and cloudOperator, which is different from localOperator who needs to be created by admin users. Enter information such as user

name, password, e-mail, and phone number to create a user account and put into use.

4. (Optional) Add cloud users in a batch: POST /ISAPI/Security/CloudUserManage/usersBatch?format=json. For installerEmployee users that has been created on HPC, use this URL to apply them to the device and synchronize user information.
5. Get information of all cloud users: GET /ISAPI/Security/CloudUserManage/users?format=json
6. Get information of a single cloud user: GET /ISAPI/Security/CloudUserManage/users/?format=json
7. Search for information of a single cloud user (by type): /ISAPI/Security/CloudUserManage/users/byType?format=json. Support searching by e-mail, phone number, and user name.
8. Set information of a single cloud user: PUT /ISAPI/Security/CloudUserManage/users/?format=json
9. Delete a single cloud user: DELETE /ISAPI/Security/CloudUserManage/users/?format=json. Delete the user by specifying the user's indexID.
10. Delete cloud users in a batch: PUT /ISAPI/Security/CloudUserManage/users/delete?format=json. Support batch deleting by user names, user types, phone numbers, and e-mails.

5.14.4 Manage User Permissions



1. Check whether the device supports configuring permissions of a specific type of users: GET /ISAPI/Security/capabilities?username=. If the node value of isSupportInstallerCap is true, the installer users' permissions can be configured. There is no specific capability node for operator type users, and by default their permissions can be configured.
2. Get the default permission capabilities of a specific type of users.
 - For installer users: GET /ISAPI/Security/UserPermission/installer/capabilities?format=json

- For operator users: GET /ISAPI/Security/UserPermission/operatorCap
- 3. Get user permissions of all users: GET /ISAPI/Security/UserPermission
- 4. (Optional) Get user permissions of a single user: GET /ISAPI/Security/UserPermission/. The index in the URL is the user ID.
- 5. Set user permissions of a single user: PUT /ISAPI/Security/UserPermission/
- 6. (Optional) Set user permissions of all users: PUT /ISAPI/Security/UserPermission

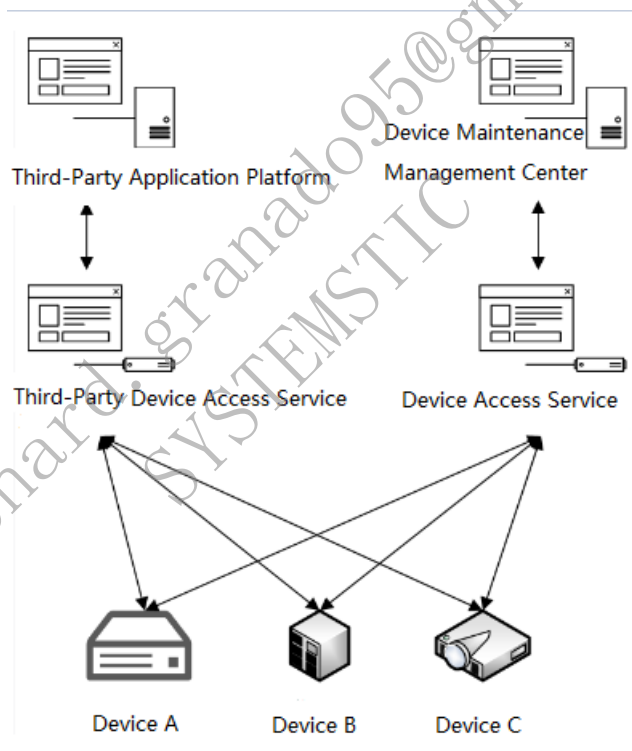
6 Network Configuration

6.1 ISUP Multi-Center Management

6.1.1 Introduction to the Function

When connected to the IoT, a device may be simultaneously connected to multiple centers. For example, the device may be connected to a third-party business center and a device maintenance management center simultaneously as shown in the figure below.

Note: A center is different from a platform. A center may contain more than one platform.



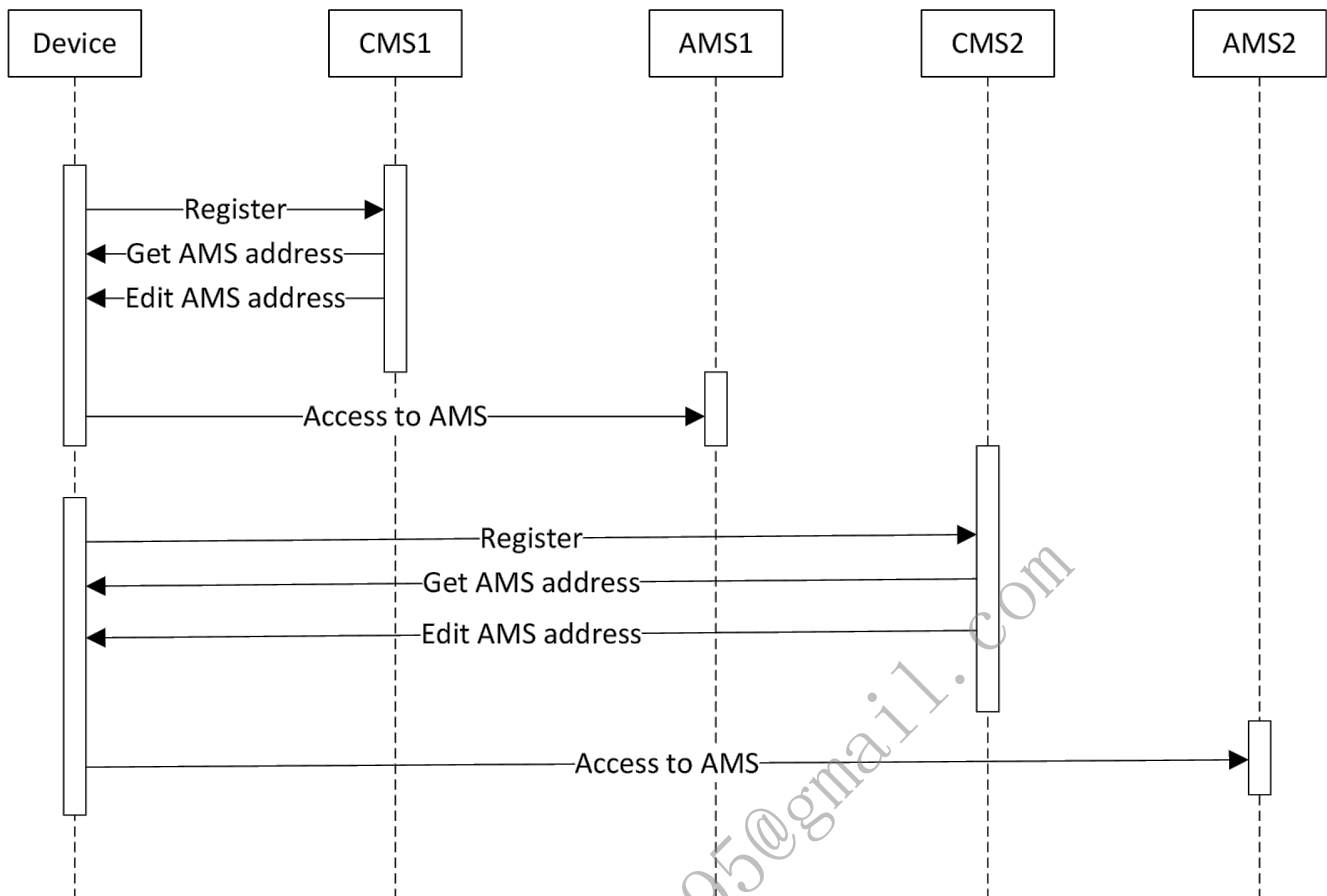
In pull mode, the center logs in to the device if the device is connected to the center via protocols like ISAPI, device network SDK, etc.

In push mode, the device registers to access the center if the device is connected to the center via EZ cloud, ISUP, etc. Here we mainly illustrate the integration process of connecting devices to ISUP multi-center via ISAPI.

6.1.2 API Calling Flow

6.1.2.1 Flow Example

The flow of the ISUP multi-center is as follows.



6.1.2.2 CMS Address Configuration

Configure the address of CMS1 and CMS2 in ISUP multi-center via ISAPI.

1. Check whether the device supports ISUP: GET /ISAPI/System/Network/capabilities. If the node isSupportCaptureFingerPrint is returned and its value is true, it indicates that the device supports ISUP.
2. Check whether the device supports multiple ISUP centers: GET /ISAPI/System/Network/Ehome/capabilities. If the node ISUPID is returned, it indicates that the device supports multiple ISUP centers.
3. Configure the address of multiple ISUP centers: [GET/PUT] /ISAPI/System/Network/Ehome/<ISUPID>.

Note:

ISUPID represents the ID of CMS 1 and CMS 2, and it is required to configure the address of each CMS respectively.

6.1.2.3 AMS Address Configuration

After the address configuration of CMS 1 and CMS 2 is finished and the device is connected to CMS 1 and CMS 2 via ISUP, configure the AMS address for CMS 1 and CMS 2 respectively via ISUP.

1. Call GET /ISAPI/Event/notification/httpHosts to get all configured HTTP/HTTPS listening center addresses and their AMS addresses via the corresponding communication link with the CMS.

Example:

(1) id=1 represents the AMS 1 address of the CMS 1 (the protocolType is EHome), id=2 represents the AMS 2 address of the CMS 2 (the protocolType is EHome), and id=3 represents the address of HTTP listening center (protocolType is HTTP).

(2) When the CMS 1 requests to get the address information, the device will return the AMS 1 address of the CMS 1 and the address of HTTP listening center.

(3) When the CMS 2 requests to get the address information, the device will return the AMS 2 address of the CMS 2 and the address of HTTP listening center.

2. For the obtained address list:

- (1) If an AMS address already exists (the protocolType contains EHome), call PUT `/ISAPI/Event/notification/httpHosts/<ID>` to edit the AMS address.
- (2) If an AMS address does not exist (the protocolType does not contain EHome), call POST `/ISAPI/Event/notification/httpHosts` to create an AMS address.
3. Connect the device to the AMS.

Note:

- (1) When calling the API (DELETE `/ISAPI/Event/notification/httpHosts`) via ISAPI, delete the HTTP/HTTPS center only and do not perform any operations on the ISUP AMS center.
- (2) When calling the API (PUT `/ISAPI/Event/notification/httpHosts`) via ISUP, if there are two or above AMS addresses (protocolType is EHome) in the list, it indicates that a parameter configuration error occurs and the API will report the error.
- (3) When calling the API (PUT `/ISAPI/Event/notification/httpHosts/<ID>`) via ISUP, and the protocolType is EHome, the returned represents the AMS address of corresponding link.

6.1.2.4 Get ISUP Information

Get the CMS and AMS address for the device to access the ISUP multi-center via ISAPI.

1. Check whether the device supports getting ISUP information: GET `/ISAPI/System/Network/Ehome/capabilities`. If `IsSupportISUPServerInfo` is returned and its value is true, it indicates that the device supports getting ISUP information.
2. Get ISUP information (including the address list of the configured CMS and AMS): GET `/ISAPI/System/Network/ISUPServerInfo?format=json`.

6.1.3 Message Format and Example

URL: `/ISAPI/System/Network/ISUPServerInfo?format=json`.

Method: GET.

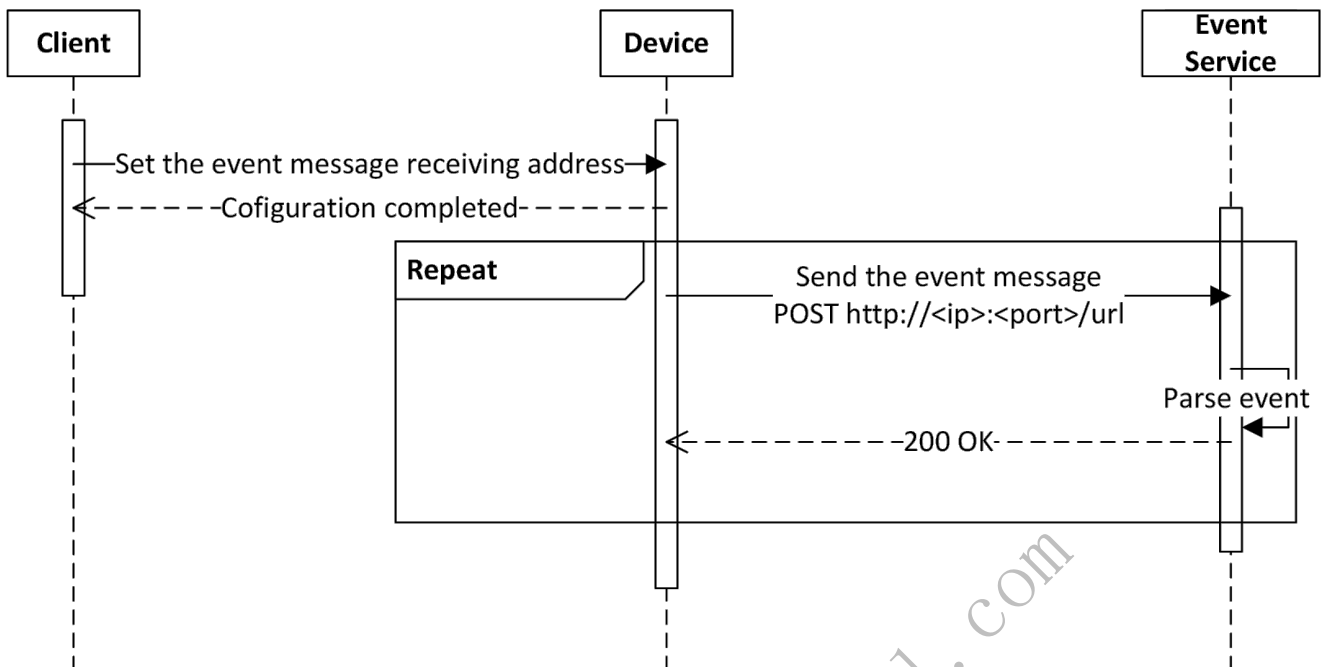
Message Example:

```
{
  "ISUPServerInfoList": [{
    "ID": 1, //req, int, ID, range:[1,2]
    "enabled": true, //req, boolean, whether it is enabled
    "CMSInfo": {
      "addressingFormatType": "ipaddress", //opt, string, address type: ("ipaddress", "hostname")
      "hostName": "", //opt, string, domain name (it is valid when the addressingFormatType is hostname)
      "ipV4Address": "", //opt, string, IPv4 address (it is valid when the addressingFormatType is ipaddress)
      "ipV6Address": "", //opt, string, IPv6 address (it is valid when the addressingFormatType is ipaddress)
      "portNo": 1, //opt, integer, port No., range:[0,65535]
      "deviceID": "", //opt, string, device ID, range:[1,64]
      "protocolVersion": "v5.0" //opt, string, protocol version ("v2.0"-v2.0 version, "v2.6"-v2.6 version, "v4.0"-v4.0 version, "v5.0"-v5.0 version)
    },
    "AMSInfo": {
      "addressingFormatType": "ipaddress", //opt, string, address type: ("ipaddress", "hostname")
      "hostName": "", //opt, string, domain name (it is valid when the addressingFormatType is hostname)
      "ipV4Address": "", //opt, string, IPv4 address (it is valid when the addressingFormatType is ipaddress)
      "ipV6Address": "", //opt, string, IPv6 address (it is valid when the addressingFormatType is ipaddress)
      "portNo": 1 //opt, integer, port No., range:[0,65535]
    }
  ]
}
```

6.2 Listening Service

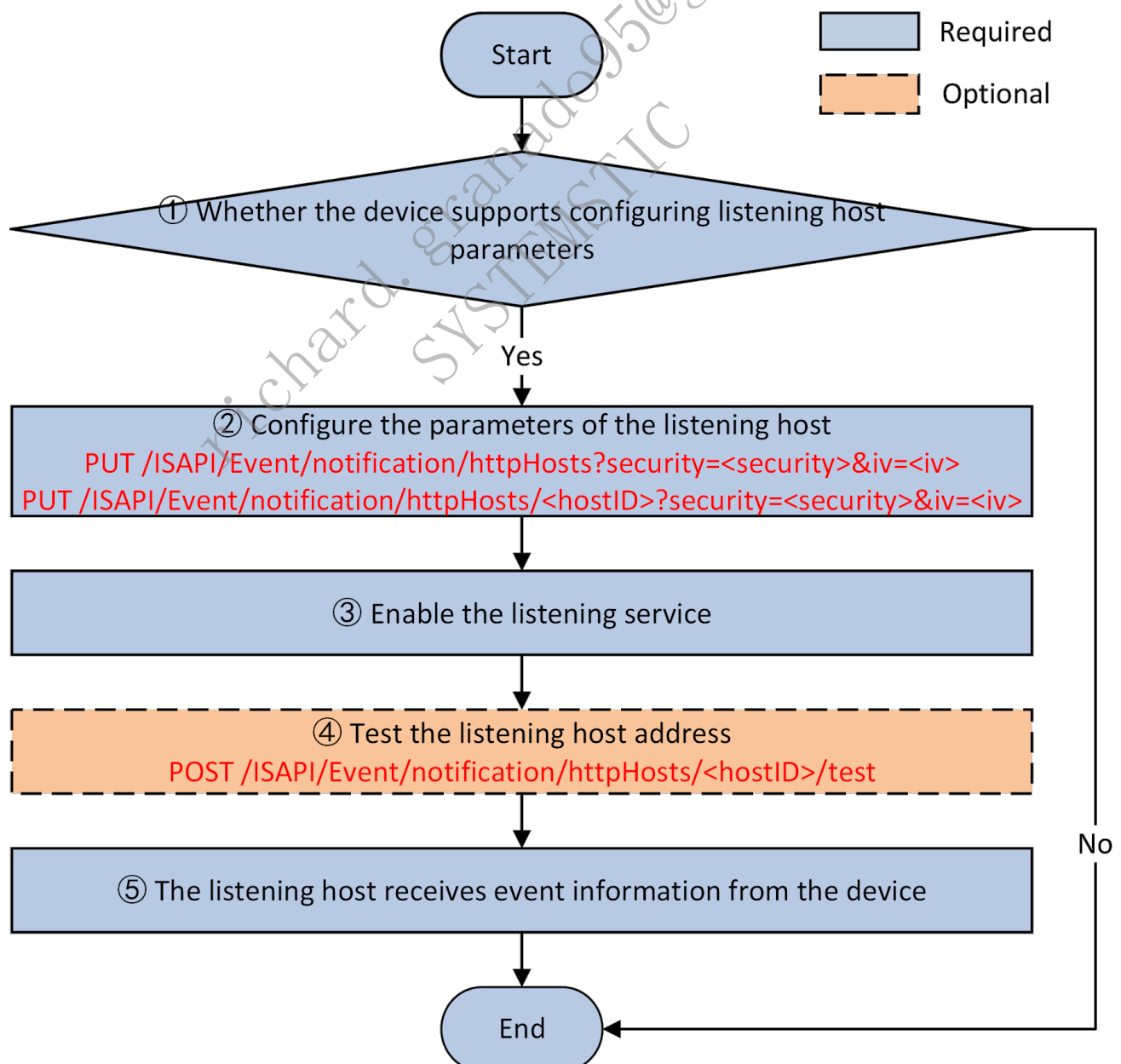
6.2.1 Introduction to the Function

When an event occurs, the device creates connection with the client and uploads alarm information. Meanwhile, the listening host receives data from the device. The IP address and the port No. of the listening host should be configured for the device. The HTTP listening service supports subscribing to specific events when adding or editing the listening host. Only the specified events will be uploaded by the device.(currently not available for the device)



6.2.2 API Calling Flow

6.2.2.1 Listening Service



1. Check whether the device supports configuring listening host parameters:

Get the configuration capability of the listening host: GET /ISAPI/Event/notification/httpHosts/capabilities.
If the node <HttpHostNotificationCap> exists in the returned message and its value is true, it indicates that the device supports configuring listening host parameters.

2. Configure the parameters of the listening host:

Configure the parameters of all listening hosts: PUT /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>;

Get the parameters of all listening hosts: GET /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>;

Configure the parameters of a listening host: PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>;

Get the parameters of a listening host: GET /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>;

3. Enable the listening service:

The user needs to enable the listening service of the listening host.

4. (Optional) Test the listening service:

The platform applies the command to the device to test whether the listening host is available for the device: POST /ISAPI/Event/notification/httpHosts/<hostID>/test.

5. The listening host receives event information from the device:

When an event occurs, the device creates connection with the client and uploads alarm information. Meanwhile, the listening host receives data from the device. See details in **Event Message Grammar**.

Remark: You can also configure the listening parameters such as the timeout.

6.2.2.2 Event Message Grammar

When an event occurs or an alarm is triggered, the event/alarm information can be with binary data (such as pictures) and without binary data.

1. Without Binary Data:

The Content-Type in the Headers of the HTTP request sent by the device is usually application/xml or application/json as follows:

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--/Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: data_gateway_ip:port <!--Host: HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml; <!--content type, which is used for the upper layer to distinguish different formats when parsing the message-->
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

<EventNotificationAlert/>
```

Response by the Listening Host Notes:

- Content-Length is required and its value is 0.
- Connection is recommended to use keep-alive to reduce the port usage.

```
HTTP/1.1 200 OK
Content-Length: 0
Connection: keep-alive
```

2. With Binary Data

The format of the data sent by the device is HTTP form (multipart/form-data). The Content-Type in the Headers of the HTTP request sent by the device is usually multipart/form-data, boundary=<frontier>: boundary is a variable, which is used to divide the HTTP Body into multiple units and each unit has its Headers and Body. See details in [RFC 1867 \(Form-based File Upload in HTML\)](#). Please note the -- before and after the boundary.

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: device_ip:port <!--Host: HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml <!--Some alarms use JSON format, so the upper-layer application can distinguish based on the Content-Type during parsing-->

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[picture data]
--<frontier>--
```

Response by the Listening Host

Notes:

1. Content-Length is required and its value is 0.
2. Connection is recommended to use keep-alive to reduce the port usage.

```
HTTP/1.1 200 OK
Content-Length: 0
Connection: keep-alive
```

Here are the descriptions of the main keywords.

keyword	example	description
Content-Type	multipart/form-data; boundary=frontier	Content type, multipart/form-data refers to data in form format.
boundary	frontier	Separator of the form message. A form message starts with --boundary and ends with --boundary--.
Content-Disposition	form-data; name="Picture_Name";	Content description. form-data refers to data in the form format.
filename	"Picture_Name"	File name. The file refers to the form message.
Content-Length	10	Content length. The length of the content which starts from \r\n to the next --boundary.

6.2.3 Exception Handling

6.2.3.1 Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Remarks
6	Invalid Content	eventNotSupport	0x60001024		Event subscription is not supported.	

6.3 Wi-Fi Network Management

6.3.1 Introduction to the Function

The user can plug a network cable into a wireless router and sets the working mode of the router as AP mode to enable Wi-Fi. The device under Wi-Fi network management is used as the AP to provide a wireless network for the Station. It can provide functions include wireless server information management of internet APIs and device list information accessed by the wireless server.

6.3.2 API Calling Flow

1. Wi-Fi parameter configuration:

Check whether the device supports configuring Wi-Fi parameters: GET /ISAPI/System/capabilities; if the node is returned and its value is true, it means the device supports wireless network and Wi-Fi parameter configuration.

Get the capability of configuring Wi-Fi parameters: GET

/ISAPI/System/Network/interfaces/<interfaceID>/wireless/capabilities;

Get Wi-Fi parameters: GET /ISAPI/System/Network/interfaces/<interfaceID>/wireless;

Configure Wi-Fi parameters: PUT /ISAPI/System/Network/interfaces/<interfaceID>/wireless;

Get Wi-Fi quality parameters: GET /ISAPI/System/Network/wifiQuality?format=json;

2. Wi-Fi area parameter configuration:

Check whether the device supports Wi-Fi area parameters: GET /ISAPI/System/capabilities; if <isSupportWifiRegion> is returned and its value is true, it indicates that device supports Wi-Fi area configuration.

Get the capability of configuring Wi-Fi area parameters: GET

/ISAPI/System/Network/wirelessServer/Region/Capabilities;

Get Wi-Fi area parameters: GET /ISAPI/System/Network/wirelessServer/Region;

Configure Wi-Fi area parameters: PUT /ISAPI/System/Network/wirelessServer/Region;

3. Wi-Fi connection parameters:

Get the capability of configuring Wi-Fi connection parameters: GET

/ISAPI/System/Network/interfaces/<interfaceID>/wirelessStatus/capabilities;

Get Wi-Fi connection parameters of single network interface: GET

/ISAPI/System/Network/interfaces/<interfaceID>/wirelessStatus;

Configure Wi-Fi connection parameters of single network interface: PUT

/ISAPI/System/Network/interfaces/<interfaceID>/wirelessStatus;

Get Wi-Fi connection status of single network interface: GET

/ISAPI/System/Network/interfaces/<interfaceID>/wireless/connectStatus;

4. Wi-Fi transmission status:

Get Wi-Fi transmission status of single network interface: GET

/ISAPI/System/Network/interfaces/<interfaceID>/wireless/transmission/capabilities;

Get Wi-Fi transmission status of single network interface: GET

/ISAPI/System/Network/interfaces/<interfaceID>/wireless/transmission;

Get the Wi-Fi transmission status of the specific network interface: GET

/ISAPI/System/Network/interfaces/<interfaceID>/wireless/transmission/<indexID>;

Get Wi-Fi transmission status of all network interfaces: GET

/ISAPI/System/Network/interfaces/<interfaceID>/wireless/transmissionList;

7 Video (General)

7.1 . Compression of Stream Transcoding

7.1.1 Introduction to the Function

When a device is deployed in a single network access point and needs to access the platform through the mobile network, the uplink network bandwidth of the device will be insufficient, which cannot meet the requirement of high-definition live view. The above problem can be solved via stream transcoding. The bit rate of network transmitted video stream is reduced to improve the resolution of the video stream, so as to achieve the high-definition live view.

7.1.2 API Calling Flow

1. Check whether the device supports compression configurations of stream transcoding: GET `/ISAPI/System/capabilities`. If the field `isSupportCompressedTranscode` is returned and its value is `true`, it indicates that the function is supported.
2. Get the capability of compression parameters of stream transcoding for a specified channel: GET `/ISAPI/Streaming/channels/<channelID>/TranscodeCfg/capabilities?format=json`.
3. Get the compression parameters of stream transcoding for a specified channel: GET `/ISAPI/Streaming/channels/<channelID>/TranscodeCfg?format=json`.
4. Set the compression parameters of stream transcoding for a specified channel: PUT `/ISAPI/Streaming/channels/<channelID>/TranscodeCfg?format=json`.

Remarks:

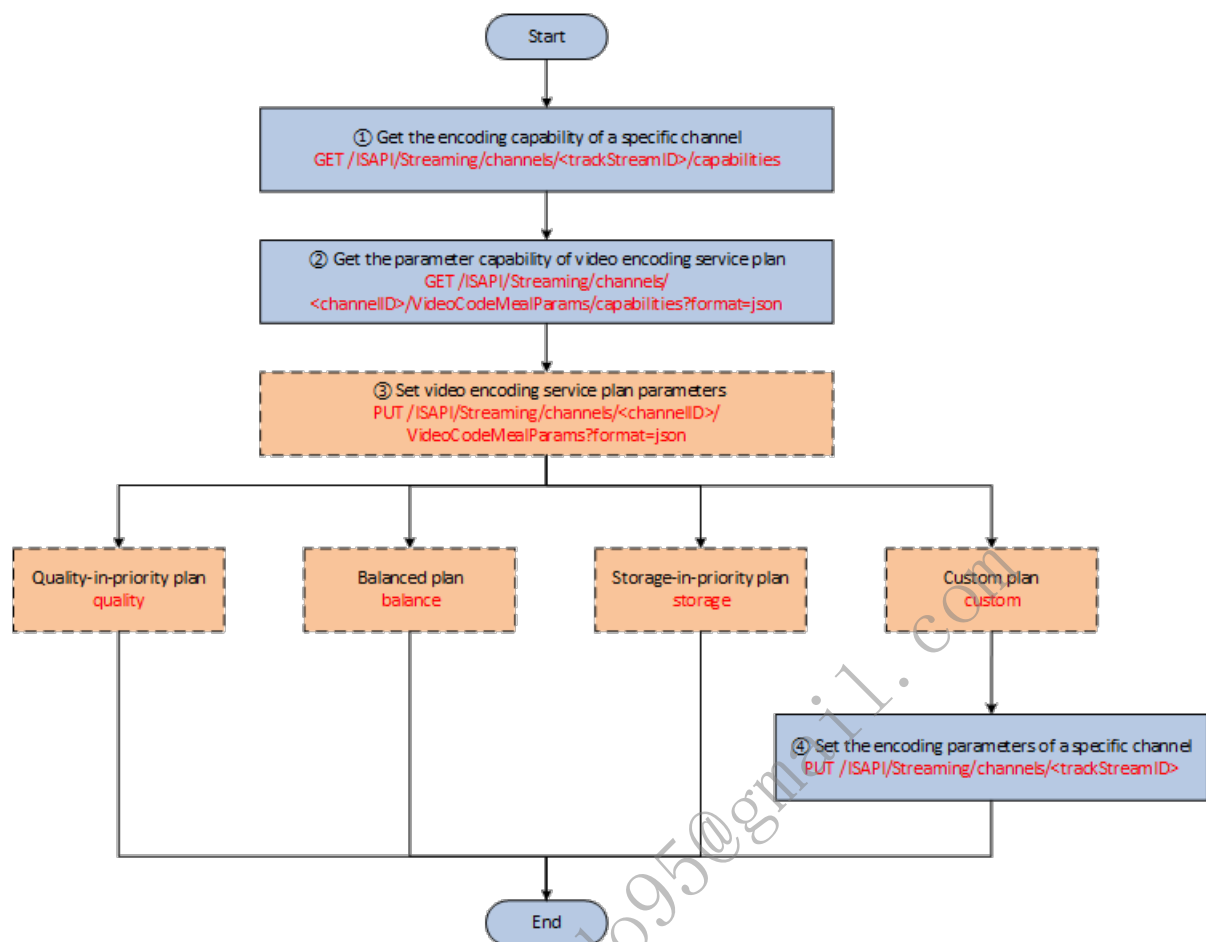
- The transcoding compression parameters of the live view stream and the playback stream can be different.
- The compression parameters configuration of stream transcoding has no influence on encoding parameters of normal videos/audios.

7.2 Configuration of Video Encoding Service Plan

7.2.1 Introduction to the Function

The existing video encoding parameters are complicated to some degree, and might not be user-friendly. Users' main focus is on image quality and the required storage capacity. Therefore, video encoding service plan is introduced to facilitate user configuration by organizing the parameters into different plans. If you configure a plan, the previous manual configurations will be overwritten. If you keep the previous manual configurations, they will be recognized as in a custom plan.

7.2.2 API Calling Flow



1. Get the encoding parameter capability of a specific channel: GET /ISAPI/Streaming/channels/<trackStreamID>/capabilities
2. Get the parameter capability of video encoding service plan: GET /ISAPI/Streaming/channels/<channelID>/VideoCodeMealParams/capabilities?format=json
3. (Optional) Set video encoding service plan parameters: PUT /ISAPI/Streaming/channels/<channelID>/VideoCodeMealParams?format=json. The client can select a service plan. The parameters in the plans cannot be edited, except the custom plan.
4. Set the encoding parameters of a specific channel: PUT /ISAPI/Streaming/channels/<trackStreamID>. The parameters in the custom plan can be edited.

Note:

Get the parameter content of the current video encoding service plan:

- Get the parameters of the video encoding service plan: POST /ISAPI/Streaming/channels/<channelID>/SearchVideoCodeMealParams?format=json.
- Get the capability of getting video encoding service plan parameters: GET /ISAPI/Streaming/channels/<channelID>/SearchVideoCodeMealParams/capabilities?format=json.

Restore to the default parameters: PUT /ISAPI/Streaming/RestoreDefaultParam?format=json. The parameters can be restored to default in custom plan.

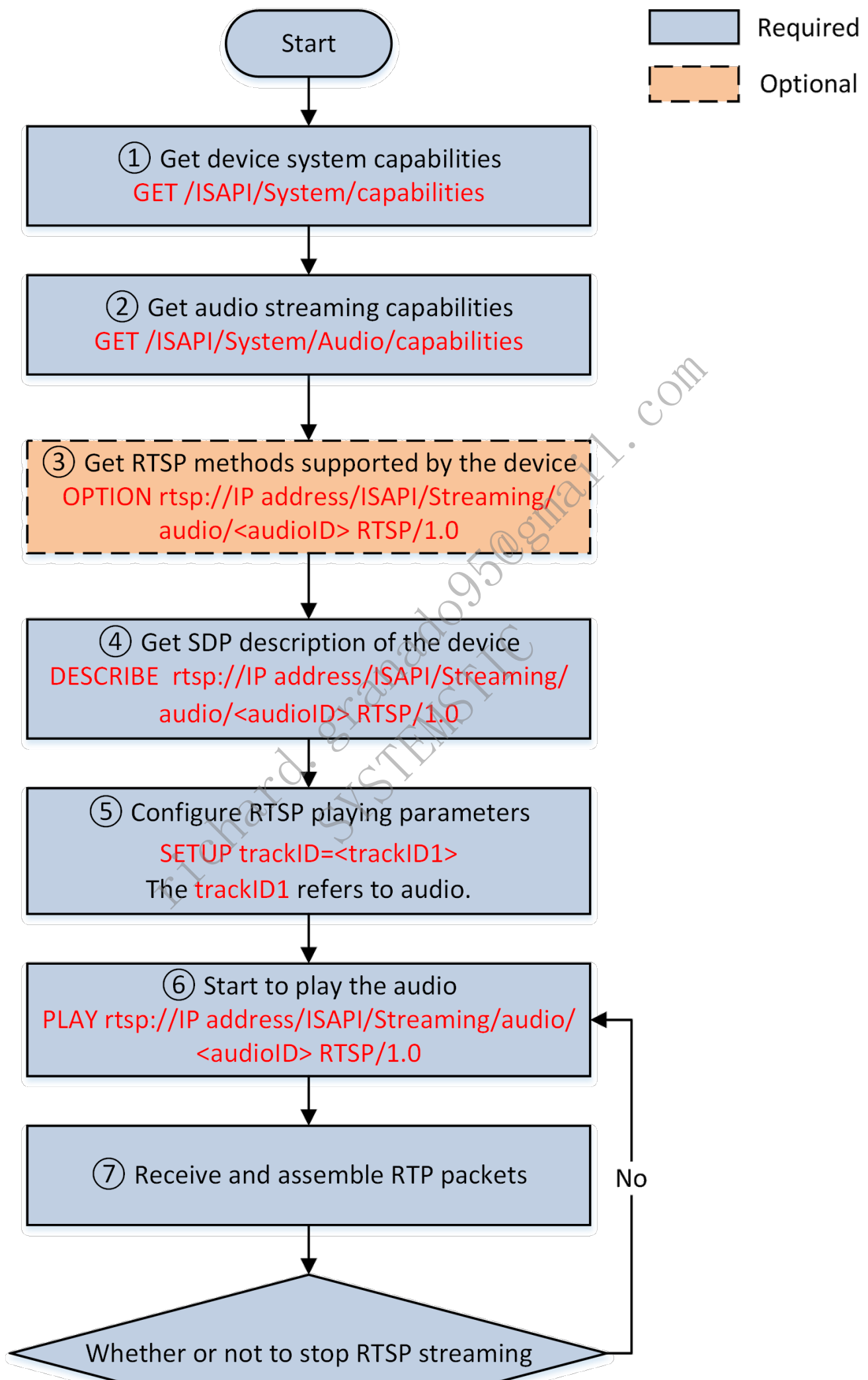
8 Audio (General)

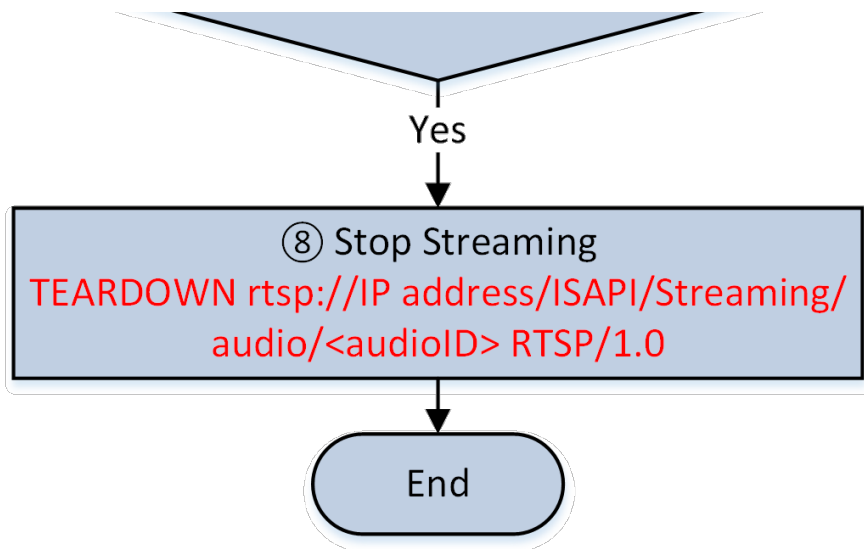
8.1 Real-Time Audio Streaming

8.1.1 Introduction to the Function

The client can get real-time audio data via RTSP to listen to the device.

8.1.2 API Calling Flow





1. Get system capabilities: `GET /ISAPI/System/capabilities`. If `isSupportRTSPAudio` is returned and its value is true, then the device supports real-time audio streaming via RTSP.
2. Get audio streaming capabilities: `GET /ISAPI/System/Audio/capabilities`. If `isSupportRTSPAudio` is returned and its value is true, the device supports RTSP real-time audio streaming.
3. (Optional) Send an RTSP OPTION command to get RTSP methods supported by the device: `OPTION rtsp://IP address/ISAPI/Streaming/audio/<audioID> RTSP/1.0`.
4. Send an RTSP DESCRIBE command: `DESCRIBE rtsp://IP address/ISAPI/Streaming/audio/<audioID> RTSP/1.0`. Digest-MD5 authentication is completed in this step.
5. Use the trackID parsed from SDP information to set RTSP playing parameters. trackID=1 indicates audio.
6. Send an RTSP PLAY command: `PLAY rtsp://IP address/ISAPI/Streaming/audio/<audioID> RTSP/1.0`, and the device starts to push audio stream.
7. Receive the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed.
8. Send an RTSP TEARDOWN command to stop streaming.

Note: `<audioID>` represents audio channel No.

8.1.3 Message Example

8.1.3.1 Send an RTSP OPTION Request

```
OPTIONS rtsp://10.17.114.136/ISAPI/Streaming/audio/1 RTSP/1.0
CSeq: 0
User-Agent: LibVLC/2.2.4 (LIVE555 Streaming Media v2016.02.22)
```

Response from the device

```
RTSP/1.0 200 OK
CSeq: 0
Public: OPTIONS, DESCRIBE, PLAY, SETUP, TEARDOWN, SET_PARAMETER, GET_PARAMETER
```

8.1.3.2 Send an RTSP DESCRIBE Request

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1 RTSP/1.0
CSeq: 1
Accept: application/sdp
User-Agent: NKPlayer-1.00.00.081112
```

```
RTSP/1.0 401 Unauthorized
CSeq: 1
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1 RTSP/1.0
CSeq:2
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
CSeq: 2
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1633703084091356 1633703084091356 IN IP4 10.17.114.136
s=Media Presentation
e=NONE
b=AS:5100
t=0 0
a=control:rtsp://10.17.114.136/ISAPI/Streaming/audio/1/
m=audio 0 RTP/AVP 8
c=IN IP4 0.0.0.0
b=AS:5000
a=recvonly
a=x-dimensions:2560,1440
m=audio 0 RTP/AVP 8
c=IN IP4 0.0.0.0
b=AS:50
a=recvonly
a=control:rtsp://10.17.114.136/ISAPI/Streaming/audio/1/trackID=1
a=rtpmap:8 PCMA/8000
a=Media_header:MEDIAINFO=494D4B48010300000400000111710110401F000000FA000000000000000000000000000000;
a=appversion:1.0
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1/trackID=1 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1/",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

The device pushes audio and video streams.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. ....d1.w....C....".T....g....).i.....a....7.S...~J.....X....X.
```

8.1.3.5 Send an RTSP TEARDOWN Request

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/audio/1",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

Response from the device

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

9 Access Control (General)

9.1 Arming Information

9.1.1 Introduction to the Function

The device supports getting the arming information, such as the armed device IP and port, arming type, and protocol type.

9.1.2 API Calling Flow

1. Get the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportDeployInfo is returned and its value is "true", it indicates that the device supports getting arming information.
2. Get arming information capability: GET /ISAPI/AccessControl/DeployInfo/capabilities.
3. Get arming information: GET /ISAPI/AccessControl/DeployInfo.

9.2 Event and Card Linkage Parameters

9.2.1 Introduction to the Function

The device supports linking specific actions when access control events triggered. The linkage actions include event, card No., MAC address, and employee No. Take event for example, if a person authenticated by the device, the door will open for the person.

9.2.2 API Calling Flow

9.2.2.1 Configure Parameters of Event and Card Linkage

1. Get the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportEventCardLinkageCfg is returned and its value is "true", it indicates that the device supports configuring the parameters of event and card linkage.
2. Get the configuration capability of the event and card linkage: GET /ISAPI/AccessControl/EventCardLinkageCfg/capabilities?format=json.

3. Get and set the parameters of event and card linkage: GET|PUT
/ISAPI/AccessControl/EventCardLinkageCfg/<ACEID>?format=json.

9.2.2.2 Search for Parameters of Event and Card Linkage

1. Get the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportEventCardLinkageCfgSearch is returned and its value is "true", it indicates that the device supports searching for the parameters of event and card linkage.
2. Get the capability of searching for parameters of event and card linkage: GET
/ISAPI/AccessControl/EventCardLinkageCfg/search/capabilities?format=json.
3. Search for parameters of event and card linkage: POST /ISAPI/AccessControl/EventCardLinkageCfg/search?format=json.

9.2.2.3 Delete Parameters of Specific Event and Card Linkage

1. Get the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportEventCardLinkageCfgDelete is returned and its value is "true", it indicates that the device supports deleting parameters of event and card linkage.
2. Delete parameters of specific event and card linkage: PUT /ISAPI/AccessControl/EventCardLinkageCfgDelete?format=json.

9.2.2.4 Get List of Event and Card Linkage ID

1. Call the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportEventCardNoList is returned and its value is true, it indicates that the device supports getting events and the card linkage ID list.
2. Get the capability of the list of event and card linkage ID: GET
/ISAPI/AccessControl/EventCardNoList/capabilities?format=json.
3. Get the list of event and card linkage ID: GET /ISAPI/AccessControl/EventCardNoList?format=json.

9.2.2.5 Optimize Event

1. Get the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportEventOptimizationCfg is returned and its value is "true", it indicates that the device supports event optimization.
2. Get the configuration capability of event optimization: GET
/ISAPI/AccessControl/EventOptimizationCfg/capabilities?format=json.
3. Get the event optimization configuration parameters: GET|PUT /ISAPI/AccessControl/EventOptimizationCfg?format=json.

9.3 Store and Search for Access Control Event

9.3.1 Introduction to the Function

The device supports configuring storage parameters of access control event, searching for access control events, and searching for the amount of access control events. There are three storage modes: deleting old events periodically, deleting old events by specified time and overwriting.

9.3.2 API Calling Flow

9.3.2.1 Configure Storage Parameters of Access Control Events

1. Call the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportEventStorageCfg is returned and its value is "true", it indicates that the device supports configuring the storage parameters of access control events.
2. Get the configuration capability of storing access control events: GET /ISAPI/AccessControl/AcsEvent/StorageCfg/capabilities?format=json.
3. Get and set storage parameters of access control events: GET|PUT /ISAPI/AccessControl/AcsEvent/StorageCfg?format=json.

9.3.2.2 Search for Access Control Events

1. Call the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportAcsEvent is returned and its value is "true", it indicates that the device supports searching for access control events.
2. Get the capability of searching for access control events: GET /ISAPI/AccessControl/AcsEvent/capabilities?format=json.
3. Search for access control events: POST /ISAPI/AccessControl/AcsEvent?format=json.

9.3.2.3 Get Total Number of Access Control Events

1. Call the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportAcsEventTotalNum is returned and its value is "true", it indicates that the device supports getting the total number of access control events.
2. Get the capability of getting total number of access control events by specific conditions: GET /ISAPI/AccessControl/AcsEventTotalNum/capabilities?format=json.
3. Get the total number of access control events by specific conditions: POST /ISAPI/AccessControl/AcsEventTotalNum?format=json.

10 Credentials Collection

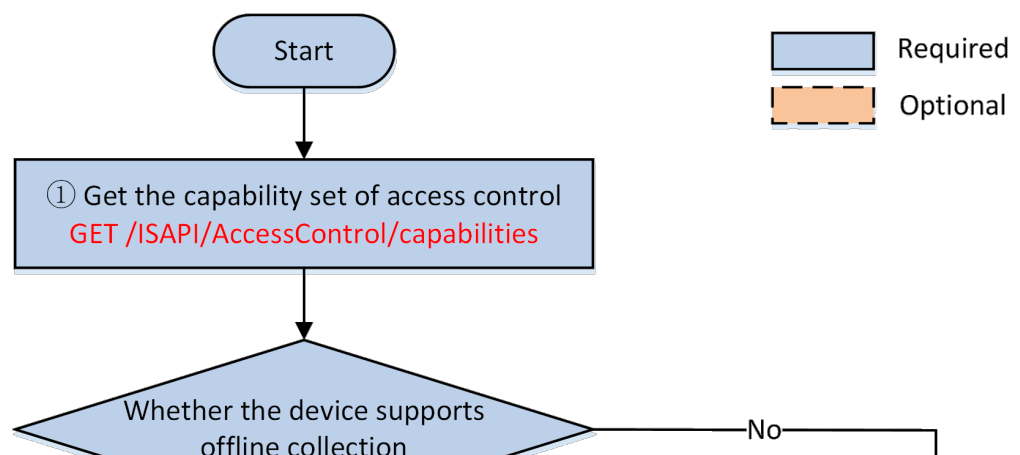
10.1 Offline Collection

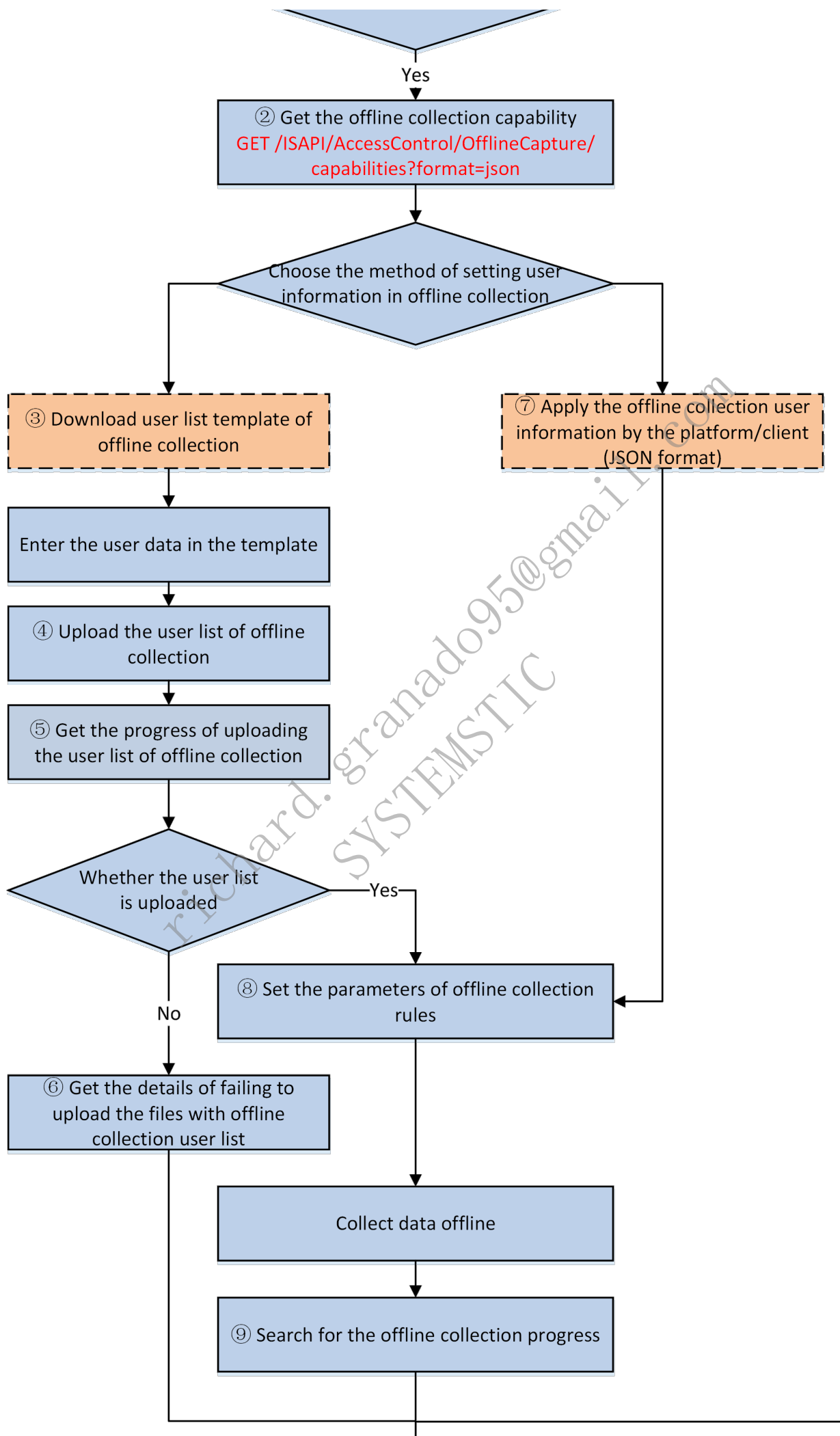
10.1.1 Introduction to the Function

After the related user information to be collected is imported to the device, the device can collect and store the data for uploading to the upper platform/client when the network restores. This function can meet the needs of data collection in non-real-time mode in specific situations.

10.1.2 API Calling Flow

10.1.2.1 Offline Data Collection





End

1. Get the functional capability of access control: GET /ISAPI/AccessControl/capabilities; if the node isSupportOfflineCapture is returned and its value is "true", it indicates that the device supports offline data collection.
2. Get the offline collection capability: GET /ISAPI/AccessControl/OfflineCapture/capabilities?format=json; get the relevant APIs and parameters.
3. Download user list template of offline collection: POST /ISAPI/AccessControl/OfflineCapture/InfoFileTemplateDownload?format=json.
4. Upload the user list of offline collection: POST /ISAPI/AccessControl/OfflineCapture/InfoFile?format=json.
5. Get the progress of uploading the user list of offline collection: GET /ISAPI/AccessControl/OfflineCapture/InfoFile/progress?format=json.
6. Get the details of failing to upload the files with offline collection user list: GET /ISAPI/AccessControl/OfflineCapture/uploadFailedDetails?format=json.
7. Apply the offline collection user information (JSON format): add offline collection user information (JSON format): POST /ISAPI/AccessControl/OfflineCapture/userInfo?format=json; edit offline collection user information (JSON format): PUT /ISAPI/AccessControl/OfflineCapture/userInfo?format=json.
8. Get the parameters of offline collection rules: GET /ISAPI/AccessControl/OfflineCapture/ruleInfo?format=json; set the parameters of offline collection rules: PUT /ISAPI/AccessControl/OfflineCapture/ruleInfo?format=json.
9. Search for the offline collection progress: GET /ISAPI/AccessControl/OfflineCapture/progress?format=json.

10.1.2.2 Collected Data Management

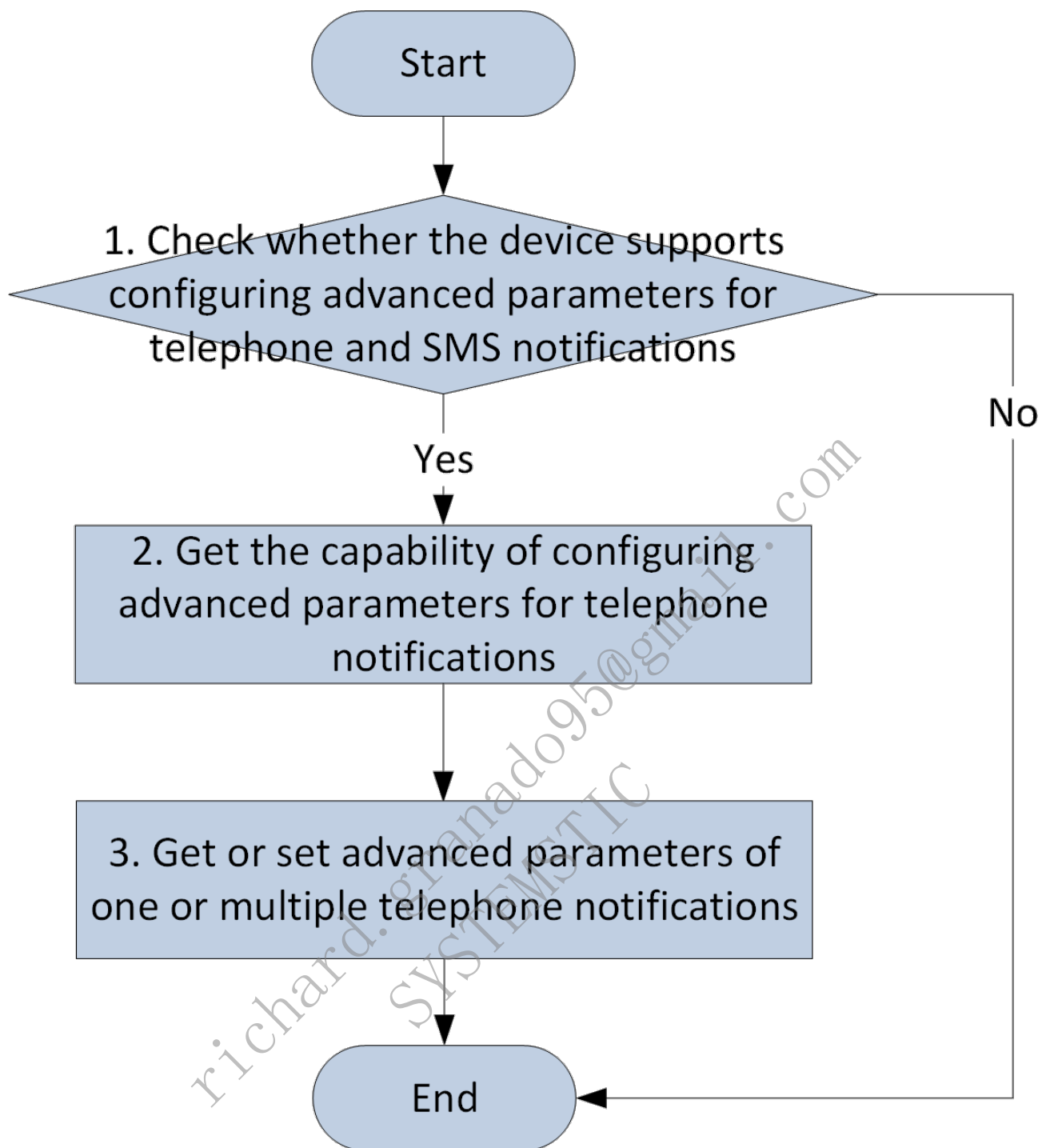
1. Search for the collected data: POST /ISAPI/AccessControl/OfflineCapture/DataCollections/searchTask?format=json.
2. Download data collected offline: POST /ISAPI/AccessControl/OfflineCapture/DataCollections/downloadTask?format=json.
3. Export the offline collected data: PUT /ISAPI/AccessControl/OfflineCapture/dataOutput?format=json&security=<security>&iv=<iv>; Get the progress of exporting the offline collected data: GET /ISAPI/AccessControl/OfflineCapture/dataOutput/progress?format=json.
4. Deleted offline collected data. Delete a specific piece of offline collected data: DELETE /ISAPI/AccessControl/OfflineCapture/DataCollections/<captureNo>?format=json; delete offline collected data by conditions: POST /ISAPI/AccessControl/OfflineCapture/deleteDataCollections?format=json; delete all offline collected data: DELETE /ISAPI/AccessControl/OfflineCapture/DataCollections?format=json.

11 Security Control Device (General)

11.1 Advanced Configuration for Telephone Notification

You can configure advanced parameters for telephone notifications, including effective time period, arming / disarming / alarm clearing permission for each telephone number, and event/alarm type (emergency alarm, medical alarm, and gas alarm).

API Calling Flow:



API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports configuring advanced parameters for telephone and SMS notifications: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node `isSptPhoneAnvanced` is true, the device supports this function.
2. Get the capability of configuring advanced parameters for telephone notifications: `GET /ISAPI/SecurityCP/Configuration/messageSendPhoneAnvanced/capabilities?format=json`.
3. Get advanced parameters of all telephone notifications: `GET /ISAPI/SecurityCP/Configuration/messageSendPhoneAnvanced?format=json`; set advanced parameters of a single telephone notification: `PUT /ISAPI/SecurityCP/Configuration/messageSendPhoneAnvanced/<phoneID>?format=json`. The `phoneID` in the URL is the telephone number ID. Up to 8 telephone numbers are supported for a wireless security control panel.

11.2 ARC Management

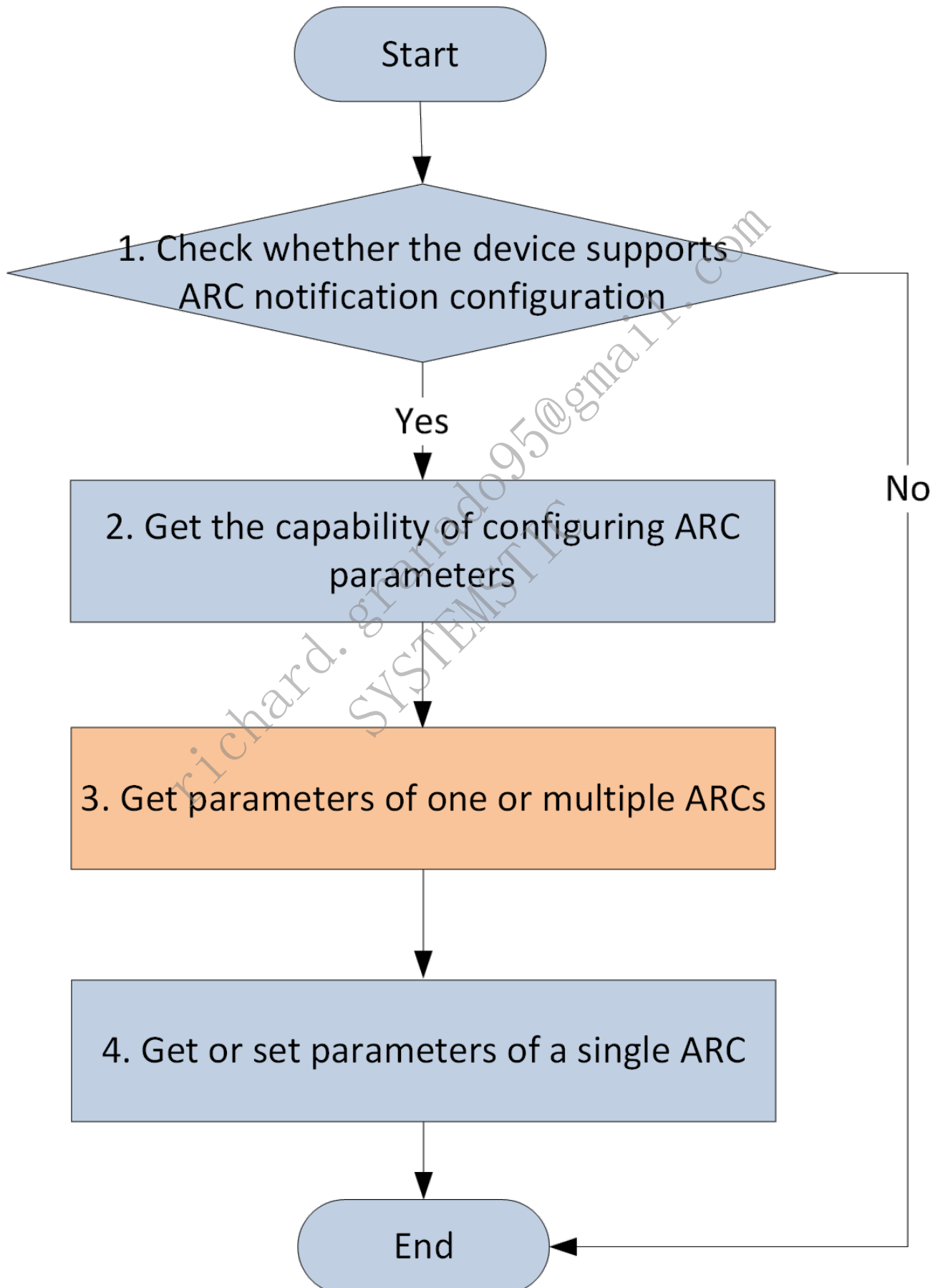
11.2.1 Introduction to the Function

If you have enabled the ARC function and configured events to be received, when the events occur, the corresponding

alarm messages will be pushed to the ARC platform. Currently up to 4 ARCs are supported. By default, No. 1 and 3 are the main ARCs and No. 2 and 4 are the spare ARCs. Only when the main ARCs fail to push alarm messages, will the spare ARCs take the charge.

11.2.2 ARC Parameter Configuration

API Calling Flow:



API Calling Steps: 1. Get the configuration capability of the security control panel to check whether the device supports ARC notification configuration: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node `isSptARC` is true, the device supports this function.

2. Get the capability of configuring ARC parameters: `GET /ISAPI/SecurityCP/Configuration/ARC/capabilities?format=json`.

3. Get parameters of one or multiple ARCs: `GET /ISAPI/SecurityCP/Configuration/ARC?format=json&security=<security>&iv=<iv>`. The `userName` and `password` nodes should be encrypted.

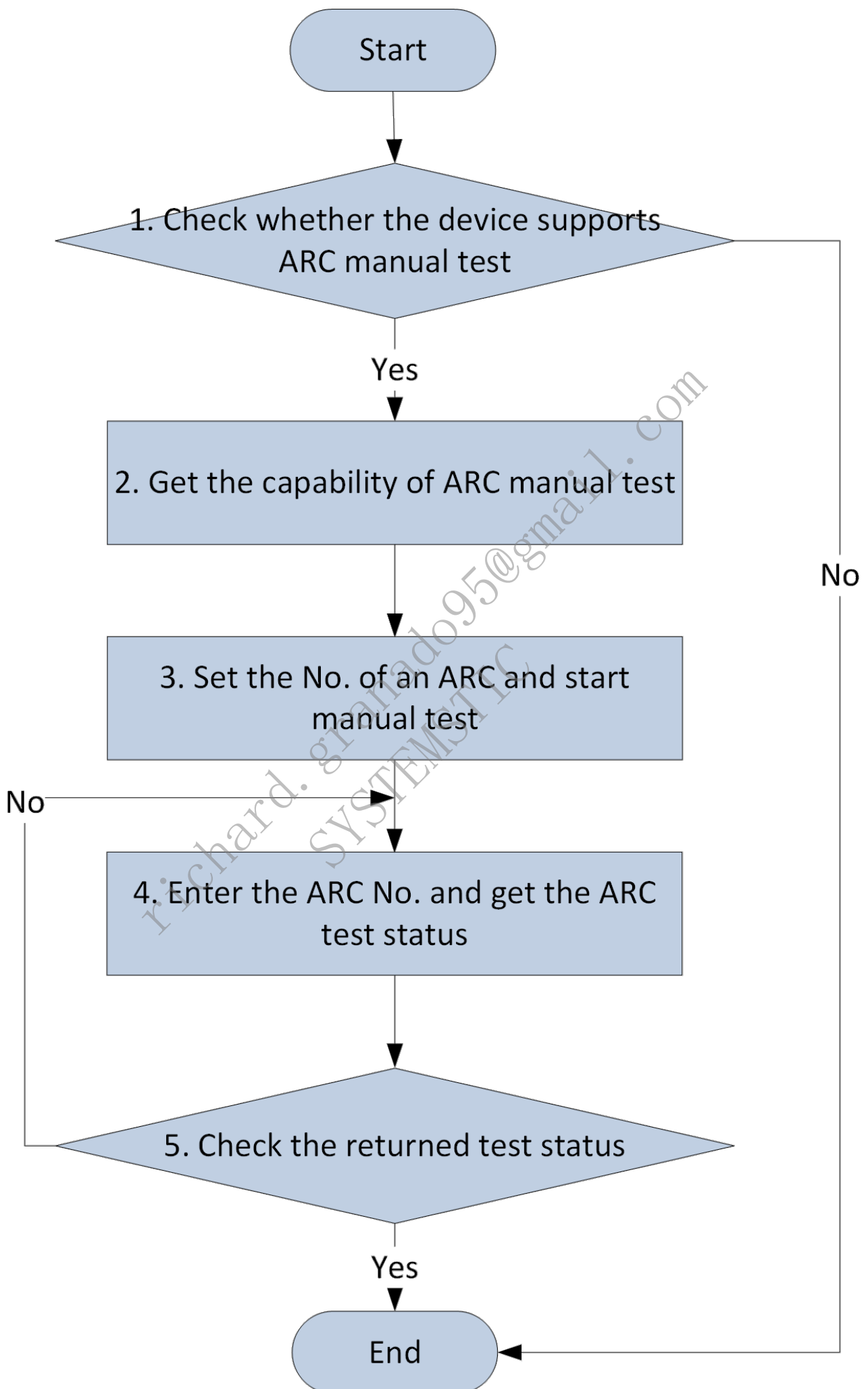
4. Get or set parameters of a single ARC: `PUT /ISAPI/SecurityCP/Configuration/ARC/<indexID>?format=json&security=<security>&iv=<iv>`. The nodes `userName` and `password` should be encrypted. The `indexID` in the URL is the ARC No.

11.2.3 Manual Test of ARC

After configuring ARC parameters, you can check whether the parameter values are valid by testing the link.

API Calling Flow:

richard.granado95@gmail.com
SYSTEMSTIC



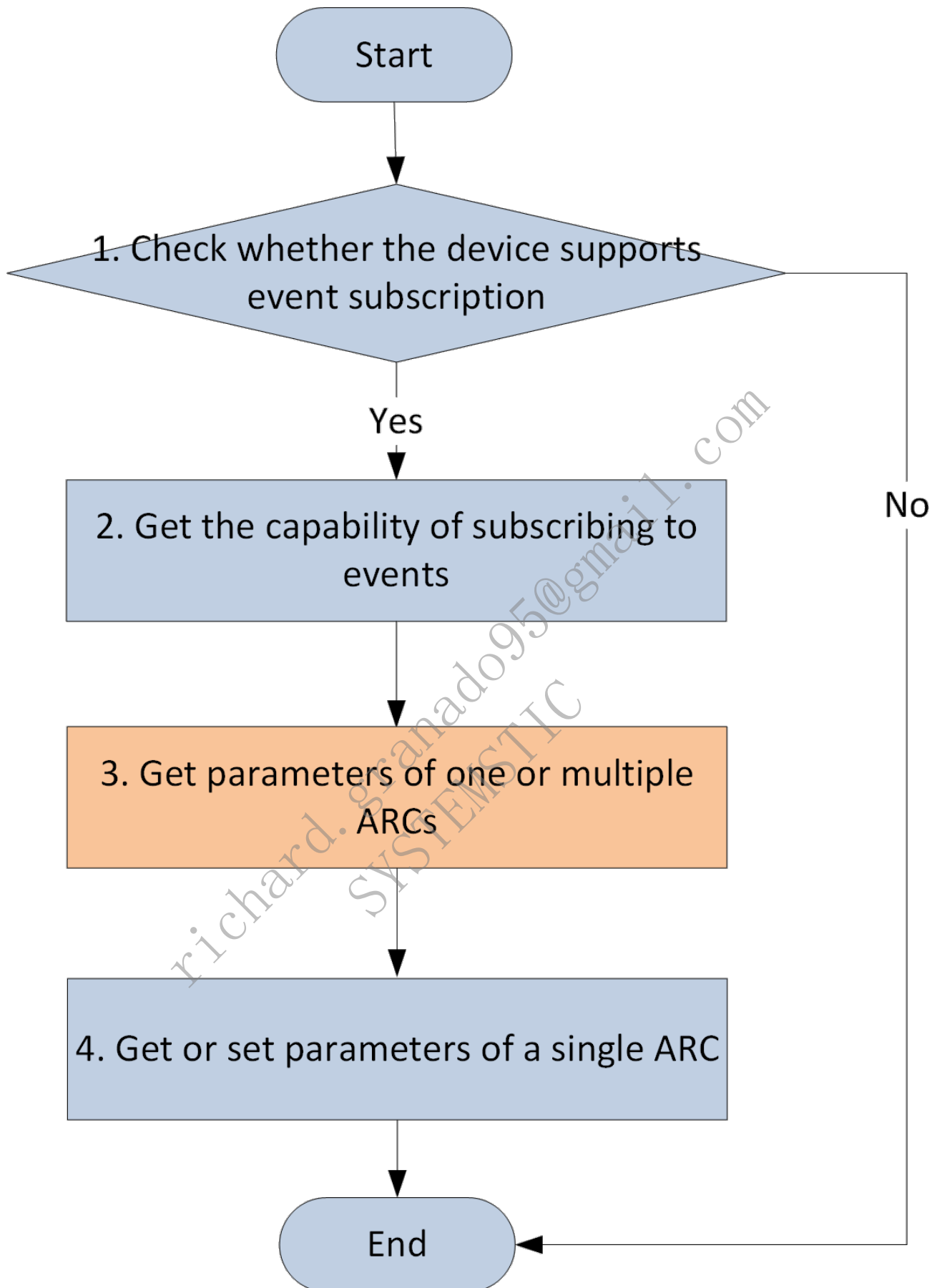
- API Calling Steps:**
1. Get the configuration capability of the security control panel to check whether the device supports manual test of ARC: GET /ISAPI/SecurityCP/Configuration/capabilities?format=json. If the returned value of the node isSptARC is true, the device supports this function.
 2. Get the capability of manual test of ARC: GET /ISAPI/SecurityCP/Configuration/ARC/manualTest/capabilities?format=json.
 3. Set the No. of an ARC and start manual test: PUT /ISAPI/SecurityCP/Configuration/ARC/manualTest?format=json.
 4. Enter the ARC No. and get the ARC test status: POST /ISAPI/SecurityCP/Configuration/ARC/manualTest/status?format=json.
 5. Check the returned test status: success for success and failed for failure. If processing is returned within the configured timeout threshold, you need to call the API in the previous step again.

11.2.4 Event Subscription

Event/alarm types include tampering event, life security event, system status event, operation event, emergency alarm, medical alarm, and gas alarm. When the alarms are triggered or the events occur, the corresponding alarm/event details will be uploaded by the device.

API Calling Flow:

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SYSTEMSTIC



API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports event subscription: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node `isSptARC` is true, the device supports this function.
2. Get the capability of subscribing to events: `GET /ISAPI/SecurityCP/Configuration/messageSendARC/capabilities?format=json`.

3. Get parameters of one or multiple ARCs: GET /ISAPI/SecurityCP/Configuration/ARC?format=json&security=<security>&iv=<iv>.

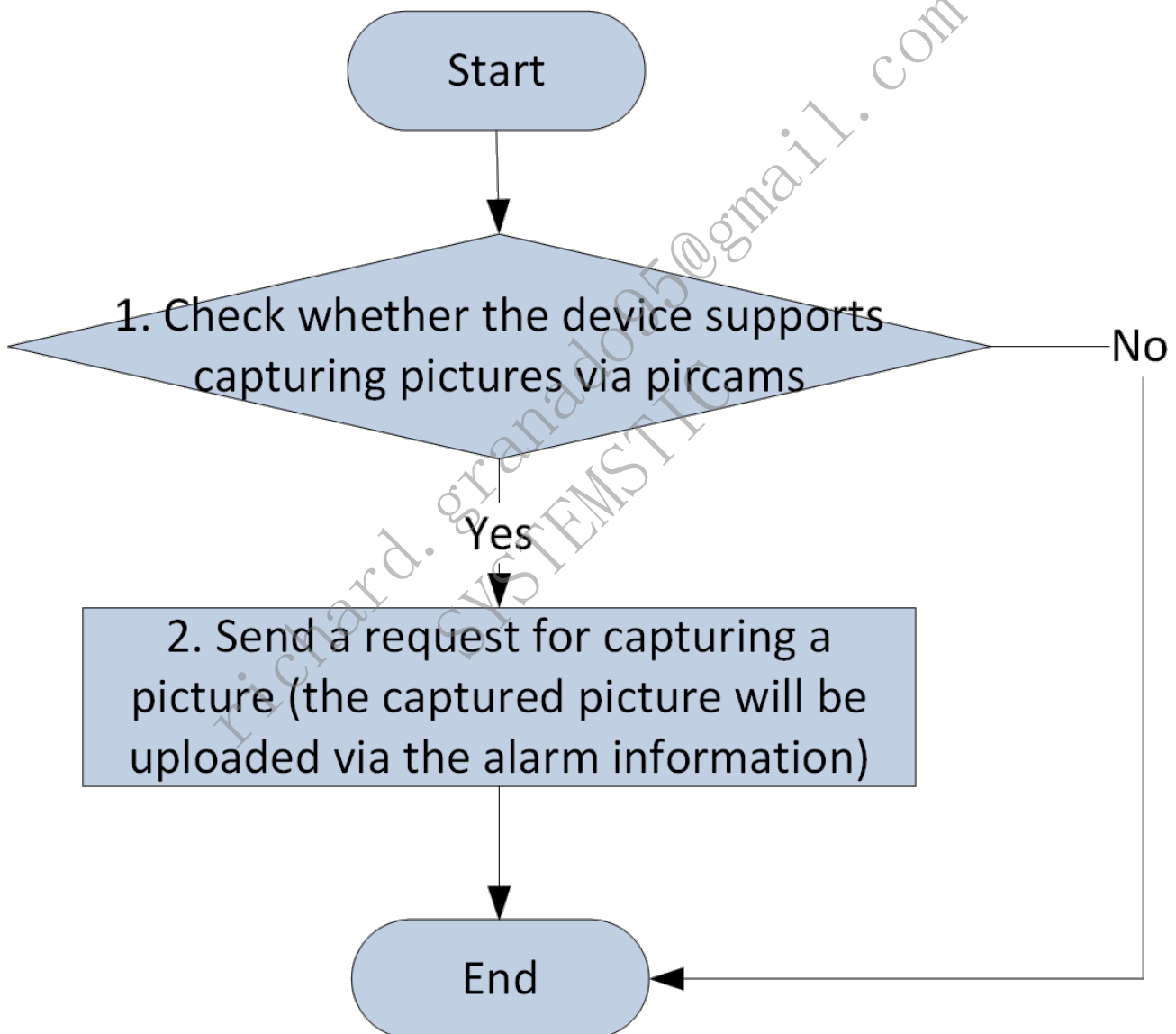
4. Get parameters of a single ARC: GET /ISAPI/SecurityCP/Configuration/messageSendARC?id=<indexID>&format=json; set parameters of a single ARC: PUT /ISAPI/SecurityCP/Configuration/messageSendARC?id=<indexID>&format=json. The indexID in the URL is the ARC No.

11.3 Capture Pictures of Specific Zones

There are two ways to get the picture captured by pircams: 1) send a request for capturing a picture and get the picture URL in the uploaded alarm (CID event) information; 2) send a request for capturing a picture and call another API to get the picture URL.

11.3.1 Upload Pictures by Pircams

After the pircam captures a picture, the picture will be uploaded via the alarm (CID event) information.

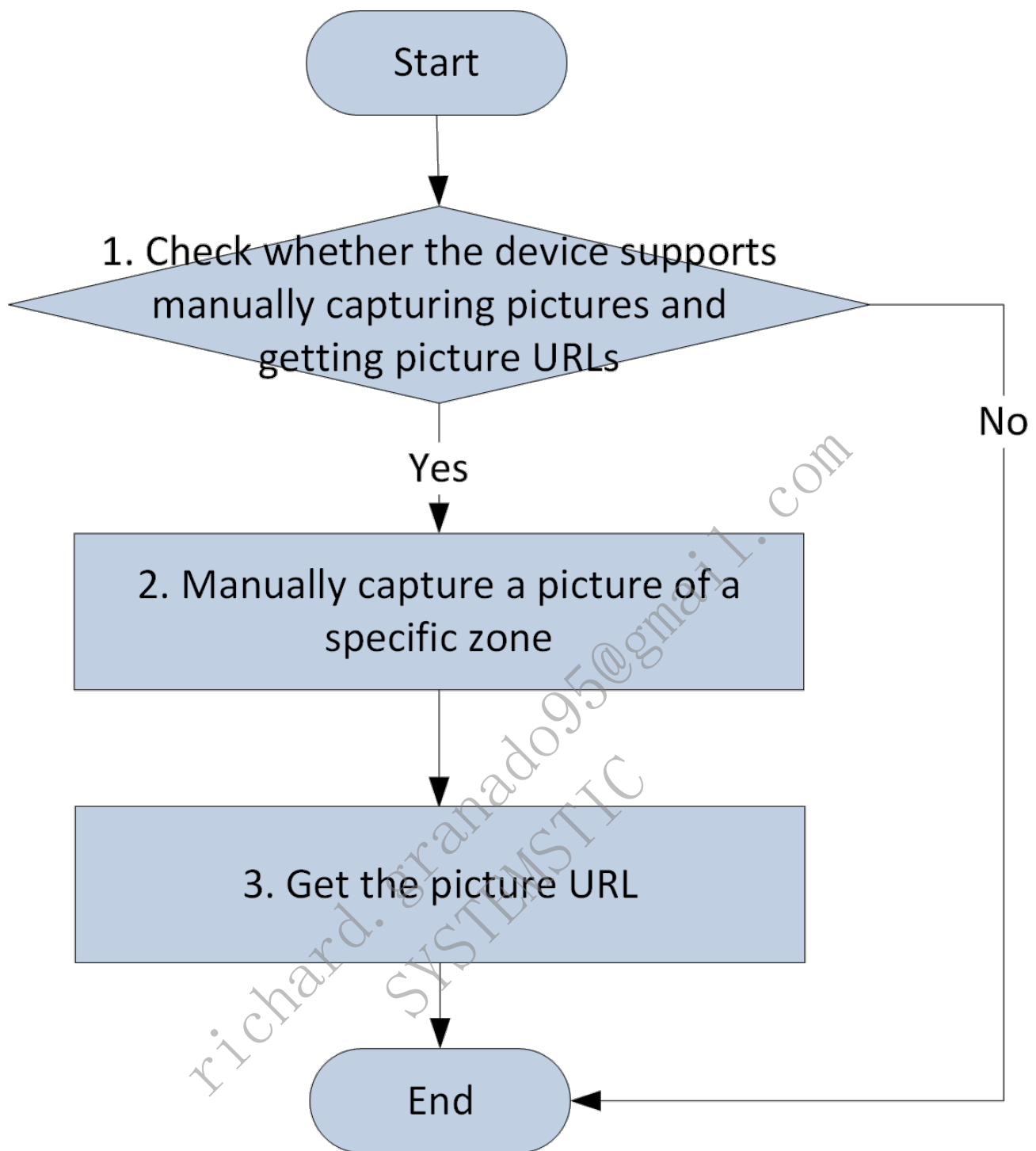


API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports capturing pictures via pircams: GET /ISAPI/SecurityCP/Configuration/capabilities?format=json. If the returned value of the node isSptPircamCapture is true, the device supports this function.

2. Send a request for capturing a picture (the captured picture will be uploaded via the alarm information): GET /ISAPI/Streaming/channels/<zoneID>/picture/devicePush. The zoneID in the URL refers to zone No.

11.3.2 Get Captured Pictures in Asynchronous Mode



API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports manually capturing pictures and getting picture URLs: `GET /ISAPI/SecurityCP/capabilities?format=json`. If the returned values of the node `isSptManualControlCapture` and `isSptGetPictureByUrl` is true, the device supports the functions above.

2. Manually capture a picture of a specific zone: `PUT /ISAPI/SecurityCP/Zone/<zoneID>/Capture/ManualControlCapture?format=json`. The `zoneID` in the URL refers to zone No.

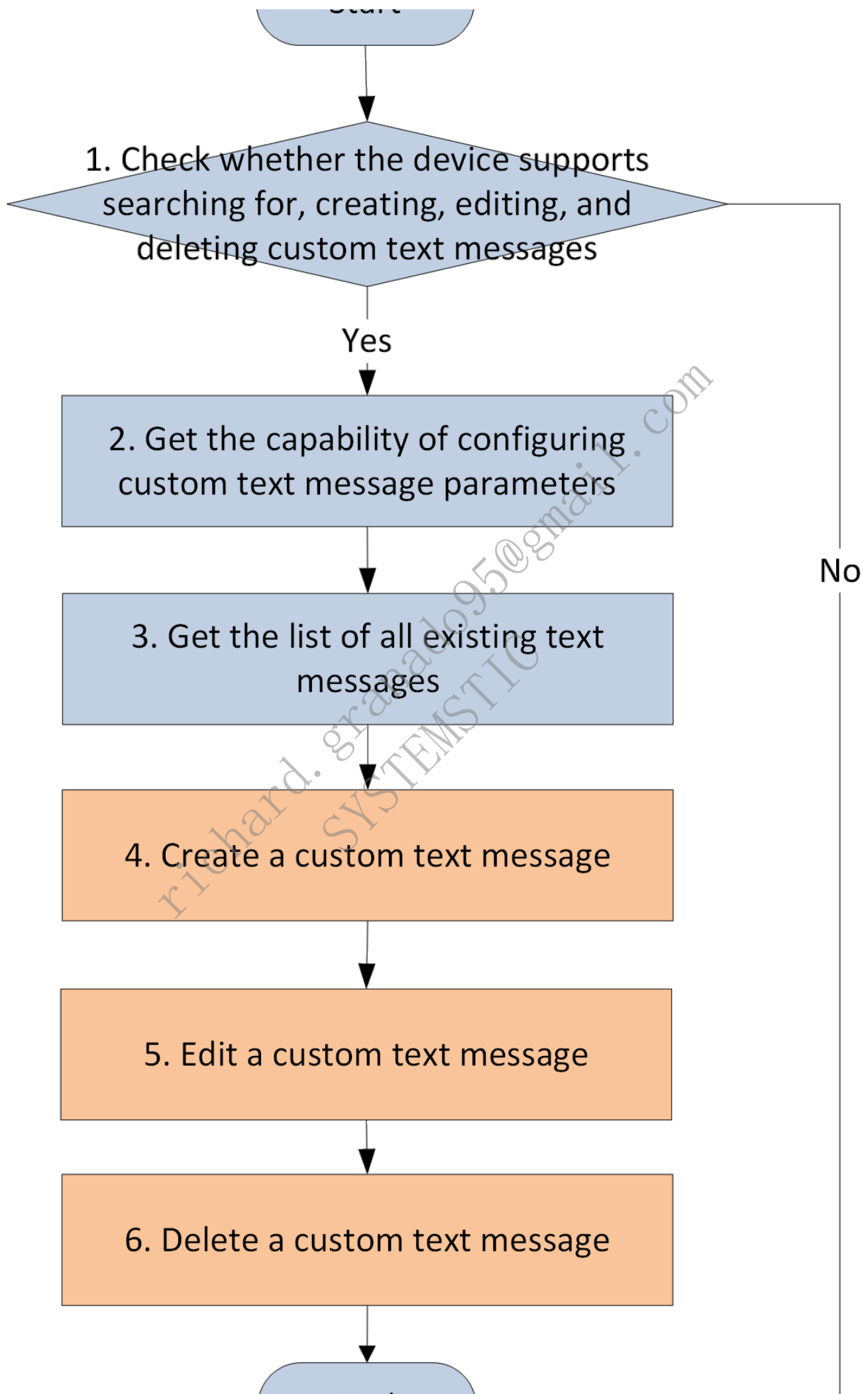
3. Get the picture URL: `GET /ISAPI/SecurityCP/Zone/<zoneID>/Capture/GetPictureByUrl?format=json`.

11.4 Custom Text Message Management

Telephone Notifications support both custom audios and custom message texts.

API Calling Flow:





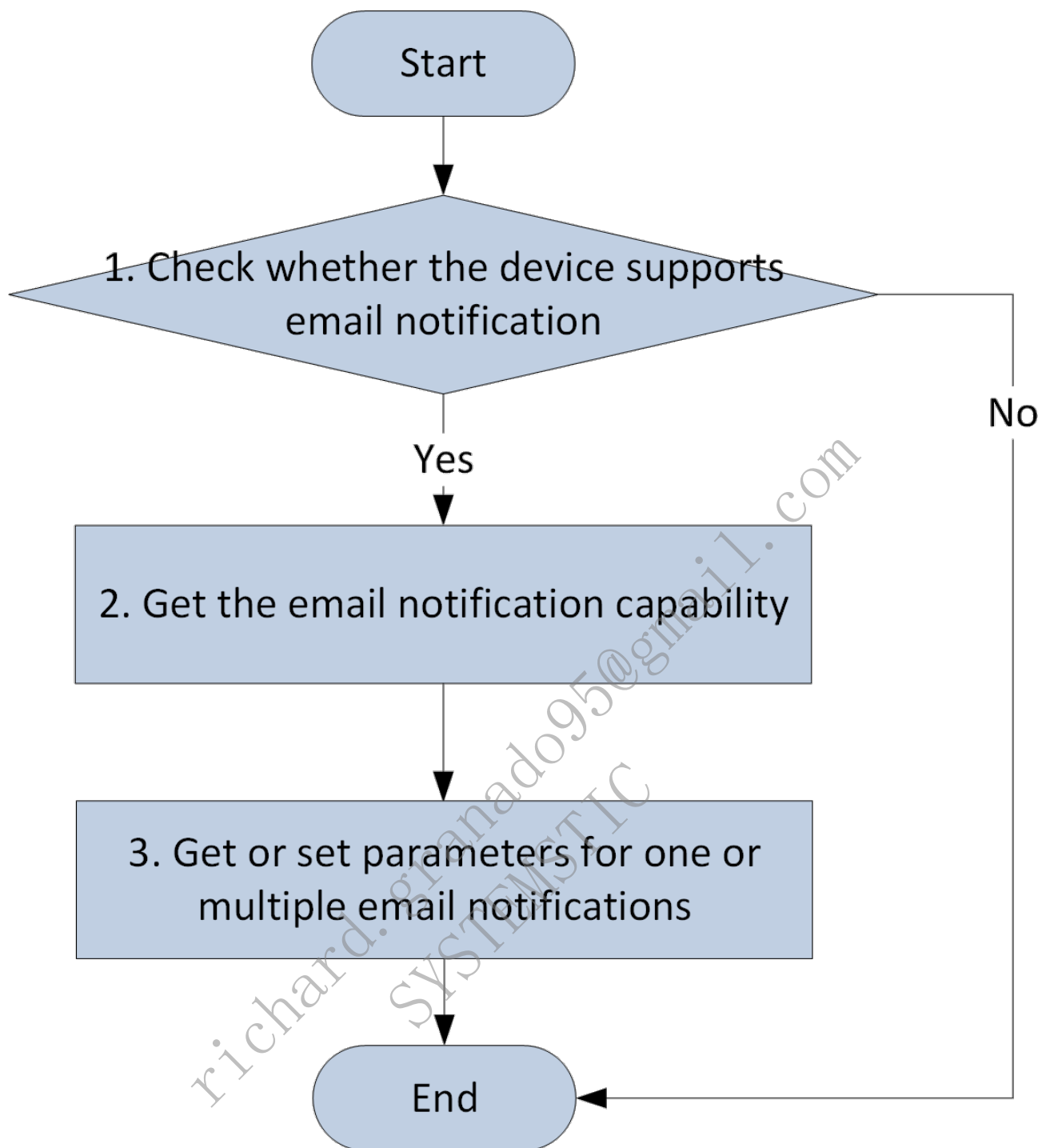
API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports searching for, creating, editing, and deleting custom text messages: GET /ISAPI/SecurityCP/Configuration/capabilities?format=json. If the returned values of isSupportSearchCustomAudioList, isSupportAddCustomMessage, isSupportModifyCustomMessage, and isSupportDeleteCustomMessage are all true, the device supports all the functions above.
2. Get the capability of configuring custom text message parameters: GET /ISAPI/SecurityCP/customMessage/capabilities?format=json.
3. Get the list of all existing text messages: GET /ISAPI/SecurityCP/customMessage/searchCustomMessageList?format=json&security=<security>&iv=<iv>. The node customMessageContent should be encrypted.
4. (Optional) Create a custom text message: POST /ISAPI/SecurityCP/customMessage/addCustomMessage?format=json&security=<security>&iv=<iv>.
5. (Optional) Edit a custom text message: PUT /ISAPI/SecurityCP/customMessage/modifyCustomMessage?format=json&security=<security>&iv=<iv>.
6. (Optional) Delete a custom text message: POST /ISAPI/SecurityCP/customMessage/deleteCustomMessage?format=json.

11.5 Email Notification Management

If the device supports email notification, you can enable the event notification and configure related email notification parameters.

API Calling Flow:



API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports email notification: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node `isSptMail` is true, the device supports this function.

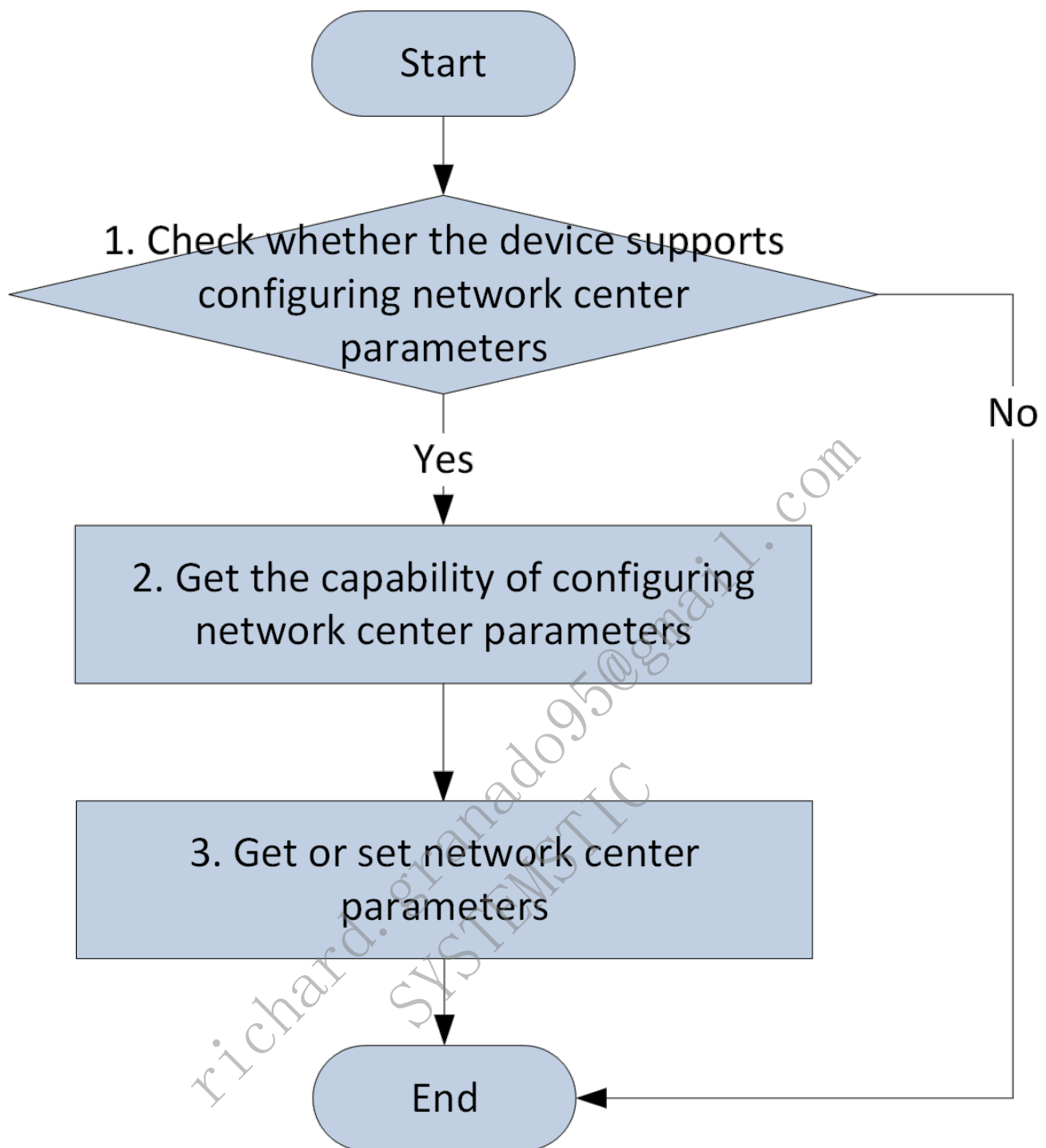
2. Get the email notification capability: `GET /ISAPI/SecurityCP/Configuration/messageSendMail/capabilities?format=json`.

3. Get parameters for multiple email notifications: `GET /ISAPI/SecurityCP/Configuration/messageSendMail?format=json`; set parameters for a single email notification: `PUT /ISAPI/SecurityCP/Configuration/messageSendMail/<mailID>?format=json`. The `mailID` in the URL is the email ID.

11.6 Network Center Parameter Configuration

For each NIC of the device, you can configure parameters of one or multiple network centers (one NIC can correspond to multiple network centers). Private protocol, NAL2300 protocol, and ISUP are supported currently.

API Calling Flow:



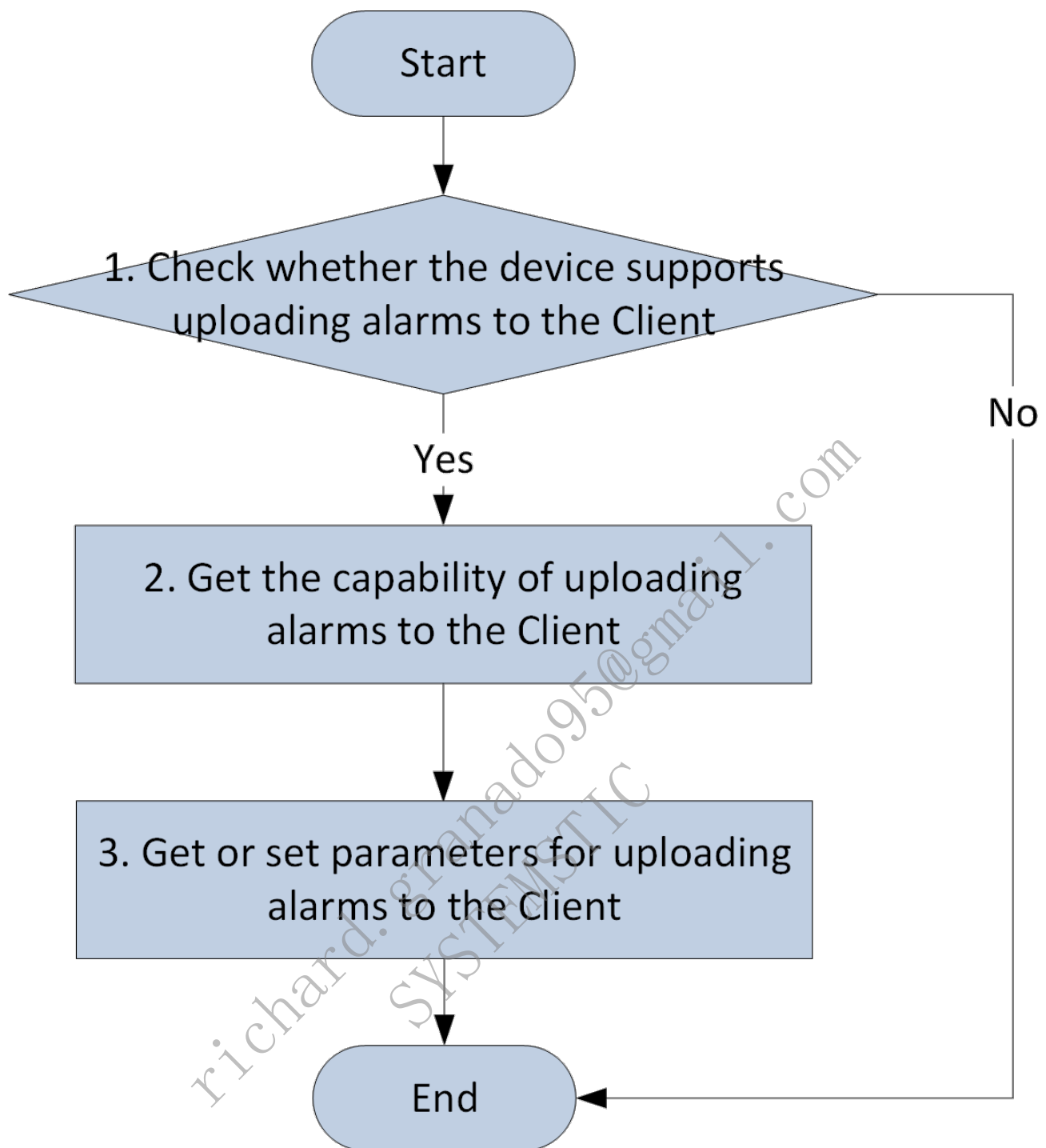
API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports configuring network center parameters: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node `isSptNetCfg` is true, the device supports this function.
2. Get the capability of configuring network center parameters: `GET /ISAPI/SecurityCP/NetCfg/capabilities?format=json`.
3. Get network center parameters: `GET /ISAPI/SecurityCP/NetCfg/<interfaceID>?format=json`; configure network center parameters: `PUT /ISAPI/SecurityCP/NetCfg/<interfaceID>?format=json`. The `interfaceID` in the URL refers to the NIC No.: 1 for the main NIC and 2 for the extended NIC.

11.7 Upload Alarms to Client

Events/alarms that can be uploaded to the Client (such as iVMS-4200) include tampering event, life security event, system status event, operation event, emergency alarm, medical alarm, and gas alarm.

API Calling Flow:



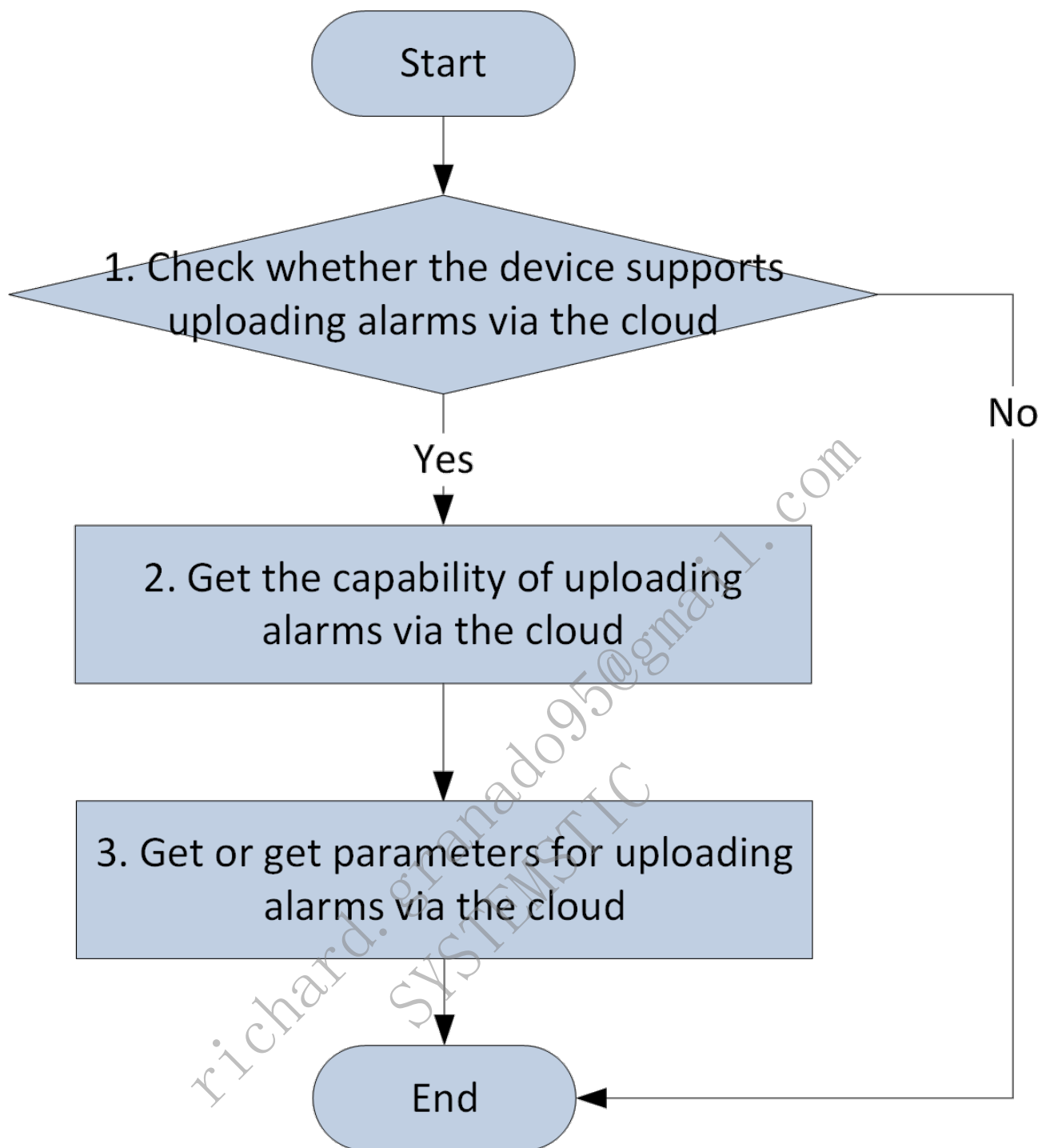
API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports uploading alarms to the Client: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node `isSptDirect` is true, the device supports this function.
2. Get the capability of uploading alarms to the Client: `GET /ISAPI/SecurityCP/Configuration/messageSendDirect/capabilities?format=json`.
3. Get parameters for uploading alarms to the Client: `GET /ISAPI/SecurityCP/Configuration/messageSendDirect?format=json`; set parameters: `PUT /ISAPI/SecurityCP/Configuration/messageSendDirect?format=json`.

11.8 Upload Alarms via Cloud

Events/alarms that can be uploaded to apps via the cloud include tampering event, life security event, system status event, operation event, emergency alarm, medical alarm, and gas alarm.

API Calling Flow:



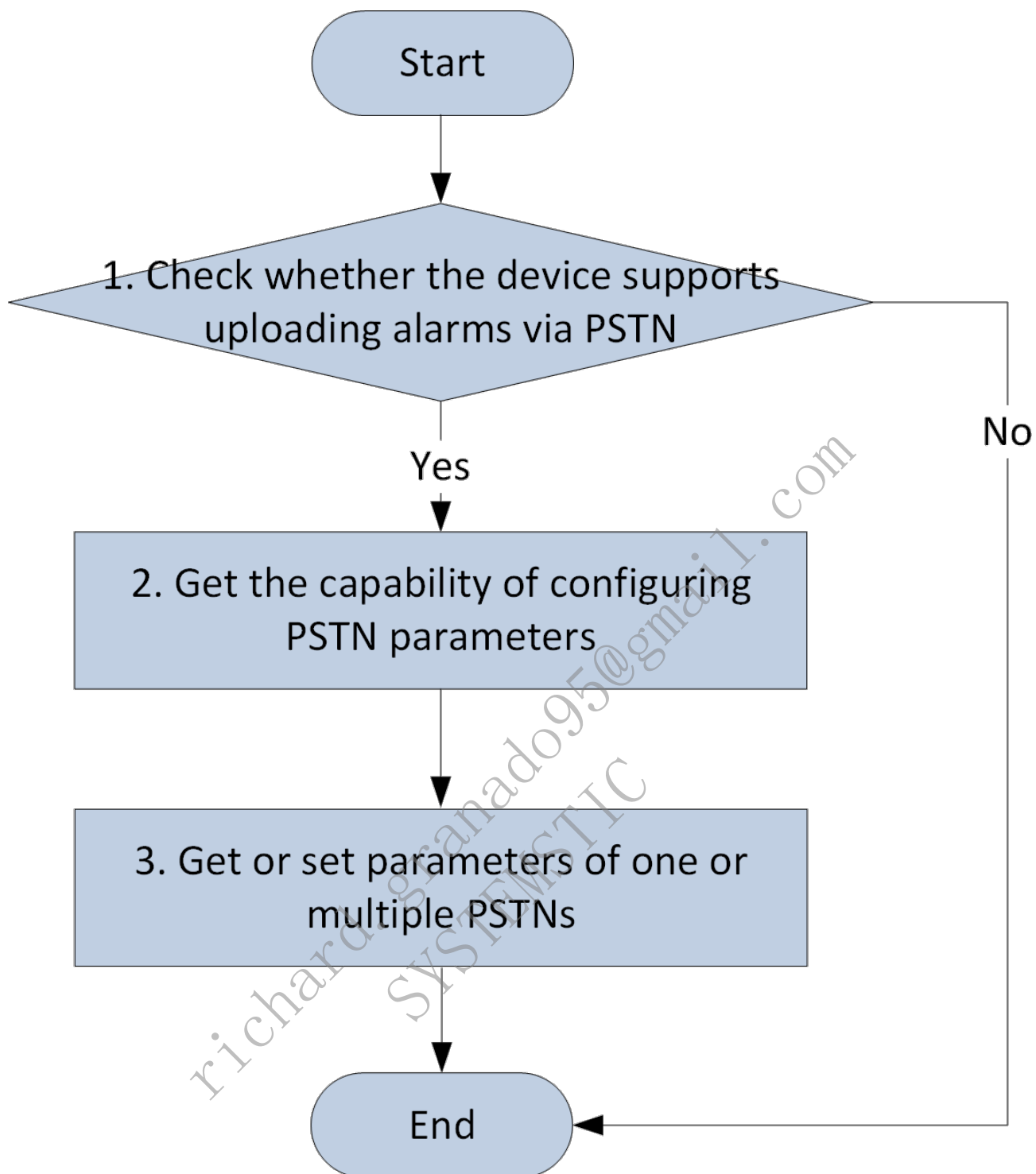
API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports uploading alarms via the cloud: GET /ISAPI/SecurityCP/Configuration/capabilities?format=json. If the returned value of the node isSptCloud is true, the device supports this function.
2. Get the capability of uploading alarms via the cloud: GET /ISAPI/SecurityCP/Configuration/messageSendCloud/capabilities?format=json.
3. Get parameters for uploading alarms via the cloud: GET /ISAPI/SecurityCP/Configuration/messageSendCloud?format=json; set parameters: PUT /ISAPI/SecurityCP/Configuration/messageSendCloud?format=json.

11.9 Upload Alarms via PSTN

If the device supports Public Switched Telephone Network (PSTN), the alarms can be uploaded by dial-up and the dial-up target can be an ARC or a person.

API Calling Flow:



API Calling Steps:

1. Get the configuration capability of the security control panel to check whether the device supports uploading alarms via PSTN: `GET /ISAPI/SecurityCP/Configuration/capabilities?format=json`. If the returned value of the node `isSptPSTNCfg` is true, the device supports this function.
2. Get the capability of configuring PSTN parameters: `GET /ISAPI/SecurityCP/Configuration/PSTNCfg/capabilities?format=json`.
3. Get parameters of all PSTNs: `GET /ISAPI/SecurityCP/Configuration/PSTNCfg?format=json`; set parameters of a single PSTN: `PUT /ISAPI/SecurityCP/Configuration/PSTNCfg/<indexID>?format=json`. Available parameters include whether to enable PSTN link, telephone of the target, communication protocol (CID), and transmission method (0#DTMF 5/S, 1#DTMF 10/S). The `indexID` in the URL is the PSTN No.

12 API Reference

12.1 Device Management (General)

12.1.1 Device Basic Information Management

12.1.1.1 Get configuration capability of the device information parameters

Request URL

GET /ISAPI/System/deviceInfo/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName min="0" max="132">
    <!--ro, req, string, device name, range:[1,132], attr:min{opt, string},max{opt, string}-->test
  </deviceName>
  <deviceId min="1" max="128">
    <!--ro, opt, string, device ID, range:[1,128], attr:min{opt, string},max{opt, string}-->123456
  </deviceId>
  <deviceDescription min="1" max="128">
    <!--ro, opt, string, device description, range:[1,128], attr:min{opt, string},max{opt, string}-->test
  </deviceDescription>
  <deviceLocation opt="STD-CGI">
    <!--ro, opt, string, device location, range:[1,32], attr:opt{opt, string}-->hangzhou
  </deviceLocation>
  <systemContact opt="STD-CGI">
    <!--ro, opt, string, manufacturer, range:[1,32], attr:opt{opt, string}-->test
  </systemContact>
  <model min="1" max="64">
    <!--ro, opt, string, device model, range:[1,64], attr:min{opt, string},max{opt, string}-->iDS-9632NX-I8/X
  </model>
  <serialNumber min="1" max="48">
    <!--ro, opt, string, device serial No., range:[1,48], attr:min{opt, string},max{opt, string}-->iDS-9632NX-I8/X1620181209CCRR77605411WCVU
  </serialNumber>
  <macAddress min="1" max="64">
    <!--ro, opt, string, MAC address, range:[1,64], attr:min{opt, string},max{opt, string}-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion min="1" max="64">
    <!--ro, opt, string, device firmware version, range:[1,64], attr:min{opt, string},max{opt, string}-->V4.1.40
  </firmwareVersion>
  <firmwareReleaseDate min="1" max="64">
    <!--ro, opt, string, release date of the device firmware version, range:[1,64], attr:min{opt, string},max{opt, string}-->build 181129
  </firmwareReleaseDate>
  <encoderVersion min="1" max="64">
    <!--ro, opt, string, device description, range:[1,64], attr:min{opt, string},max{opt, string}-->V7.3
  </encoderVersion>
  <encoderReleaseDate min="1" max="64">
    <!--ro, opt, string, release date, range:[1,64], attr:min{opt, string},max{opt, string}-->build 20161223
  </encoderReleaseDate>
  <bootVersion min="1" max="16">
    <!--ro, opt, string, boot version, range:[1,16], attr:min{opt, string},max{opt, string}-->V1.3.4
  </bootVersion>
  <bootReleaseDate min="1" max="16">
    <!--ro, opt, string, release date of boot, range:[1,16], attr:min{opt, string},max{opt, string}-->100316
  </bootReleaseDate>
  <panelVersion min="1" max="64">
    <!--ro, opt, string, range:[1,64], attr:min{opt, string},max{opt, string}-->V1.0
  </panelVersion>
  <hardwareVersion min="1" max="24">
    <!--ro, opt, string, hardware version, range:[1,24], attr:min{opt, string},max{opt, string}-->0x0
  </hardwareVersion>
  <subDeviceType
    opt="accessControlTerminal,attendanceCheckDevice,multiChannelAccessController,multiChannelTraditionalAccessController,multiChannelWirelessAccessController,personAndIdCardDevice,smartAnalysisTerminal,doorStation,indoor,mainStation,dropGate,wingGate,threeRollerGate,swingGate,elevatorSecurityGW,visitTerminal,doorTerminal,bedheadTerminal,bedsideTerminal,outerDoorStation,doorPhone,villaDoorStation,twoWireDoorStation,personnelChannelTerminal,smartPad,entrywayStation,TSS900,NormalEntrance,FaceEntrance,HighSpeedEntrance,DirPassEntrance,TCLParking,TCPParking,VPEParking,TCRParking,NormalParking,VISDecoder,FPLinkageCamera,FPTempCamera,FPSmartCamera,thermalFireDetector,visualSmokeDetector,imageFireDetector,visualInfraredFireDetector,slidingGateOpener,indoorScreen,outdoorScreen,videoPanicAlarmPanel,nonVideoPanicAlarmPanel,panicAlarmStation,panicAlarmBox">
    <!--ro, opt, string, device sub type., attr:opt{req, string}-->accessControlTerminal
  </subDeviceType>
  <telecontrolID min="1" max="255">
    <!--ro, opt, int, remote control ID, range:[1,255], attr:min{opt, string},max{opt, string}-->8
  </telecontrolID>
  <subChannelEnabled>
    <!--ro, opt, bool, whether to enable two channels-->true
  </subChannelEnabled>
  <thrChannelEnabled>
    <!--ro, opt, bool, whether to enable three channels-->true
  </thrChannelEnabled>
  <powerOnMode opt="button,adapter" def="button">
    <!--ro, opt, enum, device startup mode, subType:string, attr:opt{opt, string},def{opt, string}, desc:"button", "adapter"-->button
  </powerOnMode>
```

```

</customizeInfo>
<customizeInfo min="0" max="128">
  <!--ro, opt, string, device custom No., range:[0,128], attr:min{req, int},max{req, int}, desc:for baseline devices, returning this node is not required-->
</customizeInfo>
</customizeInfo>
<languageType
opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech,brazilianPortuguese,slovenian,swedish,norwegian,romanian,danish,german,polish,dutch,hungarian,slovak,serbian,southAmericanSpanish,ukrainian,croatian,irish,bulgarian,hebrew,thai,indonesian,arabic,traditionalChinese,lithuanian,anglicism,estonian,albanian">
  <!--ro, opt, string, language type, range:[1,32], attr:opt{opt, string}, desc:"chinese", "english", "spanish", "portuguese", "italian", "french", "russian", "turkish", "greek", "czech", "brazilianPortuguese", "slovenian", "swedish", "norwegian", "romanian", "danish", "german", "polish", "dutch", "hungarian", "slovak", "serbian", "southAmericanSpanish", "ukrainian", "croatian", "irish", "bulgarian", "hebrew", "thai", "indonesian", "arabic", "traditionalChinese", "lithuanian", "anglicism", "estonian", "albanian"-->chinese
</languageType>
<localZoneNum min="0" max="16">
  <!--ro, opt, int, number of local zones, range:[0,16], attr:min{opt, string},max{opt, string}-->1
</localZoneNum>
<alarmOutNum min="0" max="16">
  <!--ro, opt, int, number of alarm outputs, range:[0,16], attr:min{opt, string},max{opt, string}-->1
</alarmOutNum>
<relayNum min="0" max="16">
  <!--ro, opt, int, number of local relays, range:[0,16], attr:min{opt, string},max{opt, string}-->1
</relayNum>
<electroLockNum min="0" max="16">
  <!--ro, opt, int, number of local electronic locks, range:[0,16], attr:min{opt, string},max{opt, string}-->1
</electroLockNum>
<sirenNum min="0" max="16">
  <!--ro, opt, int, number of sirens, range:[0,16], attr:min{opt, string},max{opt, string}-->1
</sirenNum>
<alarmLamp min="0" max="16">
  <!--ro, opt, int, number of alarm lights, range:[0,16], attr:min{opt, string},max{opt, string}-->1
</alarmLamp>
<RS485Num min="0" max="16">
  <!--ro, opt, int, number of local RS-485, range:[0,16], attr:min{opt, string},max{opt, string}-->1
</RS485Num>
<DockStation>
  <!--ro, opt, object, dock station information configuration-->
  <Platform>
    <!--ro, opt, object, information configuration of the dock station accessing the platform-->
    <ip>
      <!--ro, opt, string, IP address, range:[1,32]-->0.0.0.0
    </ip>
    <port>
      <!--ro, opt, int, port No., range:[1,65535]-->1
    </port>
    <userName>
      <!--ro, req, string, user name, range:[1,32]-->test
    </userName>
    <password>
      <!--ro, req, string, password, range:[1,16]-->test
    </password>
  </Platform>
  <centralStorageBackupEnabled opt="true, false">
    <!--ro, opt, bool, whether to enable center storage backup, attr:opt{opt, string}, desc:when it is enabled, the device uploads all data to the center storage for backup before uploading them to the platform; when it is disabled, the device uploads data to the platform directly, and for those important data, the platform will apply the file list to the device for center storage (related URI: /ISAPI/ContentMgmt/record/control/centralStorageBackup), then the device will upload the data with storage URI to the platform-->true
  </centralStorageBackupEnabled>
  <isSupportManufacturer>
    <!--ro, opt, bool, whether it supports configuring third-party manufacturer for the body camera, desc:related URI: /ISAPI/Traffic/dockStation/slot/Manufacturer/capabilities?format=json-->true
  </isSupportManufacturer>
</DockStation>
<webVersion>
  <!--ro, opt, string, Web version No., range:[1,32]-->1.0
</webVersion>
<deviceRFProgramVersion>
  <!--ro, opt, string, device RF program version No., range:[1,32]-->1.0
</deviceRFProgramVersion>
<securityModuleSerialNo>
  <!--ro, opt, string, security module serial No., range:[1,48]-->1.0
</securityModuleSerialNo>
<securityModuleVersion>
  <!--ro, opt, string, security module version No., range:[1,32]-->1.0
</securityModuleVersion>
<securityChipVersion>
  <!--ro, opt, string, security chip version No., range:[1,32]-->1.0
</securityChipVersion>
<securityModuleKeyVersion>
  <!--ro, opt, string, security module key version No., range:[1,32]-->1.0
</securityModuleKeyVersion>
<UIDLampRecognition>
  <!--ro, opt, object, recognize the device information by UID-->
  <enabled opt="true, false">
    <!--ro, opt, bool, whether to enable, attr:opt{opt, string}-->true
  </enabled>
</UIDLampRecognition>
<confDeviceIdPrefix opt="true, false">
  <!--ro, opt, bool, whether the meeting uses the device name as its prefix, attr:opt{opt, string}-->true
</confDeviceIdPrefix>
<simpleAlgorithmVersion min="0" max="10">
  <!--ro, opt, string, single algorithm version, attr:min{req, int},max{req, int}-->test
</simpleAlgorithmVersion>
<bootTime>

```



```

    <!--ro, opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
</bootTime>
<!--ro, opt, bool, whether it supports switching languages for the new protocol version-->true
</isSupportNewVersionDevLanguageSwitch>
<intelligentAnalysisEngineModel>
    <!--ro, opt, string, engine model, range:[1,32]-->K81
</intelligentAnalysisEngineModel>
<marketType opt="0,1,2">
    <!--ro, opt, enum, market type, subType:int, attr:opt{req, string}, desc:0 (invalid), 1 (distribution type), 2 (industry type)-->0
</marketType>
<protocolFileURL min="1" max="32">
    <!--ro, opt, string, protocol notice URI, attr:min{opt, string},max{opt, string}-->test
</protocolFileURL>
<recycleRecordEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable overwritten recording, attr:opt{req, string}-->>false
</recycleRecordEnabled>
<decordChannelNums>
    <!--ro, opt, int, number of decoding channels-->0
</decordChannelNums>
<VGAEnums>
    <!--ro, opt, int, number of VGA ports-->0
</VGAEnums>
<USBNums>
    <!--ro, opt, int, number of USB ports-->0
</USBNums>
<auxoutNums>
    <!--ro, opt, int, number of auxiliary ports-->0
</auxoutNums>
<expansionBoardVersion min="0" max="32">
    <!--ro, opt, string, extension board version information, range:[0,32], attr:min{req, int},max{req, int}-->test
</expansionBoardVersion>
<initWizzardDisplay opt="true,false">
    <!--ro, opt, bool, whether it displays initialization wizard, attr:opt{req, string}, desc:it is only used for displaying configuration of applications
    integrated to the wizard such as Local GUI. By default, the value is true; it is false after the wizard is configured on the Local GUI-->true
</initWizzardDisplay>
<beaconID min="0" max="32">
    <!--ro, opt, string, device RF program version No., range:[0,32], attr:min{req, int},max{req, int}, desc:by default, it is the current value-->test
</beaconID>
<isResetDeviceLanguage opt="true,false">
    <!--ro, opt, bool, whether to reset the Language, attr:opt{req, string}, desc:only Admin and Installer have the permission to switch the device
    Language. If the value is true, the Hik-Connect Mobile Client and the device's web page display device information in the target Language; if the value is
    false, the Language switching takes no effect on the Hik-Connect Mobile Client and the device's web page-->>false
</isResetDeviceLanguage>
<dispalyNum>
    <!--ro, opt, int, number of device screens-->2
</dispalyNum>
<bspVersion min="1" max="1">
    <!--ro, opt, string, BSP version, attr:min{req, int},max{req, int}-->test
</bspVersion>
<dspVersion min="1" max="1">
    <!--ro, opt, string, DSP version, attr:min{req, int},max{req, int}-->test
</dspVersion>
<localUIVersion min="1" max="1">
    <!--ro, opt, string, Local UI version, attr:min{req, int},max{req, int}-->test
</localUIVersion>
<OPCASubType opt="PDA,pump,column,autoSampler,SOC">
    <!--ro, opt, enum, the sub type of optical physical and chemical analyzer, subType:string, attr:opt{req, string}, desc:"detector"; "pump"; "columnOven"
    (column oven); "autoSampler" (auto sampler)-->detector
</OPCASubType>
<wiegandOutNum min="0" max="1">
    <!--ro, opt, int, number of Wiegand outputs, range:[0,1], attr:min{opt, string},max{opt, string}-->1
</wiegandOutNum>
<ChipVersionInfoList size="6">
    <!--ro, opt, object, chip version information list, attr:size{req, int}-->
    <ChipVersionInfo>
        <!--ro, opt, object, chip version information-->
        <ID min="0" max="1">
            <!--ro, req, int, chip No., range:[0,1], attr:min{req, int},max{req, int}, desc:the No.s of chips of different types can be duplicate. Users can
            distinguish them by chip type and chip No. together-->1
        </ID>
        <firmwareVersion min="0" max="64">
            <!--ro, req, string, chip firmware version, range:[0,64], attr:min{req, int},max{req, int}-->1.0.0
        </firmwareVersion>
        <chipName min="0" max="32">
            <!--ro, opt, string, chip name, range:[0,32], attr:min{req, int},max{req, int}-->test
        </chipName>
    </ChipVersionInfo>
</ChipVersionInfoList>
<personBagLinkAlgoEngineVersion min="0" max="64">
    <!--ro, opt, string, engine version of the person & package Linkage module, range:[0,64], attr:min{req, int},max{req, int}, desc:this node is for
    analyzer (security inspection)-->1.0.0
</personBagLinkAlgoEngineVersion>
<BIOSVersion min="0" max="16">
    <!--ro, opt, string, BIOS version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
</BIOSVersion>
<contactInformation min="0" max="64">
    <!--ro, opt, string, contact information, range:[0,64], attr:min{req, int},max{req, int}-->test
</contactInformation>
<PedestrianWarningModuleVersion>
    <!--ro, opt, object, pedestrian warning module information, desc:this node is returned only for pedestrian warning devices-->
    <pedestrianWarningMCUVersion min="1" max="32">
        <!--ro, opt, string, MCU program version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningMCUVersion>
    <pedestrianWarningPDAVersion min="1" max="32">
        <!--ro, opt, string, PDA program version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningPDAVersion>
    <pedestrianWarningColumnVersion min="1" max="32">
        <!--ro, opt, string, Column program version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningColumnVersion>
    <pedestrianWarningAutoSamplerVersion min="1" max="32">
        <!--ro, opt, string, AutoSampler program version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningAutoSamplerVersion>
    <pedestrianWarningSOCVersion min="1" max="32">
        <!--ro, opt, string, SOC program version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningSOCVersion>
    <pedestrianWarningPumpVersion min="1" max="32">
        <!--ro, opt, string, Pump program version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningPumpVersion>
    <pedestrianWarningDetectorVersion min="1" max="32">
        <!--ro, opt, string, Detector program version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningDetectorVersion>
    <pedestrianWarningOtherVersion min="1" max="32">
        <!--ro, opt, string, Other program version, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
    </pedestrianWarningOtherVersion>
</PedestrianWarningModuleVersion>

```

```

</pedestrianWarningModuleVersion>
<pedestrianWarningRadarVersion min="1" max="32">
  <!--ro, opt, string, the program version of pedestrian warning radar, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
</pedestrianWarningRadarVersion>
<pedestrianRangingModuleVersion min="1" max="32">
  <!--ro, opt, string, the program version of the pedestrian distance measurement module, range:[0,16], attr:min{req, int},max{req, int}-->V1.3.4
</pedestrianRangingModuleVersion>
</PedestrianWarningModuleVersion>
<encryptionModel min="0" max="16">
  <!--ro, opt, string, device encryption model, range:[0,16], attr:min{req, int},max{req, int}, desc:it cannot be modified and is a hexadecimal digit
currently-->test
</encryptionModel>
<UmbVersion min="1" max="32">
  <!--ro, opt, string, Umb version, range:[1,32], attr:min{req, int},max{req, int}-->V1.3.4
</UmbVersion>
<audioBoard>
  <!--ro, opt, object, audio board information-->
  <audioBoardModel min="0" max="32">
    <!--ro, opt, string, audio board model, range:[0,32], attr:min{req, int},max{req, int}, desc:currently only C8 and R8 are supported-->C8
  </audioBoardModel>
  <audioBoardVersion min="0" max="32">
    <!--ro, opt, string, audio board version, range:[0,32], attr:min{req, int},max{req, int}-->V1.0.0 build 211210
  </audioBoardVersion>
</audioBoard>
<materialScanAlgorithmVersion min="0" max="64">
  <!--ro, opt, string, material scan algorithm version, range:[0,64], attr:min{req, int},max{req, int}, desc:algorithm version of imaging radar panel for
material scan-->1.0.0
</materialScanAlgorithmVersion>
<productionDate>
  <!--ro, opt, date, date of manufacture-->2022-04-02
</productionDate>
<wifiModuleMACAddress min="1" max="48">
  <!--ro, opt, string, Mac address of Wi-Fi module, range:[1,48], attr:min{req, int},max{req, int}-->44:47:cc:c8:d9:e4
</wifiModuleMACAddress>
<isSupportExportDeviceFingerFile>
  <!--ro, opt, bool, whether the device supports exporting device fingerprint file, desc:POST /ISAPI/System/deviceInfo/ExportDeviceFingerFile?format=json-
-->true
</isSupportExportDeviceFingerFile>
<isSupportPromptMessage>
  <!--ro, opt, bool, whether the device supports configuring the device prompt information, desc:GET/PUT /ISAPI/System/deviceInfo/PromptMessage?
format=json-->true
</isSupportPromptMessage>
<shortSerialNumber min="1" max="9">
  <!--ro, opt, string, short serial No., range:[1,9], attr:min{req, int},max{req, int}-->test
</shortSerialNumber>
<OSCoreVersionInfo>
  <!--ro, opt, object, OS/core component version information, desc:read-only-->
  <OSCoreVersion min="1" max="32">
    <!--ro, req, string, OS/core component version No., range:[1,32], attr:min{req, int},max{req, int}-->1.0.0
  </OSCoreVersion>
  <minisysVersion min="1" max="32">
    <!--ro, opt, string, minisys version No., range:[1,32], attr:min{req, int},max{req, int}-->1.0.0
  </minisysVersion>
  <networkServiceVersion min="1" max="32">
    <!--ro, opt, string, network service version No., range:[1,32], attr:min{req, int},max{req, int}-->1.0.0
  </networkServiceVersion>
  <upgradeServiceVersion min="1" max="32">
    <!--ro, opt, string, upgrade service version No., range:[1,32], attr:min{req, int},max{req, int}-->1.0.0
  </upgradeServiceVersion>
</OSCoreVersionInfo>
<isSupportExportDeviceIcon>
  <!--ro, opt, bool, whether the device supports exporting device icons, desc:POST /ISAPI/System/deviceInfo/ExportDeviceIcon?format=json-->true
</isSupportExportDeviceIcon>
<xTransVersion min="1" max="32">
  <!--ro, opt, string, version No. of X-Trans chip communication component, range:[1,32], attr:min{req, int},max{req, int}, desc:the version number of
analyzer's chip communication component, which implements communication among network, USB, and PCIe-->1.0.0
</xTransVersion>
<isSupportSearchHardwareAssets>
  <!--ro, opt, bool, whether the device supports searching for device hardware asset data, desc:POST /ISAPI/System/deviceInfo/SearchHardwareAssets?
format=json-->true
</isSupportSearchHardwareAssets>
<isSupportExportDeviceAssets>
  <!--ro, opt, bool, whether the device supports exporting device asset data, desc:GET /ISAPI/System/deviceInfo/ExportDeviceAssets?format=json-->true
</isSupportExportDeviceAssets>
<screenVersionInfo>
  <!--ro, opt, object, device display version information, desc:device (such as wall-mounted dock station) that has dual systems, that is embedded device
and Android system screen-->
  <screenAndroidOSVersion min="1" max="64">
    <!--ro, opt, string, version number of device screen's Android system, range:[1,64], attr:min{req, int},max{req, int}, desc:it is the Android kernel
version but not the Android version-->1.2.0 build20230101
  </screenAndroidOSVersion>
  <screenAndroidAPPVersion min="1" max="64">
    <!--ro, opt, string, device screen's Android app version number, range:[1,64], attr:min{req, int},max{req, int}-->1.0.0 build20230101
  </screenAndroidAPPVersion>
</screenVersionInfo>
<isSupportDeviceWebInfo>
  <!--ro, opt, bool, whether the device web information is supported, desc:GET or PUT /ISAPI/System/deviceInfo/DeviceWebInfo?format=json-->true
</isSupportDeviceWebInfo>
<ptzVersion min="1" max="32">
  <!--ro, opt, string, PTZ version No., attr:min{req, int},max{req, int}-->test
</ptzVersion>
<movementSoftVersion min="1" max="32">
  <!--ro, opt, string, module software version, attr:min{req, int},max{req, int}-->test
</movementSoftVersion>

```

```

<movementHardVersion min="1" max="32">
  <!--ro, opt, string, module hardware version, attr:min{req, int},max{req, int}-->test
</movementHardVersion>
<isSupportPipeID>
  <!--ro, opt, bool, whether building pipeline (PIPEID), desc:GET /ISAPI/System/deviceInfo/PipeID?format=json-->true
</isSupportPipeID>
<isSupportSerialNumberInURL>
  <!--ro, opt, bool, whether URL supports containing serialNumber, desc:if the node is not returned, it indicates no. If true is returned, serialNumber
can be passed in URL GET or PUT /ISAPI/System/deviceInfo-->true
</isSupportSerialNumberInURL>
<loginPassword min="1" max="16">
  <!--ro, opt, string, confirm password, attr:min{req, int},max{req, int}-->test
</loginPassword>
<isSupportDeviceGuideParams>
  <!--ro, opt, bool, whether the device supports device guidance parameters, desc:GET or PUT /ISAPI/System/deviceInfo/DeviceGuideParams?format=json-->true
</isSupportDeviceGuideParams>
<deviceMode opt="factoryMode,maintenanceMode">
  <!--ro, opt, enum, device mode, subType:string, attr:opt{req, string}, desc:"factoryMode" (factory mode), "maintenanceMode" (maintenance mode)--
>factoryMode
</deviceMode>
<maintenanceDuration min="1" max="30">
  <!--ro, opt, int, range:[1,30], unit:d, attr:min{req, int},max{req, int}-->1
</maintenanceDuration>
<pirAlgorithmVersion min="0" max="32">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->V1.0
</pirAlgorithmVersion>
<isSupportGetLensParams>
  <!--ro, opt, bool-->true
</isSupportGetLensParams>
<deviceCabinID min="0" max="128">
  <!--ro, opt, int, range:[0,128], attr:min{req, int},max{req, int}-->1
</deviceCabinID>
<detectorVersion>
  <!--ro, opt, string, range:[1,32]-->test
</detectorVersion>
<isSupportSearchDeviceDistribution>
  <!--ro, opt, bool-->true
</isSupportSearchDeviceDistribution>
<isSupportDeviceSubVersion>
  <!--ro, opt, bool-->true
</isSupportDeviceSubVersion>
<isSupportSendDevOnPlatformInfo>
  <!--ro, opt, bool-->true
</isSupportSendDevOnPlatformInfo>
<isSupportGetPortLocationInfo>
  <!--ro, opt, bool-->true
</isSupportGetPortLocationInfo>
<isSupportDeviceServiceDescription>
  <!--ro, opt, bool-->true
</isSupportDeviceServiceDescription>
<storageType opt="TF,HD">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->TF
</storageType>
<isSupportExternalDeviceInfo>
  <!--ro, opt, bool-->true
</isSupportExternalDeviceInfo>
<isSupportDeviceLongitudeLatitude>
  <!--ro, opt, bool-->true
</isSupportDeviceLongitudeLatitude>
</DeviceInfo>

```

12.1.1.2 Get device information parameters

Request URL

GET /ISAPI/System/deviceInfo?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--ro, req, string, device name, range:[0,132]-->test
  </deviceName>
  <deviceID>
    <!--ro, req, string, device No., range:[1,128]-->test
  </deviceID>

```

```

</deviceID>
<deviceDescription>
  <!--ro, opt, string, device description, range:[1,128]-->test
</deviceDescription>
<deviceLocation>
  <!--ro, opt, string, device location, range:[1,128]-->hangzhou
</deviceLocation>
<deviceStatus>
  <!--ro, opt, enum, device status, subType:string, desc:"normal", "abnormal"-->normal
</deviceStatus>
<DetailAbnormalStatus>
  <!--ro, opt, object, error status details, desc:it is valid only when deviceStatus is "abnormal"-->
  <hardDiskFull>
    <!--ro, opt, bool, disk full-->true
  </hardDiskFull>
  <hardDiskError>
    <!--ro, opt, bool, disk error-->true
  </hardDiskError>
  <ethernetBroken>
    <!--ro, opt, bool, network disconnected-->true
  </ethernetBroken>
  <ipaddrConflict>
    <!--ro, opt, bool, IP address conflict-->true
  </ipaddrConflict>
  <illegalAccess>
    <!--ro, opt, bool, illegal access-->true
  </illegalAccess>
  <recordError>
    <!--ro, opt, bool, recording exception-->true
  </recordError>
  <raidLogicDiskError>
    <!--ro, opt, bool, virtual disk exception in the array-->true
  </raidLogicDiskError>
  <spareWorkDeviceError>
    <!--ro, opt, bool, hot spare active device exception-->true
  </spareWorkDeviceError>
</DetailAbnormalStatus>
<systemContact>
  <!--ro, opt, string, manufacturer, range:[1,32]-->STD-CGI
</systemContact>
<model>
  <!--ro, req, string, device model, range:[1,64]-->IDS-9632NX-I8/X
</model>
<serialNumber>
  <!--ro, req, string, device serial No., range:[1,48]-->IDS-9632NX-I8/X1620181209CCRRC77605411WCVU
</serialNumber>
<macAddress>
  <!--ro, req, string, MAC address, range:[1,64]-->44:47:cc:c8:d9:e4
</macAddress>
<firmwareVersion>
  <!--ro, req, string, device firmware version, range:[1,64]-->V4.1.40
</firmwareVersion>
<firmwareReleasedDate>
  <!--ro, opt, string, release date of the device firmware version-->2019-11-01
</firmwareReleasedDate>
<encoderVersion>
  <!--ro, opt, string, device encoder version No., range:[1,32]-->V7.3
</encoderVersion>
<encoderReleasedDate>
  <!--ro, opt, string, release date of the device encoder version-->2019-11-02
</encoderReleasedDate>
<bootVersion>
  <!--ro, opt, string, boot version, range:[1,16]-->V1.3.4
</bootVersion>
<bootReleasedDate>
  <!--ro, opt, string, release date of boot-->2019-11-03
</bootReleasedDate>
<panelVersion>
  <!--ro, opt, string, panel version, range:[1,32]-->V1.0
</panelVersion>
<hardwareVersion>
  <!--ro, opt, string, hardware version, range:[1,24]-->0x0
</hardwareVersion>
<decoderVersion>
  <!--ro, opt, string, device decoder version, range:[1,32]-->V1.0
</decoderVersion>
<decoderReleasedDate>
  <!--ro, opt, string, release date of the device decoder version-->2019-01-01
</decoderReleasedDate>
<softwareVersion>
  <!--ro, opt, string, software version, range:[1,32]-->V1.23
</softwareVersion>
<capacity>
  <!--ro, opt, int, device capacity, range:[1,10240], desc:unit: MB-->1
</capacity>
<upgradePackageMD5>
  <!--ro, opt, string-->2f003d3a733bf219b7afc538ff16943d
</upgradePackageMD5>
<usedCapacity>
  <!--ro, opt, int, used capacity of the device, range:[1,10240], desc:unit: MB-->1
</usedCapacity>
<subDeviceType>
  <!--ro, opt, enum, device sub type, subType:string, desc:"accessControlTerminal" (access control terminal; it is valid when the value of deviceType is

```

"ACS"), "attendanceCheckDevice" (attendance check devices; it is valid when the value of deviceType is "ACS"), "multiChannelAccessController" (multi-channel access controller; it is valid when the value of deviceType is "ACS"), "personAndIdCardDevice" (person and ID card device; it is valid when the value of deviceType is "ACS"), "doorStation" (door station; it is valid when the value of deviceType is "VIS"), "indoor" (indoor station; it is valid when the value of deviceType is "VIS"), "mainStation" (main station; it is valid when the value of deviceType is "VIS"), "dropGate" (swing barrier; it is valid when the value of deviceType is "PersonnelChannel"), "wingGate" (flap barrier; it is valid when the value of deviceType is "PersonnelChannel"), "threeRollerGate" (tripod turnstile; it is valid when the value of deviceType is "PersonnelChannel")-->accessControlTerminal

```
</subDeviceType>
<telecontrolID>
  <!--ro, opt, int, remote control ID, range:[1,255]-->1
</telecontrolID>
<supportBeep>
  <!--ro, opt, bool, whether it supports buzzer-->true
</supportBeep>
<supportVideoLoss>
  <!--ro, opt, bool, whether it supports video Loss-->true
</supportVideoLoss>
<firmwareVersionInfo>
  <!--ro, opt, string, firmware version information, range:[1,32]-->B-R-H5-0
</firmwareVersionInfo>
<actualFloorNum>
  <!--ro, opt, int, actual number of floors, range:[1,128]-->1
</actualFloorNum>
<localZoneNum>
  <!--ro, opt, int, number of Local zones, range:[0,16]-->1
</localZoneNum>
<alarmOutNum>
  <!--ro, opt, int, number of alarm outputs, range:[0,16]-->1
</alarmOutNum>
<alarmInNum>
  <!--ro, opt, int, number of alarm inputs, range:[0,16]-->1
</alarmInNum>
<distanceResolution>
  <!--ro, opt, float, distance resolution, range:[0.000,0.999]-->0.000
</distanceResolution>
<angleResolution>
  <!--ro, opt, float, angle resolution, range:[0.000,0.999]-->0.000
</angleResolution>
<speedResolution>
  <!--ro, opt, float, speed resolution, range:[0.000,0.999]-->0.000
</speedResolution>
<detectDistance>
  <!--ro, opt, float, detection distance, range:[0.000,0.999]-->0.000
</detectDistance>
<relayNum>
  <!--ro, opt, int, number of relays (Local), range:[0,16]-->1
</relayNum>
<electroLockNum>
  <!--ro, opt, int, number of Local electronic Locks, range:[0,16]-->1
</electroLockNum>
<sirenNum>
  <!--ro, opt, int, number of sirens, range:[0,16]-->1
</sirenNum>
<alarmLamp>
  <!--ro, opt, int, number of alarm lights, range:[0,16]-->1
</alarmLamp>
<RS485Num>
  <!--ro, opt, int, number of Local RS-485, range:[0,16]-->1
</RS485Num>
<radarVersion>
  <!--ro, opt, string, radar version, range:[1,32]-->test
</radarVersion>
<cameraModuleVersion>
  <!--ro, opt, string, camera module version, range:[1,32]-->test
</cameraModuleVersion>
<mainVersion>
  <!--ro, opt, int, main version No., range:[1,255]-->1
</mainVersion>
<subVersion>
  <!--ro, opt, int, sub version No., range:[1,255]-->1
</subVersion>
<upgradeVersion>
  <!--ro, opt, int, updated version No., range:[1,255]-->1
</upgradeVersion>
<customizeVersion>
  <!--ro, opt, int, customized version No., range:[1,255]-->1
</customizeVersion>
<companyName>
  <!--ro, opt, string, manufacturer name, range:[1,32]-->test
</companyName>
<copyright>
  <!--ro, opt, string, copyright Information, range:[1,32]-->test
</copyright>
<systemName>
  <!--ro, opt, enum, storage system name, subType:string, desc:"storageManagement" (storage management system), "distributedStorageManagement"
(distributed storage management system)-->storageManagement
</systemName>
<systemStatus>
  <!--ro, opt, enum, system status, subType:string, desc:"configured", "unConfigured" (not configured)-->configured
</systemStatus>
<isLeaderDevice>
  <!--ro, opt, bool, whether it is the corresponding device of the resource IP address-->true
</isLeaderDevice>
<clusterVersion>
  <!--ro, opt, string, custom cluster version, range:[1,32], desc:this node is valid when the value of isLeaderDevice is true, start
```

```

<!--ro, opt, string, system cluster version, range:[1,32], desc: this node is valid when the value of isLeaderDevice is true-->test
</clusterVersion>
<centralStorageVersion>
  <!--ro, opt, string, center storage version, range:[1,16]-->test
</centralStorageVersion>
<powerOnMode>
  <!--ro, opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
</powerOnMode>
<customizedInfo>
  <!--ro, opt, string, customized project No., range:[1,32], desc:no value will be returned if it is baseline device. Custom project No. will be returned
if is custom device-->test
</customizedInfo>
<verificationCode>
  <!--ro, opt, string, device verification code-->test
</verificationCode>
<supportUrl>
  <!--ro, opt, string, service portal-->test
</supportUrl>
<subSerialNumber>
  <!--ro, opt, string, sub serial No.-->test
</subSerialNumber>
<languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech,brazilianPortuguese">
  <!--ro, opt, enum, language type, subType:string, attr:opt{req, string}, desc:"chinese", "english", "spanish", "portuguese", "italian", "french",
"russian", "turkish", "greek", "czech", "brazilianPortuguese", "slovenian", "swedish", "norwegian", "romanian", "danish", "german", "polish", "dutch",
"hungarian", "slovak", "serbian", "southAmericanSpanish", "ukrainian", "croatian", "irish", "bulgarian", "hebrew", "thai", "indonesian", "arabic",
"traditionalChinese", "lithuanian", "anglicism", "estonian"-->chinese
</languageType>
<DockStation>
  <!--ro, opt, object, dock station information configuration-->
  <Platform>
    <!--ro, opt, object, information configuration of the dock station accessing the platform-->
    <ip>
      <!--ro, opt, string, IP address, range:[1,32]-->test
    </ip>
    <port>
      <!--ro, opt, int, communication port, range:[1,65535]-->1
    </port>
    <userName>
      <!--ro, req, string, user name, range:[1,32]-->test
    </userName>
  </Platform>
  <centralStorageBackupEnabled>
    <!--ro, opt, bool, whether to enable center storage backup-->true
  </centralStorageBackupEnabled>
</DockStation>
<webVersion>
  <!--ro, opt, string, Web version No., range:[1,32]-->test
</webVersion>
<deviceRFProgramVersion>
  <!--ro, opt, string, device RF program version, range:[1,32]-->test
</deviceRFProgramVersion>
<securityModuleSerialNo>
  <!--ro, opt, string, security module serial No., range:[1,32]-->test
</securityModuleSerialNo>
<securityModuleVersion>
  <!--ro, opt, string, security module version, range:[1,32]-->test
</securityModuleVersion>
<securityChipVersion>
  <!--ro, opt, string, security chip version, range:[1,32]-->test
</securityChipVersion>
<securityModuleKeyVersion>
  <!--ro, opt, string, security module key version, range:[1,32]-->test
</securityModuleKeyVersion>
<UIDLampRecognition>
  <!--ro, opt, object, recognize the device information by UID-->
  <enabled>
    <!--ro, opt, bool, whether to enable the function-->true
  </enabled>
</UIDLampRecognition>
<confDeviceIdPrefix>
  <!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
</confDeviceIdPrefix>
<OEMCode>
  <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard device), 0 (neutral device)-->1
</OEMCode>
<simpleAlgorithmVersion>
  <!--ro, opt, string, single algorithm version, desc:for fire control analyzer, it only supports one recognition algorithm if this node is returned to
the device information-->test
</simpleAlgorithmVersion>
<bootTime>
  <!--ro, opt, datetime, system boot time, desc:ISO8601 time (TD format, Local time and time difference)-->1970-01-01T08:00:00+08:00
</bootTime>
<intelligentAnalysisEngineModel>
  <!--ro, opt, string, engine model, range:[1,32]-->test
</intelligentAnalysisEngineModel>
<marketType>
  <!--ro, opt, enum, market type, subType:int, desc:0 (invalid), 1 (distribution type), 2 (industry type)-->0
</marketType>
<algorithmVersion>
  <!--ro, opt, string, HCNN algorithm version, desc:its Data Dictionary is provided by research institute for matching the algorithm model-->test
</algorithmVersion>
<firmware>
  <!--ro, opt, string, firmware version, desc:its Data Dictionary is provided by research institute for matching the algorithm model-->test
</firmware>

```

```

<engineList>
  <!--ro, opt, object, List of device computing power, desc:it returns engine No. by number of GPU chips. For example, there are 3 chips whose numbers are
  1, 5, and 11, then the returned values are 1, 5, 11-->
  <engine>
    <!--ro, opt, int, returned engine No.-->1
  </engine>
</engineList>
<platform>
  <!--ro, opt, enum, platform type, subType:int, desc:1 (TX1), 2 (P4), 3 (3559), 4 (3519), 5 (3516)-->1
</platform>
<platformName>
  <!--ro, opt, string, the platform where the algorithm runs, desc:"TX1/K81" (NVR), "KT" (data center), "H5" (front-end device), "H7" (front-end device),
  "K82" (NVR), "G3" (front-end device)-->test
</platformName>
<touchScreenVersionInfo>
  <!--ro, opt, string, touch screen module version-->test
</touchScreenVersionInfo>
<protocolFileURL>
  <!--ro, opt, string, protocol notice URI, range:[1,32]-->test
</protocolFileURL>
<recycleRecordEnabled>
  <!--ro, opt, bool, whether to enable overwritten recording-->>false
</recycleRecordEnabled>
<decordChannelNums>
  <!--ro, opt, int, number of decoding channels-->0
</decordChannelNums>
<VGA_nums>
  <!--ro, opt, int, number of VGA ports-->0
</VGA_nums>
<USB_nums>
  <!--ro, opt, int, number of USB ports-->0
</USB_nums>
<auxoutNums>
  <!--ro, opt, int, number of auxiliary ports-->0
</auxoutNums>
<expansionBoardVersion>
  <!--ro, opt, string, extension board version information, range:[1,32]-->test
</expansionBoardVersion>
<initWizzardDisplay>
  <!--ro, opt, bool, whether it displays initialization wizard, desc:it is only used for displaying configuration of applications integrated to the wizard
  such as Local GUI. By default, the value is true; it is false after the wizard is configured on the Local GUI-->>true
</initWizzardDisplay>
<beaconID>
  <!--ro, opt, string, device RF program version No., range:[0,32], desc:by default, it is the current value-->test
</beaconID>
<isResetDeviceLanguage>
  <!--ro, opt, bool, whether to reset the language, desc:only Admin and Installer have the permission to switch the device language. If the value is true,
  the Hik-Connect Mobile Client and the device's web page display device information in the target language; if the value is false, the language switching
  takes no effect on the Hik-Connect Mobile Client and the device's web page-->>false
</isResetDeviceLanguage>
<dispalNum>
  <!--ro, opt, int, number of windows-->0
</dispalNum>
<bspVersion>
  <!--ro, opt, string, BSP software version-->test
</bspVersion>
<dspVersion>
  <!--ro, opt, string, DSP software version-->test
</dspVersion>
<localUIVersion>
  <!--ro, opt, string, Local UI version-->test
</localUIVersion>
<OPCASubType>
  <!--ro, opt, enum, sub type of optical physical and chemical analyzer, subType:string, desc:"pump", "autoSampler" (auto sampler), "detector",
  "columnOven" (column oven), "SOC" (gas chromatography)-->detector
</OPCASubType>
<wiegandOutNum>
  <!--ro, opt, int, number of Wiegand outputs, range:[0,1]-->1
</wiegandOutNum>
<ChipVersionInfoList>
  <!--ro, opt, array, chip version information list, subType:object, range:[1,6]-->
  <ChipVersionInfo>
    <!--ro, opt, object, chip version information-->
    <ID>
      <!--ro, req, int, chip No., range:[0,1], desc:the No.s of chips of different types can be duplicate. Users can distinguish them by chip type and
      chip No. together-->1
    </ID>
    <firmwareVersion>
      <!--ro, req, string, chip firmware version, range:[0,64]-->1.0.0
    </firmwareVersion>
    <algorithmVersion>
      <!--ro, opt, string, algorithm version, desc:according to the chip name to determine the specific algorithm type, such as the chip name is
      temperature control system, then it is the temperature control system algorithm-->1.0.0
    </algorithmVersion>
    <chipName>
      <!--ro, opt, string, chip name, range:[0,32], desc:N/A-->test
    </chipName>
  </ChipVersionInfo>
</ChipVersionInfoList>
<personBagLinkAlgoEngineVersion>
  <!--ro, opt, string, engine version of the person & package Linkage module, range:[0,64], desc:this node is for analyzer (security inspection)-->1.0.0
</personBagLinkAlgoEngineVersion>
<BIOSVersion>

```



```

<!--ro, opt, string, BIOS version, range:[0,16]-->V1.3.4
</BIOSVersion>
<contactInformation>
  <!--ro, opt, string, contact information, range:[0,64]-->test
</contactInformation>
<temperatureModuleVersionInfo>
  <!--ro, opt, string, temperature measurement component version-->test
</temperatureModuleVersionInfo>
<PedestrianWarningModuleVersion>
  <!--ro, opt, object, pedestrian warning module information, desc:this node is returned only for pedestrian warning devices-->
  <pedestrianWarningMCUVersion>
    <!--ro, opt, string, MCU program version, range:[0,32]-->V1.3.4
  </pedestrianWarningMCUVersion>
  <pedestrianWarningRadarVersion>
    <!--ro, opt, string, the program version of pedestrian warning radar, range:[0,32]-->V1.3.4
  </pedestrianWarningRadarVersion>
  <pedestrianRangingModuleVersion>
    <!--ro, opt, string, the program version of the pedestrian distance measurement module, range:[0,32]-->V1.3.4
  </pedestrianRangingModuleVersion>
</PedestrianWarningModuleVersion>
<encryptionModel>
  <!--ro, opt, string, device encryption model, range:[0,16], desc:it cannot be modified and is a hexadecimal digit currently-->test
</encryptionModel>
<UMBVersion>
  <!--ro, opt, string, UMB version, range:[1,32]-->V1.3.0
</UMBVersion>
<audioBoard>
  <!--ro, opt, object, audio board information-->
  <audioBoardModel>
    <!--ro, opt, string, audio board model, range:[0,32], desc:currently only C8 and R8 are supported-->C8
  </audioBoardModel>
  <audioBoardVersion>
    <!--ro, opt, string, audio board version, range:[0,32]-->V1.0.0 build 211210
  </audioBoardVersion>
</audioBoard>
<materialScanAlgorithmVersion>
  <!--ro, opt, string, material scanning algorithm version, range:[0,64], desc:material scanning algorithm version-->test
</materialScanAlgorithmVersion>
<regionVersion>
  <!--ro, opt, enum, N/A, subType:string, desc:N/A-->basic
</regionVersion>
<productionDate>
  <!--ro, opt, date, date of manufacture-->2022-04-02
</productionDate>
<wifiModuleMACAddress>
  <!--ro, opt, string, Mac address of Wi-Fi module, range:[1,48]-->44:47:cc:c8:d9:e4
</wifiModuleMACAddress>
<displayInterfaceSize>
  <!--ro, opt, float, display interface size, range:[0.0,16.0], unit:in, desc:diagonal length of display interface for video intercom devices-->0
</displayInterfaceSize>
<releaseRegion>
  <!--ro, opt, enum, N/A, subType:string, desc:N/A-->China
</releaseRegion>
<shortSerialNumber>
  <!--ro, opt, string, short serial No., range:[1,9]-->test
</shortSerialNumber>
<audioVersion>
  <!--ro, opt, string, audio version, range:[1,32]-->test
</audioVersion>
<communicationFrequency>
  <!--ro, opt, enum, device wireless communication frequency, subType:string, desc:"433MHz", "868MHz"-->433MHz
</communicationFrequency>
<OSCoreVersionInfo>
  <!--ro, opt, object, OS/core component version information-->
  <OSCoreVersion>
    <!--ro, req, string, OS/core component version No., range:[1,32]-->1.0.0
  </OSCoreVersion>
  <minisysVersion>
    <!--ro, opt, string, minisys version No., range:[1,32]-->1.0.0
  </minisysVersion>
  <networkServiceVersion>
    <!--ro, opt, string, network service version No., range:[1,32]-->1.0.0
  </networkServiceVersion>
  <upgradeServiceVersion>
    <!--ro, opt, string, upgrade service version No., range:[1,32]-->1.0.0
  </upgradeServiceVersion>
</OSCoreVersionInfo>
<xTransVersion>
  <!--ro, opt, string, version No. of xtrans chip communication component, range:[1,32], desc:this node is for analyzer-->1.0.0
</xTransVersion>
<deviceMaintenanceInfoQRCode>
  <!--ro, opt, string, Get the QR code of the device maintenance information., range:[0,64], desc:scan to get more device information, including
  maintenance and documentation materials-->test
</deviceMaintenanceInfoQRCode>
<batteryFirmwareVersion>
  <!--ro, opt, string, battery firmware version No., range:[1,32]-->1.0.0
</batteryFirmwareVersion>
<radarVideoMapInfoList>
  <!--ro, opt, array, information list of radar-assisted video coordinate conversion matrix, subType:object, range:[1,32], desc:multiple cameras are added
  as sub-devices by terminal device(the transmission solution of sub-devices is adopted, refer to Sub-device Transparent Transmission in the Video Channel
  Resource Management domain). Each camera has its own radar-assisted video coordinate conversion matrix table, while the terminal device has multiple ones
  corresponding to each camera. When getting the terminal device information parameters via GET /ISAPI/System/deviceInfo, an information List of multiple
  matrix tables of the terminal device is returned. When getting the camera information parameters via GET /ISAPI/System/deviceInfo?devIndex=<devIndex>, the
  matrix table information of the specified camera is returned-->

```



```

<radarVideoMapInfo>
  <!--ro, opt, object, radar-assisted video coordinate conversion matrix information-->
  <devIndex>
    <!--ro, opt, string, sub-device ID, range:[0,64], desc:used for corresponding sub device-->123456789
  </devIndex>
  <radarVideoMapVersion>
    <!--ro, opt, string, radar-assisted video coordinate conversion matrix version, range:[1,32], desc:video coordinate and map GPS conversion matrix
version-->1.0.0
  </radarVideoMapVersion>
</radarVideoMapInfo>
</radarVideoMapInfoList>
<screenVersionInfo>
  <!--ro, opt, object, device screen version information, desc:the device (such as wall-mounted dock station) is a dual-system device, that is, an
embedded device and an Android system screen-->
  <screenAndroidOSVersion>
    <!--ro, opt, string, Android system version number of device screen, range:[1,64], desc:system program based on Android kernel,-->1.2.0 build20230101
  </screenAndroidOSVersion>
  <screenAndroidAPPVersion>
    <!--ro, opt, string, Android application version number of device screen, range:[1,64]-->1.0.0 build20230101
  </screenAndroidAPPVersion>
</screenVersionInfo>
<isSupportBattery>
  <!--ro, opt, bool, whether the device supports battery installing, desc:if the node is not returned or the returned value is false, it indicates no--
>true
  </isSupportBattery>
  <networkConfigSettingEnabled>
    <!--ro, opt, bool, whether to enter the network configuration interface after login, desc:application scenario: the default device IP address is
192.168.0.100, which is different from the network segment or IP address of the field environment. This is because it is a static IP address, and the
default IP address cannot be accessed on the Local area network. The network cable is directly connected to the device. The technical support can open the
device web page to check if the IP address is the default IP address 192.168.0.100. If it is default IP address, the network configuration interface will be
auto displayed for IP address configuration. When it is true, it indicates that you will enter the configuration page after login. When it is false or does
not exist, the login follows the previous logic-->true
  </networkConfigSettingEnabled>
  <ptzVersion>
    <!--ro, opt, string, PTZ version number, range:[1,32]-->test
  </ptzVersion>
  <movementSoftVersion>
    <!--ro, opt, string, module software version number, range:[1,32]-->test
  </movementSoftVersion>
  <movementHardVersion>
    <!--ro, opt, string, module hardware version number, range:[1,32]-->test
  </movementHardVersion>
  <loraModuleList>
    <!--ro, opt, array, Lora module information list, subType:object-->
    <loraModule>
      <!--ro, opt, object, Lora module information-->
      <id>
        <!--ro, opt, int, sequence number, desc:currently only one-->1
      </id>
      <firmwareVersion>
        <!--ro, opt, string, firmware version, range:[1,64]-->test
      </firmwareVersion>
    </loraModule>
  </loraModuleList>
  <deviceMode>
    <!--ro, opt, enum, device mode, subType:string, desc:"factoryMode", "maintenanceMode"-->factoryMode
  </deviceMode>
  <maintenanceDuration>
    <!--ro, opt, int, maintenance mode duration, range:[1,30], unit:d, dep:and,{$.DeviceInfo.deviceMode,eq,maintenance}-->1
  </maintenanceDuration>
  <pirAlgorithmVersion>
    <!--ro, opt, string, PIR algorithm version No., range:[1,32]-->V1.0
  </pirAlgorithmVersion>
  <customSerialNumber>
    <!--ro, opt, string, range:[1,9]-->B00000021
  </customSerialNumber>
  <deviceCabinID>
    <!--ro, opt, int, range:[0,128]-->1
  </deviceCabinID>
  <detectorVersion>
    <!--ro, opt, string, range:[1,32]-->test
  </detectorVersion>
  <internalModel>
    <!--ro, opt, string, range:[1,64]-->DS-AT1000S-2U4U
  </internalModel>
  <batteryFirmwareCodeVersion>
    <!--ro, opt, string, range:[1,32]-->12345
  </batteryFirmwareCodeVersion>
  <batteryFirmwareSerialNumber>
    <!--ro, opt, string, range:[1,32]-->DS-2FGHASKGHHKWHBGKAHSGK
  </batteryFirmwareSerialNumber>
  <mainModulePictureUrl>
    <!--ro, opt, string-->test
  </mainModulePictureUrl>
  <mainModuleDialUpPictureUrl>
    <!--ro, opt, string-->test
  </mainModuleDialUpPictureUrl>
  <controlInfo>
    <!--ro, opt, object-->
    <controlType>
      <!--ro, req, enum, subType:string-->single
    </controlType>
    <singleControlInfo>

```

```

<!--ro, opt, object, dep:or,{$.DeviceInfo.controlInfo.controlType,eq,single}-->
<serialMatchState>
  <!--ro, opt, enum, subType:string-->match
</serialMatchState>
</singleControlInfo>
<doubleControlInfo>
  <!--ro, opt, object, dep:or,{$.DeviceInfo.controlInfo.controlType,eq,double}-->
  <controlInfoList>
    <!--ro, req, array, subType:object, range:[2,2]-->
    <controlInfo>
      <!--ro, opt, object-->
      <controlID>
        <!--ro, req, string-->12345
      </controlID>
    </controlInfo>
  </controlInfoList>
  <supportModuleList>
    <!--ro, opt, array, subType:object-->
    <supportModule>
      <!--ro, opt, enum, subType:string-->interfaces
    </supportModule>
  </supportModuleList>
</doubleControlInfo>
</controlInfo>
<opticalFiberNumber>
  <!--ro, opt, int, range:[1,10]-->10
</opticalFiberNumber>
<devMaterialNumber>
  <!--ro, opt, string-->336800009
</devMaterialNumber>
<middlePlateMCUVersion>
  <!--ro, opt, string-->test
</middlePlateMCUVersion>
<panelMCUVersion>
  <!--ro, opt, string-->test
</panelMCUVersion>
<middlePlateCPLDVersion>
  <!--ro, opt, string-->test
</middlePlateCPLDVersion>
<backboardCPLDVersion>
  <!--ro, opt, string-->test
</backboardCPLDVersion>
<DFDVersion>
  <!--ro, opt, string-->test
</DFDVersion>
<CPUModel>
  <!--ro, opt, enum, subType:string-->K51
</CPUModel>
<deviceAssembleType>
  <!--ro, opt, enum, subType:string-->endDevice
</deviceAssembleType>
<deviceCategory>
  <!--ro, opt, enum, subType:string-->thermometryDevice
</deviceCategory>
<storageType>
  <!--ro, opt, enum, subType:string-->TF
</storageType>
</DeviceInfo>

```

12.1.1.3 Set device information parameters

Request URL

PUT /ISAPI/System/deviceInfo?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--req, string, device name, range:[1,132]-->test
  </deviceName>
  <deviceID>
    <!--opt, string, device No., range:[1,128]-->test
  </deviceID>
  <deviceDescription>
    <!--opt, string, device description, range:[1,128]-->test
  </deviceDescription>
  <deviceLocation>
    <!--opt, string, device location, range:[1,128]-->hangzhou
  </deviceLocation>

```

```

</deviceLocation>
<deviceStatus>
  <!--ro, opt, enum, device status, subType:string, desc:"normal", "abnormal"-->normal
</deviceStatus>
<DetailAbnormalStatus>
  <!--ro, opt, object, error status details, desc:it is valid only when deviceStatus is "abnormal"-->
  <hardDiskFull>
    <!--ro, opt, bool, disk full-->true
  </hardDiskFull>
  <hardDiskError>
    <!--ro, opt, bool, disk error-->true
  </hardDiskError>
  <ethernetBroken>
    <!--ro, opt, bool, network disconnected-->true
  </ethernetBroken>
  <ipaddrConflict>
    <!--ro, opt, bool, IP address conflict-->true
  </ipaddrConflict>
  <illegalAccess>
    <!--ro, opt, bool, illegal access-->true
  </illegalAccess>
  <recordError>
    <!--ro, opt, bool, recording exception-->true
  </recordError>
  <raidLogicDiskError>
    <!--ro, opt, bool, virtual disk exception in the array-->true
  </raidLogicDiskError>
  <spareWorkDeviceError>
    <!--ro, opt, bool, hot spare active device exception-->true
  </spareWorkDeviceError>
</DetailAbnormalStatus>
<systemContact>
  <!--ro, opt, string, manufacturer, range:[1,32]-->STD-CGI
</systemContact>
<model>
  <!--ro, req, string, device model, range:[1,64]-->iDS-9632NX-I8/X
</model>
<serialNumber>
  <!--ro, req, string, device serial No., range:[1,48]-->iDS-9632NX-I8/X1620181209CCRRC7605411WCVU
</serialNumber>
<macAddress>
  <!--ro, req, string, MAC Address, range:[1,64]-->44:47:cc:c8:d9:e4
</macAddress>
<firmwareVersion>
  <!--ro, req, string, device firmware version, range:[1,64]-->V4.1.40
</firmwareVersion>
<firmwareReleasedDate>
  <!--ro, opt, string, release date of the device firmware version-->2019-11-01
</firmwareReleasedDate>
<encoderVersion>
  <!--ro, opt, string, device encoder version No., range:[1,32]-->V7.3
</encoderVersion>
<encoderReleasedDate>
  <!--ro, opt, string, release date of the device encoder version-->2019-11-02
</encoderReleasedDate>
<bootVersion>
  <!--ro, opt, string, boot version, range:[1,16]-->V1.3.4
</bootVersion>
<bootReleasedDate>
  <!--ro, opt, string, release date of boot-->2019-11-03
</bootReleasedDate>
<panelVersion>
  <!--ro, opt, string, panel version, range:[1,32]-->V1.0
</panelVersion>
<hardwareVersion>
  <!--ro, opt, string, hardware version, range:[1,16]-->0x0
</hardwareVersion>
<decoderVersion>
  <!--ro, opt, string, device decoder version, range:[1,32]-->V1.0
</decoderVersion>
<decoderReleasedDate>
  <!--ro, opt, string, release date of the device decoder version-->2019-01-01
</decoderReleasedDate>
<softwareVersion>
  <!--ro, opt, string, software version, range:[1,32]-->V1.23
</softwareVersion>
<capacity>
  <!--ro, opt, int, device capacity, range:[1,10240], desc:unit: MB-->1
</capacity>
<usedCapacity>
  <!--ro, opt, int, used capacity of the device, range:[1,10240], desc:unit: MB-->1
</usedCapacity>
<telecontrolID>
  <!--opt, int, remote control ID, range:[1,255]-->1
</telecontrolID>
<supportBeep>
  <!--ro, opt, bool, whether to support buzzer-->true
</supportBeep>
<supportVideoLoss>
  <!--ro, opt, bool, whether the device supports video loss-->true
</supportVideoLoss>
<firmwareVersionInfo>
  <!--ro, opt, string, firmware version information, range:[1,32]-->test
</firmwareVersionInfo>

```

```

<actualFloorNum>
  <!--ro, opt, int, actual number of floors, range:[1,128]-->1
</actualFloorNum>
<localZoneNum>
  <!--ro, opt, int, number of zones (local), range:[0,16]-->1
</localZoneNum>
<alarmOutNum>
  <!--ro, opt, int, number of alarm outputs, range:[0,16]-->1
</alarmOutNum>
<distanceResolution>
  <!--ro, opt, float, distance resolution, range:[0.000,0.999]-->0.000
</distanceResolution>
<angleResolution>
  <!--ro, opt, float, angle resolution, range:[0.000,0.999]-->0.000
</angleResolution>
<speedResolution>
  <!--ro, opt, float, speed resolution, range:[0.000,0.999]-->0.000
</speedResolution>
<detectDistance>
  <!--ro, opt, float, detection distance, range:[0.000,0.999]-->0.000
</detectDistance>
<relayNum>
  <!--ro, opt, int, number of relays (local), range:[0,16]-->1
</relayNum>
<electroLockNum>
  <!--ro, opt, int, number of locks (local), range:[0,16]-->1
</electroLockNum>
<sirenNum>
  <!--ro, opt, int, number of sirens, range:[0,16]-->1
</sirenNum>
<alarmLamp>
  <!--ro, opt, int, number of alarm lamps, range:[0,16]-->1
</alarmLamp>
<RS485Num>
  <!--ro, opt, int, number of Local 485, range:[0,16]-->1
</RS485Num>
<radarVersion>
  <!--ro, opt, string, radar version, range:[1,32]-->test
</radarVersion>
<cameraModuleVersion>
  <!--ro, opt, string, camera module version, range:[1,32]-->test
</cameraModuleVersion>
<mainversion>
  <!--ro, opt, int, main version No., range:[1,255]-->1
</mainversion>
<subversion>
  <!--ro, opt, int, sub version No., range:[1,255]-->1
</subversion>
<upgradeversion>
  <!--ro, opt, int, update version No., range:[1,255]-->1
</upgradeversion>
<customizeversion>
  <!--ro, opt, int, custom version No., range:[1,255]-->1
</customizeversion>
<companyName>
  <!--ro, opt, string, manufacture name, range:[1,32]-->test
</companyName>
<copyright>
  <!--ro, opt, string, version information, range:[1,32]-->test
</copyright>
<systemName>
  <!--ro, opt, enum, storage system name, subType:string, desc:"storageManagement" (storage management system), "distributedStorageManagement"
(distributed storage management system)-->storageManagement
</systemName>
<systemStatus>
  <!--ro, opt, enum, system status, subType:string, desc:"configured", "unConfigured" (not configured)-->configured
</systemStatus>
<isLeaderDevice>
  <!--ro, opt, bool, whether it is the corresponding device of the resource IP address-->true
</isLeaderDevice>
<clusterVersion>
  <!--ro, opt, string, cluster version, range:[1,32], desc:it is valid when isLeaderDevice returns true-->test
</clusterVersion>
<centralStorageVersion>
  <!--ro, opt, string, center storage version, range:[1,16]-->test
</centralStorageVersion>
<powerOnMode>
  <!--ro, opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
</powerOnMode>
<customizedInfo>
  <!--ro, opt, string, custom project No., range:[1,32], desc:no value will be returned if it is baseline device. Custom project No. will be returned if
is custom device-->test
</customizedInfo>
<verificationCode>
  <!--ro, opt, string, device verification code-->test
</verificationCode>
<supportUrl>
  <!--ro, opt, string, service portal-->test
</supportUrl>
<subSerialNumber>
  <!--ro, opt, string, sub serial No.-->test
</subSerialNumber>
<languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech,brazilianPortuguese">

```

```
<!--ro, opt, enum, language type, subType:string, attr:opt{req, string}, desc:"chinese", "english", "spanish", "portuguese", "italian", "french",
"russian", "turkish", "greek", "czech", "brazilianPortuguese", "slovenian", "swedish", "norwegian", "romanian", "danish", "german", "polish", "dutch",
"hungarian", "slovak", "serbian", "southAmericanSpanish", "ukrainian", "croatian", "irish", "bulgarian", "hebrew", "thai", "indonesian", "arabic",
"traditionalChinese"-->chinese
```

```
</languageType>
```

```
<DockStation>
```

```
<!--ro, opt, object, dock station configuration-->
```

```
<Platform>
```

```
<!--ro, opt, object, information configuration of the dock station accessing the platform-->
```

```
<ip>
```

```
<!--ro, opt, string, IP address, range:[1,32]-->test
```

```
</ip>
```

```
<port>
```

```
<!--ro, opt, int, communication port, range:[1,65535]-->1
```

```
</port>
```

```
<userName>
```

```
<!--ro, req, string, user name, range:[1,32]-->test
```

```
</userName>
```

```
<password>
```

```
<!--ro, req, string, password, range:[1,16]-->test
```

```
</password>
```

```
</Platform>
```

```
<centralStorageBackupEnabled>
```

```
<!--ro, opt, bool, whether to enable central storage backup-->true
```

```
</centralStorageBackupEnabled>
```

```
</DockStation>
```

```
<webVersion>
```

```
<!--ro, opt, string, Web version No., range:[1,32]-->test
```

```
</webVersion>
```

```
<deviceRFProgramVersion>
```

```
<!--ro, opt, string, device RF program version, range:[1,32]-->test
```

```
</deviceRFProgramVersion>
```

```
<securityModuleSerialNo>
```

```
<!--ro, opt, string, security module serial No., range:[1,32]-->test
```

```
</securityModuleSerialNo>
```

```
<securityModuleVersion>
```

```
<!--ro, opt, string, security module version, range:[1,32]-->test
```

```
</securityModuleVersion>
```

```
<securityChipVersion>
```

```
<!--ro, opt, string, security chip version, range:[1,32]-->test
```

```
</securityChipVersion>
```

```
<securityModuleKeyVersion>
```

```
<!--ro, opt, string, security module key version, range:[1,32]-->test
```

```
</securityModuleKeyVersion>
```

```
<UIDLampRecognition>
```

```
<!--ro, opt, object, recognize the device information by UID-->
```

```
<enabled>
```

```
<!--ro, opt, bool, whether to enable the function-->true
```

```
</enabled>
```

```
</UIDLampRecognition>
```

```
<confDeviceIdPrefix>
```

```
<!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
```

```
</confDeviceIdPrefix>
```

```
<OEMCode>
```

```
<!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard device), 0 (neutral device)-->1
```

```
</OEMCode>
```

```
<simpleAlgorithmVersion>
```

```
<!--ro, opt, string, single algorithm version, desc:for fire control analyzer, it only supports one recognition algorithm if this node is returned to
the device information-->test
```

```
</simpleAlgorithmVersion>
```

```
<bootTime>
```

```
<!--ro, opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
```

```
</bootTime>
```

```
<intelligentAnalysisEngineModel>
```

```
<!--ro, opt, string, engine model, range:[1,32]-->test
```

```
</intelligentAnalysisEngineModel>
```

```
<marketType>
```

```
<!--ro, opt, enum, market type, subType:int, desc:0 (invalid), 1 (distribution type), 2 (industry type)-->0
```

```
</marketType>
```

```
<algorithmVersion>
```

```
<!--ro, opt, string, HCNN algorithm version, desc:its Data Dictionary is provided by research institute for matching the algorithm model-->test
```

```
</algorithmVersion>
```

```
<firmware>
```

```
<!--ro, opt, string, firmware version, desc:its Data Dictionary is provided by research institute for matching the algorithm model-->test
```

```
</firmware>
```

```
<engineList>
```

```
<!--ro, opt, object, List of device computing power, desc:it returns engine No. by the number of GPU chips. For example, there are 3 chips whose numbers
are 1, 5, and 11, then the returned values are 1, 5, 11-->
```

```
<engine>
```

```
<!--ro, opt, int, returned engine No.-->1
```

```
</engine>
```

```
</engineList>
```

```
<platform>
```

```
<!--ro, opt, enum, platform type, subType:int, desc:1 (TX1), 2 (P4), 3 (3559), 4 (3519), 5 (3516)-->1
```

```
</platform>
```

```
<platformName>
```

```
<!--ro, opt, string, the platform where the algorithm runs, desc:TX1/K81: NVR, KT: data center, H5: front-end device, H7: front-end device, K82: NVR, G3:
front-end device-->test
```

```
</platformName>
```

```
<touchScreenVersionInfo>
```

```
<!--ro, opt, string, touch screen module version-->test
```

```
</touchScreenVersionInfo>
```

```
<protocolFileURL>
```

```
<!--ro, opt, string, protocol notice URL, range:[1,32]-->test
```

```

<!--ro, opt, string, provide module url, range:[1,32]-->test
</protocolFileURL>
<recycleRecordEnabled>
  <!--ro, opt, bool, whether to enable overwritten recording-->false
</recycleRecordEnabled>
<decordChannelNums>
  <!--ro, opt, int, number of decoding channels-->0
</decordChannelNums>
<VGANums>
  <!--ro, opt, int, number of VGA ports-->0
</VGANums>
<USBNums>
  <!--ro, opt, int, number of USB ports-->0
</USBNums>
<auxoutNums>
  <!--ro, opt, int, number of auxiliary ports-->0
</auxoutNums>
<expansionBoardVersion>
  <!--ro, opt, string, extension board version information, range:[1,32]-->test
</expansionBoardVersion>
<initWizzardDisplay>
  <!--ro, opt, bool, whether it displays initialization wizard, desc:it is only used for displaying configuration of applications integrated to the
  wizard, such as Local GUI and Hik-Connect app by default, the value is true. it is false after the Local GUI configures the wizard-->true
</initWizzardDisplay>
<beaconID>
  <!--ro, opt, string, device RF program version No., range:[0,32], desc:by default, it is the current value-->test
</beaconID>
<isResetDeviceLanguage>
  <!--ro, opt, bool, whether it supports resetting the device language, desc:only Admin and Installer have the permission to switch the device language.
  If the value is true, the Hik-Connect Mobile Client and the device's web page display device information in the target language; if the value is false, the
  Language switching takes no effect on the Hik-Connect Mobile Client and the device's web page-->false
</isResetDeviceLanguage>
<dispalyNum>
  <!--ro, opt, int, number of windows-->0
</dispalyNum>
<bspVersion>
  <!--ro, opt, string, BSP software version-->test
</bspVersion>
<dspVersion>
  <!--ro, opt, string, DSP software version-->test
</dspVersion>
<localUIVersion>
  <!--ro, opt, string, Local UI version-->test
</localUIVersion>
<detectorType>
  <!--ro, opt, enum, detector type, subType:string, desc:"PDA" (photo-diode array)-->PDA
</detectorType>
<wiegandOutNum>
  <!--ro, opt, int, number of Wiegand outputs, range:[0,1]-->1
</wiegandOutNum>
<ChipVersionInfoList>
  <!--ro, opt, array, subType:object, range:[1,6]-->
  <ChipVersionInfo>
    <!--ro, opt, object-->
    <ID>
      <!--ro, req, int, range:[0,1]-->1
    </ID>
    <firmwareVersion>
      <!--ro, req, string, range:[0,64]-->1.0.0
    </firmwareVersion>
  </ChipVersionInfo>
</ChipVersionInfoList>
<personBagLinkAlgoEngineVersion>
  <!--ro, opt, string, engine version of the person & package Linkage module, range:[0,64], desc:this node is for analyzer (security inspection)-->1.0.0
</personBagLinkAlgoEngineVersion>
<BIOSVersion>
  <!--ro, opt, string, BIOS version, range:[0,16]-->test
</BIOSVersion>
<contactInformation>
  <!--ro, opt, string, contact information, range:[0,64]-->test
</contactInformation>
<materialScanAlgorithmVersion>
  <!--opt, string, material scan algorithm version, range:[0,64], desc:algorithm version of imaging radar panel for material scan-->test
</materialScanAlgorithmVersion>
<audioVersion>
  <!--ro, opt, string, range:[1,32]-->test
</audioVersion>
<loginPassword>
  <!--opt, string, range:[1,16]-->test
</loginPassword>
<deviceMode>
  <!--opt, enum, subType:string-->factoryMode
</deviceMode>
<maintenanceDuration>
  <!--opt, int, range:[1,30], unit:d, dep:and,{$.DeviceInfo.deviceMode,eq,maintenance}-->1
</maintenanceDuration>
<deviceCabinID>
  <!--opt, int, range:[0,128]-->1
</deviceCabinID>
</DeviceInfo>

```

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

12.1.2 Event Subscription Management

12.1.2.1 Set alarm/event subscription parameters

Request URL

PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all"(upload all alarms/events), "list"(upload specified alarm/event)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, "all"-upload all alarms/events, "list"-upload specified alarm/event, subType:string, desc:"all"(upload all alarms/events), "list"(upload specified alarm/event)-->list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, uploading mode of specified alarm/event-->
      <type>
        <!--req, enum, alarm/event types, subType:string, desc:see details in event types: "ADAS"(advanced driving assistance system), "ADASAlarm"(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent", "CDsStatus", "DBD" (driving behavior detection) "GPSUpload", "HFDP"(frequently appeared person), "IO"(I/O Alarm), "IOTD", "LES", "LFDP"(Low frequency person detection), "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioException", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fieldDetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hddBadBlock"(HDD bad sector detection event), "hddImpact" (HDD impact detection event), "heatmap"(heat map alarm), "highHDDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "ilLaccess"(invalid access), "ipConflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "LineDetection"(Line crossing detection), "ListSyncException"(List synchronization exception), "Loitering"(Loitering detection), "LowHDDTemperature"(HDD Low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modeError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry" (staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"(people overstay detection), "rollover", "running"(people running), "safetHelmetDetection"(hard hat detection), "scanChangeDetection", "sensorAlarm"(angular acceleration alarm), "sensorFailure"(HDD major fault
```

```

superSynchroDetection"(thermal detection), "searchingDetection", "sensorAlarm"(angular acceleration alarm), "severeStructure"(non major fault
detection), "shelterAlarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO",
"spacingChange"(distance exception), "sysStorFull"(storing full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection),
"targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime
detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list
alarms), "vehicleRecogResult", "versionAbnormal"(cluster version exception), "videoException", "videoLoss", "violationAlarm", "violentMotion"(violent motion
detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code
event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent",
"ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals),
"ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"-->mixedTargetDetection
</type>
<minorAlarm>
  <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401,0x402,0x403
</minorAlarm>
<minorException>
  <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
</minorException>
<minorOperation>
  <!--opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401,0x402,0x403
</minorOperation>
<minorEvent>
  <!--opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x01,0x02,0x03,0x04
</minorEvent>
<pictureURLType>
  <!--opt, enum, alarm picture format, subType:string, desc:"binary"(binary), "LocalURL"(device Local URL), "CloudStorageURL"(cloud storage URL)--
>cloudStorageURL
</pictureURLType>
<channels>
  <!--opt, string, channel information linked to event, desc:event linked channel information,and multiple channels can be linked,each channel is
separated by comma,e.g., "1,2,3,4..."-->1,2,3,4
</channels>
<ConferenceIDList>
  <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
<ConferenceID>
  <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
</ConferenceID>
</ConferenceIDList>
<uploadAllTarget>
  <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for MVR to get
all targets detected by the network camera-->>false
</uploadAllTarget>
<uploadMEFData>
  <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->>true
</uploadMEFData>
<minorType>
  <!--opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
</minorType>
<childDevIDList>
  <!--opt, array, subType:object-->
<childDevID>
  <!--opt, string, range:[1,128]-->test
</childDevID>
</childDevIDList>
<radarChannels>
  <!--opt, string-->1,2,3,4
</radarChannels>
<audioChannels>
  <!--opt, string-->1,2,3,4
</audioChannels>
</Event>
</EventList>
<channels>
  <!--opt, string, event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g., "1,2,3,4...", desc:event
linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g., "1,2,3,4..."-->1,2,3,4
</channels>
<pictureURLType>
  <!--opt, enum, alarm picture format, subType:string, desc:"binary"(binary), "LocalURL"(device Local URL), "CloudStorageURL"(cloud storage URL)--
>cloudStorageURL
</pictureURLType>
<ChangedUploadSub>
  <!--opt, object, subscribe to the uploaded information of device status changed event-->
<interval>
  <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:5 minute (default). If GUID is not reconnected in the internal, a new GUID
will be generated since the device starts a new arm period-->300
</interval>
<StatusSub>
  <!--opt, object, status description-->
<all>
  <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->>false
</all>
<channel>
  <!--opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->>true
</channel>
<hd>
  <!--opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->>true
</hd>
<capability>
  <!--opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->>true
</capability>
<ChanStatus>

```



```

    <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When
this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform
focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
    <online>
    <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-
-->true
    </online>
    <record>
    <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will
be no related alarm-->true
    </record>
    <recordStatus>
    <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm--
>true
    </recordStatus>
    <signal>
    <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
    </signal>
    <arming>
    <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
    </arming>
    <ChanStatus>
    </StatusSub>
    </ChangedUploadSub>
    <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
    </identityKey>
    <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
    <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
    </ConferenceIDList>
    <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "Low"-->high
    </level>
    <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
    </middleLevelPictureEnabeld>
    <childDevIDList>
    <!--opt, array, subType:object-->
    <childDevID>
    <!--opt, string, range:[1,128]-->test
    </childDevID>
    </childDevIDList>
    <radarChannels>
    <!--opt, string-->1,2,3,4
    </radarChannels>
    <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
    <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
    <!--ro, opt, array, subType:object-->
    <event>
    <!--ro, opt, object-->
    <eventType>
    <!--ro, req, string, range:[0,64]-->test
    </eventType>
    <level>
    <!--ro, req, enum, subType:int-->1
    </level>
    <channels>
    <!--ro, opt, string, range:[0,256]-->0,1,2
    </channels>
    </event>
    </eventList>
    </qosLevel>
    <audioChannels>
    <!--opt, string-->1,2,3,4
    </audioChannels>
    </SubscribeEvent>

```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

12.1.2.2 Get alarm/event subscription parameters

Request URL

GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), the (subscription ID) will be returned in message SubscribeEventResponse. When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), and if subscribeEventID is applied, it can be used to replace the in SubscribeEventResponse.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--ro, opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "List" (subscribe to channels according to channel list)-->list
  </channelMode>
  <eventMode>
    <!--ro, req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "List" (subscribe to specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "List"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:alarm/event types, which are obtained from the capability, refer to Alarm/Event Types (eventType)-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
```

<!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04

```
</minorEvent>
<pictureURLType>
  <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "LocalURL" (device Local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
</pictureURLType>
<channels>
  <!--ro, opt, string, channel information Linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
</channels>
<ConferenceIDList>
  <!--ro, opt, array, video conference ID List, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
  <ConferenceID>
    <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
  </ConferenceID>
</ConferenceIDList>
<uploadAllTarget>
  <!--ro, opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for MVR to get all targets detected by the network camera-->>false
</uploadAllTarget>
<uploadMEFData>
  <!--ro, opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->>true
</uploadMEFData>
<minorType>
  <!--ro, opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
</minorType>
<childDevIDList>
  <!--ro, opt, array, subType:object-->
  <childDevID>
    <!--ro, opt, string, range:[1,128]-->test
  </childDevID>
</childDevIDList>
<radarChannels>
  <!--ro, opt, string-->1,2,3,4
</radarChannels>
<audioChannels>
  <!--ro, opt, string-->1,2,3,4
</audioChannels>
<childDeviceFormat>
  <!--ro, opt, bool-->true
</childDeviceFormat>
</Event>
</EventList>
<channels>
  <!--ro, opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
</channels>
<pictureURLType>
  <!--ro, opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "LocalURL" (device Local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, the <pictureURLType> of the <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is Local URL of the device-->cloudStorageURL
</pictureURLType>
<ChangedUploadSub>
  <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
  <interval>
    <!--ro, opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
  </interval>
  <StatusSub>
    <!--ro, opt, object, status description-->
    <all>
      <!--ro, opt, bool, whether to subscribe to all channels, HDDs, and capability set changes-->>false
    </all>
    <channel>
      <!--ro, opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
    </channel>
    <hd>
      <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
    </hd>
    <capability>
      <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
    </capability>
    <ChanStatus>
      <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
      <online>
        <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
      </online>
      <record>
        <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
      </record>
      <recordStatus>
        <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
      </recordStatus>
      <signal>
        <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
      </signal>
```

```

    <arming>
    <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
    </arming>
    </ChanStatus>
    </StatusSub>
    </ChangedUploadSub>
    <identityKey>
    <!--ro, opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
    </identityKey>
    <ConferenceIDList>
    <!--ro, opt, array, video conference ID list, subType:object-->
    <ConferenceID>
    <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
    </ConferenceIDList>
    <level>
    <!--ro, opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
    </level>
    <middleLevelPictureEnabeld>
    <!--ro, opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
    </middleLevelPictureEnabeld>
    <childDevIDList>
    <!--ro, opt, array, subType:object-->
    <childDevID>
    <!--ro, opt, string, range:[1,128]-->test
    </childDevID>
    </childDevIDList>
    <radarChannels>
    <!--ro, opt, string-->1,2,3,4
    </radarChannels>
    <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
    <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
    <!--ro, opt, array, subType:object-->
    <event>
    <!--ro, opt, object-->
    <eventType>
    <!--ro, req, string, range:[0,64]-->test
    </eventType>
    <level>
    <!--ro, req, enum, subType:int-->1
    </level>
    <channels>
    <!--ro, opt, string, range:[0,256]-->0,1,2
    </channels>
    </event>
    </eventList>
    </qosLevel>
    <audioChannels>
    <!--ro, opt, string-->1,2,3,4
    </audioChannels>
    <retransmissionID>
    <!--ro, opt, string, range:[1,64]-->123456
    </retransmissionID>
    <retransmissionTimeoutDuration>
    <!--ro, opt, int, range:[1,108], unit:h-->1
    </retransmissionTimeoutDuration>
    <SubscribeISAPIMessage>
    <!--ro, opt, object-->
    <EventTypeList>
    <!--ro, req, array, subType:object-->
    <eventType>
    <!--ro, req, enum, subType:string-->AIOP_Video
    </eventType>
    <uploadPicEnabled>
    <!--ro, opt, bool-->>false
    </uploadPicEnabled>
    </EventTypeList>
    <EventList>
    <!--ro, req, array, subType:object, range:[0,200]-->
    <Event>
    <!--ro, opt, object-->
    <eventType>
    <!--ro, req, enum, subType:string-->AIOP_Video
    </eventType>
    <uploadPicEnabled>
    <!--ro, opt, bool-->>false
    </uploadPicEnabled>
    </Event>
    </EventList>
    </SubscribeISAPIMessage>
    </SubscribeEvent>

```

12.1.2.3 Subscribe to events/alarms in arming mode

Request URL

POST /ISAPI/Event/notification/subscribeEvent?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "List" (subscribe to channels according to channel list)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "List" (subscribe to specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "List"-->
    <Event>
      <!--opt, object, subscription of a specified alarm/event-->
      <type>
        <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm" (advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (Logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch" (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioException", "audioException", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "fLowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "ilLaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(Line crossing detection), "listSyncException"(List synchronization exception), "Loitering"(Loitering detection), "LowHDTemperature"(HDD Low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nichroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "pos"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal" (insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention" (people overstay detection), "rollLayer", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm" (angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelterAlarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storing full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event), "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "certificateRevocation" (certificate expiry), "humanbodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, minor operation type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:this node is required when type is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "LocalURL" (device Local URL), "CloudStorageURL" (Cloud storage URL)-->CloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
    </Event>
  </EventList>
</SubscribeEvent>
```

```

<uploadAllTarget>
  <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get
all targets detected by the network camera-->false
</uploadAllTarget>
<uploadMEFData>
  <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
</uploadMEFData>
<minorType>
  <!--opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.
3. When this node is not applied, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
</minorType>
<childDevIDList>
  <!--opt, array, subType:object-->
  <childDevID>
    <!--opt, string, range:[1,128]-->test
  </childDevID>
</childDevIDList>
<radarChannels>
  <!--opt, string-->1,2,3,4
</radarChannels>
<audioChannels>
  <!--opt, string-->1,2,3,4
</audioChannels>
</Event>
</EventList>
<channels>
  <!--opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is
all, this node should not be applied-->1,2,3,4
</channels>
<pictureURLType>
  <!--opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "LocalURL" (device local URL), "cloudStorageURL" (cloud
storage URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is
not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end
devices is Local URL of the device-->cloudStorageURL
</pictureURLType>
<ChangedUploadSub>
  <!--opt, object, subscribe to the uploaded information of device status changed event-->
  <interval>
    <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client
software does not reconnect to the device, a new GUID will be generated by the device-->300
  </interval>
  <StatusSub>
    <!--opt, object, status information-->
    <all>
      <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
    </all>
    <channel>
      <!--opt, bool, status of subscribed channels, desc:it is not required if the value of <all> is true-->true
    </channel>
    <hd>
      <!--opt, bool, subscribe to HDD status, desc:it is not required if the value of <all> is true-->true
    </hd>
    <capability>
      <!--opt, bool, subscribe to changing status of capability set, desc:it is not required if the value of <all> is true-->true
    </capability>
    <ChanStatus>
      <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When
this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform
focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
      <online>
        <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-
-->true
      </online>
      <record>
        <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will
be no related alarm-->true
      </record>
      <recordStatus>
        <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-
-->true
      </recordStatus>
      <signal>
        <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
      </signal>
      <arming>
        <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
      </arming>
    </ChanStatus>
  </StatusSub>
</ChangedUploadSub>
<identityKey>
  <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
</identityKey>
<ConferenceIDList>
  <!--opt, array, video conference ID list, subType:object-->
  <ConferenceID>
    <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
  </ConferenceID>
</ConferenceIDList>
<subscribeEventID>

```

<!--opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of events will be enabled by default after the node is applied, range:[1,128], desc:when an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message SubscribeEventResponse-->test

```
</subscribeEventID>
<level>
  <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
</level>
<middleLevelPictureEnabeld>
  <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
</middleLevelPictureEnabeld>
<childDevIDList>
  <!--opt, array, subType:object-->
  <childDevID>
    <!--opt, string, range:[1,128]-->test
  </childDevID>
</childDevIDList>
<SubscribeISAPIMessage>
  <!--opt, object-->
  <EventTypeList>
    <!--req, array, subType:object-->
    <eventType>
      <!--req, enum, subType:string-->AIOP_Video
    </eventType>
    <uploadPicEnabled>
      <!--opt, bool-->false
    </uploadPicEnabled>
  </EventTypeList>
</SubscribeISAPIMessage>
<radarChannels>
  <!--opt, string-->1,2,3,4
</radarChannels>
<qosLevel>
  <!--ro, opt, object-->
  <defaultLevel>
    <!--ro, req, enum, subType:int-->0
  </defaultLevel>
  <eventList>
    <!--ro, opt, array, subType:object-->
    <event>
      <!--ro, opt, object-->
      <eventType>
        <!--ro, req, string, range:[0,64]-->test
      </eventType>
      <level>
        <!--ro, req, enum, subType:int-->1
      </level>
      <channels>
        <!--ro, opt, string, range:[0,256]-->0,1,2
      </channels>
    </event>
  </eventList>
</qosLevel>
<audioChannels>
  <!--opt, string-->1,2,3,4
</audioChannels>
</SubscribeEvent>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventResponse xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, subscription result, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, subscription ID, which is returned by the device. The device should guarantee the uniqueness of ID, range:[1,64]-->1
  </id>
  <FailedEventList>
    <!--ro, opt, array, list of events, of which subscription failed, subType:object, desc:when subscription of one or multiple events failed, this node should be returned. The client software can check if all events are subscribed to by checking whether the node <FailedEventList> is returned-->
    <Event>
      <!--ro, opt, object, event, of which the subscription failed. The event/alarm type name can be obtained via the capability set-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:event type-->ADAS
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "LocalURL" (Local URL), "cloudStorageURL" (cloud storage URL)-->binary
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel No., desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <subStatusCode>
        <!--ro, req, enum, error code, subType:string, desc:refer to the error code dictionary-->activeNumMax
      </subStatusCode>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string,
        desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
        2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel check-in.
        3. When this node is not returned, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
    </Event>
  </FailedEventList>
</SubscribeEventResponse>
```

12.1.2.4 Get the alarm/event subscription capability

Request URL

GET /ISAPI/Event/notification/subscribeEventCap

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <format opt="xml,json">
    <!--ro, opt, string, attr:opt{req, string}-->xml
  </format>
  <heartbeat min="1" max="180">
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
  </heartbeat>
  <channelMode opt="all,list">
    <!--ro, opt, enum, channel subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel list)-->list
  </channelMode>
  <eventMode opt="all,list">
    <!--ro, req, enum, event subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
```



```

<Event>
  <!--ro, opt, object, subscription of a specified alarm/event-->
  <type>
    <!--ro, req, enum, event type, subType:string, desc:refer to event type List (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
    (advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
    controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
    person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (Logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
    (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
    "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
    "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
    spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
    detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
    "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fieldddetection"(intrusion detection), "fireDetection",
    "fireEscapedetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hddBadBlock"(HDD bad sector
    detection event), "hddImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDDTemperature"(HDD high temperature detection event),
    "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "iLLaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
    conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "linedetection"(line crossing detection),
    "ListSyncException"(List synchronization exception), "Loitering"(Loitering detection), "LowHDDTemperature"(HDD Low temperature detection event),
    "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
    "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
    "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
    "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
    detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
    "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
    "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
    (people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
    (angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelterAlarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
    detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storing full alarm of cluster system),
    "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
    alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
    "unattendedBaggage", "vehicleMatchResult"(uploading List alarms), "vehicleRecogResult", "versionAbnormal"(cluster version exception), "videoException",
    "videoLoss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent",
    "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison
    event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
    "TransactionRecordEvent", "HealthInfoSyncQuery" (health information search event), "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"
    (searching the consumption status), "certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison),
    "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
  </type>
  <minorAlarm opt="0x400,0x401,0x402,0x403">
    <!--ro, opt, string, minor alarm type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"--
    >0x400,0x401
  </minorAlarm>
  <minorException opt="0x400,0x401,0x402,0x403">
    <!--ro, opt, string, minor exception type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
    "AccessControllerEvent"-->0x400,0x401
  </minorException>
  <minorOperation opt="0x400,0x401,0x402,0x403">
    <!--ro, opt, string, minor operation type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
    "AccessControllerEvent"-->0x400,0x401
  </minorOperation>
  <minorEvent opt="0x01,0x02,0x03,0x04">
    <!--ro, opt, string, minor event type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"--
    >0x400,0x401
  </minorEvent>
  <pictureURLType opt="binary,localURL,cloudStorageURL,multipart" def="cloudStorageURL">
    <!--ro, opt, enum, alarm picture format, subType:string, attr:opt{req, string},def{req, string}, desc:"binary" (binary), "LocalURL" (device Local
    URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
  </pictureURLType>
  <channels>
    <!--ro, opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
  </channels>
  <conferenceIDList size="1">
    <!--ro, opt, array, video conference ID List, subType:object, attr:size{req, int}, desc:when this node does not exist, it indicates to subscribe to
    all conferences-->
    <conferenceID min="1" max="32">
      <!--ro, opt, string, ID of subscribed video conference, range:[1,32], attr:min{req, int},max{req, int}-->test
    </conferenceID>
  </conferenceIDList>
  <uploadAllTarget opt="true,false">
    <!--ro, opt, bool, whether to upload all detected targets, attr:opt{req, string}, desc:the default value is false. In perimeter protection, this
    node is used for NVR to get all targets detected by the network camera-->>false
  </uploadAllTarget>
  <uploadMEFData opt="true,false">
    <!--ro, opt, bool, whether to upload the electronic fence data related to the event, attr:opt{req, string}, desc:true by default-->>true
  </uploadMEFData>
  <minorType opt="success,failure">
    <!--ro, opt, string, event sub-type, attr:opt{req, string},
    desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
    2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
    check-in.-->success
  </minorType>
  <childDevIDList min="1" max="8">
    <!--ro, opt, array, subType:object, attr:min{req, int},max{req, int}-->
    <childDevID min="1" max="128">
      <!--ro, opt, string, range:[1,128], attr:min{req, int},max{req, int}-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--ro, opt, string-->1,2,3,4
  </radarChannels>
  <audioChannels>
    <!--ro, opt, string-->1,2,3,4
  </audioChannels>
</Event>

```

```

</EventList>
<channels>
  <!--ro, opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode>
  is all, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
</channels>
  <pictureURLType opt="binary,localURL,cloudStorageURL,multipart" def="cloudStorageURL">
    <!--ro, opt, enum, format unified configuration of alarm pictures, subType:string, attr:opt{req, string},def{req, string}, desc:"binary", "LocalURL"
    (device Local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, the
    <pictureURLType> of the <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of
    uploaded pictures for front-end devices is binary, and for back-end devices is Local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--ro, opt, int, the Lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client
      software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--ro, opt, bool, status information-->true
    <all>
      <!--ro, opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
    </all>
    <channel>
      <!--ro, opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
    </channel>
    <hd>
      <!--ro, opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
    </hd>
    <capability>
      <!--ro, opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
    </capability>
    <ChanStatus>
      <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When
      this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform
      focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
      <online opt="true,false">
        <!--ro, opt, bool, whether the channel online status is subscribed, attr:opt{req, string}, desc:if this node does not exist, it indicates there
        will be no related alarm-->true
      </online>
      <record opt="true,false">
        <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, attr:opt{req, string}, desc:if this node does not exist,
        it indicates there will be no related alarm-->true
      </record>
      <recordStatus opt="true,false">
        <!--ro, opt, bool, whether the recording status is subscribed, attr:opt{req, string}, desc:if this node does not exist, it indicates there will be no
        related alarm-->true
      </recordStatus>
      <signal opt="true,false">
        <!--ro, opt, bool, whether the signal status is subscribed, attr:opt{req, string}, desc:if this node does not exist, it indicates there will be no
        related alarm-->true
      </signal>
      <arming opt="true,false">
        <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, attr:opt{req, string}, desc:if this node does not exist, it
        indicates there will be no related alarm-->true
      </arming>
    </ChanStatus>
  </StatusSub>
</ChangedUploadSub>
<identityKey max="64">
  <!--ro, opt, string, subscription connection interaction key, range:[1,64], attr:max{req, int}, desc:N/A-->test
</identityKey>
<ConferenceIDList size="1">
  <!--ro, opt, array, video conference ID list, subType:object, attr:size{req, int}-->
  <ConferenceID min="1" max="32">
    <!--ro, opt, string, ID of subscribed video conference, range:[1,32], attr:min{req, int},max{req, int}-->test
  </ConferenceID>
</ConferenceIDList>
<isSupportModifySubscribeEvent>
  <!--ro, opt, bool, whether the device supports the management of arming subscription, desc:related API: /ISAPI/Event/notification/subscribeEvent/<ID>--
  >true
</isSupportModifySubscribeEvent>
<subscribeEventID min="1" max="128">
  <!--ro, opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of
  events will be enabled by default after the node is applied, range:[1,128], attr:min{req, int},max{req, int}, desc:when an event/alarm is subscribed
  (related URL: POST /ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message
  SubscribeEventResponse-->test
</subscribeEventID>
<level opt="high,middle,low">
  <!--ro, opt, enum, arming level, subType:string, attr:opt{req, string}, desc:"high" (default value), "middle", "low"-->high
</level>
<middleLevelPictureEnabeld opt="true,false">
  <!--ro, opt, bool, whether to enable picture uploading for middle-level arming, attr:opt{req, string}, desc:the default value is true-->true
</middleLevelPictureEnabeld>
<deployID opt="1,2">
  <!--ro, opt, enum, arming type, subType:int, attr:opt{req, string}, desc:0 (by client software), 1 (real-time)-->0
</deployID>
<childDevIDList min="1" max="8">
  <!--ro, opt, array, subType:object, attr:min{req, int},max{req, int}-->
  <childDevID min="1" max="128">
    <!--ro, opt, string, range:[1,128], attr:min{req, int},max{req, int}-->test
  </childDevID>
</childDevIDList>
<SubscribeISAPIMessage>

```

```

<!--ro, opt, object-->
<eventType opt="AIOP_Video,AIOP_Polling_Video,AIOP_Picture,AIOP_Polling_Snap,TPS">
  <!--ro, req, enum, subType:string, attr:opt{req, string}-->AIOP_Video
</eventType>
<eventPicUploadType opt="AIOP_Picture">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->AIOP_Picture
</eventPicUploadType>
</SubscribeISAPIMessage>
<radarChannels>
  <!--ro, opt, string-->1,2,3,4
</radarChannels>
<qosLevel>
  <!--ro, opt, object-->
  <defaultLevel opt="0,1,2">
    <!--ro, req, enum, subType:int, attr:opt{req, string}-->0
  </defaultLevel>
  <eventList>
    <!--ro, opt, array, subType:object-->
    <event>
      <!--ro, opt, object-->
      <eventType>
        <!--ro, req, string, range:[0,64]-->test
      </eventType>
      <level opt="0,1,2">
        <!--ro, req, enum, subType:int, attr:opt{req, string}-->1
      </level>
      <channels>
        <!--ro, opt, string, range:[0,256]-->0,1,2
      </channels>
    </event>
  </eventList>
</qosLevel>
<isSupportUnSubscribeEvent>
  <!--ro, opt, bool-->true
</isSupportUnSubscribeEvent>
<audioChannels>
  <!--ro, opt, string-->1,2,3,4
</audioChannels>
</SubscribeEventCap>

```

12.1.2.5 Cancel subscribing alarm/event

Request URL

PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	Subscription event ID, which should match the id field in the SubscribeEventResponse of the response message from the POST /ISAPI/Event/notification/subscribeEvent?deployID=.This field is optional. When the protocol is /ISAPI/Event/notification/unSubscribeEvent, it indicates that all subscription links are to be unsubscribed. When the protocol is /ISAPI/Event/notification/unSubscribeEvent?ID=XXX, it indicates that a specific subscription link is to be unsubscribed.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

12.1.2.6 Event subscription heartbeat

EventType:heartBeat

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, alarm message, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetsDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type-->heartBeat
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->heartBeat
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
</EventNotificationAlert>
```

12.1.3 Log Management

12.1.3.1 Search for log information

Request URL

POST /ISAPI/ContentMgmt/logSearch

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, Log search, attr:version{opt, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, range:[1,32], desc:UUID/GUID for record search. It is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device for faster search next time.-->test
  </searchID>
  <trackIDList>
    <!--opt, array, channel list, subType:object-->
    <trackID>
      <!--req, int, channel No., desc:channel No.-->1
    </trackID>
  </trackIDList>
  <metaId>
    <!--req, enum, Log type, subType:string, desc:{Log.std-cgi.com/<majorType>/<minorType>}, <majorType>-major type of Log, <minorType>-minor type of Log, see remarks for detailed Log type definitions; "Log.std-cgi.com/Alarm/methaneConcentrationException", "Log.std-cgi.com/Alarm/methaneLightIntensityException", "Log.std-cgi.com/Alarm/fishingShipDetection"-->Log.std-cgi.com/Alarm/methaneConcentrationException
  </metaId>
  <timeSpanList>
    <!--opt, array, List of time periods, subType:object, desc:List of time periods-->
    <timeSpan>
      <!--opt, object, time period, desc:time period-->
      <startTime>
        <!--req, datetime, start time-->1970-01-01T00:00:00Z
      </startTime>
      <endTime>
        <!--req, datetime, end time-->1970-01-01T01:00:00Z
      </endTime>
    </timeSpan>
  </timeSpanList>
  <searchResultPosition>
    <!--opt, int, the start position of the search result, desc:the start position of the search result in the result list-->100
  </searchResultPosition>
  <maxResults>
    <!--opt, int, the maximum number of records supported in this search, range:[0,100]-->0
  </maxResults>
  <onlySmart>
    <!--opt, bool, whether to search for Logs with smart information, desc:the value of this node is false by default-->false
  </onlySmart>
  <logLevel>
    <!--opt, enum, subType:string-->emergency
  </logLevel>
  <alarmLevel>
    <!--opt, enum, subType:string-->high
  </alarmLevel>
</CMSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CMSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, Log search result, attr:version{opt, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search ID, range:[1,32], desc:UUID/GUID for record search. It is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device for faster search next time-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool, search status-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, search status string, subType:string, desc:"OK" (search completed), "NO MATCHES" (no matched data), "MORE" (more data waiting to be searched)-->OK
  </responseStatusStrg>
  <totalMatches>
    <!--ro, opt, int, total number of matched results-->12
  </totalMatches>
  <numOfMatches>
    <!--ro, opt, int, number of results returned this time, range:[0,100]-->24
  </numOfMatches>
  <matchList>
    <!--ro, opt, array, List of matched data records, subType:object-->
    <searchMatchItem>
      <!--ro, opt, object, single matched record-->
      <logDescriptor>
        <!--ro, opt, object, Log description-->
        <metaId>
```

```

    <!--ro, req, string, log type for searchn, desc:log.sta-cgi.com/<majorType>/<minorType>, <majorType> (major type of log), <minorType> (minor
type of log), see remarks for detailed log type definitions-->log.std-cgi.com/Alarm/24HZoneAlarm
</metaId>
<StartDateTime>
    <!--ro, req, datetime, log time-->1970-01-01T00:30:00Z
</StartDateTime>
<localID>
    <!--ro, opt, string, channel No.-->1
</localID>
<paraType>
    <!--ro, opt, string, parameter type-->test
</paraType>
<userName>
    <!--ro, opt, string, user name-->Admin
</userName>
<infoContent>
    <!--ro, opt, string, log information content-->test
</infoContent>
<logInfo>
    <!--ro, opt, object, log information-->
    <OpenDoorRecord>
        <!--ro, opt, object, video intercom unlocking record-->
        <type>
            <!--ro, req, enum, unlocking type, subType:string, desc:"password" (unlocking by password), "hijack" (unlocking by duress), "card" (unlocking
by card), "resident" (unlocking by resident), "center" (unlocking by management center)-->password
        </type>
    </OpenDoorRecord>
    <VisAlarmRecord>
        <!--ro, opt, object, video intercom alarm record-->
        <type>
            <!--ro, req, enum, alarm type, subType:string, desc:"zone" (zone alarm), "dismantle" (tampering alarm), "hijack" (duress alarm), "passwordErr"
(password error alarm), "doorNotOpen" (unlocking-door-failed alarm), "doorNotClose" (locking-door-failed alarm), "SOS" (SOS alarm), "callReq" (device call
request alarm), "smartLockHijackFingerPrint" (smart lock duress alarm (fingerprint)), "smartLockHijackPassword" (smart lock duress alarm (password)),
"smartLockBreaking" (forced-open door alarm), "smartLockBeLocked" (lock-up door alarm), "smartLockLowBattery" (smart-lock-battery-low alarm)-->zone
        </type>
    </VisAlarmRecord>
</logInfo>
<ipAddress>
    <!--ro, opt, string, device IP address, range:[1,32]-->10.20.30.10
</ipAddress>
<object>
    <!--ro, opt, enum, operation object, subType:string, desc:"network", "keypad", "remoteCtrl" (remote control), "card"-->network
</object>
<params>
    <!--ro, opt, string, log parameters, desc:parameters (zone No., etc.)-->test
</params>
<seq>
    <!--ro, opt, string, device serial No., range:[1,32]-->test
</seq>
<additionInformation>
    <!--ro, opt, string, additional information, range:[1,128], desc:additional information of log-->test
</additionInformation>
<panelUser>
    <!--ro, opt, string, user name of the operation panel, range:[0,16]-->admin
</panelUser>
<diskNumber>
    <!--ro, opt, int, HDD No., range:[0,10000]-->0
</diskNumber>
<alarmInPort>
    <!--ro, opt, int, alarm input port, range:[0,1000]-->0
</alarmInPort>
<alarmOutPort>
    <!--ro, opt, int, alarm output port, range:[0,10000]-->0
</alarmOutPort>
<remoteHostIPAddress>
    <!--ro, opt, string, remote host IP address, range:[1,32]-->10.12.25.23
</remoteHostIPAddress>
<logLevel>
    <!--ro, opt, enum, subType:string-->emergency
</logLevel>
<alarmLevel>
    <!--ro, opt, enum, subType:string-->high
</alarmLevel>
<moduleName>
    <!--ro, opt, string, range:[1,32]-->test
</moduleName>
</logDescriptor>
</searchMatchItem>
</matchList>
</CMSearchResult>

```

12.1.4 Serial Port Management

12.1.4.1 Get the capability of a specific serial port

Request URL

GET /ISAPI/System/Serial/ports/<portID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port No., attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <encryptKey min="0" max="10">
    <!--ro, opt, string, key, attr:min{req, int},max{req, int}, desc:key-->test
  </encryptKey>
  <serialNumber opt="1,2,3,4,5">
    <!--ro, opt, int, serial No., attr:opt{req, string}-->0
  </serialNumber>
  <serialPortType opt="RS485,RS422,RS232">
    <!--ro, opt, string, serial port type, attr:opt{req, string}-->RS485
  </serialPortType>
  <serialAddress min="0" max="10">
    <!--ro, opt, int, serial port address, attr:min{req, int},max{req, int}, desc:serial port address-->1
  </serialAddress>
  <duplexMode opt="half,full">
    <!--ro, opt, string, duplex mode of the serial port, attr:opt{req, string}-->half
  </duplexMode>
  <direction opt="monodirectional,bidirectional">
    <!--ro, opt, string, attr:opt{req, string}-->monodirectional
  </direction>
  <baudRate opt="1200,2400,4800,9600,19200,38400,57600,115200">
    <!--ro, opt, int, attr:opt{req, string}-->1200
  </baudRate>
  <dataBits opt="6,7,8">
    <!--ro, opt, int, attr:opt{req, string}-->6
  </dataBits>
  <parityType opt="none,even,odd,mark,space">
    <!--ro, opt, string, attr:opt{req, string}-->none
  </parityType>
  <stopBits opt="1,1.5,2">
    <!--ro, opt, string, stop bit, attr:opt{req, string}-->1
  </stopBits>
  <workMode opt="console,transparent,audiomix,screenCtrl,ptzCtrl,keyboard,matrix,audioMixers">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->console
  </workMode>
  <flowCtrl opt="none,software,hardware">
    <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
  </flowCtrl>
  <deviceName min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </deviceName>
  <deviceProtocol min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </deviceProtocol>
  <mode opt="readerMode,clientMode,externMode,stairsControl,accessControlHost,disabled,custom,cardReceiver,QRCodeReader">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->readerMode
  </mode>
  <modeChangeReboot>
    <!--ro, opt, bool-->true
  </modeChangeReboot>
  <outputDataType opt="cardNo,employeeNo,auto">
    <!--ro, opt, string, output data type, attr:opt{req, string}, desc:data type output from the door station to the elevator controller: floorNumber (floor No,default),cardNo (card No.)-->cardNo
  </outputDataType>
  <copyToAll opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </copyToAll>
  <isVariable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isVariable>
  <PTZInfo>
    <!--ro, opt, object-->
    <bindChannelNo min="0" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->0
    </bindChannelNo>
    <PTZProtocolList>
      <!--ro, req, object-->
      <PTZProtocol>
        <!--ro, opt, object-->
      </PTZProtocol>
    </PTZProtocolList>
  </PTZInfo>
  <id>
```

```

        <!--ro, req, int, protocol index-->0
    </id>
    <protoDesc>
        <!--ro, req, string, protocol description-->test
    </protoDesc>
</PTZProtocol>
</PTZProtocolList>
</PTZInfo>
<ScreenCtrlInfo>
    <!--ro, opt, object, ScreenCtrlInfo, desc:this field is valid when the value of workMode is "screenCtrl"-->
    <ScreenCtrlProtoList>
        <!--ro, req, object, ScreenCtrlProtoList-->
        <ScreenCtrlProto>
            <!--ro, opt, object, ScreenCtrlProto-->
            <id>
                <!--ro, req, int, protocol index-->0
            </id>
            <protoDesc>
                <!--ro, req, string, protocol description-->test
            </protoDesc>
        </ScreenCtrlProto>
    </ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo>
    <!--ro, opt, object, MatrixCtrlInfo-->
    <MatrixCtrlProtoList>
        <!--ro, req, object, MatrixCtrlProtoList-->
        <MatrixCtrlProto>
            <!--ro, opt, object, MatrixCtrlProto-->
            <id>
                <!--ro, req, int, protocol index-->0
            </id>
            <protoDesc>
                <!--ro, req, string, protocol description-->test
            </protoDesc>
        </MatrixCtrlProto>
    </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyboardCtrlInfo>
    <!--ro, opt, object, KeyboardCtrlInfo-->
    <KeyboardCtrlProtoList>
        <!--ro, req, object, KeyboardCtrlProtoList-->
        <KeyboardCtrlProto>
            <!--ro, opt, object, KeyboardCtrlProto-->
            <id>
                <!--ro, req, int, protocol index-->0
            </id>
            <protoDesc>
                <!--ro, req, string, protocol description-->test
            </protoDesc>
        </KeyboardCtrlProto>
    </KeyboardCtrlProtoList>
</KeyboardCtrlInfo>
<ControlAddress>
    <!--ro, opt, object-->
    <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <address min="0" max="255">
        <!--ro, req, int, IP address, attr:min{req, int},max{req, int}-->0
    </address>
</ControlAddress>
<rs485WorkMode>
    opt="Led,sensor,CaptureTrigger,transparent,GPS,ModbusRTU,supplementLight,audibleVisualAlarm,columnSpeaker,commModeUsing433MHz,signalLampControl">
    <!--ro, opt, string, working mode of RS-485 serial port, dep:and,{$.SerialPortList.SerialPort.serialPortType,eq,RS485}, attr:opt{req, string},
    desc:working mode of RS-485 serial port,which is used for LED display or triggering transmission of captured pictures. This node is valid only when
    <serialPortType> is "RS485"-->sensor
</rs485WorkMode>
<SensorCtrlList>
    <!--ro, opt, object-->
    <Sensor>
        <!--ro, opt, object-->
        <id min="1" max="3">
            <!--ro, req, int, index, attr:min{req, int},max{req, int}-->1
        </id>
        <name min="1" max="32">
            <!--ro, req, string, attr:min{req, int},max{req, int}-->visibilitySensor
        </name>
        <enabled opt="true,false">
            <!--ro, req, bool, whether to enable, attr:opt{req, string}, desc:whether to enable the serial port: "true,false"-->false
        </enabled>
    </Sensor>
</SensorCtrlList>
<stairsControl>
    <!--ro, opt, object, elevator control parameters-->
    <outputDataType opt="floorNumber,cardNo">
        <!--ro, opt, enum, data type output from the door station to the elevator controller, subType:string, attr:opt{req, string}, desc:floorNumber (floor
        No.,default),cardNo (card No.)-->floorNumber
    </outputDataType>
</stairsControl>
<comMode opt="active,passive">
    <!--ro, opt, enum, communication mode, subType:string, attr:opt{req, string}, desc:communication mode-->active
</comMode>
<ModbusRTU>

```



```

<!--ro, opt, object-->
<ModbusRTUBaudRate opt="1200,2400,4800,9600,19200" def="9600">
  <!--ro, opt, int, attr:opt{req, string},def{req, string}-->9600
</ModbusRTUBaudRate>
<ModbusRTUDataBits opt="5,6,7,8" def="8">
  <!--ro, opt, int, attr:opt{req, string},def{req, string}-->6
</ModbusRTUDataBits>
<ModbusRTUParityType opt="none,even,odd" def="none">
  <!--ro, opt, string, attr:opt{req, string},def{req, string}-->none
</ModbusRTUParityType>
<ModbusRTUStopBits opt="1,2" def="1">
  <!--ro, opt, string, stop bit, attr:opt{req, string},def{req, string}-->1
</ModbusRTUStopBits>
<ModbusRTUFlowCtrl opt="none,software,hardware">
  <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
</ModbusRTUFlowCtrl>
<decoderAddress min="0" max="255" def="0">
  <!--ro, opt, int, decoder IP address, attr:min{req, int},max{req, int},def{req, int}-->1
</decoderAddress>
<online opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</online>
</ModbusRTU>
<accessName min="1" max="32">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</accessName>
<serialPortUse opt="acquAndCtrl,swipe,strongElectric">
  <!--ro, opt, string, attr:opt{req, string}-->acquisition
</serialPortUse>
<acquisitionProtocol opt="modbus,custom">
  <!--ro, opt, string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}, attr:opt{req, string}-->modbus
</acquisitionProtocol>
<serialProtocol opt="FECbus,GB26875">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB26875
</serialProtocol>
<SupplementLightList size="5">
  <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}, attr:size{req, int}-->
<SupplementLight>
  <!--ro, opt, object-->
  <SupplementLightID min="1" max="5">
    <!--ro, req, int, range:[1,5], attr:min{req, int},max{req, int}-->1
  </SupplementLightID>
</SupplementLight>
</SupplementLightList>
<permissionController opt="0,1">
  <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
</permissionController>
<deviceModel min="0" max="32">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</deviceModel>
<deviceSoftwareVersion min="0" max="32">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</deviceSoftwareVersion>
<audibleVisualAlarmType opt="CIRCLE,AFE80,AFE82,XY">
  <!--ro, opt, enum, CIRCLE; AFE80; AFE82; XY, subType:string, attr:opt{req, string}, desc:CIRCLE; AFE80; AFE82; XY-->CIRCLE
</audibleVisualAlarmType>
<cardReaderType opt="K1108AM">
  <!--ro, opt, string, attr:opt{req, string}-->test
</cardReaderType>
<strongElectricType opt="19M04-RO/T,19M08-RO/T">
  <!--ro, opt, string, attr:opt{req, string}-->test
</strongElectricType>
<signalLampControl>
  <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,signalLampControl}-->
  <protocolType opt="GB920,Hisense,LESER,JERE">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB920
  </protocolType>
</signalLampControl>
</SerialPort>

```

12.1.4.2 Set the parameters of a specific serial port supported by the device

Request URL

PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	--
childDevID	string	--
deviceIndex	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, serial port ID-->0
  </id>
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <encryptKey>
    <!--opt, string-->test
  </encryptKey>
  <serialNumber>
    <!--opt, int, serial No.-->0
  </serialNumber>
  <serialPortType>
    <!--opt, enum, serial port type: "RS485","RS422","RS232", subType:string, desc:serial port type: "RS485","RS422","RS232"-->RS485
  </serialPortType>
  <serialAddress>
    <!--opt, int-->1
  </serialAddress>
  <duplexMode>
    <!--opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <direction>
    <!--opt, enum, "monodirectional,bidirectional", subType:string, desc:"monodirectional, bidirectional"-->monodirectional
  </direction>
  <baudRate>
    <!--opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--opt, int-->6
  </dataBits>
  <parityType>
    <!--opt, enum, parity type, subType:string, desc:"none, even, odd, mark, space"-->none
  </parityType>
  <stopBits>
    <!--opt, string, stop bit: "1,1.5,2"-->1
  </stopBits>
  <workMode>
    <!--opt, enum, work mode, subType:string, desc:working mode: "console","transparent","audiomixer","stairsControl"-elevator control,"cardReader"-card reader,"disabled","custom". This mode is required only when <serialPortType> is set to "RS232"-->console
  </workMode>
  <flowCtrl>
    <!--opt, enum, "none,software,hardware", subType:string, desc:"none, software, hardware"-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--opt, enum, working mode of RS-485 serial port, which is used for LED display or triggering transmission of captured pictures: "Led, CaptureTrigger". This node is valid only when <serialPortType> is "RS485", subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial port,which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is "RS485"-->sensor
  </rs485WorkMode>
  <copyToAll>
    <!--opt, bool-->true
  </copyToAll>
  <deviceName>
    <!--ro, opt, string-->test
  </deviceName>
  <deviceProtocol>
    <!--ro, opt, int-->1
  </deviceProtocol>
  <mode>
    <!--opt, enum, work mode, subType:string, desc:deq,working mode: "readerMode,clientMode,externMode,accessControlHost,disabled",this node is valid only when <serialPortType> is "RS485"-->readerMode
  </mode>
  <outputDataType>
    <!--opt, enum, output data type, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}, desc:"cardNo,employeeNo", this node is valid when <mode>is "accessControlHost"-->cardNo
  </outputDataType>
  <isVariable>
    <!--opt, bool-->true
  </isVariable>
  <PTZInfo>
    <!--opt, object, PTZ information-->
  <bindChannelNo>
    <!--req, int-->0
  </bindChannelNo>
</SerialPort>
```

```

</bindChannelNo>
<PTZProcollList>
  <!--req, object-->
  <PTZProtocol>
    <!--opt, object-->
    <id>
      <!--req, int, serial port ID-->0
    </id>
    <protoDesc>
      <!--req, string-->test
    </protoDesc>
  </PTZProtocol>
</PTZProcollList>
</PTZInfo>
<ScreenCtrlInfo>
  <!--opt, object-->
  <ScreenCtrlProtoList>
    <!--req, object-->
    <ScreenCtrlProto>
      <!--opt, object-->
      <id>
        <!--req, int, serial port ID-->0
      </id>
      <protoDesc>
        <!--req, string-->test
      </protoDesc>
    </ScreenCtrlProto>
  </ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo>
  <!--opt, object-->
  <MatrixCtrlProtoList>
    <!--req, object-->
    <MatrixCtrlProto>
      <!--opt, object-->
      <id>
        <!--req, int, serial port ID-->0
      </id>
      <protoDesc>
        <!--req, string-->test
      </protoDesc>
    </MatrixCtrlProto>
  </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyBoardCtrlInfo>
  <!--opt, object-->
  <KeyBoardCtrlProtoList>
    <!--req, object-->
    <KeyBoardCtrlProto>
      <!--opt, object-->
      <id>
        <!--req, int, serial port ID-->0
      </id>
      <protoDesc>
        <!--req, string-->test
      </protoDesc>
    </KeyBoardCtrlProto>
  </KeyBoardCtrlProtoList>
</KeyBoardCtrlInfo>
<ControlAddress>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <address>
    <!--req, int, address, range:[0,255]-->1
  </address>
</ControlAddress>
<SensorCtrlList>
  <!--opt, object-->
  <Sensor>
    <!--opt, object-->
    <id>
      <!--req, int, ID, range:[1,3]-->1
    </id>
    <name>
      <!--req, string, sensor name, range:[1,32]-->visibilitySensor
    </name>
    <enabled>
      <!--req, bool, whether to enable the function, desc:whether to enable the serial port: "true,false"-->>false
    </enabled>
  </Sensor>
</SensorCtrlList>
<stairsControl>
  <!--opt, object, elevator control parameters, this node is valid when the value of the node <mode> is stairsControl, dep:and,
  {$.SerialPort.mode,eq,stairsControl}-->
  <outputDataType>
    <!--opt, enum, output data type, subType:string, desc:"cardNo,employeeNo", this node is valid when <mode>is "accessControlHost"-->floorNumber
  </outputDataType>
</stairsControl>
<comMode>
  <!--opt, enum, subType:string-->active
</comMode>

```

```

<ModbusRTU>
  <!--opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <ModbusRTUBaudRate>
    <!--opt, enum, subType:string-->9600
  </ModbusRTUBaudRate>
  <ModbusRTUDataBits>
    <!--opt, enum, subType:int-->6
  </ModbusRTUDataBits>
  <ModbusRTUParityType>
    <!--opt, enum, subType:string-->none
  </ModbusRTUParityType>
  <ModbusRTUStopBits>
    <!--opt, enum, subType:string-->1
  </ModbusRTUStopBits>
  <ModbusRTUFlowCtrl>
    <!--opt, enum, subType:string-->none
  </ModbusRTUFlowCtrl>
  <decoderAddress>
    <!--opt, int, range:[0,255]-->0
  </decoderAddress>
</ModbusRTU>
<accessName>
  <!--opt, string, range:[1,32]-->test
</accessName>
<serialPortUse>
  <!--opt, enum, subType:string-->acquisition
</serialPortUse>
<acquisitionProtocol>
  <!--opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}-->modbus
</acquisitionProtocol>
<serialProtocol>
  <!--opt, enum, subType:string-->GB26875
</serialProtocol>
<SupplementLightList>
  <!--opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
  <SupplementLight>
    <!--opt, object-->
    <SupplementLightID>
      <!--req, int, range:[1,5]-->1
    </SupplementLightID>
  </SupplementLight>
</SupplementLightList>
<cardReaderType>
  <!--opt, enum, subType:string-->K1108AM
</cardReaderType>
</SerialPort>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>

```

12.1.4.3 Get the parameters of a specific port supported by the device

Request URL

GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	0 (main permission controller), 1 (sub permission controller) When this parameter does not exist, it indicates getting the serial port information of the main permission controller (i.e., the device serial port information).
childDevID	string	--
deviceIndex	string	For C60S devices connected to multiple LED sending cards, the URL needs to include deviceIndex= to distinguish which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, serial port ID-->0
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the serial port-->true
  </enabled>
  <encryptKey>
    <!--ro, opt, string-->test
  </encryptKey>
  <serialNumber>
    <!--ro, opt, int, serial No.-->0
  </serialNumber>
  <serialPortType>
    <!--ro, opt, enum, serial port type, subType:string, desc:"RS485", "RS422", "RS232"-->RS485
  </serialPortType>
  <serialAddress>
    <!--ro, opt, int-->1
  </serialAddress>
  <duplexMode>
    <!--ro, opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <direction>
    <!--ro, opt, enum, "monodirectional,bidirectional", subType:string, desc:"monodirectional", "bidirectional"-->monodirectional
  </direction>
  <baudRate>
    <!--ro, opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--ro, opt, int-->6
  </dataBits>
  <parityType>
    <!--ro, opt, enum, parity type, subType:string, desc:parity type: "none,even,odd,mark,space"-->none
  </parityType>
  <stopBits>
    <!--ro, opt, enum, stop bit, subType:string, desc:stop bit-->1
  </stopBits>
  <workMode>
    <!--ro, opt, enum, working mode, subType:string, desc:"console", "transparent", "audiomixer", "screenCtrl", "ptzCtrl", "keyboard", "matrix",
    "audioMixers"-->console
  </workMode>
  <flowCtrl>
    <!--ro, opt, enum, "none,software,hardware", subType:string, desc:"none", "software", "hardware"-->none
  </flowCtrl>
  <rs485WorkMode>
    <!--ro, opt, enum, working mode of RS-485 serial port, subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}, desc:working mode of RS-485 serial
    port,which is used for LED display or triggering transmission of captured pictures: "Led,CaptureTrigger". This node is valid only when <serialPortType> is
    "RS485"-->sensor
  </rs485WorkMode>
  <copyToAll>
    <!--ro, opt, bool-->true
  </copyToAll>
  <deviceName>
    <!--ro, opt, string-->test
  </deviceName>
  <deviceProtocol>
    <!--ro, opt, int-->1
  </deviceProtocol>
  <mode>
```

```

    <!--ro, opt, enum, working mode, subType:string, desc:deg,working mode: "readerMode,clientMode,externMode,accessControlHost,disabled",this node is valid
only when <serialPortType> is "RS485"-->readerMode
</mode>
<outputDataType>
    <!--ro, opt, enum, output data type, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}, desc:output data type: "cardNo,employeeNo",this
node is valid when <mode>is "accessControlHost"-->cardNo
</outputDataType>
<isVariable>
    <!--ro, opt, bool-->true
</isVariable>
<PTZInfo>
    <!--ro, opt, object, PTZ information-->
    <bindChannelNo>
        <!--ro, req, int-->0
    </bindChannelNo>
    <PTZProtocolList>
        <!--ro, req, object-->
        <PTZProtocol>
            <!--ro, opt, object-->
            <id>
                <!--ro, req, int, serial port ID-->0
            </id>
            <protoDesc>
                <!--ro, req, string-->test
            </protoDesc>
        </PTZProtocol>
    </PTZProtocolList>
</PTZInfo>
<ScreenCtrlInfo>
    <!--ro, opt, object-->
    <ScreenCtrlProtoList>
        <!--ro, req, object-->
        <ScreenCtrlProto>
            <!--ro, opt, object-->
            <id>
                <!--ro, req, int, serial port ID-->0
            </id>
            <protoDesc>
                <!--ro, req, string-->test
            </protoDesc>
        </ScreenCtrlProto>
    </ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo>
    <!--ro, opt, object-->
    <MatrixCtrlProtoList>
        <!--ro, req, object-->
        <MatrixCtrlProto>
            <!--ro, opt, object-->
            <id>
                <!--ro, req, int, serial port ID-->0
            </id>
            <protoDesc>
                <!--ro, req, string-->test
            </protoDesc>
        </MatrixCtrlProto>
    </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyboardCtrlInfo>
    <!--ro, opt, object-->
    <KeyboardCtrlProtoList>
        <!--ro, req, object-->
        <KeyboardCtrlProto>
            <!--ro, opt, object-->
            <id>
                <!--ro, req, int, serial port ID-->0
            </id>
            <protoDesc>
                <!--ro, req, string-->test
            </protoDesc>
        </KeyboardCtrlProto>
    </KeyboardCtrlProtoList>
</KeyboardCtrlInfo>
<ControlAddress>
    <!--ro, opt, object-->
    <enabled>
        <!--ro, req, bool, whether to enable the serial port-->true
    </enabled>
    <address>
        <!--ro, req, int, address, range:[0,255]-->0
    </address>
</ControlAddress>
<SensorCtrlList>
    <!--ro, opt, object-->
    <Sensor>
        <!--ro, opt, object-->
        <id>
            <!--ro, req, int, serial port ID, range:[1,10]-->1
        </id>
        <name>
            <!--ro, req, enum, sensor name, subType:string, desc:sensor name-->visibilitySensor
        </name>
        <enabled>
            <!--ro, req, bool, whether to enable the function. desc:whether to enable the serial port: "true,false"-->>false
        </enabled>
    </Sensor>
</SensorCtrlList>

```

```

    </enabled>
    <senorOnlineStatus>
      <!--ro, opt, enum, subType:string-->online
    </senorOnlineStatus>
  </Sensor>
</SensorCtrlList>
<stairsControl>
  <!--ro, opt, object, elevator control parameters, dep:and,{$.SerialPort.mode,eq,stairsControl}-->
  <outputDataType>
    <!--ro, opt, enum, output data type, subType:string, desc:output data type: "cardNo,employeeNo",this node is valid when <mode>is "accessControlHost"-->
  </outputDataType>
  <stairsControl>
  <comMode>
    <!--ro, opt, enum, subType:string-->active
  </comMode>
  <ModbusRTU>
    <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
    <ModbusRTUBaudRate>
      <!--ro, opt, enum, subType:string-->9600
    </ModbusRTUBaudRate>
    <ModbusRTUDataBits>
      <!--ro, opt, enum, subType:int-->6
    </ModbusRTUDataBits>
    <ModbusRTUParityType>
      <!--ro, opt, enum, subType:string-->none
    </ModbusRTUParityType>
    <ModbusRTUStopBits>
      <!--ro, opt, enum, subType:string-->1
    </ModbusRTUStopBits>
    <ModbusRTUFlowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </ModbusRTUFlowCtrl>
    <decoderAddress>
      <!--ro, opt, int, range:[0,255]-->0
    </decoderAddress>
    <online>
      <!--ro, opt, bool-->true
    </online>
  </ModbusRTU>
  <accessName>
    <!--ro, opt, string, range:[1,32]-->test
  </accessName>
  <serialPortUse>
    <!--ro, opt, enum, subType:string-->acQuAndCtrl
  </serialPortUse>
  <serialProtocol>
    <!--ro, opt, enum, subType:string-->GB26875
  </serialProtocol>
  <SupplementLightList>
    <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
    <SupplementLight>
      <!--ro, opt, object-->
      <SupplementLightID>
        <!--ro, req, int, range:[1,5]-->1
      </SupplementLightID>
    </SupplementLight>
  </SupplementLightList>
  <cardReaderType>
    <!--ro, opt, enum, subType:string-->K1108AM
  </cardReaderType>
  <strongElectricType>
    <!--ro, opt, enum, subType:string-->19M04-RO/T
  </strongElectricType>
  <signalLampControl>
    <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,signalLampControl}-->
    <protocolType>
      <!--ro, opt, enum, subType:string-->GB920
    </protocolType>
  </signalLampControl>
  <passingMode>
    <!--ro, opt, enum, subType:string, dep:or,{$.SerialPort.mode,eq,fingerprintReader}-->entrance
  </passingMode>
  <modbusProtocol>
    <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
    <checkBitOrder>
      <!--ro, opt, enum, subType:string-->bigEndian
    </checkBitOrder>
    <deviceMode>
      <!--ro, opt, enum, subType:string-->subordinate
    </deviceMode>
  </modbusProtocol>
  <serialPortName>
    <!--ro, opt, string, range:[1,32]-->test
  </serialPortName>
  <accessProtocolType>
    <!--ro, opt, enum, subType:string-->PELCO
  </accessProtocolType>
  <weightSenseInfo>
    <!--ro, opt, object-->
    <countTimes>
      <!--ro, opt, enum, subType:float-->0.01
    </countTimes>

```

```

    <countUnit>
    <!--ro, opt, enum, subType:string-->kg
    </countUnit>
</weightSenseInfo>
<PDUCtrlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PDUCtrl}-->DS-65A2T0008
</PDUCtrlType>
<paperLessOfficeType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,paperLess}-->DS-65PL0015-T
</paperLessOfficeType>
<videoWallController1Type>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,spellingCtrl}-->matrixProtocol
</videoWallController1Type>
<audioCtrlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,audioCtrl}-->DS-QAPDM16081C
</audioCtrlType>
<interactiveTabletType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,interactiveTablet}-->interactiveTabletProtocol
</interactiveTabletType>
<humanSensorType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,humanSensor}-->HM-TD1017-2/QW-HS121
</humanSensorType>
<transparentOverWebsocket>
    <!--ro, opt, bool-->true
</transparentOverWebsocket>
<PTZCtrlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PTZCtrl}-->VISCA
</PTZCtrlType>
<airConditionerCtrlType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,airConditionerCtrl}-->GL
</airConditionerCtrlType>
<acquisitionLinkageType>
    <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisitionLinkage}-->general
</acquisitionLinkageType>
<externalInterfaceBoard>
    <!--ro, opt, object, dep:or,{$.SerialPort.mode,eq,externalInterfaceBoard}-->
    <serialPortList>
    <!--ro, opt, array, subType:object-->
    <serialPort>
    <!--ro, opt, object-->
    <id>
    <!--ro, req, int-->0
    </id>
    <serialPortType>
    <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <RS485serialProtocol>
    <!--ro, opt, enum, subType:string-->private485
    </RS485serialProtocol>
    <private485Mode>
    <!--ro, opt, enum, subType:string-->candReceiver
    </private485Mode>
    <RS232Mode>
    <!--ro, opt, enum, subType:string-->readerMode
    </RS232Mode>
    </serialPort>
    </serialPortList>
</externalInterfaceBoard>
</SerialPort>

```

12.1.4.4 Get the list of serial ports supported by the device

Request URL

GET /ISAPI/System/Serial/ports?permissionController= <indexID> &deviceIndex= <deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	0 (main permission controller), 1 (sub permission controller) If this parameter does not exist, it indicates getting the serial port information of main permission controller, that is, the serial port information of the device.
deviceIndex	string	For C60S device connecting multiple LED sending cards, the URL needs to include the parameter deviceIndex= to specify which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPortList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SerialPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <encryptKey>
      <!--ro, opt, string-->test
    </encryptKey>
    <serialNumber>
      <!--ro, opt, int-->0
    </serialNumber>
    <serialPortType>
      <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <serialAddress>
      <!--ro, opt, int-->1
    </serialAddress>
    <duplexMode>
      <!--ro, opt, enum, subType:string-->half
    </duplexMode>
    <direction>
      <!--ro, opt, enum, subType:string-->monodirectional
    </direction>
    <baudRate>
      <!--ro, opt, enum, subType:int-->2400
    </baudRate>
    <dataBits>
      <!--ro, opt, int-->6
    </dataBits>
    <parityType>
      <!--ro, opt, enum, subType:string-->none
    </parityType>
    <stopBits>
      <!--ro, opt, enum, subType:string-->1
    </stopBits>
    <workMode>
      <!--ro, opt, enum, subType:string-->console
    </workMode>
    <flowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </flowCtrl>
    <rs485WorkMode>
      <!--ro, opt, enum, subType:string, dep:or,{$.SerialPort.serialPortType,eq,RS485}-->sensor
    </rs485WorkMode>
    <copyToAll>
      <!--ro, opt, bool-->true
    </copyToAll>
    <deviceName>
      <!--ro, opt, string-->test
    </deviceName>
    <deviceProtocol>
      <!--ro, opt, int-->1
    </deviceProtocol>
    <mode>
      <!--ro, opt, enum, subType:string-->readerMode
    </mode>
    <outputDataType>
      <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.mode,eq,accessControlHost}-->cardNo
    </outputDataType>
    <isVariable>
      <!--ro, opt, bool-->true
    </isVariable>
    <PTZInfo>
      <!--ro, opt, object-->
      <bindChannelNo>
        <!--ro, req, int-->0
      </bindChannelNo>
      <PTZProtocolList>
        <!--ro, req, object-->
        <PTZProtocol>
          <!--ro, opt, object-->
          <id>
            <!--ro, req, int-->0
          </id>
          <protoDesc>
            <!--ro, req, string-->test
          </protoDesc>
        </PTZProtocol>
      </PTZProtocolList>
    </PTZInfo>
    <ScreenCtrlInfo>
      <!--ro, opt, object-->
      <ScreenCtrlProtoList>
        <!--ro, req, object-->
```

```

    <ScreenCtrlProto>
    <!--ro, opt, object-->
    <id>
    <!--ro, req, int-->0
    </id>
    <protoDesc>
    <!--ro, req, string-->test
    </protoDesc>
  </ScreenCtrlProto>
</ScreenCtrlProtoList>
</ScreenCtrlInfo>
<MatrixCtrlInfo>
  <!--ro, opt, object-->
  <MatrixCtrlProtoList>
    <!--ro, req, object-->
    <MatrixCtrlProto>
      <!--ro, opt, object-->
      <id>
      <!--ro, req, int-->0
      </id>
      <protoDesc>
      <!--ro, req, string-->test
      </protoDesc>
    </MatrixCtrlProto>
  </MatrixCtrlProtoList>
</MatrixCtrlInfo>
<KeyboardCtrlInfo>
  <!--ro, opt, object-->
  <KeyboardCtrlProtoList>
    <!--ro, req, object-->
    <KeyboardCtrlProto>
      <!--ro, opt, object-->
      <id>
      <!--ro, req, int-->0
      </id>
      <protoDesc>
      <!--ro, req, string-->test
      </protoDesc>
    </KeyboardCtrlProto>
  </KeyboardCtrlProtoList>
</KeyboardCtrlInfo>
<ControlAddress>
  <!--ro, opt, object-->
  <enabled>
  <!--ro, req, bool-->true
  </enabled>
  <address>
  <!--ro, req, int, range:[0,255]-->1
  </address>
</ControlAddress>
<SensorCtrlList>
  <!--ro, opt, object-->
  <Sensor>
    <!--ro, opt, object-->
    <id>
    <!--ro, req, int, range:[1,10]-->1
    </id>
    <name>
    <!--ro, req, enum, subType:string-->visibilitySensor
    </name>
    <enabled>
    <!--ro, req, bool-->false
    </enabled>
    <senorOnlineStatus>
    <!--ro, opt, enum, subType:string-->online
    </senorOnlineStatus>
  </Sensor>
</SensorCtrlList>
<stairsControl>
  <!--ro, opt, object, dep:and,{$.SerialPort.mode,eq,stairsControl}-->
  <outputDataType>
  <!--ro, opt, enum, subType:string-->floorNumber
  </outputDataType>
</stairsControl>
<comMode>
  <!--ro, opt, enum, subType:string-->active
</comMode>
<ModbusRTU>
  <!--ro, opt, object, dep:and,{$.SerialPortList.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <ModbusRTUBaudRate>
  <!--ro, opt, enum, subType:string-->9600
  </ModbusRTUBaudRate>
  <ModbusRTUDataBits>
  <!--ro, opt, enum, subType:int-->6
  </ModbusRTUDataBits>
  <ModbusRTUParityType>
  <!--ro, opt, enum, subType:string-->none
  </ModbusRTUParityType>
  <ModbusRTUStopBits>
  <!--ro, opt, enum, subType:string-->1
  </ModbusRTUStopBits>
  <ModbusRTUFlowCtrl>
  <!--ro, opt, enum, subType:string-->none
  </ModbusRTUFlowCtrl>

```

```

</ModbusRTUFlowCtrl>
<decoderAddress>
  <!--ro, opt, int, range:[0,255]-->0
</decoderAddress>
<online>
  <!--ro, opt, bool-->true
</online>
</ModbusRTU>
<accessName>
  <!--ro, opt, string, range:[1,32]-->test
</accessName>
<serialPortUse>
  <!--ro, opt, enum, subType:string-->acquAndCtrl
</serialPortUse>
<serialProtocol>
  <!--ro, opt, enum, subType:string-->GB26875
</serialProtocol>
<accessProtocolType>
  <!--ro, opt, enum, subType:string-->PELCO
</accessProtocolType>
<SupplementLightList>
  <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}-->
  <SupplementLight>
    <!--ro, opt, object-->
    <SupplementLightID>
      <!--ro, req, int, range:[1,5]-->1
    </SupplementLightID>
  </SupplementLight>
</SupplementLightList>
<cardReaderType>
  <!--ro, opt, enum, subType:string-->K1108AM
</cardReaderType>
<strongElectricType>
  <!--ro, opt, enum, subType:string-->19M04-RO/T
</strongElectricType>
<passingMode>
  <!--ro, opt, enum, subType:string, dep:or,{$.SerialPortList.SerialPort.mode,eq,fingerprintReader}-->entrance
</passingMode>
<modbusProtocol>
  <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,ModbusRTU}-->
  <checkBitOrder>
    <!--ro, opt, enum, subType:string-->bigEndian
  </checkBitOrder>
  <deviceMode>
    <!--ro, opt, enum, subType:string-->subordinate
  </deviceMode>
</modbusProtocol>
<serialPortName>
  <!--ro, opt, string, range:[1,32]-->test
</serialPortName>
<PDUCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PDUCtrl}-->DS-65A2T0008
</PDUCtrlType>
<paperLessOfficeType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,paperLess}-->DS-65PL0015-T
</paperLessOfficeType>
<videoWallControllerType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,spellingCtrl}-->matrixProtocol
</videoWallControllerType>
<audioCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,audioCtrl}-->DS-QAPDM16081C
</audioCtrlType>
<interactiveTabletType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,interactiveTablet}-->interactiveTabletProtocol
</interactiveTabletType>
<humanSensorType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,humanSensor}-->HM-TD1017-2/QW-HS121
</humanSensorType>
<transparentOverWebsocket>
  <!--ro, opt, bool-->true
</transparentOverWebsocket>
<PTZCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,PTZCtrl}-->VISCA
</PTZCtrlType>
<airConditionerCtrlType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,airConditionerCtrl}-->GL
</airConditionerCtrlType>
<acquisitionLinkageType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SerialPort.serialPortUse,eq,acquisitionLinkage}-->general
</acquisitionLinkageType>
<externalInterfaceBoard>
  <!--ro, opt, object, dep:or,{$.SerialPort.mode,eq,externalInterfaceBoard}-->
  <serialPortList>
    <!--ro, opt, array, subType:object-->
    <serialPort>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->0
      </id>
      <serialPortType>
        <!--ro, opt, enum, subType:string-->RS485
      </serialPortType>
      <RS485serialProtocol>
        <!--ro, opt, enum, subType:string-->private485
      </RS485serialProtocol>
    </serialPort>
  </serialPortList>

```

```

</RS485SerialProtocol>
<private485Mode>
  <!--ro, opt, enum, subType:string-->cardReceiver
</private485Mode>
<RS232Mode>
  <!--ro, opt, enum, subType:string-->readerMode
</RS232Mode>
</serialPort>
</serialPortList>
</externalInterfaceBoard>
</SerialPort>
</SerialPortList>

```

12.1.5 System Maintenance

12.1.5.1 Get the system security capability

Request URL

GET /ISAPI/Security/capabilities?username=<userName>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
userName	string	user name
deviceIndex	string	For centralized splicing controllers connecting to LED sending cards, the URL needs to include the parameter deviceIndex= to specify LED sending card The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<SecurityCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, system security capability, attr:version(req, string, protocolVersion)-->
  <supportUserNums>
    <!--ro, opt, int, supported maximum number of users-->0
  </supportUserNums>
  <userBondIpNums>
    <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->0
  </userBondIpNums>
  <userBondMacNums>
    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->0
  </userBondMacNums>
  <isSupCertificate>
    <!--ro, opt, bool, whether the device supports authentication-->true
  </isSupCertificate>
  <issupIllegalLoginLock>
    <!--ro, opt, bool, whether the device supports Locking Login-->true
  </issupIllegalLoginLock>
  <isSupportOnlineUser>
    <!--ro, opt, bool, whether the device supports the online user configuration-->true
  </isSupportOnlineUser>
  <isSupportAnonymous>
    <!--ro, opt, bool, whether the device supports anonymous Login-->true
  </isSupportAnonymous>
  <isSupportStreamEncryption>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
  </isSupportStreamEncryption>
  <securityVersion opt="1,2,3,4">
    <!--ro, opt, int, encryption capability set, attr:opt(req, string), desc:the encryption capability of each version consists of two parts: encryption algorithm and the range of encrypted nodes currently 1 refers to AES128 encryption and 2 refers to AES256 encryption, the range of encrypted nodes is described in each protocol-->1
  </securityVersion>
  <keyIterateNum>
    <!--ro, opt, int, secret key iteration times, dep:or,{$.SecurityCap.securityVersion,eq,1},{$.SecurityCap.securityVersion,eq,2}, desc:this node depends on the node securityVersion, the range is between 100 and 1000-->100
  </keyIterateNum>
  <isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the Login password when editing (editing/adding/deleting) user parameters, dep:or,{$.SecurityCap.securityVersion,eq,0},{$.SecurityCap.securityVersion,eq,1}, desc:it is an added capability, which indicates that whether supporting the Login password verification for editing/adding/deleting user parameters, this node depends on the node securityVersion, which means that it is only valid for the versions that support encrypting the sensitive information-->true
  </isSupportUserCheck>

```

```

<isSupportGUIDFileDataExport>
  <!--ro, opt, bool, whether the device supports exporting the device's GUID file, desc:related URI: /ISAPI/Security/GUIDFileData-->true
</isSupportGUIDFileDataExport>
<isSupportSecurityQuestionConfig>
  <!--ro, opt, bool, whether the device supports answering security questions, desc:related URI: /ISAPI/Security/questionConfiguration-->true
</isSupportSecurityQuestionConfig>
<supportSecurityNode opt="wirelessServer,wirelessDial">
  <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt{req, string},
desc:in the earlier version of the solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI),
some URIs containing sensitive information are left out now these URIs need to be encrypted for transmission, this capability node is added to be compatible
with old devices _x000D_
    if wirelessServer in this node is returned, it indicates that the following sensitive information should be encrypted _x000D_
    /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities, /ISAPI/System/Network/interfaces/<ID>/wirelessServer, if wirelessDial in this
node is returned, it indicates that the following sensitive information should be encrypted _x000D_
    /ISAPI/System/Network/WirelessDial/Interfaces/_x000D_
    /ISAPI/System/Network/WirelessDial/Interfaces/<ID>-->wirelessServer
</supportSecurityNode>
<isSupportGetOnlineUserListSC>
  <!--ro, opt, bool, whether the device supports searching the online user list, desc:related URI: /ISAPI/Security/onlineUser-->true
</isSupportGetOnlineUserListSC>
<SecurityLimits>
  <!--ro, opt, object, capability of configuring security limit parameters-->
  <LoginPasswordLenLimit min="1" max="16">
    <!--ro, opt, string, length limit of the user's login password, attr:min{req, int},max{req, int}-->1
  </LoginPasswordLenLimit>
  <SecurityAnswerLenLimit min="1" max="128">
    <!--ro, opt, string, length limit of the security question's answer, attr:min{req, int},max{req, int}-->1
  </SecurityAnswerLenLimit>
</SecurityLimits>
<RSAKeyLength opt="512,1024,2048" def="2048">
  <!--ro, opt, enum, HTTPS certificate length, subType:int, attr:opt{req, string},def{req, string}, desc:512, 1024, 2048-->2048
</RSAKeyLength>
<isSupportONVIFUserManagement>
  <!--ro, opt, bool, whether the device supports user management of Open Network Video Interface Protocol-->true
</isSupportONVIFUserManagement>
<WebCertificateCap>
  <!--ro, opt, object, HTTP authentication capability, desc:if this node is not returned, it indicates that device supports basic and digest
authentication-->
  <CertificateType opt="basic,digest,digest/basic">
    <!--ro, req, enum, certificate type, subType:string, attr:opt{req, string}, desc:"basic" (basic authentication), "digest" (digest authentication),
"digest/basic" (digest/basic authentication)-->basic
  </CertificateType>
  <SecurityAlgorithm>
    <!--ro, opt, object, security algorithm, dep:or,{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest},
{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256">
      <!--ro, opt, enum, algorithm type, subType:string, attr:opt{req, string}, desc:"MD5", "SHA256", "MD5/SHA256"-->MD5
    </algorithmType>
  </SecurityAlgorithm>
</WebCertificateCap>
<isSupportConfigFileImport>
  <!--ro, opt, bool, whether the device supports importing the configuration file, desc:true (support), this node is not returned (not support)-->true
</isSupportConfigFileImport>
<isSupportConfigFileExport>
  <!--ro, opt, bool, whether the device supports exporting the configuration file, desc:true (support), this node is not returned (not support)-->true
</isSupportConfigFileExport>
<cfgFileSecretKeyLenLimit min="0" max="16">
  <!--ro, opt, int, length limit of the configuration file's verification key, attr:min{req, int},max{req, int}-->1
</cfgFileSecretKeyLenLimit>
<supportIPCActivatePassword>
  <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
</supportIPCActivatePassword>
<isIrreversible>
  <!--ro, opt, bool, whether the device supports irreversible password storage, desc:If this function is not supported, the plaintext password of the user
information will be stored in the device; otherwise, the password will be hashed for storage in the device.-->true
</isIrreversible>
<salt>
  <!--ro, opt, string, the specific salt used by the user to log in-->test
</salt>
<isSupportOnvifInfo>
  <!--ro, opt, bool, whether the device supports getting the Open Network Video Interface Protocol, desc:true (support), this node is not returned (not
support)-->true
</isSupportOnvifInfo>
<isSupportPictureURLCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{req, string}, desc:if this node is not
returned, it indicates the digest authentication for picture URL by default-->true
</isSupportPictureURLCertificate>
<isSupportGB35114Certificate>
  <!--ro, opt, bool, whether supports GB35114 protocol and CA/SIP authentication certificate-->true
</isSupportGB35114Certificate>
<isSupportAuthenticationMode opt="uKey,private">
  <!--ro, opt, enum, verification mode, subType:string, attr:opt{req, string}, desc:"uKey", "private"-->private
</isSupportAuthenticationMode>
<isSupportUnloggedUserPermissionConfig opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{req, string}-->true
</isSupportUnloggedUserPermissionConfig>
<isSupportUserNamePasswordCheckUpgrade>
  <!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password, desc:if this node is not returned, it indicates
that device does not support this function-->true
</isSupportUserNamePasswordCheckUpgrade>
<isSupportDeviceCertificatesManagement>
  <!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not
support this function-->true

```

```

</isSupportDeviceCertificatesManagement>
<isSupportSIPCertificatesManagement>
  <!--ro, opt, bool, whether the device supports SIP certificate management, desc:this node is returned if the device supports this function and not
  returned if device does not support this function, related URI: /ISAPI/Security/SIPCertificate, additional node: <SIPCertificatesManagementCap>-->true
</isSupportSIPCertificatesManagement>
<isSupportDeviceSelfSignCertExport>
  <!--ro, opt, bool, whether the device supports exporting self-signed certificate-->true
</isSupportDeviceSelfSignCertExport>
<isSupportSecurityEmail>
  <!--ro, opt, bool, whether the device supports configuring the security email, desc:if this node is not returned, it indicates that device does not
  support this function-->true
</isSupportSecurityEmail>
<isSupportRTSPCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports RTSP certificate management, attr:opt{req, string}-->true
</isSupportRTSPCertificate>
<maxIllegalLoginTimes min="3" max="20" def="5">
  <!--ro, opt, int, maximum failed illegal login attempts, range:[3,20], attr:min{req, int},max{req, int},def{req, int}, desc:this node is valid when the
  value of the node issupportIllegalLoginLock is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
  limit-->3
</maxIllegalLoginTimes>
<SecurityAdvanced>
  <!--ro, opt, object, security advanced parameters-->
  <securityEnhanced>
    <!--ro, opt, bool, whether the device supports security reinforcement-->true
  </securityEnhanced>
  <noOperationEnabled>
    <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
  </noOperationEnabled>
  <noOperationTime min="1" max="60" def="15">
    <!--ro, req, int, control timeout when there is no operation, range:[1,60], unit:min, attr:min{req, int},max{req, int},def{req, int}-->15
  </noOperationTime>
  <isSupportDigestStatus>
    <!--ro, opt, bool, digest status-->true
  </isSupportDigestStatus>
  <passwordLimitLevel opt="middling,high">
    <!--ro, opt, enum, password strength limit level, subType:string, attr:opt{req, string},
    desc:the configured password is required by the following limit _x0000_
    "middling" (middle level), "high" (high level), middle level password contains 8 to 16 characters, including at least 2 types of the following characters:
    digits, lower case letters, upper case letters, and special characters; high level password contains 8 to 16 characters, including at least 3 types of the
    following characters: digits, lower case letters, upper case letters, and special characters-->middling
  </passwordLimitLevel>
</SecurityAdvanced>
<LoginLinkNum>
  <!--ro, opt, object, number of controls logging in at the same time-->
  <maxLinkNum min="1" max="128" def="50">
    <!--ro, req, int, maximum number of controls logging in at the same time, range:[1,128], attr:min{req, int},max{req, int},def{req, int}-->50
  </maxLinkNum>
</LoginLinkNum>
<isSupportCCClientCertificate>
  <!--ro, opt, bool, whether the device supports the CC (Client Certificate) function-->true
</isSupportCCClientCertificate>
<passwordValidity min="0" max="365">
  <!--ro, opt, string, range of the password validity period, range:[0,365], attr:min{req, int},max{req, int}, desc:if this node is returned, it indicates
  that the device supports configuring the password validity period, and only the administrator can edit the configuration-->1
</passwordValidity>
<isSupportResetChannelPassword>
  <!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
</isSupportResetChannelPassword>
<DoubleVerificationCap>
  <!--ro, opt, object, secondary authentication capability-->
  <isSupportUsersConfig>
    <!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
  </isSupportUsersConfig>
  <isSupportUsersPermissionConfig>
    <!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
  </isSupportUsersPermissionConfig>
  <isSupportFactoryReset>
    <!--ro, opt, bool-->true
  </isSupportFactoryReset>
</DoubleVerificationCap>
<isSupportUserSearch>
  <!--ro, opt, bool, whether the device supports searching users-->true
</isSupportUserSearch>
<keypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length, attr:min{req, int},max{req, int}, desc:When the password length of different user types is different, this
  node only indicates the keypad password length of the administrator; otherwise, the keypad password length of all user types is the same and defined by the
  value of this node-->1
</keypadPassword>
<installerKeypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length of the installer, attr:min{req, int},max{req, int}-->1
</installerKeypadPassword>
<operatorKeypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length of the operator, attr:min{req, int},max{req, int}-->1
</operatorKeypadPassword>
<userOperateType opt="1,2,3">
  <!--ro, opt, enum, user type, subType:int, attr:opt{req, string}, desc:1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
</userOperateType>
<isSptUserEnabled>
  <!--ro, opt, bool, whether the device supports configuring parameters for enabling the user-->true
</isSptUserEnabled>
<isSptAdminCap>
  <!--ro, opt, bool, whether the device supports getting the admin permission capability-->true
</isSptAdminCap>
<isSupportBackLogCfg>

```

```

<!--ro, opt, bool, whether the device supports configuring basic parameters of the background Log-->true
</isSupportBackLogCfg>
<isSupportBackLogExport>
<!--ro, opt, bool, whether the device supports exporting the background Log-->true
</isSupportBackLogExport>
<isSupportModifyInnerPassword opt="true,false">
<!--ro, opt, bool, whether the device supports editing its built-in passwords,, attr:opt{req, string}, desc:such as the database password, system
account password-->true
</isSupportModifyInnerPassword>
<isSupportEncryptCertificate>
<!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
</isSupportEncryptCertificate>
<maxIllegalLoginLockTime min="1" max="120" def="30.00">
<!--ro, opt, int, Lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
</maxIllegalLoginLockTime>
<cloudSensitiveInfoEncryptType>
<!--ro, opt, string, encryption type of the cloud sensitive information, desc:the current device returns the value "2.1", which indicates that the
method of generating the key for encrypting the sensitive information is SHA256 (calc_sha256 (user, salt, SHA256 (user+salt+SHA256 (user+salt+password))) +
AaBbCcDd1234!@#%)-->2.1
</cloudSensitiveInfoEncryptType>
<isSupportCertificateCustomID>
<!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
</isSupportCertificateCustomID>
<isSupportResetLANPassword>
<!--ro, opt, bool, whether the device supports resetting the device's password in the LAN, desc:related URI:
/ISAPI/Security/resetPassword/certification?format=json-->true
</isSupportResetLANPassword>
<isSupportPicHttpsCertificatesManagement>
<!--ro, opt, bool, whether the device supports managing the HTTPS certificate for pictures, desc:true (support), this node is not returned (not
support)-->true
</isSupportPicHttpsCertificatesManagement>
<isSupportAKSKImport>
<!--ro, opt, bool, whether the device supports importing AKSK, desc:true (support), this node is not returned (not support)-->true
</isSupportAKSKImport>
<isSupportAKSKExport>
<!--ro, opt, bool, whether the device supports exporting AKSK, desc:true (support), this node is not returned (not support)-->true
</isSupportAKSKExport>
<isSupportAKSKReset>
<!--ro, opt, bool, whether the device supports resetting AKSK, desc:true (support), this node is not returned (not support)-->true
</isSupportAKSKReset>
<ExportCertificateFile>
<!--ro, opt, object, whether the device supports exporting the certificate, desc:related URI: POST
/ISAPI/Security/serverCertificate/exportCertificateFile-->
<type opt="GB35114,GB35114Service">
<!--ro, req, enum, types of certificates that can be exported, subtype:string, attr:opt{req, string}, desc:"GB35114" (support exporting GB35114
certificate for the client), "GB35114Service" (support exporting GB35114 certificate for the server)-->GB35114
</type>
</ExportCertificateFile>
<isSupportGB35114FDWSFCertificate>
<!--ro, opt, bool, whether the device supports FDWSF certificate management, desc:true (support), this node is not returned (not support), related URIs:
/ISAPI/Security/GB35114/FDWSFCertificate/capabilities?format=json, POST /ISAPI/Security/GB35114/FDWSFCertificate?format=json, GET
/ISAPI/Security/GB35114/FDWSFCertificate?format=json, and DELETE /ISAPI/Security/GB35114/FDWSFCertificate?format=json-->true
</isSupportGB35114FDWSFCertificate>
<isSupportMediaObjectEncryption>
<!--ro, opt, bool, whether supports setting encryption parameters of the private target in media data for the specified video channel, desc:if this
function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI:
/ISAPI/Security/MediaObjectEncryption/capabilities?format=json-->true
</isSupportMediaObjectEncryption>
<isSupportDisclaimer>
<!--ro, opt, bool, whether support configuring the product disclaimer, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/disclaimer/capabilities?format=json-->true
</isSupportDisclaimer>
<isSupportSoftwareLicense>
<!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
</isSupportSoftwareLicense>
<isSupportInstallerCap>
<!--ro, opt, bool, whether the device supports getting the installer permission, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/UserPermission/installer/capabilities?format=json-->true
</isSupportInstallerCap>
<isSupportResetPasswordFile>
<!--ro, opt, bool, whether the device supports exporting and importing the secret key file of device in the LAN, desc:Related URI:
/ISAPI/Security/resetPassword/keyFile/capabilities?format=json-->true
</isSupportResetPasswordFile>
<isNotSupportLogSeverCertificate>
<!--ro, opt, bool, whether the device support importing Log server certificate, desc:related URI: /ISAPI/Security/serverCertificate/certificate-->true
</isNotSupportLogSeverCertificate>
<isSupportLocalLoginCfg>
<!--ro, opt, bool, whether it supports the capability of device local login parameters, desc:GET/PUT /ISAPI/Security/LocalLoginCfg?format=json-->true
</isSupportLocalLoginCfg>
<isSupportGetProtocolFirewallRulesFileImportStatus>
<!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
</isSupportGetProtocolFirewallRulesFileImportStatus>
<isSupportImportProtocolFirewallRulesFile>
<!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
</isSupportImportProtocolFirewallRulesFile>
<isSupportGetProtocolFirewallEngineUpgradeStatus>
<!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
</isSupportGetProtocolFirewallEngineUpgradeStatus>
<isSupportUpgradeProtocolFirewallEngine>
<!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
</isSupportUpgradeProtocolFirewallEngine>
<isSupportProtocolFirewallVersionInfo>

```



```

<!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
<isSupportProtocolFirewallVersionInfo>
<isSupportProtocolFirewallParams>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
</isSupportProtocolFirewallParams>
<isSupportResetPasswordBySecurityPhoneNo>
  <!--ro, opt, bool, whether it supports resetting device password with phone number verification,
desc:1. attendance check devices support applying a preset phone number during device activation. The user can reset the device password with the preset
phone number, which is similar with password reset via email. The phone number can be changed on the website or Local UI of the device _x0000_
2. if the node is not returned or the returned value is false, it indicates "no"-->true
</isSupportResetPasswordBySecurityPhoneNo>
<isSupportGetIntrusionPreventionEngineUpgradeStatus>
  <!--ro, opt, bool-->true
</isSupportGetIntrusionPreventionEngineUpgradeStatus>
<isSupportIntrusionPreventionEngineVersionInfo>
  <!--ro, opt, bool-->true
</isSupportIntrusionPreventionEngineVersionInfo>
<isSupportIntrusionPreventionParams>
  <!--ro, opt, bool-->true
</isSupportIntrusionPreventionParams>
<isSupportUpgradeIntrusionPreventionEngine>
  <!--ro, opt, bool-->true
</isSupportUpgradeIntrusionPreventionEngine>
<notSupportImportCertificateType opt="CA,CCClientCertificate,GB35114,GB35114Service">
  <!--ro, opt, string, attr:opt{req, string}-->test
</notSupportImportCertificateType>
<isSupportPasswordProtection>
  <!--ro, opt, bool-->true
</isSupportPasswordProtection>
<isSupportGMT0028SecretStandardParam>
  <!--ro, opt, bool-->true
</isSupportGMT0028SecretStandardParam>
<isSupportCipherCardRunningPolicy>
  <!--ro, opt, bool-->true
</isSupportCipherCardRunningPolicy>
</SecurityCap>

```

12.1.5.2 Get device system capability

Request URL

GET /ISAPI/System/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
  <isSupportPreview>
    <!--ro, opt, bool, whether the device supports live view-->true
  </isSupportPreview>
  <isSupportSystemMaintain>
    <!--ro, opt, bool, whether the device supports system maintenance, desc:for traffic devices, this node is required and must be true-->true
  </isSupportSystemMaintain>
  <isSupportReboot>
    <!--ro, opt, bool, whether the device supports rebooting, desc:for traffic devices, this node is required and must be true-->true
  </isSupportReboot>
  <isSupportFactoryReset>
    <!--ro, opt, bool, whether the device supports restoring to default settings, desc:for traffic devices, this node is required and must be true-->true
  </isSupportFactoryReset>
  <isSupportConfigurationData>
    <!--ro, opt, bool, whether the device supports importing and exporting the configuration files safely, desc:for traffic devices, this node is required
and must be true-->true
  </isSupportConfigurationData>
  <isSupportUpdateFirmware>
    <!--ro, opt, bool, whether the device supports upgrading, desc:for traffic devices, this node is required and must be true-->true
  </isSupportUpdateFirmware>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the device information, desc:for traffic devices, this node is required and must be true-->true
  </isSupportDeviceInfo>
  <isSupportTime>
    <!--ro, opt, bool, whether the device supports time configuration, desc:for traffic device, this node is required and must be true. For other devices,
the isSupportTimeCap is used to configure time-->true
  </isSupportTime>
  <SysCap>
    <!--ro, opt, object, system capability-->
    <isSupportDst>
      <!--ro, opt, bool, whether the device supports DST (DayLight Saving Time)-->true
    </isSupportDst>
    <NetworkCap>
      <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
    </NetworkCap>
  </SysCap>
</DeviceCap>

```



```

<isSupportWireless>
  <!--ro, req, bool, whether the device supports wireless network-->true
</isSupportWireless>
<isSupportWired>
  <!--ro, opt, bool, whether it supports wired network-->true
</isSupportWired>
<isSupportWAN>
  <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
</isSupportWAN>
<isSupportPPPoE>
  <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
</isSupportPPPoE>
<isSupportBond>
  <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
</isSupportBond>
<isSupport802_1x>
  <!--ro, req, bool, whether the device supports 802.1x protocol-->true
</isSupport802_1x>
<isSupportNtp>
  <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
</isSupportNtp>
<isSupportFtp>
  <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
</isSupportFtp>
<isSupportUpnp>
  <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
</isSupportUpnp>
<isSupportPNP>
  <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
</isSupportPNP>
<isSupportDdns>
  <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
</isSupportDdns>
<isSupportHttps>
  <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
</isSupportHttps>
<isSupport28181>
  <!--ro, opt, bool, N/A-->true
</isSupport28181>
<SnmpCap>
  <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
  <isSupport>
    <!--ro, req, bool, whether the device supports SNMP-->true
  </isSupport>
</SnmpCap>
<isSupportExtNetCfg>
  <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
</isSupportExtNetCfg>
<isSupportIPFilter>
  <!--ro, opt, bool, whether the device supports IP filtering-->true
</isSupportIPFilter>
<isSupportSSH>
  <!--ro, opt, bool, whether the device supports SSH-->true
</isSupportSSH>
<isSupportNetPreviewStrategy>
  <!--ro, opt, bool, whether the device supports network preview strategy-->true
</isSupportNetPreviewStrategy>
<isSupportEZVIZ>
  <!--ro, opt, bool, whether the device supports EZ protocol-->true
</isSupportEZVIZ>
<isSupportEhome>
  <!--ro, opt, bool, whether the device supports ISUP server configuration-->true
</isSupportEhome>
<isSupportWirelessDial>
  <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
</isSupportWirelessDial>
<isSupportWirelessServer>
  <!--ro, opt, bool, whether the device supports wireless server-->true
</isSupportWirelessServer>
<isSupportWPS>
  <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
</isSupportWPS>
<isWirelessMutexWithWirelessServer>
  <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
</isWirelessMutexWithWirelessServer>
<isSupportMACFilter>
  <!--ro, opt, bool, whether the device supports physical address filtering-->true
</isSupportMACFilter>
<GB28181Cap>
  <!--ro, opt, object, N/A-->
  <isSupportGB28181Service>
    <!--ro, opt, bool, N/A-->true
  </isSupportGB28181Service>
</GB28181Cap>
<WPS>
  <!--ro, opt, object, Wi-Fi Protected Setup-->
  <NetworkInterfaceList size="2">
    <!--ro, opt, array, List of network nodes, subType:object, attr:size{opt, string}-->
    <NetworkInterface>
      <!--ro, opt, object, network node-->
      <id>
        <!--ro, req, string, NIC index-->1
      </id>
    </NetworkInterface>
  </NetworkInterfaceList>

```

```

    <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <isSupportAutoConnect>
    <!--ro, opt, bool, whether the device supports automatic connection-->true
  </isSupportAutoConnect>
  <isSupportDevicePinCode>
    <!--ro, opt, bool, whether the device supports PIN code-->true
  </isSupportDevicePinCode>
  <isSupportDevicePinCodeUpdate>
    <!--ro, opt, bool, whether the device supports updating the PIN code-->true
  </isSupportDevicePinCodeUpdate>
  <ApPinCode>
    <!--ro, opt, object, PIN code of the access point-->
    <ssid min="1" max="1">
      <!--ro, opt, string, SSID (Service Set Identifier), attr:min{opt, string},max{opt, string}-->test
    </ssid>
    <pinCode min="1" max="1">
      <!--ro, opt, string, PIN code, attr:min{opt, string},max{opt, string}-->test
    </pinCode>
  </ApPinCode>
</NetworkInterface>
</NetworkInterfaceList>
</WPS>
<isSupportRFIDData>
  <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
</isSupportRFIDData>
<isSupportWifiProbeSSID>
  <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
</isSupportWifiProbeSSID>
<isSupportWifiProbe>
  <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
</isSupportWifiProbe>
<verificationCode min="1" max="1">
  <!--ro, opt, string, verification code, attr:min{opt, string},max{opt, string}-->test
</verificationCode>
<WPSCap>
  <!--ro, opt, object, capability of configuring Wi-Fi Protected Setup-->
  <isSupport>
    <!--ro, req, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupport>
  <isSupportAutoConnect>
    <!--ro, req, bool, whether the device supports automatic connection-->true
  </isSupportAutoConnect>
</WPSCap>
<NetworkMode>
  <!--ro, opt, object, network mode configuration-->
  <workMode>
    <!--ro, opt, string, network mode-->true
  </workMode>
</NetworkMode>
<VerificationCodeModification>
  <!--ro, opt, object, device verification code configuration-->
  <verificationCodeType opt="normal,empty">
    <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
  </verificationCodeType>
  <isSupportDeclarationURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
  </isSupportDeclarationURL>
  <isSupportPrivacyPolicyURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
  </isSupportPrivacyPolicyURL>
  <verificationCodeModify opt="true,false">
    <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{opt, string}-->true
  </verificationCodeModify>
  <Hyperlinks>
    <!--ro, opt, object, hyperlink configuration-->
    <declarationURL>
      <!--ro, opt, string, hyperlink to the Terms of Service-->test
    </declarationURL>
    <privacyPolicyURL>
      <!--ro, opt, string, hyperlink to the Privacy Policy-->test
    </privacyPolicyURL>
  </Hyperlinks>
  <isSupportVerificationCodeCheck>
    <!--ro, opt, bool, whether the device supports verifying the verification code-->true
  </isSupportVerificationCodeCheck>
  <isSupportOldVerificationCode>
    <!--ro, opt, bool, whether the device supports the EZ password configuration of old version (the password of old version consists of six
characters)-->true
  </isSupportOldVerificationCode>
  <maintenanceVerificationCodeType opt="normal,empty">
    <!--ro, opt, string, attr:opt{opt, string}-->empty
  </maintenanceVerificationCodeType>
  <maintenanceVerificationCodeModify opt="true,false">
    <!--ro, opt, bool, attr:opt{opt, string}-->true
  </maintenanceVerificationCodeModify>
</VerificationCodeModification>
<isSupportIntegrate>
  <!--ro, opt, bool, whether the device supports access protocol configuration-->true
</isSupportIntegrate>
<isSupportPlatformAccess>
  <!--ro, opt, bool, whether the device supports platform access, desc:whether the device supports platform access-->true
</isSupportPlatformAccess>

```

```

<isSupportIntelligentBoost>
  <!--ro, opt, bool, whether the device supports intelligent acceleration-->true
</isSupportIntelligentBoost>
<isSupportResourceStatistics>
  <!--ro, opt, bool, whether the device supports getting the network resource information-->true
</isSupportResourceStatistics>
<isSupportBandwidthLimit>
  <!--ro, opt, bool, whether the device supports bandwidth Limit-->true
</isSupportBandwidthLimit>
<isSupportPOEPortsDisableAdaptiveServer>
  <!--ro, opt, bool, whether the device supports the POE protocol-->true
</isSupportPOEPortsDisableAdaptiveServer>
<isSupportPOEConfiguration>
  <!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
</isSupportPOEConfiguration>
<isSupportGetLinkSocketIP>
  <!--ro, opt, bool, whether the device supports getting the SocketIP of the current Link-->true
</isSupportGetLinkSocketIP>
<isSupportWebSocket>
  <!--ro, opt, bool, whether the device supports WebSocket-->true
</isSupportWebSocket>
<isSupportWebSockets>
  <!--ro, opt, bool, whether the device supports WebSockets-->true
</isSupportWebSockets>
<Adaption>
  <!--ro, opt, object, network self-adaption configuration-->
  <streamType opt="0,1,2,3,4,5,7,8,9,10">
    <!--ro, opt, string, stream type that supports network self-adaption during Live view, attr:opt{opt, string}, desc:0 (main stream), 1 (sub-
    stream), 2 (third stream), 3 (fourth stream/virtual stream), 4 (fifth stream), 5 (sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10
    (tenth stream), and so forth. The index value here is the same as that in the Live view structure NET_DVR_PREVIEWINFO of Device Network SDK. Note that 6 is
    an invalid value, and from the seventh stream (index value: 7), the index value is the same as that for Live view via RTSP-->test
  </streamType>
  <isSupportPlayback>
    <!--ro, opt, bool, whether the device supports playback-->true
  </isSupportPlayback>
</Adaption>
<isSupportVideoImgDB>
  <!--ro, opt, bool, whether the device supports image and video Library configuration-->true
</isSupportVideoImgDB>
<isSupportVideoImgLibService>
  <!--ro, opt, bool-->true
</isSupportVideoImgLibService>
<isSupportWirelessWifi>
  <!--ro, opt, bool-->true
</isSupportWirelessWifi>
<isSupportPrivatePlatom>
  <!--ro, opt, bool-->true
</isSupportPrivatePlatom>
<isSupportGantryETCServer>
  <!--ro, opt, bool-->true
</isSupportGantryETCServer>
<isSupportEventDataOverWebSocket opt="true,false">
  <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{opt, string}-->true
</isSupportEventDataOverWebSocket>
<isSupportDynamicHostName>
  <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
</isSupportDynamicHostName>
<isSupportTCPAcceleration>
  <!--ro, opt, bool, whether the device supports TCP acceleration-->true
</isSupportTCPAcceleration>
<isSupportFlowControl>
  <!--ro, opt, bool, whether the device supports traffic control-->true
</isSupportFlowControl>
<isSupportEZVIZITiming>
  <!--ro, opt, bool, whether the device supports synchronizing time with EZ platform, desc:related URI: enabledTiming in the message of
  /ISAPI/System/Network/EZVIZ-->true
</isSupportEZVIZITiming>
<isSupportTrafficMonitor>
  <!--ro, opt, bool, whether the device supports traffic monitoring, desc:related URI:
  /ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json-->true
</isSupportTrafficMonitor>
<isSupportDMSAuthInfo>
  <!--ro, opt, bool, whether the device supports DMS authentication information management-->true
</isSupportDMSAuthInfo>
<isSupportWiredandWirelessTrafficMonitor>
  <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC), desc:related URI:
  /ISAPI/System/Network/Interfaces/<ID>/trafficMonitor?format=json-->true
</isSupportWiredandWirelessTrafficMonitor>
<isSupport4G>
  <!--ro, opt, bool, whether the device supports 4G module-->true
</isSupport4G>
<isSupport4GConfig>
  <!--ro, opt, bool, whether the device supports 4G configuration-->true
</isSupport4GConfig>
<isSupportGRIDServer>
  <!--ro, opt, bool, N/A-->true
</isSupportGRIDServer>
<isSupportGRIDInfo>
  <!--ro, opt, bool, N/A-->true
</isSupportGRIDInfo>
<isSupportSipTransmit>
  <!--ro, opt, bool, whether the device supports configuring SIP transmission parameters-->true
</isSupportSipTransmit>

```

```

<isSupportSip>
  <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
</isSupportSip>
<isSupportWifiCascade>
  <!--ro, opt, bool, whether the device supports enabling automatic Wi-Fi cascading, desc:related URI: /ISAPI/System/Network/wifiCascade?format=json-->true
>true
</isSupportWifiCascade>
<isSupportRelativeInfo>
  <!--ro, opt, bool, whether the device supports getting the information about the connected devices, desc:related URI: /ISAPI/System/Network/wifiCascade/relativeInfo?format=json-->true
</isSupportRelativeInfo>
<isSupportEmailEncrypt>
  <!--ro, opt, bool, whether the device supports email encryption configuration-->true
</isSupportEmailEncrypt>
<isSupportPortFilter>
  <!--ro, opt, bool, whether the device supports port filtering-->true
</isSupportPortFilter>
<isSupportPingDeny>
  <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
</isSupportPingDeny>
<isSupportDOSDeny>
  <!--ro, opt, bool, whether the device supports configuring parameters to prevent the DoS attack-->true
</isSupportDOSDeny>
<isSupportNetworkStatus>
  <!--ro, opt, bool, whether the device supports searching for the network status, desc:related URI: /ISAPI/System/Network/status?format=json-->true
</isSupportNetworkStatus>
<isSupportEZVIZQRCode>
  <!--ro, opt, bool, whether the device supports getting the EZ QR code, desc:related URI: /ISAPI/System/Network/EZVIZ/QRCode?format=json-->true
</isSupportEZVIZQRCode>
<isSupportEZVIZUnbind>
  <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
</isSupportEZVIZUnbind>
<isSupportPingTest>
  <!--ro, opt, bool, whether the device supports ping test, desc:Related URI: /ISAPI/System/Network/pingtest/capabilities-->true
</isSupportPingTest>
<isSupportMOTCloudPlatform>
  <!--ro, opt, bool, whether the device supports configuring parameters for accessing the cloud platform service, desc:related URI: /ISAPI/System/Network/Traffic/MOTCloudPlatform/capabilities?format=json-->true
</isSupportMOTCloudPlatform>
<isSupportSpectrumScan>
  <!--ro, opt, bool, whether the device supports frequency spectrum scanning, desc:related URI: /ISAPI/System/Network/interface/<netCardID>/spectrumScan/capabilities?format=json-->true
</isSupportSpectrumScan>
<isSupportFRP>
  <!--ro, opt, bool, whether the device supports FRP (Fast Reverse Proxy), desc:related URI: /ISAPI/Syetem/Network/FRP?format=json-->true
</isSupportFRP>
<isSupport5GModule>
  <!--ro, opt, bool, whether the device supports 5G module configuration, desc:related URI: /ISAPI/System/Network/5GModule/capabilities?format=json-->true
>true
</isSupport5GModule>
<isSupportWirelessDialRoute>
  <!--ro, opt, bool, whether the device supports configuring parameters of route forwarding rules of wireless dial-up, desc:related URI: /ISAPI/System/Network/WirelessDia/route/capabilities?format=json-->true
</isSupportWirelessDialRoute>
<isSupportNPlusOne>
  <!--ro, opt, bool, whether the device supports N+1 protocol-->true
</isSupportNPlusOne>
<supportEnvironmentalStandards opt="HJ212">
  <!--ro, opt, string, environmental monitoring standard, attr:opt{req, string}, desc:N/A-->true
</supportEnvironmentalStandards>
<isSupportIntranetSegment>
  <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI: /ISAPI/System/Network/intranetSegment/capabilities?format=json-->true
</isSupportIntranetSegment>
<isSupportDHCPServer>
  <!--ro, opt, bool, whether the device supports DHCP service configuration, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI: /ISAPI/System/Network/DHCPServer/capabilities?format=json-->true
</isSupportDHCPServer>
<isSupportClusterIntercom>
  <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will be returned and its value is true or not be returned.-->true
</isSupportClusterIntercom>
<isSupportHydrologicMonitoring>
  <!--ro, opt, bool, whether the device supports hydrologic monitoring data communication, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI: /ISAPI/System/Network/HydrologicMonitoring/capabilities?format=json-->true
</isSupportHydrologicMonitoring>
<isWirelessDialMuxedWithWlanAndWifi>
  <!--ro, opt, bool, whether the wireless dial-up is mutually exclusive with the WLAN hotspot and Wi-Fi, desc:before a function is enabled, another one must be disabled first-->true
</isWirelessDialMuxedWithWlanAndWifi>
<isSupportSpeechTranscription>
  <!--ro, opt, bool, whether the device supports accessing the speech-to-text transcription platform, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI: /ISAPI/System/Network/speechTranscriptionPlatform/capabilities?format=json-->true
</isSupportSpeechTranscription>
<isSupportAsynPingTest>
  <!--ro, opt, bool, whether the device supports asynchronous network detection, desc:network detection means testing the network stability and packet loss rate of the network using the ping command for knowing about the network condition. Related URI: /ISAPI/System/Network/asynPingTest/capabilities?format=json-->true
</isSupportAsynPingTest>
<isSupportBasebandAntenna>

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<!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and
the base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
</isSupportBasebandCapture>
<isSupportCellularNetworkSpeedTest>
<!--ro, opt, bool, whether the device supports testing the connection speed between the device and the cellular network, desc:you can test the
connection speed of the network where the current device is. Related URI: /ISAPI/System/Network/cellularNetwork/speedTest/capabilities?format=json-->true
</isSupportCellularNetworkSpeedTest>
<isSupportCellularNetworkCellInfo>
<!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the
cellular network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&security=<security>&iv=<iv>-->true
</isSupportCellularNetworkCellInfo>
<isSupportMultiBandFrequencyScanning>
<!--ro, opt, bool, whether the device supports scanning all frequency bands of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/multiBandFrequencyScanning/capabilities?format=json-->true
</isSupportMultiBandFrequencyScanning>
<isSupportCellularNetworkLockBand>
<!--ro, opt, bool, whether the device accessing the Locked frequency band of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/LockBand/capabilities?format=json-->true
</isSupportCellularNetworkLockBand>
<isSupportRadioFrequencyFluctuation>
<!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
</isSupportRadioFrequencyFluctuation>
<isSupportBasebandBlock>
<!--ro, opt, bool, whether the device supports the unauthorized List of the base station, desc:related URI:
/ISAPI/System/Network/cellularNetwork/basebandBlock/capabilities?format=json-->true
</isSupportBasebandBlock>
<isSupportRFIDTagCheckTest>
<!--ro, opt, bool, whether the device supports the RFID tag disk storage performance test, desc:related URI: GET
/ISAPI/System/Network/RFIDData/GetTagCheckTestInfo?format=json-->true
</isSupportRFIDTagCheckTest>
<isSupportQoSdscptolocalPriorityMapping>
<!--ro, opt, bool, whether the device supports DSCP and Local mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/DSCPTolocalPriorityMapping/capabilities?format=json-->true
</isSupportQoSdscptolocalPriorityMapping>
<isSupportQoS802.1pt0802.1pPriorityMapping>
<!--ro, opt, bool, whether the device supports 802.1p mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/802.1pt0802.1pPriorityMapping/capabilities?format=json-->true
</isSupportQoS802.1pt0802.1pPriorityMapping>
<isSupportLinkTestStart>
<!--ro, opt, bool, whether the network Link test is enabled, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
</isSupportLinkTestStart>
<isSupportLinkTestResult>
<!--ro, opt, bool, whether the device supports getting the result of network Link test, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
</isSupportLinkTestResult>
<isSupportNetworkModuleUpgrade>
<!--ro, opt, bool, whether it supports upgrading network module, desc:including importing upgrade package
(/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
(/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrade
(/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json)-->true
</isSupportNetworkModuleUpgrade>
<isSupportNetworkModuleVersionInfo>
<!--ro, opt, bool, whether it supports getting the version information of network module, desc:GET
/ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
</isSupportNetworkModuleVersionInfo>
<isSupportGreenShieldPlatform>
<!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/GreenShieldPlatform/GreenShieldPlatform?format=json-->true
</isSupportGreenShieldPlatform>
<isSupportFusionUniversalApplicationPlatCfg>
<!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/FusionUniversalApplicationPlatform/FusionUniversalApplicationPlatCfg?format=json-->true
</isSupportFusionUniversalApplicationPlatCfg>
<isSupportModbusAccess>
<!--ro, opt, bool, whether the device supports Modbus accessing parameters, desc:GET/PUT /ISAPI/System/Network/ModbusAccess?format=json-->true
</isSupportModbusAccess>
<isSupportModbusTestSubordinateStatus>
<!--ro, opt, bool, whether it supports Modbus testing specified client status, desc:POST /ISAPI/System/Network/ModbusTestSubordinateStatus?
format=json-->true
</isSupportModbusTestSubordinateStatus>
<isSupportGetModbusAndRegisterMap>
<!--ro, opt, bool, whether it supports getting the mapping relationship between Modbus uploaded content and register quantity, desc:GET
/ISAPI/System/Network/GetModbusAndRegisterMap?format=json-->true
</isSupportGetModbusAndRegisterMap>
<isSupportTSOverWebsocket>
<!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
</isSupportTSOverWebsocket>
<isSupportRequestOutlookAuth>
<!--ro, opt, bool-->true
</isSupportRequestOutlookAuth>
<isSupportOutlookEmailList>
<!--ro, opt, bool-->true
</isSupportOutlookEmailList>
<isSupportDSTPServerCfg>
<!--ro, opt, bool-->true
</isSupportDSTPServerCfg>
<isSupportDSTPProtocolsCap>
<!--ro, opt, bool-->true
</isSupportDSTPProtocolsCap>
</NetworkCap>
<IOCap>
<!--ro, opt, object, I/O capability, desc:related URI: /ISAPI/System/IO/capabilities-->
<IOInputPortNums>

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    <!--ro, opt, int, number of Local input ports of the device-->true
  </IOInputPortNums>
  <IOOutputPortNums>
    <!--ro, opt, int, number of Local output ports of the device-->true
  </IOOutputPortNums>
  <isSupportStrobeLamp>
    <!--ro, opt, bool, whether the device supports configuring the output of the gate's Light, desc:related URI:
/ISAPI/System/IO/outputs/strobeLampConf-->true
  </isSupportStrobeLamp>
  <SoftIOInputPortNums>
    <!--ro, opt, int, number of soft input ports of the device, desc:this node depends on whether the URI /ISAPI/System/IO/softInputs/capabilities?
format=json is supported-->true
  </SoftIOInputPortNums>
  <isSupportIOOutputAdvanceParameter>
    <!--ro, opt, bool, whether the device supports configuring advanced parameters of the output port, desc:this node determines whether the URI
/ISAPI/System/IO/outputs/advanceParameter/capabilities?format=json is supported-->true
  </isSupportIOOutputAdvanceParameter>
  <isSupportSetAllIOOutput>
    <!--ro, opt, bool, whether the device supports configuring all output ports, desc:this node determines whether the PUT method of the URI
/ISAPI/System/IO/outputs is supported-->true
  </isSupportSetAllIOOutput>
  <isSupportCombinationAlarm>
    <!--ro, opt, bool, whether the device supports the combined alarm, desc:CombinationAlarmCap in /ISAPI/System/IO/inputs/capabilities-->true
  </isSupportCombinationAlarm>
  <enabledIOOutputPortNums>
    <!--ro, opt, int-->0
  </enabledIOOutputPortNums>
  <isSupportAlarmKeyParam>
    <!--ro, opt, bool, whether supports configuring the physical keys of one-push alarm, desc:related URI:
/ISAPI/System/IO/inputs/<IOInputID>/AlarmKeyParam/capabilities?format=json-->true
  </isSupportAlarmKeyParam>
  <isSupportWebDisplayAlarmStatus>
    <!--ro, opt, bool, whether the web supports displaying the alarm output status, desc:used for web interface display, true indicates that the web
supports displaying the alarm output status; otherwise, it does not support-->true
  </isSupportWebDisplayAlarmStatus>
</IOCap>
<SerialCap>
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
  </rs232PortNums>
  <rs485WorkMode opt="Led,CaptureTrigger">
    <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:Led (external LED screen, which is used for LED
display), CaptureTrigger (external device, which is used to trigger the transmission of the captured data)-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, string, range of RS-232 serial port numbers supported by the device, attr:opt{opt, string)-->test
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, string, range of RS-485 serial port numbers supported by the device, attr:opt{opt, string)-->test
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the Linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>
<VideoCap>
  <!--ro, opt, object, video encoding capability, desc:related URI: /ISAPI/System/Video/capabilities-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->true
  </videoInputPortNums>
  <videoOutputPortNums>
    <!--ro, opt, int, number of video output ports-->true
  </videoOutputPortNums>
  <menuNums>
    <!--ro, opt, int, number of Local menus that can be displayed-->true
  </menuNums>
  <externalChannelNum>
    <!--ro, opt, bool, number of extended anaLog channels-->true
  </externalChannelNum>
  <isSupportHeatmap>
    <!--ro, opt, bool, whether the device supports heat map-->true
  </isSupportHeatmap>
  <isSupportCounting>
    <!--ro, opt, bool, whether the device supports people counting-->true
  </isSupportCounting>
  <countingType>
    <!--ro, opt, string, statistics type, desc:human (people counting); object (counting)-->test
  </countingType>
  <isSupportPicture>

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<!--ro, opt, bool, whether the device supports OSD picture overlay-->true
</isSupportPicture>
<isSupportCapturePicOverlays>
  <!--ro, opt, bool-->true
</isSupportCapturePicOverlays>
<isSupportMergePicOverlays>
  <!--ro, opt, bool-->true
</isSupportMergePicOverlays>
<isSupportPreviewSwitch>
  <!--ro, opt, bool, whether the device supports switching live view-->true
</isSupportPreviewSwitch>
<isSupportMultiChannelCounting>
  <!--ro, opt, bool, whether the device supports multichannel statistics-->true
</isSupportMultiChannelCounting>
<isSupportCountingCollection>
  <!--ro, opt, bool, whether the device supports people counting ANR-->true
</isSupportCountingCollection>
<isSupportHeatmapCollection>
  <!--ro, opt, bool, whether the device supports heat map data ANR-->true
</isSupportHeatmapCollection>
<channelFlexible opt="name,online,linknum">
  <!--ro, opt, string, N/A, attr:opt{opt, string}-->true
</channelFlexible>
<isSupportMixedChannel>
  <!--ro, opt, bool, N/A-->true
</isSupportMixedChannel>
<isSupportMixedChannelStatus>
  <!--ro, opt, bool, N/A-->true
</isSupportMixedChannelStatus>
<isSupportRecodStatus>
  <!--ro, opt, bool, whether the device supports searching for the recording status-->true
</isSupportRecodStatus>
<isSupportPrivacyMask>
  <!--ro, opt, bool, whether the device supports privacy mask-->true
</isSupportPrivacyMask>
<isSupportBinocularPreviewSwitch>
  <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
</isSupportBinocularPreviewSwitch>
<isSupportCalibCheck>
  <!--ro, opt, bool, whether the device supports calibration verification-->true
</isSupportCalibCheck>
<isSupportPIP>
  <!--ro, opt, bool, whether the device supports PIP-->true
</isSupportPIP>
<isSupportFocusVideoMode>
  <!--ro, opt, bool, whether the device supports the focus mode of the video-->true
</isSupportFocusVideoMode>
<isSupportExternalChannel>
  <!--ro, opt, bool, whether the device supports extended analog channels-->true
</isSupportExternalChannel>
<isSupportCompressedTranscode>
  <!--ro, opt, bool, whether the device supports configuring the encoding compression of transcoding stream, desc:related URL:
  /ISAPI/Streaming/channels/<channelID>/TranscodeCfg/capabilities?format=json-->true
</isSupportCompressedTranscode>
<isSupportCompressedStream>
  <!--ro, opt, bool-->true
</isSupportCompressedStream>
<isSupportLocalStreamCfg>
  <!--ro, opt, bool, whether the device supports configuring the streaming parameters of local device, desc:related URL:
  /ISAPI/System/Video/inputs/channels/<channelID>/LocalStreamCfg/capabilities?format=json-->true
</isSupportLocalStreamCfg>
<isSupportMenuStatus>
  <!--ro, opt, bool, whether it supports getting the menu status, desc:corresponding URI: /ISAPI/System/Video/Menu/<indexID>/status?format=json, the
  range of indexID comes from VideoCap.menuNums-->true
</isSupportMenuStatus>
<isSupportVideoOutputMode>
  <!--ro, opt, bool-->true
</isSupportVideoOutputMode>
<pictureChannelList>
  <!--ro, opt, array, subType:object-->
  <channelID>
    <!--ro, opt, int-->0
  </channelID>
</pictureChannelList>
<transcodeStreamType opt="previewTranscode,recordTranscode">
  <!--ro, opt, string, attr:opt{req, string}-->true
</transcodeStreamType>
<isSupportImportPanoramaIOOverlayPicture>
  <!--ro, opt, bool-->true
</isSupportImportPanoramaIOOverlayPicture>
<isSupportPanoramaIOOverlayPictureParams>
  <!--ro, opt, bool-->true
</isSupportPanoramaIOOverlayPictureParams>
</VideoCap>
<AudioCap>
  <!--ro, opt, object, audio encoding capability, desc:related URL: /ISAPI/System/Audio/capabilities-->
  <audioInputNums>
    <!--ro, req, int, number of audio inputs-->1
  </audioInputNums>
  <audioOutputNums>
    <!--ro, req, int, number of audio outputs-->1
  </audioOutputNums>
  <mixAudioInSet>

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    <!--ro, opt, bool, N/A-->true
</mixAudioInSet>
<mixAudioOutSet>
    <!--ro, opt, bool, N/A-->true
</mixAudioOutSet>
<isSupportAudioMixing>
    <!--ro, opt, bool, whether the device supports audio mixing-->true
</isSupportAudioMixing>
<isSupportRemoveFrequencyPoint>
    <!--ro, opt, bool, whether the device supports clearing the howling frequency point-->true
</isSupportRemoveFrequencyPoint>
<isSupportAudioInConfig>
    <!--ro, opt, bool, whether the device supports configuring parameters of all audio inputs, desc:related URI:
/ISAPI/System/Audio/AudioIn/capabilities-->true
</isSupportAudioInConfig>
<muteDuringPanTilt opt="MicIn">
    <!--ro, opt, string, whether the device supports muting during motion, attr:opt{opt, string}, desc:muteDuringPanTilt in
/ISAPI/System/TwoWayAudio/channels/ID-->true
</muteDuringPanTilt>
<isSupportAudioOutConfig>
    <!--ro, opt, bool, whether the device supports configuring parameters of all audio outputs, desc:related URI:
/ISAPI/System/Audio/AudioOut/capabilities?format=json-->true
</isSupportAudioOutConfig>
<isSupportSAIP>
    <!--ro, opt, bool, whether the device supports connecting to the independent audio track channel-->true
</isSupportSAIP>
<SAIPChannelsNum>
    <!--ro, opt, int, number of devices that supports connecting to the independent audio track channel, dep:and,{$.AudioCap.isSupportSAIP,eq,true}-->1
</SAIPChannelsNum>
<isSupportDataStorage>
    <!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URL: /ISAPI/System/Audio/dataStorage?format=json-->true
</isSupportDataStorage>
<isSupportDataRelease>
    <!--ro, opt, bool, whether the device supports releasing the audio file, desc:related URL: /ISAPI/System/Audio/dataRelease?format=json-->true
</isSupportDataRelease>
<isSupportAudioPlan>
    <!--ro, opt, bool, whether the device supports the audio scheme plan, desc:related URL: /ISAPI/System/Audio/plan?format=json-->true
</isSupportAudioPlan>
<isSupportAudioTask>
    <!--ro, opt, bool, whether the device supports the audio scheme task, desc:related URL: /ISAPI/System/Audio/plan/<PlanID>/task?format=json-->true
</isSupportAudioTask>
<isSupportAudioCollection>
    <!--ro, opt, bool, whether the device supports collecting audio files, desc:related URL: /ISAPI/System/Audio/audioCollection/capabilities?
format=json; maintenance function, it is used for collecting audio data to analyze audio questions-->true
</isSupportAudioCollection>
<isSupportWirelessPickUp>
    <!--ro, opt, bool, whether the device supports connecting wireless pickup, desc:related URL: /ISAPI/System/Audio/wirelessPickUp/capabilities?
format=json-->true
</isSupportWirelessPickUp>
<audioOutputBroadcastNums opt="1,2,3,4">
    <!--ro, opt, int, number of audio broadcast outputs, attr:opt{opt, string}, desc:at present, this node is only supported by broadcast audio devices
when opt="1,2,3,4" is returned and , the output depends on the actual node; when not returned, the output starts from 1 by default according to business
type, the audio devices classify the output ports into broadcast output port and audio output port considering the start No. of every output starts from 1,
it should return the specific number of supported broadcast outputs if this node is returned, it indicates that the device supports the audio output volume
configuration; otherwise, the device does not support. Corresponding URI: /ISAPI/System/Audio/AudioOut/broadcast/capabilities?format=json-->0
</audioOutputBroadcastNums>
<isSupportAudioOutMixing>
    <!--ro, opt, bool, whether the device supports outputting audio mix, desc:related URL: /ISAPI/System/Audio/AudioOut/mixing/capabilities?format=json-
-->true
</isSupportAudioOutMixing>
<isSupportRTSPAudio>
    <!--ro, opt, bool, whether the device supports listening to the audio transmitted via RTSP in real time, desc:RTSP URI:
/ISAPI/System/Audio/AudioIn/channels/<audioID>-->true
</isSupportRTSPAudio>
<isSupportRTSPAudioPlaybackByFile>
    <!--ro, opt, bool, whether the device supports playing back audio transmitted via RTSP according to files, desc:the RTSP streaming URL is
playbackURI returned by /ISAPI/ContentMgmt/recordAudio/search?format=json-->true
</isSupportRTSPAudioPlaybackByFile>
<isSupportDeviceAudioCapture>
    <!--ro, opt, bool-->true
</isSupportDeviceAudioCapture>
</AudioCap>
<isSupportHolidy>
    <!--ro, opt, bool, this node is located after the node <AudioCap>-->true
</isSupportHolidy>
<RebootConfigurationCap>
    <!--ro, opt, object, capability of configuring parameters for rebooting the device, desc:related URI: /ISAPI/System/reboot/capabilities. This node is
located before the node <isSupportSubscribeEvent> and after the node <isSupportHolidy>-->
<Genetec>
    <!--ro, opt, bool, whether to enable the Genetec protocol-->true
</Genetec>
<ONVIF>
    <!--ro, opt, bool, whether to enable the ONVIF protocol-->true
</ONVIF>
<RTSP>
    <!--ro, opt, bool, whether to enable the RTSP protocol-->true
</RTSP>
<HTTP>
    <!--ro, opt, bool, whether to enable the HTTP protocol-->true
</HTTP>
<SADP>
    <!--ro, opt, object, SADP (Search Active Devices Protocol, including the Link Layer and multicast discovery)-->
<ISDiscoveryMode>

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    <!--ro, opt, bool, mode of the SADP multicast discovery protocol-->true
    </ISDiscoveryMode>
    <PcapMode>
    <!--ro, opt, bool, mode of the SADP Link Layer discovery protocol-->true
    </PcapMode>
    </SADP>
    <IPCAddStatus>
    <!--ro, opt, bool, IP address status-->true
    </IPCAddStatus>
    </RebootConfigurationCap>
    <isSupportExternalDevice>
    <!--ro, opt, bool, whether the device supports connecting to peripherals: true (support), desc:related URI: /ISAPI/System/externalDevice/capabilities-
->true
    </isSupportExternalDevice>
    <isSupportSoundCfg>
    <!--ro, opt, bool, whether the device supports the sound configuration, desc:related URI: /ISAPI/System/SoundCfg/capabilities-->true
    </isSupportSoundCfg>
    <isSupportSubscribeEvent>
    <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
    </isSupportSubscribeEvent>
    <isSupportSubscribeIOTInfo>
    <!--ro, opt, bool, whether the device supports subscribing to the IoT information-->true
    </isSupportSubscribeIOTInfo>
    <isSupportDiagnosedData>
    <!--ro, opt, bool, whether the device supports exporting the device diagnosis data, desc:true (support), this node is not returned (not support).
Related URI: /ISAPI/System/diagnosedData-->true
    </isSupportDiagnosedData>
    <isSupportChangedUpload>
    <!--ro, opt, bool, whether the device supports uploading changes-->true
    </isSupportChangedUpload>
    <isSupportGettingWorkingStatus>
    <!--ro, opt, bool, whether the device supports getting the device status-->true
    </isSupportGettingWorkingStatus>
    <isSupportGettingChannelInfoByCondition>
    <!--ro, opt, bool, whether the device supports getting the channel information by condition-->true
    </isSupportGettingChannelInfoByCondition>
    <isSupportDiagnosedDataParameter>
    <!--ro, opt, bool, whether the device supports enabling or disabling exporting the diagnosis information-->true
    </isSupportDiagnosedDataParameter>
    <isSupportSimpleDevStatus>
    <!--ro, opt, bool, whether the device supports getting the device's simple status, desc:related URI: /ISAPI/System/workingstatus?format=json-->true
    </isSupportSimpleDevStatus>
    <isSupportFlexible>
    <!--ro, opt, bool, whether the device supports the flexible combination configuration. Related URI: /ISAPI/System/Video/inputs/channels?format=json--
>true
    </isSupportFlexible>
    <isSupportPTZChannels>
    <!--ro, opt, bool, whether the device supports returning the PTZ channel (it is distinguished from the video channel)-->true
    </isSupportPTZChannels>
    <isSupportMetadata>
    <!--ro, opt, bool, whether the device supports metadata, desc:related URI: /ISAPI/Streaming/channels/<ID>/metadata/capabilities-->true
    </isSupportMetadata>
    <isSupportTimeCap>
    <!--ro, opt, bool, whether the device supports the time configuration, desc:related URI: /ISAPI/System/time/capabilities-->true
    </isSupportTimeCap>
    <isSupportThermalStreamData>
    <!--ro, opt, bool, whether the device supports uploading thermal imaging stream data in real time-->true
    </isSupportThermalStreamData>
    <isSupportShutdown>
    <!--ro, opt, bool, whether the device supports the shutdown configuration, desc:related URI: /ISAPI/System/shutdown?format=json-->true
    </isSupportShutdown>
    <isSupportConsumptionMode>
    <!--ro, opt, bool, whether the device supports switching the consumption mode, desc:related URI: /ISAPI/System/consumptionMode/capabilities?
format=json-->true
    </isSupportConsumptionMode>
    <isSupportManualPowerConsumption>
    <!--ro, opt, bool, whether the device supports controlling the device consumption manually-->true
    </isSupportManualPowerConsumption>
    <isSupportPostUpdateFirmware>
    <!--ro, opt, bool, whether the device supports upgrading firmware by POST method-->true
    </isSupportPostUpdateFirmware>
    <isSupportPostConfigData>
    <!--ro, opt, bool, whether the device supports importing and exporting the configuration file by POST method-->true
    </isSupportPostConfigData>
    <isSupportUserLock>
    <!--ro, opt, bool, whether the device supports locking the user, desc:related URI: /ISAPI/System/userLock/config/capabilities?format=json-->true
    </isSupportUserLock>
    <isSupportModuleLock>
    <!--ro, opt, bool, whether the device supports locking the module, desc:related URI: /ISAPI/System/moduleLock-->true
    </isSupportModuleLock>
    <supportSmartOverlapChannels opt="1">
    <!--ro, opt, bool, whether the device supports configuring the smart event stream, attr:opt[opt, string], desc:if this function is supported, this
node and the corresponding channel ID will be returned; otherwise, this node will not be returned-->true
    </supportSmartOverlapChannels>
    <isSupportNetworkCapture>
    <!--ro, opt, bool, whether the device supports capturing network packets, desc:this node is used to determine whether the device supports capturing
communication packets transmitted between the device and the external. Related URI: /ISAPI/System/networkCapture/capabilities?format=json-->true
    </isSupportNetworkCapture>
    <isSupportOperationRecord>
    <!--ro, opt, bool, whether the device supports the device operation records, desc:the charging voltage and monthly flow can be recorded in the device
in real-time low-consumption mode and full-consumption mode. Related URI: /ISAPI/System/operationRecord/capabilities?format=json-->true
    </isSupportOperationRecord>
    <isSupportSearchModuleLock>
    <!--ro, opt, bool, whether the device support search module Locking parameters. desc:/ISAPI/System/moduleLock/searchconfia?format=json-->true

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</isSupportSearchModuleLock>
<devSupportUpdateVersion>
  <!--ro, opt, string-->1
</devSupportUpdateVersion>
<notSupportCatalogUpgrade>
  <!--ro, opt, bool-->true
</notSupportCatalogUpgrade>
</SysCap>
<voicetalkNums>
  <!--ro, opt, int, number of two-way audio channels-->2
</voicetalkNums>
<isSupportSnapshot>
  <!--ro, opt, bool, whether the device supports capturing pictures-->true
</isSupportSnapshot>
<SecurityCap>
  <!--ro, opt, object, encryption capability set-->
  <supportUserNums>
    <!--ro, opt, int, supported maximum number of users-->1
  </supportUserNums>
  <userBondIpNums>
    <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->1
  </userBondIpNums>
  <userBondMacNums>
    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->1
  </userBondMacNums>
  <isSupCertificate>
    <!--ro, opt, bool, whether the device supports authentication-->true
  </isSupCertificate>
  <issupIllegalLoginLock>
    <!--ro, opt, bool, whether the device supports Locking Login-->true
  </issupIllegalLoginLock>
  <isSupportOnlineUser>
    <!--ro, opt, bool, whether the device supports the online user configuration-->true
  </isSupportOnlineUser>
  <isSupportAnonymous>
    <!--ro, opt, bool, whether the device supports anonymous Login-->true
  </isSupportAnonymous>
  <isSupportStreamEncryption>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
  </isSupportStreamEncryption>
  <securityVersion opt="1,2,3,4,7">
    <!--ro, opt, string, encryption capability, attr:opt{opt, string}, desc:the encryption capability of each version consists of two parts: encryption
    algorithm and the range of encrypted nodes. Currently 1 refers to AES128 encryption and 2 refers to AES256 encryption. The range of encrypted nodes is
    described in each protocol-->test
  </securityVersion>
  <keyIterateNum>
    <!--ro, opt, int, iteration times, desc:the value is usually between 100 and 1000-->100
  </keyIterateNum>
  <isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, desc:this node
    depends on the node <securityVersion>, which means that it is only valid for the versions that support encrypting the sensitive information-->true
  </isSupportUserCheck>
  <isSupportGUIDFileDataExport>
    <!--ro, opt, bool, whether the device supports exporting the device's GUID file-->true
  </isSupportGUIDFileDataExport>
  <isSupportSecurityQuestionConfig>
    <!--ro, opt, bool, whether the device supports answering security questions-->true
  </isSupportSecurityQuestionConfig>
  <supportSecurityNode opt="wirelessServer,wirelessDial">
    <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt{opt, string}, desc:in the earlier version of the
    solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI), some URIs containing sensitive
    information are left out. Now these URIs need to be encrypted for transmission. This capability node is added to be compatible with old devices-->test
  </supportSecurityNode>
  <isSupportGetOnlineUserListSC>
    <!--ro, opt, bool, whether the device supports searching the online user list-->true
  </isSupportGetOnlineUserListSC>
  <SecurityLimits>
    <!--ro, opt, object, capability of configuring security limit parameters-->
    <LoginPasswordLenLimit min="1" max="16">
      <!--ro, opt, string, length limit of the user's login password, attr:min{opt, string},max{opt, string}-->test
    </LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128">
      <!--ro, opt, string, length limit of the security question's answer, attr:min{opt, string},max{opt, string}-->test
    </SecurityAnswerLenLimit>
  </SecurityLimits>
  <RSAKeyLength opt="512,1024,2048" def="2048">
    <!--ro, opt, enum, HTTPS certificate length, subType:string, attr:opt{opt, string},def{opt, string}, desc:512, 1024, 2048-->2048
  </RSAKeyLength>
  <isSupportONVIFUserManagement>
    <!--ro, opt, bool, whether the device supports ONVIF user management-->true
  </isSupportONVIFUserManagement>
  <WebCertificateCap>
    <!--ro, opt, object, HTTP authentication capability-->
    <CertificateType opt="basic,digest,digest/basic">
      <!--ro, req, string, certificate type: basic (authentication), attr:opt{opt, string}-->test
    </CertificateType>
    <SecurityAlgorithm>
      <!--ro, opt, object, encryption algorithm, dep:or,{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest},
      {$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
      <algorithmType opt="MD5,SHA256,MD5/SHA256">
        <!--ro, opt, string, algorithm type, attr:opt{req, string}-->test
      </algorithmType>
    </SecurityAlgorithm>

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</WebCertificateCap>
<isSupportConfigFileImport>
  <!--ro, opt, bool, whether the device supports importing the configuration file-->true
</isSupportConfigFileImport>
<isSupportConfigFileExport>
  <!--ro, opt, bool, whether the device supports exporting the configuration file-->true
</isSupportConfigFileExport>
<cfgFileSecretKeyLenLimit min="0" max="16">
  <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{opt, string},max{opt, string}-->0
</cfgFileSecretKeyLenLimit>
<supportIPCActivatePassword>
  <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
</supportIPCActivatePassword>
<isIrreversible>
  <!--ro, opt, bool, whether the device supports irreversible password storage, desc:if this function is not supported, the plaintext password of the
user information will be stored in the device; otherwise, the password will be hashed for storage in the device-->true
</isIrreversible>
<salt>
  <!--ro, opt, string, the specific salt used by the user to log in-->test
</salt>
<isSupportOnvifInfo>
  <!--ro, opt, bool, whether the device supports getting the ONVIF protocol information-->true
</isSupportOnvifInfo>
<isSupportUserNamePasswordCheckUpgrade>
  <!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password-->true
</isSupportUserNamePasswordCheckUpgrade>
<isSupportDeviceCertificatesManagement>
  <!--ro, opt, bool, whether the device supports certificate management-->true
</isSupportDeviceCertificatesManagement>
<isSupportSIPCertificatesManagement>
  <!--ro, opt, bool, whether the device supports SIP certificate management-->true
</isSupportSIPCertificatesManagement>
<isSupportDeviceSelfSignCertExport>
  <!--ro, opt, bool, whether the device supports exporting the self-signed certificate-->true
</isSupportDeviceSelfSignCertExport>
<isSupportSecurityEmail>
  <!--ro, opt, bool, whether the device supports configuring the security email-->true
</isSupportSecurityEmail>
<isSupportRTSPCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports RTSP authentication, attr:opt{opt, string}-->true
</isSupportRTSPCertificate>
<isSupportPictureURLCertificate opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{opt, string}-->true
</isSupportPictureURLCertificate>
<isSupportGB35114Certificate>
  <!--ro, opt, bool, N/A-->true
</isSupportGB35114Certificate>
<isSupportAuthenticationMode opt="uKey,private">
  <!--ro, opt, enum, verification mode, subType:string, attr:opt{req, string}, desc:"uKey", "private"-->private
</isSupportAuthenticationMode>
<isSupportUnloggedUserPermissionConfig opt="true,false">
  <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{opt, string}-->true
</isSupportUnloggedUserPermissionConfig>
<maxIllegalLoginTimes min="3" max="20" def="5">
  <!--ro, opt, string, maximum failed illegal login attempts, attr:min{opt, string},max{opt, string},def{opt, string}, desc:this node is valid when the
value of the node <issupportIllegalLoginLock> is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
limit-->1
</maxIllegalLoginTimes>
<SecurityAdvanced>
  <!--ro, opt, object, advanced security configuration-->
  <securityEnhanced>
    <!--ro, opt, bool, whether the device supports security reinforcement-->true
  </securityEnhanced>
  <noOperationEnabled>
    <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
  </noOperationEnabled>
  <noOperationTime min="1" max="60" def="15">
    <!--ro, req, int, control timeout when there is no operation, unit:分钟, attr:min{opt, string},max{opt, string},def{opt, string}-->1
  </noOperationTime>
  <isSupportDigestStatus>
    <!--ro, opt, bool, digest status-->true
  </isSupportDigestStatus>
</SecurityAdvanced>
<LoginLinkNum>
  <!--ro, opt, object, number of users logging in at the same time-->
  <maxLinkNum min="1" max="128" def="50">
    <!--ro, req, int, number of controls logging in at the same time, attr:min{opt, string},max{opt, string},def{opt, string}-->64
  </maxLinkNum>
</LoginLinkNum>
<isSupportCCClientCertificate>
  <!--ro, opt, bool, whether the device supports the CC (Client Certificate) function-->true
</isSupportCCClientCertificate>
<passwordValidity min="0" max="365">
  <!--ro, opt, string, range of the password validity period, attr:min{opt, string},max{opt, string}, desc:unit: day. If this node is returned, it
indicates that the device supports configuring the password validity period, and only the administrator can edit the configuration-->1
</passwordValidity>
<isSupportResetChannelPassword>
  <!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
</isSupportResetChannelPassword>
<DoubleVerificationCap>
  <!--ro, opt, object, secondary authentication capability-->
  <isSupportUsersConfig>
    <!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
  </isSupportUsersConfig>

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</isSupportUserSearch>
<isSupportUsersPermissionConfig>
  <!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
</isSupportUsersPermissionConfig>
<isSupportFactoryReset>
  <!--ro, opt, bool, whether the device supports restoring secondary authentication to defaults-->true
</isSupportFactoryReset>
<isSupportQuestionConfiguration>
  <!--ro, opt, bool, whether it supports configuring a single security question for secondary authentication, desc:PUT
/ISAPI/Security/questionConfiguration/<indexID>-->true
</isSupportQuestionConfiguration>
</DoubleVerificationCap>
<isSupportUserSearch>
  <!--ro, opt, bool, whether the device supports searching users-->true
</isSupportUserSearch>
<keypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length, attr:min{req, int},max{req, int}, desc:when the password length of different user types is different, this
node only indicates the keypad password length of the administrator; otherwise, the keypad password length of all user types is the same and defined by the
value of this node-->1
</keypadPassword>
<installerKeypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length of the installer, attr:min{req, int},max{req, int}-->1
</installerKeypadPassword>
<operatorKeypadPassword min="1" max="10">
  <!--ro, opt, int, keypad password length of the operator, attr:min{req, int},max{req, int}-->1
</operatorKeypadPassword>
<userOperateType opt="1,2,3">
  <!--ro, opt, enum, user type: 1 (network user), subType:int, attr:opt{req, string}, desc:1 (network user), 2 (keypad user), 3 (network user and keypad
user)-->1
</userOperateType>
<isSptUserEnabled>
  <!--ro, opt, bool, whether the device supports configuring parameters for enabling the user-->true
</isSptUserEnabled>
<isSptAdminCap>
  <!--ro, opt, bool, whether the device supports getting the admin permission-->true
</isSptAdminCap>
<isSupportBackLogCfg>
  <!--ro, opt, bool, whether the device supports configuring basic parameters of the background log-->true
</isSupportBackLogCfg>
<isSupportBackLogExport>
  <!--ro, opt, bool, whether the device supports exporting the background log-->true
</isSupportBackLogExport>
<isSupportModifyInnerPassword opt="true,false">
  <!--ro, opt, bool, whether the device supports editing the built-in password, attr:opt{req, string}, desc:the device's built-in passwords include the
database password, system account password and so on-->true
</isSupportModifyInnerPassword>
<isSupportEncryptCertificate>
  <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
</isSupportEncryptCertificate>
<maxIllegalLoginLockTime min="1" max="120" def="30">
  <!--ro, opt, int, lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
</maxIllegalLoginLockTime>
<cloudSensitiveInfoEncryptType>
  <!--ro, opt, string, encryption type of the cloud sensitive information, desc:the current device returns the value "2.1", which means that the method
of generating the key for encrypting the sensitive information is SHA256(calc_sha256(user, salt, SHA256(user+salt+SHA256(user+salt+password))))+
AaBbCcDd1234!@#$%&'.-->2.1
</cloudSensitiveInfoEncryptType>
<isSupportCertificateCustomID>
  <!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
</isSupportCertificateCustomID>
<isSupportResetLANPassword>
  <!--ro, opt, bool, whether the device supports resetting the device's password in the LAN, desc:related URI:
/ISAPI/Security/resetPassword/certification?format=json-->true
</isSupportResetLANPassword>
<isSupportPicHttpsCertificatesManagement>
  <!--ro, opt, bool, whether the device supports managing the HTTPS certificate for pictures, desc:if this function is supported, this node will be
returned and its value is true; otherwise, this node will not be returned-->true
</isSupportPicHttpsCertificatesManagement>
<isSupportAKSKImport>
  <!--ro, opt, bool, whether the device supports importing AKSK, desc:if this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned-->true
</isSupportAKSKImport>
<isSupportAKSKExport>
  <!--ro, opt, bool, whether the device supports exporting AKSK, desc:If this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned-->true
</isSupportAKSKExport>
<isSupportAKSKReset>
  <!--ro, opt, bool, whether the device supports resetting AKSK, desc:If this function is supported, this node will be returned and its value is true;
otherwise, this node will not be returned-->true
</isSupportAKSKReset>
<ExportCertificateFile>
  <!--ro, opt, object, whether the device supports exporting the certificate, desc:related URI: POST
/ISAPI/Security/serverCertificate/exportCertificateFile-->
</ExportCertificateFile>
<isSupportGB351114FDWSFCertificate>
  <!--ro, opt, bool, whether the device supports FDWSF certificate management, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URIs: /ISAPI/Security/GB351114/FDWSFCertificate/capabilities?format=json, POST
/ISAPI/Security/GB351114/FDWSFCertificate?format=json, GET /ISAPI/Security/GB351114/FDWSFCertificate?format=json, and DELETE
/ISAPI/Security/GB351114/FDWSFCertificate?format=json-->true
</isSupportGB351114FDWSFCertificate>
<isSupportMediaObjectEncryption>
  <!--ro, opt, bool, whether the device supports setting encryption parameters of the private target in media data for the specified video channel,
desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI:
/ISAPI/Security/MediaObjectEncryption/capabilities?format=json-->true

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</isSupportMediaObjectEncryption>
<isSupportDisclaimer>
  <!--ro, opt, bool, whether support configuring the product disclaimer, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/disclaimer/capabilities?format=json-->true
</isSupportDisclaimer>
<isSupportSoftwareLicense>
  <!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
</isSupportSoftwareLicense>
<isSupportInstallerCap>
  <!--ro, opt, bool, whether the device supports getting the installer permission, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/UserPermission/installer/capabilities?format=json-->true
</isSupportInstallerCap>
<isSupportResetPasswordFile>
  <!--ro, opt, bool, whether the device supports exporting and importing the secret key file of device in the LAN, desc:Related URI: /ISAPI/Security/resetPassword/keyFile/capabilities?format=json-->true
</isSupportResetPasswordFile>
<isNotSupportLogSeverCertificate>
  <!--ro, opt, bool, whether the device support importing Log server certificate, desc:related URI: /ISAPI/Security/serverCertificate/certificate-->true
</isNotSupportLogSeverCertificate>
<isSupportLocalLoginCfg>
  <!--ro, opt, bool, whether it supports the capability of device Local Login parameters, desc:GET/PUT /ISAPI/Security/LocalLoginCfg?format=json-->true
</isSupportLocalLoginCfg>
<isSupportGetProtocolFirewallRulesFileImportStatus>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
</isSupportGetProtocolFirewallRulesFileImportStatus>
<isSupportImportProtocolFirewallRulesFile>
  <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
</isSupportImportProtocolFirewallRulesFile>
<isSupportGetProtocolFirewallEngineUpgradeStatus>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
</isSupportGetProtocolFirewallEngineUpgradeStatus>
<isSupportUpgradeProtocolFirewallEngine>
  <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
</isSupportUpgradeProtocolFirewallEngine>
<isSupportProtocolFirewallVersionInfo>
  <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
</isSupportProtocolFirewallVersionInfo>
<isSupportProtocolFirewallParams>
  <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
</isSupportProtocolFirewallParams>
<isSupportResetPasswordBySecurityPhoneNo>
  <!--ro, opt, bool, whether it supports resetting device password with phone number verification, desc:1. attendance check devices support applying a preset phone number during device activation. The user can reset the device password with the preset phone number, which is similar with password reset via email. The phone number can be changed on the website or local UI of the device
2. if the node is not returned or the returned value is false, it indicates "no"-->true
</isSupportResetPasswordBySecurityPhoneNo>
<isSupportGetIntrusionPreventionEngineUpgradeStatus>
  <!--ro, opt, bool, whether the device supports getting the upgrade status of intrusion prevention engine library, desc:GET /ISAPI/Security/GetIntrusionPreventionEngineUpgradeStatus?format=json-->true
</isSupportGetIntrusionPreventionEngineUpgradeStatus>
<isSupportIntrusionPreventionEngineVersionInfo>
  <!--ro, opt, bool, whether the device supports version information of intrusion prevention engine, desc:GET /ISAPI/Security/IntrusionPreventionEngineVersionInfo?format=json-->true
</isSupportIntrusionPreventionEngineVersionInfo>
<isSupportIntrusionPreventionParams>
  <!--ro, opt, bool, whether the device supports intrusion prevention parameters, desc:GET/PUT /ISAPI/Security/IntrusionPreventionParams?format=json-->true
</isSupportIntrusionPreventionParams>
<isSupportUpgradeIntrusionPreventionEngine>
  <!--ro, opt, bool, whether the device supports upgrading intrusion prevention engine library, desc:PUT /ISAPI/Security/UpgradeIntrusionPreventionEngine?format=json-->true
</isSupportUpgradeIntrusionPreventionEngine>
<notSupportImportCertificateType opt="CA,CClntCertificate,GB35114,GB35114Service,https">
  <!--ro, opt, string, file type that cannot be imported, attr:opt(req, string), desc:if this node does not exist, all supported certificate types can be imported-->test
</notSupportImportCertificateType>
<isSupportPasswordProtection>
  <!--ro, opt, bool, whether it supports parameter configuration of SDK protocol authentication mode, desc:PUT/GET /ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
</isSupportPasswordProtection>
</SecurityCap>
<EventCap>
  <!--ro, opt, object, event capability-->
  <isSupportHDFull>
    <!--ro, opt, bool, whether the device supports HDD full alarm-->true
  </isSupportHDFull>
  <isSupportHDError>
    <!--ro, opt, bool, whether the device supports HDD error alarm-->true
  </isSupportHDError>
  <isSupportNicBroken>
    <!--ro, opt, bool, whether the device supports the network disconnected event-->true
  </isSupportNicBroken>
  <isSupportIpConflict>
    <!--ro, opt, bool, whether the device supports the IP address conflict event-->true
  </isSupportIpConflict>
  <isSupportIllaAccess>
    <!--ro, opt, bool, whether the device supports the illegal Login event-->true
  </isSupportIllaAccess>
  <isSupportViException>
    <!--ro, opt, bool, whether the device supports video input exception (only for analog channels)-->true
  </isSupportViException>
  <isSupportViMismatch>
    <!--ro, opt, bool, whether the device supports the event of video standard mismatch-->true
  </isSupportViMismatch>

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<!--ro, opt, bool, whether the device supports the event of video standard mismatch-->true
</isSupportViMismatch>
<isSupportRecordException>
  <!--ro, opt, bool, whether the device supports the recording exception event-->true
</isSupportRecordException>
<isSupportRaidException>
  <!--ro, opt, bool, whether the device supports the array exception event-->true
</isSupportRaidException>
<isSupportViResMismatch>
  <!--ro, opt, bool, whether the device supports the event of video resolution mismatch-->true
</isSupportViResMismatch>
<isSupportPOCException>
  <!--ro, opt, bool, whether the device supports the event of powering over coaxial exception-->true
</isSupportPOCException>
<isSupportSmartDetection>
  <!--ro, opt, bool, whether the device supports the Smart event-->true
</isSupportSmartDetection>
<isSupportSpareException>
  <!--ro, opt, bool, whether the device supports hot spare exception-->true
</isSupportSpareException>
<isSupportPoePowerException>
  <!--ro, opt, bool, whether the device supports the POE exception event-->true
</isSupportPoePowerException>
<isSupportTriggerFocus>
  <!--ro, opt, bool, whether the device supports defocus exception-->true
</isSupportTriggerFocus>
<isSupportMotionDetection>
  <!--ro, opt, bool, whether the device supports motion detection-->true
</isSupportMotionDetection>
<isSupportVideoLoss>
  <!--ro, opt, bool, whether the device supports video Loss events-->true
</isSupportVideoLoss>
<isSupportTamperDetection>
  <!--ro, opt, bool, whether the device supports tampering detection-->true
</isSupportTamperDetection>
<isSupportFireDetection>
  <!--ro, opt, bool, whether the device supports fire detection events-->true
</isSupportFireDetection>
<isSupportRegionEntrance>
  <!--ro, opt, bool, whether the device supports region entrance events-->true
</isSupportRegionEntrance>
<isSupportRegionExiting>
  <!--ro, opt, bool, whether the device supports region exiting events-->true
</isSupportRegionExiting>
<isSupportLoitering>
  <!--ro, opt, bool, whether the device supports loitering events-->true
</isSupportLoitering>
<isSupportGroup>
  <!--ro, opt, bool, whether the device supports people gathering events-->true
</isSupportGroup>
<isSupportRapidMove>
  <!--ro, opt, bool, whether the device supports fast moving events-->true
</isSupportRapidMove>
<isSupportParking>
  <!--ro, opt, bool, whether the device supports parking detection events-->true
</isSupportParking>
<isSupportUnattendedBaggage>
  <!--ro, opt, bool, whether the device supports unattended baggage events-->true
</isSupportUnattendedBaggage>
<isSupportAttendedBaggage>
  <!--ro, opt, bool, whether the device supports object removal events-->true
</isSupportAttendedBaggage>
<isSupportHumanAttribute>
  <!--ro, opt, bool, whether the device supports human body features-->true
</isSupportHumanAttribute>
<isSupportHumanRecognition>
  <!--ro, opt, bool, whether the device supports target capture events-->true
</isSupportHumanRecognition>
<isSupportFaceContrast>
  <!--ro, opt, bool, whether the device supports face picture comparison events-->true
</isSupportFaceContrast>
<isSupportFaceLib>
  <!--ro, opt, bool, Front-end devices support linkage by face picture Library., desc:related URI: /ISAPI/Event/schedules/faceLib/<ID>/<FDID>-->true
</isSupportFaceLib>
<isSupportBlackListFaceContrast>
  <!--ro, opt, bool, whether the device supports events of blacklist face picture comparison-->true
</isSupportBlackListFaceContrast>
<isSupportWhiteListFaceContrast>
  <!--ro, opt, bool, whether the device supports events of allowlist face picture comparison-->true
</isSupportWhiteListFaceContrast>
<isSupportIllegalParking>
  <!--ro, opt, bool, whether the device supports illegal parking detection-->true
</isSupportIllegalParking>
<isSupportPedestrian>
  <!--ro, opt, bool, whether the device supports pedestrian detection-->true
</isSupportPedestrian>
<isSupportTrafficAccident>
  <!--ro, opt, bool, whether the device supports traffic incident detection-->true
</isSupportTrafficAccident>
<isSupportConstruction>
  <!--ro, opt, bool, whether the device supports construction detection-->true
</isSupportConstruction>
<isSupportRoadBlock>
  <!--ro, opt, bool, whether the device supports roadblock detection-->true
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</isSupportRoadBlock>
<isSupportAbandonedObject>
  <!--ro, opt, bool, whether the device supports fallen object detection-->true
</isSupportAbandonedObject>
<isSupportParallelParking>
  <!--ro, opt, bool, whether the device supports parallel parking detection-->true
</isSupportParallelParking>
<isSupportParkingState>
  <!--ro, opt, bool, whether the device supports parking space status detection-->true
</isSupportParkingState>
<isSupportTrafficJam>
  <!--ro, opt, bool, whether the device supports traffic jam detection (old version)-->true
</isSupportTrafficJam>
<isSupportCongestion>
  <!--ro, opt, bool, whether the device supports traffic jam detection-->true
</isSupportCongestion>
<isSupportStudentsStoodUp>
  <!--ro, opt, bool, whether the device supports the student standing up detection event-->true
</isSupportStudentsStoodUp>
<isSupportFramesPeopleCounting>
  <!--ro, opt, bool, whether the device supports the regional people counting event-->true
</isSupportFramesPeopleCounting>
<isSupportFaceSnap>
  <!--ro, opt, bool, whether the device supports face capture events-->true
</isSupportFaceSnap>
<isSupportPersonDensityDetection>
  <!--ro, opt, bool, whether the device supports people density detection-->true
</isSupportPersonDensityDetection>
<HFPD>
  <!--ro, opt, bool, whether the device supports frequently appeared person detection-->true
</HFPD>
<isSupportPersonQueueDetection>
  <!--ro, opt, bool, whether the device supports queue detection events-->true
</isSupportPersonQueueDetection>
<isSupportVehicleStatistics>
  <!--ro, opt, bool, whether the device supports vehicle statistics events-->true
</isSupportVehicleStatistics>
<isSupportFaceThermometry>
  <!--ro, opt, bool, whether the device supports face temperature screening-->true
</isSupportFaceThermometry>
<isSupportSafetyHelmetDetection>
  <!--ro, opt, bool, whether the device supports hard hat detection-->true
</isSupportSafetyHelmetDetection>
<isSupportPowerSupplyCabinet>
  <!--ro, opt, bool, whether the device supports cabinet powering detection-->true
</isSupportPowerSupplyCabinet>
<isSupportSensorDetection>
  <!--ro, opt, bool, whether the device supports sensor detection-->true
</isSupportSensorDetection>
<isSupportSignAbnormal>
  <!--ro, opt, bool, whether the device supports the physical sign exception detection-->true
</isSupportSignAbnormal>
<isSupportTeacherBehaviorDetect>
  <!--ro, opt, bool, whether the device supports teacher's behavior detection-->true
</isSupportTeacherBehaviorDetect>
<isSupportCityManagement>
  <!--ro, opt, bool, whether the device supports the city management function-->true
</isSupportCityManagement>
<isSupportMixedTargetDetection>
  <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
</isSupportMixedTargetDetection>
<isSupportFaceSnapModeling>
  <!--ro, opt, bool, whether the device supports face modeling events-->true
</isSupportFaceSnapModeling>
<isSupportIntersectionAnalysis>
  <!--ro, opt, bool, whether the device supports the intersection analysis-->true
</isSupportIntersectionAnalysis>
<isSupportTriggerCapCheck>
  <!--ro, opt, bool, whether the device supports verifying the capability of triggering linkage actions, desc:this node indicates that the capability
message returned by calling the URI GET /ISAPI/Event/triggersCap has been verified, and the event linkage actions can be displayed exactly according to the
capability-->true
</isSupportTriggerCapCheck>
<isSupportLuma>
  <!--ro, opt, bool, VQD brightness exception-->true
</isSupportLuma>
<isSupportChroma>
  <!--ro, opt, bool, VQD image color cast-->true
</isSupportChroma>
<isSupportSnow>
  <!--ro, opt, bool, VQD snow noise-->true
</isSupportSnow>
<isSupportStreak>
  <!--ro, opt, bool, VQD stripe noise-->true
</isSupportStreak>
<isSupportFreeze>
  <!--ro, opt, bool, VQD image freeze-->true
</isSupportFreeze>
<isSupportSigLose>
  <!--ro, opt, bool, VQD signal loss-->true
</isSupportSigLose>
<isSupportClarity>
  <!--ro, opt, bool, VQD resolution exception-->true
</isSupportClarity>

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<isSupportJitter>
  <!--ro, opt, bool, VQD image flutter-->true
</isSupportJitter>
<isSupportBlock>
  <!--ro, opt, bool, VQD video tampering-->true
</isSupportBlock>
<isSupportFlowers>
  <!--ro, opt, bool, VQD blurred image alarm-->true
</isSupportFlowers>
<isSupportNoise>
  <!--ro, opt, bool, VQD blurred screen-->true
</isSupportNoise>
<isSupportGhost>
  <!--ro, opt, bool, VQD image noise-->true
</isSupportGhost>
<isSupportPurple>
  <!--ro, opt, bool, VQD purple fringing-->true
</isSupportPurple>
<isSupportICR>
  <!--ro, opt, bool, VQD ICP-->true
</isSupportICR>
<isSupportProtectiveFilm>
  <!--ro, opt, bool, VQD protective film unremoved-->true
</isSupportProtectiveFilm>
<isSupportVoltageInstable>
  <!--ro, opt, bool, whether the device supports the voltage instable exception detection-->true
</isSupportVoltageInstable>
<isSupportCertificateRevocation>
  <!--ro, opt, bool, whether the device supports the certificate expiry alarm-->true
</isSupportCertificateRevocation>
<isSupportSysStorFull>
  <!--ro, opt, bool, whether the device supports the storage full alarm of the cluster system-->true
</isSupportSysStorFull>
<isSupportVersionAbnormal>
  <!--ro, opt, bool, whether the device supports cluster version exception-->true
</isSupportVersionAbnormal>
<isSupportNodeOffline>
  <!--ro, opt, bool, whether the device supports device offline detection-->true
</isSupportNodeOffline>
<isSupportRecordCycleAbnormal>
  <!--ro, opt, bool, whether the device supports the event of insufficient recording period-->true
</isSupportRecordCycleAbnormal>
<isSupportIpcTransferAbnormal>
  <!--ro, opt, bool, whether the device supports the event of network camera migration failure-->true
</isSupportIpcTransferAbnormal>
<isSupportDrivingBehaviorDetect>
  <!--ro, opt, bool, whether the device supports driving behavior detection-->true
</isSupportDrivingBehaviorDetect>
<isSupportWasteGasDetection>
  <!--ro, opt, bool, whether the device supports waste gas monitoring-->true
</isSupportWasteGasDetection>
<isSupportGreyScaleAlarm>
  <!--ro, opt, bool, whether the device supports the gray scale alarm-->true
</isSupportGreyScaleAlarm>
<isSupportPTLocking>
  <!--ro, opt, bool, whether the device supports the PTZ locking exception-->true
</isSupportPTLocking>
<isSupportNoMaskDetection>
  <!--ro, opt, bool, whether the device supports not wearing mask detection-->true
</isSupportNoMaskDetection>
<isSupportTMPA>
  <!--ro, opt, bool, whether the device supports the warning linkage of face temperature screening-->true
</isSupportTMPA>
<RuleScheduleCap>
  <!--ro, opt, object, capability of configuring the arming schedule by rule-->
  <isSupportCityManagement>
    <!--ro, opt, bool, whether the device supports configuring the arming schedule by rule for the city management event-->true
  </isSupportCityManagement>
</RuleScheduleCap>
<RegionScheduleCap>
  <!--ro, opt, object, capability of configuring the arming schedule by area-->
  <isSupportPersonDensityDetection>
    <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the people density detection-->true
  </isSupportPersonDensityDetection>
  <isSupportPersonQueueDetection>
    <!--ro, opt, bool, whether the device supports configuring the arming schedule by area for the queue detection (area popularity analytics)-->true
  </isSupportPersonQueueDetection>
</RegionScheduleCap>
<isSupportThermalCalibrationFileException>
  <!--ro, opt, bool, whether the device supports the temperature calibration file exception-->true
</isSupportThermalCalibrationFileException>
<isSupportTemperatureIntervalMeasurement>
  <!--ro, opt, bool, whether the device supports the temperature range detection-->true
</isSupportTemperatureIntervalMeasurement>
<isSupportThermalVehicleDetection>
  <!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
</isSupportThermalVehicleDetection>
<isSupportWaterQualityDetection>
  <!--ro, opt, bool, whether the device supports water quality detection-->true
</isSupportWaterQualityDetection>
<isSupportWaterLevelDetection>
  <!--ro, opt, bool, whether the device supports water level detection-->true
</isSupportWaterLevelDetection>
<isSupportDeviceTemperatureAbnormal>
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    <!--ro, opt, bool, whether the device supports temperature exception detection-->true
</isSupportDeviceTemperatureAbnormal>
<isSupportObjectAccumulationDetection>
    <!--ro, opt, bool, whether the device supports object accumulation detection-->true
</isSupportObjectAccumulationDetection>
<isSupportAppAbnormal>
    <!--ro, opt, bool, whether the device supports the App exception alarm or event-->true
</isSupportAppAbnormal>
<isSupportPTEventCfg>
    <!--ro, opt, bool, whether the device supports transparent transmission event configuration event, desc:/ISAPI/Event/PTEventCfg/capabilities?
format=json-->true
</isSupportPTEventCfg>
<isSupportLensTemperatureException>
    <!--ro, opt, bool, whether the device supports uploading Lens abnormal temperature event-->true
</isSupportLensTemperatureException>
<isSupportPTZFXFinish>
    <!--ro, opt, bool, the event of railway robot and PTZ going to the preset, desc:PTZFXFinish-->true
</isSupportPTZFXFinish>
<isSupportNormalTemperatureAlarm>
    <!--ro, opt, bool, whether the device supports normal temperature alarm-->true
</isSupportNormalTemperatureAlarm>
<isSupportLowTemperatureAlarm>
    <!--ro, opt, bool, whether the device supports low temperature alarm-->true
</isSupportLowTemperatureAlarm>
<isSupportLowCoolerLifeEvent>
    <!--ro, opt, bool, whether it supports short life exception event of refrigerating machine-->true
</isSupportLowCoolerLifeEvent>
<isSupportObstacleAvoidanceAlarm>
    <!--ro, opt, bool, whether it supports obstacle avoidance alarm-->true
</isSupportObstacleAvoidanceAlarm>
<isSupportPTZRotationAbnormal>
    <!--ro, opt, bool, whether it supports PTZ rotation exception event-->true
</isSupportPTZRotationAbnormal>
<isSupportBlackListVehContrast>
    <!--ro, opt, bool-->true
</isSupportBlackListVehContrast>
<isSupportWhiteListVehContrast>
    <!--ro, opt, bool-->true
</isSupportWhiteListVehContrast>
</EventCap>
<ITCCap>
    <!--ro, opt, object, related node and URIs: the node <isSupportVehicleDetection> in /ISAPI/ITC/capability and /ISAPI/ITC/capabilities-->
<isSupportITC>
    <!--ro, opt, bool, whether the device supports ITC-->true
</isSupportITC>
<isSupportIllegalDictionary>
    <!--ro, opt, bool, whether the device supports the violation dictionary-->true
</isSupportIllegalDictionary>
<isSupportVehicleDetection>
    <!--ro, opt, bool, whether the device supports vehicle detection-->true
</isSupportVehicleDetection>
<isSupportHMTVehicleDetection>
    <!--ro, opt, bool, whether the device supports mixed-traffic vehicle detection-->true
</isSupportHMTVehicleDetection>
<isSupportLicencePlateAuditData>
    <!--ro, opt, bool, whether the device supports importing or exporting the license plate blocklist and allowlist data-->true
</isSupportLicencePlateAuditData>
<isSupportSearchPLListAudit>
    <!--ro, opt, bool, whether the device supports searching for the license plate blocklist and allowlist-->true
</isSupportSearchPLListAudit>
<isSupportVehicleDetectionManualTrigger opt="false,true">
    <!--ro, opt, bool, whether the device supports manually triggering vehicle detection, attr:opt{opt, string}-->true
</isSupportVehicleDetectionManualTrigger>
<isSupportSyncSignalOutput>
    <!--ro, opt, bool, whether the device supports configuring output parameters-->true
</isSupportSyncSignalOutput>
</ITCCap>
<ImageCap>
    <!--ro, opt, object, capability of configuring image parameters-->
<isSupportRegionalExposure>
    <!--ro, opt, bool, whether the device supports regional exposure-->true
</isSupportRegionalExposure>
<isSupportRegionalFocus>
    <!--ro, opt, bool, whether the device supports regional focus-->true
</isSupportRegionalFocus>
</ImageCap>
<RacmCap>
    <!--ro, opt, object, UI before picture search-->
<isSupportZeroChan>
    <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
</isSupportZeroChan>
<inputProxyNums>
    <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->1
</inputProxyNums>
<eSATANums>
    <!--ro, opt, int, supported number of eSATA interfaces-->1
</eSATANums>
<miniSASNums>
    <!--ro, opt, int, supported number of miniSAS interfaces-->1
</miniSASNums>
<nasNums>
    <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->1
</nasNums>

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<ipSanNums>
  <!--ro, opt, int, supported number of IPSANs-->1
</ipSanNums>
<isSupportRaid>
  <!--ro, opt, bool, whether the device supports RAID-->true
</isSupportRaid>
<isSupportExtHdCfg>
  <!--ro, opt, bool, whether the device supports advanced HDD management-->true
</isSupportExtHdCfg>
<isSupportTransCode>
  <!--ro, opt, bool, whether the device supports Live view transcoding-->true
</isSupportTransCode>
<isSupportIpcImport>
  <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
</isSupportIpcImport>
<NasMountType>
  <!--ro, opt, object, NAS mounting type-->
  <isNFSSupportAuthentication>
    <!--ro, opt, bool, whether the device supports CIFS authentication-->true
  </isNFSSupportAuthentication>
  <isCIFSsupportAuthentication>
    <!--ro, opt, bool, whether the device supports CIFS authentication-->true
  </isCIFSsupportAuthentication>
</NasMountType>
<isSupportIpcStreamType>
  <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
</isSupportIpcStreamType>
<isSupportIOInputProxy>
  <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
</isSupportIOInputProxy>
<isSupportIOOutputProxy>
  <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
</isSupportIOOutputProxy>
<isSupportPTZRs485Proxy>
  <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
</isSupportPTZRs485Proxy>
<isSupportSrcIDSearch>
  <!--ro, opt, bool, whether the device supports stream ID search-->true
</isSupportSrcIDSearch>
<isSupportReversePlayback>
  <!--ro, opt, bool, whether the device supports reverse playback-->true
</isSupportReversePlayback>
<isSupportSMARTTest>
  <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
</isSupportSMARTTest>
<isSupportDownloadByTime>
  <!--ro, opt, bool, whether the device supports downloading by time, desc:related URI: /ISAPI/ContentMgmt/download/capabilities-->true
</isSupportDownloadByTime>
<pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection>manual,manualSnapshot,playSnapshot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminate,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad, illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,alarmResult">
  <!--ro, opt, string, picture search type, attr:opt{opt, string}-->test
</pictureSearchType>
<recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection>manual,manualSnapshot,playSnapshot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
">
  <!--ro, opt, string, video search type, attr:opt{opt, string}, desc:"LLEvent" (all events), "CMR" (capture by schedule), "MOTION" (capture motion
detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures
and motion detection pictures), "Command", "pir" (pir alarm), "wlsensor" (wireless alarm), "callhelp" (panic alarm), "facedetection" (face detection),
"FieldDetection" (intrusion detection), "scenechangedetection" (scene change detection), "LineDetection" (line crossing detection)-->test
</recordSearchType>
<isSupportActivateIpc>
  <!--ro, opt, bool, whether the device supports activating the network camera-->true
</isSupportActivateIpc>
<isSupportCheckIpcSecurity>
  <!--ro, opt, object, device password detection-->
  <isSupportCheckPassword>
    <!--ro, opt, bool, whether the device supports password detection-->true
  </isSupportCheckPassword>
</isSupportCheckIpcSecurity>
<isSupportMainAndSubRecord>
  <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
</isSupportMainAndSubRecord>
<isSupportSyncIPCPassword>
  <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true

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    <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
  </isSupportSyncIPCPassword>
  <isSupportTransferIPC>
    <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
  </isSupportTransferIPC>
  <isSupportSmartSearch>
    <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
  </isSupportSmartSearch>
  <isSupportPOS>
    <!--ro, opt, bool, whether the device supports POS-->true
  </isSupportPOS>
  <isSupportLinkNum>
    <!--ro, opt, bool, whether the device supports getting the number of Links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/LinkNum--
  >true
  </isSupportLinkNum>
  <isSupportPassBackBasicCfg>
    <!--ro, opt, bool, whether the device supports configuring basic parameters of the CVR copy-back function, desc:related URI:
  /ISAPI/ContentMgmt/record/passback/basicCfg/capabilities-->true
  </isSupportPassBackBasicCfg>
  <PassBackTaskConfig>
    <!--ro, opt, object, capability of managing tasks of the CVR copy-back function-->
  <isSupportAddTask>
    <!--ro, opt, bool, whether the device supports adding a task, desc:related URI: /ISAPI/ContentMgmt/record/passback/addTask/capabilities-->true
  </isSupportAddTask>
  <isSupportSearchTask>
    <!--ro, opt, bool, whether the device supports searching tasks, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchTask/capabilities-->true
  </isSupportSearchTask>
  <isSupportControlTask>
    <!--ro, opt, bool, whether the device supports controlling the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/controlTask/capabilities--
  >true
  </isSupportControlTask>
  <isSupportDeleteTask>
    <!--ro, opt, bool, whether the device supports deleting the task, desc:related URI: /ISAPI/ContentMgmt/record/passback/deleteTask/capabilities--
  >true
  </isSupportDeleteTask>
  </PassBackTaskConfig>
  <PassBackPlanConfig>
    <!--ro, opt, object, capability of configuring the CVR copy-back schedule-->
  <isSupportAddPlan>
    <!--ro, opt, bool, whether the device supports adding a schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/addPlan/capabilities-->true
  </isSupportAddPlan>
  <isSupportSearchPlan>
    <!--ro, opt, bool, whether the device supports searching schedules, desc:related URI: /ISAPI/ContentMgmt/record/passback/searchPlan/capabilities--
  >true
  </isSupportSearchPlan>
  <isSupportDeletePlan>
    <!--ro, opt, bool, whether the device supports deleting the schedule, desc:related URI: /ISAPI/ContentMgmt/record/passback/deletePlan/capabilities--
  >true
  </isSupportDeletePlan>
  </PassBackPlanConfig>
  <IRAIDCap>
    <!--ro, opt, object, network array capability-->
  <isSupportIRAIDRapidConfiguration>
    <!--ro, opt, bool, whether the device supports the rapid IRAID configuration, desc:related URI:
  /ISAPI/ContentMgmt/Storage/IRAID/RapidConfiguration/capabilities-->true
  </isSupportIRAIDRapidConfiguration>
  </IRAIDCap>
  <isSupportStorageExtraInfo>
    <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
  /ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
  </isSupportStorageExtraInfo>
  <isSupportRecordStatus>
    <!--ro, opt, bool, whether the device supports getting the recording status, desc:related URI: /ISAPI/ContentMgmt/channels/<ID>/record/capabilities--
  >true
  </isSupportRecordStatus>
  <isSupportRacmChannelsCap opt="true,false">
    <!--ro, opt, bool, whether the device supports the Racm capability by channel, attr:opt{opt, string}, desc:related URI:
  /ISAPI/ContentMgmt/channels/<ID>/capabilities-->true
  </isSupportRacmChannelsCap>
  <LockCap>
    <!--ro, opt, object, video locking and unlocking, desc:related URI: /ISAPI/ContentMgmt/record/control/Locks/capabilities-->
  </LockCap>
  <isSupportFormatAll>
    <!--ro, opt, bool, whether the device supports formatting all HDDs, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/format-->true
  </isSupportFormatAll>
  <isSupportExtendCabinetCfg>
    <!--ro, opt, bool, whether the device supports configuring parameters of the extended cabinet. Related URI:
  /ISAPI/ContentMgmt/Storage/ExtendCabinet/capabilities-->true
  </isSupportExtendCabinetCfg>
  <diskGroupNums>
    <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->1
  </diskGroupNums>
  <ExternalDevice>
    <!--ro, opt, object, external device storage-->
  <USB>
    <!--ro, opt, object, USB information-->
  <debugLogOutput>
    <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
  </debugLogOutput>
  <isSupportCapturePackage>
    <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
  /ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
  </isSupportCapturePackage>
  <isSupportStreamStorage>

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    <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
    </isSupportStreamStorage>
    </USB>
  </ExternalDevice>
  <customProtocolNums>
    <!--ro, opt, int, number of custom protocols connected via IPC, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols, supported number of
custom protocols-->1
    </customProtocolNums>
    <isSupportIPCTiming>
      <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <enableTiming> node supports-->true
    </isSupportIPCTiming>
    <isSupportLogConfig>
      <!--ro, opt, bool, whether the device supports Log configuration, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
    </isSupportLogConfig>
    <SecurityLog>
      <!--ro, opt, object, device's capability of security Log-->
      <isSupportSecurityLog>
        <!--ro, opt, bool, whether the device supports the security Log-->true
      </isSupportSecurityLog>
      <isSupportLogServer>
        <!--ro, opt, bool, whether it supports security Log server configuration-->true
      </isSupportLogServer>
      <isSupportLogServerTest>
        <!--ro, opt, bool, whether it supports security Log server test-->true
      </isSupportLogServerTest>
      <SecurityLogTypeList>
        <!--ro, req, array, List of security log types, subType:object-->
        <SecurityLogType>
          <!--ro, opt, object, security Log type-->
          <primaryType>
            <!--ro, req, string, main Log type, desc:"Event", "Operation", "Other", "ALL"-->test
          </primaryType>
          <secondaryType opt="all">
            <!--ro, req, string, minor Log type, attr:opt{opt, string}, desc:for other types, refer to the appendix-->test
          </secondaryType>
        </SecurityLogType>
      </SecurityLogTypeList>
      <isSupportLogServerList>
        <!--ro, opt, bool, whether the device supports batch configuring security Log server-->true
      </isSupportLogServerList>
    </SecurityLog>
    <isSupportGetBackupDevice>
      <!--ro, opt, bool, whether the device supports getting the backup device, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/device?format=json--
>true
    </isSupportGetBackupDevice>
    <isSupportAutoBackup>
      <!--ro, opt, bool, whether the device supports automatic backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/auto/capabilities?format=json--
>true
    </isSupportAutoBackup>
    <certificateValidationEnabled>
      <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
    </certificateValidationEnabled>
    <defaultAdminPortEnabled>
      <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
    </defaultAdminPortEnabled>
    <isSupportLogDataPackage>
      <!--ro, opt, bool, whether the device supports exporting the Log files, desc:related URI: /ISAPI/ContentMgmt/LogSearch/dataPackage-->true
    </isSupportLogDataPackage>
    <LogSearchTimeSpanNums>
      <!--ro, opt, int, number of time periods for Log search, desc:related URI: /ISAPI/ContentMgmt/LogSearch; corresponding node: timeSpanList in
CMSearchDescription-->1
    </LogSearchTimeSpanNums>
    <CloudStorageServerCap>
      <!--ro, opt, object, capability of configuring the cloud storage server-->
      <isSupportCloudStorageParameter>
        <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
      </isSupportCloudStorageParameter>
      <isSupportCloudStoragePool>
        <!--ro, opt, bool, whether the device supports configuring the cloud storage pool, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/pool/capabilities?format=json-->true
      </isSupportCloudStoragePool>
      <supportChannelsNum>
        <!--ro, opt, int, number of channels supported by the cloud storage, desc:if this node is not returned, all channels of the device support; if this
node is returned, several channels of the device supports cloud storage (at present, it will not return the specific number of the channels supporting cloud
storage)-->1
      </supportChannelsNum>
    </CloudStorageServerCap>
    <isSupportWebPrivatePlaybackByUTC>
      <!--ro, opt, bool, whether it supports playback (via private protocol) by UTC-->true
    </isSupportWebPrivatePlaybackByUTC>
    <isSupportFindCommonFileByUTC>
      <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
    </isSupportFindCommonFileByUTC>
    <isSupportFindEventFileByUTC>
      <!--ro, opt, bool, whether it supports searching for event files (private protocols) by UTC time-->true
    </isSupportFindEventFileByUTC>
    <isSupportSmartSearchRecordByUTC>
      <!--ro, opt, bool, whether it supports smart video search by UTC time-->true
    </isSupportSmartSearchRecordByUTC>

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</isSupportSmartSearchRecordByUTC>
<isSupportMRDSearchByTimeZone>
  <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
</isSupportMRDSearchByTimeZone>
<isSupportSearchRecordLabelByUTC>
  <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
</isSupportSearchRecordLabelByUTC>
<isSupportSearchPictureByUTC>
  <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
</isSupportSearchPictureByUTC>
<isSupportSmartSearchPictureByUTC>
  <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
</isSupportSmartSearchPictureByUTC>
<isSupportFindLogByUTC>
  <!--ro, opt, bool, whether it supports searching log by UTC time-->true
</isSupportFindLogByUTC>
<isSupportUploadRecordByUTC>
  <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
</isSupportUploadRecordByUTC>
<isSupportPlaybackByUTC>
  <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
</isSupportPlaybackByUTC>
<isSupportFaceDetectionAlarmByTimeZone>
  <!--ro, opt, bool, no file is found-->true
</isSupportFaceDetectionAlarmByTimeZone>
<isSupportSourceModify>
  <!--ro, opt, bool, whether it supports configuring the parameters of the LAN IPC, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/sourceModify/capabilities-->true
</isSupportSourceModify>
<isSupportCountingSearchByUTC>
  <!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
</isSupportCountingSearchByUTC>
<isSupportCluster>
  <!--ro, opt, bool, whether the device supports the cluster function, desc:related URI: /ISAPI/ContentMgmt/Cluster-->true
</isSupportCluster>
<supportAIDTFSType opt="illegalParking,wrongdirection,crosslane,vehicleexist,lanchange,turnround,evidence">
  <!--ro, opt, string, the event picture search type supported by AID and TFS, attr:opt{opt, string}-->test
</supportAIDTFSType>
<isSupportMisinfoStateSearch>
  <!--ro, opt, bool, whether it supports searching by false alarm status-->true
</isSupportMisinfoStateSearch>
<isSupportOneKeyOperationStorageData>
  <!--ro, opt, bool, whether it supports one-touch configuration for storage data, desc:related URI: /ISAPI/ContentMgmt/Storage/oneKey/capabilities?
format=json-->true
</isSupportOneKeyOperationStorageData>
<eventRecordSearch>
  <!--ro, opt, object, whether the device supports searching for videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
  <eventType opt="faceContrast,faceSnap,vehicleMatchResult,hawkResult">
    <!--ro, opt, string, event type, attr:opt{opt, string}-->test
  </eventType>
  <auxEventType opt="faceContrastFailure,faceContrastSuccess">
    <!--ro, opt, string, search sub type of the event, attr:opt{opt, string}-->test
  </auxEventType>
</eventRecordSearch>
<isSupportManualRecord>
  <!--ro, opt, bool, whether the device supports manual recording, desc:related URI: /ISAPI/ContentMgmt/record/control/manual/capabilities?format=json-->true
</isSupportManualRecord>
<isSupportRemark>
  <!--ro, opt, bool, whether it supports file remarks-->true
</isSupportRemark>
<FileUpload>
  <!--ro, opt, object, automatically upload file-->
  <enabled>
    <!--ro, req, bool, whether it supports uploading the file-->true
  </enabled>
</FileUpload>
<PoliceInfoUpload>
  <!--ro, opt, object, automatically upload police officer information-->
  <enabled>
    <!--ro, req, bool, whether the device supports uploading the police officer information-->true
  </enabled>
</PoliceInfoUpload>
<PoliceInfo>
  <!--ro, opt, object, police officer information-->
  <policeID>
    <!--ro, opt, bool, whether the device supports police officer ID-->true
  </policeID>
  <policeCode>
    <!--ro, opt, bool, whether the device supports police officer number-->true
  </policeCode>
  <policeName>
    <!--ro, opt, bool, whether the device supports police officer name-->true
  </policeName>
  <password>
    <!--ro, opt, bool, whether the device supports police officer password-->true
  </password>
  <deviceID>
    <!--ro, opt, bool, whether the device supports device ID-->true
  </deviceID>
</PoliceInfo>
<isSupportAssignChannelID>
  <!--ro, opt, bool, whether the device supports specifying the access channel No. when adding the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true

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</isSupportAssignChannelID>
<isSupportAssignStreamID>
  <!--ro, opt, bool, whether it supports specifying stream ID of the channel when adding the network camera in channel mode, desc:related URI:
  /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <defaultAdminPortEnabled> node supports-->true
</isSupportAssignStreamID>
<PictureSearchSubTypeList>
  <!--ro, opt, array, List of picture search sub type, subType:object-->
  <PictureSearchSubType>
    <!--ro, opt, object, the sub type of picture search-->
    <mainType>
      <!--ro, req, string, main event type of the searched picture-->test
    </mainType>
    <subType opt="all,motorVehicle,nonMotorVehicle,pedestrian">
      <!--ro, req, string, sub type of each main event type, attr:opt{opt, string}-->test
    </subType>
  </PictureSearchSubType>
</PictureSearchSubTypeList>
<isSupportPlaybackStreamIDByUTC>
  <!--ro, opt, bool, whether it supports playback by stream ID according to UTC time-->true
</isSupportPlaybackStreamIDByUTC>
<isSupportTimeLockStreamIDByUTC>
  <!--ro, opt, bool, whether it supports Locking and unLocking the video segment by steam ID according to UTC time-->true
</isSupportTimeLockStreamIDByUTC>
<isSupportTimeSearch>
  <!--ro, opt, bool, whether it supports searching for recording start and end time by channel, desc:related URI: /ISAPI/ContentMgmt/time/search?
  format=json-->true
</isSupportTimeSearch>
<behaviorEventPicSearch opt="allBehaviorEvent,running,group,violentMotion,failDown,playCellphone,peopleNumChange,leavePosition,retention,sleepOnduty">
  <!--ro, opt, string, abnormal event detection supported by picture search, attr:opt{opt, string}-->test
</behaviorEventPicSearch>
<perimeterEventPicSearch opt="allPerimeterEvent,linedetection,fielddetection,regionEntrance,regionExiting,loitering">
  <!--ro, opt, string, perimeter protection events supported by picture search, attr:opt{opt, string}-->test
</perimeterEventPicSearch>
<isSupportSourceStatus>
  <!--ro, opt, bool, whether it supports getting the status of access source-->true
</isSupportSourceStatus>
<isSupportSSDSMARTTest>
  <!--ro, opt, bool, whether the device supports S.M.A.R.T. detection on SSD, desc:related URI: /ISAPI/ContentMgmt/Storage/ssd/<ID>/SMARTTest/start--
  >true
</isSupportSSDSMARTTest>
<MobileStorage>
  <!--ro, opt, object, storage capability of the mobile HDD-->
  <isSupportGetFileTime>
    <!--ro, opt, bool, whether it supports getting the file time of the mobile HDD, desc:it supports mobile HDD currently. Related URI:
    /ISAPI/ContentMgmt/mobileStorage/fileTime/capabilities?format=json-->true
  </isSupportGetFileTime>
</MobileStorage>
<isSupportGetChannelList>
  <!--ro, opt, bool, whether it supports getting channels of the HDD, desc:it supports mobile HDD currently. Related URI:
  /ISAPI/ContentMgmt/Storage/hdd/channelList/capabilities?format=json-->true
</isSupportGetChannelList>
<isSupportCascade>
  <!--ro, opt, bool, whether it supports configuring parameters for NVR cascading topology-->true
</isSupportCascade>
<isSupportPictureInfo>
  <!--ro, opt, bool, whether it supports picture search on the SD card of the capture camera-->true
</isSupportPictureInfo>
<isSupportCloudStorageModel>
  <!--ro, opt, bool, whether the device supports configuring the cloud storage mode,
  desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>/capabilities?format=json-->true
</isSupportCloudStorageModel>
<isSupportCloudStorageModelList>
  <!--ro, opt, bool, whether the device supports configuring cloud storage modes in a batch,
  desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>?format=json-->true
</isSupportCloudStorageModelList>
<isSupportRecordAudio>
  <!--ro, opt, bool, whether the device supports recording the audio, desc:only the audio will be recorded and the video will not be recorded. Related
  URI: /ISAPI/ContentMgmt/recordAudio/<audioID>/capabilities?format=json-->true
</isSupportRecordAudio>
<isSupportSearchAudio>
  <!--ro, opt, bool, whether the device supports searching for the audio, desc:related URI: /ISAPI/ContentMgmt/recordAudio/search/capabilities?
  format=json-->true
</isSupportSearchAudio>
<isSupportAudioDailyDistribution>
  <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI:
  /ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
</isSupportAudioDailyDistribution>
<isSupportDownloadAudio>
  <!--ro, opt, bool, whether the device supports downloading the audio by file name, desc:related URI:
  /ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
</isSupportDownloadAudio>
<isSupportDeleteAudio>
  <!--ro, opt, bool, whether the device supports deleting the recorded audio file, desc:related URI: /ISAPI/ContentMgmt/recordAudio/deleteFile?
  format=json-->true
</isSupportDeleteAudio>
<isSupportSearchRemoteAlbum>
  <!--ro, opt, bool, whether the device supports remotely searching for album information, desc:related URI: /ISAPI/ContentMgmt/searchRemoteAlbum?
  format=json-->true
</isSupportSearchRemoteAlbum>
<isSupportStreamParam>
  <!--ro, opt, bool, whether the device supports configuring the access stream parameters of all digital channels, desc:related URI:
  /ISAPI/ContentMgmt/InputProxy/channels/streamParam/capabilities?format=json-->true
</isSupportStreamParam>

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<isSupportFusionChan>
  <!--ro, opt, bool, whether the device supports fusion channel, desc:if this node does not exist, it indicates that the device does not support the
function; interaction IP channel: it is a concept from the education sharing system; the education sharing system will allocate IP channels to images of the
connected sub classrooms or meeting rooms. fusion channel: a channel generated when adding the NVR locally or remotely; it supports live view, playback,
storage, and recording schedule/linkage configuration; the channel will take effect by enabling the fusion channel-->true
</isSupportFusionChan>
<isSupportRecordMultipleDownload>
  <!--ro, opt, bool, whether the device supports downloading videos in multiples, desc:if this node does not exist, it indicates that the device does
not support the function by default-->true
</isSupportRecordMultipleDownload>
</RacmCap>
<isSupportMI3704>
  <!--ro, opt, bool-->true
</isSupportMI3704>
<isSupportLogServer>
  <!--ro, opt, bool-->true
</isSupportLogServer>
<isSupportLogServerTest>
  <!--ro, opt, bool-->true
</isSupportLogServerTest>
<PTZCtrlCap>
  <!--ro, opt, object, device's PTZ capability-->
<isSupportPatrols>
  <!--ro, opt, bool, whether the device supports patrol-->true
</isSupportPatrols>
<isSupportCombinedPath>
  <!--ro, opt, bool, whether the device supports the PTZ combined path-->true
</isSupportCombinedPath>
<isSupportRailwayRobot>
  <!--ro, opt, bool, whether the device supports railway robot, desc:/ISAPI/PTZCtrl/channels/<channelID>/RailwayRobot/capabilities?format=json-->true
</isSupportRailwayRobot>
</PTZCtrlCap>
<SmartCap>
  <!--ro, opt, object, compact and smart capability, desc:/ISAPI/Smart/capabilities-->
<isSupportROI>
  <!--ro, opt, bool, whether the device supports ROI encoding configuration-->true
</isSupportROI>
<isSupportFaceDetect>
  <!--ro, opt, bool, whether the device supports face detection-->true
</isSupportFaceDetect>
<isSupportIntelliTrace>
  <!--ro, opt, bool, whether the device supports smart tracking-->true
</isSupportIntelliTrace>
<isSupportFieldDetection>
  <!--ro, opt, bool, whether the device supports intrusion detection-->true
</isSupportFieldDetection>
<isSupportDefocusDetection>
  <!--ro, opt, bool, whether the device supports defocus detection-->true
</isSupportDefocusDetection>
<isSupportAudioDetection>
  <!--ro, opt, bool, whether the device supports audio detection-->true
</isSupportAudioDetection>
<isSupportSceneChangeDetection>
  <!--ro, opt, bool, whether the device supports scene change detection-->true
</isSupportSceneChangeDetection>
<isSupportLineDetection>
  <!--ro, opt, bool, whether the device supports line crossing detection-->true
</isSupportLineDetection>
<isSupportRegionEntrance>
  <!--ro, opt, bool, whether the device supports region entrance detection-->true
</isSupportRegionEntrance>
<isSupportRegionExiting>
  <!--ro, opt, bool, whether the device supports region exiting detection-->true
</isSupportRegionExiting>
<isSupportLoitering>
  <!--ro, opt, bool, whether the device supports loitering detection-->true
</isSupportLoitering>
<isSupportGroup>
  <!--ro, opt, bool, whether the device supports people gathering detection-->true
</isSupportGroup>
<isSupportRapidMove>
  <!--ro, opt, bool, whether the device supports fast moving detection-->true
</isSupportRapidMove>
<isSupportParking>
  <!--ro, opt, bool, whether the device supports parking detection-->true
</isSupportParking>
<isSupportUnattendedBaggage>
  <!--ro, opt, bool, whether the device supports unattended baggage detection-->true
</isSupportUnattendedBaggage>
<isSupportAttendedBaggage>
  <!--ro, opt, bool, whether the device supports object removal detection-->true
</isSupportAttendedBaggage>
<isSupportPeopleDetection>
  <!--ro, opt, bool, whether the device supports human body detection-->true
</isSupportPeopleDetection>
<isSupportSmartCalibration>
  <!--ro, opt, bool, whether the device supports calibration filter size-->true
</isSupportSmartCalibration>
<isSupportStorageDetection>
  <!--ro, opt, bool, whether the device supports health monitoring storage-->true
</isSupportStorageDetection>
<isSupportShipsDetection>
  <!--ro, opt, bool, whether the device supports ship detection-->true
</isSupportShipsDetection>

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<isSupportShield>
  <!--ro, opt, bool, whether the device supports configuring the compact and smart shielded regions-->true
</isSupportShield>
<isSupportAlgVersion>
  <!--ro, opt, bool, whether the device supports algorithm version-->true
</isSupportAlgVersion>
<isSupportVideoOverlap>
  <!--ro, opt, bool, whether the device supports configuring information overLay for video stream-->true
</isSupportVideoOverlap>
<isSupportParkingState>
  <!--ro, opt, bool, whether the device supports parking status-->true
</isSupportParkingState>
<isSupportChannelResource>
  <!--ro, opt, bool, whether the device supports channel resources-->true
</isSupportChannelResource>
<isSupportAnalysisUnitSwitch opt="true,false">
  <!--ro, opt, bool, whether the device supports switching analysis unit, attr:opt{opt, string}-->true
</isSupportAnalysisUnitSwitch>
<isSupportHFPD>
  <!--ro, opt, bool, whether the device supports frequently appeared person detection, desc:related URI: /ISAPI/SDT/HFPD?format=json-->true
</isSupportHFPD>
<isSupportImageROI>
  <!--ro, opt, bool, whether the device supports smartJpeg (image ROI)-->true
</isSupportImageROI>
<isSupportLFPD>
  <!--ro, opt, bool, whether the device supports rarely appeared person detection, desc:related URI: /ISAPI/SDT/LFPD?format=json-->true
</isSupportLFPD>
<isSupportSmartOverLapParams>
  <!--ro, opt, bool, whether it supports smart overLay parameters, desc:related URI: /ISAPI/Smart/channels/<channelID>/SmartOverLapParams?format=json--
>true
</isSupportSmartOverLapParams>
</SmartCap>
<isSupportEhome>
  <!--ro, opt, bool, whether the device supports ISUP functions-->true
</isSupportEhome>
<isSupportStreamingEncrypt>
  <!--ro, opt, bool, whether the device supports stream encryption-->true
</isSupportStreamingEncrypt>
<TestCap>
  <!--ro, opt, object, test capability-->
  <isSupportFTPTest>
    <!--ro, opt, bool, whether it supports FTP test-->true
  </isSupportFTPTest>
  <isSupportPingTest>
    <!--ro, opt, bool, whether it supports PING test-->true
  </isSupportPingTest>
  <isSupportNTPTest>
    <!--ro, opt, bool, whether it supports NTP test-->true
  </isSupportNTPTest>
  <isSupportNASTest>
    <!--ro, opt, bool, whether it supports NAS test-->true
  </isSupportNASTest>
  <isSupportEmailTest>
    <!--ro, opt, bool, whether the device supports email test-->true
  </isSupportEmailTest>
</TestCap>
<ImageMiscCap>
  <!--ro, opt, object, DPC capability-->
  <isSupportDPC>
    <!--ro, opt, bool, whether the device supports DPC-->true
  </isSupportDPC>
</ImageMiscCap>
<PanoramaCap>
  <!--ro, opt, object, capability of panoramic channels, desc:capability of panoramic channels-->
  <isSupportGeneratePanorama>
    <!--ro, opt, bool, whether the device supports generating panoramic images-->true
  </isSupportGeneratePanorama>
  <isSupportPanoramaPosition3D>
    <!--ro, opt, bool, whether the device supports panoramic 3D position-->true
  </isSupportPanoramaPosition3D>
  <isSupportPreset>
    <!--ro, opt, bool, whether the device supports configuring panoramic preset-->true
  </isSupportPreset>
  <isSupportGetPanoramicSplicingStatus>
    <!--ro, opt, bool-->true
  </isSupportGetPanoramicSplicingStatus>
  <isSupportStopPanoramicSplicing>
    <!--ro, opt, bool-->true
  </isSupportStopPanoramicSplicing>
  <isSupportSplicingChannelMotionControl>
    <!--ro, opt, bool-->true
  </isSupportSplicingChannelMotionControl>
  <isSupportGetPanoramaSplicingPictures>
    <!--ro, opt, bool-->true
  </isSupportGetPanoramaSplicingPictures>
  <isSupportImportPanoramaSplicingParamFile>
    <!--ro, opt, bool-->true
  </isSupportImportPanoramaSplicingParamFile>
  <isSupportExportPanoramaSplicingParamFile>
    <!--ro, opt, bool-->true
  </isSupportExportPanoramaSplicingParamFile>
  <isSupportImportPanoramicSplicingPicture>
    <!--ro, opt, bool-->true
  </isSupportImportPanoramicSplicingPicture>

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</isSupportImportPanoramicSplicingPicture>
<isSupportExportPanoramicSplicingPicture>
  <!--ro, opt, bool-->true
</isSupportExportPanoramicSplicingPicture>
</PanoramaCap>
<ThermalCap>
  <!--ro, opt, object, thermal capability, desc:related URI: /ISAPI/Thermal/capabilities-->
  <isSupportFireDetection>
    <!--ro, opt, bool, whether the device supports fire detection, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities-->true
  </isSupportFireDetection>
  <isFireFocusZoomSupport>
    <!--ro, opt, bool, whether the device supports visible light lens zooming, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireFocusZoom-->true
  </isFireFocusZoomSupport>
  <isSupportThermometry>
    <!--ro, opt, bool, whether the device supports thermography, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities-->true
  </isSupportThermometry>
  <isSupportRealtimeThermometry>
    <!--ro, opt, bool, whether the device supports uploading real-time temperature measurement data-->true
  </isSupportRealtimeThermometry>
  <isSupportThermIntell>
    <!--ro, opt, bool, whether the device supports configuring smart thermal mutual exclusion, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermIntell/capabilities-->true
  </isSupportThermIntell>
  <isSupportNVR>
    <!--ro, opt, bool, it is used for client to check-->true
  </isSupportNVR>
  <isSupportThermometrySchedule>
    <!--ro, opt, bool, whether the device supports the arming schedule of temperature measurement, desc:related URI: /ISAPI/Event/schedules/thermometry--
>true
  </isSupportThermometrySchedule>
  <isSupportTemperatureSchedule>
    <!--ro, opt, bool, whether the device supports the arming schedule of temperature difference, desc:related URI: /ISAPI/Event/schedules/temperature--
>true
  </isSupportTemperatureSchedule>
  <isSupportFireDetectionSchedule>
    <!--ro, opt, bool, whether the device supports the arming schedule of fire detection, desc:related URI: /ISAPI/Event/schedules/fireDetections-->true
  </isSupportFireDetectionSchedule>
  <isSupportPower>
    <!--ro, opt, bool, whether the device supports power on/off capability, desc:related URI: /ISAPI/Thermal/Power/capabilities-->true
  </isSupportPower>
  <isSupportRealtimeTempHumi>
    <!--ro, opt, bool, whether the device supports real-time detection of temperature and humidity, desc:related URI:
/ISAPI/Thermal/TempHumi/capabilities-->true
  </isSupportRealtimeTempHumi>
  <ManualThermCap>
    <!--ro, opt, object, the capability of manual temperature measurement-->
    <manualThermRuleNum>
      <!--ro, opt, int, if this node is not returned, it indicates manual temperature measurement is not supported, desc:related URI:
/ISAPI/Thermal/manualThermometry/capabilities-->true
    </manualThermRuleNum>
    </ManualThermCap>
    <isSupportManualThermBasic>
      <!--ro, opt, bool, whether the device supports basic configuration of manual temperature measurement, desc:related URI:
/ISAPI/Thermal/manualThermBasic/capabilities-->true
    </isSupportManualThermBasic>
    <isSupportFireShieldMask>
      <!--ro, opt, bool, whether the device supports configuring the parameters of fire shielded regions, desc:related URI:
/ISAPI/Thermal/channels/<ID>/fireShieldMask/capabilities-->true
    </isSupportFireShieldMask>
    <isSupportSmokeShieldMask>
      <!--ro, opt, bool, whether the device supports configuring the parameters of smoke shielded regions, desc:related URI:
/ISAPI/Thermal/channels/<ID>/smokeShieldMask/capabilities-->true
    </isSupportSmokeShieldMask>
    <isSupportThermometryMode>
      <!--ro, opt, bool, whether the device supports configuring temperature measurement mode, desc:related URI:
/ISAPI/Thermal/channels/<ID>/thermometryMode/capabilities-->true
    </isSupportThermometryMode>
    <isSupportThermalPip>
      <!--ro, opt, bool, whether the device supports the PIP configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/pip/capabilities-->true
    </isSupportThermalPip>
    <isSupportThermalIntelRuleDisplay>
      <!--ro, opt, bool, whether the device supports the PIP configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/IntelRuleDisplay/capabilities--
>true
    </isSupportThermalIntelRuleDisplay>
    <AlgVersionInfo>
      <!--ro, opt, object, whether the device supports getting the version information of thermal algorithms Library, desc:related URI:
/ISAPI/Thermal/AlgVersion-->
      <thermometryAlgName min="1" max="128">
        <!--ro, opt, string, version information of temperature measurement algorithms Library, attr:min{opt, string},max{opt, string}-->test
      </thermometryAlgName>
      <shipsAlgName min="1" max="128">
        <!--ro, opt, string, version name of ship detection algorithms Library, attr:min{opt, string},max{opt, string}-->test
      </shipsAlgName>
      <fireAlgName min="1" max="128">
        <!--ro, opt, string, algorithm Library version name of fire detection, attr:min{opt, int},max{opt, int}-->test
      </fireAlgName>
      <smokeAlgName min="1" max="128">
        <!--ro, opt, string, algorithm Library version name of smoke detection, attr:min{opt, int},max{opt, int}-->test
      </smokeAlgName>
      <wasteGasAlgName min="1" max="128">
        <!--ro, opt, string, algorithm Library version name of waste gas detection, attr:min{opt, string},max{opt, string}-->test
      </wasteGasAlgName>
      <greyScaleAlarmAlgName min="1" max="128">
        <!--ro, opt, string, algorithm Library version name of greyscale alarm, attr:min{opt, string},max{opt, string}-->test
      </greyScaleAlarmAlgName>

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        </greyScaleAlarmAlgName>
    </AlgVersionInfo>
    <isSupportFaceThermometry>
        <!--ro, opt, bool, whether supports temperature screening configuration, desc:related URI: /ISAPI/Thermal/channels/<ID>/faceThermometry/capabilities-->
    >true
    </isSupportFaceThermometry>
    <isSupportThermalBlackBody>
        <!--ro, opt, bool, whether the device supports configuring black body parameters of thermal imaging, desc:related URI:
    /ISAPI/Thermal/channels/<ID>/blackBody/capabilities-->true
    </isSupportThermalBlackBody>
    <isSupportThermalStreamParam>
        <!--ro, opt, bool, whether the device supports configuring stream parameters of thermal imaging, desc:related URI:
    /ISAPI/Thermal/channels/<ID>/streamParam/capabilities-->true
    </isSupportThermalStreamParam>
    <isSupportBodyTemperatureCompensation>
        <!--ro, opt, bool, whether the device supports temperature compensation configuration, desc:related URI:
    /ISAPI/Thermal/channels/<ID>/bodyTemperatureCompensation/capabilities-->true
    </isSupportBodyTemperatureCompensation>
    <isSupportClickToThermometry>
        <!--ro, opt, bool, whether the device supports click-to-thermometry, desc:related URI: ISAPI/Thermal/channels/<ID>/clickToThermometry/initialization-->
    >true
    </isSupportClickToThermometry>
    <isSupportThermometryHistorySearch>
        <!--ro, opt, bool, whether the device supports searching for historical temperature data, desc:related URI:
    /ISAPI/Thermal/channels/<ID>/historyTemperature/capabilities-->true
    </isSupportThermometryHistorySearch>
    <isSupportThermometryShieldMask>
        <!--ro, opt, bool, whether the device supports configuring shield parameters of thermography region, desc:related URI:
    /ISAPI/Thermal/channels/<ID>/thermometryShieldMask/capabilities-->true
    </isSupportThermometryShieldMask>
    <isSupportTemperatureCorrection>
        <!--ro, opt, bool, whether device supports temperature measurement correction, desc:related URI:
    /ISAPI/Thermal/channels/<ID>/CorrectionParam/capabilities-->true
    </isSupportTemperatureCorrection>
    <isSupportBurningPrevention>
        <!--ro, opt, bool, whether device supports burning prevention, desc:related URI: /ISAPI/Thermal/channels/<ID>/burningPrevention/capabilities-->true
    </isSupportBurningPrevention>
    <isSupportTemperatureCollection>
        <!--ro, opt, bool, whether device supports temperature ANR, desc:related URI: ISAPI/Thermal/temperature/collection/capabilities?format=json-->true
    </isSupportTemperatureCollection>
    <isSupportJpegPicWithAppendData>
        <!--ro, opt, bool, whether the device supports capturing pictures, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/jpegPicWithAppendData?
    format=json-->true
    </isSupportJpegPicWithAppendData>
    <isSupportRealTimeThermometryForHTTP>
        <!--ro, opt, bool, whether device supports real-time temperature measurement, desc:related URI:
    /ISAPI/Thermal/channels/<ID>/thermometry/realTimeThermometry/rules?format=json-->true
    </isSupportRealTimeThermometryForHTTP>
    <isSupportThermometryOfflineCapture>
        <!--ro, opt, bool, whether the device supports capturing pictures offline, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/OfflineCapture?
    format=json-->true
    </isSupportThermometryOfflineCapture>
    <isSupportShipsDetectionWithScene>
        <!--ro, opt, bool, whether the device supports ship detection by scene, desc:related URI: /ISAPI/Thermal/channels/<ID>/shipsDetection/capabilities?
    format=json-->true
    </isSupportShipsDetectionWithScene>
    <isSupportWasteGasDetection>
        <!--ro, opt, bool, whether the device supports waste gas monitoring-->true
    </isSupportWasteGasDetection>
    <isSupportGreyScaleAlarm>
        <!--ro, opt, bool, whether the device supports the gray scale alarm-->true
    </isSupportGreyScaleAlarm>
    <isSupportFaceSnapThermometry>
        <!--ro, opt, bool, whether the device supports temperature screening when capturing face pictures-->true
    </isSupportFaceSnapThermometry>
    <isSupportThermalTemperatureCorrect>
        <!--ro, opt, bool, whether device supports temperature calibration, desc:related URI: /ISAPI/Thermal/channels/<ID>/temperatureCorrect?format=json-->
    >true
    </isSupportThermalTemperatureCorrect>
    <isSupportThermalVehicleDetection>
        <!--ro, opt, bool, whether the device supports thermal imaging vehicle detection-->true
    </isSupportThermalVehicleDetection>
    <isSupportSensorParam>
        <!--ro, opt, bool, whether the device supports configuring the sensor temperature measurement parameters, desc:related URI:
    /ISAPI/Thermal/channels/<channelID>/thermometry/sensorParam/capabilities?format=json-->true
    </isSupportSensorParam>
    <isSupportTemperatureIntervalMeasurement>
        <!--ro, opt, bool, whether the device supports temperature range detection-->true
    </isSupportTemperatureIntervalMeasurement>
    <isSupportIndustryThermometryTemperatureRise>
        <!--ro, opt, bool, whether the device supports setting temperature rise compensation in thermal industrial temperature measurement, desc:related URI:
    /ISAPI/Thermal/channels/<ID>/industryThermometry/temperatureRise?format=json-->true
    </isSupportIndustryThermometryTemperatureRise>
    <isSupportThermalDataLinkInfo>
        <!--ro, opt, bool, whether the device supports getting the temperature data link information, desc:related URI: /ISAPI/Thermal/thermalDataLinkInfo?
    format=json; /ISAPI/Thermal/thermalDataLinkStatus?format=json-->true
    </isSupportThermalDataLinkInfo>
    <isSupportTemperatureCompensationCalib>
        <!--ro, opt, bool, whether the device supports temperature compensation, desc:related URI:
    /ISAPI/Thermal/channels/<channelID>/temperatureCompensationCalibration?format=json-->true
    </isSupportTemperatureCompensationCalib>
    <isSupportWebDisplayPixelToPixelParam>
        <!--ro, opt, bool, whether the web needs to display full-screen temperature parameters, desc:related URI:

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/ISAPI/Thermal/channels/<channelID>/thermometry/pixelToPixelParam/capabilities-->true
</isSupportWebDisplayPixelToPixelParam>
</isSupportThermalInspection>
<!--ro, opt, bool, whether the device supports thermal patrol, desc:related URI: /ISAPI/Thermal/inspection/capabilities?format=json-->true
</isSupportThermalInspection>
</isSupportBlackBodyTemperature>
<!--ro, opt, bool, whether the device supports searching for black body center point temperature, desc:related URI:
/ISAPI/Thermal/channels/<channelID>/blackBodyTemperature?format=json-->true
</isSupportBlackBodyTemperature>
</isSupportExportThermalCalibrationFile>
<!--ro, opt, bool, whether the device supports exporting the temperature calibration file, desc:POST /ISAPI/Thermal/ExportThermalCalibrationFile?
format=json-->true
</isSupportExportThermalCalibrationFile>
</isSupportSearchSmokeAndFireDetection>
<!--ro, opt, bool, whether the device supports searching for history records of smoke and fire detection, desc:POST
/ISAPI/Thermal/fireDetection/SearchSmokeAndFireDetection?format=json-->true
</isSupportSearchSmokeAndFireDetection>
</isSupportColumnOven>
<!--ro, opt, bool, whether the device supports column oven parameters, desc:GET/PUT /ISAPI/OPCA/GC/ColumnOven?format=json-->true
</isSupportColumnOven>
</isSupportImportThermalCalibrationFile>
<!--ro, opt, bool, whether the device supports importing the temperature calibration file, desc:POST /ISAPI/Thermal/ImportThermalCalibrationFile?
format=json-->true
</isSupportImportThermalCalibrationFile>
</isSupportThermalCalibrationFileCreatParam>
<!--ro, opt, bool, whether the device supports the parameters of generating temperature calibration file, desc:GET
/ISAPI/Thermal/ThermalCalibrationFileCreatParam?format=json-->true
</isSupportThermalCalibrationFileCreatParam>
</isSupportNormalRuleParams>
<!--ro, opt, bool, whether the device supports the parameters of smoke and fire detection normal rules, desc:GET/PUT
/ISAPI/Thermal/fireDetection/channels/<channelID>/NormalRuleParams?format=json-->true
</isSupportNormalRuleParams>
</isSupportEnvironmentTemperatureMonitorScheduleParams>
<!--ro, opt, bool, whether the device supports the parameters of environment temperature monitor arming schedule, desc:GET/PUT
/ISAPI/Thermal/EnvironmentTemperatureMonitorScheduleParams?format=json-->true
</isSupportEnvironmentTemperatureMonitorScheduleParams>
</isSupportEnvironmentTemperatureMonitorParam>
<!--ro, opt, bool, whether the device supports the parameters of environment temperature monitoring, desc:GET/PUT
/ISAPI/Thermal/EnvironmentTemperatureMonitorParam?format=json-->true
</isSupportEnvironmentTemperatureMonitorParam>
</isSupportEnvironmentTemperatureMonitorTriggerParams>
<!--ro, opt, bool, whether the device supports the parameters of environment temperature monitoring linkage action, desc:GET/PUT
/ISAPI/Thermal/EnvironmentTemperatureMonitorTriggerParams?format=json-->true
</isSupportEnvironmentTemperatureMonitorTriggerParams>
</isSupportSearchAsynThermalInspectionTaskTestPoint>
<!--ro, opt, bool, whether the device supports searching for information of specified asynchronous thermal inspection point, desc:POST
/ISAPI/Thermal/SearchAsynThermalInspectionTaskTestPoint?format=json-->true
</isSupportSearchAsynThermalInspectionTaskTestPoint>
</isSupportSearchAsynThermalInspectionTask>
<!--ro, opt, bool, whether the device supports searching for specified asynchronous thermal inspection task, desc:POST
/ISAPI/Thermal/SearchAsynThermalInspectionTask?format=json-->true
</isSupportSearchAsynThermalInspectionTask>
</isSupportAddAsynThermalInspectionTaskTestPoint>
<!--ro, opt, bool, whether the device supports adding information of specified asynchronous thermal inspection point, desc:POST
/ISAPI/Thermal/AddAsynThermalInspectionTaskTestPoint?format=json-->true
</isSupportAddAsynThermalInspectionTaskTestPoint>
</isSupportAddAsynThermalInspectionTask>
<!--ro, opt, bool, whether the device supports adding asynchronous thermal inspection task, desc:POST /ISAPI/Thermal/AddAsynThermalInspectionTask?
format=json-->true
</isSupportAddAsynThermalInspectionTask>
</isSupportImportAsynThermalInspectionTaskBaseMap>
<!--ro, opt, bool, whether the device supports importing asynchronous thermal inspection task base map, desc:POST
/ISAPI/Thermal/ImportAsynThermalInspectionTaskBaseMap?format=json-->true
</isSupportImportAsynThermalInspectionTaskBaseMap>
</isSupportShutterFreeze>
<!--ro, opt, bool, whether the device supports freezing the shutter, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ShutterFreeze?format=json-->true
</isSupportShutterFreeze>
</ThermalCap>
<WLAlarmCap>
<!--ro, opt, object, wireless alarm capability-->
</isSupportTeleControl>
<!--ro, opt, bool, whether the device supports remote control-->true
</isSupportTeleControl>
</isSupportPIR>
<!--ro, opt, bool, whether the device supports PIR configuration-->true
</isSupportPIR>
</isSupportWLSensors>
<!--ro, opt, bool, N/A-->true
</isSupportWLSensors>
</isSupportCallHelp>
<!--ro, opt, bool, whether the device supports calling for help-->true
</isSupportCallHelp>
</WLSensorsNum>
<!--ro, opt, int, N/A-->1
</WLSensorsNum>
</WLAlarmCap>
</isSupportGIS>
<!--ro, opt, bool, whether the device supports GIS, desc:related URI: /ISAPI/GIS/channels-->true
</isSupportGIS>
</isSupportCompass>
<!--ro, opt, bool, whether the device supports compass configuration, desc:related URI: /ISAPI/Compass/channels/<ID>/capabilities-->true
</isSupportCompass>
</isSupportRoadInfoOverlays>
<!--ro, opt, bool, whether the device supports evaluating the lane information-->true

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<!--ro, opt, bool, whether the device supports overlaying the time information-->true
</isSupportRoadInfoOverlays>
<isSupportFaceCaptureStatistics>
  <!--ro, opt, bool, whether the device supports face capture statistics, desc:related URI: /ISAPI/Intelligent/channels/<ID>/faceCaptureStatistics/search-
->true
  </isSupportFaceCaptureStatistics>
  <isSupportElectronicsEnlarge>
    <!--ro, opt, bool, whether the device supports electronics enlarge-->true
  </isSupportElectronicsEnlarge>
  <isSupportRemoveStorage>
    <!--ro, opt, bool, whether the device supports remote storage-->true
  </isSupportRemoveStorage>
  <isSupportCloud>
    <!--ro, opt, bool, whether the device supports cloud storage-->true
  </isSupportCloud>
  <isSupportRecordHost>
    <!--ro, opt, bool, whether the device supports configuring the education sharing server, desc:related URI: /ISAPI/RecordHost/capabilities-->true
  </isSupportRecordHost>
  <isSupportViewshedOverlay>
    <!--ro, opt, bool, whether the device supports overlaying viewshed camera target, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/viewshedOverlay-->true
  </isSupportViewshedOverlay>
  <isSupportGPSLabelTracking>
    <!--ro, opt, bool, whether the device supports GPS tag tracking, desc:related URI: /ISAPI/GIS/channels/<ID>/GPSLabelTracking?format=json-->true
  </isSupportGPSLabelTracking>
  <isSupportEagleEye>
    <!--ro, opt, bool, whether the device supports the PanoVu series camera-->true
  </isSupportEagleEye>
  <isSupportPanorama>
    <!--ro, opt, bool, whether the device supports panoramic mode-->true
  </isSupportPanorama>
  <isSupportFirmwareVersionInfo>
    <!--ro, opt, bool, whether the device supports displaying the firmware version information-->true
  </isSupportFirmwareVersionInfo>
  <isSupportExternalWirelessServer>
    <!--ro, opt, bool, whether the device supports external wireless servers: true (support)-->true
  </isSupportExternalWirelessServer>
  <isSupportSetupCalibration>
    <!--ro, opt, bool, whether the device supports configuring calibration-->true
  </isSupportSetupCalibration>
  <isSupportGetmutexFuncErrMsg>
    <!--ro, opt, bool, whether the device supports the mutual exclusion information-->true
  </isSupportGetmutexFuncErrMsg>
  <isSupportTokenAuthenticate>
    <!--ro, opt, bool, whether the device supports token authentication-->true
  </isSupportTokenAuthenticate>
  <isSupportStreamDualVCA>
    <!--ro, opt, bool, whether the device supports overlaying intelligent private data on the stream, desc:related URI: in the
/ISAPI/Intelligent/channels/<ID>/intelliResource; corresponding node: <isOverlapIntelli-->true
  </isSupportStreamDualVCA>
  <isSupportLaserSpotManual>
    <!--ro, opt, bool, whether the device supports configuring Laser Light spot-->true
  </isSupportLaserSpotManual>
  <isSupportRTMP>
    <!--ro, opt, bool, whether the device supports RTMP configuration, desc:related URI: /ISAPI/Streaming/channels/<ID>/RTMPCfg/capabilities-->true
  </isSupportRTMP>
  <isSupportTraffic>
    <!--ro, opt, bool, traffic service capability-->true
  </isSupportTraffic>
  <isSupportLaserSpotAdjustment>
    <!--ro, opt, bool, adjust the size of Laser spot, desc:related URI: /ISAPI/Image/channels/<ID>/LaserLight/LaserSpotAdjustment/capabilities-->true
  </isSupportLaserSpotAdjustment>
  <VideoIntercomCap>
    <!--ro, opt, object, video intercom capability, desc:related URI: /ISAPI/VideoIntercom/capabilities-->
    <isSupportSceneManage>
      <!--ro, opt, bool, scene management (the configuration port needs to be switched according to various scenes)-->true
    </isSupportSceneManage>
    <isSupportDeviceId>
      <!--ro, opt, bool, whether the device supports device ID-->true
    </isSupportDeviceId>
    <isSupportOperationTime>
      <!--ro, opt, bool, whether the device supports the operation time-->true
    </isSupportOperationTime>
    <isSupportCallerInfo>
      <!--ro, opt, bool, whether the device supports the information management of the calling persons-->true
    </isSupportCallerInfo>
    <isSupportCallSignal>
      <!--ro, opt, bool, whether the device supports the call signaling interaction, desc:related URI: /ISAPI/VideoIntercom/callSignal?format=json-->true
    </isSupportCallSignal>
    <isSupportCallStatus>
      <!--ro, opt, bool, whether the device supports getting call status, desc:related URI: /ISAPI/VideoIntercom/callStatus?format=json-->true
    </isSupportCallStatus>
    <isSupportCallElevator>
      <!--ro, opt, bool, whether the device supports calling for elevator-->true
    </isSupportCallElevator>
    <isSupportRelatedDeviceAddress>
      <!--ro, opt, bool, whether the device supports linking device address-->true
    </isSupportRelatedDeviceAddress>
    <isSupportCardSectorCheck>
      <!--ro, opt, bool, sector encryption verification-->true
    </isSupportCardSectorCheck>
    <isSupportRemoteOpenDoor>
      <!--ro, opt, bool, whether it supports remote unlocking-->true
    </isSupportRemoteOpenDoor>

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<isSupportKeyCfg>
  <!--ro, opt, bool, configuration for pressing button to call for resident-->true
</isSupportKeyCfg>
<isSupportWorkModeCfg>
  <!--ro, opt, bool, whether the device supports configuring the working mode, desc:related URI: URL/ISAPI/VideoIntercom/workMode/capabilities-->true
</isSupportWorkModeCfg>
<isSupportSystemSwitchCfg>
  <!--ro, opt, bool, configuring the switch of video intercom system-->true
</isSupportSystemSwitchCfg>
<isSupportPrivilegePasswordStatus>
  <!--ro, opt, bool, getting the status of the device's permission password-->true
</isSupportPrivilegePasswordStatus>
<isSupportSmartHomeRoomsCfg>
  <!--ro, opt, bool, whether the device supports configuring room information-->true
</isSupportSmartHomeRoomsCfg>
<isSupportSmartHomeDevices>
  <!--ro, opt, bool, whether the device supports intelligent devices management-->true
</isSupportSmartHomeDevices>
<isSupportScenesCfg>
  <!--ro, opt, bool, whether the device supports scene configuration (adding, deleting, editing and searching for)-->true
</isSupportScenesCfg>
<isSupportNameModify>
  <!--ro, opt, bool, whether the device supports editing the zone name-->true
</isSupportNameModify>
<isSupportElevatorControlCfg>
  <!--ro, opt, bool, whether the device supports elevator control-->true
</isSupportElevatorControlCfg>
<isSupportPrivilegePassword>
  <!--ro, opt, bool, whether the device supports configuring passwords for different scenes-->true
</isSupportPrivilegePassword>
<isSupportSubModules>
  <!--ro, opt, bool, whether the device supports sub-module management-->true
</isSupportSubModules>
<isSupportIssueCards>
  <!--ro, opt, bool, whether the device supports configuring the card issuing function of the master card-->true
</isSupportIssueCards>
<isSupportSendCardCfg>
  <!--ro, opt, bool, whether the door station supports configuring card issuing function-->true
</isSupportSendCardCfg>
<isSupportRingBackTone>
  <!--ro, opt, bool, whether the device supports importing the ring back tone-->true
</isSupportRingBackTone>
<isSupportCallPriority>
  <!--ro, opt, bool, whether the device supports calling permission-->true
</isSupportCallPriority>
<isSupportPhoneCfg>
  <!--ro, opt, bool, whether the device supports configuring resident phones-->true
</isSupportPhoneCfg>
<isSupportWatchPoint>
  <!--ro, opt, bool, whether the device supports watch point, desc:/ISAPI/VideoIntercom/WatchPoint-->true
</isSupportWatchPoint>
<isSupportVoiceConfiguration>
  <!--ro, opt, bool, whether the device supports the configuration of enabling the voice recognition and prompt-->true
</isSupportVoiceConfiguration>
<isSupportSubModulesCfg>
  <!--ro, opt, bool, whether the device supports configuring editing the sub module parameters-->true
</isSupportSubModulesCfg>
<isSupportDisplayRegion>
  <!--ro, opt, bool, whether the device supports configuring the area to be displayed-->true
</isSupportDisplayRegion>
<isSupportUpdateData>
  <!--ro, opt, bool, whether the device supports importing upgrade files-->true
</isSupportUpdateData>
<isSupportUpdateStrategy>
  <!--ro, opt, bool, whether the device supports upgrade configuration-->true
</isSupportUpdateStrategy>
<isSupportMetaInfo>
  <!--ro, opt, bool, whether the device supports getting the information of upgrade files-->true
</isSupportMetaInfo>
<isSupportStreamAbility>
  <!--ro, opt, bool, whether the device supports IPC management-->true
</isSupportStreamAbility>
<platform opt="R0,R1,S29,N10,N8,N4">
  <!--ro, opt, enum, platform type, subType:string, attr:opt{req, string}, desc:"R0", "R1"; it will be returned if the device supports-->R0
</platform>
<isSupportSecurityMode>
  <!--ro, opt, bool, whether the device supports switching the version of security protocols-->true
</isSupportSecurityMode>
<isSupportDeviceCommunication>
  <!--ro, opt, bool, whether the device supports configuring the parameters of device group-net communication-->true
</isSupportDeviceCommunication>
<isSupportAutoSync>
  <!--ro, opt, bool, whether the device supports automatically synchronous parameters configuration-->true
</isSupportAutoSync>
<isSupportDifferentKeystrokes>
  <!--ro, opt, bool, whether the device supports distinguishing long press and short press-->true
</isSupportDifferentKeystrokes>
<isSupportKeyCfgWeekPlan>
  <!--ro, opt, bool, the week schedule configuration for pressing to call-->true
</isSupportKeyCfgWeekPlan>
<isSupportKeyCfgHolidayPlan>
  <!--ro, opt, bool, the holiday schedule configuration for pressing to call-->true
</isSupportKeyCfgHolidayPlan>

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<isSupportKeyCfgHolidayGroup>
  <!--ro, opt, bool, the holiday group configuration for pressing to call-->true
</isSupportKeyCfgHolidayGroup>
<isSupportKeyCfgPlanTemplate>
  <!--ro, opt, bool, the schedule template configuration for pressing to call-->true
</isSupportKeyCfgPlanTemplate>
<isSupportCallCfg>
  <!--ro, opt, bool, device call configuration-->true
</isSupportCallCfg>
<isSupportAccessParams>
  <!--ro, opt, bool, the application access parameters-->true
</isSupportAccessParams>
<isSupportTransmitRequestEventConfirm>
  <!--ro, opt, bool, response to the transparent transmission request-->true
</isSupportTransmitRequestEventConfirm>
<isSupportIncomingCallLinkage>
  <!--ro, opt, bool, whether the device supports configuring incoming calls Linkage-->true
</isSupportIncomingCallLinkage>
<isSupportAdbControl>
  <!--ro, opt, bool, whether the device supports remotely enabling and disabling ADB-->true
</isSupportAdbControl>
<isSupportProtocolFileManager>
  <!--ro, opt, bool, whether the device supports the management of the protocol notice files, desc:/ISAPI/VideoIntercom/protocolFileManager/capabilities?
format=json-->true
</isSupportProtocolFileManager>
<isSupportStreamSwitch>
  <!--ro, opt, bool, whether the device supports stream switch, desc:/ISAPI/VideoIntercom/streamSwitch/capabilities?format=json-->true
</isSupportStreamSwitch>
<isSupportKeywordAudioRecognition>
  <!--ro, opt, bool, whether the device supports keyword voice recognition, desc:/ISAPI/VideoIntercom/keywordAudioRecognition/capabilities?format=json--
>true
</isSupportKeywordAudioRecognition>
<isSupportOneWaySpeaking>
  <!--ro, opt, bool, whether the device supports speaking Loud, desc:true (support), this node is not returned (not support)-->true
</isSupportOneWaySpeaking>
<isSupportPositionInfo>
  <!--ro, opt, bool, whether the device supports configuring the device location information, desc:/ISAPI/VideoIntercom/positionInfo?format=json-->true
</isSupportPositionInfo>
<isSupportElevatorControlDisplay>
  <!--ro, opt, bool, whether the device supports configuring displaying elevator control, desc:/ISAPI/VideoIntercom/elevatorControlDisplay?format=json--
>true
</isSupportElevatorControlDisplay>
<isSupportRingManage>
  <!--ro, opt, bool, whether the device supports ring tone management, desc:related URI: /ISAPI/VideoIntercom/ringTone?format=json;
/ISAPI/VideoIntercom/ringTone/<ringID>?format=json-->true
</isSupportRingManage>
<isSupportPasswordAuthentication>
  <!--ro, opt, bool, whether the device supports verifying the video intercom password, desc:related URI:
/ISAPI/VideoIntercom/passwordAuthentication/capabilities-->true
</isSupportPasswordAuthentication>
<isSupportAlarmControlByPhone>
  <!--ro, opt, bool, whether the device supports phone arming control, desc:related URI: /ISAPI/SecurityCP/AlarmControlByPhone/capabilities-->true
</isSupportAlarmControlByPhone>
<isSupportZoneCfgByScene>
  <!--ro, opt, bool, whether the device supports configuring the zones parameter by scene, desc:related URI:
/ISAPI/SecurityCP/BasicParam/sceneZoneCfg/capabilities-->true
</isSupportZoneCfgByScene>
<isSupportAppKeyConfiguration>
  <!--ro, opt, bool, whether the device supports configuring APPKey, desc:related URI: /ISAPI/VideoIntercom/AppKeyConfiguration/capabilities-->true
</isSupportAppKeyConfiguration>
<isSupportIntercomProtocolChange>
  <!--ro, opt, bool, whether the device supports configuring the switch of device VoIP protocols, desc:related URI:
/ISAPI/VideoIntercom/IntercomProtocolType/capabilities-->true
</isSupportIntercomProtocolChange>
<isSupportGetAnnouncementMessage>
  <!--ro, opt, bool, whether the device supports getting announcement message, desc:related URI: /ISAPI/VideoIntercom/AnnouncementMessage/capabilities--
>true
</isSupportGetAnnouncementMessage>
<isSupportGetDeployInfo>
  <!--ro, opt, bool, whether the device supports getting arming information, desc:related URI: /ISAPI/AccessControl/DeployInfo/capabilities-->true
</isSupportGetDeployInfo>
<detectorType opt="panicButton,magneticContact,smokeDetector,activeInfraredDetector,passiveInfraredDetector,combustibleGasDetector,doorbell">
  <!--ro, opt, string, detector types supported by the device, attr:opt(req, string), desc:"panicButton" (panic button), "magneticContact" (magnetic
contact), "smokeDetector" (smoke detector), "activeInfraredDetector" (active infrared detector), "passiveInfraredDetector" (passive infrared detector),
"combustibleGasDetector" (combustible gas detector)-->test
</detectorType>
<isSupportBroadcastRealtime>
  <!--ro, opt, bool, whether the device supports real-time audio broadcast-->true
</isSupportBroadcastRealtime>
<isSupportOrganizationNodes>
  <!--ro, opt, bool, whether the device supports organization architecture management-->true
</isSupportOrganizationNodes>
<isSupportAudioPlayStatus>
  <!--ro, opt, bool, whether the device supports uploading the audio playing status-->true
</isSupportAudioPlayStatus>
<isSupportDeviceIdAnalysisRuleCfg>
  <!--ro, opt, bool, whether the device supports configuring the analysis rule of the device No. (related URI:
/ISAPI/VideoIntercom/AnalysisRuleCfg/capabilities?format=json)-->true
</isSupportDeviceIdAnalysisRuleCfg>
<isSupportOperationTimeByChannel>
  <!--ro, opt, bool, whether the device supports configuring the operation time by channel-->true
</isSupportOperationTimeByChannel>
<isSupportCallAuthorization>
  <!--ro, opt, bool, whether the device supports configuring a permission for incoming calls, desc:related URI:

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/ISAPI/VideoIntercom/CallAuthorization/capabilities?format=json-->true
</isSupportCallAuthorization>
<isSupportRecordControl>
  <!--ro, opt, bool, whether the device supports audio/video control, desc:related URI: /ISAPI/VideoIntercom/recordControl/capabilities?format=json-->true
>true
  </isSupportRecordControl>
  <isSupportTrusteeship>
    <!--ro, opt, bool, whether the device supports configuring trusteeship, desc:related URI: /ISAPI/VideoIntercom/trusteeship/capabilities?format=json-->true
>true
    </isSupportTrusteeship>
    <isSupportAnalogIndoorStation>
      <!--ro, opt, bool, whether the device supports the management of the analog indoor station, desc:related URI:
/ISAPI/VideoIntercom/analogIndoorStation/capabilities?format=json-->true
    </isSupportAnalogIndoorStation>
    <isSupportImportPartitions>
      <!--ro, opt, bool, whether the device supports importing the terminal List, desc:(partition List for the paging microphone devices distributed by the
platform) (Related URI: /ISAPI/VideoIntercom/ImportTerminals/capabilities?format=json)-->true
    </isSupportImportPartitions>
    <isSupportBroadcastPlan>
      <!--ro, opt, bool, whether the device supports full application of schedule broadcast, desc:related URI:
ISAPI/VideoIntercom/broadcast/plan/capabilities?format=json-->true
    </isSupportBroadcastPlan>
    <isSupportBroadcastPlanSearch>
      <!--ro, opt, bool, whether the device supports searching for full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/planSearch?
format=json-->true
    </isSupportBroadcastPlanSearch>
    <isSupportBroadcastPlanPause>
      <!--ro, opt, bool, whether the device supports pausing full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/pause?
format=json-->true
    </isSupportBroadcastPlanPause>
    <isSupportBroadcastPlanResume>
      <!--ro, opt, bool, whether the device supports restoring full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/resume?
format=json-->true
    </isSupportBroadcastPlanResume>
    <isSupportBroadcastPlanTest>
      <!--ro, opt, bool, whether the device supports the test run of full schedule broadcast, desc:related URI: /ISAPI/VideoIntercom/broadcast/plan/test?
format=json-->true
    </isSupportBroadcastPlanTest>
    <isSupportBroadcastPlanEx>
      <!--ro, opt, bool, whether the device supports incremental application of schedule broadcast, desc:related URI:
/ISAPI/VideoIntercom/broadcast/planEx/capabilities?format=json-->true
    </isSupportBroadcastPlanEx>
    <isSupportBroadcastPlanExSearch>
      <!--ro, opt, bool, whether the device supports searching for incremental schedule broadcast, desc:related URI:
/ISAPI/VideoIntercom/broadcast/planExSearch?format=json-->true
    </isSupportBroadcastPlanExSearch>
    <isSupportBroadcastPlanModify>
      <!--ro, opt, bool, whether the device supports editing a single schedule broadcast, desc:it is valid only during incremental application-->true
    </isSupportBroadcastPlanModify>
    <isSupportBroadcastPlanDelete>
      <!--ro, opt, bool, whether the device supports deleting a single schedule broadcast, desc:it is valid only during incremental application-->true
    </isSupportBroadcastPlanDelete>
    <isSupportAudioInPlayCfg>
      <!--ro, opt, bool, whether the device supports the configuration of playing the audio input sources of the broadcast, desc:URI for getting the
capability: /ISAPI/VideoIntercom/broadcast/AudioInPlayCfg/capabilities?format=json-->true
    </isSupportAudioInPlayCfg>
    <isSupportSearchWardBedInfo>
      <!--ro, opt, bool, whether the device supports searching for ward bed information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/searchWardBedInfo/capabilities?format=json. It is used for the main station to search for whether the bed connected to the sub machine
in the ward is already occupied-->true
    </isSupportSearchWardBedInfo>
    <isSupportPatient>
      <!--ro, opt, bool, whether the device supports applying patient information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/medical/patient/capabilities?format=json-->true
    </isSupportPatient>
    <isSupportIntroduction>
      <!--ro, opt, bool, whether the device supports applying the medical introduction, desc:URI for getting the capability:
/ISAPI/VideoIntercom/medical/introduction/capabilities?format=json-->true
    </isSupportIntroduction>
    <isSupportConstantVoltage>
      <!--ro, opt, bool, whether the device supports constant voltage broadcast, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/ConstantVoltage/capabilities?format=json-->true
    </isSupportConstantVoltage>
    <isSupportRecordMode>
      <!--ro, opt, bool, whether it supports configuring recording mode, desc:URI for getting the capability: /ISAPI/VideoIntercom/RecordMode/capabilities?
format=json-->true
    </isSupportRecordMode>
    <isSupportCallForwarding>
      <!--ro, opt, bool, whether the device supports active call transfer, desc:URI for getting the capability:
/ISAPI/VideoIntercom/callForwarding/capabilities?format=json-->true
    </isSupportCallForwarding>
    <isSupportMediaLibrary>
      <!--ro, opt, bool, whether the device supports applying media library information, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/MediaLibrary/capabilities?format=json-->true
    </isSupportMediaLibrary>
    <isSupportBroadcastControl>
      <!--ro, opt, bool, whether the device supports configuring broadcast control parameters, desc:URI for getting the capability:
/ISAPI/VideoIntercom/broadcast/control/capabilities?format=json-->true
    </isSupportBroadcastControl>
    <isSupportGetImportPartitions>
      <!--ro, opt, bool, whether the device supports getting partition List, desc:POST /ISAPI/VideoIntercom/broadcast/GetImportPartitions?format=json-->true
    </isSupportGetImportPartitions>
    <isSupportSearchMediaLibrary>

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    <!--ro, opt, bool, whether the device supports searching media library, desc:POST /ISAPI/VideoIntercom/broadcast/SearchMediaLibrary?format=json-->true
  </isSupportSearchMediaLibrary>
  <isSupportImportPartitionsBatch>
    <!--ro, opt, bool, whether the device supports batch importing of partitions, desc:PUT /ISAPI/VideoIntercom/broadcast/ImportPartitionsBatch?
format=json-->true
  </isSupportImportPartitionsBatch>
  <isSupportMediaLibraryBatch>
    <!--ro, opt, bool, whether the device supports batch applying of media library information, desc:PUT /ISAPI/VideoIntercom/broadcast/MediaLibraryBatch?
format=json-->true
  </isSupportMediaLibraryBatch>
  <isSupportPlanScheme>
    <!--ro, opt, bool, whether the device supports broadcast schedules, desc:GET /ISAPI/VideoIntercom/broadcast/PlanScheme/capabilities?format=json-->true
  </isSupportPlanScheme>
  <isSupportRealTimeInsertion>
    <!--ro, opt, bool, whether the device supports real-time cut-in, desc:POST /ISAPI/VideoIntercom/broadcast/RealTimeInsertion?format=json-->true
  </isSupportRealTimeInsertion>
</VideoIntercomCap>
<SecurityCPCapabilities>
  <!--ro, opt, object, [TBC], desc:related URI: /ISAPI/SecurityCP/capabilities?format=json-->true
</SecurityCPCapabilities>
<isSupportSafetyCabin>
  <!--ro, opt, bool, safety cabinet controller, desc:related URI: /ISAPI/SafetyCabin/capabilities-->true
</isSupportSafetyCabin>
<isSupportAcUpdate>
  <!--ro, opt, bool, whether the device supports peripheral module upgrade (true, support), desc:related URI: /ISAPI/System/AcsUpdate/capabilities-->true
</isSupportAcUpdate>
<isSupportSecurityCP>
  <!--ro, opt, bool, N/A-->true
</isSupportSecurityCP>
<isSupportPEA>
  <!--ro, opt, bool, the host capability of one-push alarm-->true
</isSupportPEA>
<isSupportCurrentLock>
  <!--ro, opt, bool, whether the device supports the configuration of locking current-->true
</isSupportCurrentLock>
<isSupportGuardAgainstTheft>
  <!--ro, opt, bool, whether the device supports configuring device guard against theft, desc:related URI: /ISAPI/System/guardAgainstTheft/capabilities--
>true
  </isSupportGuardAgainstTheft>
  <isSupportPicInfoOverlap>
    <!--ro, opt, bool, whether the device supports overlying pictures and texts-->true
  </isSupportPicInfoOverlap>
  <isSupportPlay>
    <!--ro, opt, bool, shielded live view-->true
  </isSupportPlay>
  <isSupportPlayback>
    <!--ro, opt, bool, shielded playback-->true
  </isSupportPlayback>
<UHFRFIDReader>
  <!--ro, opt, object, capability supported by the RFID reader-->
  <isSupportBasicInformation>
    <!--ro, opt, bool, whether the device supports the basic parameters of the reader-->true
  </isSupportBasicInformation>
  <isSupportHardDiskStorageTest>
    <!--ro, opt, bool, whether the device supports disk storage performance test of the reader-->true
  </isSupportHardDiskStorageTest>
  <isSupportCarReaderConfig>
    <!--ro, opt, bool, whether the device supports configuring the parameters of the electronic license plate reader, desc:related URI:
/ISAPI/Traffic/UHFRFIDReader/carReaderConfig/capabilities?format=json-->true
  </isSupportCarReaderConfig>
</UHFRFIDReader>
<isSupportMasterSlaveConfig>
  <!--ro, opt, bool, whether the device supports configuring main and sub device, desc:related URI: /ISAPI/System/masterSlaveConfig?format=json-->true
</isSupportMasterSlaveConfig>
<isSupportIntelligentStructureAnalysis>
  <!--ro, opt, bool, whether the device supports intelligent structuralization analysis, desc:related URI:
/ISAPI/Intelligent/structureAnalysis/capabilities-->true
  </isSupportIntelligentStructureAnalysis>
  <isSupportIntelligentAnalysisEngines>
    <!--ro, opt, bool, whether the device supports configuring the intelligent engine, desc:related URI: /ISAPI/Intelligent/analysisEngines/capabilities--
>true
  </isSupportIntelligentAnalysisEngines>
  <PreviewDisplayNum>
    <!--ro, opt, int, number of the browser live view window-->1
  </PreviewDisplayNum>
  <isSupportBoard opt="true,false">
    <!--ro, opt, bool, whether the device supports related sub-board protocols, attr:opt{opt, string}-->true
  </isSupportBoard>
<ResourceSwitch>
  <!--ro, opt, object, configuration of resource conversion-->
  <workMode opt="4KPreview,educationRecord">
    <!--ro, req, string, the working mode: education record mode, attr:opt{opt, string}-->test
  </workMode>
</ResourceSwitch>
<isSupportCustomStream>
  <!--ro, opt, bool, whether the device supports custom stream, desc:related URI: /ISAPI/Streaming/channels/<ID>/customStream/capabilities-->true
</isSupportCustomStream>
<isSupportTriggerCapCheck>
  <!--ro, opt, bool, it indicates that the capability of the URL (/ISAPI/Event/triggersCap) is already verified-->true
</isSupportTriggerCapCheck>
<isSupportChannelEventCap>
  <!--ro, opt, bool, whether the device supports getting the event capability by channel, desc:related URI: /ISAPI/Event/channels/<ID>/capabilities-->true
</isSupportChannelEventCap>
<SensorCan>

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<!--ro, opt, object, capability of configuring sensor parameters, desc:related URI: /ISAPI/System/Sensor/capabilities-->
<SensorNums>
  <!--ro, opt, int, the maximum number of sensors that can be added-->1
</SensorNums>
<isSupportSensorPortConfiguration>
  <!--ro, opt, bool, supports configuring the detection parameters-->true
</isSupportSensorPortConfiguration>
<isSupportSensorPortControl>
  <!--ro, opt, bool, supports detection control-->true
</isSupportSensorPortControl>
<isSupportAcquireSensorStatus>
  <!--ro, opt, bool, supports getting the real-time status of the sensor-->true
</isSupportAcquireSensorStatus>
<isSupportSearchSensorData>
  <!--ro, opt, bool, supports searching the sensor data-->true
</isSupportSearchSensorData>
</SensorCap>
<isSupportEncryption>
  <!--ro, opt, bool, stream encryption capability-->true
</isSupportEncryption>
<isSupportTrial>
  <!--ro, opt, bool, whether the device supports configuring inquest device-->true
</isSupportTrial>
<isSupportSignInstrument>
  <!--ro, opt, bool, the capability of physical sign monitor-->true
</isSupportSignInstrument>
<isSupportLocate>
  <!--ro, opt, bool, whether the device supports location configuration-->true
</isSupportLocate>
<isSupportVibration>
  <!--ro, opt, bool, whether the device supports configuring vibration parameters-->true
</isSupportVibration>
<isSupportMixedTargetDetection>
  <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
</isSupportMixedTargetDetection>
<SDCardCtrlCap>
  <!--ro, opt, object, the remote control capability of the memory card-->
  <isSupportIOCtrl>
    <!--ro, opt, bool, whether the device supports the capability of control and command upgrade-->true
  </isSupportIOCtrl>
  <isSupportReadCardStatus>
    <!--ro, opt, bool, whether the device supports the capability of card recognition operation-->true
  </isSupportReadCardStatus>
  <isSupportStartUpdate>
    <!--ro, opt, bool, whether the device supports the capability of upgrade start control-->true
  </isSupportStartUpdate>
  <isSupportEndUpdate>
    <!--ro, opt, bool, whether the device supports the capability of upgrade end control-->true
  </isSupportEndUpdate>
  <isSupportPowerCtrl>
    <!--ro, opt, bool, whether the device supports the capability of power on and off control-->true
  </isSupportPowerCtrl>
</SDCardCtrlCap>
<isSupportMultiChannelCalibration>
  <!--ro, opt, bool, whether the device supports multi-channel calibration-->true
</isSupportMultiChannelCalibration>
<isSupportElevatorFloorCalibration>
  <!--ro, opt, bool, whether the device supports configuring floor display-->true
</isSupportElevatorFloorCalibration>
<isSupportPictureCaptureComparison>
  <!--ro, opt, bool, whether the device supports 1V1 comparison of the pictures in the library and from camera-->true
</isSupportPictureCaptureComparison>
<isSupportGPSCalibration>
  <!--ro, opt, bool, whether the device supports the GPS calibration function-->true
</isSupportGPSCalibration>
<isSupportCityManagement>
  <!--ro, opt, bool, whether the device supports city management detection event-->true
</isSupportCityManagement>
<isSupportBVCorrect>
  <!--ro, opt, bool, whether the device supports configuring the parameters of camera correction-->true
</isSupportBVCorrect>
<isSupportStreamingCodingModeSwitch>
  <!--ro, opt, bool, encoding output mode (true-support), desc:related URI: /ISAPI/System/streamingCoding/mode/capabilities?format=json-->true
</isSupportStreamingCodingModeSwitch>
<supportImageChannel opt="1,2,3,4">
  <!--ro, opt, string, whether the device supports configuring the image channel, attr:opt(req, string),
  desc:the configuration of actual image channel supported by the device (URI for picture parameters: /ISAPI/Image/channels/<channelID>), if this node is not
  returned, the image channel is the same with the encoding channel (/ISAPI/Streaming/channels) by default
  this node is mainly for the situation when the image channel is inconsistent with the actual stream channel. For example, some device, which altogether
  has four sensors and they are all spliced into one encoding channel for output. Every sensor supports configuring image parameters; by default, the image
  channel 1 is only valid for the images of sensor1-->test
</supportImageChannel>
<isSupportPanoramaCalibration>
  <!--ro, opt, bool, whether the device supports configuring the calibration of the panoramic camera-->true
</isSupportPanoramaCalibration>
<isSupportPanoramaMode>
  <!--ro, opt, bool, whether the device supports the mode switch of the panoramic camera-->true
</isSupportPanoramaMode>
<supportSnapshotChannel opt="1">
  <!--ro, opt, string, the capability supported by the capture schedule channel, attr:opt(opt, string)-->test
</supportSnapshotChannel>
<isSupportPanoramaCorrections>
  <!--ro, opt, bool, whether the device supports configuring the correction algorithm parameters-->true

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</isSupportPanoramaCorrections>
<isSupportActiveMulticast>
  <!--ro, opt, bool, whether the device supports multicast configuration (/ISAPI/Streaming/channels/<ID>/capabilities)-->true
</isSupportActiveMulticast>
<isSupportChannelEventListCap>
  <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
</isSupportChannelEventListCap>
<VCAResourceChannelsCap>
  <!--ro, opt, array, subType:object-->
  <ChannelsList>
    <!--ro, opt, object-->
    <channelsID>
      <!--ro, req, int-->1
    </channelsID>
  </ChannelsList>
</VCAResourceChannelsCap>
<isSupportTimerVCAResource>
  <!--ro, opt, bool, whether the device supports scheduled switch of intelligent resources-->true
</isSupportTimerVCAResource>
<MixedTargetDetection>
  <!--ro, opt, object, the capability of recognizing specified target among mixed targets supported by the device-->
  <isSupportFaceRecognition>
    <!--ro, opt, bool, whether the device supports facial recognition-->true
  </isSupportFaceRecognition>
  <isSupportHumanRecognition>
    <!--ro, opt, bool, whether the device supports human body recognition-->true
  </isSupportHumanRecognition>
  <isSupportVehicleRecognition>
    <!--ro, opt, bool, whether the device supports vehicle recognition-->true
  </isSupportVehicleRecognition>
</MixedTargetDetection>
<isSupportDiscoveryMode>
  <!--ro, opt, bool, whether the device supports discovery mode, desc:related URI: /ISAPI/System/discoveryMode/capabilities-->true
</isSupportDiscoveryMode>
<isSupportDumpData>
  <!--ro, opt, bool, whether the device supports exporting the Dump file-->true
</isSupportDumpData>
<isSupportCameraAccessConfiguration>
  <!--ro, opt, bool, whether the device supports the camera access configuration, desc:related URI: /ISAPI/System/cameraAccess/configuration/capabilities?
format=json-->true
</isSupportCameraAccessConfiguration>
<isSupportIntelligentSearch>
  <!--ro, opt, bool, whether the device supports intelligent search, desc:related URI: /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json--
>true
</isSupportIntelligentSearch>
<isSupportAutoMaintenance>
  <!--ro, opt, bool, whether the device supports automatic maintenance, desc:related URI: /ISAPI/System/autoMaintenance/capabilities?format=json-->true
</isSupportAutoMaintenance>
<isSupportRtspOverHTTPS>
  <!--ro, opt, bool, supports tspOverHTTPS-->true
</isSupportRtspOverHTTPS>
<isSupportIntersectionAnalysis>
  <!--ro, opt, bool, whether the device supports the intersection analysis, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/intersectionAnalysis/capabilities?format=json-->true
</isSupportIntersectionAnalysis>
<isSupportAudioInteract>
  <!--ro, opt, bool, whether the device supports audio interaction, desc:related URI: /ISAPI/Streaming/interact/capabilities?format=json-->true
</isSupportAudioInteract>
<isSupportChannelFullEventListCap>
  <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
</isSupportChannelFullEventListCap>
<isSupportAUXInfoCap>
  <!--ro, opt, bool, it indicates that whether the device supports getting the capability of getting the attributes of device's all channels-->true
</isSupportAUXInfoCap>
<supportPanoramaChannel opt="2">
  <!--ro, opt, string, the panoramic channels supported by the device, attr:opt{opt, string}-->test
</supportPanoramaChannel>
<regionClip>
  <!--ro, opt, object, the channels and streams that supports area cropping-->
  <channels opt="101">
    <!--ro, opt, string, the channels and streams that supports area cropping, attr:opt{opt, string}, desc:101: channel No.*100+stream type-->test
  </channels>
</regionClip>
<AIDEventSupport
opt="abandonedObject,pedestrian,congestion,roadBlock,construction,trafficAccident,fogDetection,wrongDirection,illegalParking,SSharpDriving,lowSpeed,dragRaci
ng,obstacle,vehNoYieldPedest,illegalMannedVeh,illegalMannedNonMotorVeh,umbrellaTentInstall,nonMotorVehOnVehLane,wearingNoHelmet,pedestRedLightRunning,pedest
OnNonMotorVehLane,pedestOnVehLane,conflagration,smoke,VehicleSpeedDrop,lowVisibility,animalsOccupy,targetSpeedDetection,emergencyLane,gasser,changeLaneConti
nuously">
  <!--ro, opt, string, supported AID event, attr:opt{opt, string}-->test
</AIDEventSupport>
<TFSEventSupport opt="illegalParking
,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,notSlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,chang
eLaneContinuously,SSharpDriving,largeVehicleOccupyLine,jamCrossLine,obstacle,blackSmokeVehicle,turnRightStop,occupyDedicatedLane,notDriveInDedicatedLane,TFS
_nonZipperMerge,overSpeed,emergencyLane,gasser">
  <!--ro, opt, string, supported TFS, attr:opt{opt, string}-->test
</TFSEventSupport>
<isVehicleStatisticsSupport>
  <!--ro, opt, bool, whether the device supports configuring the parameters for traffic data statistics-->true
</isVehicleStatisticsSupport>
<isSupportElevatorFloorName>
  <!--ro, opt, bool, whether the device supports configuring floor custom name-->true
</isSupportElevatorFloorName>
<isSupportVehicleFaceRecognition>
  <!--ro, opt, bool, whether the device supports checkpoint vehicle recognition, desc:background: the traffic devices support "facial recognition"

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<!--ro, opt, bool, whether the device supports checkpoint vehicle recognition, desc:background, the traffic devices support facial recognition
(driver), however, the corresponding node is not added to the device to notify the integration of this capability-->true
</isSupportVehicleFaceRecognition>
<VQDEventSupport opt="luma,chroma,snow,streak,freeze,siglose,clarity,jitter,block,flowers,noise,ghost,purple,ICR,protectiveFilm,blur,contrast">
  <!--ro, opt, string, VQD event List, attr:opt{opt, string}, desc:"Luma" (brightness exception); "Chroma" (image color cast); "Snow" (snow noise);
  "Streak" (strip noise); "Freeze" (frozen screen); "Siglose" (signal loss); "Clarity" (resolution exception); "Jitter" (image flutter); "Block"
  (video tampering); "Flowers" (blurred screen); "Noise" (image noise); "Ghost" (ghost image); "Purple" (purple fringing); "ICR" (ICP); "protectiveFilm"
  (protective film unremoved)-->test
</VQDEventSupport>
<IOTCap>
  <!--ro, opt, object, supports searching for the device status list by page, desc:not returned (not support, get the device status list via GET)-->
  <supportChannelNum opt="5000">
    <!--ro, opt, int, the number of supported IOT device channels, attr:opt{opt, string}-->1
  </supportChannelNum>
  <startChannelNo>
    <!--ro, opt, int, the channel No. of start IOT device-->1
  </startChannelNo>
  <isSupportLinkageChannelsSearch>
    <!--ro, opt, bool, true (support), desc:related URI: (POST /ISAPI/System/IOT/LinkageChannels?format=json)-->true
  </isSupportLinkageChannelsSearch>
  <isSupportIOTChannelCap>
    <!--ro, opt, bool, whether the device supports the channel configuration capability, desc:it returns true for support, related URI: (GET
    /ISAPI/System/IOT/channels/<IoTChannelID>/capabilities?format=json)-->true
  </isSupportIOTChannelCap>
  <satusSearchTaskCap>
    <!--ro, opt, object, task search capability-->
    <maxResults>
      <!--ro, opt, int, the maximum number of returned results-->1
    </maxResults>
    <filterType opt="all,online,offline,updateFailed,notPartitioned">
      <!--ro, opt, enum, filtering type, subType:string, attr:opt{req, string}, desc:all, online, offline, updateFailed (upgrading failed)-->all
    </filterType>
    <deviceStatus opt="calling,noCall,medicationChangePrompt,alarming,offline,free,callWaiting">
      <!--ro, opt, string, the value range for the device status search condition, range:[0,32], attr:opt{req, string}, desc:"calling" (calling), "noCall"
      (non-calling status), "medicationChangePrompt", "alarming", "offline", "free"; if multiple conditions are specified, their relationship is "and" by default-
      >calling
    </deviceStatus>
    <logicRelationship opt="and,or">
      <!--ro, opt, string, the value range for the Logic relationship of device status search condition, range:[0,16], attr:opt{req, string}, desc:and-
      the relationship between them is "and" (by default), or- the relationship between them is "or"; it is valid when deviceStatus exists-->and
    </logicRelationship>
    <alarmType opt="SOS,toilet">
      <!--ro, opt, string, value range of alarm type, range:[0,16], attr:opt{req, string}, desc:SOS-SOS call, toilet-toilet alarm. It is valid when the
      deviceStatus contains Alarming-->test
    </alarmType>
    <callWaitingType opt="normal,cisit">
      <!--ro, opt, string, alarm type, attr:opt{req, string}, desc:normal-normal call waiting, visit-visit call waiting. It is valid when the deviceStatus
      contains callWaiting-->test
    </callWaitingType>
  </satusSearchTaskCap>
  <channelConfigList>
    <!--ro, opt, array, capability List returned according to unitType, subType:object-->
    <channelConfig>
      <!--ro, opt, object, device type capability-->
      <unitType>
        <!--ro, opt, enum, unit type, subType:string, desc:"indoor" (indoor station), "villa" (villa door station), "confirm" (doorphone), "outdoor" (door
        station), "fence" (outer door station), "doorbell" (doorbell), "manage" (main station), "acs" (access control devices), "wardStation", "bedheadExtension",
        "bedsideExtension", "terminal", "netAudio" (network sound box), "analogIndoor" (analog indoor station), "decoder" (decoder), "networkFireAlarmMatrix",
        "amplifier" (network amplifier), "pagingMicrophone" (SIP phone)-->outdoor
      </unitType>
      <floorNumber min="1" max="999">
        <!--ro, opt, int, floor No., attr:min{req, int},max{req, int}-->0
      </floorNumber>
      <roomNumber min="0" max="999">
        <!--ro, opt, int, room No., attr:min{req, int},max{req, int}-->0
      </roomNumber>
      <isSupportSlaveDevice>
        <!--ro, opt, bool, whether the device supports sub-devices-->true
      </isSupportSlaveDevice>
      <isSupportZeroFloorNumber>
        <!--ro, opt, bool, whether the zero floor is supported, desc:it is used together with node floorNumber-->true
      </isSupportZeroFloorNumber>
      <periodNumber min="1" max="999">
        <!--ro, opt, int, community number, attr:min{req, int},max{req, int}-->0
      </periodNumber>
      <buildingNumber min="1" max="999">
        <!--ro, opt, int, building number, attr:min{req, int},max{req, int}-->0
      </buildingNumber>
      <unitNumber min="0" max="999">
        <!--ro, opt, int, unit number, attr:min{req, int},max{req, int}-->0
      </unitNumber>
      <deviceIndex min="1" max="32">
        <!--ro, opt, string, device No. Length range, attr:min{req, int},max{req, int}-->test
      </deviceIndex>
      <industryType opt="buildings,prison,medicalTreatment,broadcasting,general">
        <!--ro, opt, string, industry (scene) type, range:[0,32], attr:opt{req, string}, desc:"buildings" (building), "prison", "medicalTreatment"
        (medical), "broadcasting", "general"-->test
      </industryType>
      <alarmEnabled opt="true,false">
        <!--ro, opt, bool, whether to enable the arming function of IoT devices, attr:opt{req, string}, desc:this field is used to check if arming is
        supported when adding devices. (Related field: alarmEnabled; related URI: /ISAPI/System/IOT/channels?format=json)-->true
      </alarmEnabled>
      <priority min="1" max="15">
        <!--ro, opt, int, calling priority, attr:min{req, int},max{req, int}-->0
      </priority>
    </channelConfig>
  </channelConfigList>

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</priority>
<transportType opt="unicast,multicast">
  <!--ro, opt, string, broadcasting method, attr:opt{req, string}, desc:unicast, multicast-->test
</transportType>
<audioVolume min="0" max="15">
  <!--ro, opt, int, volume, attr:min{req, int},max{req, int}-->0
</audioVolume>
<callForwardingEnabled>
  <!--ro, opt, bool, whether to enable call transfer, desc:true (enable), false (disable). The default value is false-->true
</callForwardingEnabled>
<callForwardingType opt="unconditional,busy,noReply,noReachable,calling">
  <!--ro, opt, string, call transfer type, range:[0,32], attr:opt{req, string}, desc:"unconditional" (call transfer under any condition), "busy"
(call transfer when busy), "noReply" (call transfer when there is no reply), "noReachable" (call transfer when it is unreachable), "calling" (call transfer
when calling)-->test
</callForwardingType>
<isSupportCallTerminalNumber>
  <!--ro, opt, bool, whether the device supports configuring the target master station No. for call forwarding-->true
</isSupportCallTerminalNumber>
<customCommunicationName min="0" max="128">
  <!--ro, opt, string, custom contact name, range:[0,128], attr:min{req, int},max{req, int}, desc:when this node is configured, the custom contact
name will be displayed during device calling prior to the terminal number-->test
</customCommunicationName>
<isSupportMoreFeatures>
  <!--ro, opt, bool, whether the device supports more functions, desc:it is not supported by default if this node is not returned-->true
</isSupportMoreFeatures>
<isNotSupportRegPassword>
  <!--ro, opt, bool, whether the password registration configuration is not supported, desc:it is supported if this node is not returned or false is
returned-->true
</isNotSupportRegPassword>
<mappingRoomNumber min="0" max="5">
  <!--ro, opt, int, mapping room No., attr:min{req, int},max{req, int}, desc:the mapping room No. is unique and can be called without exposing the
actual room No., for privacy requirement for overseas users-->0
</mappingRoomNumber>
<deviceID min="0" max="64">
  <!--ro, opt, string, device ID, attr:min{req, int},max{req, int}, desc:supported for ISUP protocol access-->test
</deviceID>
<ISUPKey min="1" max="32">
  <!--ro, opt, string, ISUP Key, attr:min{req, int},max{req, int}, desc:supported for ISUP protocol access-->test
</ISUPKey>
<dialUpAddress min="1" max="1024">
  <!--ro, opt, string, dial address, attr:min{req, int},max{req, int}, desc:this node is required when <unitType> is "analogIndoor"-->test
</dialUpAddress>
</channelConfig>
<channelConfigList>
<isSupportConfigTemplate>
  <!--ro, opt, bool, whether the device supports exporting the device List template, desc:whether the device supports exporting the device List
template-->true
</isSupportConfigTemplate>
<isSupportConfigReport>
  <!--ro, opt, bool, whether the device supports getting the importing details, desc:whether the device supports getting the importing details-->true
</isSupportConfigReport>
<unitType
opt="indoor,villa,confirm,outdoor,fence,doorbell,manage,acs,interactive,wardStation,bedheadExtension,bedsideExtension,analogIndoor,decoder,networkFireAlarmM
atrix,amplifier,pagingMicrophone,PDA,netSwitch,entrywayStation,alarmManage,alarmTerminal">
  <!--ro, opt, string, device type, range:[0,32], attr:opt{req, string}, desc:terminal, netAudio (network sound box)-->test
</unitType>
<serialNumber min="1" max="32">
  <!--ro, opt, string, the value range of device serial No. Length, range:[0,32], attr:min{req, int},max{req, int}-->test
</serialNumber>
<isSupportAsyncStatus>
  <!--ro, opt, bool, whether the device supports asynchronously getting the status of non-video channels, desc:URI for device update: PUT
/ISAPI/System/IOT/channels/updateStatusTask?format=json; URI for getting the update progress: GET /ISAPI/System/IOT/channels/updateStatusProgress?
format=json-->true
</isSupportAsyncStatus>
<isSupportAddIOTChannelList>
  <!--ro, opt, bool, whether the device supports batch adding channels of IoT devices, desc:URI for batch adding channels of IoT devices: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
</isSupportAddIOTChannelList>
<isSupportSearchMoreIOTSource>
  <!--ro, opt, bool, whether the device supports searching for the IoT devices which can be added (by multiple conditions), desc:related URI: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
</isSupportSearchMoreIOTSource>
<channelConfigReportMode>
  <!--ro, opt, enum, mode of importing the List of IoT devices, subType:string, desc:different devices (main station and door station) process the
imported device List in a different way, which should be returned to the users for prompt. URL for importing the device List:
/ISAPI/System/IOT/channelConfig?format=json; cover; add-->cover
</channelConfigReportMode>
<isSupportEncryptImportIOTChannelFile>
  <!--ro, opt, bool, whether it supports encrypting the imported non-video channel files, desc:corresponding URI: PUT /ISAPI/System/IOT/channelConfig?
format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned-->true
</isSupportEncryptImportIOTChannelFile>
<isSupportEncryptExportIOTChannelFile>
  <!--ro, opt, bool, whether it supports encrypting exported non-video channel files, desc:corresponding URI: GET /ISAPI/System/IOT/channelConfig?
format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned-->true
</isSupportEncryptExportIOTChannelFile>
<isSupportExportAbnormalReport>
  <!--ro, opt, bool, whether it supports exporting error reports for non-video channel devices, desc:corresponding URI: GET
/ISAPI/System/IOT/ChannelConfig/ExportAbnormalReport?format=json&security=<security>&iv=<iv>, it is not supported by default if this filed is not returned--
>true
</isSupportExportAbnormalReport>
</IOTCap>
<isSupportLCDScreen>
  <!--ro, opt, bool, whether the device supports LCD display screen-->true

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</isSupportLCDScreen>
<isSupportBluetooth>
  <!--ro, opt, bool, whether the device supports bluetooth, desc:related URI: /ISAPI/System/Bluetooth/capabilities-->true
</isSupportBluetooth>
<isSupportNetScreen>
  <!--ro, opt, bool, whether the device supports network screen (Nova screen), desc:related URI: /ISAPI/System/netScreens/capabilities?format=json-->true
</isSupportNetScreen>
<isSupportLocation>
  <!--ro, opt, bool, whether the device supports getting the device location information, desc:related URI: /ISAPI/System/Location?format=json-->true
</isSupportLocation>
<isSupportDebugInfoExport>
  <!--ro, opt, bool, whether the device supports exporting the debugging information, desc:related URI: /ISAPI/System/debugInfoExport/capabilities?
format=json-->true
</isSupportDebugInfoExport>
<isSupportIoTGateway>
  <!--ro, opt, bool, whether the device supports IoT gateway functions, desc:related URI: /ISAPI/System/IoTGateway/capabilities?format=json-->true
</isSupportIoTGateway>
<isSupportImageCap>
  <!--ro, opt, bool, whether the device supports configuring image parameters-->true
</isSupportImageCap>
<isSupportLogToUSB opt="true,false">
  <!--ro, opt, bool, whether the device supports exporting logs to USB, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/exportLogToUSB/capabilities?format=json-->true
</isSupportLogToUSB>
<isSupportWiresharkToUSB opt="true,false">
  <!--ro, opt, bool, whether the device supports exporting packet capture information to USB, attr:opt{opt, string}, desc:related URI:
ISAPI/System/wiresharkToUSB/capabilities?format=json-->true
</isSupportWiresharkToUSB>
<CBurningStreamEncryptionChannels opt="1,2">
  <!--ro, opt, string, return the capability of device recording stream encryption channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/Streaming/channels/<ID>/capabilities-->true
</CBurningStreamEncryptionChannels>
<isSupportOCR>
  <!--ro, opt, bool, whether the device supports OCR detection, desc:related URI: /ISAPI/Intelligent/channels/<ID>/OCR/capabilities?format=json-->true
</isSupportOCR>
<isSupportInstallationAngleCalibration>
  <!--ro, opt, bool, whether the device supports the capability of installation angle calibration, desc:related URI:
/ISAPI/System/installationAngleCalibration/channels/<ID>/capabilities-->true
</isSupportInstallationAngleCalibration>
<isSupportZeroBiasCalibration>
  <!--ro, opt, bool, whether the device supports the zero angle calibration capability node, desc:related URI:
/ISAPI/System/zeroBiasCalibration/channels/<ID>/capabilities?format=json-->true
</isSupportZeroBiasCalibration>
<isSupportCalibrationFile>
  <!--ro, opt, bool, whether the device supports importing the calibration file, desc:related URI: /ISAPI/System/configurationData-->true
</isSupportCalibrationFile>
<isSupportDisplayTrajectory>
  <!--ro, opt, bool, whether the device supports target track overlay (playing library)-->true
</isSupportDisplayTrajectory>
<maximumSuperPositionTime opt="5,10,20,30">
  <!--ro, opt, string, the maximum overlay time, attr:opt{opt, string}, desc:unit: second; it is only returned when the target track overlay is supported-
-->true
</maximumSuperPositionTime>
<isSupportUnitConfig>
  <!--ro, opt, bool, whether the device supports the unit configuration, desc:related URI: /ISAPI/System/unitConfig/capabilities?format=json-->true
</isSupportUnitConfig>
<isSupportShipsDetectionWithScene>
  <!--ro, opt, bool, whether the device supports ship detection by scene-->true
</isSupportShipsDetectionWithScene>
<isSupportPictureServer>
  <!--ro, opt, bool, whether the device supports configuring the picture storage server, desc:related URI: /ISAPI/System/PictureServer/capabilities?
format=json-->true
</isSupportPictureServer>
<isSupportEmmcInfo>
  <!--ro, opt, bool, the capability of supporting the EMMC configuration, desc:related URI: /ISAPI/System/emmcInfo/capabilities?format=json-->true
</isSupportEmmcInfo>
<isSupportVehicleIllegalType opt="1,2...29">
  <!--ro, opt, string, the vehicle violation types supported by the device, attr:opt{opt, string}, desc:the content value is the same as the index value,
refer to NET_ITS_PLATE_RESULT.NET_DVR_VEHICLE_INFO.byIllegalType the capture camera is connected to the new platform, and the device supports "traffic
violation alarm"; however, the corresponding node is not added to the device to notify the integration of this capability; that is, the traffic violation
alarm capability lacks details. If later the platform needs to develop capability with details, the current device will be incompatible with the capability;
and detailed capability is hidden in the configuration of all trigger modes, which will influence overall access-->test
</isSupportVehicleIllegalType>
<isSupportRFIDChannels>
  <!--ro, opt, bool, whether the device supports getting the RFID channel, desc:related URI: /ISAPI/System/RFIDChannels?format=json-->true
</isSupportRFIDChannels>
<isSupportRfidCollection>
  <!--ro, opt, bool, whether the device supports the event of uploading RFID tag information, desc:related URI:
/ISAPI/System/Network/RFIDData/<channelID>/capabilities, eventType: rfidCollection-->true
</isSupportRfidCollection>
<DevInfoSearch opt="tiltAngle">
  <!--ro, opt, string, whether the device supports searching for the device information, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/workingstatus/devInfoSearch?format=json-->test
</DevInfoSearch>
<isSupportAIOpenPlatform>
  <!--ro, opt, bool, whether the device supports AI open platform device capability, desc:related URI: /ISAPI/Intelligent/AIOpenPlatform/capabilities?
format=json-->true
</isSupportAIOpenPlatform>
<isSupportPictureDownloadError>
  <!--ro, opt, bool, whether the device supports uploading the exception that downloading the picture failed, desc:eventType: pictureDownloadError-->true
</isSupportPictureDownloadError>
<isSupportStudentBehavior>
  <!--ro, opt, bool, whether the device supports uploading the data of students in-class behavior, desc:eventType: studentBehavior-->true
</isSupportStudentBehavior>

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<guideEventSupport opt="linkageCapture">
  <!--ro, opt, string, the list of web quick configuration guide, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/guideConfig/guideEvent/capabilities?format=json-->test
</guideEventSupport>
<supportRemoteCtrl opt="up,down,left,right,enter,menu,num,power,esc,edit,F1,.prev,rec,play,stop,notSupport">
  <!--ro, opt, string, whether the device supports remote control, attr:opt{opt, string}, desc:related URI: /ISAPI/System/remoteCtrl/XXX-->test
</supportRemoteCtrl>
<isSupportSSDFileSystemUpgrade>
  <!--ro, opt, bool, whether the device supports updating the SSD file system, desc:related URI: /ISAPI/System/SSDFileSystem/upgrade?format=json-->true
</isSupportSSDFileSystemUpgrade>
<isSupportSSDFileSystemFormat>
  <!--ro, opt, bool, whether the device supports formatting SSD, desc:related URI: /ISAPI/System/SSDFileSystem/format?format=json-->true
</isSupportSSDFileSystemFormat>
<FaceCounting>
  <!--ro, opt, object, whether the device supports the customer traffic (face)-->
  <ChannelsList>
    <!--ro, opt, array, channel list, subType:object-->
    <channelsID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelsID>
  </ChannelsList>
</FaceCounting>
<isSupportCluster>
  <!--ro, opt, bool, related URI:, desc:/ISAPI/ContentMgmt/Cluster/capabilities-->true
</isSupportCluster>
<DeviceConfigurationSupport opt="time,network,accessProtocol,recordSchedule,channel,customProtocol">
  <!--ro, opt, string, content supported by device parameters configuration, attr:opt{opt, string}, desc:related URI: /ISAPI/System/deviceConfiguration--
>test
</DeviceConfigurationSupport>
<behaviorEventSupport opt="getUp,toiletTarry,playCellphone">
  <!--ro, opt, string, abnormal event detection types, attr:opt{opt, string}-->test
</behaviorEventSupport>
<isSupportBVCorrectPassthrough>
  <!--ro, opt, bool, whether the device supports transparent transmission of camera correction data, desc:related URI:
/ISAPI/System/BVCorrect/Passthrough/channels/<ID>/capabilities?format=json-->true
</isSupportBVCorrectPassthrough>
<isSupportChangedCallStatus>
  <!--ro, opt, bool-->true
</isSupportChangedCallStatus>
<VehiclePositionControl>
  <!--ro, opt, object, whether the device supports vehicle position arming-->
  <license min="0" max="16">
    <!--ro, req, string, license plate number, attr:min{opt, string},max{opt, string}-->test
  </license>
  <intervalTime min="0" max="65535" def="1">
    <!--ro, req, int, uploading interval, unit:s, attr:min{opt, string},max{opt, string},def{opt, string}-->1
  </intervalTime>
</VehiclePositionControl>
<supportGISEvent opt="AID,TPS,ANPR,mixedTargetDetection">
  <!--ro, opt, string, the capability of supporting getting the GIS information, attr:opt{opt, string}-->test
</supportGISEvent>
<isSupportGPSPTZCalibration>
  <!--ro, opt, bool, whether the device supports GPS PTZ calibration, desc:related URI: /ISAPI/System/GPSPTZCalibration/channels/<ID>/points?
format=json-->true
</isSupportGPSPTZCalibration>
<isSupportRadar>
  <!--ro, opt, bool, whether the device supports radar functions, desc:related URI: /ISAPI/Radar/capabilities?format=json-->true
</isSupportRadar>
<isSupportISUPHttpPassthrough>
  <!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
</isSupportISUPHttpPassthrough>
<isSupportMaterialListQuery>
  <!--ro, opt, bool, whether the device supports uploading the material list, desc:event: materialListQuery-->true
</isSupportMaterialListQuery>
<isSupportDebugLogAsync>
  <!--ro, opt, bool, whether the device supports exporting the debugging log asynchronously-->true
</isSupportDebugLogAsync>
<isSupportAlgorithmsInfo>
  <!--ro, opt, bool, whether the device supports getting the algorithm version of the device-->true
</isSupportAlgorithmsInfo>
<isSupportSecurityChecking>
  <!--ro, opt, bool, whether the device supports configuring security checking, desc:true (support); otherwise, not support-->true
</isSupportSecurityChecking>
<isSupportSecurityCheckException>
  <!--ro, opt, bool, whether the device supports the security checking exception event, desc:eventType: securityCheckException-->true
</isSupportSecurityCheckException>
<isSupportPictureManage>
  <!--ro, opt, bool, whether the device supports picture management function, desc:related URI: /ISAPI/System/isSupportPictureManage/capabilities?
format=json-->true
</isSupportPictureManage>
<isSupportAutoSwitch>
  <!--ro, opt, bool, whether the device supports automatically powering on and powering off, desc:related URI: /ISAPI/System/autoSwitch/capabilities?
format=json-->true
</isSupportAutoSwitch>
<SHMCap>
  <!--ro, opt, object, SHM alarm-->
  <isSupportHighHDTemperature>
    <!--ro, opt, bool, whether the device supports HDD high temperature detection, desc:eventType: highHDTemperature-->true
  </isSupportHighHDTemperature>
  <isSupportLowHDTemperature>
    <!--ro, opt, bool, whether the device supports HDD Low temperature detection, desc:eventType: LowHDTemperature-->true
  </isSupportLowHDTemperature>
  <isSupportHDImpact>

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    <!--ro, opt, bool, whether the device supports HDD impact detection, desc:eventType: hdImpact-->true
  </isSupportHDImpact>
  <isSupportHDBadBlock>
    <!--ro, opt, bool, whether the device supports HDD bad sector detection, desc:eventType: hdBadBlock-->true
  </isSupportHDBadBlock>
  <isSupportSevereHDFailure>
    <!--ro, opt, bool, whether the device supports HDD major fault detection, desc:eventType: severeHDFailure-->true
  </isSupportSevereHDFailure>
</SHMcap>
<isSupportLinkStatusEvent>
  <!--ro, opt, bool, whether the device supports uploading the information about the network connection status of the device, desc:eventType: LinkEvent--
>true
  </isSupportLinkStatusEvent>
  <isSupportVideoDealStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the video data processing status, desc:eventType: videoDealStatus-->true
  </isSupportVideoDealStatus>
  <isSupportAudioDealStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the audio data processing status, desc:eventType: audioDealStatus-->true
  </isSupportAudioDealStatus>
  <isSupportNetPackLossStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the network lost package status, desc:eventType: netPacketLoss-->true
  </isSupportNetPackLossStatus>
  <isSupportVideoConferenceStatus>
    <!--ro, opt, bool, whether the device supports uploading the information about the video conference status, desc:eventType: videoConference-->true
  </isSupportVideoConferenceStatus>
  <isSupportDataPrealarm>
    <!--ro, opt, bool, whether the device supports network traffic pre-alarm event, desc:eventType: dataPrealarm-->true
  </isSupportDataPrealarm>
  <isSupportClearCache>
    <!--ro, opt, bool, whether the device supports clearing the cache, desc:related URI: /ISAPI/System/clearCache?format=json-->true
  </isSupportClearCache>
  <isSupportIntelligentMode>
    <!--ro, opt, bool, whether the device supports switching the intelligent mode, desc:related URI: /ISAPI/System/IntelligentMode?format=json-->true
  </isSupportIntelligentMode>
  <isSupportFaceTemperatureMeasurementEvent>
    <!--ro, opt, bool, whether the device supports the skin-surface temperature measurement event, desc:eventType: FaceTemperatureMeasurementEvent-->true
  </isSupportFaceTemperatureMeasurementEvent>
  <isSupportQRCodeEvent>
    <!--ro, opt, bool, whether the device supports the QR code event, desc:eventType: QRCodeEvent-->true
  </isSupportQRCodeEvent>
  <isSupportVisitorEvent>
    <!--ro, opt, bool, whether the device supports the visitor event, desc:eventType: VisitorEvent-->true
  </isSupportVisitorEvent>
  <isSupportEmployeeAuthEvent>
    <!--ro, opt, bool, whether the device supports the employee authorization event, desc:eventType: EmployeeAuthEvent-->true
  </isSupportEmployeeAuthEvent>
  <isSupportVisitorInfoSearchEvent>
    <!--ro, opt, bool, whether the device supports the visitor information search event, desc:eventType: VisitorInfoSearchEvent-->true
  </isSupportVisitorInfoSearchEvent>
  <isSupportConsumptionEvent>
    <!--ro, opt, bool, whether the device supports the transaction record event, desc:eventType: ConsumptionEvent-->true
  </isSupportConsumptionEvent>
  <isSupportTransactionRecordEvent>
    <!--ro, opt, bool, whether the device supports the transaction record event, desc:eventType: TransactionRecordEvent-->true
  </isSupportTransactionRecordEvent>
  <isSupportFaceContrastResultDisplay>
    <!--ro, opt, bool, whether the device supports displaying the face picture comparison result, desc:related URI:
/ISAPI/Intelligent/faceContrast/resultDisplay/capabilities?format=json-->true
  </isSupportFaceContrastResultDisplay>
  <isSupportPatchUpgrade>
    <!--ro, opt, bool, whether it supports incremental upgrade, desc:related URI: /ISAPI/Custom/OpenPlatform/patchUpgrade/capabilities?format=json-->true
  </isSupportPatchUpgrade>
  <isSupportSnapshotAsync>
    <!--ro, opt, bool, whether the device supports capturing pictures asynchronously (the picture URL will be returned), desc:related URI:
/ISAPI/Streaming/channels/<ID>/picture/async?format=json-->true
  </isSupportSnapshotAsync>
  <isSupportLogExportByUSB>
    <!--ro, opt, bool, exporting logs to PC from USB flash, desc:related URI: /ISAPI/System/exportUSBLog?format=json-->true
  </isSupportLogExportByUSB>
  <isSupportShellCmd>
    <!--ro, opt, bool, whether the device supports shell command control, desc:related URI: /ISAPI/System/shellCmd?format=json-->true
  </isSupportShellCmd>
  <faceMonitor opt="dailyFaceMonitor,dailyFaceTracking,VCAFaceMonitor,VCAFaceTracking>manualFaceMonitor,advanceConfiguration,">
    <!--ro, opt, bool, whether the device supports face arming, attr:opt{opt, string}, desc:related URI: /ISAPI/Intelligent/channels/<ID>/dailyFaceMonitor?
format=json-->true
  </faceMonitor>
  <isSupportRelatedVCAResource>
    <!--ro, opt, bool, whether the device supports the system-level intelligent resources configuration protocol, desc:related URI:
/ISAPI/System/RelatedVCAResource/capabilities?format=json; this protocol conflicts with the previous intelligent resources switch protocol-->true
  </isSupportRelatedVCAResource>
  <isSupportVehicleMonitor>
    <!--ro, opt, bool, whether the device supports intelligent arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/capabilities?format=json-->true
  </isSupportVehicleMonitor>
  <isSupportManualVehicleMonitor>
    <!--ro, opt, bool, whether the device supports manual arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/manual/capabilities?format=json-->true
  </isSupportManualVehicleMonitor>
  <isSupportDailyVehicleMonitor>
    <!--ro, opt, bool, whether the device supports daily arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleDetect/capabilities-->true
  </isSupportDailyVehicleMonitor>
  <isSupportInputProxyDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the information of the devices added to the proxy device channel, desc:related URI:

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    <!--ro, opt, bool, whether the device supports getting the information of the devices added to the proxy device channel, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities-->true
  </isSupportInputProxyDeviceInfo>
  <isSupportAccessControlCap>
    <!--ro, opt, bool, access control capability, desc:related URI: /ISAPI/AccessControl/capabilities-->true
  </isSupportAccessControlCap>
  <isSupportDebugLog>
    <!--ro, opt, bool, whether the device supports exporting the debugging log, desc:related URI: /ISAPI/System/debugLog?format=json-->true
  </isSupportDebugLog>
  <isSupportFaceAppear>
    <!--ro, opt, bool, whether the device supports face appearing detection event, desc:eventType: faceAppear-->true
  </isSupportFaceAppear>
  <isSupportFaceDisappear>
    <!--ro, opt, bool, whether the device supports face disappearing detection event, desc:eventType: faceDisappear-->true
  </isSupportFaceDisappear>
  <isSupportTeacherBehavior>
    <!--ro, opt, bool, whether the device supports the abnormal event detection of teachers, desc:eventType: teacherBehavior-->true
  </isSupportTeacherBehavior>
  <isSupportHeadRiseRate>
    <!--ro, opt, bool, whether the device supports face appearing detection event, desc:eventType: headRiseRate-->true
  </isSupportHeadRiseRate>
  <isSupportIPCUpgrade>
    <!--ro, opt, bool, whether the platform supports upgrading network cameras via NVR and other devices, desc:related URI: /ISAPI/System/IPCUpgrade/<ID>?format=json-->true
  </isSupportIPCUpgrade>
  <isSupportFaceLibRebuildEvent>
    <!--ro, opt, bool, whether the device supports uploading the modeling rebuilding of the face picture library event, desc:eventType: faceLibRebuild-->true
  </isSupportFaceLibRebuildEvent>
  <isSupportCalibrationStream>
    <!--ro, opt, bool, supported capability of calibration stream, desc:related URI: /ISAPI/Streaming/calibStream/capabilities?format=json-->true
  </isSupportCalibrationStream>
  <isSupportChannelOccupy>
    <!--ro, opt, bool, whether the device supports outdoor fire engine access occupied by vehicles, desc:eventType: channelOccupy-->true
  </isSupportChannelOccupy>
  <isSupportOffDuty>
    <!--ro, opt, bool, whether to support fire control room person absence detection, desc:eventType: offDuty-->true
  </isSupportOffDuty>
  <isSupportNoCertificate>
    <!--ro, opt, bool, whether to support fire control room person without certificate detection, desc:eventType: noCertificate-->true
  </isSupportNoCertificate>
  <isSupportSmokeAlarm>
    <!--ro, opt, bool, whether the device supports smoke alarm, desc:eventType: smokeAlarm-->true
  </isSupportSmokeAlarm>
  <isSupportBatteryCarDisobey>
    <!--ro, opt, bool, whether the device supports illegal parking detection of electric moped, desc:eventType: batteryCarDisobey-->true
  </isSupportBatteryCarDisobey>
  <isSupportNoFireExtinguisherRecog>
    <!--ro, opt, bool, whether the device supports fire extinguisher loss detection, desc:eventType: noFireExtinguisherRecog-->true
  </isSupportNoFireExtinguisherRecog>
  <isSupportIndoorPasswayBlock>
    <!--ro, opt, bool, whether the device supports indoor access blocking detection, desc:eventType: indoorPasswayBlock-->true
  </isSupportIndoorPasswayBlock>
  <isSupportFireSmartFireDetect>
    <!--ro, opt, bool, whether the device supports fire detection, desc:eventType: fireSmartFireDetect-->true
  </isSupportFireSmartFireDetect>
  <isSupportDetectorRunningStatus>
    <!--ro, opt, bool, whether the device supports getting the detector running status, desc:eventType: detectorRunningStatus-->true
  </isSupportDetectorRunningStatus>
  <isSupportDetectorOperationStatus>
    <!--ro, opt, bool, whether the device supports getting the detector operation status, desc:eventType: detectorOperationStatus-->true
  </isSupportDetectorOperationStatus>
  <isSupportDetectorTemperatureAlarm opt="highTemperature, riseTemperature, flame">
    <!--ro, opt, bool, whether the device supports the detector temperature alarm, attr:opt{req, string}-->true
  </isSupportDetectorTemperatureAlarm>
  <isSupportDetectorShelterAlarm>
    <!--ro, opt, bool, whether the device supports detector sheltering alarm, desc:eventType: detectorShelterAlarm-->true
  </isSupportDetectorShelterAlarm>
  <isSupportDetectorMotionAlarm>
    <!--ro, opt, bool, whether the device supports the detector moving alarm, desc:eventType: detectorMotionAlarm-->true
  </isSupportDetectorMotionAlarm>
  <isSupportDetectorTamperAlarm>
    <!--ro, opt, bool, whether the device supports detector tampering alarm, desc:eventType: detectorTamperAlarm-->true
  </isSupportDetectorTamperAlarm>
  <isSupportDetectorEmergencyAlarm>
    <!--ro, opt, bool, whether the device supports the detector emergency alarm, desc:eventType: detectorEmergencyAlarm-->true
  </isSupportDetectorEmergencyAlarm>
  <isSupportDetectorSmokeAlarm>
    <!--ro, opt, bool, whether the device supports the detector smoke alarm, desc:eventType: detectorSmokeAlarm-->true
  </isSupportDetectorSmokeAlarm>
  <isSupportDetectorCombustibleGasAlarm>
    <!--ro, opt, bool, whether the device supports the detector combustible gas alarm, desc:eventType: detectorCombustibleGasAlarm-->true
  </isSupportDetectorCombustibleGasAlarm>
  <isSupportFireControlData>
    <!--ro, opt, bool, whether the device supports uploading real-time fire control data, desc:eventType: FireControlData-->true
  </isSupportFireControlData>
  <isSupportFireNoRegulation>
    <!--ro, opt, bool, whether to support people leaving event uploading, desc:eventType: fireNoRegulation-->true
  </isSupportFireNoRegulation>
  <isSupportSmokeFireRecognize>
    <!--ro, opt, bool, whether the device supports smoke and fire recognition event uploading, desc:eventType: smokeFireRecognize-->true
  </isSupportSmokeFireRecognize>
  <isSupportClientProxyWEB>
    <!--ro, opt, bool, whether to support WEB page jumping,

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desc:the function of going to the web page

package the web program and plug-in into the client (for example, iVMS-4200) for going to the web page it is mainly used in such a scenario: the device removes the web-related resources as the limit of resources, so it does not support operations on the web browser-->true

</isSupportClientProxyWEB>

<WEBLocation>

<!--ro, opt, string, WEB page placement Location, range:[1,32], desc:Local-Local server, remote-remote server; (if this node is not returned, the WEB page will be placed in the local server by default)-->local

</WEBLocation>

<isSupportWEBLocalPlug>

<!--ro, opt, bool, whether to support local download of web plug-in, desc:true=yes (by default), false=no; when this node is not returned, it is supported by default-->true

</isSupportWEBLocalPlug>

<deviceId>

<!--ro, opt, string, device ID, range:[1,32], desc:if WEBLocation is "remote", the unique device ID should be returned to identify the unique corresponding web package. This device ID is managed by the Web configuration package. For a specified device of different versions, the same deviceID will be returned-->test

</deviceId>

<isSupportTimeZone>

<!--ro, opt, bool, whether the device supports configuring the DST (DayLight Saving Time), desc:related URI: /ISAPI/System/time/timeZone-->true

</isSupportTimeZone>

<isSptHealthDetection>

<!--ro, opt, bool, whether the device supports health monitoring, desc:related URI: /ISAPI/System/healthDetection-->true

</isSptHealthDetection>

<isSptDiagnosis>

<!--ro, opt, bool, whether the device supports diagnosis, desc:related URI: /ISAPI/System/diagnosis-->true

</isSptDiagnosis>

<isSptSerialLogCfg>

<!--ro, opt, bool, whether the device supports configuring parameters of serial port log redirection, desc:related URI: /ISAPI/System/serialLogCfg?format=json-->true

</isSptSerialLogCfg>

<isSptFileExport>

<!--ro, opt, bool, whether the device supports exporting the device file, desc:related URI: /ISAPI/System/fileExport?format=json-->true

</isSptFileExport>

<isSptCertificationStandard>

<!--ro, opt, bool, whether the device supports configuring host verification standard, desc:related URI: /ISAPI/SecurityCP/certificationStandard-->true

</isSptCertificationStandard>

<isSptKeypadLock>

<!--ro, opt, bool, whether the device supports locking the keypad, desc:related URI: /ISAPI/System/keypadLock/config-->true

</isSptKeypadLock>

<isSupportVibrationDetection>

<!--ro, opt, bool, whether the device supports the vibration detection, desc:related URI: /ISAPI/System/channels/<ID>/vibrationDetection?format=json-->true

</isSupportVibrationDetection>

<isSupportSmokingDetectAlarm>

<!--ro, opt, bool, whether the device supports the smoking alarm, desc:eventType: smokeDetectAlarm-->true

</isSupportSmokingDetectAlarm>

<isSupportRadarChannels>

<!--ro, opt, bool, whether the device supports getting the radar channel, desc:related URI: /ISAPI/System/RadarChannels?format=json-->true

</isSupportRadarChannels>

<radarIPDForm>

<!--ro, opt, enum, radar-camera status, subType:string, desc:"single" (single radar); "double_diagonal" (double radars arranged diagonally); "double_vertical" (double radars arranged vertically)-->single

</radarIPDForm>

<isSupportRadarFieldDetection>

<!--ro, opt, bool, whether the device supports radar's area intrusion detection, desc:related URI: /ISAPI/Smart/RadarChannels/<radarChannelID>/radarFieldDetection/capabilities?format=json-->true

</isSupportRadarFieldDetection>

<isSupportRadarLineDetection>

<!--ro, opt, bool, whether the device supports radar line crossing detection, desc:related URI: /ISAPI/Smart/RadarChannels/<radarChannelID>/radarLineDetection/capabilities?format=json-->true

</isSupportRadarLineDetection>

<mixedTargetDetectionWebNoDisplay>

<!--ro, opt, bool, multiple targets recognition is not displayed-->true

</mixedTargetDetectionWebNoDisplay>

<isSupportRemovePlaybackRecord>

<!--ro, opt, bool, whether to support removing playback (the device does not support playback), desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. This node will be returned when the value of isSupportRemoveStorage is true; otherwise, this node will not be returned-->true

</isSupportRemovePlaybackRecord>

<isSupportAudioUpload>

<!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URI: /ISAPI/System/Audio/audioUpload?format=json-->true

</isSupportAudioUpload>

<VCAResourceList>

<!--ro, opt, array, supported smart resources List, subType:object-->

<VCAResource>

<!--ro, opt, object, smart resources-->

<VCAResourceType>

<!--ro, opt, string, smart resources type-->test

</VCAResourceType>

<ChannelVCAResourceList>

<!--ro, opt, array, supported smart resources List, subType:object-->

<ChannelVCAResource>

<!--ro, opt, object, smart channel resources-->

<channel>

<!--ro, opt, int, channel No.-->1

</channel>

<channelVCAResourceType>

<!--ro, opt, string, smart channel resources type-->test

</channelVCAResourceType>

</ChannelVCAResource>

</ChannelVCAResourceList>

</VCAResource>

</VCAResourceList>

<isSupportEventDetection>

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<!--ro, opt, bool, whether to support slot plugging in and out, desc:eventType: slotInsertion-->true
</isSupportSlotInsertion>
<isSupportNetworkLoop>
<!--ro, opt, bool, whether to support network loop alarm, desc:eventType: networkLoop-->true
</isSupportNetworkLoop>
<isSupportApOnOffline>
<!--ro, opt, bool, whether to support AP online and offline alarm event uploading, desc:eventType: apOnOffline-->true
</isSupportApOnOffline>
<isSupportAbnormalWANPort>
<!--ro, opt, bool, whether to support WAN abnormal alarm event uploading, desc:eventType: abnormalWANPort-->true
</isSupportAbnormalWANPort>
<isSupportAPException>
<!--ro, opt, bool, whether to support AP exception alarm information uploading, desc:eventType: APException-->true
</isSupportAPException>
<isSupportACException>
<!--ro, opt, bool, whether to support AC exception alarm information uploading, desc:eventType: ACException-->true
</isSupportACException>
<isSupportCartridgeCapacity>
<!--ro, opt, bool, whether the device supports the configuration of notifying cartridge capacity, desc:related URL:
/ISAPI/System/printer/cartridgeCapacity/capabilities?format=json-->true
</isSupportCartridgeCapacity>
<isSupportCDCakeBoxCapacity>
<!--ro, opt, bool, whether the device supports notifying CD capacity, desc:related URL: /ISAPI/System/printer/CDCakeBoxCapacity/capabilities?
format=json-->true
</isSupportCDCakeBoxCapacity>
<isSupportSleep>
<!--ro, opt, bool, whether the device supports configuring sleep, desc:related URL: /ISAPI/System/sleep/capabilities?format=json-->true
</isSupportSleep>
<isSupportContainerDetection>
<!--ro, opt, bool, whether the device supports container detection, desc:/ISAPI/Intelligent/containerDetection/capabilities?format=json (for devices
that does not return this node, you can check according to the node returned by the following protocol:
/ISAPI/Traffic/ContentMgmt/InputProxy/channels/<ID>/ocrScene/capabilities)-->true
</isSupportContainerDetection>
<isSupportRadarVideoDetection>
<!--ro, opt, bool, related URL: /ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json, desc:related URL:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
</isSupportRadarVideoDetection>
<isSupportRadarVideoDetectionRealtime>
<!--ro, opt, bool, whether the device supports radar video detection, desc:related URL:
/ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
</isSupportRadarVideoDetectionRealtime>
<isSupportBatteryStatusUpload>
<!--ro, opt, bool, whether the device supports uploading the battery status, desc:eventType: batteryStatus-->true
</isSupportBatteryStatusUpload>
<isSupportPowerConsumptionStatusUpload>
<!--ro, opt, bool, whether the device supports uploading device consumption status-->true
</isSupportPowerConsumptionStatusUpload>
<isSupportOnlineUpgradeTask>
<!--ro, opt, bool, whether device online upgrading is supported-->true
</isSupportOnlineUpgradeTask>
<isSupportPersonArmingTrack>
<!--ro, opt, bool, whether the device supports person arming configuration, desc:related URL:
/ISAPI/Intelligent/channels/<ID>/personArmingTrack/capabilities?format=json-->true
</isSupportPersonArmingTrack>
<isSupportManualPersonArmingTrack>
<!--ro, opt, bool, whether the device supports manual person arming, desc:related URL: /ISAPI/Intelligent/channels/<ID>/manualPersonArmingTrack?
format=json-->true
</isSupportManualPersonArmingTrack>
<isSupportGPSVerification>
<!--ro, opt, bool, whether the device supports GPS verification points, desc:related URL: /ISAPI/System/GPSVerification/channels/<ID>/points?
format=json-->true
</isSupportGPSVerification>
<isSupportHBDLib>
<!--ro, opt, bool, whether the device supports the human body library, desc:related URL: /ISAPI/Intelligent/HBDLib/capabilities?format=json-->true
</isSupportHBDLib>
<isSupportDisinfectionInfo>
<!--ro, opt, bool, whether the device supports uploading disinfection information, desc:eventType: disinfectionInfo-->true
</isSupportDisinfectionInfo>
<isSupportTempQueryDelete>
<!--ro, opt, bool, whether the device supports deleting temperature query, desc:eventType: tempQueryDelete-->true
</isSupportTempQueryDelete>
<isSupportMobilePayment>
<!--ro, opt, bool, whether the device supports mobile payment, desc:/ISAPI/System/mobilePayment?format=json-->true
</isSupportMobilePayment>
<isSupportFileSearch>
<!--ro, opt, bool, whether the device supports searching files, desc:related URL: /ISAPI/System/fileSearch?format=json-->true
</isSupportFileSearch>
<isSupportFileExportAsync>
<!--ro, opt, bool, whether the device supports exporting files asynchronously, desc:related URL: /ISAPI/System/fileExport/async?format=json-->true
</isSupportFileExportAsync>
<isSupportSSDFileSystemCapacity>
<!--ro, opt, bool, whether the device supports getting the capacity distribution of the SSD system, desc:related URL:
/ISAPI/System/SSDFileSystem/capacity?format=json-->true
</isSupportSSDFileSystemCapacity>
<isSupport4GSafetyHelmetInfo>
<!--ro, opt, bool, whether the device supports the hard hat event, desc:eventType: 4GSafetyHelmetInfo-->true
</isSupport4GSafetyHelmetInfo>
<isSupportICCID>
<!--ro, opt, bool, whether the device supports getting the ICCID information, desc:related URL: /ISAPI/System/deviceInfo/ICCID?format=json-->true
</isSupportICCID>
<isSupportFireEscapeDetection>
<!--ro, opt, bool, whether the device supports fire escape detection, desc:related URL:
/ISAPI/Intelligent/channels/<ID>/fireEscapeDetection/capabilities?format=json-->true

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</isSupportFireEscapeDetection>
<isSupportTakingElevatorDetection>
  <!--ro, opt, bool, whether the device supports taking elevator detection, desc:related URI:
  /ISAPI/Intelligent/channels/<ID>/takingElevatorDetection/capabilities?format=json-->true
</isSupportTakingElevatorDetection>
<isSupportVehicleMatchResult>
  <!--ro, opt, bool, whether the device supports license plate comparison, desc:related URI:
  /ISAPI/Intelligent/channels/<ID>/licensePlateContrast/capabilities?format=json-->true
</isSupportVehicleMatchResult>
<isSupportDataAware>
  <!--ro, opt, bool, the data aware capability, desc:related URI: /ISAPI/System/dataAware/capabilities?format=json-->true
</isSupportDataAware>
<isSupportAbnormalReboot>
  <!--ro, opt, bool, whether the device supports abnormal reboot event, desc:eventType: abnormalReboot-->true
</isSupportAbnormalReboot>
<isSupportVehicleThermometry>
  <!--ro, opt, bool, whether the device supports vehicle temperature measurement detection, desc:related URI:
  /ISAPI/Traffic/channels/<ID>/vehicleDetect/thermometry/capabilities?format=json-->true
</isSupportVehicleThermometry>
<AssociationScenesList>
  <!--ro, opt, array, the list of scene adaptive linkage, subType:object, desc:it indicates the device supports the scene adaptive linkage between
  channels. The current device has two channels: main channel (box camera) and sub channel (radar camera), however, the device only opens a main channel to
  users for scene configuration; the scene configuration of the sub channel is not open for the users; the scene of the sub channel is automatically connected
  to the main channel scene; currently, the front-end returns this capability-->
  <AssociationScenes>
    <!--ro, req, object, the channel list of scene adaptive linkage between channels-->
    <channelID>
      <!--ro, req, int, supported scene channel ID, desc:supported scene channel ID-->1
    </channelID>
    <isSupportScene>
      <!--ro, opt, bool, whether the device supports the scene configuration, desc:whether the current channel is the main channel and supports scene
      configuration, true (main channel and support), this node is not returned (sub channel and not support)-->true
    </isSupportScene>
  </AssociationScenes>
</AssociationScenesList>
<supportRelationVideoEvent opt="PIR,cityManagement,roadMaint">
  <!--ro, opt, string, supports uploading the video of linked triggering event, attr:opt(req, string), desc:"PIR" (PIR event), "cityManagement" (city
  management event), "roadMaint" (road maintenance event)-->test
</supportRelationVideoEvent>
<isSupportVCAResourceList>
  <!--ro, opt, bool, whether the device supports configuring intelligent resources in a batch, desc:related URI:
  /ISAPI/System/Video/inputs/channels/VCAResourceList?format=json-->true
</isSupportVCAResourceList>
<isSupportConvergenceCloud>
  <!--ro, opt, bool, whether the device supports the URI for EZ interconnection channels, desc:URI for capability of all channels:
  /ISAPI/Event/convergenceCloud/channels/capabilities?format=json; URI for capability of specified channel:
  /ISAPI/Event/convergenceCloud/channels/<ID>/capabilities?format=json-->true
</isSupportConvergenceCloud>
<isSupportInterconnection>
  <!--ro, opt, bool, whether the device supports the URI for EZ interconnection devices, desc:related URI: /ISAPI/Event/convergenceCloud/capabilities?
  format=json-->true
</isSupportInterconnection>
<isSupportHPVehicleStatus>
  <!--ro, opt, bool, whether the device supports the platform vehicle status event, desc:eventType: HPVehicleStatus-->true
</isSupportHPVehicleStatus>
<isSupportHPWorkingStatus>
  <!--ro, opt, bool, whether the device supports the scheduled platform working status event, desc:eventType: HPWorkingStatus-->true
</isSupportHPWorkingStatus>
<isSupportHPPeopleDetection>
  <!--ro, opt, bool, whether the device supports the platform people detection event, desc:eventType: HPPeopleDetection-->true
</isSupportHPPeopleDetection>
<isSupportHPVehicleDoorDetection>
  <!--ro, opt, bool, whether the device supports the platform door detection event, desc:eventType: HPVehicleDoorDetection-->true
</isSupportHPVehicleDoorDetection>
<isSupportRoadMaint>
  <!--ro, opt, bool, whether the device supports road maintenance, desc:related URI: /ISAPI/Intelligent/channels/<ID>/roadMaint?format=json-->true
</isSupportRoadMaint>
<isSupportDockStationInfo>
  <!--ro, opt, bool, whether the device supports the event of uploading the dock station information to the media, desc:eventType: dockStationInfo; this
  node is not returned if not supported-->true
</isSupportDockStationInfo>
<isSupportBodyCameraManualForensics>
  <!--ro, opt, bool, whether the device supports the event of uploading the manual evidence information of the body camera, desc:eventType:
  bodyCameraManualForensics; this node is not returned if not supported-->true
</isSupportBodyCameraManualForensics>
<isSupportBodyCameraLogin>
  <!--ro, opt, bool, whether the device supports the event of police officer logging in to the body camera, desc:eventType: bodyCameraLogin; this node is
  not returned if not supported-->true
</isSupportBodyCameraLogin>
<isSupportBodyCameraRunStatus>
  <!--ro, opt, bool, whether the device supports the status of body camera, desc:eventType: bodyCameraRunStatus; this node is not returned if not
  supported-->true
</isSupportBodyCameraRunStatus>
<isSupportBodyCameraSignIn>
  <!--ro, opt, bool, whether the device supports the event of police officer signing in the body camera, desc:eventType: bodyCameraSignIn; this node is
  not returned if not supported-->true
</isSupportBodyCameraSignIn>
<isSupportAlertDistribution>
  <!--ro, opt, bool, whether the device supports the alert distribution, desc:eventType: alertDistribution; this node is not returned if not supported--
  >true
</isSupportAlertDistribution>
<isSupportImageCapture>
  <!--ro, opt, bool, whether the device supports the event of scheduled capture, desc:related URI: /ISAPI/System/Video/inputs/channels/<ID>/imageCapture?

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format=json-->true
</isSupportImageCapture>
<isSupportScheduledRecordUpload>
  <!--ro, opt, bool, whether the device supports the event of channel scheduled recording, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/scheduledRecordUpload?format=json-->true
</isSupportScheduledRecordUpload>
<isSupportTargetPatternCollect>
  <!--ro, opt, bool, whether the device supports the target track collection event, desc:eventType: targetPatternCollect-->true
</isSupportTargetPatternCollect>
<isSupportAudioAnalysis>
  <!--ro, opt, bool, whether the device supports the audio analysis event, desc:eventType: audioAnalysis; this node is not returned if not supported-->true
</isSupportAudioAnalysis>
<isSupportDebuggingDataExport>
  <!--ro, opt, bool, whether the device supports importing the parameters (in-out parameters, rule configuration parameters), desc:related URI:
/ISAPI/System/debuggingData/parameters?format=json; this node is not returned if not supported-->true
</isSupportDebuggingDataExport>
<isSupportInternalParaAdjust>
  <!--ro, opt, bool, whether the device supports trinocular device; whether the device supports editing in-parameters, desc:related URI:
/ISAPI/System/internalParameters/adjustment?format=json; this node is not returned if not supported-->true
</isSupportInternalParaAdjust>
<isSupportManualImageCapture>
  <!--ro, opt, bool, whether the device supports manual capture of the video channel, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/manualImageCapture?format=json; this node is not returned if not supported-->true
</isSupportManualImageCapture>
<characteristicCode min="1" max="128">
  <!--ro, opt, string, device attribute code, range:[1,128], attr:min{opt, string},max{opt, string}, desc:related URI:
/ISAPI/System/deviceInfo/characteristicCode?format=json-->test
</characteristicCode>
<isSupportSIMCardStatus>
  <!--ro, opt, bool, whether the device supports getting the information of SIM card status, desc:related URI: /ISAPI/System/SIMCardStatus?format=json-->true
</isSupportSIMCardStatus>
<OpenPlatformCap>
  <!--ro, opt, object, detailed HEOP capability supported by the device, desc:related URI: /ISAPI/Custom/OpenPlatform/capabilities-->
<isSupportOpenPlatform>
  <!--ro, req, bool, whether the device supports HEOP (Embedded Open Platform)-->true
</isSupportOpenPlatform>
<maxAPPNum>
  <!--ro, opt, int, the maximum number of third-party application packages allowed to be installed-->0
</maxAPPNum>
</OpenPlatformCap>
<isSupportLogException>
  <!--ro, opt, bool, whether the device supports the Log exception alarm, desc:eventType: LogException-->true
</isSupportLogException>
<isSupportMEF>
  <!--ro, opt, bool, whether the device supports Mobile electronic fence - mobile electronic fence data detection, desc:related URI:
/ISAPI/System/Network/MEF/<ID>?format=json-->true
</isSupportMEF>
<manualEvent
opt="fieldDetection,linedetection,ANPR,faceCapture,mixedTargetDetection,studentsStoodUp,headRiseRate,teacherBehavior,ElevatorRunningDetectionEvent">
  <!--ro, opt, string, whether the device supports URI for manual triggering event, attr:opt{req, string}, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/manualEvent/<eventType>?format=json-->test
</manualEvent>
<isSupportAppCfg>
  <!--ro, opt, bool, whether the device supports application management, desc:related URI: /ISAPI/System/app/capabilities?format=json-->true
</isSupportAppCfg>
<isSupportCleanup>
  <!--ro, opt, bool, whether the device supports one-click clearing, desc:related URI: /ISAPI/System/cleanup?format=json-->true
</isSupportCleanup>
<isSupportEndUserCheckMsg opt="sleep,reboot,factoryReset,cleanup,parentPickUpQuery,schoolAttendanceQuery,screenShare,screenMirror">
  <!--ro, opt, bool, whether the device supports uploading the confirmation information of end users, attr:opt{req, string}, desc:eventType:
endUserCheckMsg-->true
</isSupportEndUserCheckMsg>
<isSupportScreenConfig>
  <!--ro, opt, bool, whether the device supports configuring screen parameters, desc:related URI: /ISAPI/System/screenConfig/capabilities?format=json-->true
</isSupportScreenConfig>
<isSupportTemperatureMonitor>
  <!--ro, opt, bool, whether the device supports temperature monitoring configuration, desc:related URI: /ISAPI/System/temperatureMonitor/capabilities?
format=json-->true
</isSupportTemperatureMonitor>
<isSupportObjectServer>
  <!--ro, opt, bool, whether the device supports the object storage service, desc:related URI: /ISAPI/System/objectServer/capabilities?format=json-->true
</isSupportObjectServer>
<isSupportCDsStatus>
  <!--ro, opt, bool, whether the device supports notifying disk status, desc:related URI for enabling the function: the bEnableCDsStatus node of
/ISAPI/Trial/roompara/capabilities; corresponding event: CDsStatus-->true
</isSupportCDsStatus>
<isSupportCoalGangueDetection>
  <!--ro, opt, bool, whether the device supports coal gangue detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/coalGangueDetection;
this node is not returned if not supported-->true
</isSupportCoalGangueDetection>
<isSupportVacantSeatRate>
  <!--ro, opt, bool, whether the device supports seat vacancy detection, desc:related URI: /ISAPI/Intelligent/channels/<channelID>/vacantSeatRate?
format=json; corresponding event: VacantSeatRate; this node is not returned if not supported-->true
</isSupportVacantSeatRate>
<isSupportObjectsThrownDetection>
  <!--ro, opt, bool, whether the device supports objects thrown detection, desc:related URI:
/ISAPI/Intelligent/channels/<channelID>/objectsThrownDetection?format=json; corresponding event: objectsThrownDetection; this node is not returned if not
supported-->true
</isSupportObjectsThrownDetection>
<isSupportBVCorrectDataSearch>
  <!--ro, opt, bool, whether the device supports searching for the parameters of camera correction, desc:related URI: /ISAPI/System/BVCorrectData/search?

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format=json; this node is not returned if not supported-->true
</isSupportBVCorrectDataSearch>
<isSupportSlagTruckWashingDetection>
  <!--ro, opt, bool, whether the device supports dump truck cleaning detection event, desc:eventType: slagTruckWashingDetection-->true
</isSupportSlagTruckWashingDetection>
<isSupportWOL>
  <!--ro, opt, bool, whether the device supports configuring the LAN wakeup parameters, desc:it can be realized via the WOL (Wake On-Lan) function.
Related URL: /ISAPI/System/WOL/capabilities?format=json-->true
</isSupportWOL>
<isSupportPersonQueueTimingStatistics>
  <!--ro, opt, bool, whether the device supports counting the number of persons in each queue, desc:eventType: isSupportPersonQueueTimingStatistics-->true
</isSupportPersonQueueTimingStatistics>
<isSupportLensParamFile>
  <!--ro, opt, bool, whether the device supports importing and exporting camera parameters files-->true
</isSupportLensParamFile>
<isSupportVehiclePassingInParkingLot>
  <!--ro, opt, bool, whether the device supports the detection of vehicle passing in parking lot entrance and exit, desc:eventType:
vehiclePassingInParkingLot-->true
</isSupportVehiclePassingInParkingLot>
<isSupportPowerOutput>
  <!--ro, opt, bool, whether power supply output configuration is supported, desc:related URL: /ISAPI/System/powerOutput/capabilities?format=json-->true
</isSupportPowerOutput>
<isSupportIPCUpgradeByFTP>
  <!--ro, opt, bool, it indicates that the device supports upgrading the proxy device channel, such as IPC upgrade via NVR, desc:related URI:
/ISAPI/System/IPCUpgrade/<channelID>/FTP?format=json&security=<security>&iv=<iv>-->true
</isSupportIPCUpgradeByFTP>
<isSupportMethaneConcentrationException>
  <!--ro, opt, bool, whether the device supports methane concentration exception detection, desc:eventType: methaneConcentrationException-->true
</isSupportMethaneConcentrationException>
<isSupportMethaneLightIntensityException>
  <!--ro, opt, bool, whether the device supports methane light intensity exception detection, desc:eventType: methaneLightIntensityException-->true
</isSupportMethaneLightIntensityException>
<isSupportLocalButtonPermission>
  <!--ro, opt, bool, whether the device supports controlling local button permission, desc:related URI: /ISAPI/System/LocalButtonPermission/capabilities?
format=json-->true
</isSupportLocalButtonPermission>
<isSupportLocalButtonRecord>
  <!--ro, opt, bool, whether the device supports button triggering record, desc:eventType: LocalButtonRecord-->true
</isSupportLocalButtonRecord>
<isSupportLocalButtonSnapshot>
  <!--ro, opt, bool, whether the device supports button triggering snapshot, desc:eventType: LocalButtonSnapshot-->true
</isSupportLocalButtonSnapshot>
<isSupportVoiceTalkEvent>
  <!--ro, opt, bool, whether the device supports voice talk, desc:eventType: voiceTalkEvent-->true
</isSupportVoiceTalkEvent>
<isSupportPowerSupply>
  <!--ro, opt, bool, whether the device supports external power supply, desc:related URI: /ISAPI/System/powerSupply/capabilities?format=json-->true
</isSupportPowerSupply>
<isSupportBarrierGateStatus>
  <!--ro, opt, bool, whether the device supports barrier gate status, desc:eventType: barrierGateStatus-->true
</isSupportBarrierGateStatus>
<DeployCap>
  <!--ro, opt, object, the capability of arming non-subscription events-->
  <deployID opt="1,2">
    <!--ro, opt, enum, arming type, subType:int, attr:opt{req, string}, desc:0 (client arming); 1 (real-time arming)-->0
  </deployID>
</DeployCap>
<isSupportSignalControllerStatus>
  <!--ro, opt, bool, whether the device supports uploading the signal controller status, desc:eventType: signalControllerStatus-->true
</isSupportSignalControllerStatus>
<isSupportSignalControllerFault>
  <!--ro, opt, bool, whether the device supports uploading the signal controller fault, desc:eventType: signalControllerFault-->true
</isSupportSignalControllerFault>
<isSupportNTPConfig>
  <!--ro, opt, bool, whether the device supports configuring NTP parameters, desc:/ISAPI/System/time/ntp/capabilities?format=json-->true
</isSupportNTPConfig>
<isSupportConsumptionQuery>
  <!--ro, opt, bool, whether the device supports uploading the request for searching for consumption, desc:eventType: ConsumptionQuery-->true
</isSupportConsumptionQuery>
<StreamingCodingRegionPositionCap>
  <!--ro, opt, object, whether the device supports getting all stream splice area positions information capability,
desc:/ISAPI/System/streamingCoding/channels/<channelID>/regionPosition-->
  <ChannelsList>
    <!--ro, opt, array, channel list, subType:object-->
    <channelsID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelsID>
  </ChannelsList>
</StreamingCodingRegionPositionCap>
<isSupportLowPowerPlan>
  <!--ro, opt, bool, whether the device supports configuring low consumption mode schedule, desc:related URI:
/ISAPI/System/consumptionMode/LowPowerPlan/capabilities?format=json-->true
</isSupportLowPowerPlan>
<isSupportPlateQuaAlarm>
  <!--ro, opt, bool, whether the device supports plate quality detection, desc:eventType: plateQuaAlarm-->true
</isSupportPlateQuaAlarm>
<isSupportWiegand>
  <!--ro, opt, bool, whether the device supports the Wiegand protocol, desc:related URI: /ISAPI/System/Wiegand/<wiegandID>/capabilities?format=json-->true
</isSupportWiegand>
<zoomFocusWebDisplay
opt="ROI,roadTrafficDetection,SMD,mixedTargetDetection,faceCapture,personCounting,multidimensionalPeopleCounting,AIOP,heelPDC,packingSpaceRecognition,tunnel
Detection,cityManagement">
  <!--ro, opt, string, whether the web supports displaying the zoomfocus page, attr:opt{req, string},

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desc:ROI page path: Configuration-Vedio/Audio-ROI
roadTrafficDetection:Configuration-Road Traffic-Detection Configuration; SMD; mixedTargetDetection; faceCapture-->ROI
</zoomFocusWebDisplay>
<isSupportInternalParamsAdjustment>
  <!--ro, opt, bool, whether the device supports in-parameters correction by region, desc:true (support), false or this node is not returned (not support). Related URI: /ISAPI/System/internalParameters/regionAdjustment/capabilities?format=json-->true
</isSupportInternalParamsAdjustment>
<isSupportDiagnosticReport>
  <!--ro, opt, bool, whether the device supports diagnostic report function, desc:true (support), false or this node is not returned (not support). Related URI: /ISAPI/System/diagnosticReport/capabilities?format=json-->true
</isSupportDiagnosticReport>
<isSupportWaterQualitySearch>
  <!--ro, opt, bool, whether the device supports the function of searching for water quality, desc:true (support), false or this node is not returned (not support). Related URI: /ISAPI/Event/waterQualityDetection/search?format=json-->true
</isSupportWaterQualitySearch>
<isSupportExhaustDetection>
  <!--ro, opt, bool, whether the device supports configuring gas detection parameters, desc:true (support), false or this node is not returned (not support). Related URI: /ISAPI/Event/exhaustDetection/channels/<channelID>/capabilities?format=json-->true
</isSupportExhaustDetection>
<isSupportTPMSecretKey>
  <!--ro, opt, bool, whether the device supports enabling stream encryption for all TPM keys Lists, desc:true (support), false or this node is not returned (not support). Related URI: /ISAPI/Streaming/encryption/TPMSecretKeyList?format=json&security=<security>&iv=<iv>-->true
</isSupportTPMSecretKey>
<isSupportWeatherStation>
  <!--ro, opt, bool, whether the device supports uploading weather station information, desc:eventType: weatherStation; this node is not returned if not supported-->true
</isSupportWeatherStation>
<isSupportRoadVisibilityAnalysis>
  <!--ro, opt, bool, whether the device supports road visibility detection analysis, desc:eventType: weatherStation; this node is not returned if not supported-->true
</isSupportRoadVisibilityAnalysis>
<isSupportRoadWeatherAnalysis>
  <!--ro, opt, bool, whether the device supports road weather detection analysis, desc:eventType: roadWeatherAnalysis; this node is not returned if not supported-->true
</isSupportRoadWeatherAnalysis>
<isSupportRoadVisibilityAbnormal>
  <!--ro, opt, bool, whether the device supports uploading road visibility detection exception, desc:eventType: roadVisibilityAbnormal; this node is not returned if not supported-->true
</isSupportRoadVisibilityAbnormal>
<isSupportRoadWeatherAbnormal>
  <!--ro, opt, bool, whether the device supports uploading road weather detection exception, desc:eventType: roadWeatherAbnormal; this node is not returned if not supported-->true
</isSupportRoadWeatherAbnormal>
<isSupportChannelsTimeSync>
  <!--ro, opt, bool, whether the device supports batch device time synchronization of the IPC/IOT channel, desc:related URL: /ISAPI/System/time/channelsTimeSync/capabilities?format=json-->true
</isSupportChannelsTimeSync>
<isSupport433Sensor>
  <!--ro, opt, bool, whether the device supports configuring 433 sensor transmission, desc:related URLs: /ISAPI/System/Sensor/433sensor/433wirelessTransfer/capabilities?format=json
/ISAPI/System/Sensor/433sensor/capabilities?format=json
/ISAPI/System/Sensor/433sensor/sensorDetection/capabilities?format=json
/ISAPI/System/Sensor/433sensor/sensorPicOverlay/capabilities?format=json
/ISAPI/System/Sensor/433sensor/rfParamTemplate/capabilities?format=json-->true
</isSupport433Sensor>
<isSupportSignalSource>
  <!--ro, opt, bool, whether the device supports configuring signal source parameters, desc:related URL: /ISAPI/System/signalSource/capabilities?format=json-->true
</isSupportSignalSource>
<isSupportUSBConfig>
  <!--ro, opt, bool, whether the device supports configuring USB parameters, desc:related URL: /ISAPI/System/USBConfig/capabilities?format=json-->true
</isSupportUSBConfig>
<isSupportStandbyStrategy>
  <!--ro, opt, bool, whether the device supports configuring standby strategy parameters, desc:related URL: /ISAPI/System/standbyStrategy/capabilities?format=json-->true
</isSupportStandbyStrategy>
<isSupportWakeupStrategy>
  <!--ro, opt, bool, whether the device supports configuring wakeup strategy parameters, desc:related URL: /ISAPI/System/wakeupStrategy/capabilities?format=json-->true
</isSupportWakeupStrategy>
<isSupportSeachMutexFunction>
  <!--ro, opt, bool, whether the device supports searching for disabling the mutual exclusion function, desc:related URL: GET /ISAPI/System/mutexFunction/capabilities?format=json; function description: whether the device supports searching for mutual exclusion function-->true
</isSupportSeachMutexFunction>
<isSupportMultiDeviceID>
  <!--ro, opt, bool, whether the device supports multiple Long numbers configuration, desc:related URL: /ISAPI/VideoIntercom/deviceId/batchCfg/capabilities?format=json-->true
</isSupportMultiDeviceID>
<isNotSupportMultiDeviceIDCustomName>
  <!--ro, opt, bool, whether the device supports configuring custom name for multiple Long numbers, desc:if this function is not supported, the user cannot configure custom name for the device via the Long number configuration API; true (not support), false or this node is not returned (support)-->true
</isNotSupportMultiDeviceIDCustomName>
<isSupportMixedSecurityChecking>
  <!--ro, opt, bool, whether the device supports uploading the event of mixed security checking, desc:eventType: mixedSecurityChecking-->true
</isSupportMixedSecurityChecking>
<isSupportVisiblePackageEvent>
  <!--ro, opt, bool, whether the device supports visible package event, desc:eventType: visiblePackageEvent-->true
</isSupportVisiblePackageEvent>
<isSupportBarCodeEvent>
  <!--ro, opt, bool, whether the device supports the barcode event-->true
</isSupportBarCodeEvent>
<isSupportGaugeReading>
  <!--ro, opt, bool, whether the device supports the gauge reading function, desc:related URL: GET /ISAPI/Event/channels/<channelID>/gaugeReading/capabilities?format=json; this node is not returned if not supported-->true

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</isSupportGaugeReading>
<isSupportUncertificateCompareEvent>
  <!--ro, opt, bool, whether the device supports the UncertificateCompareEvent event, desc:eventType: UncertificateCompareEvent; this node is not returned if not supported-->true
</isSupportUncertificateCompareEvent>
<isSupportAudioDiacritical>
  <!--ro, opt, bool, whether the device supports audio voice changing function, desc:related URL:
/ISAPI/System/Video/inputs/channels/<channelID>/audioDiacritical/capabilities?format=json-->true
</isSupportAudioDiacritical>
<isSupportMEFReportSearch>
  <!--ro, opt, bool, whether the device supports searching for electronic fence data statistics report, desc:URL for getting the capability:
/ISAPI/System/Network/MEF/reportSearch/capabilities?format=json; this node is not returned if not supported-->true
</isSupportMEFReportSearch>
<isSupportBroadcastTriggerEvent>
  <!--ro, opt, bool, whether the device supports broadcast triggering event, desc:after the event is uploaded to the platform via the paging microphone,
the platform will broadcast and check according to the uploaded terminal No. List; eventType: broadcastTriggerEvent. URL for checking:
/ISAPI/VideoIntercom/broadcastTriggerConfirm?format=json-->true
</isSupportBroadcastTriggerEvent>
<isSupportFishingShipDetection>
  <!--ro, opt, bool, whether the device supports fishing ship detection event, desc:eventType: fishingShipDetection; related URL:
/ISAPI/Thermal/channels/<channelID>/fishingShipDetection/<SID>?format=json-->true
</isSupportFishingShipDetection>
<isSupportHPVehicleLoadingRate>
  <!--ro, opt, bool, whether the device supports the platform vehicle Loading rate event, desc:eventType: vehicleLoadingRate, this node is not returned if
not supported. Related URL: /ISAPI/Intelligent/channels/<channelID>/HPDetection/capabilities?format=json-->true
</isSupportHPVehicleLoadingRate>
<isSupportVideoBarrierGateDetection>
  <!--ro, opt, bool, whether the device supports port and gate video detection, desc:eventType: videoBarrierGateDetection, this node is not returned if
not supported. Related URL: /ISAPI/Traffic/ContentMgmt/videoBarrierGate/capabilities?format=json-->true
</isSupportVideoBarrierGateDetection>
<isSupportLogicalResource>
  <!--ro, opt, bool, whether the device supports camera organization and management, desc:related URL: /ISAPI/System/LogicalResource/capabilities?
format=json, this node is not returned if not supported-->true
</isSupportLogicalResource>
<isSupportSlagTruckCoverPlate>
  <!--ro, opt, bool, whether the device supports detecting whether back cover the slag truck is covered, desc:eventType: slagTruckCoverPlateDetection,
this node is not returned if not supported. Related URL: /ISAPI/Event/channels/<channelID>/slagTruckCoverPlate/capabilities?format=json-->true
</isSupportSlagTruckCoverPlate>
<isSupportTrainDetection>
  <!--ro, opt, bool, whether the device supports train detection, desc:If train detection is not supported, this node will not be returned. Corresponding
eventType: trainDetection-->true
</isSupportTrainDetection>
<isNotSupportVideoOverlays>
  <!--ro, opt, bool, whether the device does not support configuring OSD overlay on the device web configuration page, desc:this node is only valid for
checking whether the OSD overlay configuration of the image menu on the person and vehicle verification station is supported or not (this configuration is
supported by default in the protocol and it does not have node control capability). If true is returned, it indicates that the OSD overlay is not supported;
if false or no node is returned, it indicates the OSD overlay function is supported-->true
</isNotSupportVideoOverlays>
<isSupportScheduledCaptureTask>
  <!--ro, opt, bool, whether the device supports scheduled capturing tasks, desc:related URI: /ISAPI/Sytem/Video/scheduledCapture/capabilities-->true
</isSupportScheduledCaptureTask>
<isSupportGuideCfg>
  <!--ro, opt, bool, whether the device supports the wizard configuration, desc:this node is only valid for the indoor station to check whether to enable
wizard configuration, which is provided for the users to perform basic function configuration after activating and logging in to the device (including
Internet parameters configuration, dialog configuration and batch adding analog indoor stations). If true is returned, it indicates wizard configuration is
supported; if false or no node is returned, it indicates the function is not supported-->true
</isSupportGuideCfg>
<isSupportCameraGroupPeopleCounting>
  <!--ro, opt, bool, whether the device supports passenger flow statistics of multiple cameras, desc:/ISAPI/Intelligent/counting/cameraGroup/capabilities?
format=json-->true
</isSupportCameraGroupPeopleCounting>
<isSupportAngleDeviationDetection>
  <!--ro, opt, bool, whether the device supports tilting detection, desc:related URL:
/ISAPI/Event/channels/<channelID>/angleDeviationDetection/capabilities?format=json; this node is not returned if not supported-->true
</isSupportAngleDeviationDetection>
<isSupportCityCode>
  <!--ro, opt, bool, whether the device supports getting city code, desc:related URI: /ISAPI/System/cityCode?format=json-->true
</isSupportCityCode>
<isSupportAlcoholDetectionResult>
  <!--ro, opt, bool, whether the device supports uploading alcohol test, desc:eventType: alcoholDetectionResult-->true
</isSupportAlcoholDetectionResult>
<isSupportAlcoholDetectionRodBind>
  <!--ro, opt, bool, whether the device supports the request of binding alcohol test stick, desc:eventType: alcoholDetectionRodBind-->true
</isSupportAlcoholDetectionRodBind>
<isSupportAlcoholDetectionRodBindQuery>
  <!--ro, opt, bool, whether the device supports the query for the request of binding alcohol test stick, desc:eventType: alcoholDetectionRodBindQuery--
>true
</isSupportAlcoholDetectionRodBindQuery>
<isSupportAlcoholDetectionRodUnbind>
  <!--ro, opt, bool, whether the device supports the event of unbinding alcohol test stick, desc:eventType: alcoholDetectionRodUnbind-->true
</isSupportAlcoholDetectionRodUnbind>
<isSupportAbnormalDriving>
  <!--ro, opt, bool, whether the device supports abnormal driving behaviors detection, desc:eventType: abnormalDriving-->true
</isSupportAbnormalDriving>
<isSupportEmanationsEvent>
  <!--ro, opt, bool, whether the device supports the event of emanation test, desc:eventType: emanationsEvent; the security inspection system is equipped
with an emanation tester, and when the emanation exceeds the threshold, an alarm event will be uploaded. This event is only triggered by the emanation
tester, and its configuration is not required-->true
</isSupportEmanationsEvent>
<isSupportClientUpdatePackage>
  <!--ro, opt, bool, whether the device supports the client upgrade package management, desc:related URL: /ISAPI/System/clientUpdatePackage/capabilities?
format=json-->true
</isSupportClientUpdatePackage>

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<isSupportDeviceStatusUploadStrategy>
  <!--ro, opt, bool, whether the device supports configuring the parameters of device status uploading strategy, desc:related URL:
  /ISAPI/System/status/uploadStrategy/capabilities?format=json-->true
</isSupportDeviceStatusUploadStrategy>
<isSupportDialIndicatorReading>
  <!--ro, opt, bool, whether the devices supports dial indicator reading, desc:related URL:
  /ISAPI/Intelligent/channels/<channelID>/dialIndicatorReading/capabilities?format=json; the function refers to reading single pointer dial and uploading the
  result (currently, it can only recognize dials that do not accumulate the readings such as thermometer and barometer)-->true
</isSupportDialIndicatorReading>
<isNotSupportAudioVideoCompress>
  <!--ro, opt, bool, whether the device does not support audio/video encoding, desc:this node is only valid for the person and vehicle verification
  station; if true is returned, the video/audio page will not be displayed on the device web page. If false or no node is returned, the video/audio page will
  be displayed on the device web page by default-->true
</isNotSupportAudioVideoCompress>
<isSupportUPLCExceptionEvent>
  <!--ro, opt, bool, whether the device supports the event of liquid chromatograph system exception, desc:eventType: UPLCException-->true
</isSupportUPLCExceptionEvent>
<isSupportUrgentTurn>
  <!--ro, opt, bool, whether the device supports the sharp turn event, desc:eventType: urgentTurn-->true
</isSupportUrgentTurn>
<isSupportUrgentBrake>
  <!--ro, opt, bool, whether the device supports the sudden brake event, desc:eventType: urgentBrake-->true
</isSupportUrgentBrake>
<isSupportUrgentAcceleration>
  <!--ro, opt, bool, whether the device supports the rapid acceleration event, desc:eventType: urgentAcceleration-->true
</isSupportUrgentAcceleration>
<isSupportVehiclePanicButton>
  <!--ro, opt, bool, whether the device supports the alarm/event of mobile emergency button, desc:eventType: vehiclePanicButton-->true
</isSupportVehiclePanicButton>
<isSupportOverSpeed>
  <!--ro, opt, bool, whether the device supports the vehicle speeding alarm/event, desc:eventType: overSpeed-->true
</isSupportOverSpeed>
<isSupportAbnormalAcceleration>
  <!--ro, opt, bool, whether the device supports the event of abnormal vehicle custom acceleration alarm, desc:eventType: abnormalAcceleration-->true
</isSupportAbnormalAcceleration>
<isSupportRollover>
  <!--ro, opt, bool, whether the device supports the vehicle turnover alarm/event, desc:eventType: rollover-->true
</isSupportRollover>
<isSupportCollision>
  <!--ro, opt, bool, whether the device supports the vehicle collision alarm/event, desc:eventType: collision-->true
</isSupportCollision>
<isSupportVehicleOBUInfo>
  <!--ro, opt, bool, whether the device supports uploading the vehicle OBU information, desc:eventType: vehicleOBUInfo-->true
</isSupportVehicleOBUInfo>
<isSupportPowerUpPicture>
  <!--ro, opt, bool, whether the device supports importing and exporting pictures displayed during startup,
  desc:/ISAPI/System/powerUpPicture/capabilities?format=json-->true
</isSupportPowerUpPicture>
<isSupportPowerUpPictureCfg>
  <!--ro, opt, bool, whether the device supports configuring parameters of pictures displayed during startup,
  desc:/ISAPI/System/powerUpPictureCfg/capabilities?format=json-->true
</isSupportPowerUpPictureCfg>
<isSupportAlgoPackageScheduling>
  <!--ro, opt, bool, whether the device supports algorithm package scheduling, desc:Related URIs: /ISAPI/System/AlgoPackageScheduling/capabilities?
  format=json and /ISAPI/System/AlgoPackageScheduling/task/capabilities?format=json-->true
</isSupportAlgoPackageScheduling>
<isSupportVoiceTalkRequestingAlarm>
  <!--ro, opt, bool, whether the device supports the two-way audio request alarm, desc:eventType: voiceTalkRequestingAlarm-->true
</isSupportVoiceTalkRequestingAlarm>
<isSupportSetupStructure>
  <!--ro, opt, bool, whether the device supports configuring construction structure parameters, desc:related URI: /ISAPI/System/setupStructure?
  format=json-->true
</isSupportSetupStructure>
<isSupportPassNumInfoEvent>
  <!--ro, opt, bool, whether the device supports uploading the number of people passed, desc:eventType: passNumInfoEvent-->true
</isSupportPassNumInfoEvent>
<isShowClientRemote>
  <!--ro, opt, bool, whether the device supports displaying remote configuration on the client,
  desc:1. If this node is not returned or it is returned and its value is true, it indicates that the device supports this function.
  2. If this node is returned and its value is false, it indicates that the device does not support this function.
  iVMS-4200 client software supports remote configuration by default. However, there are still some devices that do not support remote configuration via iVMS-
  4200 and they can only be configured remotely via the web browser. The capability determined by this node is used to control whether to display the remote
  configuration on iVMS-4200 client software-->true
</isShowClientRemote>
<isSupportUserManualQRCode>
  <!--ro, opt, bool, whether the device supports getting the QR code of the user manual, desc:related URI: /ISAPI/System/userManualQRCode/capabilities?
  format=json-->true
</isSupportUserManualQRCode>
<isSupportUnclosedFrontPanel>
  <!--ro, opt, bool, whether the device supports uploading the event of the front panel not closed, desc:related URI for getting the capability:
  /ISAPI/Trial/roompara/capabilities; corresponding node: < unclosedFrontPanelEnabled>. Related event: unclosedFrontPanel-->true
</isSupportUnclosedFrontPanel>
<isSupportVehicleWeight>
  <!--ro, opt, bool, whether the device supports configuring parameters for vehicle weight and axle detection,
  desc:/ISAPI/Traffic/channels/<channelID>/vehicleWeight?format=json-->true
</isSupportVehicleWeight>
<isSupportWeightInterface>
  <!--ro, opt, bool, whether it supports integrated overLoad control parameters, desc:/ISAPI/Traffic/ContentMgmt/WeightInterface?format=json-->true
</isSupportWeightInterface>
<isSupportSecurityPanicEvent>
  <!--ro, opt, bool, whether the device supports emergency events of security inspection, desc:eventType: securityPanicEvent-->true
</isSupportSecurityPanicEvent>
<isSupportTrafficLightMalfunction>
  <!--ro, opt, bool, whether the device supports the event of detecting faults of traffic lights, desc:eventType: trafficLightMalfunction-->true
</isSupportTrafficLightMalfunction>

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<!--ro, opt, bool, whether the device supports the event of detecting faults of traffic lights, desc:eventType: trafficLightMalfunction-->true
</isSupportTrafficLightMalfunction>
<isSupportTOFShelterAlarm>
  <!--ro, opt, bool, whether the device supports TOF tampering alarm, desc:/ISAPI/Event/channels/<channelID>/TOFShelterAlarm/capabilities?format=json-->true
</isSupportTOFShelterAlarm>
<isSupportSetMealQuery>
  <!--ro, opt, bool, whether the device supports uploading the event of searching set meals, desc:eventType: SetMealQuery-->true
</isSupportSetMealQuery>
<isSupportMultiScreenControlAlarm>
  <!--ro, opt, bool, whether the device supports uploading the multi-screen controller alarm, desc:eventType: multiScreenControlAlarm-->true
</isSupportMultiScreenControlAlarm>
<isSupportKeyFrameLoss>
  <!--ro, opt, bool, whether the device supports uploading the event of key frame loss, desc:related URI: /ISAPI/Event/keyFrameLoss/capabilities?format=json; eventType: keyFrameLoss.-->true
</isSupportKeyFrameLoss>
<isSupportWorkStatusCfg>
  <!--ro, opt, bool, whether the device supports working status configuration, desc:/ISAPI/System/WorkStatusCfg?format=json-->true
</isSupportWorkStatusCfg>
<isSupportHealthInfoSyncQuery>
  <!--ro, opt, bool, whether the device supports the event of searching for health information, desc:eventType: HealthInfoSyncQuery-->true
</isSupportHealthInfoSyncQuery>
<isSupportVisitorPictureSearchEvent>
  <!--ro, opt, bool, whether the device supports the event of searching for visitor profiles, desc:eventType: VisitorPictureSearchEvent-->true
</isSupportVisitorPictureSearchEvent>
<isSupportEZVIZParameterTest>
  <!--ro, opt, bool, whether the device supports testing EZ parameters, desc:related URI: /ISAPI/System/Network/EZVIZ/parameterTest?format=json-->true
</isSupportEZVIZParameterTest>
<isSupportUnattendedPackageEvent>
  <!--ro, opt, bool, whether the device supports the unattended package event, desc:eventType: unattendedPackageEvent-->true
</isSupportUnattendedPackageEvent>
<isSupportPackageFragmentEvent>
  <!--ro, opt, bool, whether the device supports the event of uploading package fragment data, desc:eventType: packageFragmentEvent-->true
</isSupportPackageFragmentEvent>
<isSupportPackageBarcodeMismatchEvent>
  <!--ro, opt, bool, whether the device supports the event of package and barcode mismatching, desc:eventType: packageBarcodeMismatchEvent-->true
</isSupportPackageBarcodeMismatchEvent>
<isSupportCheckingModeSwitchEvent>
  <!--ro, opt, bool, whether the device supports the event of switching the image checking mode, desc:eventType: checkingModeSwitchEvent-->true
</isSupportCheckingModeSwitchEvent>
<FaceSnapPrivacyMask>
  <!--ro, opt, object, whether the device supports privacy mask on faces, desc:based on face picture capture, all faces will be masked to protect the privacy. Related URI: /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json-->
  <ChannelsList>
    <!--ro, opt, array, channel list, subType:object-->
    <channelID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelID>
  </ChannelsList>
</FaceSnapPrivacyMask>
<FaceContrastPrivacyMask>
  <!--ro, opt, object, whether the device supports privacy mask on faces of key persons, desc:based on face picture comparison, the faces of key persons will be masked to protect the privacy. Related URI: /ISAPI/Intelligent/channels/<channelID>/faceContrast/privacyMask/capabilities?format=json-->
  <ChannelsList>
    <!--ro, opt, array, channel list, subType:object-->
    <channelID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelID>
  </ChannelsList>
</FaceContrastPrivacyMask>
<isSupportHumanBodyComparison>
  <!--ro, opt, bool, whether the device supports human body comparison events, desc:related URI: /ISAPI/Event/channels/<channelID>/humanBodyComparison/capabilities?format=json; eventType: humanBodyComparison-->true
</isSupportHumanBodyComparison>
<isSupportMedicalInfoSearchEvent>
  <!--ro, opt, bool, whether the device supports the event of searching for medical information, desc:medicalInfoSearchEvent-->true
</isSupportMedicalInfoSearchEvent>
<isSupportConsumptionStatusQuery>
  <!--ro, opt, bool, whether the device supports uploading the request for searching for consumption status, desc:eventType: ConsumptionStatusQuery-->true
</isSupportConsumptionStatusQuery>
<isSupportShipChannelAbnormal>
  <!--ro, opt, bool, whether the device supports uploading ship channel exception detection event, desc:eventType: shipChannelAbnormal-->true
</isSupportShipChannelAbnormal>
<isSupportDepositRetrieveEvent>
  <!--ro, opt, bool, whether the device supports the event of object access in the intelligent cabinet-->true
</isSupportDepositRetrieveEvent>
<isSupportDepositTimeoutEvent>
  <!--ro, opt, bool, whether the device supports the event of overtime object deposit in the intelligent cabinet-->true
</isSupportDepositTimeoutEvent>
<isSupportCabinetStatusEvent>
  <!--ro, opt, bool, whether the device supports the event of intelligent cabinet's changed status-->true
</isSupportCabinetStatusEvent>
<isSupportCabinetAuthorityExpired>
  <!--ro, opt, bool, whether the device supports the event of intelligent cabinet's permission expiration-->true
</isSupportCabinetAuthorityExpired>
<isSupportT1Test>
  <!--ro, opt, bool, whether the device supports the configuration of controlling T1 tool-->true
</isSupportT1Test>
<isSupportDrivingDirectionAtIntersection>
  <!--ro, opt, bool, whether the device supports intersection driving direction detection, desc:eventType: drivingDirectionAtIntersection; this node is not returned if not supported-->true
</isSupportDrivingDirectionAtIntersection>
<isSupportPlaceSceneExecuteStatus>
  <!--ro, opt, bool, whether the device supports the event of scene execution status, desc:eventType: placeSceneExecuteStatus; this node is not returned

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if not supported-->true
</isSupportPlaceSceneExecuteStatus>
<isSupportSceneRecordingCtrl>
  <!--ro, opt, bool, whether the device supports the event of controlling scene recording, desc:eventType: sceneRecordingCtrl; this node is not returned
if not supported-->true
</isSupportSceneRecordingCtrl>
<isSupportStreamRecordStatus>
  <!--ro, opt, bool, whether the device supports getting the stream ID record status, desc:/ISAPI/System/Video/inputs/streams/recordStatus-->true
</isSupportStreamRecordStatus>
<isSupportManualSnapPicture>
  <!--ro, opt, bool, whether the device supports manual picture capturing, desc:related URI:
/ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json-->true
</isSupportManualSnapPicture>
<isSupportVerificationResultEvent>
  <!--ro, opt, bool, whether the device supports the event of human and vehicle verification result, desc:eventType: verificationResultEvent-->true
</isSupportVerificationResultEvent>
<OTAP>
  <!--ro, opt, object, the capability of Open Things Access Protocol, desc:OTAP capability-->
  <isSupportOTAPModel>
    <!--ro, opt, bool, whether the device supports getting the virtual No. and device No. of the OTAP, desc:GET /ISAPI/System/Network/OTAP/Model?
format=json-->true
  </isSupportOTAPModel>
  </OTAP>
  <isSupportForeignObjectInRoadDetection>
    <!--ro, opt, bool, whether the device supports the detection of foreign object in road, desc:eventType: foreignObjectInRoadDetection-->true
  </isSupportForeignObjectInRoadDetection>
  <isSupportUPLCMethodOperatingStatus>
    <!--ro, opt, bool, whether the device supports the event of UPLC method operating status, desc:eventType: UPLCMethodOperatingStatus-->true
  </isSupportUPLCMethodOperatingStatus>
  <isSupportMaterialScanTaskStatus>
    <!--ro, opt, bool, whether the device supports material scanning task status, desc:eventType: materialScanTaskStatus-->true
  </isSupportMaterialScanTaskStatus>
  <isSupportSearchInputProxy>
    <!--ro, opt, bool, whether the device supports searching for accessible IPC of device in the LAN, desc:related URI: GET
/ISAPI/ContentMgmt/InputProxy/search?security=<security>&iv=<iv>; true (support), this node is not returned (call the specific API to confirm whether it
supports or not)-->true
  </isSupportSearchInputProxy>
  <isSupportPostureDetection>
    <!--ro, opt, bool, whether the device supports people posture detection, desc:eventType: MeetingLock-->true
  </isSupportPostureDetection>
  <isSupportFans>
    <!--ro, opt, bool, whether the device supports configuring fan, desc:/ISAPI/System/Fans?format=json-->true
  </isSupportFans>
  <pickUpCalibration>
    <!--ro, opt, object, whether the device supports wireless pickup calibration, desc:related URI:
/ISAPI/System/PickUpCalibration/channels/<channelID>/capabilities?format=json-->
  <ChannelsList>
    <!--ro, opt, array, channel List, subType:object-->
    <channelsID>
      <!--ro, req, int, No. of channels supported by the device-->1
    </channelsID>
  </ChannelsList>
  </pickUpCalibration>
  <isSupportSerialLogUpload>
    <!--ro, opt, bool, whether the device supports uploading serial port log, desc:true (support), false (not support)-->true
  </isSupportSerialLogUpload>
  <isSupportLiquidDetectionEvent>
    <!--ro, opt, bool, whether the device supports liquid detection, desc:eventType: LiquidDetectionEvent-->true
  </isSupportLiquidDetectionEvent>
  <isSupportETCRSU>
    <!--ro, opt, bool, whether the device supports ETCRSU configuration, desc:related URI: /ISAPI/Traffic/ETCRSU Level protocol-->true
  </isSupportETCRSU>
  <isSupportVideoCodeConvertTool>
    <!--ro, opt, bool, whether the device supports getting the video encoding conversion tool, desc:related URI: /ISAPI/System/videoCodeConvertTool?
format=json protocol-->true
  </isSupportVideoCodeConvertTool>
  <supportQuickConfigType opt="icr,dayNightGate,syncSignalOutput">
    <!--ro, opt, string, the list of supported functions of the quick configuration pop-up on the web, attr:opt(opt, string),
desc:the web quick configuration does not distinguish the configuration order, and it displays all supported function lists on the pop-up on the tab page
icr refers to ICR switch configuration (related URL: /ISAPI/Image/channels/<channelID>/); icr
dayNightGate refers to day and night mode switching (related URL: /ISAPI/Image/channels/<channelID>/dayNightGate);
IOOutputs refers to synchronous signal output (related URL: /ISAPI/System/syncSignalOutput)-->syncSignalOutput
  </supportQuickConfigType>
  <isSupportVisualLocation>
    <!--ro, opt, bool, whether the device supports configuring visual location service, desc:related URI:
/ISAPI/System/LocationMgr/visualLocation/capabilities?format=json-->true
  </isSupportVisualLocation>
  <isSupportVideoAccessAbnormal>
    <!--ro, opt, bool, whether the device supports abnormal video access, desc:eventType: VideoAccessAbnormal; this node is not returned if not supported--
>true
  </isSupportVideoAccessAbnormal>
  <isSupportSubSystemAbnormal>
    <!--ro, opt, bool, whether the device supports abnormal subsystem, desc:eventType: SubSystemAbnormal; this node is not returned if not supported-->true
  </isSupportSubSystemAbnormal>
  <isSupportSearchFileList>
    <!--ro, opt, bool, whether the device supports searching for list of device files, desc:related URI: /ISAPI/System/fileExport/SearchFileList?
format=json-->true
  </isSupportSearchFileList>
  <isSupportDistanceDetectionEvent>
    <!--ro, opt, bool, whether the device supports distance detection, desc:eventType: distanceDetectionEvent; currently it is only used for devices test
tools-->true
  </isSupportDistanceDetectionEvent>
  <isSupportMeetingQueryEvent>

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<!--ro, opt, bool, whether the device supports searching for meetings, desc:eventType: MeetingElementCancel-->true
</isSupportMeetingQueryEvent>
<isSupportMeetingLockEvent>
  <!--ro, opt, bool, whether the device supports locking meetings, desc:eventType: MeetingLock-->true
</isSupportMeetingLockEvent>
<isSupportMeetingAddEvent>
  <!--ro, opt, bool, whether the device supports reserving meetings, desc:eventType: MeetingAdd-->true
</isSupportMeetingAddEvent>
<isSupportMeetingModifyEvent>
  <!--ro, opt, bool, whether the device supports modifying meetings, desc:eventType: MeetingModify-->true
</isSupportMeetingModifyEvent>
<isSupportMeetingCancelEvent>
  <!--ro, opt, bool, whether the device supports canceling meetings, desc:eventType: ScreenBindIPC-->true
</isSupportMeetingCancelEvent>
<isSupportScreenBindIPCEvent>
  <!--ro, opt, bool, whether the device supports the screen binding IPC, desc:eventType: ScreenBindIPC-->true
</isSupportScreenBindIPCEvent>
<isSupportSensorDetection>
  <!--ro, opt, bool, whether the device supports uploading sensor data, desc:eventType: sensorDetection-->true
</isSupportSensorDetection>
<isSupportClearLogCache>
  <!--ro, opt, bool, whether the device supports clearing the log cache, desc:related URI: /ISAPI/ContentMgmt/ClearLogCache?format=json-->true
</isSupportClearLogCache>
<isSupportRoadOccupyDetection>
  <!--ro, opt, bool, whether the device supports the entrance and exit block detection, desc:related URI:
/ISAPI/ITC/Entrance/channels/<channelID>/roadOccupyDetection/capabilities?format=json-->true
</isSupportRoadOccupyDetection>
<isSupportMeetingRoomSignInEvent>
  <!--ro, opt, bool, whether the device supports the MeetingRoomSignIn event, desc:eventType: MeetingRoomSignIn-->true
</isSupportMeetingRoomSignInEvent>
<isSupportMeetingRoomSignOutEvent>
  <!--ro, opt, bool, whether the device supports the MeetingRoomSignOut event, desc:eventType: MeetingRoomSignOut-->true
</isSupportMeetingRoomSignOutEvent>
<isSupportTimeTypeSTD>
  <!--ro, opt, bool, whether the device supports the time type of ISO8601, desc:if "true" is returned, it indicates that all the devices containing
ISO8601 parameters support adding the timeType=STD, such as /ISAPI/ContentMgmt/search?timeType=STD, the time will be implemented according to the standard-->true
</isSupportTimeTypeSTD>
<isSupportScanner>
  <!--ro, opt, bool, whether the device supports scanning events and configuration, desc:if true is returned, it indicates the device supports the
/ISAPI/System/scanner/capabilities?format=json and scannerInfo event-->true
</isSupportScanner>
<isSupportRegionTargetNumberCounting>
  <!--ro, opt, bool, whether the device supports the region target counting, desc:related URI:
/ISAPI/Event/channels/<channelID>/RegionTargetNumberCounting/Capabilities?format=json-->true
</isSupportRegionTargetNumberCounting>
<depthMapMaintainMode>
  <!--ro, opt, object, the maintenance mode of depth image, desc:it will influence the smart functions when using depth images for tests, such as vertical
customer traffic, object accumulation detection and so on enabling the maintenance mode will change the audio and video parameters of the channels related
URI: /ISAPI/Intelligent/channels/<channelID>/DepthMapMaintainModeCfg/capabilities?format=json-->
<ChannelsList>
  <!--ro, opt, array, channel list, subType:object-->
  <channelID>
    <!--ro, req, int, No. of channels supported by the device-->1
  </channelID>
</ChannelsList>
</depthMapMaintainMode>
<isSupportExportPipCalibrationFile>
  <!--ro, opt, bool, whether it supports exporting visual and thermal image fusion calibration file, desc:related URI:
/ISAPI/System/ExportPipCalibrationFile?format=json-->true
</isSupportExportPipCalibrationFile>
<isSupportCourseUploadInfo>
  <!--ro, opt, bool, whether it supports notifying the course uploading completion, desc:eventType: courseUploadInfo-->true
</isSupportCourseUploadInfo>
<isSupportPresetSwitchingInformation>
  <!--ro, opt, bool, whether the device supports uploading the preset switching information, desc:eventType: presetSwitchingInformation-->true
</isSupportPresetSwitchingInformation>
<isSupportCabinetChargingQuery>
  <!--ro, opt, bool, whether it supports the request of searching for the charging of intelligent cabinet, desc:eventType: cabinetChargingQuery-->true
</isSupportCabinetChargingQuery>
<isSupportSearchVoltageAndPowerConsumptionReport>
  <!--ro, opt, bool, whether the device supports searching for voltage and consumption report, desc:related URI:
/ISAPI/System/SearchVoltageAndPowerConsumptionReport/capabilities?format=json-->true
</isSupportSearchVoltageAndPowerConsumptionReport>
<isSupportDeviceDrop>
  <!--ro, opt, bool, whether the device supports the deviceDrop event, desc:corresponding event: deviceDrop-->true
</isSupportDeviceDrop>
<isSupportDeviceNotMoved>
  <!--ro, opt, bool, whether the device supports the deviceNotMoved event, desc:corresponding event: deviceNotMoved-->true
</isSupportDeviceNotMoved>
<isSupportWalkWrongWayOnEscalator>
  <!--ro, opt, bool, whether the device supports the walkWrongWayOnEscalator event, desc:corresponding event: walkWrongWayOnEscalator-->true
</isSupportWalkWrongWayOnEscalator>
<isSupportFallOnEscalator>
  <!--ro, opt, bool, whether the device supports the fallOnEscalator event, desc:corresponding event: fallOnEscalator-->true
</isSupportFallOnEscalator>
<isSupportCarryLargeLuggageOnEscalator>
  <!--ro, opt, bool, whether the device supports the carryLargeLuggageOnEscalator event, desc:corresponding event: carryLargeLuggageOnEscalator-->true
</isSupportCarryLargeLuggageOnEscalator>
<isSupportPushStrollerOnEscalator>
  <!--ro, opt, bool, whether the device supports the pushStrollerOnEscalator event, desc:corresponding event: pushStrollerOnEscalator-->true
</isSupportPushStrollerOnEscalator>
<isSupportTeachingSpeechDetection>
  <!--ro, opt, bool, whether it supports teachingSpeechDetection event, desc:corresponding event: teachingSpeechDetection-->true

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</isSupportTeachingSpeechDetection>
<updateFirmwareType opt="warningScreenFirmware,warningScreenFont,warningScreenVoice,ledReceiverCard">
  <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:corresponding to <type> in /ISAPI/System/updateFirmware?
type=<type>; "warningScreenFirmware", "warningScreenFont", "warningScreenVoice"; note that <isSupportUpdateFirmware> is supported when this node is
supported-->warningScreenFirmware
</updateFirmwareType>
<isSupportPaymentQRCodeQuery>
  <!--ro, opt, bool, whether the device supports the paymentQRCodeQuery event, desc:corresponding event: paymentQRCodeQuery-->true
</isSupportPaymentQRCodeQuery>
<isSupportTopUpEvent>
  <!--ro, opt, bool, whether the device supports the topUpEvent event, desc:corresponding event: topUpEvent-->true
</isSupportTopUpEvent>
<isSupportTopUpRequest>
  <!--ro, opt, bool, whether the device supports the topUpRequest event, desc:corresponding event: topUpRequest-->true
</isSupportTopUpRequest>
<isSupportTopUpStatusQuery>
  <!--ro, opt, bool, whether the device supports the topUpStatusQuery event, desc:corresponding event: topUpStatusQuery-->true
</isSupportTopUpStatusQuery>
<isSupportLogOffRefundRequest>
  <!--ro, opt, bool, whether the device supports the LogOffRefundRequest event, desc:corresponding event: LogOffRefundRequest-->true
</isSupportLogOffRefundRequest>
<isSupportLogOffRefundStatusQuery>
  <!--ro, opt, bool, whether the device supports searching for the refund status of logged-off persons, desc:corresponding event: LogOffRefundStatusQuery-
-->true
</isSupportLogOffRefundStatusQuery>
<isSupportAuthenticationTimeoutEvent>
  <!--ro, opt, bool, whether the device supports the LogOffRefundStatusQuery event, desc:corresponding event: LogOffRefundStatusQuery-->true
</isSupportAuthenticationTimeoutEvent>
<isSupportCertificateCapture>
  <!--ro, opt, bool, whether the device supports the certificateCapture events, desc:corresponding event: certificateCapture-->true
</isSupportCertificateCapture>
<isSupportDeviceBatchActivation>
  <!--ro, opt, bool, whether the device supports batch activation, desc:POST /ISAPI/System/activate/DeviceBatchActivation?format=json-->true
</isSupportDeviceBatchActivation>
<isSupportPlaceEnvironmentMonitoringRule>
  <!--ro, opt, bool, whether the device supports place environment monitoring rule parameters,
desc:/ISAPI/Event/channels/<channelID>/PlaceEnvironmentMonitoringRule?format=json-->true
</isSupportPlaceEnvironmentMonitoringRule>
<isSupportEngineResource>
  <!--ro, opt, bool, whether the device supports engine resources, desc:/ISAPI/System/EngineResource?format=json-->true
</isSupportEngineResource>
<isSupportPTModulePowerCfg>
  <!--ro, opt, bool, whether the device supports modifying the power supply parameters of PTZ base, desc:PUT /ISAPI/System/ModifyPTModulePowerCfg?
format=json POST /ISAPI/System/GetPTModulePowerCfg?format=json-->true
</isSupportPTModulePowerCfg>
<isSupportGetBoxDoorDetectionCfg>
  <!--ro, opt, bool, whether the device supports getting chassis door status detection configuration, desc:POST /ISAPI/System/GetBoxDoorDetectionCfg?
format=json-->true
</isSupportGetBoxDoorDetectionCfg>
<isSupportModifyBoxDoorDetectionCfg>
  <!--ro, opt, bool, whether the device supports modifying chassis door status configuration, desc:PUT /ISAPI/System/ModifyBoxDoorDetectionCfg?
format=json-->true
</isSupportModifyBoxDoorDetectionCfg>
<isSupportSmartBoxStatus>
  <!--ro, opt, bool, whether the device supports intelligent chassis status, desc:/ISAPI/System/SmartBoxStatus?format=json-->true
</isSupportSmartBoxStatus>
<isSupportBatteryStatusCtrl>
  <!--ro, opt, bool, whether the device supports battery status event upload control parameters, desc:/ISAPI/System/BatteryStatusCtrl?format=json-->true
</isSupportBatteryStatusCtrl>
<isSupportCooler>
  <!--ro, opt, bool, whether the device supports cooling management, desc:/ISAPI/System/Cooler/-->true
</isSupportCooler>
<isSupportBoxDoorDetectionCfg>
  <!--ro, opt, bool, whether the device supports configuring chassis door status detection, desc:/ISAPI/System/BoxDoorDetectionCfg?format=json-->true
</isSupportBoxDoorDetectionCfg>
<isSupportBoxDoorStatus>
  <!--ro, opt, bool, whether the device supports chassis door status, desc:/ISAPI/System/BoxDoorStatus?format=json-->true
</isSupportBoxDoorStatus>
<isSupportLightningProtectionStatus>
  <!--ro, opt, bool, whether the device supports lightning protection status, desc:/ISAPI/System/LightningProtectionStatus?format=json-->true
</isSupportLightningProtectionStatus>
<isSupportDeviceVisualLocalizationDetectionEvent>
  <!--ro, opt, bool, whether the device supports the DeviceVisualLocalizationDetectionEvent event, desc:corresponding event:
DeviceVisualLocalizationDetectionEvent-->true
</isSupportDeviceVisualLocalizationDetectionEvent>
<isSupportDeviceVisualLocalizationAnalysisEvent>
  <!--ro, opt, bool, whether the device supports the DeviceVisualLocalizationAnalysisEvent event, desc:corresponding event:
DeviceVisualLocalizationAnalysisEvent-->true
</isSupportDeviceVisualLocalizationAnalysisEvent>
<isSupportGetPowerPortWorkParamList>
  <!--ro, opt, bool, whether the device supports getting the working parameter list of power supply interface, desc:POST
/ISAPI/System/GetPowerPortWorkParamList?format=json-->true
</isSupportGetPowerPortWorkParamList>
<isSupportUsersLoginNotification>
  <!--ro, opt, bool, whether the device supports parameters of notifications on user login, desc:/ISAPI/System/UsersLoginNotification?format=json-->true
</isSupportUsersLoginNotification>
<isSupportLiquidLevelDetectorInfo>
  <!--ro, opt, bool, whether the device supports uploading information of liquid level detector, desc:eventType: LiquidLevelDetectorInfo-->true
</isSupportLiquidLevelDetectorInfo>
<isSupportWaterPressureDetectorInfo>
  <!--ro, opt, bool, whether the device supports uploading information of water pressure detector, desc:eventType: WaterPressureDetectorInfo-->true
</isSupportWaterPressureDetectorInfo>
<isSupportEncryptionFileUpgradeMgr>

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    <!--ro, opt, bool, whether the device supports advanced management of encrypted files, desc:GET /ISAPI/System/EncryptionFileUpgradeMgr/capabilities?
format=json-->true
    </isSupportEncryptionFileUpgradeMgr>
    <isSupportInformedConsentTextParam>
    <!--ro, opt, bool, whether the device supports text parameters of consent confirmation document,
desc:/ISAPI/System/InformedConsentMgr/InformedConsentTextParam?format=json-->true
    </isSupportInformedConsentTextParam>
    <isSupportSendConsolePassword>
    <!--ro, opt, bool, whether the device supports applying control panel password, desc:POST /ISAPI/System/SendConsolePassword?format=json-->true
    </isSupportSendConsolePassword>
    <isSupportAutoLaserRanging>
    <!--ro, opt, bool, whether to support auto laser ranging parameter configuration, desc:/ISAPI/System/AutoLaserRanging?format=json-->true
    </isSupportAutoLaserRanging>
    <isSupportDelayedReboot>
    <!--ro, opt, bool, whether the device supports delayed reboot after power disconnection, desc:POST /ISAPI/System/DelayedReboot?format=json-->true
    </isSupportDelayedReboot>
    <isSupportFalseAlarmMaterialLibMgr>
    <!--ro, opt, bool, whether the device supports false alarm library management, desc:GET /ISAPI/Intelligent/falseAlarmMaterialLibMgr/capabilities?
format=json-->true
    </isSupportFalseAlarmMaterialLibMgr>
    <isSupportGetServerRandom>
    <!--ro, opt, bool, whether the device supports getting random number of Chinese cryptographic server, desc:POST
/ISAPI/Security/ProtocolCertificate/GetServerRandom?format=json-->true
    </isSupportGetServerRandom>
    <isSupportCustomPreviewSplitCfg>
    <!--ro, opt, bool-->true
    </isSupportCustomPreviewSplitCfg>
    <isSupportSendTelnetTestCmd>
    <!--ro, opt, bool, whether the device supports applying telnet testing command, desc:POST /ISAPI/System/Network/NetworkAdvancedCfg/SendTelnetTestCmd?
format=json-->true
    </isSupportSendTelnetTestCmd>
    <isSupportTelnetTestCmdStatus>
    <!--ro, opt, bool, whether the device supports the execution status of telnet testing command, desc:GET
/ISAPI/System/Network/NetworkAdvancedCfg/TelnetTestCmdStatus?format=json-->true
    </isSupportTelnetTestCmdStatus>
    <isSupportImportUSBKeyAuthInfo>
    <!--ro, opt, bool, whether the device supports Chinese cryptographic USBKey authentication information, desc:POST
/ISAPI/Security/ProtocolCertificate/ImportUSBKeyAuthInfo?format=json-->true
    </isSupportImportUSBKeyAuthInfo>
    <isSupportSendSupervisionRecordCollection>
    <!--ro, opt, bool, whether the device supports sending completing status of supervision recording replenishment, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SendSupervisionRecordCollection?format=json-->true
    </isSupportSendSupervisionRecordCollection>
    <isSupportSearchPersonnelOperationSupervisionTask>
    <!--ro, opt, bool, whether it supports searching for operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchPersonnelOperationSupervisionTask?format=json-->true
    </isSupportSearchPersonnelOperationSupervisionTask>
    <isSupportDeletePersonnelOperationSupervisionTask>
    <!--ro, opt, bool, whether it supports deleting operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/DeletePersonnelOperationSupervisionTask?format=json-->true
    </isSupportDeletePersonnelOperationSupervisionTask>
    <isSupportAddPersonnelOperationSupervisionTask>
    <!--ro, opt, bool, whether it supports adding operation supervision tasks, desc:POST
/ISAPI/Intelligent/channels/<channelID>/AddPersonnelOperationSupervisionTask?format=json-->true
    </isSupportAddPersonnelOperationSupervisionTask>
    <isSupportUserLoginStatusChangeEvent>
    <!--ro, opt, bool, whether the device supports the UserLoginStatusChangeEvent event, desc:eventType: UserLoginStatusChangeEvent-->true
    </isSupportUserLoginStatusChangeEvent>
    <isSupportObstacleAvoidance>
    <!--ro, opt, bool, whether it supports obstacle avoidance management, desc:GET/PUT /ISAPI/System/ObstacleAvoidance/-->true
    </isSupportObstacleAvoidance>
    <isSupportManualSnapFishEyeWallPanoramaView>
    <!--ro, opt, bool, whether the device supports manually capturing expanded panorama images by wall-mounted fisheye, desc:GET
/ISAPI/Streaming/channels/<channelID>/ManualSnapFishEyeWallPanoramaView?format=json-->true
    </isSupportManualSnapFishEyeWallPanoramaView>
    <isSupportPreviewSplitSwitchCfg>
    <!--ro, opt, bool, whether the device supports custom parameters of divided window switch, desc:GET/PUT /ISAPI/System/Video/PreviewSplitSwitchCfg?
format=json-->true
    </isSupportPreviewSplitSwitchCfg>
    <isSupportCountingTargetParam>
    <!--ro, opt, bool, whether the device supports target parameters of people counting, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/counting/CountingTargetParam?format=json-->true
    </isSupportCountingTargetParam>
    <isSupportDebugLogModuleType opt="playService,communicationService,attendanceService,faceService,LCD">
    <!--ro, opt, bool, whether the device supports exporting debug log by module, attr:opt(req, string)-->true
    </isSupportDebugLogModuleType>
    <isSupportLoginPasswordCfg>
    <!--ro, opt, bool, whether the device supports login password parameters, desc:GET/PUT /ISAPI/Security/users/LoginPasswordCfg?format=json-->true
    </isSupportLoginPasswordCfg>
    <isSupportUserDataFile>
    <!--ro, opt, bool, whether the device supports importing and exporting the user data file, desc:POST /ISAPI/Security/users/UserDataFile/capabilities?
format=json-->true
    </isSupportUserDataFile>
    <isSupportNtripAccessCfg>
    <!--ro, opt, bool, whether the device supports NTRIP protocol access parameters, desc:GET/PUT /ISAPI/System/Network/Ntrip/NtripAccessCfg?format=json--
>true
    </isSupportNtripAccessCfg>
    <isSupportNetworkWatchdogParams>
    <!--ro, opt, bool, whether the device supports parameters of watchdog for network performance, desc:GET/PUT /ISAPI/System/NetworkWatchdogParams?
format=json-->true
    </isSupportNetworkWatchdogParams>
    <isSupportAudioRecordPlan>
    <!--ro, opt, bool, whether it supports recording schedule, desc:GET/PUT /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json-->true
    </isSupportAudioRecordPlan>

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</isSupportStartVoiceMonitor>
  <!--ro, opt, bool, whether it supports enabling audio listening, desc:POST
/ISAPI/System/TwoWayAudio/channels/channels/<twoWayAudioChannelID>/StartVoiceMonitor?format=json-->true
</isSupportStartVoiceMonitor>
<isSupportWebPTZCfg>
  <!--ro, opt, bool, whether it supports displaying the web page for PTZ configuration, desc:this node is to determine whether to display the web page for
PTZ configuration and is irrelevant to whether PTZ is supported; default process logic: when this node does not exist, whether to display the web page for
PTZ configuration is subject to the device type-->true
</isSupportWebPTZCfg>
<isSupportGaugeReadingEvent>
  <!--ro, opt, bool, whether the device supports the GaugeReadingEvent event, desc:eventType: GaugeReadingEvent-->true
</isSupportGaugeReadingEvent>
<isSupportVNCServerCfg>
  <!--ro, opt, bool, whether it supports VNC service configuration, desc:GET/PUT /ISAPI/System/VNCServerCfg?format=json-->true
</isSupportVNCServerCfg>
<isSupportResources>
  <!--ro, opt, bool, whether it supports resources, desc:GET /ISAPI/System/DynamicResource/Resources?format=json, Resources-->true
</isSupportResources>
<isSupportDeviceGroupMgr>
  <!--ro, opt, bool, whether it supports device application resources group management, desc:GET /ISAPI/System/DeviceGroupMgr/capabilities?format=json--
>true
</isSupportDeviceGroupMgr>
<isSupportAsyncResetSubDevicePassword>
  <!--ro, opt, bool, whether the device supports resetting the password of sub-device asynchronously,
desc:/ISAPI/Security/resetPassword/AsyncResetSubDevicePassword?format=json-->true
</isSupportAsyncResetSubDevicePassword>
<isSupportAsyncResetSubDevicePasswordStatusList>
  <!--ro, opt, bool, whether the device supports the status list of asynchronously resetting sub-devices' passwords,
desc:/ISAPI/Security/resetPassword/AsyncResetSubDevicePasswordStatusList?format=json-->true
</isSupportAsyncResetSubDevicePasswordStatusList>
<isSupportEDRAccessList>
  <!--ro, opt, bool, whether the device supports accessing address parameters of EDR service, desc:GET/PUT /ISAPI/System/Network/EDRAccessList?
format=json-->true
</isSupportEDRAccessList>
<isSupportBatteryCarInfoEvent>
  <!--ro, opt, bool, whether to support the BatteryCarInfoEvent event, desc:eventType: BatteryCarInfoEvent-->true
</isSupportBatteryCarInfoEvent>
<isSupportChildmanage>
  <!--ro, opt, bool, whether the device supports gateway sub-device topology management, desc:GET /ISAPI/IoTGateway/Childmanage/capabilities?format=json--
>true
</isSupportChildmanage>
<isSupportSearchResources>
  <!--ro, opt, bool, whether the device supports searching the resource list, desc:POST /ISAPI/System/DynamicResource/SearchResources?format=json-->true
</isSupportSearchResources>
<isSupportResetPasswordPhoneNo>
  <!--ro, opt, bool, whether the device supports the phone No. for resetting password, desc:GET/PUT /ISAPI/Security/resetPassword/ResetPasswordPhoneNo?
format=json-->true
</isSupportResetPasswordPhoneNo>
<isSupportGetPhoneNoQRCode>
  <!--ro, opt, bool, whether the device supports getting the QR code to be scanned by phone for resetting password, desc:GET
/ISAPI/Security/resetPassword/GetPhoneNoQRCode?format=json-->true
</isSupportGetPhoneNoQRCode>
<isSupportResetPasswordByPhoneNoSecurityCode>
  <!--ro, opt, bool, whether it supports resetting device password with phone number verification, desc:PUT
/ISAPI/Security/resetPassword/ResetPasswordByPhoneNoSecurityCode?format=json-->true
</isSupportResetPasswordByPhoneNoSecurityCode>
<isSupportGaugeReadingOverlapParams>
  <!--ro, opt, bool, whether the device supports water level gauge detection information overlay parameters, desc:GET/PUT
/ISAPI/Event/gaugeReading/channels/<channelID>/GaugeReadingOverlapParams?format=json-->true
</isSupportGaugeReadingOverlapParams>
<isSupportGaugeReadingRainfallEvent>
  <!--ro, opt, bool, whether the device supports the GaugeReadingRainfallEvent event, desc:eventType: GaugeReadingRainfallEvent-->true
</isSupportGaugeReadingRainfallEvent>
<isSupportSearchTrafficLightMalfunctionResult>
  <!--ro, opt, bool, whether the device supports searching for traffic light fault detection result, desc:POST
/ISAPI/Event/TrafficLightMalfunctionDetection/SearchTrafficLightMalfunctionResult?format=json-->true
</isSupportSearchTrafficLightMalfunctionResult>
<isSupportCheckPhoneNoSecurityCode>
  <!--ro, opt, bool, whether the device supports phone security code verification, desc:PUT /ISAPI/Security/resetPassword/CheckPhoneNoSecurityCode?
format=json-->true
</isSupportCheckPhoneNoSecurityCode>
<isSupportUserValidExpired>
  <!--ro, opt, bool, whether the device supports the UserValidExpired event, desc:eventType: UserValidExpired-->true
</isSupportUserValidExpired>
<isSupportSearchWaterLevelData>
  <!--ro, opt, bool, whether the device supports searching for water level detection data, desc:POST
/ISAPI/Event/waterQualityDetection/SearchWaterLevelData?format=json-->true
</isSupportSearchWaterLevelData>
<isSupportSearchRecommendPerspective>
  <!--ro, opt, bool, whether the device supports searching for recommended FoV, desc:POST
/ISAPI/Streaming/channels/channels/<channelID>/SearchRecommendPerspective?format=json-->true
</isSupportSearchRecommendPerspective>
<isSupportGetLegendaryPerspectiveInfo>
  <!--ro, opt, bool, whether the device supports getting the information of camera FoV, desc:GET
/ISAPI/Streaming/channels/channels/<channelID>/GetLegendaryPerspectiveInfo?format=json-->true
</isSupportGetLegendaryPerspectiveInfo>
<configurationDataType opt="OSD,MODE,IMAGE,IOOUT,OTHERS">
  <!--ro, opt, string, "OSD"-import video and image OSD settings, "MODE"-import application mode, "IMAGE"-import image parameters, "IOOUT"-import
supplement Light Parameters, "OTHERS", attr:opt{req, string)-->test
</configurationDataType>
<isSupportRestoreVideoAdjustmentParam>
  <!--ro, opt, bool, whether the device supports restoring video image setting parameters, desc:GET/PUT
/ISAPI/System/Video/channels/<channelID>/RestoreVideoAdjustmentParam?format=json-->true

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</isSupportRestoreVideoAdjustmentParam>
<isSupportWeatherStationInfo>
  <!--ro, opt, bool, whether the device supports weather station information, desc:GET/PUT /ISAPI/Traffic/channels/<channelID>/WeatherStationInfo?
format=json-->true
</isSupportWeatherStationInfo>
<isSupportRadarPerimeterAlgCap>
  <!--ro, opt, bool, whether the device supports the capability of radar perimeter algorithm parameters, desc:GET
/ISAPI/Smart/RadarPerimeterAlg/capabilities?format=json-->true
</isSupportRadarPerimeterAlgCap>
<isSupportBodyCameraOneKeyAlarmEvent>
  <!--ro, opt, bool, whether the device supports the BodyCameraOneKeyAlarmEvent event, desc:eventType: BodyCameraOneKeyAlarmEvent; this node is not
returned if not supported-->true
</isSupportBodyCameraOneKeyAlarmEvent>
<isSupportGetOnlineTrafficLight>
  <!--ro, opt, bool, whether the device supports getting online traffic light information, desc:GET
/ISAPI/Event/TrafficLightMalfunctionDetection/GetOnlineTrafficLight?format=json-->true
</isSupportGetOnlineTrafficLight>
<isSupportTrafficLightMalfunctionDefaultParams>
  <!--ro, opt, bool, whether the device supports default parameters of traffic light fault detection rules, desc:GET/PUT
/ISAPI/Event/TrafficLightMalfunctionDetection/TrafficLightMalfunctionDefaultParams?format=json-->true
</isSupportTrafficLightMalfunctionDefaultParams>
<isSupportFallenObjectsOnTracksDetection>
  <!--ro, opt, bool, whether the device supports detection of fallen objects on tracks, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FallenObjectsOnTracksDetection/capabilities?format=json-->true
</isSupportFallenObjectsOnTracksDetection>
<isSupportMonitorAuthorization>
  <!--ro, opt, bool, whether the device supports device service monitoring authorization information, desc:GET
/ISAPI/Security/DeviceServiceMonitor/MonitorAuthorization?format=json-->true
</isSupportMonitorAuthorization>
<isSupportBodyCameraTakeEvent>
  <!--ro, opt, bool, whether the device supports the BodyCameraTakeEvent event, desc:eventType: BodyCameraTakeEvent; this node is not returned if not
supported-->true
</isSupportBodyCameraTakeEvent>
<isSupportCollaborationModeCfg>
  <!--ro, opt, bool, whether the device supports configuring multi-lens coordination mode, desc:GET/PUT /ISAPI/Intelligent/CollaborationModeCfg?
format=json-->true
</isSupportCollaborationModeCfg>
<isSupportAudioChannelVCAResource>
  <!--ro, opt, bool, whether the device supports switching intelligent resources running on audio channels, desc:GET/PUT
/ISAPI/System/VCAResource/channels/<audioID>/AudioChannelVCAResource?format=json-->true
</isSupportAudioChannelVCAResource>
<isSupportGasTankMonitoringEvent>
  <!--ro, opt, bool, whether the device supports the GasTankMonitoringEvent event, desc:eventType: GasTankMonitoringEvent-->true
</isSupportGasTankMonitoringEvent>
<isSupportChannelClearEvent>
  <!--ro, opt, bool, whether the device supports the ChannelClearEvent event, desc:eventType: ChannelClearEvent-->true
</isSupportChannelClearEvent>
<isSupportBackDuty>
  <!--ro, opt, bool, whether the device supports the backDuty event, desc:eventType: backDuty-->true
</isSupportBackDuty>
<isSupportSleepDuty>
  <!--ro, opt, bool, whether the device supports the sleepDuty event, desc:eventType: sleepDuty-->true
</isSupportSleepDuty>
<isSupportSearchChildDevResourcesStatus>
  <!--ro, opt, bool, whether the device supports searching for sub-device resource status, desc:POST
/ISAPI/System/DynamicResource/SearchChildDevResourcesStatus?format=json-->true
</isSupportSearchChildDevResourcesStatus>
<isSupportDeleteResources>
  <!--ro, opt, bool, whether the device supports deleting resources, desc:PUT /ISAPI/System/DynamicResource/DeleteResources?format=json-->true
</isSupportDeleteResources>
<isSupportMenuFunctionList>
  <!--ro, opt, bool, whether the device supports the menu list, desc:GET /ISAPI/Thermal/MenuFunctionList?format=json-->true
</isSupportMenuFunctionList>
<isSupportKeyFunctionList>
  <!--ro, opt, bool, whether the device supports the button list, desc:GET /ISAPI/Thermal/KeyFunctionList?format=json-->true
</isSupportKeyFunctionList>
<isSupportKeyValueList>
  <!--ro, opt, bool, whether the device supports button key list, desc:GET /ISAPI/Thermal/KeyValueList?format=json-->true
</isSupportKeyValueList>
<isSupportGetPseudoColorBackgroundPic>
  <!--ro, opt, bool, whether the device supports 3-channel palette background picture, desc:GET
/ISAPI/Streaming/channels/<channelID>/GetPseudoColorBackgroundPic?format=json-->true
</isSupportGetPseudoColorBackgroundPic>
<isSupportElevatorRunningDetection>
  <!--ro, opt, bool, whether the device supports elevator running status detection, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ElevatorRunningDetection/capabilities?format=json-->true
</isSupportElevatorRunningDetection>
<isSupportSearchPersonQueueRealTimeResult>
  <!--ro, opt, bool, whether the device supports searching for history records of person waiting detection event, desc:POST
/ISAPI/Intelligent/personQueueDetection/SearchPersonQueueRealTimeResult?format=json-->true
</isSupportSearchPersonQueueRealTimeResult>
<isSupportSearchLeavePositionEventResult>
  <!--ro, opt, bool, whether the device supports searching for history records of absence detection event, desc:POST
/ISAPI/Intelligent/BehaviorAnalysis/SearchLeavePositionEventResult?format=json-->true
</isSupportSearchLeavePositionEventResult>
<isSupportImportElectronicBadgeBackgroundImage>
  <!--ro, opt, bool, whether the device supports importing the background picture of nameplate, desc:POST
/ISAPI/Traffic/ImportElectronicBadgeBackgroundImage?format=json-->true
</isSupportImportElectronicBadgeBackgroundImage>
<isSupportVoiceMonitoringEvent>
  <!--ro, opt, bool, whether the device supports uploading the VoiceMonitoringEvent event, desc:eventType: VoiceMonitoringEvent-->true
</isSupportVoiceMonitoringEvent>
<isSupportPseudoColorBackgroundParam>
  <!--ro, opt, bool, whether the device supports 3-channel palette background parameters, desc:GET/PUT
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<!--ro, opt, bool, whether the device supports 3-channel picture background parameters, desc:GET/PUT
/ISAPI/Streaming/channels/<channelID>/PseudoColorBackgroundParam?format=json-->true
</isSupportPseudoColorBackgroundParam>
<isSupportGotoPerimeterShieldMaskScene>
<!--ro, opt, bool, whether the device supports jumping to shielded scenes of specified radar perimeter, desc:POST
/ISAPI/Smart/RadarPerimeterAlg/GotoPerimeterShieldMaskScene?format=json-->true
</isSupportGotoPerimeterShieldMaskScene>
<isSupportSearchPersonQueueTimingStatistics>
<!--ro, opt, bool, whether the device supports the SearchPersonQueueTimingStatistics event, desc:POST
/ISAPI/Intelligent/personQueueDetection/SearchPersonQueueTimingStatistics?format=json-->true
</isSupportSearchPersonQueueTimingStatistics>
<isSupportLocalUserChange>
<!--ro, opt, bool, whether the device supports the LocalUserChange event, desc:eventType: LocalUserChange-->true
</isSupportLocalUserChange>
<isSupportSendClientTask>
<!--ro, opt, bool, whether the device supports applying tasks configured on the client, desc:POST /ISAPI/System/RailwayPatrol/SendClientTask?
format=json-->true
</isSupportSendClientTask>
<isSupportGetCurrentClientTaskStatus>
<!--ro, opt, bool, whether the device supports getting the current execution progress of the client task, desc:GET
/ISAPI/System/RailwayPatrol/GetCurrentClientTaskStatus?format=json-->true
</isSupportGetCurrentClientTaskStatus>
<isSupportTestEDRServerConnect>
<!--ro, opt, bool, whether the device supports testing the connection status of the EDR platform, desc:POST /ISAPI/System/Network/TestEDRServerConnect?
format=json-->true
</isSupportTestEDRServerConnect>
<isSupportRealTimeStreamInfo>
<!--ro, opt, bool, whether the device supports getting the real-time stream information, desc:GET /ISAPI/Streaming/RealTimeStreamInfo?format=json-->true
</isSupportRealTimeStreamInfo>
<isSupportIrcutFilterStatusChange>
<!--ro, opt, bool, whether the device supports the IrcutFilterStatusChange event, desc:eventType: IrcutFilterStatusChange-->true
</isSupportIrcutFilterStatusChange>
<isSupportSendRoadMaintPictureAnalysisTask>
<!--ro, opt, bool, whether the device supports applying tasks of road maintenance picture analysis, desc:POST
/ISAPI/Intelligent/SendRoadMaintPictureAnalysisTask?format=json-->true
</isSupportSendRoadMaintPictureAnalysisTask>
<isSupportFDRemoveDuplicateParams>
<!--ro, opt, bool, whether the device supports deduplication parameters of custom face picture library, desc:GET/PUT
/ISAPI/System/Video/inputs/counting/FDRemoveDuplicateParams?format=json-->true
</isSupportFDRemoveDuplicateParams>
<isSupportRadarMixedTargetSensitivityParams>
<!--ro, opt, bool, whether the device supports sensitivity parameters of multi-target-type detection via radar, desc:GET/PUT
/ISAPI/Smart/RadarMixedTargetAlg/RadarMixedTargetSensitivityParams?format=json-->true
</isSupportRadarMixedTargetSensitivityParams>
<isSupportMixedTargetRadarSensitivityParams>
<!--ro, opt, bool, whether the device supports sensitivity parameters of multi-target-type detection via video, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/MixedTargetRadarSensitivityParams?format=json-->true
</isSupportMixedTargetRadarSensitivityParams>
<isSupportMixedTargetRadarDebugParams>
<!--ro, opt, bool, whether the device supports algorithm debugging parameters of multi-target-type detection via video, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/MixedTargetRadarDebugParams?format=json-->true
</isSupportMixedTargetRadarDebugParams>
<isSupportHttpPreview>
<!--ro, opt, bool, whether the device supports HTTP live view,
desc:related URI: GET /ISAPI/Streaming/channels/<trackStreamID>/httppreview;
there are 2 steps as follows:
1. check if this node is returned by device and if the returned value is true. If yes, the HTTP live view function is supported by device.
2. if this node is not returned by device, you should go through the stream ID in the API (GET /ISAPI/Streaming/channels/<trackStreamID>/httppreview) of all
channels. If notSupport is returned for all channels, HTTP live view is not supported by device; if at least the success code is returned for at least one
channel, HTTP live view is supported by device.
Note: whether the stream ID (trackStreamID) is supported is determined by the response parameters StreamingChannelList, StreamingChannel, and id from GET
/ISAPI/Streaming/channels-->true
</isSupportHttpPreview>
<isSupportGetRecordThumbnail>
<!--ro, opt, bool, whether the device supports getting the video file thumbnail, desc:POST /ISAPI/ContentMgmt/search/GetRecordThumbnail?format=json--
>true
</isSupportGetRecordThumbnail>
<isSupportProxyAIOP>
<!--ro, opt, bool, whether the device supports proxy access of AIOP (AIOP devices connected with network cameras via NVR transparent transmission),
desc:when this node is supported, full AIOP functions (device AIOP capability, model package management, task rule management, detection data, etc.) can be
accessed via proxy transmission of ISAPI sub-device-->true
</isSupportProxyAIOP>
<isSupportStaticObjectPositionOffsetComparison>
<!--ro, opt, bool, whether the device supports position deviation comparison of static objects, desc:GET
/ISAPI/Intelligent/channels/<channelID>/StaticObjectPositionOffsetComparison/capabilities?format=json-->true
</isSupportStaticObjectPositionOffsetComparison>
<isSupportClearFDRemoveDuplicateFDLib>
<!--ro, opt, bool, whether the device supports manually clearing the custom face picture library on device, desc:POST
/ISAPI/System/Video/inputs/counting/ClearFDRemoveDuplicateFDLib?format=json-->true
</isSupportClearFDRemoveDuplicateFDLib>
<isSupportDevPowerStatusReportCtrl>
<!--ro, opt, bool, whether the device supports control parameters of device battery status upLoad, desc:GET/PUT /ISAPI/System/DevPowerStatusReportCtrl?
format=json-->true
</isSupportDevPowerStatusReportCtrl>
<isSupportIndicatorLight>
<!--ro, opt, bool, whether the device supports indicator light configuration, desc:GET /ISAPI/System/IndicatorLight/capabilities?format=json-->true
</isSupportIndicatorLight>
<isSupportEnvironmentTemperatureMonitorEvent>
<!--ro, opt, bool, whether the device supports the EnvironmentTemperatureMonitorEvent event, desc:eventType: EnvironmentTemperatureMonitorEvent-->true
</isSupportEnvironmentTemperatureMonitorEvent>
<isSupportPhysicalTraining>
<!--ro, opt, bool, whether the device supports physical training, desc:GET /ISAPI/Intelligent/channels/<channelID>/PhysicalTraining/capabilities?
format=json-->true
</isSupportPhysicalTraining>

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<isSupportPullUp>
  <!--ro, opt, bool, whether the device supports the PullUp event, desc:eventType: PullUp-->true
</isSupportPullUp>
<isSupportSitUpWithHandsBehindHead>
  <!--ro, opt, bool, whether the device supports the SitUpWithHandsBehindHead event, desc:eventType: SitUpWithHandsBehindHead-->true
</isSupportSitUpWithHandsBehindHead>
<isSupportSitUpWithArmsOnChest>
  <!--ro, opt, bool, whether the device supports the SitUpWithArmsOnChest event, desc:eventType: SitUpWithArmsOnChest-->true
</isSupportSitUpWithArmsOnChest>
<isSupportPushUp>
  <!--ro, opt, bool, whether the device supports the PushUp event, desc:eventType: PushUp-->true
</isSupportPushUp>
<isSupportPullUpHold>
  <!--ro, opt, bool, whether the device supports the PullUpHold event, desc:eventType: PullUpHold-->true
</isSupportPullUpHold>
<isSupportVerificationPlatformParams>
  <!--ro, opt, bool, whether the device supports accessing parameters for the verification and recording platform, desc:GET/PUT
  /ISAPI/System/Network/VerificationPlatformAccess/VerificationPlatformParams?format=json-->true
</isSupportVerificationPlatformParams>
<isSupportVehicleMaintenanceDetection>
  <!--ro, opt, bool, whether the device supports vehicle maintenance detection, desc:GET /ISAPI/Intelligent/VehicleMaintenanceDetection/capabilities?
  format=json; detects the status of vehicles in the maintenance shop and triggers the alarm when the vehicle is Lifted (when detecting chassis) or Lowered,
  or the hood is opened-->true
</isSupportVehicleMaintenanceDetection>
<isSupportPhysicalTrainingStatus>
  <!--ro, opt, bool, whether the device supports the PhysicalTrainingStatus event, desc:eventType: PhysicalTrainingStatus-->true
</isSupportPhysicalTrainingStatus>
<isSupportDeviceAbnormalDebug>
  <!--ro, opt, bool, whether the device supports device operation exception debugging, desc:GET /ISAPI/System/DeviceAbnormalDebug/capabilities?
  format=json; during device operation exception, the device operation debugging data (debugging log, dump file, etc.) will be collected by technical support-
  -->true
</isSupportDeviceAbnormalDebug>
<isSupportSetWaterLevelResetOfWaterTank>
  <!--ro, opt, bool, whether the device supports manually filling up the water tank and uploading the filled tank information for calculating remaining
  water via Lens wiper usage, desc:PUT /ISAPI/System/Hardware/SetWaterLevelResetOfWaterTank?format=json-->true
</isSupportSetWaterLevelResetOfWaterTank>
<isSupportGetRemoteAlbumPicture>
  <!--ro, opt, bool, whether the device supports getting pictures from the remote album, desc:POST /ISAPI/ContentMgmt/GetRemoteAlbumPicture?format=json--
  -->true
</isSupportGetRemoteAlbumPicture>
<isSupportUltravioletImagingFusionParams>
  <!--ro, opt, bool, whether the device supports UV imaging fusion parameters, desc:GET/PUT
  /ISAPI/System/Video/channels/<channelID>/UltravioletImagingFusionParams?format=json-->true
</isSupportUltravioletImagingFusionParams>
<isSupportAudioAnalysisCfg>
  <!--ro, opt, bool, whether it supports audio analysis parameters, desc:GET/PUT
  /ISAPI/Intelligent/SensitiveInfoAnalysis/AudioIn/<AudioInID>/AudioAnalysisCfg?format=json-->true
</isSupportAudioAnalysisCfg>
<isSupportRunningLogPlan>
  <!--ro, opt, bool, whether the device supports device operation log schedule, desc:GET/PUT /ISAPI/ContentMgmt/RunningLogPlan?format=json-->true
</isSupportRunningLogPlan>
<isSupportSearchRunningLogData>
  <!--ro, opt, bool, whether the device supports searching for device operation log data, desc:POST /ISAPI/ContentMgmt/SearchRunningLogData?format=json--
  -->true
</isSupportSearchRunningLogData>
<isSupportExportRunningLogData>
  <!--ro, opt, bool, whether the device supports exporting device operation log data, desc:POST /ISAPI/ContentMgmt/ExportRunningLogData?format=json-->true
</isSupportExportRunningLogData>
<isSupportCertificateCaptureEvent>
  <!--ro, opt, bool, whether the device supports the CertificateCaptureEvent event, desc:corresponding event: CertificateCaptureEvent-->true
</isSupportCertificateCaptureEvent>
<isSupportUpgradeKeywordModel>
  <!--ro, opt, bool, whether the device supports upgrading the sensitive information keyword model, desc:PUT
  /ISAPI/Intelligent/SensitiveInfoAnalysis/UpgradeKeywordModel?format=json-->true
</isSupportUpgradeKeywordModel>
<isSupportGetKeywordModelStatus>
  <!--ro, opt, bool, whether the device supports getting algorithm running status of the sensitive information keyword model, desc:GET
  /ISAPI/Intelligent/SensitiveInfoAnalysis/GetKeywordModelStatus?format=json-->true
</isSupportGetKeywordModelStatus>
<isSupportCloseLensPrivacyCover>
  <!--ro, opt, bool, whether the device supports manually closing the lens privacy cap, desc:PUT /ISAPI/System/channels/<channelID>/CloseLensPrivacyCover?
  format=json-->true
</isSupportCloseLensPrivacyCover>
<isSupportOpenLensPrivacyCover>
  <!--ro, opt, bool, whether the device supports manually opening the lens privacy cap, desc:PUT /ISAPI/System/channels/<channelID>/OpenLensPrivacyCover?
  format=json-->true
</isSupportOpenLensPrivacyCover>
<isSupportLensPrivacyCoverModeParams>
  <!--ro, opt, bool, whether the device supports working mode parameters of the lens privacy cap, desc:GET/PUT
  /ISAPI/System/channels/<channelID>/LensPrivacyCoverModeParams?format=json-->true
</isSupportLensPrivacyCoverModeParams>
<isSupportOTAPService>
  <!--ro, opt, bool, whether the device supports OTAP service configuration, desc:/ISAPI/System/Network/OTAPService/capabilities?format=json-->true
</isSupportOTAPService>
<isSupportGetPlaybackStreamInfo>
  <!--ro, opt, bool, whether the device supports getting video playback stream information, desc:GET /ISAPI/Streaming/GetPlaybackStreamInfo?format=json--
  -->true
</isSupportGetPlaybackStreamInfo>
<isSupportNBioTPlatformAccessStatus>
  <!--ro, opt, bool, whether the device supports the NB-IoT platform access status, desc:GET /ISAPI/System/Network/NBioTPlatformAccessStatus?format=json-
  -->true
</isSupportNBioTPlatformAccessStatus>
<isSupportScannedListEvent>

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    <!--ro, opt, bool, whether the device supports Bluetooth scanning List event, desc:eventType: ScannedListEvent-->true
  </isSupportScannedListEvent>
  <isSupportDownloadAudioRecordByTime>
    <!--ro, opt, bool, whether the device supports downloading audio files by time, desc:POST /ISAPI/System/RecordAudio/DownloadAudioRecordByTime?
format=json-->true
  </isSupportDownloadAudioRecordByTime>
  <isSupportBackUpConfigurationFile>
    <!--ro, opt, bool, whether the device supports local backup of the configuration file, desc:PUT /ISAPI/System/BackUpConfigurationFile?format=json-->true
  </isSupportBackUpConfigurationFile>
  <isSupportRestoreConfigurationFile>
    <!--ro, opt, bool, whether the device supports restoring the device Local configuration file, desc:PUT /ISAPI/System/RestoreConfigurationFile?
format=json-->true
  </isSupportRestoreConfigurationFile>
  <isSupportGetCfgFileBackupInfo>
    <!--ro, opt, bool, whether the device supports getting the Local configuration file backup information, desc:GET /ISAPI/System/GetCfgFileBackupInfo?
format=json-->true
  </isSupportGetCfgFileBackupInfo>
  <isSupportSearchVideoCodeMealParams>
    <!--ro, opt, bool, whether the device supports searching for video encoding plan parameters, desc:POST
/ISAPI/Streaming/channels/<channelID>/SearchVideoCodeMealParams?format=json-->true
  </isSupportSearchVideoCodeMealParams>
  <isSupportVideoCodeMealParams>
    <!--ro, opt, bool, whether the device supports video encoding plan parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/VideoCodeMealParams?
format=json-->true
  </isSupportVideoCodeMealParams>
  <isSupportFlameCombustionStatusRuleParams>
    <!--ro, opt, bool, whether the device supports flame burning status detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusRuleParams?format=json-->true
  </isSupportFlameCombustionStatusRuleParams>
  <isSupportFlameCombustionStatusOverlapParams>
    <!--ro, opt, bool, whether the device supports flame burning status detection information overlay parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusOverlapParams?format=json-->true
  </isSupportFlameCombustionStatusOverlapParams>
  <isSupportAcousticLeakDetection>
    <!--ro, opt, bool, whether the device supports acoustic leak detection capability, desc:related URL: /ISAPI/System/AcousticLeakDetection/capabilities?
format=json-->true
  </isSupportAcousticLeakDetection>
  <isSupportFlameCombustionStatusSceneList>
    <!--ro, opt, bool, whether the device supports flame burning status detection scene list, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusSceneList?format=json-->true
  </isSupportFlameCombustionStatusSceneList>
  <isSupportGasLeakageDetection>
    <!--ro, opt, bool, whether the device supports gas leak event, desc:eventType: GasLeakageDetection-->true
  </isSupportGasLeakageDetection>
  <isSupportDischargeDetection>
    <!--ro, opt, bool, whether the device supports discharging event, desc:eventType: DischargeDetection-->true
  </isSupportDischargeDetection>
  <isSupportSearchBaseMapIDList>
    <!--ro, opt, bool, whether the device supports searching for base map ID list, desc:POST /ISAPI/ContentMgmt/SearchBaseMapIDList?format=json-->true
  </isSupportSearchBaseMapIDList>
  <isSupportRestoreDefaultParam>
    <!--ro, opt, bool, whether the device supports restoring default parameters, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?format=json-->true
  </isSupportRestoreDefaultParam>
  <isSupportSearchEventTargetInfo>
    <!--ro, opt, bool, whether the device supports searching for event detection target information in FoV, desc:GET
/ISAPI/System/Video/inputs/channels/<channelID>/SearchEventTargetInfo?format=json-->true
  </isSupportSearchEventTargetInfo>
  <isSupportElevatorRunningDetectionEvent>
    <!--ro, opt, bool, whether the device supports elevator running event, desc:eventType: ElevatorRunningDetectionEvent-->true
  </isSupportElevatorRunningDetectionEvent>
  <isSupportElevatorOperationStatistics>
    <!--ro, opt, bool, whether the device supports elevator running statistics event, desc:eventType: ElevatorOperationStatistics-->true
  </isSupportElevatorOperationStatistics>
  <isSupportStandingLongJump>
    <!--ro, opt, bool, whether the device supports standing long jump event, desc:eventType: StandingLongJump-->true
  </isSupportStandingLongJump>
  <isSupportJumpRope>
    <!--ro, opt, bool, whether the device supports jumping rope event, desc:eventType: JumpRope-->true
  </isSupportJumpRope>
  <isSupportUserAvailablePwQuery>
    <!--ro, opt, bool, whether the device supports person query through password, desc:eventType:UserAvailablePwQuery-->true
  </isSupportUserAvailablePwQuery>
  <isSupportAddHistoryEventSubscribeCfg>
    <!--ro, opt, bool, whether the device supports adding history event subscription, desc:POST /ISAPI/Event/notification/AddHistoryEventSubscribeCfg?
format=json-->true
  </isSupportAddHistoryEventSubscribeCfg>
  <isSupportHistoryEventSubscribeUploadCtrl>
    <!--ro, opt, bool, whether the device supports history event subscription uploading control, desc:PUT
/ISAPI/Event/notification/HistoryEventSubscribeUploadCtrl?format=json-->true
  </isSupportHistoryEventSubscribeUploadCtrl>
  <isSupportHLSStreamCfg>
    <!--ro, opt, bool, whether the device supports HLS streaming address parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/HLSStreamCfg?
format=json-->true
  </isSupportHLSStreamCfg>
  <isSupportGetKeywordModelVersionInfo>
    <!--ro, opt, bool, whether the device supports getting sensitive information keyword model version information, desc:GET
/ISAPI/Intelligent/SensitiveInfoAnalysis/GetKeywordModelVersionInfo?format=json-->true
  </isSupportGetKeywordModelVersionInfo>
  <isSupportManualTriggerRemoteAlbumSnapshot>
    <!--ro, opt, bool, whether the device supports manually triggering remote album capture, desc:PUT /ISAPI/ContentMgmt/ManualTriggerRemoteAlbumSnapshot?
format=json-->true
  </isSupportManualTriggerRemoteAlbumSnapshot>
  <isSupportRadarVideoAssistedFocusParams>
    <!--ro, opt, bool, whether the device supports radar-assisted focusing parameters, desc:GET/PUT

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/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusParams?format=json-->true
</isSupportRadarVideoAssistedFocusParams>
<isSupportCollectionResultUpload>
  <!--ro, opt, bool, whether the device supports the uploading collected result event, desc:eventType: CollectionResultUpload-->true
</isSupportCollectionResultUpload>
<isSupportDeveloperMode>
  <!--ro, opt, bool, whether the device supports developer mode, desc:GET/PUT /ISAPI/System/DeveloperMode?format=json-->true
</isSupportDeveloperMode>
<isSupportGetDetailsFunction>
  <!--ro, opt, bool, whether the device supports getting detailed functions supported in T1 test, desc:GET /ISAPI/System/T1Test/DetailsFunction?
format=json-->true
</isSupportGetDetailsFunction>
<isSupportSearchDeviceIANAList>
  <!--ro, opt, bool, whether the device supports searching for supported IANA List, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
</isSupportSearchDeviceIANAList>
<isSupportModifySpecificROIStreamJointParams>
  <!--ro, opt, bool, whether the device supports setting ROI stream jointing parameters, desc:PUT
/ISAPI/Streaming/channels/<channelID>/ModifySpecificROIStreamJointParams?format=json-->true
</isSupportModifySpecificROIStreamJointParams>
<isSupportDeviceTamperParams>
  <!--ro, opt, bool, whether the device supports tampering parameters, desc:GET/PUT /ISAPI/System/DeviceTamperParams?format=json-->true
</isSupportDeviceTamperParams>
<isSupportDeviceTamperEvent>
  <!--ro, opt, bool, whether the device supports tampering event, desc:eventType: DeviceTamperEvent-->true
</isSupportDeviceTamperEvent>
<isSupportRoadVariableLaneCtrlParams>
  <!--ro, opt, bool, whether the device supports dynamic management parameters of changeable lane, desc:GET/PUT
/ISAPI/Traffic/RoadVariableLaneCtrl/channels/<channelID>/RoadVariableLaneCtrlParams?format=json-->true
</isSupportRoadVariableLaneCtrlParams>
<isSupportImportModelFile>
  <!--ro, opt, bool, whether the device supports importing the water quality detection model file, desc:POST
/ISAPI/Event/waterQualityDetection/ImportModelFile?format=json-->true
</isSupportImportModelFile>
<isSupportSearchImportStatus>
  <!--ro, opt, bool, whether the device supports getting water quality detection model file importing progress, desc:GET
/ISAPI/Event/waterQualityDetection/SearchImportStatus?format=json-->true
</isSupportSearchImportStatus>
<isSupportSafetyWorkwearDetectionOverlapParams>
  <!--ro, opt, bool, whether the device supports parameters of wearing safety work clothes detection information overlay, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionOverlapParams?format=json-->true
</isSupportSafetyWorkwearDetectionOverlapParams>
<isSupportSafetyWorkwearDetectionRuleParams>
  <!--ro, opt, bool, whether the device supports detection rule parameters of wearing safety work clothes, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionRuleParams?format=json-->true
</isSupportSafetyWorkwearDetectionRuleParams>
<isSupportUpdateFirmwareMode>
  <!--ro, opt, bool, whether the device supports the upgrade mode, desc:GET/PUT /ISAPI/System/UpdateFirmwareMode?format=json-->true
</isSupportUpdateFirmwareMode>
<isSupportIndoorPasswayRecovery>
  <!--ro, opt, bool, whether the device supports indoor passway block alarm restoring event, desc:eventType: IndoorPasswayRecovery#-->true
</isSupportIndoorPasswayRecovery>
<isSupportBatteryCarRecovery>
  <!--ro, opt, bool, whether the device supports electric moped parking violation alarm restoring event, desc:eventType: BatteryCarRecovery-->true
</isSupportBatteryCarRecovery>
<isSupportDetectorTemperatureRecovery>
  <!--ro, opt, bool, whether the device supports detector temperature alarm restoring event, desc:eventType: DetectorTemperatureRecovery-->true
</isSupportDetectorTemperatureRecovery>
<isSupportGetRFIDReaderWorkStatus>
  <!--ro, opt, bool, whether the device supports getting RFID read/write working status, desc:GET /ISAPI/Traffic/UHFRFIDReader/GetRFIDReaderWorkStatus?
format=json-->true
</isSupportGetRFIDReaderWorkStatus>
<isSupportRFIDReaderWorkStatusCtrl>
  <!--ro, opt, bool, whether the device supports RFID read/write working status control, desc:PUT /ISAPI/Traffic/UHFRFIDReader/RFIDReaderWorkStatusCtrl?
format=json-->true
</isSupportRFIDReaderWorkStatusCtrl>
<isSupportFlameCombustionStatusAlgorithmsVersion>
  <!--ro, opt, bool, whether the device supports getting flame burning status detection algorithm version, desc:GET
/ISAPI/Intelligent/channels/<channelID>/FlameCombustionStatusAlgorithmsVersion?format=json-->true
</isSupportFlameCombustionStatusAlgorithmsVersion>
<isSupportTestResultUpload>
  <!--ro, opt, bool, whether the device supports test result uploading event, desc:eventType: TestResultUpload-->true
</isSupportTestResultUpload>
<isSupportAutoTestParams>
  <!--ro, opt, bool, whether the device supports automation testing parameters, desc:GET/PUT /ISAPI/System/AutoTest/AutoTestParams?format=json-->true
</isSupportAutoTestParams>
<isSupportExportATInstructRecord>
  <!--ro, opt, bool, whether the device supports exporting AT command records, desc:POST /ISAPI/System/ExportATInstructRecord?format=json-->true
</isSupportExportATInstructRecord>
<isSupportOfflineVideoFileManagement>
  <!--ro, opt, bool, whether the device supports offline video file management, desc:GET /ISAPI/ContentMgmt/OfflineVideoFileManagement/capabilities?
format=json-->true
</isSupportOfflineVideoFileManagement>
<isSupportChannelOccupyScenelist>
  <!--ro, opt, bool, whether the device supports getting the scene list of fire escape occupation, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyScenelist?format=json-->true
</isSupportChannelOccupyScenelist>
<isSupportChannelOccupyAlgorithmsVersion>
  <!--ro, opt, bool, whether the device supports getting the algorithm version of fire escape occupation, desc:GET
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyAlgorithmsVersion?format=json-->true
</isSupportChannelOccupyAlgorithmsVersion>
<isSupportChannelOccupyOverlapParams>
  <!--ro, opt, bool, whether the device supports overlay parameters of fire escape occupation, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyOverlapParams?format=json-->true
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</isSupportChannelOccupancyOverlapParams>
</isSupportChannelOccupancyRuleParams>
  <!--ro, opt, bool, whether the device supports rule parameters of fire escape occupation, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupancyRuleParams?format=json-->true
</isSupportChannelOccupancyRuleParams>
</isSupportGotoChannelOccupancyScene>
  <!--ro, opt, bool, whether the device supports jumping to the specified fire escape occupation scene, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/GotoChannelOccupancyScene?format=json-->true
</isSupportGotoChannelOccupancyScene>
</isSupportChannelPassingEvent>
  <!--ro, opt, bool, whether the device supports vehicles passing outdoor fire escape, desc:eventType:channelPassingEvent-->true
</isSupportChannelPassingEvent>
</isSupportGunsHandoverRuleParams>
  <!--ro, opt, bool, whether the device supports object handover rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/GunsHandoverRuleParams?format=json-->true
</isSupportGunsHandoverRuleParams>
</isSupportGunsHandover>
  <!--ro, opt, bool, whether the device supports object handover event, desc:eventType:GunsHandover-->true
</isSupportGunsHandover>
</isSupportBulkUpgradeChildDevice>
  <!--ro, opt, bool, whether the device supports batch upgrading sub-devices, desc:POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json-->true
</isSupportBulkUpgradeChildDevice>
</isSupportSearchBulkUpgradeChildDeviceProgress>
  <!--ro, opt, bool, whether the device supports checking the progress of batch upgrading sub-devices, desc:PUT
/ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json-->true
</isSupportSearchBulkUpgradeChildDeviceProgress>
</isSupportDash>
  <!--ro, opt, bool, whether the device supports the dash event, desc:eventType:Dash-->true
</isSupportDash>
</isSupportStopRadarVideoAssistedFocusCalibrating>
  <!--ro, opt, bool, whether the device supports stopping the calibration of radar-assisted focusing, desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StopRadarVideoAssistedFocusCalibrating?format=json-->true
</isSupportStopRadarVideoAssistedFocusCalibrating>
</isSupportRadarVideoAssistedFocusCalibrating>
  <!--ro, opt, bool, whether the device supports the calibration of radar-assisted focusing, desc:GET/PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusCalibrating?format=json-->true
</isSupportRadarVideoAssistedFocusCalibrating>
</isSupportRadarVideoAssistedFocusCalibratingStatus>
  <!--ro, opt, bool, whether the device supports displaying the calibration status of radar-assisted focusing, desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoAssistedFocusCalibratingStatus?format=json-->true
</isSupportRadarVideoAssistedFocusCalibratingStatus>
</isSupportETCMatchResultUpload>
  <!--ro, opt, bool, whether the device supports uploading ETC vehicle comparison results, desc:eventType:ETCMatchResultUpload-->true
</isSupportETCMatchResultUpload>
</isSupportPartsLifeRemind>
  <!--ro, opt, bool, whether the device supports prompting the life span of device component, desc:corresponding event: PartsLifeRemind-->true
</isSupportPartsLifeRemind>
</isSupportAuthorizeMicAcquisitionParam>
  <!--ro, opt, bool, whether the device supports audio authorization information, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/AuthorizeMicAcquisitionParam?format=json-->true
</isSupportAuthorizeMicAcquisitionParam>
</isSupportTOFDetectEvent>
  <!--ro, opt, bool, whether the device supports TOF detection event, desc:eventType:TOFDetectEvent-->true
</isSupportTOFDetectEvent>
</isSupportAudioAnalysisResultOSDOverlay>
  <!--ro, opt, bool, whether the device supports OSD parameters of audio analysis results, desc:GET/PUT
/ISAPI/Intelligent/SensitiveInfoAnalysis/AudioAnalysisResultOSDOverlay?format=json-->true
</isSupportAudioAnalysisResultOSDOverlay>
</isSupportChildPasswordStatus>
  <!--ro, opt, bool, whether the device supports getting the activation password status of sub-devices, desc:GET
/ISAPI/System/activate/ChildPasswordStatus?format=json-->true
</isSupportChildPasswordStatus>
</isSupportMutexFunctionClose>
  <!--ro, opt, bool, whether the device supports quickly disabling exclusive functions, desc:POST /ISAPI/System/mutexFunctionClose?format=json-->true
</isSupportMutexFunctionClose>
</isSupportFirePointAlarmRecovery>
  <!--ro, opt, bool, whether the device supports fire detection alarm restoring event, desc:eventType:FirePointAlarmRecovery-->true
</isSupportFirePointAlarmRecovery>
</isSupportSmokeAlarmRecovery>
  <!--ro, opt, bool, whether the device supports smoke alarm restoring event, desc:eventType:FirePointAlarmRecovery-->true
</isSupportSmokeAlarmRecovery>
</isSupportTemperatureMeasurementPreviewAlarmRecovery>
  <!--ro, opt, bool, whether the device supports temperature measurement pre-alarm restoring event, desc:eventType:TemperatureMeasurementPreviewRecovery-->true
</isSupportTemperatureMeasurementPreviewAlarmRecovery>
</isSupportTemperatureMeasurementAlarmRecovery>
  <!--ro, opt, bool, whether the device supports temperature measurement alarm restoring event, desc:eventType:TemperatureMeasurementRecovery-->true
</isSupportTemperatureMeasurementAlarmRecovery>
</isSupportUVPhotonDetection>
  <!--ro, opt, bool, whether the device supports UV photon detection event, desc:eventType:UVPhotonDetection-->true
</isSupportUVPhotonDetection>
</isSupportUVPhotonTimeDetection>
  <!--ro, opt, bool, whether the device supports UV photon scheduled-detection event, desc:eventType:UVPhotonTimeDetection-->true
</isSupportUVPhotonTimeDetection>
</isSupportPatientExpensesInfoQuery>
  <!--ro, opt, bool, whether the device supports the patient cost search event, desc:eventType:PatientExpensesInfoQuery-->true
</isSupportPatientExpensesInfoQuery>
</isSupportReinforcementCallEvent>
  <!--ro, opt, bool, whether the device supports reinforcement call event, desc:eventType:ReinforcementCallEvent-->true
</isSupportReinforcementCallEvent>
</isSupportChargeMode>
  <!--ro, opt, bool, whether the device supports charging mode, desc:GET/PUT /ISAPI/System/ChargeMode?format=json-->true
</isSupportChargeMode>
</isSupportCombinationRuleParams>

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<!--ro, opt, bool, whether the device supports combination rule parameters of perimeter detection, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterCombinationDetection/CombinationRuleParams?format=json-->true
</isSupportCombinationRuleParams>
<isSupportSwitchHolidayPlanCfg>
<!--ro, opt, bool, whether the device supports the regular startup/shutdown on holiday, desc:GET/PUT /ISAPI/System/SwitchHolidayPlanCfg?format=json--
>true
</isSupportSwitchHolidayPlanCfg>
<isSupportSwitchIrregularHolidayPlanCfg>
<!--ro, opt, bool, whether the device supports irregular startup/shutdown on holiday, desc:GET/PUT /ISAPI/System/SwitchIrregularHolidayPlanCfg?
format=json-->true
</isSupportSwitchIrregularHolidayPlanCfg>
<isSupportSearchPresetConfigInfo>
<!--ro, opt, bool, whether the device supports searching for the configuration of specified presets, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchPresetConfigInfo?format=json-->true
</isSupportSearchPresetConfigInfo>
<isSupportSwitchPlanCfg>
<!--ro, opt, bool, whether the device supports startup/shutdown schedule, desc:GET/PUT /ISAPI/System/SwitchPlanCfg?format=json-->true
</isSupportSwitchPlanCfg>
<isSupportVegetationSpectrumDetection>
<!--ro, opt, bool, whether the device supports vegetation spectrum detection, desc:eventType:VegetationSpectrumDetection-->true
</isSupportVegetationSpectrumDetection>
<isSupportCaptureResultAttributeSearch>
<!--ro, opt, bool, whether the device supports searching for attributes of human body capture, desc:POST
/ISAPI/Intelligent/CaptureResultAttributeSearch?format=json-->true
</isSupportCaptureResultAttributeSearch>
<isSupportHealthEducationInfoReadStatus>
<!--ro, opt, bool, whether the device supports the event of uploading the reading status of health education information,
desc:eventType:HealthEducationInfoReadStatus-->true
</isSupportHealthEducationInfoReadStatus>
<isSupportNurseLocationEvent>
<!--ro, opt, bool, whether the device supports nurse locating event, desc:eventType:NurseLocationEvent-->true
</isSupportNurseLocationEvent>
<isSupportAddChildDevResource>
<!--ro, opt, bool-->true
</isSupportAddChildDevResource>
<isSupportModifyChildDevResource>
<!--ro, opt, bool, whether the device supports editing gateway resource information, desc:POST /ISAPI/System/DynamicResource/ModifyChildDevResource?
format=json-->true
</isSupportModifyChildDevResource>
<isSupportDiagnosisDataCollectionParams>
<!--ro, opt, bool, whether the device supports diagnosis data statistics parameters, desc:GET/PUT /ISAPI/System/DiagnosisDataCollectionParams?
format=json-->true
</isSupportDiagnosisDataCollectionParams>
<isSupportServicePlatformAccountAuth>
<!--ro, opt, bool, whether the device supports platform account authentication, desc:GET /ISAPI/Security/ServicePlatformAccountAuth/capabilities?
format=json-->true
</isSupportServicePlatformAccountAuth>
<isSupportSyncPatientInfoEvent>
<!--ro, opt, bool, whether the device supports synchronizing patient information, desc:eventType:SyncPatientInfoEvent-->true
</isSupportSyncPatientInfoEvent>
<isSupportAuthenticationNetworkCfg>
<!--ro, opt, bool, whether the device supports the authentication parameters of networking method, desc:GET/PUT
/ISAPI/Security/ProtocolCertificate/AuthenticationNetworkCfg?format=json-->true
</isSupportAuthenticationNetworkCfg>
<isSupportPersonnelChannelLogParams>
<!--ro, opt, bool, whether the device supports turnstile log parameters, desc:GET/PUT /ISAPI/ContentMgmt/PersonnelChannelLogParams?format=json-->true
</isSupportPersonnelChannelLogParams>
<isSupportElectricSaveModeParams>
<!--ro, opt, bool, whether the device supports power-saving mode parameters, desc:GET/PUT /ISAPI/System/ElectricSaveModeParams?format=json-->true
</isSupportElectricSaveModeParams>
<isSupportChannelOccupyPOSPParams>
<!--ro, opt, bool, whether the device supports overlying POS information on the outdoor access blocking detection data, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ChannelOccupyPOSPParams?format=jsonGET or PUT
/ISAPI/Intelligent/channels/<channelID>/SafetyWorkwearDetectionRuleParams?format=json-->true
</isSupportChannelOccupyPOSPParams>
<isSupportChildPowerTimePlanParams>
<!--ro, opt, bool, whether the device supports power-off schedule parameters of sub-devices, desc:GET/PUT /ISAPI/System/ChildPowerTimePlanParams?
format=json-->true
</isSupportChildPowerTimePlanParams>
<isSupportPowerIndicatorLightCtrlParams>
<!--ro, opt, bool, whether the device supports parameters of device battery indicator control, desc:GET/PUT /ISAPI/System/PowerIndicatorLightCtrlParams?
format=json-->true
</isSupportPowerIndicatorLightCtrlParams>
<isSupportRadarVideoMapDetectionParams>
<!--ro, opt, bool, whether the device supports parameters of radar-assisted map detection, desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/RadarVideoMapDetectionParams?format=json-->true
</isSupportRadarVideoMapDetectionParams>
<isSupportRadarPOSPParams>
<!--ro, opt, bool, whether the device supports overlying POS information on the radar detection data, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/RadarPOSPParams?format=json-->true
</isSupportRadarPOSPParams>
<isSupportVideoIntercomModeEvent>
<!--ro, opt, bool, whether the device supports the two-way audio mode event, desc:eventType:VideoIntercomModeEvent-->true
</isSupportVideoIntercomModeEvent>
<isSupportSoftwareAndHardwareCtrlParams>
<!--ro, opt, bool, whether the device supports parameters of hardware and software service control, desc:GET/PUT
/ISAPI/System/FunctionSwitchCtrl/SoftwareAndHardwareCtrlParams?format=json-->true
</isSupportSoftwareAndHardwareCtrlParams>
<isSupportUserLoginAbnormal>
<!--ro, opt, bool, whether the device supports user login exception information, desc:GET/PUT /ISAPI/Security/users/UserLoginAbnormal?format=json-->true
</isSupportUserLoginAbnormal>
<isSupportIndustryThermometryAdvancedParam>
<!--ro, opt, bool, whether the device supports the advanced parameters of industrial temperature measurement, desc:GET/PUT

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/ISAPI/Thermal/channels/<channelID>/IndustryThermometryAdvancedParam?format=json-->true
</isSupportIndustryThermometryAdvancedParam>
<isSupportGotoVideoRecheckScene>
  <!--ro, opt, bool, whether the device supports jumping to the specified scene of water level recheck detection, desc:POST
/ISAPI/Event/channels/<channelID>/GotoVideoRecheckScene?format=json-->true
</isSupportGotoVideoRecheckScene>
<isSupportVideoRecheckRuleParams>
  <!--ro, opt, bool, whether the device supports rule parameters of water level recheck scene, desc:PUT
/ISAPI/Event/channels/<channelID>/VideoRecheckRuleParams?format=json-->true
</isSupportVideoRecheckRuleParams>
<isSupportVideoRecheckScenelist>
  <!--ro, opt, bool, whether the device supports getting the list of water level recheck detection scenes, desc:GET
/ISAPI/Event/channels/<channelID>/VideoRecheckScenelist?format=json-->true
</isSupportVideoRecheckScenelist>
<isSupportPeopleSedentary>
  <!--ro, opt, bool, whether the device supports sedentary detection, desc:eventType:PeopleSedentary-->true
</isSupportPeopleSedentary>
<isSupportPeopleFallingDownDetection>
  <!--ro, opt, bool, whether the device supports person fall detection, desc:eventType: PeopleFallingDownDetection-->true
</isSupportPeopleFallingDownDetection>
<isSupportBridgeDeflectionTargetCalibration>
  <!--ro, opt, bool, whether the device supports target calibration of bridge deflection, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetCalibration?format=json-->true
</isSupportBridgeDeflectionTargetCalibration>
<isSupportGetBridgeDeflectionTargetCalibrationStatus>
  <!--ro, opt, bool, whether the device supports getting the target calibration status of bridge deflection, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetBridgeDeflectionTargetCalibrationStatus?format=json-->true
</isSupportGetBridgeDeflectionTargetCalibrationStatus>
<isSupportBridgeDeflectionMonitoringRuleParams>
  <!--ro, opt, bool, whether the device supports the rule parameters of bridge deflection monitoring, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionMonitoringRuleParams?format=json-->true
</isSupportBridgeDeflectionMonitoringRuleParams>
<isSupportHardwareReboot>
  <!--ro, opt, bool, whether the device supports hardware rebooting, desc:PUT /ISAPI/System/HardwareReboot?format=json-->true
</isSupportHardwareReboot>
<isSupportSearchSceneCalibrationStatus>
  <!--ro, opt, bool, whether the device supports searching for calibration status of multiple channels by scene, desc:POST
/ISAPI/System/setupCalibration/channels/<channelID>/SearchSceneCalibrationStatus?format=json-->true
</isSupportSearchSceneCalibrationStatus>
<isSupportSearchSceneCalibration>
  <!--ro, opt, bool, whether the device supports searching for calibration status of multiple channels by scene, desc:POST
/ISAPI/System/setupCalibration/channels/<channelID>/SearchSceneCalibration?format=json-->true
</isSupportSearchSceneCalibration>
<isSupportSoundSourceLocationRuleParams>
  <!--ro, opt, bool, whether the device supports rule parameters of sound source locating, desc:GET/PUT
/ISAPI/System/SoundSourceLocation/AudioIn/<AudioInID>/SoundSourceLocationRuleParams?format=json-->true
</isSupportSoundSourceLocationRuleParams>
<isSupportPersonnelChannelLogDebug>
  <!--ro, opt, bool, whether the device supports debugging commands of turnstile logs, desc:GET /ISAPI/ContentMgmt/PersonnelChannelLogDebug/capabilities?
format=json-->true
</isSupportPersonnelChannelLogDebug>
<isSupportCheckDeviceExistNUCFile>
  <!--ro, opt, bool, whether the device supports checking the locally stored the NUC file, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/CheckDeviceExistNUCFile?format=json-->true
</isSupportCheckDeviceExistNUCFile>
<isSupportExportSmartDataData>
  <!--ro, opt, bool, whether the device supports exporting smart data, desc:GET /ISAPI/System/VideoQualityMeasurementCollection/ExportSmartDataData?
format=json-->true
</isSupportExportSmartDataData>
<isSupportGetCollectImageCalculateResult>
  <!--ro, opt, bool, whether the device supports getting the calculation result of image collection, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/GetCollectImageCalculateResult?format=json-->true
</isSupportGetCollectImageCalculateResult>
<isSupportAIOPAlarmTrans>
  <!--ro, opt, bool, whether the device supports uploading AIOP-related events via ISAPI message transmitted by COMM_VCA_ALARM,
desc:AIOP-related events include the following binary alarm types defined in HCNetSDK:
AIOP-accessed video detection alarm information (binary alarm: COMM_UPLOAD_AIOP_VIDEO, ISAPI alarm: AIOP_Video), AIOP-accessed picture detection alarm
information (binary alarm: COMM_UPLOAD_AIOP_PICTURE, ISAPI alarm: AIOP_Picture), AIOP-accessed capture polling detection alarm information (binary alarm:
COMM_UPLOAD_AIOP_POLLING_SNAP, ISAPI alarm: AIOP_Polling_Snap), AIOP-accessed video polling detection alarm information (binary alarm:
COMM_UPLOAD_AIOP_POLLING_VIDEO, ISAPI alarm: AIOP_Polling_Video)) -->true
</isSupportAIOPAlarmTrans>
<isSupportLongDistanceRace>
  <!--ro, opt, bool, whether the device supports long-distance race event, desc:eventType:LongDistanceRace-->true
</isSupportLongDistanceRace>
<isSupportIndicatorLightAbnormal>
  <!--ro, opt, bool, whether the device supports abnormal alarm indicator event, desc:eventType:IndicatorLightAbnormal-->true
</isSupportIndicatorLightAbnormal>
<isSupportVideoLostPictureStatus>
  <!--ro, opt, bool, whether the device supports the status of custom pictures displayed when video loss occurred, desc:GET
/ISAPI/System/Video/videoLosses/VideoLostPictureStatus?format=json-->true
</isSupportVideoLostPictureStatus>
<DualVCA>
  <!--ro, opt, object, intelligent information overlay-->
  <isSupportdualVCAList>
    <!--ro, opt, bool, whether it supports setting the parameters for overlaying intelligent information of channels in a batch,
desc:/ISAPI/Streaming/channels/dualVCA?format=json-->true
  </isSupportdualVCAList>
  <enabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable the function, attr:opt{req, string},def{req, bool}, desc:"ruleRegion" (overlay only rule area information)-->true
  </enabled>
  <channelID min="1" max="256">
    <!--ro, opt, int, channel No., attr:min{req, int},max{req, int}-->1
  </channelID>
  <ruleRegionType opt="ruleRegion_fullScreen" def="ruleRegion">

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<overlapRangeType opt= ruleRegion,fullScreen det= ruleRegion >
  <!--ro, opt, string, type of overlay area, attr:opt{req, string},def{req, string}-->ruleRegion
</overlapRangeType>
</DualVCA>
<isSupportGetScopeCalibrationStatus>
  <!--ro, opt, bool, whether the device supports getting the scope calibration status, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetScopeCalibrationStatus?format=json-->true
</isSupportGetScopeCalibrationStatus>
<isSupportSearchShootTableUserGroupParams>
  <!--ro, opt, bool, whether the device supports searching for the parameters of specified zeroing profiles, desc:POST
/ISAPI/Intelligent/channels/<channelID>/SearchShootTableUserGroupParams?format=json-->true
</isSupportSearchShootTableUserGroupParams>
<isSupportModifyCurrentShootTableParams>
  <!--ro, opt, bool, whether the device supports editing the information of current valid zeroing profiles, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/ModifyCurrentShootTableParams?format=json-->true
</isSupportModifyCurrentShootTableParams>
<isSupportCalibrateScopeTargetPointPosition>
  <!--ro, opt, bool, whether the device supports calibrating the absolute position of the sight cross and drop point, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/CalibrateScopeTargetPointPosition?format=json-->true
</isSupportCalibrateScopeTargetPointPosition>
<isSupportGetCurrentShootTableParams>
  <!--ro, opt, bool, whether the device supports getting the information of the current valid zeroing profiles, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetCurrentShootTableParams?format=json-->true
</isSupportGetCurrentShootTableParams>
<isSupportVideoGunsightRuleParam>
  <!--ro, opt, bool, whether the device supports rule parameters of video scope, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/VideoGunsightRuleParam?format=json-->true
</isSupportVideoGunsightRuleParam>
<isSupportFinishCollectImageData>
  <!--ro, opt, bool, whether it supports completing the collection of image data, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/FinishCollectImageData?format=json-->true
</isSupportFinishCollectImageData>
<isSupportStartCollectImageData>
  <!--ro, opt, bool, whether it supports starting collecting image data, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/StartCollectImageData?format=json-->true
</isSupportStartCollectImageData>
<isSupportSearchBridgeDeflectionMonitoringEvent>
  <!--ro, opt, bool, whether it supports searching for the event result of bridge deflection detection, desc:POST
/ISAPI/Intelligent/SearchBridgeDeflectionMonitoringEvent?format=json-->true
</isSupportSearchBridgeDeflectionMonitoringEvent>
<isSupportGetAllPIPConfigParams>
  <!--ro, opt, bool, whether it supports getting all picture-in-picture configuration parameters, desc:GET /ISAPI/System/Video/GetAllPIPCofigParams?
format=json-->true
</isSupportGetAllPIPCofigParams>
<isSupportManualDetectionCarriageClearStatus>
  <!--ro, opt, bool, whether it supports manual detection of the clearing status of the carriage, desc:POST
/ISAPI/System/channels/<channelID>/ManualDetectionCarriageClearStatus?format=json-->true
</isSupportManualDetectionCarriageClearStatus>
<isSupportGetCarriageClearCalibrationStatus>
  <!--ro, opt, bool, whether it supports getting the reference calibration carriage clearing status, desc:GET
/ISAPI/System/channels/<channelID>/GetCarriageClearCalibrationStatus?format=json-->true
</isSupportGetCarriageClearCalibrationStatus>
<isSupportModifyCarriageClearCalibration>
  <!--ro, opt, bool, whether it supports setting the reference calibration of carriage clearing, desc:POST
/ISAPI/System/channels/<channelID>/ModifyCarriageClearCalibration?format=json-->true
</isSupportModifyCarriageClearCalibration>
<isSupportCarriageClearRuleParams>
  <!--ro, opt, bool, whether it supports setting rule parameters of carriage clearing, desc:GET/PUT
/ISAPI/System/channels/<channelID>/CarriageClearRuleParams?format=json-->true
</isSupportCarriageClearRuleParams>
<isSupportRadarFallenObjectsOnTracks>
  <!--ro, opt, bool, whether it supports radar detection of fallen objects on tracks, desc:eventType:RadarFallenObjectsOnTracks-->true
</isSupportRadarFallenObjectsOnTracks>
<isSupportTemperatureRiseCalibrationParams>
  <!--ro, opt, bool, whether it supports temperature rise calibration parameters, desc:GET/PUT /ISAPI/Thermal/TemperatureRiseCalibrationParams?
format=json-->true
</isSupportTemperatureRiseCalibrationParams>
<isSupportStartTemperatureRiseCalibration>
  <!--ro, opt, bool, whether it supports starting temperature rise calibration, desc:PUT /ISAPI/Thermal/StartTemperatureRiseCalibration?format=json-->true
</isSupportStartTemperatureRiseCalibration>
<isSupportGetTemperatureRiseCalibrationStatus>
  <!--ro, opt, bool, whether it supports getting temperature rise calibration status, desc:GET /ISAPI/Thermal/GetTemperatureRiseCalibrationStatus?
format=json-->true
</isSupportGetTemperatureRiseCalibrationStatus>
<isSupportExportTemperatureRiseCalibrationFile>
  <!--ro, opt, bool, whether it supports exporting temperature rise calibration file, desc:GET /ISAPI/Thermal/ExportTemperatureRiseCalibrationFile?
format=json-->true
</isSupportExportTemperatureRiseCalibrationFile>
<isSupportGetTemperatureRiseCorrectionVersion>
  <!--ro, opt, bool, whether it supports getting temperature rise calibration version information, desc:GET
/ISAPI/Thermal/GetTemperatureRiseCorrectionVersion?format=json-->true
</isSupportGetTemperatureRiseCorrectionVersion>
<isSupportSearchImportDeviceDataTaskProgress>
  <!--ro, opt, bool, whether it supports searching for the progress of device data importing task, desc:POST
/ISAPI/System/SearchImportDeviceDataTaskProgress?format=json-->true
</isSupportSearchImportDeviceDataTaskProgress>
<isSupportSearchExportDeviceDataTaskProgress>
  <!--ro, opt, bool, whether it supports searching for the progress of device data exporting task, desc:POST
/ISAPI/System/SearchExportDeviceDataTaskProgress?format=json-->true
</isSupportSearchExportDeviceDataTaskProgress>
<isSupportAddImportDeviceDataTask>
  <!--ro, opt, bool, whether it supports creating device data importing task, desc:POST /ISAPI/System/AddImportDeviceDataTask?format=json-->true
</isSupportAddImportDeviceDataTask>
<isSupportAddExportDeviceDataTask>

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    <!--ro, opt, bool, whether it supports creating device data exporting task, desc:POST /ISAPI/System/AddExportDeviceDataTask?format=json-->true
</isSupportAddExportDeviceDataTask>
<isSupportSearchRecordStorage>
    <!--ro, opt, bool, whether it supports searching for record storage HDD information, desc:POST /ISAPI/ContentMgmt/search/SearchRecordStorage?
format=json-->true
</isSupportSearchRecordStorage>
<isSupportDetectionPathRuleParams>
    <!--ro, opt, bool, whether it supports hazardous gas detection patrol path rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/DetectionPathRuleParams?format=json-->true
</isSupportDetectionPathRuleParams>
<isSupportSearchObjectDistanceFocusInfoList>
    <!--ro, opt, bool, whether it supports searching for the list of object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/SearchObjectDistanceFocusInfoList?format=json-->true
</isSupportSearchObjectDistanceFocusInfoList>
<isSupportMoveObjectDistanceFocusLocation>
    <!--ro, opt, bool, whether it supports moving to the point where the focus of the object distance is calibrated, desc:PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/MoveObjectDistanceFocusLocation?format=json-->true
</isSupportMoveObjectDistanceFocusLocation>
<isSupportAddObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports adding object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/AddObjectDistanceFocusInfo?format=json-->true
</isSupportAddObjectDistanceFocusInfo>
<isSupportModifyObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports editing object distance focus calibration information, desc:PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ModifyObjectDistanceFocusInfo?format=json-->true
</isSupportModifyObjectDistanceFocusInfo>
<isSupportObjectDistanceCalibrateAutoFocusParams>
    <!--ro, opt, bool, whether it supports auto-focus parameters of object distance calibration, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
</isSupportObjectDistanceCalibrateAutoFocusParams>
<isSupportExpresswayNetworkTollAccessParams>
    <!--ro, opt, bool, whether it supports protocol access parameters of network parking fee charging on highway, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceCalibrateAutoFocusParams?format=json-->true
</isSupportExpresswayNetworkTollAccessParams>
<isSupportControlCameraDetectAction>
    <!--ro, opt, bool, whether it supports controlling the patrol path of camera, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/methaneDetection/ControlCameraDetectAction?format=json-->true
</isSupportControlCameraDetectAction>
<isSupportObjectDistanceAccessWay>
    <!--ro, opt, bool, whether it supports the method of getting the object distance, desc:GET/PUT
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/ObjectDistanceAccessWay?format=json-->true
</isSupportObjectDistanceAccessWay>
<isSupportCancelTemperatureRiseCalibration>
    <!--ro, opt, bool, whether it supports canceling temperature rise calibration, desc:PUT /ISAPI/Thermal/CancelTemperatureRiseCalibration?format=json--
>true
</isSupportCancelTemperatureRiseCalibration>
<isSupportRTCTimeTrigger>
    <!--ro, opt, bool, whether the device supports RTC time exception linkage, desc:GET/PUT /ISAPI/Event/RTCTimeTrigger?format=json-->true
</isSupportRTCTimeTrigger>
<isSupportChargingGunAbnormalEvent>
    <!--ro, opt, bool, whether it supports charging station inserted gun exception event, desc:eventType: ChargingGunAbnormalEvent-->true
</isSupportChargingGunAbnormalEvent>
<isSupportVideoChannelEnhance>
    <!--ro, opt, bool, whether it supports enhancement configuration of adding video channel, desc:GET/PUT
/ISAPI/ContentMgmt/InputProxy/VideoChannelEnhance?format=json-->true
</isSupportVideoChannelEnhance>
<isSupportDeleteObjectDistanceFocusInfo>
    <!--ro, opt, bool, whether it supports deleting the object distance focus calibration information, desc:POST
/ISAPI/Image/ObjectDistanceFocusCalibrate/channels/<channelID>/DeleteObjectDistanceFocusInfo?format=json-->true
</isSupportDeleteObjectDistanceFocusInfo>
<isSupportGetStartTemperatureRiseCalibrationParams>
    <!--ro, opt, bool, whether the device supports getting parameter of starting temperature rise calibration, desc:GET
/ISAPI/Thermal/GetStartTemperatureRiseCalibrationParams?format=json-->true
</isSupportGetStartTemperatureRiseCalibrationParams>
<isSupportSearchAudioRecordPlan>
    <!--ro, opt, bool, whether the device supports searching all recording schedules, desc:POST /ISAPI/System/RecordAudio/SearchAudioRecordPlan?format=json-
>true
</isSupportSearchAudioRecordPlan>
<isSupportCleanerDetectionOverlapParams>
    <!--ro, opt, bool, whether the device supports cleaning liquid detection information overlay parameters, desc:GET/PUT
/ISAPI/System/Hardware/channels/<channelID>/CleanerDetectionOverlapParams?format=json-->true
</isSupportCleanerDetectionOverlapParams>
<isSupportWaterLevelVerticalAreaDisplayParams>
    <!--ro, opt, bool, whether it supports the display parameters of water level vertical view monitoring area, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/WaterLevelVerticalAreaDisplayParams?format=json-->true
</isSupportWaterLevelVerticalAreaDisplayParams>
<isSupportGetRecordStatusList>
    <!--ro, opt, bool, whether the device supports getting the recording status of all channels, desc:POST /ISAPI/ContentMgmt/search/GetRecordStatusList?
format=json-->true
</isSupportGetRecordStatusList>
<isSupportWiperClean>
    <!--ro, opt, bool, whether the device supports wiper cleaning event, desc:eventType: WiperClean-->true
</isSupportWiperClean>
<isSupportGetNATStatus>
    <!--ro, opt, bool, whether the device supports getting NAT status, desc:GET /ISAPI/System/Network/NetworkAdvancedCfg/GetNATStatus?format=json-->true
</isSupportGetNATStatus>
<isSupportRestoreSensitivityDefaultParams>
    <!--ro, opt, bool, whether the device supports restoring radar perimeter sensitivity parameters to the default, desc:PUT
/ISAPI/Smart/RadarPerimeterAlg/RestoreSensitivityDefaultParams?format=json-->true
</isSupportRestoreSensitivityDefaultParams>
<isSupportCarryPassengersPricingStatus>
    <!--ro, opt, bool, whether the device supports displaying the status change of meter, desc:eventType: CarryPassengersPricingStatus-->true
</isSupportCarryPassengersPricingStatus>

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<isSupportCarryPassengersNoPricing>
  <!--ro, opt, bool, whether to support event of not using the meter when passengers are on board, desc:eventType: CarryPassengersNoPricing-->true
</isSupportCarryPassengersNoPricing>
<isSupportNPlusMultiple>
  <!--ro, opt, bool, whether the device supports N+M mode of host and standby device, desc:whether the device supports setting workMode to N+M. Related
API: PUT /ISAPI/System/Network/NPlusOne/capabilities?format=json-->true
</isSupportNPlusMultiple>
<ifPopUpNetworkCfgSetting>
  <!--ro, opt, bool, whether to pop up network configuration interface, desc:PUT/GET /ISAPI/System/Network/interfaces/<interfaceID>. true (if the default
IP address of the device is 192.168.0.100, the device will pop up the network configuration interface), false or node does not exist (the network
configuration interface will not pop up)-->true
</ifPopUpNetworkCfgSetting>
<AudioOutputChannelCfg>
  <!--ro, opt, object, support linking audio output channel with audio file-->
  <AudioOutputChannelIDList>
    <!--ro, opt, array, audio output channel List, subType:object-->
    <AudioOutputChannelID>
      <!--ro, req, int, audio output channel No.-->1
    </AudioOutputChannelID>
  </AudioOutputChannelIDList>
</AudioOutputChannelCfg>
<isSupportImportTemperatureRiseCalibrationFile>
  <!--ro, opt, bool, whether the device supports importing the temperature rise calibration file, desc:PUT
/ISAPI/Thermal/ImportTemperatureRiseCalibrationFile?format=json-->true
</isSupportImportTemperatureRiseCalibrationFile>
<isSupportGetAPPQRCodePic>
  <!--ro, opt, bool, whether it supports getting QR code picture of the app accessed by the device, desc:GET /ISAPI/System/GetAPPQRCodePic?format=json--
>true
</isSupportGetAPPQRCodePic>
<isSupportPersionQueueRealTimeData>
  <!--ro, opt, bool, whether it supports real-time counting of the number of people in the queuing area, desc:GET
/ISAPI/Intelligent/personQueueDetection/channels/<channelID>/PersionQueueRealTimeData?format=json-->true
</isSupportPersionQueueRealTimeData>
<isSupportImageVignettingOptimizeParams>
  <!--ro, opt, bool, whether it supports image dark corner optimization parameters, desc:GET/PUT
/ISAPI/Streaming/channels/<channelID>/ImageVignettingOptimizeParams?format=json-->true
</isSupportImageVignettingOptimizeParams>
<isSupportStartMultispectralCalibration>
  <!--ro, opt, bool, whether it supports enabling multispectral calibration, desc:PUT
/ISAPI/System/SpectrumCalibration/channels/<channelID>/StartMultispectralCalibration?format=json-->true
</isSupportStartMultispectralCalibration>
<isSupportGetMultispectralCalibrationStatus>
  <!--ro, opt, bool, whether it supports getting the multispectral calibration status, desc:GET
/ISAPI/System/SpectrumCalibration/channels/<channelID>/GetMultispectralCalibrationStatus?format=json-->true
</isSupportGetMultispectralCalibrationStatus>
<isSupportStopRadarVideoTargetTracking>
  <!--ro, opt, bool, whether it supports stopping target tracking (radar), desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StopRadarVideoTargetTracking?format=json-->true
</isSupportStopRadarVideoTargetTracking>
<isSupportGetRadarVideoTargetTrackingStatus>
  <!--ro, opt, bool, whether it supports getting the status of target tracking (radar), desc:GET
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/GetRadarVideoTargetTrackingStatus?format=json-->true
</isSupportGetRadarVideoTargetTrackingStatus>
<isSupportStartRadarVideoTargetTracking>
  <!--ro, opt, bool, whether it supports starting target tracking (radar), desc:PUT
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/StartRadarVideoTargetTracking?format=json-->true
</isSupportStartRadarVideoTargetTracking>
<isSupportBatteryChargeCtrl>
  <!--ro, opt, bool, whether it supports battery charging control parameters, desc:GET/PUT /ISAPI/System/BatteryChargeCtrl?format=json-->true
</isSupportBatteryChargeCtrl>
<isUpgradeAuxDevice opt="true,false">
  <!--ro, opt, string, whether it supports synchronous upgrade of auxiliary device, attr:opt{opt, string}, desc:it represents support details of
isUpgradeAuxDevice in /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpgradeAuxDevice=
<isUpgradeAuxDevice>-->true
</isUpgradeAuxDevice>
<isSupportClearFile>
  <!--ro, opt, bool, whether it supports clearing files under the specified path of the device, desc:POST
/ISAPI/System/VideoQualityMeasurementCollection/ClearFile?format=json-->true
</isSupportClearFile>
<isSupportRestoreDefaultParamByChannel>
  <!--ro, opt, bool, whether it supports restoring default parameters of a single channel, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?
format=json&channelID=<channelID>-->true
</isSupportRestoreDefaultParamByChannel>
<isSupportGetMultispectralCalibrationParams>
  <!--ro, opt, bool, whether it supports getting multispectral calibration parameters, desc:GET
/ISAPI/System/SpectrumCalibration/channels/<channelID>/GetMultispectralCalibrationParams?format=json-->true
</isSupportGetMultispectralCalibrationParams>
<isSupportGetThermometryCalibrationData>
  <!--ro, opt, bool, whether it supports getting temperature calibration-related data, desc:GET /ISAPI/Thermal/GetThermometryCalibrationData?format=json--
>true
</isSupportGetThermometryCalibrationData>
<isSupportImportEncryptedComponentFile>
  <!--ro, opt, bool, whether it supports importing encrypted component files, desc:POST
/ISAPI/System/EncryptionFileUpgradeMgr/ImportEncryptedComponentFile?format=json-->true
</isSupportImportEncryptedComponentFile>
<isSupportGetFrameMeanValue>
  <!--ro, opt, bool, whether it supports getting frame average range information, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/channels/<channelID>/GetFrameMeanValue?format=json-->true
</isSupportGetFrameMeanValue>
<isSupportGetBaffleImage>
  <!--ro, opt, bool, whether it supports getting image data before and after the image is captured, desc:GET
/ISAPI/System/VideoQualityMeasurementCollection/GetBaffleImage?format=json-->true
</isSupportGetBaffleImage>
<isSupportThermometryGainMode>

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    <!--ro, opt, bool, whether it supports the temperature measurement gain parameters, desc:GET/PUT /ISAPI/Thermal/ThermometryGainMode?format=json-->true
</isSupportThermometryGainMode>
<isSupportGetChannelEnhancePreOperationTips>
    <!--ro, opt, bool, whether it supports getting the prompt information before the enhancement configuration of adding the channel, desc:POST
/ISAPI/ContentMgmt/InputProxy/GetChannelEnhancePreOperationTips?format=json-->true
</isSupportGetChannelEnhancePreOperationTips>
<isSupportGetDeviceNetworkLinkStatus>
    <!--ro, opt, bool, whether it supports getting the network connection status of the device, desc:GET /ISAPI/System/GetDeviceNetworkLinkStatus?
format=json-->true
</isSupportGetDeviceNetworkLinkStatus>
<isSupportManualRefreshOnlineUserList>
    <!--ro, opt, bool, whether the web needs to display a refresh button, desc:used for web interface display, determine whether the device needs to display
the refresh button to refresh the List of online users via this field-->true
</isSupportManualRefreshOnlineUserList>
<isSupportVCAAlarmTrans>
    <!--ro, opt, bool, whether it supports uploading events related to abnormal behavior detection through the transmission of ISAPI messages, desc:for all
events related to abnormal behavior detection, refer to HCNetsdk enumeration of VCA_RULE_EVENT_TYPE_EX-->true
</isSupportVCAAlarmTrans>
<isSupportHumidityEvent>
    <!--ro, opt, bool, whether it supports humidity alarm events, desc:HumidityEvent-->true
</isSupportHumidityEvent>
<isSupportDisplayThermometryFirePrevention>
    <!--ro, opt, bool, whether the web needs to display the temperature measurement function as
temperature measurement & fire prevention,
desc:if this field is true, the web interface display
"Temperature Measurement & Fire Prevention", otherwise, it will be displayed as "Temperature Measurement". If this field is not supported, the web interface
will display "Temperature Measurement"-->true
</isSupportDisplayThermometryFirePrevention>
<isSupportPreRemoteCheckEvent>
    <!--ro, opt, bool, whether it supports the collection of pre-verification events, desc:eventType:PreRemoteCheckEvent-->true
</isSupportPreRemoteCheckEvent>
<isSupportSearchExportDeviceStatusTaskProgress>
    <!--ro, opt, bool, whether it supports getting the progress of the task to export device information and status, desc:POST
/ISAPI/System/SearchExportDeviceStatusTaskProgress?format=json-->true
</isSupportSearchExportDeviceStatusTaskProgress>
<isSupportAddExportDeviceStatusTask>
    <!--ro, opt, bool, whether it supports creating the task to export device information and status, desc:POST /ISAPI/System/AddExportDeviceStatusTask?
format=json-->true
</isSupportAddExportDeviceStatusTask>
<isSupportInsulatorThermometryRuleParams>
    <!--ro, opt, bool, whether it supports the insulator temperature measurement rule parameters, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/InsulatorThermometryRuleParams?format=json-->true
</isSupportInsulatorThermometryRuleParams>
<isSupportInsulatorCruiseParams>
    <!--ro, opt, bool, whether it supports insulator patrol parameters, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/InsulatorCruiseParams?format=json--
>true
</isSupportInsulatorCruiseParams>
<isSupportModifyInsulatorDetectionRegion>
    <!--ro, opt, bool, whether it supports modifying the insulator group detection area, desc:PUT
/ISAPI/Thermal/channels/<channelID>/ModifyInsulatorDetectionRegion?format=json-->true
</isSupportModifyInsulatorDetectionRegion>
<isSupportMoveInsulatorPosition>
    <!--ro, opt, bool, whether it supports movement to the specified insulator string position, desc:POST
/ISAPI/Thermal/channels/<channelID>/MoveInsulatorPosition?format=json-->true
</isSupportMoveInsulatorPosition>
<isSupportDeleteInsulatorGroup>
    <!--ro, opt, bool, whether it supports deleting a specified insulator group, desc:POST /ISAPI/Thermal/channels/<channelID>/DeleteInsulatorGroup?
format=json-->true
</isSupportDeleteInsulatorGroup>
<isSupportGenerateInsulatorRuleRegion>
    <!--ro, opt, bool, whether it supports generating a specified insulator string rule area, desc:POST
/ISAPI/Thermal/channels/<channelID>/GenerateInsulatorRuleRegion?format=json-->true
</isSupportGenerateInsulatorRuleRegion>
<isSupportGetInsulatorAllName>
    <!--ro, opt, bool, whether it supports getting all the names of insulator groups, desc:GET /ISAPI/Thermal/channels/<channelID>/GetInsulatorAllName?
format=json-->true
</isSupportGetInsulatorAllName>
<isSupportSearchUploadLog>
    <!--ro, opt, bool, whether it supports searching and uploading log files, desc:PUT /ISAPI/ContentMgmt/SearchUploadLog?format=json-->true
</isSupportSearchUploadLog>
<isSupportRainfallTelemeter>
    <!--ro, opt, bool, whether it supports remotely monitoring rainfall, desc:eventType:RainfallTelemeter-->true
</isSupportRainfallTelemeter>
<isSupportPosOverlayTest>
    <!--ro, opt, bool, whether it supports the test of enabling POS information overlay configuration, desc:POST
/ISAPI/System/Video/inputs/channels/<channelID>/PosOverlayTest?format=json-->true
</isSupportPosOverlayTest>
<isSupportFullScreenPosOverlay>
    <!--ro, opt, bool, whether it supports full-screen POS information overlay parameters, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlay?format=json-->true
</isSupportFullScreenPosOverlay>
<isSupportFullScreenPosOverlayCap>
    <!--ro, opt, bool, whether to get the dynamic capability of full-screen POS information overlay parameters, desc:GET
/ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlayCap?format=json-->true
</isSupportFullScreenPosOverlayCap>
<isSupportVCAOptimizeDisplay>
    <!--ro, opt, bool, whether it supports the display of the intelligent resource usability page, desc:if the field is true, the user-friendly interface is
displayed; otherwise, the previous one will be displayed; if the field is not supported, it is determined by the device type-->true
</isSupportVCAOptimizeDisplay>
<isSupportSearchInsulatorParams>
    <!--ro, opt, bool, whether it supports searching the parameters of a specified insulator group, desc:POST
/ISAPI/Thermal/channels/<channelID>/SearchInsulatorParams?format=json-->true
</isSupportSearchInsulatorParams>

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<isSupportSetInsulatorParams>
  <!--ro, opt, bool, whether it supports setting insulator group parameters, desc:PUT /ISAPI/Thermal/channels/<channelID>/SetInsulatorParams?format=json-->
-->true
</isSupportSetInsulatorParams>
<isSupportInsulatorAbnormalAlarm>
  <!--ro, opt, bool, whether it supports insulator exception alarm, desc:eventType:InsulatorAbnormalAlarm-->true
</isSupportInsulatorAbnormalAlarm>
<isSupportGetFlashLifespanInfo>
  <!--ro, opt, bool, whether it supports getting the Lifespan information of the built-in disk storage medium, desc:GET
/ISAPI/System/GetFlashLifespanInfo?format=json-->true
</isSupportGetFlashLifespanInfo>
<isSupportDownloadAudioAnalysisFile>
  <!--ro, opt, bool, whether it supports downloading audio analysis files, desc:POST /ISAPI/Intelligent/SensitiveInfoAnalysis/DownloadAudioAnalysisFile?
format=json-->true
</isSupportDownloadAudioAnalysisFile>
<isSupportCrowdDispersing>
  <!--ro, opt, bool, whether it supports crowd dispersing, desc:corresponding event: CrowdDispersing-->true
</isSupportCrowdDispersing>
<isSupportPeripheralAbnormalEvent>
  <!--ro, opt, bool, whether it supports peripheral fault event, desc:If it is not supported, this node will not be returned. Corresponding eventType:
PeripheralAbnormalEvent-->true
</isSupportPeripheralAbnormalEvent>
<isSupportPeripheralAbnormalTrigger>
  <!--ro, opt, bool, whether it supports peripheral fault Linkage, desc:GET/PUT /ISAPI/Event/PeripheralAbnormalTrigger?format=json-->true
</isSupportPeripheralAbnormalTrigger>
<isSupportBodyCameraAbnormalEvent>
  <!--ro, opt, bool, whether it supports exception events of body cameras, desc:corresponding event: BodyCameraAbnormalEvent-->true
</isSupportBodyCameraAbnormalEvent>
<isSupportDeviceWorkModeAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports the event of enforcement device working mode asynchronous status, desc:corresponding event:
DeviceWorkModeAsyncStatusEvent-->true
</isSupportDeviceWorkModeAsyncStatusEvent>
<isSupportDeviceUserAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports asynchronous parameter status event for Law enforcement device users, desc:corresponding event:
DeviceUserAsyncStatusEvent-->true
</isSupportDeviceUserAsyncStatusEvent>
<isSupportVideoCodingAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports video encoding asynchronous parameter status event, desc:corresponding event: VideoCodingAsyncStatusEvent-->true
</isSupportVideoCodingAsyncStatusEvent>
<isSupportRecordAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports manual recording asynchronous parameter status event, desc:corresponding event: RecordAsyncStatusEvent-->true
</isSupportRecordAsyncStatusEvent>
<isSupportISUPAccessAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports asynchronous parameter status event for ISUP protocol access, desc:corresponding event:
ISUPAccessAsyncStatusEvent-->true
</isSupportISUPAccessAsyncStatusEvent>
<isSupportWirelessConfigListAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports asynchronous parameters status event for wireless configuration list, desc:corresponding event:
WirelessConfigListAsyncStatusEvent-->true
</isSupportWirelessConfigListAsyncStatusEvent>
<isSupportDeviceLocationAsyncStatusEvent>
  <!--ro, opt, bool, whether it supports asynchronous parameter status event for device location, desc:corresponding event:
DeviceLocationAsyncStatusEvent-->true
</isSupportDeviceLocationAsyncStatusEvent>
<isSupportVideoCodingAsyncParams>
  <!--ro, opt, bool, whether it supports video encoding async parameters, desc:GET/PUT /ISAPI/Streaming/VideoCodingAsyncParams?format=json-->true
</isSupportVideoCodingAsyncParams>
<isSupportMoveShield>
  <!--ro, opt, bool, whether it supports movement to a specified shielded area, desc:POST /ISAPI/Intelligent/channels/<channelID>/MoveShield?format=json--
-->true
</isSupportMoveShield>
<isSupportStopAutoGenerateShield>
  <!--ro, opt, bool, whether it supports stopping the automatic recognition of smart shielded area, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/StopAutoGenerateShield?format=json-->true
</isSupportStopAutoGenerateShield>
<isSupportStartGenerateParams>
  <!--ro, opt, bool, whether it supports starting generating smart shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/StartGenerateParams?
format=json-->true
</isSupportStartGenerateParams>
<isSupportIntelligentShieldAlgorithmsVersion>
  <!--ro, opt, bool, whether it supports getting the version information of the intelligent shielding area recognition algorithm library, desc:GET
/ISAPI/Intelligent/channels/<channelID>/IntelligentShieldAlgorithmsVersion?format=json-->true
</isSupportIntelligentShieldAlgorithmsVersion>
<isSupportConvertShield>
  <!--ro, opt, bool, whether it supports converting a specified shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/ConvertShield?format=json-
-->true
</isSupportConvertShield>
<isSupportDeleteShield>
  <!--ro, opt, bool, whether it supports deleting a specified shielded area, desc:PUT /ISAPI/Intelligent/channels/<channelID>/DeleteShield?format=json--
-->true
</isSupportDeleteShield>
<isSupportGetGenerateParams>
  <!--ro, opt, bool, whether it supports getting the parameters of generating smart shielded areas, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetGenerateParams?format=json-->true
</isSupportGetGenerateParams>
<isSupportManualAddShield>
  <!--ro, opt, bool, whether it supports manually adding shielded areas, desc:PUT /ISAPI/Intelligent/channels/<channelID>/ManualAddShield?format=json--
-->true
</isSupportManualAddShield>
<isSupportIntelligentShieldParams>
  <!--ro, opt, bool, whether it supports smart shielded area parameters, desc:GET/PUT /ISAPI/Intelligent/channels/<channelID>/IntelligentShieldParams?
format=json-->true
</isSupportIntelligentShieldParams>
<isSupportIntelligentShieldListParams>

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<!--ro, opt, bool, whether it supports the smart shielded area List parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/IntelligentShieldListParams?format=json-->true
</isSupportIntelligentShieldListParams>
<isSupportDeicingCfg>
<!--ro, opt, bool, whether it supports lens heating and de-icing control parameters, desc:GET/PUT /ISAPI/System/Hardware/DeicingCfg?format=json-->true
</isSupportDeicingCfg>
<isSupportInspectionOverdueEvent>
<!--ro, opt, bool, whether it supports annual inspection overdue event for alcohol detection enforcement device, desc:corresponding event:
InspectionOverdueEvent-->true
</isSupportInspectionOverdueEvent>
<isSupportChannelsPanoramaCalibration>
<!--ro, opt, bool, whether it supports batch configuring channel calibration distance, desc:GET/PUT /ISAPI/Panorama/channels/calibration/distance?
format=json-->true
</isSupportChannelsPanoramaCalibration>
<isSupportRadarPerimeterShieldOverlapParams>
<!--ro, opt, bool, whether it supports radar perimeter overLay parameters, desc:GET/PUT
/ISAPI/Smart/RadarPerimeterAlg/Radar/<RadarID>/RadarPerimeterShieldOverlapParams?format=json-->true
</isSupportRadarPerimeterShieldOverlapParams>
<isSupportAlarmFilterInterval>
<!--ro, opt, bool, whether it supports configuring alarm filter time interval, desc:corresponding to the "alarmFilterInterval" node in the POST
/ISAPI/IoTGateway/AlarmRules/SetAlarmRules?format=json-->true
</isSupportAlarmFilterInterval>
<isSupportImportLicense>
<!--ro, opt, bool, whether it supports importing the authorization file, desc:POST /ISAPI/System/SoftwareLicense/ImportLicense?format=json-->true
</isSupportImportLicense>
<isSupportShipStatisticsAnalysisParams>
<!--ro, opt, bool, whether it supports ship traffic statistics parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/ShipStatisticsAnalysisParams?format=json-->true
</isSupportShipStatisticsAnalysisParams>
<isSupportShipStatisticsEvent>
<!--ro, opt, bool, whether it supports ship traffic statistics event, desc:corresponding event: ShipStatisticsEvent-->true
</isSupportShipStatisticsEvent>
<isSupportPeopleVitalSignsDetection>
<!--ro, opt, bool, whether it supports vital sign detection event, desc:eventType: PeopleVitalSignsDetection-->true
</isSupportPeopleVitalSignsDetection>
<isSupportPeopleVitalSignsStatistic>
<!--ro, opt, bool, whether it supports vital sign statistics event, desc:eventType: PeopleVitalSignsStatistic-->true
</isSupportPeopleVitalSignsStatistic>
<isSupportPeopleBodyMoving>
<!--ro, opt, bool, whether it supports human body movement detection event, desc:eventType: PeopleBodyMoving-->true
</isSupportPeopleBodyMoving>
<isSupportPeopleGetUp>
<!--ro, opt, bool, whether it supports person get-up detection event, desc:eventType: PeopleGetUp (detected a person who got up from the bed during
posture detection)-->true
</isSupportPeopleGetUp>
<isSupportPersonLocationDetection>
<!--ro, opt, bool, whether it supports person location detection event, desc:eventType: PersonLocationDetection-->true
</isSupportPersonLocationDetection>
<isSupportPeopleFallingDownRecovery>
<!--ro, opt, bool, whether it supports person fall restoring event, desc:eventType: PeopleFallingDownRecovery-->true
</isSupportPeopleFallingDownRecovery>
<isSupportPersonRetentionEvent>
<!--ro, opt, bool, whether it supports people overstay event, desc:eventType: PersonRetentionEvent-->true
</isSupportPersonRetentionEvent>
<isSupportPersonRetentionRecoveryEvent>
<!--ro, opt, bool, whether it supports people overstay restoring event, desc:eventType: PersonRetentionRecoveryEvent-->true
</isSupportPersonRetentionRecoveryEvent>
<isSupportPeopleLongTimeLeaveEvent>
<!--ro, opt, bool, whether it supports person long-time off-bed event, desc:eventType: PeopleLongTimeLeaveEvent-->true
</isSupportPeopleLongTimeLeaveEvent>
<isSupportPeopleLongTimeLeaveRecoveryEvent>
<!--ro, opt, bool, whether it supports person long-time off-bed restoring event, desc:eventType: PeopleLongTimeLeaveRecoveryEvent-->true
</isSupportPeopleLongTimeLeaveRecoveryEvent>
<isSupportPeopleVitalSignsAbnormalEvent>
<!--ro, opt, bool, whether it supports vital sign exception event, desc:eventType: PeopleVitalSignsAbnormalEvent-->true
</isSupportPeopleVitalSignsAbnormalEvent>
<isSupportPeopleVitalSignsRecoveryEvent>
<!--ro, opt, bool, whether it supports vital sign exception restoring event, desc:eventType: PeopleVitalSignsRecoveryEvent-->true
</isSupportPeopleVitalSignsRecoveryEvent>
<isSupportPeopleVitalSignsExportTaskEvent>
<!--ro, opt, bool, whether it supports vital sign data exporting task result event, desc:eventType: PeopleVitalSignsExportTaskEvent-->true
</isSupportPeopleVitalSignsExportTaskEvent>
<isSupportTOFCalibrate>
<!--ro, opt, bool, whether it supports configuring TOF detection calibration, desc:PUT /ISAPI/System/Video/inputs/tamperDetection/TOFCalibrate?
format=json-->true
</isSupportTOFCalibrate>
<isSupportSearchBulkUpgradeChildDeviceTask>
<!--ro, opt, bool, whether it supports searching for tasks of batch upgrading sub-devices, desc:/ISAPI/System/SearchBulkUpgradeChildDeviceTask?
format=json-->true
</isSupportSearchBulkUpgradeChildDeviceTask>
<isSupportModifyBulkUpgradeChildDeviceTask>
<!--ro, opt, bool, whether it supports modifying the task of batch upgrading sub-devices, desc:/ISAPI/System/ModifyBulkUpgradeChildDeviceTask?
format=json-->true
</isSupportModifyBulkUpgradeChildDeviceTask>
<isSupportDeleteBulkUpgradeChildDeviceTask>
<!--ro, opt, bool, whether it supports deleting tasks of batch upgrading sub-devices, desc:/ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json--
>true
</isSupportDeleteBulkUpgradeChildDeviceTask>
<isSupportAddEventTemplate>
<!--ro, opt, bool, whether it supports adding event message templates, desc:POST /ISAPI/Event/notification/httpHosts/AddEventTemplate?format=json-->true
</isSupportAddEventTemplate>
<isSupportTestEventTemplate>
<!--ro, opt, bool, whether it supports testing event message templates, desc:POST /ISAPI/Event/notification/httpHosts/TestEventTemplate?format=json--

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>true
</isSupportTestEventTemplate>
<isSupportEventTemplateParams>
  <!--ro, opt, bool, whether it supports event message template parameters, desc:GET /ISAPI/Event/notification/httpHosts/EventTemplateParams?format=json-->
>true
</isSupportEventTemplateParams>
<isSupportAddCombineEvents>
  <!--ro, opt, bool, whether it supports adding combined event, desc:POST /ISAPI/Event/CombineEvents/AddCombineEvents?format=json-->true
</isSupportAddCombineEvents>
<isSupportModifyCombineEventsParams>
  <!--ro, opt, bool, whether it supports modifying combined event parameters, desc:POST /ISAPI/Event/CombineEvents/ModifyCombineEventsParams?format=json-->
>true
</isSupportModifyCombineEventsParams>
<isSupportSearchCombineEventsParams>
  <!--ro, opt, bool, whether it supports searching for combined event parameters, desc:POST /ISAPI/Event/CombineEvents/SearchCombineEventsParams?
format=json-->true
</isSupportSearchCombineEventsParams>
<isSupportDeleteCombineEventsList>
  <!--ro, opt, bool, whether it supports batch deleting combined event, desc:POST /ISAPI/Event/CombineEvents/DeleteCombineEventsList?format=json-->true
</isSupportDeleteCombineEventsList>
<isSupportEventRuleAndTriggerParams>
  <!--ro, opt, bool, whether it supports configuring the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/EventRuleAndTriggerParams?format=json-->true
</isSupportEventRuleAndTriggerParams>
<isSupportSearchEventRuleAndTriggerParams>
  <!--ro, opt, bool, whether it supports searching for the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/SearchEventRuleAndTriggerParams?format=json-->true
</isSupportSearchEventRuleAndTriggerParams>
<isSupportCallMessageHandlingEvent>
  <!--ro, opt, bool, whether it supports the call message handling event, desc:eventType: CallMessageHandlingEvent-->true
</isSupportCallMessageHandlingEvent>
<isSupportResetBatchChild>
  <!--ro, opt, bool, whether it supports batch restoring sub-devices, desc:PUT /ISAPI/System/ResetBatchChild?format=json-->true
</isSupportResetBatchChild>
<isSupportSearchResetBatchChildStatus>
  <!--ro, opt, bool, whether it supports searching for the status of batch restoring sub-devices, desc:POST /ISAPI/System/SearchResetBatchChildStatus?
format=json-->true
</isSupportSearchResetBatchChildStatus>
<isSupportSearchRebootBatchChildStatus>
  <!--ro, opt, bool, whether it supports searching for the status of batch rebooting sub-devices, desc:POST /ISAPI/System/SearchRebootBatchChildStatus?
format=json-->true
</isSupportSearchRebootBatchChildStatus>
<isSupportRebootBatchChild>
  <!--ro, opt, bool, whether it supports batch restarting sub-devices, desc:PUT /ISAPI/System/RebootBatchChild?format=json-->true
</isSupportRebootBatchChild>
<isSupportFlowStatisticsTelemeter>
  <!--ro, opt, bool, whether it supports traffic statistics telemetry event, desc:eventType:FlowStatisticsTelemeter-->true
</isSupportFlowStatisticsTelemeter>
<isSupportModifyFlowStatisticsTelemeterUpload>
  <!--ro, opt, bool, whether it supports modifying the parameters of traffic statistics telemetry report strategy, desc:PUT
/ISAPI/System/Network/NetworkFlowMonitor/ModifyFlowStatisticsTelemeterUpload?format=json-->true
</isSupportModifyFlowStatisticsTelemeterUpload>
<isSupportClearFlowStatisticsTelemeter>
  <!--ro, opt, bool, whether it supports clearing the flow statistics telemetry data, desc:PUT
/ISAPI/System/Network/NetworkFlowMonitor/ClearFlowStatisticsTelemeter?format=json-->true
</isSupportClearFlowStatisticsTelemeter>
<isSupportFlowStatisticsTelemeterUpload>
  <!--ro, opt, bool, whether it supports the remote telemetry reporting strategy parameters for traffic statistics, desc:GET
/ISAPI/System/Network/NetworkFlowMonitor/FlowStatisticsTelemeterUpload?format=json-->true
</isSupportFlowStatisticsTelemeterUpload>
<isSupportScreenGraphicsColorControl>
  <!--ro, opt, bool, whether it supports screen image color control, desc:POST /ISAPI/System/ScreenColorQualityDetection/ScreenGraphicsColorControl?
format=json-->true
</isSupportScreenGraphicsColorControl>
<isSupportSearchScreenColorQualityDetectionResult>
  <!--ro, opt, bool, whether it supports searching for the screen image color quality detection results, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SearchScreenColorQualityDetectionResult?format=json-->true
</isSupportSearchScreenColorQualityDetectionResult>
<isSupportSearchScreenColorQualityDetectionProcess>
  <!--ro, opt, bool, whether it supports searching for the screen image color quality detection progress, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SearchScreenColorQualityDetectionProcess?format=json-->true
</isSupportSearchScreenColorQualityDetectionProcess>
<isSupportScreenColorQualityDetectionParams>
  <!--ro, opt, bool, whether it supports the screen image color quality detection parameters, desc:GET/PUT
/ISAPI/System/ScreenColorQualityDetection/channels/<channelID>/ScreenColorQualityDetectionParams?format=json-->true
</isSupportScreenColorQualityDetectionParams>
<isSupportAddInsulatorGroup>
  <!--ro, opt, bool, whether it supports adding insulator List, desc:GET /ISAPI/Thermal/channels/<channelID>/AddInsulatorGroup?format=json-->true
</isSupportAddInsulatorGroup>
<isSupportSuperResolutionReboot>
  <!--ro, opt, bool, whether the device supports rebooting super resolution, desc:whether the device supports restart to complete the conversion: true
(yes), false (no). Different devices use different chips. Some devices convert videos/images to ones with super resolution effect when the super resolution
function is enabled, while others require a device restart before the conversion is completed-->true
</isSupportSuperResolutionReboot>
<isSupportStartAutoTest>
  <!--ro, opt, bool, whether it supports starting automatic test, desc:POST /ISAPI/System/AutoTest/StartAutoTest?format=json-->true
</isSupportStartAutoTest>
<isSupportGetUserManualLink>
  <!--ro, opt, bool, whether it supports getting the device user manual link, desc:GET /ISAPI/System/GetUserManualLink?format=json-->true
</isSupportGetUserManualLink>
<isSupportExportSignaturePublicKey>
  <!--ro, opt, bool, whether it supports exporting the signed public key, desc:GET /ISAPI/Security/ProtocolCertificate/ExportSignaturePublicKey?
format=json-->true
</isSupportExportSignaturePublicKey>

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</isSupportExportSignaturePublicKey>
<isSupportCipherCardRunningStatus>
  <!--ro, opt, bool, whether it supports the password card running status, desc:GET /ISAPI/System/CipherCardRunningStatus?format=json-->true
</isSupportCipherCardRunningStatus>
<isSupportBridgeDeflectionMonitoringCapture>
  <!--ro, opt, bool, whether it supports manual triggering of bridge deflection monitoring and capturing, desc:POST
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionMonitoringCapture?format=json-->true
</isSupportBridgeDeflectionMonitoringCapture>
<isSupportAdaptiveVideoStreamParams>
  <!--ro, opt, bool, whether it supports adaptive video stream control parameters, desc:GET/PUT /ISAPI/ContentMgmt/InputProxy/AdaptiveVideoStreamParams?
format=json-->true
</isSupportAdaptiveVideoStreamParams>
<isSupportSearchAudioDetectionTaskProcess>
  <!--ro, opt, bool, whether it supports searching for the progress of audio detection tasks, desc:POST
/ISAPI/System/DeviceAudioDetection/SearchAudioDetectionTaskProcess?format=json-->true
</isSupportSearchAudioDetectionTaskProcess>
<isSupportSendAudioDetectionTask>
  <!--ro, opt, bool, whether it supports applying audio detection tasks, desc:POST /ISAPI/System/DeviceAudioDetection/SendAudioDetectionTask?format=json--
>true
</isSupportSendAudioDetectionTask>
<isSupportMedicalImageURLExpiredEvent>
  <!--ro, opt, bool, whether it supports medical image URL expiration event, desc:eventType:MedicalImageURLExpiredEvent-->true
</isSupportMedicalImageURLExpiredEvent>
<isSupportBridgeDeflectionMonitoringEvent>
  <!--ro, opt, bool, whether it supports bridge deflection monitoring event, desc:eventType:BridgeDeflectionMonitoringEvent-->true
</isSupportBridgeDeflectionMonitoringEvent>
<isSupportBridgeDeflectionTargetLossEvent>
  <!--ro, opt, bool, whether it supports bridge deflection target loss event, desc:eventType:BridgeDeflectionTargetLossEvent-->true
</isSupportBridgeDeflectionTargetLossEvent>
<isSupportNobodyDetectionRuleParams>
  <!--ro, opt, bool, whether it supports unmanned detection rule parameters, desc:GET/PUT /ISAPI/Event/channels/<channelID>/NobodyDetectionRuleParams?
format=json-->true
</isSupportNobodyDetectionRuleParams>
<isSupportNobodyDetectionRegionShieldParams>
  <!--ro, opt, bool, whether it supports the shielding parameters for unmanned detection areas, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/NobodyDetectionRegionShieldParams?format=json-->true
</isSupportNobodyDetectionRegionShieldParams>
<isSupportReduceTemperatureFilterParams>
  <!--ro, opt, bool, whether it supports the cooling filter parameter, desc:GET/PUT /ISAPI/Thermal/channels/<channelID>/ReduceTemperatureFilterParams?
format=json-->true
</isSupportReduceTemperatureFilterParams>
<isSupportGetStreamOverlayLogDebugStatus>
  <!--ro, opt, bool, whether it supports getting the status of overlaying log debugging status on stream, desc:GET
/ISAPI/ContentMgmt/channels/<channelID>/GetStreamOverlayLogDebugStatus?format=json-->true
</isSupportGetStreamOverlayLogDebugStatus>
<isSupportStreamOverlayLogDebugParams>
  <!--ro, opt, bool, whether it supports the parameters of overlaying log debugging information on stream, desc:GET/PUT
/ISAPI/ContentMgmt/channels/<channelID>/StreamOverlayLogDebugParams?format=json-->true
</isSupportStreamOverlayLogDebugParams>
<isSupportBridgeDeflectionAbnormalRuleParams>
  <!--ro, opt, bool, whether it supports the bridge deflection exception detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionAbnormalRuleParams?format=json-->true
</isSupportBridgeDeflectionAbnormalRuleParams>
<isSupportBridgeDeflectionAbnormalEvent>
  <!--ro, opt, bool, whether it supports bridge deflection exception event, desc:eventType:BridgeDeflectionAbnormalEvent-->true
</isSupportBridgeDeflectionAbnormalEvent>
<isSupportRestoreBackup>
  <!--ro, opt, bool, whether it supports data restoring and backup, desc:POST /ISAPI/System/SystemSnapshot/RestoreBackup?format=json-->true
</isSupportRestoreBackup>
<isSupportDeleteBackupFile>
  <!--ro, opt, bool, whether it supports deleting backup files, desc:POST /ISAPI/System/SystemSnapshot/DeleteBackupFile?format=json-->true
</isSupportDeleteBackupFile>
<isSupportSearchRestoreBackupProgress>
  <!--ro, opt, bool, whether it supports searching for the progress of restoring backup files, desc:POST
/ISAPI/System/SystemSnapshot/SearchRestoreBackupProgress?format=json-->true
</isSupportSearchRestoreBackupProgress>
<isSupportGetBackupsFileList>
  <!--ro, opt, bool, whether it supports getting the list of backup files, desc:POST /ISAPI/System/SystemSnapshot/GetBackupsFileList?format=json-->true
</isSupportGetBackupsFileList>
<isSupportAutoBackupParams>
  <!--ro, opt, bool, whether it supports automatic backup parameters, desc:GET/PUT /ISAPI/System/SystemSnapshot/AutoBackupParams?format=json-->true
</isSupportAutoBackupParams>
<isSupportDeviceSoftwareRunningProgramRestart>
  <!--ro, opt, bool, whether it supports restarting the device software program, desc:POST /ISAPI/System/DeviceSoftwareRunningProgramRestart?format=json--
>true
</isSupportDeviceSoftwareRunningProgramRestart>
<isSupportSendScreenColorQualityDetectionRequest>
  <!--ro, opt, bool, whether it supports sending a screen color quality detection request, desc:POST
/ISAPI/System/ScreenColorQualityDetection/SendScreenColorQualityDetectionRequest?format=json-->true
</isSupportSendScreenColorQualityDetectionRequest>
<isSupportTelephoningDetectionRuleParams>
  <!--ro, opt, bool, whether it supports phone call detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/TelephoningDetect/TelephoningDetectionRuleParams?format=json-->true
</isSupportTelephoningDetectionRuleParams>
<isSupportTelephoningBehaviorAlarmEvent>
  <!--ro, opt, bool, whether it supports phone call alarm event, desc:eventType:TelephoningBehaviorAlarmEvent-->true
</isSupportTelephoningBehaviorAlarmEvent>
<isSupportTelephoningBehaviorRecoveryEvent>
  <!--ro, opt, bool, whether it supports phone call restoring event, desc:eventType:TelephoningBehaviorRecoveryEvent-->true
</isSupportTelephoningBehaviorRecoveryEvent>
<isSupportSleepDutyRecoveryEvent>
  <!--ro, opt, bool, whether the device supports absence or sleeping on duty restoring event, desc:eventType:sleepDutyRecoveryEvent-->true
</isSupportSleepDutyRecoveryEvent>
<isSupportGetBridgeDeflectionTargetOffsetResetStatus>

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    <!--ro, opt, bool, whether it supports getting the zeroing result of bridge deflection target displacement, desc:GET
/ISAPI/Intelligent/channels/<channelID>/GetBridgeDeflectionTargetOffsetResetStatus?format=json-->true
    </isSupportGetBridgeDeflectionTargetOffsetResetStatus>
    <isSupportBridgeDeflectionTargetOffsetReset>
    <!--ro, opt, bool, whether it supports bridge deflection target displacement reset, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetOffsetReset?format=json-->true
    </isSupportBridgeDeflectionTargetOffsetReset>
    <isSupportBridgeDeflectionTargetLossRuleParams>
    <!--ro, opt, bool, whether it supports bridge deflection target loss rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/BridgeDeflectionTargetLossRuleParams?format=json-->true
    </isSupportBridgeDeflectionTargetLossRuleParams>
    <isSupportGetAnalogOutputParams>
    <!--ro, opt, bool, whether it supports getting all analog output parameters, desc:GET /ISAPI/System/AnalogOutput/GetAnalogOutputParams?format=json--
>true
    </isSupportGetAnalogOutputParams>
    <isSupportSystemReset>
    <!--ro, opt, bool, whether the device supports restoring system to default settings, desc:POST /ISAPI/System/SystemReset?format=json-->true
    </isSupportSystemReset>
    <isSupportFakePlateDetection>
    <!--ro, opt, bool, whether it supports fake license plate detection event, desc:corresponding event: FakePlateDetection-->true
    </isSupportFakePlateDetection>
    <isSupportSafetyWorkwearDetectionEvent>
    <!--ro, opt, bool, whether it supports the event of wearing safety work clothes detection, desc:corresponding event: SafetyWorkwearDetectionEvent-->true
    </isSupportSafetyWorkwearDetectionEvent>
    <isSupportVehicleMaintenanceStatusEvent>
    <!--ro, opt, bool, whether it supports vehicle maintenance status events, desc:corresponding event: VehicleMaintenanceStatusEvent-->true
    </isSupportVehicleMaintenanceStatusEvent>
    <isSupportEventRetransmissionInfo>
    <!--ro, opt, bool, whether it supports reloading event information, desc:GET /ISAPI/System/EventRetransmissionInfo?format=json-->true
    </isSupportEventRetransmissionInfo>
    <isSupportGetEnvironmentCaptureStatus>
    <!--ro, opt, bool, whether it supports getting the environmental data collection status, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/GetEnvironmentCaptureStatus?format=json-->true
    </isSupportGetEnvironmentCaptureStatus>
    <isSupportEnvironmentCapture>
    <!--ro, opt, bool, whether it supports environmental data collection, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/EnvironmentCapture?format=json-->true
    </isSupportEnvironmentCapture>
    <isSupportSearchCPUDetailInfo>
    <!--ro, opt, bool, whether it supports searching for detailed CPU performance information, desc:POST /ISAPI/System/SearchCPUDetailInfo?format=json--
>true
    </isSupportSearchCPUDetailInfo>
    <isSupportPanelControlParams>
    <!--ro, opt, bool, whether it supports device hardware panel configuration parameters, desc:GET/PUT /ISAPI/System/PanelControlParams?format=json-->true
    </isSupportPanelControlParams>
    <isSupportSearchGoodsIncAndDecStatistics>
    <!--ro, opt, bool, whether it supports searching for cargo increase/decrease statistics, desc:POST
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/SearchGoodsIncAndDecStatistics?format=json-->true
    </isSupportSearchGoodsIncAndDecStatistics>
    <isSupportCustomAlarmHTTPParams>
    <!--ro, opt, bool, whether it supports custom alarm HTTP parameters, desc:GET/PUT /ISAPI/Event/notification/httpHosts/CustomAlarmHTTPParams?format=json-
>true
    </isSupportCustomAlarmHTTPParams>
    <isSupportTestCustomAlarmHTTP>
    <!--ro, opt, bool, whether it supports custom alarm HTTP request test, desc:PUT /ISAPI/Event/notification/httpHosts/TestCustomAlarmHTTP?format=json--
>true
    </isSupportTestCustomAlarmHTTP>
    <isSupportCustomAlarmHTTPLinkageParams>
    <!--ro, opt, bool, whether it supports customizing the alarm HTTP linkage parameters, desc:GET/PUT /ISAPI/Thermal/CustomAlarmHTTPLinkageParams?
format=json-->true
    </isSupportCustomAlarmHTTPLinkageParams>
    <isSupportAccumulationDetectionPeriodicReport>
    <!--ro, opt, bool, whether it supports scheduled reporting of pile detection events, desc:corresponding event: AccumulationDetectionPeriodicReport--
>true
    </isSupportAccumulationDetectionPeriodicReport>
    <isSupportImportDiskCompatibilityCertificate>
    <!--ro, opt, bool, whether it supports importing disk compatibility certificate, desc:PUT /ISAPI/Security/serverCertificate/certificate?type=
<type>&pPlatformNo=<pPlatformNo>, and type is DiskCompatibility-->true
    </isSupportImportDiskCompatibilityCertificate>
    <isSupportGetAccumulationDetectionAlgorithmicVersion>
    <!--ro, opt, bool, whether it supports getting the version of the stacking detection algorithm, desc:GET
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/GetAccumulationDetectionAlgorithmicVersion?format=json-->true
    </isSupportGetAccumulationDetectionAlgorithmicVersion>
    <isSupportSnapshotUpgradeDifferenceFile>
    <!--ro, opt, bool, whether it supports getting snapshot upgrade difference files, desc:GET /ISAPI/System/SystemSnapshot/SnapshotUpgradeDifferenceFile?
format=json-->true
    </isSupportSnapshotUpgradeDifferenceFile>
    <isSupportGroupRecoveryEvent>
    <!--ro, opt, bool, whether it supports people gathering restoring event-->true
    </isSupportGroupRecoveryEvent>
    <isSupportElectronicEyepieceParams>
    <!--ro, opt, bool, whether it supports electron microscope channel parameters, desc:GET/PUT /ISAPI/System/Video/ElectronicEyepieceParams?format=json--
>true
    </isSupportElectronicEyepieceParams>
    <isSupportSmokingDetectionRecoverEvent>
    <!--ro, opt, bool, whether it supports smoking detection restoring event, desc:fire alarm upload in progress, eventType: SmokingDetectionRecoverEvent--
>true
    </isSupportSmokingDetectionRecoverEvent>
    <isSupportDeviceRegionParams>
    <!--ro, opt, bool, whether it supports device area parameters, desc:GET/PUT /ISAPI/System/DeviceRegionParams?format=json-->true
    </isSupportDeviceRegionParams>
    <isSupportTemperatureEvent>
    <!--ro, opt, bool, whether it supports temperature alarm event, desc:TemperatureEvent-->true

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</isSupportTemperatureEvent>
<isSupportTemperatureRecoverEvent>
  <!--ro, opt, bool, whether it supports temperature restoring event, desc:TemperatureRecoverEvent-->true
</isSupportTemperatureRecoverEvent>
<isSupportWebSSH>
  <!--ro, opt, bool, whether it supports WebSSH, desc:if the node is not returned, it indicates no-->true
</isSupportWebSSH>
<isSupportCreateDebugChannel>
  <!--ro, opt, bool, whether it supports creating WebSSH debugging channel, desc:if the node is not returned, it indicates no-->true
</isSupportCreateDebugChannel>
<isSupportExportDebugLog>
  <!--ro, opt, bool, whether it supports exporting the debugging log, desc:if the node is not returned, it indicates no-->true
</isSupportExportDebugLog>
<isSupportAccumulationDetectionEffectiveAreaCfgParams>
  <!--ro, opt, bool, whether it supports the configuration parameters of pile detection effective area, desc:GET/PUT
/ISAPI/Event/channels/<channelID>/objectAccumulationDetection/AccumulationDetectionEffectiveAreaCfgParams?format=json-->true
</isSupportAccumulationDetectionEffectiveAreaCfgParams>
<isSupportGetGoalDryWetBallTempCurve>
  <!--ro, opt, bool, whether it supports getting the target dry-wet bulb temperature curve, desc:POST
/ISAPI/System/TobaccoBakeMgr/GetGoalDryWetBallTempCurve?format=json-->true
</isSupportGetGoalDryWetBallTempCurve>
<isSupportBakeTobaccoRuleParams>
  <!--ro, opt, bool, whether it supports tobacco baking rule parameters, desc:GET/PUT /ISAPI/System/TobaccoBakeMgr/BakeTobaccoRuleParams?format=json--
>true
</isSupportBakeTobaccoRuleParams>
<isSupportStartBakeTobacco>
  <!--ro, opt, bool, whether it supports starting baking tobacco, desc:POST /ISAPI/System/TobaccoBakeMgr/StartBakeTobacco?format=json-->true
</isSupportStartBakeTobacco>
<isSupportEndBakeTobacco>
  <!--ro, opt, bool, whether it supports stopping baking tobacco, desc:POST /ISAPI/System/TobaccoBakeMgr/EndBakeTobacco?format=json-->true
</isSupportEndBakeTobacco>
<isSupportDryWetBallTempEventRuleParams>
  <!--ro, opt, bool, whether it supports the dry-wet bulb temperature reporting rule parameters, desc:GET/PUT
/ISAPI/System/TobaccoBakeMgr/DryWetBallTempEventRuleParams?format=json-->true
</isSupportDryWetBallTempEventRuleParams>
<isSupportDevAccessServerNetworkStatus>
  <!--ro, opt, bool, whether it supports device connection status with multi-access service network, desc:GET /ISAPI/System/DevAccessServerNetworkStatus?
format=json-->true
</isSupportDevAccessServerNetworkStatus>
<isSupportMaintenanceServiceAuthorization>
  <!--ro, opt, bool, whether it supports device maintenance service authorization information, desc:GET/PUT /ISAPI/System/MaintenanceServiceAuthorization?
format=json-->true
</isSupportMaintenanceServiceAuthorization>
<isSupportUploadIndividualCallSignalOperation>
  <!--ro, opt, bool, whether it supports uploading single-mode call signaling events, desc:UploadIndividualCallSignalOperation-->true
</isSupportUploadIndividualCallSignalOperation>
<isSupportUploadGroupCallSignalOperation>
  <!--ro, opt, bool, whether it supports uploading group call signaling event, desc:UploadGroupCallSignalOperation-->true
</isSupportUploadGroupCallSignalOperation>
<isSupportDrivingBehaviorVoiceReminder>
  <!--ro, opt, bool, whether it supports configuring intelligent voice reminders for driving behavior, desc:GET/PUT
/ISAPI/Event/DrivingBehaviorVoiceReminder?format=json-->true
</isSupportDrivingBehaviorVoiceReminder>
<isSupportImportMultiCameraCalibrationFile>
  <!--ro, opt, bool, whether it supports importing image calibration files of multiple cameras, desc:POST
/ISAPI/Intelligent/radarVideoDetection/channels/<channelID>/ImportMultiCameraCalibrationFile?format=json-->true
</isSupportImportMultiCameraCalibrationFile>
<isSupportDeletePlayerFileList>
  <!--ro, opt, bool, whether it supports batch deleting player files, desc:POST /ISAPI/System/Video/DeletePlayerFileList?format=json-->true
</isSupportDeletePlayerFileList>
<isSupportPlayerInfoParams>
  <!--ro, opt, bool, whether it supports player information parameters, desc:GET /ISAPI/System/Video/PlayerInfoParams?format=json-->true
</isSupportPlayerInfoParams>
<isSupportBurnPlayerParams>
  <!--ro, opt, bool, whether it supports burning player parameters, desc:GET/PUT /ISAPI/System/Video/BurnPlayerParams?format=json-->true
</isSupportBurnPlayerParams>
<isSupportOneKeyVoiceAbnormalDetection>
  <!--ro, opt, bool, whether it supports quick voice exception detection, desc:POST
/ISAPI/System/VoiceAbnormalDetection/channels/<channelID>/OneKeyVoiceAbnormalDetection?format=json-->true
</isSupportOneKeyVoiceAbnormalDetection>
<isSupportVoiceAbnormalDetectionRuleParams>
  <!--ro, opt, bool, whether it supports quick voice exception detection parameters, desc:GET/PUT
/ISAPI/System/VoiceAbnormalDetection/channels/<channelID>/VoiceAbnormalDetectionRuleParams?format=json-->true
</isSupportVoiceAbnormalDetectionRuleParams>
<isSupportSearchOneKeyVoiceAbnormalDetectionProcess>
  <!--ro, opt, bool, whether it supports searching for the progress of quick voice exception detection, desc:POST
/ISAPI/System/VoiceAbnormalDetection/SearchOneKeyVoiceAbnormalDetectionProcess?format=json-->true
</isSupportSearchOneKeyVoiceAbnormalDetectionProcess>
<isSupportSearchOneKeyAudioDetection>
  <!--ro, opt, bool, whether it supports searching for the progress of quick audio detection, desc:POST
/ISAPI/Smart/AudioExceptionDetection/SearchOneKeyAudioDetection?format=json-->true
</isSupportSearchOneKeyAudioDetection>
<isSupportOneKeyAudioDetection>
  <!--ro, opt, bool, whether it supports quick audio detection, desc:POST /ISAPI/Smart/AudioExceptionDetection/channels/<channelID>/OneKeyAudioDetection?
format=json-->true
</isSupportOneKeyAudioDetection>
<isNotSupportScheduledRecordUpload>
  <!--ro, opt, bool, whether it supports the event of scheduled recording of channel, desc:if this node does not exist or returns false (default value),
it indicates for support; if it returns true, it indicates no-->>false
</isNotSupportScheduledRecordUpload>
<isSupportDevEnvironmentalCtrlMaterielInfoDetect>
  <!--ro, opt, bool, whether it supports the detection of equipment environmental control material information?, desc:POST
/ISAPI/System/DeviceModuleDiagnosis/DevEnvironmentalCtrlMaterielInfoDetect?format=json-->true
</isSupportDevEnvironmentalCtrlMaterielInfoDetect>

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<isSupportWaterRescueDetectionRuleParams>
  <!--ro, opt, bool, whether it supports water rescue detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/WaterRescueDetectionRuleParams?format=json-->true
</isSupportWaterRescueDetectionRuleParams>
<isSupportWaterRescueDetectionEvent>
  <!--ro, opt, bool, whether it supports water rescue detection event, desc:corresponding event: WaterRescueDetectionEvent-->true
</isSupportWaterRescueDetectionEvent>
<isSupportGetActualDryWetBallTempCurve>
  <!--ro, opt, bool, whether it supports getting the actual wet-bulb temperature curve, desc:POST
/ISAPI/System/TobaccoBakeMgr/GetActualDryWetBallTempCurve?format=json-->true
</isSupportGetActualDryWetBallTempCurve>
<isSupportPerimeterAlgoTargetFilterParams>
  <!--ro, opt, bool, whether it supports target filtering parameters of perimeter algorithm, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/PerimeterAlgoTargetFilterParams?format=json-->true
</isSupportPerimeterAlgoTargetFilterParams>
<isSupportSendDeviceMaintenanceStatus>
  <!--ro, opt, bool, whether it supports applying device maintenance status, desc:PUT /ISAPI/System/SendDeviceMaintenanceStatus?format=json-->true
</isSupportSendDeviceMaintenanceStatus>
<videoEncodeChannel opt="1,2">
  <!--ro, opt, string, channel supporting video parameter display on the web, attr:opt{req, string}, desc:channel supporting video parameter display on
the web-->1,2
</videoEncodeChannel>
<isSupportLocalButtonRecordControl>
  <!--ro, opt, bool, whether it supports controlling Local recording button on the device, desc:/ISAPI/System/LocalButtonRecord?format=json-->true
</isSupportLocalButtonRecordControl>
<isSupportLocalButtonSnapshotControl>
  <!--ro, opt, bool, whether it supports controlling Local capture button on the device, desc:/ISAPI/System/LocalButtonSnapshot?format=json-->true
</isSupportLocalButtonSnapshotControl>
<isSupportSecurityDisclaimerParams>
  <!--ro, opt, bool, whether it supports the security disclaimer parameters, desc:GET/PUT /ISAPI/System/InformedConsentMgr/SecurityDisclaimerParams?
format=json-->true
</isSupportSecurityDisclaimerParams>
<isSupportThermometryGatewayParams>
  <!--ro, opt, bool, whether it supports configuring parameters of the temperature measurement gateway, desc:GET/PUT
/ISAPI/Thermal/ThermometryGatewayParams?format=json-->true
</isSupportThermometryGatewayParams>
<isSupportSearchRegionDefaultInfo>
  <!--ro, opt, bool, whether it supports searching for the default parameters for the region, desc:POST /ISAPI/System/SearchRegionDefaultInfo?format=json
-->true
</isSupportSearchRegionDefaultInfo>
<isSupportAntiCollisionWarningEvent>
  <!--ro, opt, bool, whether it supports anti-collision pre-alarm event, desc:AntiCollisionWarningEvent-->true
</isSupportAntiCollisionWarningEvent>
<isSupportMovementWorkStatus>
  <!--ro, opt, bool, whether it supports gas cloud module working status, desc:GET /ISAPI/System/MovementMgr/MovementWorkStatus?format=json-->true
</isSupportMovementWorkStatus>
<isSupportMovementTimingPowerOffParams>
  <!--ro, opt, bool, whether it supports scheduled shutdown parameters of the gas cloud imaging module, desc:GET/PUT
/ISAPI/System/MovementMgr/MovementTimingPowerOffParams?format=json-->true
</isSupportMovementTimingPowerOffParams>
<isSupportFireExtinguisherDetectionRuleParams>
  <!--ro, opt, bool, whether it supports fire extinguisher detection rule parameters, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/FireExtinguisherDetectionRuleParams?format=json-->true
</isSupportFireExtinguisherDetectionRuleParams>
<isSupportFireExtinguisherRecoveryEvent>
  <!--ro, opt, bool, whether it supports fire extinguisher detection restoring event, desc:eventType:FireExtinguisherRecoveryEvent-->true
</isSupportFireExtinguisherRecoveryEvent>
<isSupportGasTankMonitoringRecoveryEvent>
  <!--ro, opt, bool, whether it supports gas tank detection restoring event, desc:eventType:GasTankMonitoringRecoveryEvent-->true
</isSupportGasTankMonitoringRecoveryEvent>
<isSupportSearchImportPlayerFileProcess>
  <!--ro, opt, bool, whether it supports searching for the progress of importing player files, desc:POST
/ISAPI/System/Video/SearchImportPlayerFileProcess?format=json-->true
</isSupportSearchImportPlayerFileProcess>
<isSupportStatisticalDataFlashStorageParams>
  <!--ro, opt, bool, whether it supports data statistics for Flash storage parameters, desc:GET/PUT
/ISAPI/System/dataAware/StatisticalDataFlashStorageParams?format=json-->true
</isSupportStatisticalDataFlashStorageParams>
<eventRecheckCap>
  <!--ro, opt, object, event review capability, desc:Low-end cameras depend on powerful NVRs, due to the limitation of computing power for secondary
review of Low-end camera alarms to reduce false alarms-->
<algorithmInfolist>
  <!--ro, opt, array, subType:object, range:[1,5]-->
<algorithmInfoitem>
  <!--ro, req, object-->
<algorithmType>
  <!--ro, req, enum, subType:string-->VMD
</algorithmType>
<algorithmVersion>
  <!--ro, req, string-->1B
</algorithmVersion>
</algorithmInfoitem>
</algorithmInfolist>
</eventRecheckCap>
<isSupportModifyWindowGroup>
  <!--ro, opt, bool, whether it supports modifying the specified window group of the scope, desc:PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/ModifyWindowGroup?format=json-->true
</isSupportModifyWindowGroup>
<isSupportCalibrateWindowGroup>
  <!--ro, opt, bool, whether it supports calibrating specified window group of the scope, desc:PUT
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/CalibrateWindowGroup?format=json-->true
</isSupportCalibrateWindowGroup>
<isSupportAllWindowParams>

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    <!--ro, opt, bool, whether it supports all scope window parameters, desc:GET /ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/AllWindowParams?
format=json-->true
</isSupportAllWindowParams>
<isSupportWindowCalibrationStatus>
    <!--ro, opt, bool, whether it supports getting the scope window calibration status, desc:GET
/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/WindowCalibrationStatus?format=json-->true
</isSupportWindowCalibrationStatus>
<isSupportOBUModuleInfo>
    <!--ro, opt, bool, whether it supports OBU module information, desc:GET /ISAPI/Traffic/ETC/OBUModuleInfo?format=json-->true
</isSupportOBUModuleInfo>
<isSupportExportOBUDetectionLogFile>
    <!--ro, opt, bool, whether it supports exporting the OBU Label detection module Logs, desc:POST /ISAPI/Traffic/ETC/ExportOBUDetectionLogFile?
format=json-->true
</isSupportExportOBUDetectionLogFile>
<isSupportOBUDetectionReboot>
    <!--ro, opt, bool, whether it supports restarting OBU Label detection module, desc:PUT /ISAPI/Traffic/ETC/OBUDetectionReboot?format=json-->true
</isSupportOBUDetectionReboot>
<isSupportGetOBUR rebootStatus>
    <!--ro, opt, bool, whether it supports getting the restart status of the OBU module, desc:GET /ISAPI/Traffic/ETC/GetOBUR rebootStatus?format=json-->true
</isSupportGetOBUR rebootStatus>
<isSupportSearchOBUSStatisticsReport>
    <!--ro, opt, bool, whether it supports searching for the statistical data of OBU tag detection, desc:POST /ISAPI/Traffic/ETC/SearchOBUSStatisticsReport?
format=json-->true
</isSupportSearchOBUSStatisticsReport>
<isSupportSearchTargetEnhancementInfo>
    <!--ro, opt, bool, whether it supports searching for target prominence parameters, desc:POST
/ISAPI/System/Video/TargetEnhancement/channels/<channelID>/SearchTargetEnhancementInfo?format=json-->true
</isSupportSearchTargetEnhancementInfo>
<isSupportTargetEnhancementParams>
    <!--ro, opt, bool, whether it supports target prominence parameters, desc:GET/PUT
/ISAPI/System/Video/TargetEnhancement/channels/<channelID>/TargetEnhancementParams?format=json-->true
</isSupportTargetEnhancementParams>
<isSupportPicSearchByCoordinate>
    <!--ro, opt, bool, whether the device supports configuring AcuSearch parameters, desc:GET/PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json--
>true
</isSupportPicSearchByCoordinate>
<AcuSearchCap>
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    <targetType opt="face,human,vehicle">
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<isSupportSendDevRemoteGUICtrl>
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/ISAPI/System/RemoteCtrlDevGUI/SendDevRemoteGUICtrl?format=json-->true
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<isSupportStreamTemperatureParams>
    <!--ro, opt, bool, whether the device supports configuring stream temperature parameters of thermal imaging, desc:GET/PUT
/ISAPI/Thermal/channels/<channelID>/StreamTemperatureParams?format=json-->true
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<moduleTypeSupportCap>
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        <!--ro, opt, bool, whether it supports maintenance module, desc:the wire products are embedded with the business module (businessModule) and
maintenance module (maintenanceModule), which supports independent access to EZVIZ cloud, etc.; at the same time, they support the cloud platform to send
maintenance instructions to the maintenance module-->true
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    <isSupportGetDeviceMaintenanceStatus>
        <!--ro, opt, bool, whether it supports getting device maintenance status, desc:GET /ISAPI/System/GetDeviceMaintenanceStatus?format=json-->true
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    <isSupportImportConfigurationFileByModel>
        <!--ro, opt, bool, whether it supports importing configuration files by device model, desc:POST /ISAPI/System/ImportConfigurationFileByModel?
format=json-->true
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    <isSupportRadarVideoDetectionStorageParams>
        <!--ro, opt, bool, whether it supports the parameters of saving radar-assisted target detection result, desc:GET/PUT
/ISAPI/IntelLigent/radarVideoDetection/RadarVideoDetectionStorageParams?format=json-->true
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    <isSupportBodyCameraMediaInfo>
        <!--ro, opt, bool, whether it supports body camera reporting media information, desc:eventType: BodyCameraMediaInfo-->true
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    <isSupportRadarVideoPOSOVerlayParams>
        <!--ro, opt, bool, whether it supports smart radar-assisted POS information overlay parameters, desc:GET/PUT
/ISAPI/IntelLigent/radarVideoDetection/RadarVideoPOSOVerlayParams?format=json-->true
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/ISAPI/Event/notification/httpHosts/RecoverCustomEventTemplate?format=json-->true
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/ISAPI/Event/notification/httpHosts/ModCustomEventTemplateParams?format=json-->true
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/ISAPI/Smart/VideoGunsightCalibration/channels/<channelID>/WindowRuleParams?format=json-->true
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/ISAPI/Thermal/GetMultiRemoteAlbumPicture?format=json-->true
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<isSupportGetDeviceTokenInfo>
  <!--ro, opt, bool, whether the device supports getting the device token information, desc:/ISAPI/System/ImportConfigurationFileByURL?format=json
/ISAPI/System/DeviceAuth/GetDeviceTokenInfo?format=json-->true
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<isSupportDisplayEnableCloudServicePrompt>
  <!--ro, opt, bool, whether it supports displaying the prompt for enabling cloud services, desc:false by default. When the device returns true, it
indicates that displaying a prompt of enabling cloud services is supported. Scenario example: After your device is connected to the Internet and the cloud
service is enabled, we will provide you with better cloud device management and maintenance upgrade services. Therefore, we will collect relevant device
information such as the device's IP address, device name, and device serial No. If you have any questions, you can contact customer service at any time--
>true
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  <!--ro, opt, bool, whether it supports playing mobile restoring event, desc:eventType:PlayCellphoneRecoveryEvent-->true
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</isSupportGetIntellAndTemperatureRangeMap>
<isSupportSendScreenColorEnvErrorCalibrationTask>
  <!--ro, opt, bool-->true
</isSupportSendScreenColorEnvErrorCalibrationTask>
<isSupportSearchScreenColorEnvErrorCalibrationTask>
  <!--ro, opt, bool-->true
</isSupportSearchScreenColorEnvErrorCalibrationTask>
<isSupportCurrentScreenColorEnvErrorCalibrationParams>
  <!--ro, opt, bool-->true
</isSupportCurrentScreenColorEnvErrorCalibrationParams>
<isSupportSynchronizationRulesBetweenChannels>
  <!--ro, opt, bool-->true
</isSupportSynchronizationRulesBetweenChannels>
<isSupportGestureRecognitionRuleParams>
  <!--ro, opt, bool-->true
</isSupportGestureRecognitionRuleParams>
<isSupportSolidBallSport>
  <!--ro, opt, bool-->true
</isSupportSolidBallSport>
<isSupportVolleyballSport>
  <!--ro, opt, bool-->true
</isSupportVolleyballSport>
<isSupportBasketballSport>
  <!--ro, opt, bool-->true
</isSupportBasketballSport>
<isSupportSoccerSport>
  <!--ro, opt, bool-->true
</isSupportSoccerSport>
<isSupportPersonnelGestureDetection>
  <!--ro, opt, bool-->true
</isSupportPersonnelGestureDetection>
<isSupportExportLensThermalCalibrationFile>
  <!--ro, opt, bool-->true
</isSupportExportLensThermalCalibrationFile>
<isSupportModifyPresetRatioCombination>
  <!--ro, opt, bool-->true
</isSupportModifyPresetRatioCombination>
<isSupportExportChannelConfigAbnormalReport>
  <!--ro, opt, bool-->true
</isSupportExportChannelConfigAbnormalReport>
<isSupportExportChannelConfigTemplate>
  <!--ro, opt, bool-->true
</isSupportExportChannelConfigTemplate>
<isSupportSoftStorageInterfaceParams>
  <!--ro, opt, bool-->true
</isSupportSoftStorageInterfaceParams>
<isSupportImportSceneFile>
  <!--ro, opt, bool-->true
</isSupportImportSceneFile>
<isSupportExportSceneFile>
  <!--ro, opt, bool-->true
</isSupportExportSceneFile>
<isSupportRadarWaterLevelAdvancedParams>
  <!--ro, opt, bool-->true
</isSupportRadarWaterLevelAdvancedParams>
<isSupportOpenAndCloseDoorAlarmEvent>
```

```
<!--ro, opt, bool-->true
</isSupportOpenAndCloseDoorLearningEvent>
<isSupportResetScreenColorEnvErrorCalibration>
  <!--ro, opt, bool-->true
</isSupportResetScreenColorEnvErrorCalibration>
<isSupportConsumptionPermissionSearchEvent>
  <!--ro, opt, bool-->true
</isSupportConsumptionPermissionSearchEvent>
<isSupportDeviceCap>
  <!--ro, opt, bool-->true
</isSupportDeviceCap>
<isSupportVideoPipList>
  <!--ro, opt, bool-->true
</isSupportVideoPipList>
<maxProtocolConcurrencyNum>
  <!--ro, opt, int-->10
</maxProtocolConcurrencyNum>
<isSupportReplayShotImage>
  <!--ro, opt, bool-->true
</isSupportReplayShotImage>
<isSupportColumnSpeakers>
  <!--ro, opt, bool-->true
</isSupportColumnSpeakers>
<isSupportPublish>
  <!--ro, opt, bool-->true
</isSupportPublish>
<isSupportGetDevicePartsLife>
  <!--ro, opt, bool-->true
</isSupportGetDevicePartsLife>
<isSupportVisibilityTelemeter>
  <!--ro, opt, bool-->true
</isSupportVisibilityTelemeter>
<isSupportVisibilityEvent>
  <!--ro, opt, bool-->true
</isSupportVisibilityEvent>
<isSupportRestoreRadarWaterLevelDefaultParams>
  <!--ro, opt, bool-->true
</isSupportRestoreRadarWaterLevelDefaultParams>
<isSupportDeviceCreateUserNum>
  <!--ro, opt, bool-->true
</isSupportDeviceCreateUserNum>
<isSupportSetStreamingSecretKey>
  <!--ro, opt, bool-->true
</isSupportSetStreamingSecretKey>
<isSupportSearchClickToThermometry>
  <!--ro, opt, bool-->true
</isSupportSearchClickToThermometry>
<isSupportResourceChildParams>
  <!--ro, opt, bool-->true
</isSupportResourceChildParams>
<isSupportSearchThermometryInfo>
  <!--ro, opt, bool-->true
</isSupportSearchThermometryInfo>
<isSupportSendScanInfoToCamera>
  <!--ro, opt, bool-->true
</isSupportSendScanInfoToCamera>
<isSupportRtspPreRecord>
  <!--ro, opt, bool-->true
</isSupportRtspPreRecord>
<isSupportGetClientAccessCapabilities>
  <!--ro, opt, bool-->true
</isSupportGetClientAccessCapabilities>
<isSupportISUPChileDevHttp>
  <!--ro, opt, bool-->true
</isSupportISUPChileDevHttp>
<isSupportRemoteDesktopParams>
  <!--ro, opt, bool-->true
</isSupportRemoteDesktopParams>
<isSupportPassbackRecordUpload>
  <!--ro, opt, bool-->true
</isSupportPassbackRecordUpload>
<alertDistributionAddMode opt="cover,append" def="append">
  <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->append
</alertDistributionAddMode>
<isSupportSkeletonKeypointRuleParams>
  <!--ro, opt, bool-->true
</isSupportSkeletonKeypointRuleParams>
<isSupportHumanSkeletonKeypointDetection>
  <!--ro, opt, bool-->true
</isSupportHumanSkeletonKeypointDetection>
<isSupportWaterQualityCountingRule>
  <!--ro, opt, bool-->true
</isSupportWaterQualityCountingRule>
<isSupportWaterQualityReport>
  <!--ro, opt, bool-->true
</isSupportWaterQualityReport>
<isOnlySupportAnalogChannel>
  <!--ro, opt, bool-->true
</isOnlySupportAnalogChannel>
<isSupportClarityLevel>
  <!--ro, opt, bool-->true
</isSupportClarityLevel>
<isSupportDeviceNetworkDiscoverySearchParams>
```

```
<!--ro, opt, bool-->true
</isSupportDeviceNetworkDiscoverySearchParams>
<isSupportConfigurationFileAsyncUpload>
  <!--ro, opt, bool-->true
</isSupportConfigurationFileAsyncUpload>
<isSupportSearchAsyncUploadStatus>
  <!--ro, opt, bool-->true
</isSupportSearchAsyncUploadStatus>
<isSupportSoundClassificationDetection>
  <!--ro, opt, bool-->true
</isSupportSoundClassificationDetection>
<isSupportTwoLevelPaletteFrameReductionTips>
  <!--ro, opt, bool-->false
</isSupportTwoLevelPaletteFrameReductionTips>
<isSupportSwitchToUpgradePreparationMode>
  <!--ro, opt, bool-->true
</isSupportSwitchToUpgradePreparationMode>
<isSupportMovementUpgrade>
  <!--ro, opt, bool-->true
</isSupportMovementUpgrade>
<isSupportGetMovementUpgradeStatus>
  <!--ro, opt, bool-->true
</isSupportGetMovementUpgradeStatus>
<isSupportSupplementAndWhiteAlarmConflictTips>
  <!--ro, opt, bool-->false
</isSupportSupplementAndWhiteAlarmConflictTips>
<isSupportNotifyDevRemoteGUICtrl>
  <!--ro, opt, bool-->true
</isSupportNotifyDevRemoteGUICtrl>
<isSupportExampleTerms>
  <!--ro, opt, bool-->true
</isSupportExampleTerms>
<isSupportDeleteHistoricalTerms>
  <!--ro, opt, bool-->true
</isSupportDeleteHistoricalTerms>
<isSupportGetHistoricalTerms>
  <!--ro, opt, bool-->true
</isSupportGetHistoricalTerms>
<isSupportAddModelFineTuning>
  <!--ro, opt, bool-->true
</isSupportAddModelFineTuning>
<isSupportSearchModelFineTuning>
  <!--ro, opt, bool-->true
</isSupportSearchModelFineTuning>
<isSupportDeleteModelFineTuning>
  <!--ro, opt, bool-->true
</isSupportDeleteModelFineTuning>
<isSupportAlarmTest>
  <!--ro, opt, bool-->true
</isSupportAlarmTest>
<isSupportAcuSeekImageModel>
  <!--ro, opt, bool-->true
</isSupportAcuSeekImageModel>
<isSupportRadarWaterFlowDetectRegionRestore>
  <!--ro, opt, bool-->true
</isSupportRadarWaterFlowDetectRegionRestore>
<isSupportLoRaPeriodicStatus>
  <!--ro, opt, bool-->true
</isSupportLoRaPeriodicStatus>
<isSupportEnableModelCtrl>
  <!--ro, opt, bool-->true
</isSupportEnableModelCtrl>
<isSupportStorageServiceNetwork>
  <!--ro, opt, bool-->true
</isSupportStorageServiceNetwork>
<isSupportSearchAudioDetectionEvent>
  <!--ro, opt, bool-->true
</isSupportSearchAudioDetectionEvent>
<isSupportAcuSeekInfo>
  <!--ro, opt, bool-->true
</isSupportAcuSeekInfo>
<isSupportManualTriggerDeicing>
  <!--ro, opt, bool-->true
</isSupportManualTriggerDeicing>
<isSupportSearchCameraIndex>
  <!--ro, opt, bool-->true
</isSupportSearchCameraIndex>
<isSupportGetStrategyOrchTPL>
  <!--ro, opt, bool-->true
</isSupportGetStrategyOrchTPL>
<isSupportSearchStrategyOrchResult>
  <!--ro, opt, bool-->true
</isSupportSearchStrategyOrchResult>
<isSupportAddStrategyOrch>
  <!--ro, opt, bool-->true
</isSupportAddStrategyOrch>
<isSupportModifyStrategyOrch>
  <!--ro, opt, bool-->true
</isSupportModifyStrategyOrch>
<isSupportSearchStrategyOrch>
  <!--ro, opt, bool-->true
</isSupportSearchStrategyOrch>
<isSupportDeleteStrategyOrch>
  <!--ro, opt, bool-->true
</isSupportDeleteStrategyOrch>
```

```
<!--ro, opt, bool-->true
</isSupportDeleteStrategyOrch>
<isSupportIOTAsynPingTest>
  <!--ro, opt, bool-->true
</isSupportIOTAsynPingTest>
<isSupportImportSoftActiveFile>
  <!--ro, opt, bool-->true
</isSupportImportSoftActiveFile>
<isSupportExportSoftActiveFile>
  <!--ro, opt, bool-->true
</isSupportExportSoftActiveFile>
<isSupportAcuSeekAuthorizeParam>
  <!--ro, opt, bool-->true
</isSupportAcuSeekAuthorizeParam>
<isSupportSearchSoftwareActivationStatus>
  <!--ro, opt, bool-->true
</isSupportSearchSoftwareActivationStatus>
<isSupportGetCollectTerms>
  <!--ro, opt, bool-->true
</isSupportGetCollectTerms>
<isSupportCourseKeyStatus>
  <!--ro, opt, bool-->true
</isSupportCourseKeyStatus>
<isSupportVehicleTurnBackDetectionEvent>
  <!--ro, opt, bool-->true
</isSupportVehicleTurnBackDetectionEvent>
<isSupportSearchStrategyOrchList>
  <!--ro, opt, bool-->true
</isSupportSearchStrategyOrchList>
<isSupportSearchEventInputUnit>
  <!--ro, opt, bool-->true
</isSupportSearchEventInputUnit>
<isSupportSearchOperatorUnit>
  <!--ro, opt, bool-->true
</isSupportSearchOperatorUnit>
<isSupportSearchEventOutputUnit>
  <!--ro, opt, bool-->true
</isSupportSearchEventOutputUnit>
<isSupportFDLibAsyncResults>
  <!--ro, opt, bool-->true
</isSupportFDLibAsyncResults>
<isSupportWorkingMode>
  <!--ro, opt, bool-->true
</isSupportWorkingMode>
<isSupportFlowVelocityCalibration>
  <!--ro, opt, bool-->true
</isSupportFlowVelocityCalibration>
<isSupportEditConfigFile>
  <!--ro, opt, bool-->true
</isSupportEditConfigFile>
<isSupportDisturbSchedule>
  <!--ro, opt, bool-->true
</isSupportDisturbSchedule>
<isSupportappUsageDuration>
  <!--ro, opt, bool-->true
</isSupportappUsageDuration>
<isSupportExpyNetTollServerParams>
  <!--ro, opt, bool-->true
</isSupportExpyNetTollServerParams>
<isSupportExpyNetTollStgSrvParams>
  <!--ro, opt, bool-->true
</isSupportExpyNetTollStgSrvParams>
<isSupportKeepSampleRecordEvent>
  <!--ro, opt, bool-->true
</isSupportKeepSampleRecordEvent>
<isSupportSearchWaterLevel>
  <!--ro, opt, bool-->true
</isSupportSearchWaterLevel>
<isSupportGotowaterFlowVelocityScene>
  <!--ro, opt, bool-->true
</isSupportGotowaterFlowVelocityScene>
<isSupportFlowVelocitySceneCfgList>
  <!--ro, opt, bool-->true
</isSupportFlowVelocitySceneCfgList>
<isSupportSignalTriggerSnap>
  <!--ro, opt, bool-->true
</isSupportSignalTriggerSnap>
<isSupportBakeTobaccoAlarmRuleParams>
  <!--ro, opt, bool-->true
</isSupportBakeTobaccoAlarmRuleParams>
<isSupportManualTriggerAirCurtainFan>
  <!--ro, opt, bool-->true
</isSupportManualTriggerAirCurtainFan>
<isSupportAirCurtainFanParams>
  <!--ro, opt, bool-->true
</isSupportAirCurtainFanParams>
<isSupportModifyFlowVelocityScene>
  <!--ro, opt, bool-->true
</isSupportModifyFlowVelocityScene>
<isSupportDeleteFlowVelocityScene>
  <!--ro, opt, bool-->true
</isSupportDeleteFlowVelocityScene>
<isSupportAddFlowVelocityScene>
  <!--ro, opt, bool-->true
```

```
</isSupportAddFlowVelocityScene>
<isSupportFiredoorAccessConfiguration>
  <!--ro, opt, bool-->true
</isSupportFiredoorAccessConfiguration>
<isSupportAcuSeekSpecifications>
  <!--ro, opt, bool-->true
</isSupportAcuSeekSpecifications>
<isSupportGuardAgainstTheftEvent>
  <!--ro, opt, bool-->true
</isSupportGuardAgainstTheftEvent>
<isSupportFacePassEvent>
  <!--ro, opt, bool-->true
</isSupportFacePassEvent>
<isSupportHomeTriggerParams>
  <!--ro, opt, bool-->true
</isSupportHomeTriggerParams>
<isSupportAwayTriggerParams>
  <!--ro, opt, bool-->true
</isSupportAwayTriggerParams>
<isSupportScheduleParams>
  <!--ro, opt, bool-->true
</isSupportScheduleParams>
<isSupportCurrentSceneMode>
  <!--ro, opt, bool-->true
</isSupportCurrentSceneMode>
<isSupportSearchOperatorUnitSchema>
  <!--ro, opt, bool-->true
</isSupportSearchOperatorUnitSchema>
<isSupportIntegrationGuidelines>
  <!--ro, opt, bool-->true
</isSupportIntegrationGuidelines>
<isSupportAcuSeekImageModelList>
  <!--ro, opt, bool-->true
</isSupportAcuSeekImageModelList>
<isSupportSearchAcuSeekImageModeResult>
  <!--ro, opt, bool-->true
</isSupportSearchAcuSeekImageModeResult>
<isSupportGetToolModuleUpgradeStatus>
  <!--ro, opt, bool-->true
</isSupportGetToolModuleUpgradeStatus>
<isSupportPerimeterVideoRadarFusion>
  <!--ro, opt, bool-->true
</isSupportPerimeterVideoRadarFusion>
<isSupportHeadRiseRateAdvancedCfg>
  <!--ro, opt, bool-->true
</isSupportHeadRiseRateAdvancedCfg>
<isSupportFishEyeCroppingCfg>
  <!--ro, opt, bool-->true
</isSupportFishEyeCroppingCfg>
<isSupportPLTOFRecommendation>
  <!--ro, opt, bool-->true
</isSupportPLTOFRecommendation>
<isSupportPLTOFParams>
  <!--ro, opt, bool-->true
</isSupportPLTOFParams>
<isSupportDeletePLScene>
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</isSupportDeletePLScene>
<isSupportGotoPLScene>
  <!--ro, opt, bool-->true
</isSupportGotoPLScene>
<isSupportSavePLScene>
  <!--ro, opt, bool-->true
</isSupportSavePLScene>
<isSupportPLSceneFunctionRange>
  <!--ro, opt, bool-->true
</isSupportPLSceneFunctionRange>
<isSupportPLSceneList>
  <!--ro, opt, bool-->true
</isSupportPLSceneList>
<isSupportPLVideoOverlayTOFRegion>
  <!--ro, opt, bool-->true
</isSupportPLVideoOverlayTOFRegion>
<isSupportSearchImageAttributes>
  <!--ro, opt, bool-->true
</isSupportSearchImageAttributes>
<isSupportPLTriggerSnap>
  <!--ro, opt, bool-->true
</isSupportPLTriggerSnap>
<isSupportAlcoholDetectionEvent>
  <!--ro, opt, bool-->true
</isSupportAlcoholDetectionEvent>
<isSupportGetPerimeterTargetInfo>
  <!--ro, opt, bool-->true
</isSupportGetPerimeterTargetInfo>
<isSupportModifyPLSceneName>
  <!--ro, opt, bool-->true
</isSupportModifyPLSceneName>
<isSupportAssociateVideoChannelToDevice>
  <!--ro, opt, bool-->true
</isSupportAssociateVideoChannelToDevice>
<isSupportAddGasTask>
  <!--ro, opt, bool-->true
</isSupportAddGasTask>
```

```

</isSupportAddGasTask>
<isSupportModifyGasTask>
  <!--ro, opt, bool-->true
</isSupportModifyGasTask>
<isSupportGasTaskCtrl>
  <!--ro, opt, bool-->true
</isSupportGasTaskCtrl>
<isSupportSearchGasTasksRule>
  <!--ro, opt, bool-->true
</isSupportSearchGasTasksRule>
<isSupportSearchGasTasksStatus>
  <!--ro, opt, bool-->true
</isSupportSearchGasTasksStatus>
<isSupportSearchGasTasksHistoryData>
  <!--ro, opt, bool-->true
</isSupportSearchGasTasksHistoryData>
<isSupportDeleteGasTask>
  <!--ro, opt, bool-->true
</isSupportDeleteGasTask>
<isSupportAllGasTasksInfo>
  <!--ro, opt, bool-->true
</isSupportAllGasTasksInfo>
</DeviceCap>

```

12.1.5.3 Export device configuration file

Request URL

GET /ISAPI/System/configurationData?secretkey= <secretkey>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.

Request Message

None

Response Message

Binary Data

statusCode	statusString	subStatusCode	errorCode	description
2	Device Busy	serviceUnavailable	0x20000002	--

12.1.5.4 Import device configuration file

Request URL

POST /ISAPI/System/configurationData?secretkey= <secretkey> &deviceIndex= <deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
deviceIndex	string	--

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
6	Invalid Content	badDevType	0x60000010	--

12.1.5.5 Get the languages supported by the device

Request URL

GET /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--ro, req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian, "Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian", "Uzbek", "Kazak", "Kirghiz", "Farsi", "Azerbaijdzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
</DeviceLanguage>
```

12.1.5.6 Set device language parameters

Request URL

PUT /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian", "Bulgarian",
    "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese",
    "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian" (Slovenian), "Croatian",
    "Lithuanian", "Serbian", "BrazilianPortuguese" (Brazilian Portuguese), "Indonesian", "Ukrainian", "EURSpanish"-->SimChinese
  </language>
</DeviceLanguage>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

12.1.5.7 Get the capability of configuring the device language

Request URL

GET /ISAPI/System/DeviceLanguage/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device language configuration, attr:version{req, string, protocolVersion}-->
  <language
    opt="SimChinese,TraChinese,English,Russian,Bulgarian,Hungarian,Greek,German,Italian,Czech,Slovakia,French,Polish,Dutch,Portuguese,Spanish,Romanian,Turkish,Japanese,Danish,Swedish,Norwegian,Finnish,Korean,Thai,Estonia,Vietnamese,Hebrew,Latvian,Arabic,Sovenian,Croatian,Lithuanian,Serbian,BrazilianPortuguese,Indonesian,Ukrainian,EURSpanish,Uzbek,Kazak,Kirghiz,Farsi,Azerbaidzhan,Burmese,Mongolian,Anglicism,Estonian">
    <!--ro, req, enum, language, subType:string, attr:opt{req, string}, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese),
    "English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish",
    "Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic",
    "Sovenian"-Slovenian, "Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian",
    "Uzbek", "Kazak", "Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
  <upgradeFirmwareEnabled>
    <!--ro, opt, bool, whether to enable upgrading the firmware-->true
  </upgradeFirmwareEnabled>
</DeviceLanguage>
```

12.1.5.8 Set restoring devices' default parameters

Request URL

PUT /ISAPI/System/factoryReset?mode= <mode> &childDevID= <devIndex> &loginPassword= <loginPassword> &module= <module>

Query Parameter

Parameter Name	Parameter Type	Description
mode	enum	--
devIndex	string	--
loginPassword	string	--
module	enum	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

12.1.5.9 Get the network service capability

Request URL

GET /ISAPI/System/Network/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportWired>
    <!--ro, opt, bool, whether the device supports wired network-->true
  </isSupportWired>
  <isSupportMobileNetwork>
    <!--ro, opt, bool, whether the device supports mobile network-->true
  </isSupportMobileNetwork>
  <isSupportWAN>
    <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
  </isSupportWAN>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
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<isSupportUpnp>
  <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
</isSupportUpnp>
<isSupportPnP>
  <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
</isSupportPnP>
<isSupportDdns>
  <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
</isSupportDdns>
<isSupportHttps>
  <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
</isSupportHttps>
<isSupport28181>
  <!--ro, opt, bool, N/A-->true
</isSupport28181>
<SnmpCap>
  <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
  <isSupport>
    <!--ro, req, bool, whether the device supports SNMP-->true
  </isSupport>
</SnmpCap>
<isSupportExtNetCfg>
  <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
</isSupportExtNetCfg>
<isSupportIPFilter>
  <!--ro, opt, bool, whether the device supports IP filtering-->true
</isSupportIPFilter>
<isSupportSSH>
  <!--ro, opt, bool, whether the device supports SSH-->true
</isSupportSSH>
<isSupportNetPreviewStrategy>
  <!--ro, opt, bool, whether the device supports network Live view strategy-->true
</isSupportNetPreviewStrategy>
<isSupportEZVIZ>
  <!--ro, opt, bool, whether the device supports EZ protocol-->true
</isSupportEZVIZ>
<isSupportEhome>
  <!--ro, opt, bool, whether the device supports EHome (ISUP) protocol-->true
</isSupportEhome>
<isSupportWirelessDial>
  <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
</isSupportWirelessDial>
<isSupportWirelessServer>
  <!--ro, opt, bool, whether the device supports wireless server-->true
</isSupportWirelessServer>
<isSupportWPS>
  <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
</isSupportWPS>
<isWirelessMutexWithWirelessServer>
  <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
</isWirelessMutexWithWirelessServer>
<isSupportMACFilter>
  <!--ro, opt, bool, whether the device supports physical address filtering-->true
</isSupportMACFilter>
<GB28181Cap>
  <!--ro, opt, object, N/A-->
  <isSupportGB28181Service>
    <!--ro, opt, bool, N/A-->true
  </isSupportGB28181Service>
</GB28181Cap>
<WPS>
  <!--ro, opt, object, Wi-Fi Protected Setup-->
  <NetworkInterfaceList size="2">
    <!--ro, opt, array, Internet API List, subType:object, attr:size(req, int)-->
    <NetworkInterface>
      <!--ro, opt, object, Internet API-->
      <id>
        <!--ro, req, string, NIC index-->test
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable or not-->true
      </enabled>
      <isSupportAutoConnect>
        <!--ro, opt, bool, whether the device supports automatic connection-->true
      </isSupportAutoConnect>
      <isSupportDevicePinCode>
        <!--ro, opt, bool, whether the device supports PIN code-->true
      </isSupportDevicePinCode>
      <isSupportDevicePinCodeUpdate>
        <!--ro, opt, bool, whether the device supports updating the PIN code-->true
      </isSupportDevicePinCodeUpdate>
      <ApPinCode>
        <!--ro, opt, object, PIN code of the access point-->
        <ssid min="1" max="10">
          <!--ro, opt, string, SSID (Service Set Identifier), attr:min(req, int),max(req, int)-->test
        </ssid>
        <pinCode min="1" max="10">
          <!--ro, opt, string, PIN code, attr:min(req, int),max(req, int)-->test
        </pinCode>
      </ApPinCode>
    </NetworkInterface>
  </NetworkInterfaceList>
</WPS>

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<isSupportRFIDData>
  <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
</isSupportRFIDData>
<isSupportWifiProbeSSID>
  <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
</isSupportWifiProbeSSID>
<isSupportWifiProbe>
  <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
</isSupportWifiProbe>
<verificationCode min="1" max="10">
  <!--ro, opt, string, verification code, attr:min{req, int},max{req, int}-->test
</verificationCode>
<WPSCap>
  <!--ro, opt, object, Wi-Fi Protected Setup capability-->
  <isSupport>
    <!--ro, req, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupport>
  <isSupportAutoConnect>
    <!--ro, req, bool, whether the device supports automatic connection-->true
  </isSupportAutoConnect>
</WPSCap>
<NetworkMode>
  <!--ro, opt, object, network mode-->
  <workMode>
    <!--ro, opt, enum, network mode, subType:string, desc:"close", "wifi" (wireless access), "wifiAp" (wireless access point)-->close
  </workMode>
</NetworkMode>
<VerificationCodeModification>
  <!--ro, opt, object, verification code editing-->
  <verificationCodeType opt="normal,empty">
    <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
  </verificationCodeType>
  <isSupportDeclarationURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
  </isSupportDeclarationURL>
  <isSupportPrivacyPolicyURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
  </isSupportPrivacyPolicyURL>
  <verificationCodeModify opt="true,false">
    <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{req, string}-->true
  </verificationCodeModify>
  <Hyperlinks>
    <!--ro, opt, object, hyperlink-->
    <declarationURL>
      <!--ro, opt, string, hyperlink to the Terms of Service-->test
    </declarationURL>
    <privacyPolicyURL>
      <!--ro, opt, string, hyperlink to the Privacy Policy-->test
    </privacyPolicyURL>
  </Hyperlinks>
  <isSupportVerificationCodeCheck>
    <!--ro, opt, bool, whether the device supports verifying the verification code-->true
  </isSupportVerificationCodeCheck>
  <isSupportOldVerificationCode>
    <!--ro, opt, bool, whether the device supports the EZ password configuration of old version (the password of old version consists of six characters)-->true
  </isSupportOldVerificationCode>
</VerificationCodeModification>
<isSupportIntegrate>
  <!--ro, opt, bool, whether the device supports access protocol configuration-->true
</isSupportIntegrate>
<isSupportPlatformAccess>
  <!--ro, opt, bool, whether the device supports platform access-->true
</isSupportPlatformAccess>
<isSupportIntelligentBoost>
  <!--ro, opt, bool, whether the device supports intelligent acceleration-->true
</isSupportIntelligentBoost>
<WifiRegionConfiguration>
  <!--ro, opt, object, Wi-Fi area configuration, desc:when both are not supported, the <WifiRegionConfiguration> node will not return-->
  <isSupportWifiRegion>
    <!--ro, opt, bool, whether the device supports Wi-Fi area configuration, desc:supported by overseas devices; true (support), this node is not returned (not support)-->true
  </isSupportWifiRegion>
  <isSupportWifiEnhancement>
    <!--ro, opt, bool, whether the device supports Wi-Fi enhancement, desc:N/A-->true
  </isSupportWifiEnhancement>
</WifiRegionConfiguration>
<isSupportResourceStatistics>
  <!--ro, opt, bool, whether the device supports getting the network resource information-->true
</isSupportResourceStatistics>
<isSupportBandwidthLimit>
  <!--ro, opt, bool, whether the device supports bandwidth limit-->true
</isSupportBandwidthLimit>
<isSupportPOEPortsDisableAdaptiveServer>
  <!--ro, opt, bool, whether the device supports the POE protocol-->true
</isSupportPOEPortsDisableAdaptiveServer>
<isSupportPOEConfiguration>
  <!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
</isSupportPOEConfiguration>
<isSupportGetLinkSocketIP>
  <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
</isSupportGetLinkSocketIP>
<isSupportWebSocket>
  <!--ro, opt, bool, whether the device supports WebSocket-->true

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</isSupportWebSocket>
<isSupportWebSockets>
  <!--ro, opt, bool, whether the device supports WebSocketS-->true
</isSupportWebSockets>
<Adaption>
  <!--ro, opt, object, network self-adaption-->
  <streamType opt="0,1,2,3,4,5,7,8,9,10">
    <!--ro, opt, enum, stream type that supports network self-adaption during Live view, subType:int, attr:opt{req, string}, desc:stream types that
support network self-adaption during Live view: 0 (main stream), 1 (sub-stream), 2 (third stream), 3 (fourth stream / virtual stream), 4 (fifth stream), 5
(sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10 (tenth stream), and so on-->0
  </streamType>
  <isSupportPlayback>
    <!--ro, opt, bool, whether the device supports playback-->true
  </isSupportPlayback>
</Adaption>
<isSupportVideoImgDB>
  <!--ro, opt, bool, whether the device supports image and video library configuration-->true
</isSupportVideoImgDB>
<isSupportVideoImgLibService>
  <!--ro, opt, bool, whether the device supports image and video library service configuration-->true
</isSupportVideoImgLibService>
<isSupportWirelessWifi>
  <!--ro, opt, bool, whether the device supports Wi-Fi configuration-->true
</isSupportWirelessWifi>
<isSupportPrivatePlatom>
  <!--ro, opt, bool, whether the device supports third party platform configuration-->true
</isSupportPrivatePlatom>
<isSupportEventDataOverWebSocket opt="true,false">
  <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{req, string}-->true
</isSupportEventDataOverWebSocket>
<isSupportDynamicHostName>
  <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
</isSupportDynamicHostName>
<isSupportTCPAcceleration>
  <!--ro, opt, bool, whether the device supports TCP acceleration-->true
</isSupportTCPAcceleration>
<isSupportFlowControl>
  <!--ro, opt, bool, whether the device supports traffic control-->true
</isSupportFlowControl>
<isSupportEZVIZTiming>
  <!--ro, opt, bool, whether the device supports EZ platform synchronization (enabledTiming of message returned by/ISAPI/System/Network/EZVIZ)-->true
</isSupportEZVIZTiming>
<isSupportTrafficMonitor>
  <!--ro, opt, bool, whether the device supports traffic monitoring-->true
</isSupportTrafficMonitor>
<isSupportPingWatchdog>
  <!--ro, opt, bool, whether the device supports configuring the watchdog parameters-->true
</isSupportPingWatchdog>
<isSupportAutoFrequencyHop>
  <!--ro, opt, bool, whether the device supports auto frequency hopping-->true
</isSupportAutoFrequencyHop>
<isSupportTDMACfg>
  <!--ro, opt, bool, whether the device supports configuring TDMA-->true
</isSupportTDMACfg>
<isSupportDynamicChannel>
  <!--ro, opt, bool, whether the device supports dynamic channel selection-->true
</isSupportDynamicChannel>
<isSupportSIMCardSwitch>
  <!--ro, opt, bool, whether the device supports configuring scheduled SIM card switch, desc:related URI:
/ISAPI/System/Network/WirelessDial/SIMCardSwitch?format=json-->true
</isSupportSIMCardSwitch>
<isSupportDMSAuthInfo>
  <!--ro, opt, bool, whether the device supports DMS authentication information management-->true
</isSupportDMSAuthInfo>
<isSupportWiredandWirelessTrafficMonitor>
  <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC)-->true
</isSupportWiredandWirelessTrafficMonitor>
<isSupport4G>
  <!--ro, opt, bool, whether the device supports 4G module-->true
</isSupport4G>
<isSupport4GConfig>
  <!--ro, opt, bool, whether the device supports 4G configuration-->true
</isSupport4GConfig>
<isSupportGRIDServer>
  <!--ro, opt, bool, N/A-->true
</isSupportGRIDServer>
<isSupportGRIDInfo>
  <!--ro, opt, bool, N/A-->true
</isSupportGRIDInfo>
<isSupportSipTransmit>
  <!--ro, opt, bool, whether the device supports configuring SIP transmission parameters-->true
</isSupportSipTransmit>
<isSupportSip>
  <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
</isSupportSip>
<isSupportWifiCascade>
  <!--ro, opt, bool, whether the device supports enabling automatic Wi-Fi cascading-->true
</isSupportWifiCascade>
<isSupportRelativeInfo>
  <!--ro, opt, bool, whether the device supports getting the information about the connected devices-->true
</isSupportRelativeInfo>
<isSupportANR>
  <!--ro, opt, bool, whether the device supports automatic network replenishment, desc:related URI: /ISAPI/System/Network/ANR-->true

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</isSupportANR>
<isSupportEmailEncrypt>
  <!--ro, opt, bool, whether the device supports email encryption configuration-->true
</isSupportEmailEncrypt>
<isSupportPortFilter>
  <!--ro, opt, bool, whether the device supports port filtering-->true
</isSupportPortFilter>
<isSupportPingDeny>
  <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
</isSupportPingDeny>
<isSupportDOSDeny>
  <!--ro, opt, bool, whether the device supports configuring parameters to prevent the DoS attack-->true
</isSupportDOSDeny>
<isSupportNetworkStatus>
  <!--ro, opt, bool, whether the device supports searching for the network status-->true
</isSupportNetworkStatus>
<isSupportEZVIZQRCode>
  <!--ro, opt, bool, whether the device supports getting the EZ QR code, desc:/ISAPI/System/Network/EZVIZ/QRCode?format=json-->true
</isSupportEZVIZQRCode>
<isSupportEZVIZUnbind>
  <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
</isSupportEZVIZUnbind>
<isSupportPingTest>
  <!--ro, opt, bool, whether the device supports ping test, desc:related URI: /ISAPI/System/Network/pingtest/capabilities-->true
</isSupportPingTest>
<isSupportMOTCloudPlatform>
  <!--ro, opt, bool, whether the device supports configuring parameters for accessing the cloud platform service, desc:related URI:
/ISAPI/System/Network/Traffic/MOTCloudPlatform/capabilities?format=json-->true
</isSupportMOTCloudPlatform>
<isSupportSpectrumScan>
  <!--ro, opt, bool, whether the device supports frequency spectrum scanning, desc:related URI:
/ISAPI/System/Network/interface/cnetCardID/spectrumScan/capabilities?format=json-->true
</isSupportSpectrumScan>
<isSupportFRP>
  <!--ro, opt, bool, whether the device supports FRP (Fast Reverse Proxy), desc:related URI: /ISAPI/Syetem/Network/FRP?format=json-->true
</isSupportFRP>
<isSupport5GModule>
  <!--ro, opt, bool, whether the device supports 5G module configuration, desc:related URI: /ISAPI/System/Network/5GModule/capabilities?format=json-->true
</isSupport5GModule>
<isSupportWirelessDialRoute>
  <!--ro, opt, bool, whether the device supports configuring parameters of route forwarding rules of wireless dial-up, desc:related URI:
/ISAPI/System/Network/WirelessDia/route/capabilities?format=json-->true
</isSupportWirelessDialRoute>
<isSupportNPlusOne>
  <!--ro, opt, bool, whether the device supports N+1 protocol-->true
</isSupportNPlusOne>
<supportEnvironmentalStandards opt="HJ212">
  <!--ro, opt, string, environmental monitoring standard, attr:opt{req, string}, desc:N/A-->true
</supportEnvironmentalStandards>
<isSupportIntranetSegment>
  <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/intranetSegment/capabilities?
format=json-->true
</isSupportIntranetSegment>
<isSupportDHCPServer>
  <!--ro, opt, bool, whether the device supports DHCP service configuration, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/DHCPServer/capabilities?
format=json-->true
</isSupportDHCPServer>
<isSupportClusterIntercom>
  <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/clusterIntercom/capabilities?
format=json-->true
</isSupportClusterIntercom>
<isSupportHydrologicMonitoring>
  <!--ro, opt, bool, whether the device supports hydrologic monitoring data communication, desc:if this function is supported, this node will be returned
and its value is true; otherwise, this node will be returned and its value is false or not be returned. Related URI:
/ISAPI/System/Network/HydrologicMonitoring/capabilities?format=json-->true
</isSupportHydrologicMonitoring>
<isSupportGantryETCServer>
  <!--ro, opt, bool, N/A, desc:N/A-->true
</isSupportGantryETCServer>
<RoutePolicy>
  <!--ro, opt, object, policy routing-->
  <isSupportRoutePolicy>
    <!--ro, opt, bool, whether the device supports configuring the policy routing parameters-->true
  </isSupportRoutePolicy>
  <isRebootHint>
    <!--ro, opt, bool, whether to alarm to reboot after configuration-->true
  </isRebootHint>
</RoutePolicy>
<isWirelessDialMuxWithWlanAndWifi>
  <!--ro, opt, bool, whether the wireless dial-up is mutually exclusive with the WLAN hotspot and Wi-Fi-->true
</isWirelessDialMuxWithWlanAndWifi>
<isSupportDMZCfg>
  <!--ro, opt, bool, whether the device supports configuring DMZ host, desc:related URI: /ISAPI/System/Network/DMZCfg/capabilities?format=json-->true
</isSupportDMZCfg>
<isSupportSpeechTranscription>
  <!--ro, opt, bool, whether the device supports accessing the speech-to-text transcription platform, desc:if this function is supported, this node will
be returned and its value is true; otherwise, this node will be returned and its value is false or not be returned. Related URI:
/ISAPI/System/Network/speechTranscriptionPlatform/capabilities?format=json-->true
</isSupportSpeechTranscription>
<isSupportAsymPingTest>
  <!--ro, opt, bool, whether the device supports asynchronous network detection, desc:network detection means testing the network stability and network

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<!--ro, opt, bool, whether the device supports asynchronous network detection, desc:network detection means testing the network stability and packet
loss rate of the network using the ping command for knowing about the network condition. Related URI: /ISAPI/System/Network/asynPingTest/capabilities?
format=json-->true
</isSupportAsynPingTest>
<isSupportBasebandCapture>
  <!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and the
base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
</isSupportBasebandCapture>
<isSupportCellularNetworkSpeedTest>
  <!--ro, opt, bool, whether the device supports testing the connection speed between the device and the cellular network, desc:you can test the
connection speed of the network where the current device is. Related URI: /ISAPI/System/Network/cellularNetwork/speedTest/capabilities?format=json-->true
</isSupportCellularNetworkSpeedTest>
<isSupportCellularNetworkCellInfo>
  <!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the cellular
network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&security=<security>&iv=<iv>-->true
</isSupportCellularNetworkCellInfo>
<isSupportMultiBandFrequencyScanning>
  <!--ro, opt, bool, whether the device supports scanning all frequency bands of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/multiBandFrequencyScanning/capabilities?format=json-->true
</isSupportMultiBandFrequencyScanning>
<isSupportCellularNetworkLockBand>
  <!--ro, opt, bool, whether the device supports accessing the locked frequency band of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/LockBand/capabilities?format=json-->true
</isSupportCellularNetworkLockBand>
<isSupportRadioFrequencyFluctuation>
  <!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
</isSupportRadioFrequencyFluctuation>
<isSupportBasebandBlock>
  <!--ro, opt, bool, whether the device supports the unauthorized list of the base station, desc:related URI:
/ISAPI/System/Network/cellularNetwork/basebandBlock/capabilities?format=json-->true
</isSupportBasebandBlock>
<isSupportWifiQuality>
  <!--ro, opt, bool, whether the device supports getting the device's Wi-Fi quality information, desc:related URI: /ISAPI/System/Network/wifiQuality?
format=json-->true
</isSupportWifiQuality>
<isSupportWiredNetworkQuality>
  <!--ro, opt, bool, whether the device supports getting the wired network quality information, desc:related URI:
/ISAPI/System/Network/wiredNetworkQuality?format=json-->true
</isSupportWiredNetworkQuality>
<isSupportQoSBehaviorCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS traffic behavior, desc:related URI: /ISAPI/System/Network/qos/BehaviorCfg/capabilities?
format=json-->true
</isSupportQoSBehaviorCfg>
<isSupportQoSClassifierCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS traffic types, desc:related URI: /ISAPI/System/Network/qos/ClassifierCfg/capabilities?
format=json-->true
</isSupportQoSClassifierCfg>
<isSupportQoSPolicyCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS policy, desc:related URI: /ISAPI/System/Network/qos/PolicyCfg/capabilities?format=json--
>true
</isSupportQoSPolicyCfg>
<isSupportQoSPolicyVLANCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS policy VLAN, desc:related URI: /ISAPI/System/Network/qos/PolicyVLANCfg/capabilities?
format=json-->true
</isSupportQoSPolicyVLANCfg>
<isSupportQoSPortRateLimiting>
  <!--ro, opt, bool, whether the device supports configuring port speed limit, desc:related URI: /ISAPI/System/Network/qos/portRateLimiting/capabilities?
format=json-->true
</isSupportQoSPortRateLimiting>
<isSupportQoSQueueAlgorithmCfg>
  <!--ro, opt, bool, whether the device supports configuring QoS queue algorithm, desc:related URI:
/ISAPI/System/Network/qos/queueAlgorithmCfg/capabilities?format=json-->true
</isSupportQoSQueueAlgorithmCfg>
<isSupportQoSPortPriorityCfg>
  <!--ro, opt, bool, whether the device supports QoS port priority configuration, desc:related URI:
/ISAPI/System/Network/qos/portPriorityCfg/capabilities?format=json-->true
</isSupportQoSPortPriorityCfg>
<isSupportQoS802.1pPriorityMapping>
  <!--ro, opt, bool, whether the device supports DSCP and 802.1p mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/802.1pPriorityMapping/capabilities?format=json-->true
</isSupportQoS802.1pPriorityMapping>
<isSupportQoSDSCTPriorityMapping>
  <!--ro, opt, bool, whether the device supports DSCP mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/DSCPPriorityMapping/capabilities?format=json-->true
</isSupportQoSDSCTPriorityMapping>
<isSupportQoSLoaclPriorityMapping>
  <!--ro, opt, bool, whether the device supports 802.1p and Local mapping priority configuration, desc:related URI:
/ISAPI/System/Network/qos/LoaclPriorityMapping/capabilities?format=json-->true
</isSupportQoSLoaclPriorityMapping>
<isSupportStopPingtest>
  <!--ro, opt, bool, whether to stop PingTest, desc:related URI: /ISAPI/System/Network/StopPingtest?format=json-->true
</isSupportStopPingtest>
<isSupportSNMPView>
  <!--ro, opt, bool, whether the device supports SNMP view, desc:related URI: /ISAPI/System/Network/SNMP/ViewCfg/capabilities?format=json-->true
</isSupportSNMPView>
<isSupportSNMPUserGroup>
  <!--ro, opt, bool, whether the device supports SNMP user groups, desc:related URI: /ISAPI/System/Network/SNMP/UserGroup/capabilities?format=json-->true
</isSupportSNMPUserGroup>
<isSupportSNMPBasicInfo>
  <!--ro, opt, bool, whether the device supports the basic information of SNMP, desc:related URI: /ISAPI/System/Network/SNMP/BasicInfo/capabilities?
format=json-->true
</isSupportSNMPBasicInfo>
<isSupportRFIDTagCheckTest>

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    <!--ro, opt, bool, whether the device supports the RFID tag disk storage performance test, desc:related URI:
    /ISAPI/System/Network/RFIDData/GetTagCheckTestInfo?format=json-->true
    </isSupportRFIDTagCheckTest>
    <isSupportQoS802.1pTo802.1pPriorityMapping>
    <!--ro, opt, bool, whether the device supports DSCP and Local mapping priority configuration, desc:related URI:
    /ISAPI/System/Network/qos/DSCPtoLoaclPriorityMapping/capabilities?format=json-->true
    </isSupportQoS802.1pTo802.1pPriorityMapping>
    <isSupportQoS802.1pTo802.1pPriorityMapping>
    <!--ro, opt, bool, whether the device supports 802.1p mapping priority configuration, desc:related URI:
    /ISAPI/System/Network/qos/802.1pTo802.1pPriorityMapping/capabilities?format=json-->true
    </isSupportQoS802.1pTo802.1pPriorityMapping>
    <isSupportNFCAllowList>
    <!--ro, opt, bool, whether the device supports NFC network allowlist management, desc:related URI: /ISAPI/System/Network/NFC/allowList/capabilities?
    format=json-->true
    </isSupportNFCAllowList>
    <isSupportLinkTestStart>
    <!--ro, opt, bool, whether the device supports enabling network Link test, desc:related URI:
    /ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Start?format=json-->true
    </isSupportLinkTestStart>
    <isSupportLinkTestResult>
    <!--ro, opt, bool, whether the device supports getting the result of network Link test, desc:related URI:
    /ISAPI/System/Network/WirelessDial/Interfaces/<interfaceID>/LinkTest/Result?format=json-->true
    </isSupportLinkTestResult>
    <isSupportGetAsynTestEmailResult>
    <!--ro, opt, bool, whether the device supports getting email testing result asynchronously, desc:POST
    /ISAPI/System/Network/Email/GetAsynTestEmailResult?format=json-->true
    </isSupportGetAsynTestEmailResult>
    <isSupportTestAsynEmail>
    <!--ro, opt, bool, whether the device supports asynchronously testing email, desc:POST /ISAPI/System/Network/Email/TestAsynEmail?format=json-->true
    </isSupportTestAsynEmail>
    <isSupportModbusAccess>
    <!--ro, opt, bool, whether the device supports Modbus accessing parameters, desc:GET/PUT /ISAPI/System/Network/ModbusAccess?format=json-->true
    </isSupportModbusAccess>
    <isSupportNetworkModuleUpgrade>
    <!--ro, opt, bool, whether the device supports upgrading network module, desc:including importing upgrade package
    (/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
    (/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrading
    (/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json), get upgrade status
    (/ISAPI/System/Network/WirelessDial/GetUpgradeNetworkModuleStatus?format=json)-->true
    </isSupportNetworkModuleUpgrade>
    <isSupportNetworkModuleVersionInfo>
    <!--ro, opt, bool, whether the device supports getting the version information of network module, desc:GET
    /ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
    </isSupportNetworkModuleVersionInfo>
    <isSupportGreenShieldPlatform>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/GreenShieldPlatform/GreenShieldPlatform?format=json-->true
    </isSupportGreenShieldPlatform>
    <isSupportFusionUniversalApplicationPlatCfg>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/FusionUniversalApplicationPlatform/FusionUniversalApplicationPlatCfg?format=json-->true
    </isSupportFusionUniversalApplicationPlatCfg>
    <isSupportNPQControlStrategy>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/NPQMgr/NPQControlStrategy?format=json-->true
    </isSupportNPQControlStrategy>
    <isSupportNAT>
    <!--ro, opt, bool, whether NAT is supported, desc:GET/PUT /ISAPI/System/Network/nats-->true
    </isSupportNAT>
    <isSupportGetCurrentlyUsingNetworkCardInfo>
    <!--ro, opt, bool, whether the device supports getting the current NIC information, desc:GET
    /ISAPI/System/Network/WirelessDial/GetCurrentlyUsingNetworkCardInfo?format=json-->true
    </isSupportGetCurrentlyUsingNetworkCardInfo>
    <isSupportAutoRecampParams>
    <!--ro, opt, bool, whether the device supports parameters of auto re-camping on a regular basis, desc:GET/PUT
    /ISAPI/System/Network/WirelessDial/AutoRecampParams?format=json-->true
    </isSupportAutoRecampParams>
    <isSupportDSTPServerCfg>
    <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET/PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
    format=json-->true
    </isSupportDSTPServerCfg>
    <isSupportJT1008ServerParams>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/JT1008Access/JT1008ServerParams?format=json-->true
    </isSupportJT1008ServerParams>
    <isSupportImportNARIKeyFile>
    <!--ro, opt, bool, N/A, desc:PUT /ISAPI/System/Network/NARIEncryptionKeyMgr/ImportNARIKeyFile?format=json-->true
    </isSupportImportNARIKeyFile>
    <isSupportExportNARIKeyFile>
    <!--ro, opt, bool, N/A, desc:GET /ISAPI/System/Network/NARIEncryptionKeyMgr/ExportNARIKeyFile?format=json-->true
    </isSupportExportNARIKeyFile>
    <isSupportNARIEncryptionParams>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/NARIEncryptionKeyMgr/NARIEncryptionParams?format=json-->true
    </isSupportNARIEncryptionParams>
    <isSupportDSTPProtocolCap>
    <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolCap?format=json-->true
    </isSupportDSTPProtocolCap>
    <isSupportIPAdaptiveParams>
    <!--ro, opt, bool, whether the device supports self-adaptive IP address parameters, desc:GET/PUT /ISAPI/System/Network/IPAdaptiveParams?format=json--
    >true
    </isSupportIPAdaptiveParams>
    <isSupportPowerCenterPlatform>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/System/Network/PowerCenterAccess/PowerCenterPlatform?format=json-->true
    </isSupportPowerCenterPlatform>
    <isSupportTestPowerCenterPlatformConnect>
    <!--ro, opt, bool, N/A, desc:POST /ISAPI/System/Network/PowerCenterAccess/TestPowerCenterPlatformConnect?format=json-->true
    </isSupportTestPowerCenterPlatformConnect>

```

```

<!--ro, opt, bool, whether the device supports Modbus testing the status of the specified client., desc:POST
/ISAPI/System/Network/ModbusTestSubordinateStatus?format=json-->true
</isSupportModbusTestSubordinateStatus>
<isSupportGetModbusAndRegisterMap>
<!--ro, opt, bool, whether the device supports get Modbus uploaded content and the mapping relation between the uploaded content and register quantity,
desc:GET /ISAPI/System/Network/GetModbusAndRegisterMap?format=json-->true
</isSupportGetModbusAndRegisterMap>
<isSupportAddFRPDynamicAllocationMapPort>
<!--ro, opt, bool, whether the device supports batch adding FRP dynamic allocation mapping ports, desc:POST
/ISAPI/System/Network/NATPenetration/AddFRPDynamicAllocationMapPort?format=json-->true
</isSupportAddFRPDynamicAllocationMapPort>
<isSupportGetFRPDynamicAllocationMapPort>
<!--ro, opt, bool, whether the device supports batch getting FRP dynamic allocation mapping ports, desc:GET
/ISAPI/System/Network/NATPenetration/GetFRPDynamicAllocationMapPort?format=json-->true
</isSupportGetFRPDynamicAllocationMapPort>
<isSupportUSBNetworkCard>
<!--ro, opt, bool, whether the device supports USB NIC-->true
</isSupportUSBNetworkCard>
<isSupportOptimumOperatorParams>
<!--ro, opt, bool, whether the device supports optimal carrier parameters, desc:GET/PUT /ISAPI/System/Network/WirelessDial/OptimumOperatorParams?
format=json-->true
</isSupportOptimumOperatorParams>
<isSupportERTCJoin>
<!--ro, opt, bool, whether the device supports ERTC access, desc:PUT ISAPI/System/Network/ERTCAccess/ERTCJoin?format=json-->true
</isSupportERTCJoin>
<isSupportERTCQuit>
<!--ro, opt, bool, whether the device supports ERTC quitting, desc:PUT ISAPI/System/Network/ERTCAccess/ERTCQuit?format=json-->true
</isSupportERTCQuit>
<isSupportLoRaCommunicationParams>
<!--ro, opt, bool, whether the device supports LoRa protocol communication parameters, desc:GET/PUT
/ISAPI/System/Network/LoRAAccess/LoRaCommunicationParams?format=json-->true
</isSupportLoRaCommunicationParams>
<isSupportSetSIMCardSwitch>
<!--ro, opt, bool, whether the device supports SIM NIC switch configuration, desc:PUT /ISAPI/System/Network/WirelessDial/SetSIMCardSwitch?format=json--
>true
</isSupportSetSIMCardSwitch>
<isSupportDialSMSAutoSwitchParams>
<!--ro, opt, bool, whether the device supports configuring the NIC suto-switch parameter configuration, desc:GET/PUT
/ISAPI/System/Network/WirelessDial/DialSMSAutoSwitchParams?format=json, DialSMSAutoSwitchParams-->true
</isSupportDialSMSAutoSwitchParams>
<isSupportBatchModifyDeviceIPAddress>
<!--ro, opt, bool, whether the device supports batch configuring device network addresses, desc:PUT /ISAPI/System/Network/BatchModifyDeviceIPAddress?
format=json-->true
</isSupportBatchModifyDeviceIPAddress>
<isSupportWirelessConfigListAsyncParams>
<!--ro, opt, bool, whether the device supports wireless configuration list async parameters, desc:GET/PUT
/ISAPI/System/Network/WirelessConfigListAsyncParams?format=json-->true
</isSupportWirelessConfigListAsyncParams>
<isSupportRestartPaging>
<!--ro, opt, bool, whether the device supports paging again, desc:PUT /ISAPI/System/Network/LoRAAccess/RestartPaging?format=json-->true
</isSupportRestartPaging>
<isSupportGetCurrentWirelessServerMode>
<!--ro, opt, bool, whether the device supports getting the current network hotspot mode, desc:GET
/ISAPI/System/Network/WLANServer/GetCurrentWirelessServerMode?format=json-->true
</isSupportGetCurrentWirelessServerMode>
<isSupportNetworkInterfaceMaterialInfoDetect>
<!--ro, opt, bool, whether the device supports NIC material information detection, desc:PUT /ISAPI/System/Network/NetworkInterfaceMaterialInfoDetect?
format=json-->true
</isSupportNetworkInterfaceMaterialInfoDetect>
<isSupportMQTTAccessCfg>
<!--ro, opt, bool, whether the device supports MQTT access configuration, desc:GET/PUT /ISAPI/System/Network/MQTTAccess/MQTTAccessCfg?format=json-->true
</isSupportMQTTAccessCfg>
<isSupportModifySavedWirelessList>
<!--ro, opt, bool, whether the device supports editing Wi-Fi list saved to specified network interface, desc:PUT
/ISAPI/System/Network/ModifySavedWirelessList?format=json-->true
</isSupportModifySavedWirelessList>
<isSupportGetSavedWirelessList>
<!--ro, opt, bool, whether the device supports getting Wi-Fi list saved to specified network interface, desc:POST
/ISAPI/System/Network/GetSavedWirelessList?format=json-->true
</isSupportGetSavedWirelessList>
<isSupportWirelessCommunication433AccessParams>
<!--ro, opt, bool, whether the device supports 433 protocol communication parameters, desc:GET/PUT
/ISAPI/System/Network/WirelessCommunication433Access/WirelessCommunication433AccessParams?format=json-->true
</isSupportWirelessCommunication433AccessParams>
<isSupportTSOverWebSocket>
<!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
</isSupportTSOverWebSocket>
<isSupportSouthernPowerGridServer>
<!--ro, opt, bool, N/A, desc:N/A-->true
</isSupportSouthernPowerGridServer>
<isSupportGeologicalHazardMonitoringCfg>
<!--ro, opt, bool, whether the device supports geological hazard detection service parameters, desc:GET/PUT
/ISAPI/System/Network/GeologicalHazardMonitoring/GeologicalHazardMonitoringCfg?format=json-->true
</isSupportGeologicalHazardMonitoringCfg>
<isSupportFixedNetworkInterfaceCfg>
<!--ro, opt, bool, whether the device supports network interface fixed address allocation parameters, desc:GET/PUT
/ISAPI/System/Network/FixedNetworkInterfaceCfg?format=json-->true
</isSupportFixedNetworkInterfaceCfg>
<isSupportSearchWiFiChannelDetectionState>
<!--ro, opt, bool, whether the device supports searching for Wi-Fi channel detection status, desc:POST
/ISAPI/System/Network/SearchWiFiChannelDetectionState?format=json-->true
</isSupportSearchWiFiChannelDetectionState>
<isSupportStartWiFiChannelDetection>

```

```

    <!--ro, opt, bool, whether the device supports enabling Wi-Fi channel detection, desc:POST /ISAPI/System/Network/StartWiFiChannelDetection?format=json-->true
    </isSupportStartWiFiChannelDetection>
    <isSupportStartManualChannelAdaptation>
    <!--ro, opt, bool, whether the device supports enabling manual configuration of channel self-adaption, desc:POST
    /ISAPI/System/Network/StartManualChannelAdaptation?format=json-->true
    </isSupportStartManualChannelAdaptation>
    <isSupportSearchManualChannelAdaptationState>
    <!--ro, opt, bool, whether the device supports searching for manual channel configuration self-adaption status, desc:POST
    /ISAPI/System/Network/SearchManualChannelAdaptationState?format=json-->true
    </isSupportSearchManualChannelAdaptationState>
    <isSupportLoRaWorkMode>
    <!--ro, opt, bool, whether the device supports LoRa module working mode, desc:GET/PUT /ISAPI/System/Network/LoRaAccess/LoRaWorkMode?format=json-->true
    </isSupportLoRaWorkMode>
    <isSupportPLCParams>
    <!--ro, opt, bool-->true
    </isSupportPLCParams>
    <isSupportWiFiRoamingConfigParams>
    <!--ro, opt, bool-->true
    </isSupportWiFiRoamingConfigParams>
    <isSupportPLCNetworkingStatus>
    <!--ro, opt, bool-->true
    </isSupportPLCNetworkingStatus>
    <isSupportAddPLCDiagnosis>
    <!--ro, opt, bool-->true
    </isSupportAddPLCDiagnosis>
    <isSupportGetPLCDiagnosisProgress>
    <!--ro, opt, bool-->true
    </isSupportGetPLCDiagnosisProgress>
    <isSupportSearchOutlookAuthResult>
    <!--ro, opt, bool-->true
    </isSupportSearchOutlookAuthResult>
    <isSupportRequestOutlookAuth>
    <!--ro, opt, bool-->true
    </isSupportRequestOutlookAuth>
    <isSupportOutlookEmailList>
    <!--ro, opt, bool-->true
    </isSupportOutlookEmailList>
    <isSupportNARIEncryptionPortParams>
    <!--ro, opt, bool-->true
    </isSupportNARIEncryptionPortParams>
    <isSupportAlphaPlatformParams>
    <!--ro, opt, bool-->true
    </isSupportAlphaPlatformParams>
    <isSupportSouthernUniverseIOTAccessParams>
    <!--ro, opt, bool-->true
    </isSupportSouthernUniverseIOTAccessParams>
    <isSupportModifyGeologicalHazardMonitoringParams>
    <!--ro, opt, bool-->true
    </isSupportModifyGeologicalHazardMonitoringParams>
    <isSupportGeologicalHazardMonitoringListParams>
    <!--ro, opt, bool-->true
    </isSupportGeologicalHazardMonitoringListParams>
    <isSupportServerCfg>
    <!--ro, opt, bool-->true
    </isSupportServerCfg>
    <isSupportDelimiterCfg>
    <!--ro, opt, bool-->true
    </isSupportDelimiterCfg>
    <isSupportNetworkSignalEnhancementParams>
    <!--ro, opt, bool-->true
    </isSupportNetworkSignalEnhancementParams>
</NetworkCap>

```

12.1.5.10 Reboot device

Request URL

PUT /ISAPI/System/reboot?childDevID=<devIndex>&module=<module>&loginPassword=<loginPassword>

Query Parameter

Parameter Name	Parameter Type	Description
devIndex	string	Reboot a sub-device via specifying sub-device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/loTGateway/Childmanage/SearchChild?format=json.
module	enum	The "module" specifies a module to restart, without requiring a device reboot, corresponding to the capability (/ISAPI/System/capabilities/deviceReboot->module). "algorithmProgram (traffic terminal radar-assisted algorithm program, supports independent algorithm program startup), "radar" (restart radar module of the radar-assisted camera).
loginPassword	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

12.1.5.11 Get the video analysis task capability

Request URL

GET /ISAPI/System/Video/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->0
  </videoInputPortNums>
  <videoOutputPortNums>
    <!--ro, opt, int, number of video output ports-->0
  </videoOutputPortNums>
  <menuNums>
    <!--ro, opt, int, number of Local menus that can be displayed-->0
  </menuNums>
</VideoCap>
```

```

</methods>
<externalChannelNum>
  <!--ro, opt, bool, number of extended analog channels-->true
</externalChannelNum>
<isSupportHeatmap>
  <!--ro, opt, bool, whether the device supports heat map-->true
</isSupportHeatmap>
<isSupportCounting>
  <!--ro, opt, bool, whether the device supports people counting-->true
</isSupportCounting>
<countingType>
  <!--ro, opt, enum, statistics type, subType:string, desc:statistics type-->human
</countingType>
<isSupportPicture>
  <!--ro, opt, bool, whether the device supports OSD picture overLay, desc:related URI: /ISAPI/System/Video/inputs/channels/<channelID>/image/picture--
>true
</isSupportPicture>
<isSupportPreviewSwitch>
  <!--ro, opt, bool, whether the device supports switching Live view-->true
</isSupportPreviewSwitch>
<isSupportRecodStatus>
  <!--ro, opt, bool, whether the device supports searching for the recording status-->true
</isSupportRecodStatus>
<isSupportPrivacyMask>
  <!--ro, opt, bool, whether the device supports privacy mask-->true
</isSupportPrivacyMask>
<isSupportBinocularPreviewSwitch>
  <!--ro, opt, bool, whether the device supports switching Live view of the dual-Lens camera-->true
</isSupportBinocularPreviewSwitch>
<isSupportCalibCheck>
  <!--ro, opt, bool, whether the device supports calibration verification-->true
</isSupportCalibCheck>
<isSupportPIP>
  <!--ro, opt, bool-->true
</isSupportPIP>
<isSupportFocusVideoMode>
  <!--ro, opt, bool-->true
</isSupportFocusVideoMode>
<isSupportExternalChannel>
  <!--ro, opt, bool, whether the device supports extended analog channels-->true
</isSupportExternalChannel>
<isSupportMultiChannelCounting>
  <!--ro, opt, bool-->true
</isSupportMultiChannelCounting>
<isSupportCountingCollection>
  <!--ro, opt, bool, whether the device supports people counting ANR-->true
</isSupportCountingCollection>
<isSupportHeatmapCollection>
  <!--ro, opt, bool, whether the device supports heat map data ANR-->true
</isSupportHeatmapCollection>
<channelFlexible opt="name,enable,online,linknum">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->name
</channelFlexible>
<isSupportOutputsResource>
  <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/resource?format=json, desc:related URI: /ISAPI/System/Video/outputs/resource?format=json--
>true
</isSupportOutputsResource>
<OSDLanguage opt="GBK,EUC-KR,Hebrew" def="GBK">
  <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->GBK
</OSDLanguage>
<isSupportMixedChannel>
  <!--ro, opt, bool-->true
</isSupportMixedChannel>
<isSupportMixedChannelStatus>
  <!--ro, opt, bool-->true
</isSupportMixedChannelStatus>
<isSupportOutputCourseware>
  <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/courseware/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/courseware/capabilities?format=json-->true
</isSupportOutputCourseware>
<isSupportVideoInputMode>
  <!--ro, opt, bool, related URI: /ISAPI/System/Video/inputs/mode/capabilities, desc:related URI: /ISAPI/System/Video/inputs/mode/capabilities-->true
</isSupportVideoInputMode>
<isSupportVideoOutputMode>
  <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/mode/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/mode/capabilities?format=json-->true
</isSupportVideoOutputMode>
<PreviewMode>
  <!--ro, opt, object-->
  <PIPType>
    <!--ro, opt, object-->
    <mainScreenChannelID>
      <!--ro, opt, int-->0
    </mainScreenChannelID>
    <subScreenChannelID>
      <!--ro, opt, int-->0
    </subScreenChannelID>
  </PIPType>
</PreviewMode>
<isSupportViewTag>
  <!--ro, opt, bool-->true
</isSupportViewTag>
<isSupportMenuStatus>
  <!--ro, opt, bool-->true

```

```
</isSupportMenuStatus>  
</VideoCap>
```

12.1.6 Time Management

12.1.6.1 Get device time synchronization management parameters

Request URL

GET /ISAPI/System/time

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>  
  
<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">  
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->  
  <timeMode>  
    <!--ro, req, enum, time synchronization mode, subType:string, desc:"NTP" (NTP time synchronization), "manual" (manual time synchronization), "satellite"  
    (satellite time synchronization), "platform" (platform time synchronization), "NONE" (time synchronization is not allowed or no time synchronization  
    source), "GB28181" (GB28181 time synchronization)-->NTP  
  </timeMode>  
  <localTime>  
    <!--ro, opt, string, Local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:00  
  </localTime>  
  <timeZone>  
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}-->CST-  
    8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00  
  </timeZone>  
  <satelliteInterval>  
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, desc:unit: minute-->60  
  </satelliteInterval>  
  <isSummerTime>  
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system-->true  
  </isSummerTime>  
  <platformType>  
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform--  
>EZVIZ  
  </platformType>  
  <platformNo>  
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in  
    GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1  
  </platformNo>  
</Time>
```

12.1.6.2 Set device time synchronization management parameters

Request URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message


```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:"NTP" (NTP time synchronization), "manual" (manual time synchronization), "satellite" (satellite time synchronization), "platform" (platform time synchronization), "NONE" (time synchronization is not allowed or no time synchronization source), "GB28181" (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, Local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:00
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--opt, bool, whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID-->1
  </platformNo>
</Time>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
</ResponseStatus>
```

12.1.6.3 Get the capability of device time synchronization management

Request URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允许校时或无校时源),platform">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:"NTP" (NTP time synchronization), "manual" (manual time synchronization), "satellite" (satellite time synchronization), "platform" (platform time synchronization), "NONE" (time synchronization is not allowed or no time synchronization source), "GB28181" (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:"Local" (Local time), "UTC" (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,pPlatform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device's NTP service information with the camera, desc:related URI: /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
</Time>
```

12.1.6.4 Set device local time

Request URL

PUT /ISAPI/System/time/localTime

Query Parameter

None

Request Message

Binary Data

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

12.1.6.5 Set parameters of all NTP servers

Request URL

PUT /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information List, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10000], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->>false
    </enabled>
  </NTPServer>
</NTPServerList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

12.1.6.6 Get the parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information List, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10000], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->>false
    </enabled>
  </NTPServer>
</NTPServerList>
```

12.1.6.7 Set the parameters of a NTP server

Request URL

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10000], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->>false
  </enabled>
</NTPServer>
```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

12.1.6.8 Get the parameters of a NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	NTP server No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10000], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->>false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```

12.1.6.9 Get the configuration capability of a specific NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id min="1" max="1">
    <!--ro, req, string, ID, range:[0,1], attr:min{req, int},max{req, int}-->1
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, IP address type of NTP server, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName min="0" max="68">
    <!--ro, opt, string, NTP server domain name, range:[0,68], attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="0" max="32">
    <!--ro, opt, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address min="0" max="128">
    <!--ro, opt, string, IPv6 address, range:[0,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="0" max="65535">
    <!--ro, opt, int, port No., range:[0,65535], attr:min{req, int},max{req, int}, desc:port No.-->123
  </portNo>
  <synchronizeInterval min="0" max="10800">
    <!--ro, opt, int, time synchronization interval, range:[0,10800], unit:min, attr:min{req, int},max{req, int}, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <portType opt="auto,custom" def="auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->auto
  </portType>
  <customPortNo min="1" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.NTPServer.portType,eq,custom}, attr:min{req, int},max{req, int}-->123
  </customPortNo>
</NTPServer>
```

12.1.6.10 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers/capabilities

Query Parameter

None

Request Message

None

Response Message


```

<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information List, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>

```

12.1.6.11 Get time zone

Request URL

GET /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

None

12.1.6.12 Set time zone

Request URL

PUT /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

12.1.7 Upgrade Management

12.1.7.1 Get the capabilities of peripheral module upgrade

Request URL

GET /ISAPI/System/AcsUpdate/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AcsUpdate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capabilities of peripheral module upgrade, attr:version{req, string, protocolVersion}-->
  <type>
    opt="cardReader,FPModule,securityModule,extendModule,channelController,IRModule,lampModule,elevatorController,FPAlgorithmProgram,uboot,keypad,wirelessRecv,wiredZone,sirenIndoor,sirenOutdoor,sirenAudio,peater,bluetoothModule,single,wallSwitch,smartPlug,detector,transmitter,remoteCtrl,subModule,dispModule,netReader,lockControlBoard,networkZoneModule,userInterfaceBoard,subPermissionController,electricLock,heatingModule,RS485Module,QRCodeModule,R3WirelessRecv,RXWirelessRecv,wiredOutput,electricGenie,zigbeeModule,PMRModule,networkInputAndOutputModule,infoModule,keypadAndCardReader,socialSecurityCardModule,localController"
    <!--ro, opt, enum, upgrade type: upgrade peripheral, subType:string, attr:opt{req, string}, desc:"cardReader" (485 card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "uboot" (uboot upgrade), "keypad" (keypad), "wirelessRecv" (wireless receiver module), "wiredZone" (wired zone module), "sirenIndoor" (indoor sounder), "sirenOutdoor" (outdoor sounder), "sirenAudio" (sounder two-way audio), "peater" (repeater), "bluetoothModule" (bluetooth module), "single" (single output module), "wallSwitch" (wall mounted switch), "smartPlug" (smart plug), "detector" (detector), "transmitter" (transmitter peripheral), "remoteCtrl" (keyfob), "subModule" (sub module), "dispModule" (display module), "netReader" (network card reader), "faceModule" (face module), "touchScreenModule" (touch screen module), "temperatureModule" (temperature measurement module), "lockControlBoard" (lock control board), "networkZoneModule" (network zone module), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "electricLock" (electric lock), "heatingModule" (heating module), "RS485Module" (RS-485 module), "QRCodeModule" (QR code module), "electricGenie" (electric Genie), "zigbeeModule" (zigbee module), "PMRModule" (PMR module), "networkInputAndOutputModule" (network input and output module via network TAP)-->cardReader
  </type>
  <batchType opt="electricGenie,wiredOutput,networkZoneModule,networkInputAndOutputModule">
    <!--ro, opt, enum, batch upgrade type (peripheral type), subType:string, attr:opt{req, string}, desc:peripheral types which support batch upgrading; if this node is supported, the following operations are supported: 1. Upgrade devices: /ISAPI/System/updateFirmware?type=<type>; 2. Start upgrading the specified analysis unit (device of the cluster): /ISAPI/System/upgradeStatus/startUpgrade?format=json; 3. Get the device upgrade progress: /ISAPI/System/upgradeStatus?format=json-->electricGenie
  </batchType>
  <cardReaderNo min="1" max="10" opt="1,4">
    <!--ro, opt, int, card reader No. range, it is valid when the returned value of type consists cardReader, attr:min{opt, int, step:1},max{opt, int, step:1},opt{opt, string}-->1
  </cardReaderNo>
  <FPModuleNo min="1" max="10">
    <!--ro, opt, int, fingerprint module No. range, it is valid when the returned value of type consists FPModule, attr:min{req, int},max{req, int}-->1
  </FPModuleNo>
  <securityModuleNo min="1" max="10">
    <!--ro, opt, int, security module No. range, it is valid when the returned value of type consists securityModule, attr:min{req, int},max{req, int}-->1
  </securityModuleNo>
  <extendModuleNo min="1" max="10">
    <!--ro, opt, int, No. range of I/O extended module, it is valid when the returned value of type consists extendModule, attr:min{req, int},max{req, int}-->1
  </extendModuleNo>
  <channelControllerNo min="1" max="10">
    <!--ro, opt, int, lane controller No. range, it is valid when the returned value of type consists channelController, attr:min{req, int},max{req, int}-->1
  </channelControllerNo>
  <IRModuleNo min="1" max="10">
```

```

<!--ro, opt, int, infrared module No. range, it is valid when the returned value of type consists IRModule, attr:min{req, int},max{req, int}-->1
</IRModuleNo>
<lampModuleNo min="1" max="10">
<!--ro, opt, int, indicator module No. range, it is valid when the returned value of type consists LampModule, attr:min{req, int},max{req, int}-->1
</lampModuleNo>
<elevatorControllerNo min="1" max="10">
<!--ro, opt, int, sub elevator controller No. range, it is valid when the returned value of type consists elevatorController, attr:min{req,
int},max{req, int}-->1
</elevatorControllerNo>
<FPAlgorithmProgramNo min="1" max="10">
<!--ro, opt, int, No. range of fingerprint algorithm programma of card reader, it is valid when the returned value of type consists FPAlgorithmProgram,
attr:min{req, int},max{req, int}-->1
</FPAlgorithmProgramNo>
<faceModuleNo min="1" max="10">
<!--ro, opt, int, face module No. range, it is valid when the returned value of type consists faceModule, attr:min{req, int},max{req, int}-->1
</faceModuleNo>
<touchScreenModuleNo min="1" max="10">
<!--ro, opt, int, touch screen module No. range, it is valid when the returned value of type consists touchScreenModule, attr:min{req, int},max{req,
int}-->1
</touchScreenModuleNo>
<temperatureModuleNo min="1" max="10">
<!--ro, opt, int, No. range of temperature measurement module, it is valid when the returned value of type consists temperatureModule, attr:min{req,
int},max{req, int}-->1
</temperatureModuleNo>
< keypadAddress opt="1,3,5">
<!--ro, opt, int, keypad address range, it is valid when the returned value of type consists keypad, attr:opt{req, string}-->1
</keypadAddress>
<wirelessRecvAddress opt="1,3,5">
<!--ro, opt, int, No. range of wireless receiver module, it is valid when the returned value of type consists wirelessRecv, attr:opt{req, string}-->1
</wirelessRecvAddress>
<wiredZoneAddress opt="1,3,5">
<!--ro, opt, int, No. range of wired zone module, it is valid when the returned value of type consists wiredZone, attr:opt{req, string}-->1
</wiredZoneAddress>
<sirenIndoorNo opt="1,3,5">
<!--ro, opt, int, indoor sounder No. range, it is valid when the returned value of type consists sirenIndoor, attr:opt{req, string}-->1
</sirenIndoorNo>
<sirenOutdoorNo opt="1,3,5">
<!--ro, opt, int, outdoor sounder No. range, it is valid when the returned value of type consists sirenOutdoor, attr:opt{req, string}-->1
</sirenOutdoorNo>
<repeaterNo opt="1,3,5">
<!--ro, opt, int, repeater No. range, it is valid when the returned value of type consists repeater, attr:opt{req, string}-->1
</repeaterNo>
<subModuleNo opt="1,3,5">
<!--ro, opt, int, sub module No. range, it is valid when the returned value of type consists subModule, attr:opt{req, string}-->1
</subModuleNo>
<dispModuleNo opt="1,3,5">
<!--ro, opt, int, display module No. range, it is valid when the returned value of type consists dispModule, attr:opt{req, string}-->1
</dispModuleNo>
<single opt="1,3,5">
<!--ro, opt, int, No. range of single output module, it is valid when the returned value of type consists single, attr:opt{req, string}-->1
</single>
<wallSwitch opt="1,3,5">
<!--ro, opt, int, No. range of wall mounted switch, it is valid when the returned value of type consists wallSwitch, attr:opt{req, string}-->1
</wallSwitch>
<smartPlug opt="1,3,5">
<!--ro, opt, int, smart plug No. range, it is valid when the returned value of type consists smartPlug, attr:opt{req, string}-->1
</smartPlug>
<zoneNo opt="1,3,5">
<!--ro, opt, int, zone No. range, it is valid when the returned value of type consists detector, attr:opt{req, string}, desc:get the detector via the
zone No.-->1
</zoneNo>
<transmitterNo opt="1,3,5">
<!--ro, opt, int, transmitter No. range, it is valid when the returned value of type consists transmitter, attr:opt{req, string}-->1
</transmitterNo>
<remoteCtrlNo opt="1,3,5">
<!--ro, opt, int, keyfob No. range, it is valid when the returned value of type consists remoteCtrl, attr:opt{req, string}-->1
</remoteCtrlNo>
<netReaderNo min="1" max="10">
<!--ro, opt, int, No. range of network card reader, it is valid when the returned value of type consists netReader, attr:min{req, int},max{req, int}-->1
</netReaderNo>
<lockControlBoardLNo opt="1,2,3,4,5,6,7,8">
<!--ro, opt, int, No. range of the Left part of the bracket, it is valid when the value of type returns, attr:opt{req, string}-->1
</lockControlBoardLNo>
<lockControlBoardRNo opt="1,2,3,4,5,6,7,8">
<!--ro, opt, int, No. range of the right part of the bracket, it is valid when the value of type returns, attr:opt{req, string}-->1
</lockControlBoardRNo>
<networkZoneModuleNo opt="1,3,5">
<!--ro, opt, int, No. range of network zone module, it is valid when the returned value of type consists networkZoneModule, attr:opt{req, string}-->1
</networkZoneModuleNo>
<userInterfaceBoardNo opt="0,1">
<!--ro, opt, enum, interface board No. range, it is valid when the returned value of type consists userInterfaceBoard, subType:string, attr:opt{req,
string}, desc:0-main user extended interface board, 1-sub user extended interface board-->0
</userInterfaceBoardNo>
<electricLockNo opt="1,3,5">
<!--ro, opt, int, electric lock No. range, it is valid when the returned value of type consists electricLock, attr:opt{req, string}-->1
</electricLockNo>
<heatingModuleNo opt="1,3,5">
<!--ro, opt, int, heating module No. range, it is valid when the returned value of type consists heatingModule, attr:opt{req, string}, desc:the heating
module starts to work to increase the device temperature when the devices runs under Low temperature-->1
</heatingModuleNo>
<sirenAudioNo opt="1,3,5">
<!--ro, opt, int, sounder (two-way audio) No. range, it is valid when the returned value of type consists sirenAudio, attr:opt{req, string}-->1
</sirenAudioNo>
<NRCodeModuleNo min="1" max="10">

```

```

<!--ro, opt, int, No. range of QR code module, it is valid when the returned value of type consists QRCodeModule, attr:min{req, int},max{req, int}-->1
</QRCodeModuleNo>
<R3WirelessRecvNo opt="1,2,3,4">
  <!--ro, opt, int, No. range of R3 wireless receiver module, it is valid when the returned value of type consists R3WirelessRecv, attr:opt{req, string}--
>1
  </R3WirelessRecvNo>
  <RXWirelessRecvNo opt="1,2,3,4">
    <!--ro, opt, int, No. range of RX wireless receiver module, it is valid when the returned value of type consists RXWirelessRecv, attr:opt{req, string}--
>1
    </RXWirelessRecvNo>
    <wiredOutputNo opt="1,2,3,4">
      <!--ro, opt, int, No. range of wired output module , it is valid when the returned value of type consists wiredOutput, attr:opt{req, string}-->1
    </wiredOutputNo>
    <electricGenieNo opt="1,2,3,4">
      <!--ro, opt, int, No. range of electric Genie, it is valid when the returned value of type or batchType consists electricGenie, attr:opt{req, string}--
>1
    </electricGenieNo>
    <networkInputAndOutputModuleNo opt="1,2,3,4">
      <!--ro, opt, int, No. range of network input and output module, it is valid when the returned value of type consists networkInputAndOutputModule,
      attr:opt{req, string}-->1
    </networkInputAndOutputModuleNo>
    <infoModuleNo opt="1,2,3,4">
      <!--ro, opt, int, dep:or,{$.AcsUpdate.type,eq,infoModule}, attr:opt{req, string}-->1
    </infoModuleNo>
    <keypadAndCardReaderModuleNo opt="1,2,3,4">
      <!--ro, opt, int, dep:or,{$.AcsUpdate.type,eq,keypadAndCardReaderModule}, attr:opt{req, string}-->1
    </keypadAndCardReaderModuleNo>
    <threeDimensionalStructuredLightModuleNo opt="1,2,3,4">
      <!--ro, opt, int, dep:or,{$.AcsUpdate.type,eq,threeDimensionalStructuredLightModule}, attr:opt{req, string}-->1
    </threeDimensionalStructuredLightModuleNo>
    <socialSecurityCardModuleNo opt="1">
      <!--ro, opt, int, dep:or,{$.AcsUpdate.type,eq,socialSecurityCardModule}, attr:opt{req, string}-->1
    </socialSecurityCardModuleNo>
    <localControllerNo min="1" max="62">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </localControllerNo>
  </AcsUpdate>

```

12.1.7.2 Upgrade devices

Request URL

POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>&childDevID=<devIndex>&isUpdateAuxDevice=<isUpdateAuxDevice>

Query Parameter

Parameter Name	Parameter Type	Description
type	enum	device type 1.alarm device: "keypad" (keypad), "wirelessRecv" (wireless receiver module), "wiredZone" (wired zone module), "sirenIndoor" (indoor sounder), "sirenOutdoor" (outdoor sounder), "sirenAudio" (sounder two-way audio), "repeater" (repeater), "bluetoothModule" (bluetooth module), "single" (single output module), "wallSwitch" (wall mounted switch), "smartPlug" (smart plug), "detector" (detector), "transmitter" (transmitter peripheral), "remoteCtrl" (keyfob), "subModule" (sub module), "dispModule" (display module), "netReader" (network card reader), "faceModule" (face module), "touchScreenModule" (touch screen module), "temperatureModule" (temperature measurement module), "lockControlBoard" (lock control board), "networkZoneModule" (network zone module), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "electricLock" (electric lock), "heatingModule" (heating module), "RS485Module" (RS-485 module), "QRCodeModule" (QR code module), "electricGenie" (electric Genie), "zigbeeModule" (zigbee module), "PMRModule" (PMR module), "networkInputAndOutputModule" (network input and output module via network TAP) 2. access control device: "cardReader" (485 card reader), "FPMModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAAlgorithmProgram" (fingerprint algorithm program of card reader), "bluetoothModule" (bluetooth module), "MCU"-MCU upgrade, "temperatureModule" (temperature measurement module), "faceModule" (face module), "touchScreenModule" (touch screen module), "netReader" (network card reader), "userInterfaceBoard" (interface board), "subPermissionController" (sub permission controller), "QRCodeModule" (QR code module) 3. intelligent cabinet: "lockControlBoard" (lock control board) 4. "ledReceiverCard"-receiving card; 5. "subsystem"-sub system; 6. radar-assisted device: "warningScreenFirmware"-firmware upgrade of pre-alarm screen, "warningScreenFont"-font library upgrade of pre-alarm screen, "warningScreenVoice"-audio upgrade of pre-alarm screen; 7. "electricGenie"-electric Genie
moduleAddress	string	module address is an exclusive node for Ax Hybrid example: keypad: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; relay: GET /ISAPI/SecurityCP/Configuration/outputs?format=json; sounder: GET /ISAPI/SecurityCP/Configuration/wirelessSiren?format=json; output module: GET /ISAPI/SecurityCP/Configuration/outputModules?format=json; extension module: GET /ISAPI/SecurityCP/Configuration/extensionModule?format=json; zone: GET /ISAPI/SecurityCP/Configuration/zones?format=json
indexID	string	device No. 1. AX Hybrid Pro and wireless security control upgrade by ID; 2. access control devices upgrade by ID; 3. intelligent cabinet upgrade by IS
devIndex	string	--
isUpdateAuxDevice	string	--

Request Message

Binary Data

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:it describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information, range:[0,1024], desc:this node is used for debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
2	Device Busy	noMemory	0x20000001	--
2	Device Busy	networkError	0x20000009	--
3	Device Error	badFlash	0x30000002	--
3	Device Error	RFJammingDetected	0x30008002	--
4	Invalid Operation	notSupport	0x40000001	--
4	Invalid Operation	armedStatus	0x4000800B	--
4	Invalid Operation	registerMode	0x40008015	--
4	Invalid Operation	upgradePackageTransferFailed	0x4000806A	--
4	Invalid Operation	DetectorOrPeripheralUpgrading	0x4000807D	--
6	Invalid Content	badVersion	0x6000000F	--
6	Invalid Content	badLanguage	0x60000011	--
6	Invalid Content	versionMismatch	0x60006052	--
7	Reboot Required	rebootRequired	0x70000001	--
7	Reboot Required	autoReboot	0x70000002	--

12.1.7.3 Get the device upgrading status and progress

Request URL

GET /ISAPI/System/upgradeStatus?type= <type> &childDevID= <childDevID> &id= <id>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	"usbmic" (array microphone module)
childDevID	string	--
id	string	The id indicates the peripheral No. to be upgraded.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	noMemory	0x4000202A	--

12.1.8 User Management

12.1.8.1 Log in to the device by digest

Request URL

GET /ISAPI/Security/userCheck

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<userCheck xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <statusValue>
    <!--ro, req, enum, status code, subType:int, desc:"200"(succeeded), "401"(failed)-->200
  </statusValue>
  <statusString>
    <!--ro, opt, enum, status, subType:string, desc:"OK", "Unauthorized"-->OK
  </statusString>
  <isDefaultPassword>
    <!--ro, opt, bool-->true
  </isDefaultPassword>
  <isRiskPassword>
    <!--ro, opt, bool-->true
  </isRiskPassword>
  <isActivated>
    <!--ro, opt, bool-->true
  </isActivated>
  <residualValidity>
    <!--ro, opt, int-->1
  </residualValidity>
  <lockStatus>
    <!--ro, opt, enum, locking status, subType:string, desc:"unlock", "Locked"-->unlock
  </lockStatus>
  <unlockTime>
    <!--ro, opt, int, unlocking remaining time, unit:s-->1
  </unlockTime>
  <retryLoginTime>
    <!--ro, opt, int, remaining login attempts-->1
  </retryLoginTime>
</userCheck>
```

statusCode	statusString	subStatusCode	errorCode	description
6	Invalid Content	forbiddenIP	0x60000070	--

12.1.8.2 Get the permission of a single user

Request URL

GET /ISAPI/Security/UserPermission/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UserPermission xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user permissions, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index, range:[1,32]-->test
  </id>
  <userID>
    <!--ro, req, string, user ID, range:[1,32]-->test
  </userID>
  <userType>
    <!--ro, req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission),
"operator" (operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)--
>admin
  </userType>
  <localPermission>
    <!--ro, opt, object, Local permission node-->
    <backup>
      <!--ro, opt, bool, whether to enable the backup permission (of all channels), desc:true (enable), false (disable)-->true
    </backup>
    <record>
      <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
    </record>
    <playBack>
      <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
    </playBack>
    <preview>
      <!--ro, opt, bool, whether to enable the Live view permission (of all channels), desc:true (enable), false (disable)-->true
    </preview>
    <privacyMaskClose>
      <!--ro, opt, bool-->>false
    </privacyMaskClose>
    <videoChannelPermissionList>
      <!--ro, opt, object, permission configuration List for video channel-->
      <videoChannelPermission>
        <!--ro, opt, object, permission configuration for a single video channel-->
        <id>
          <!--ro, req, string, video channel ID, range:[1,32]-->test
        </id>
        <playBack>
          <!--ro, opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
        </playBack>
        <preview>
          <!--ro, opt, bool, whether to enable the Live view permission of the current channel, desc:true (enable), false (disable)-->true
        </preview>
        <record>
          <!--ro, opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
        </record>
        <backup>
          <!--ro, opt, bool, whether to enable the backup permission of the current channel, desc:true (enable), false (disable)-->true
        </backup>
        <playBackDoubleVerification>
          <!--ro, opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)--
>true
        </playBackDoubleVerification>
        <backupDoubleVerification>
          <!--ro, opt, bool, whether to enable double verification permission for backup of the current channel, desc:true (enable), false (disable)-->true
        </backupDoubleVerification>
        <privacyMaskClose>
          <!--ro, opt, bool-->>false
        </privacyMaskClose>
      </videoChannelPermission>
    </videoChannelPermissionList>
    <ptzControl>
      <!--ro, req, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
    </ptzControl>
    <ptzChannelPermissionList>
      <!--ro, opt, object, PTZ control permission List of the video channels-->
      <ptzChannelPermission>
        <!--ro, req, object, PTZ control permission of a specified video channel-->
        <id>
          <!--ro, req, string, video channel ID, range:[1,32]-->test
        </id>
```



```

    <ptzControl>
      <!--ro, opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
    </ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
<logOrStateCheck>
  <!--ro, opt, bool, whether to enable permission for checking logs and device status, desc:true (enable), false (disable)-->true
</logOrStateCheck>
<parameterConfig>
  <!--ro, opt, bool, whether to enable the configuration permission for remote parameters, desc:true (enable), false (disable)-->true
</parameterConfig>
<restartOrShutdown>
  <!--ro, opt, bool, whether to enable the permission to reboot or power off, desc:true (enable), false (disable)-->true
</restartOrShutdown>
<upgrade>
  <!--ro, opt, bool, whether to enable the upgrade permission, desc:true (enable), false (disable)-->true
</upgrade>
<upgradeNotShow>
  <!--ro, opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure
compatibility, true (support), false or this node is not returned (not support)-->true
</upgradeNotShow>
<manageChannel>
  <!--ro, opt, bool, whether to enable the channel management permission, desc:true (enable), false (disable)-->true
</manageChannel>
<AIModelManagement>
  <!--ro, opt, bool, whether to enable permission for operation on the AI model packet, desc:true (enable), false (disable)-->true
</AIModelManagement>
<AITaskManagement>
  <!--ro, opt, bool, whether to enable permission for management and operation on the AI model packet, desc:true (enable), false (disable)-->true
</AITaskManagement>
<subSysOrZoneArm>
  <!--ro, opt, bool, permission to arm partitions (areas) / zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
  <!--ro, opt, bool, permission to disarm partitions (areas) / zones-->true
</subSysOrZoneDisarm>
<operateOutput>
  <!--ro, opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
  <!--ro, opt, array, list of the partitions (areas) that support linkage, subType:object, desc:N/A-->
  <subSystem>
    <!--ro, opt, int, the partition (area) that supports linkage, range:[1,64]-->1
  </subSystem>
</supportLinkageSubSystemList>
<subSystemList>
  <!--ro, opt, array, partition (area) list, subType:object, desc:partitions (areas) can be linked when adding or configuring operators N/A-->
  <subSystem>
    <!--ro, opt, int, partition (area), range:[1,64]-->1
  </subSystem>
</subSystemList>
<localFileMgr>
  <!--ro, opt, bool, permission for local folder management, desc:it refers to the permission of adding, deleting, editing, and getting local files--
>true
</localFileMgr>
<LEDConfig>
  <!--ro, opt, bool, permission to edit LED configuration-->true
</LEDConfig>
<logMgr>
  <!--ro, opt, bool, permission for log management-->true
</logMgr>
<TIPConfig>
  <!--ro, opt, bool, permission for TIP configuration-->true
</TIPConfig>
<securityFunctionCfg>
  <!--ro, opt, bool, permission for security check function configuration, desc:it is used for permission configuration for security check devices--
>true
</securityFunctionCfg>
<networkCfg>
  <!--ro, opt, bool, permission for network configuration, desc:including wireless and wired network-->true
</networkCfg>
<ISUPAccess>
  <!--ro, opt, bool, permission for ISUP access configuration, desc:it refers to the ISUP configuration permission for a device to access. Field: ISUP
Access Management-->true
</ISUPAccess>
<timeMgr>
  <!--ro, opt, bool, permission for time management, desc:it refers to the permission of managing system time and time zone. Field: Time Management--
>true
</timeMgr>
<HDDMgr>
  <!--ro, opt, bool, permission for HDD management, desc:field: HDD Management-->true
</HDDMgr>
<securitySystemMaintenance>
  <!--ro, opt, bool, permission for security inspection system maintenance, desc:the permission includes testing detector, ray source, and operation
keyboard, and diagnosing system, maintaining records, and performing other tests on, e.g., main control board, I/O board, and ray source light-->true
</securitySystemMaintenance>
<securityCheckingPicExport>
  <!--ro, opt, bool, permission to export baggage security check pictures-->true
</securityCheckingPicExport>
<securityCheckingPicDelete>
  <!--ro, opt, bool, permission to delete baggage security check pictures-->true
</securityCheckingPicDelete>
<operationScheduleList>
  <!--ro, opt, array, operation schedule list, subType:object, desc:

```

```

<!--ro, opt, array, operation schedule list, subtype:object, dep:or,
{$.UserPermissionList.UserPermission.LocalPermission.operationScheduleEnabled,eq,true}<!-->
<operationSchedule>
  <!--ro, opt, object, operation schedule-->
  <operationScheduleEnabled>
    <!--ro, opt, bool, whether to enable operation schedule-->true
  </operationScheduleEnabled>
  <operationDetailItem>
    <!--ro, opt, enum, operation detailed item, subType:string, desc:"all", "arm" (arming), "disarm" (disarming), "operateOutput" (operating relays)--
>arm
    </operationDetailItem>
  <timeRange>
    <!--ro, opt, object, time range-->
    <beginTime>
      <!--ro, opt, time, start time-->00:00:00
    </beginTime>
    <endTime>
      <!--ro, opt, time, end time-->00:00:00
    </endTime>
  </timeRange>
</operationSchedule>
</operationScheduleList>
</localPermission>
<remotePermission>
  <!--ro, opt, object, remote permission node-->
  <record>
    <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
  </record>
  <playBack>
    <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
  </playBack>
  <preview>
    <!--ro, opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
  </preview>
  <privacyMaskClose>
    <!--ro, opt, bool-->>false
  </privacyMaskClose>
  <videoChannelPermissionList>
    <!--ro, opt, object, permission configuration list for video channel-->
    <videoChannelPermission>
      <!--ro, opt, object, permission configuration for a single video channel-->
      <id>
        <!--ro, req, string, corresponding video channel ID, range:[1,32]<!--test
      </id>
      <preview>
        <!--ro, opt, bool, whether to enable the live view permission of the current channel, desc:true (enable), false (disable)-->true
      </preview>
      <playBack>
        <!--ro, opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
      </playBack>
      <record>
        <!--ro, opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
      </record>
      <playBackDoubleVerification>
        <!--ro, opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)--
>true
      </playBackDoubleVerification>
      <privacyMaskClose>
        <!--ro, opt, bool-->>false
      </privacyMaskClose>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl>
    <!--ro, opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
  </ptzControl>
  <ptzChannelPermissionList>
    <!--ro, opt, object, PTZ control permission list of the video channels-->
    <ptzChannelPermission>
      <!--ro, opt, object, PTZ control permission of a specified video channel-->
      <id>
        <!--ro, req, string, video channel ID, range:[1,32], desc:/ISAPI/PTZCtrl/channels/ID-->test
      </id>
      <ptzControl>
        <!--ro, opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
      </ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck>
    <!--ro, opt, bool, whether to enable permission for checking logs and device status-->true
  </logOrStateCheck>
  <parameterConfig>
    <!--ro, opt, bool, whether to enable the configuration permission for remote parameters-->true
  </parameterConfig>
  <restartOrShutdown>
    <!--ro, opt, bool, whether to enable the permission to reboot or power off-->true
  </restartOrShutdown>
  <upgrade>
    <!--ro, opt, bool, whether to enable the upgrade permission-->true
  </upgrade>
  <upgradeNotShow>
    <!--ro, opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure
compatibility, true (support), false or this node is not returned (not support)-->true
  </upgradeNotShow>
  <voiceTalk>

```

```

    <!--ro, opt, bool, whether to enable the two-way audio permission (of all channels)-->true
</voiceTalk>
<transParentChannel>
    <!--ro, opt, bool, whether to enable the transparent channel permission (of all channels), desc:the serial port control-->true
</transParentChannel>
<contorlLocalOut>
    <!--ro, opt, bool, whether to enable the Local output control permission-->true
</contorlLocalOut>
<manageChannel>
    <!--ro, opt, bool, whether to enable the channel management permission-->true
</manageChannel>
<alarmOutOrUpload>
    <!--ro, opt, bool, whether to enable permission for alarm uploading and output-->true
</alarmOutOrUpload>
<IRAID>
    <!--ro, opt, object, permission for Iraid quick configuration-->
    <rapidConfiguration>
        <!--ro, opt, bool, whether to enable permission for Iraid quick configuration-->true
    </rapidConfiguration>
    <reset>
        <!--ro, opt, bool, whether to enable permission for restoring Iraid configuration-->true
    </reset>
</IRAID>
<LEDConfig>
    <!--ro, opt, bool, whether it allows users to edit the LED configuration-->true
</LEDConfig>
<AIModelManagement>
    <!--ro, opt, bool, whether to enable permission for operation on the AI model packet-->true
</AIModelManagement>
<AITaskManagement>
    <!--ro, opt, bool, whether to enable permission for management and operation on AI tasks-->true
</AITaskManagement>
<materialReview>
    <!--ro, opt, bool, material checking permission, desc:information release-->true
</materialReview>
<programReview>
    <!--ro, opt, bool, program checking permission, desc:information release-->true
</programReview>
<uploadMaterial>
    <!--ro, opt, bool, upload material, desc:information release-->true
</uploadMaterial>
<addProgram>
    <!--ro, opt, bool, program editing permission, desc:add, delete, and edit-->true
</addProgram>
<addSchedule>
    <!--ro, opt, bool, schedule editing permission, desc:add, delete, and edit-->true
</addSchedule>
<programPlay>
    <!--ro, opt, bool, information playing permission-->true
</programPlay>
<terminalMgr>
    <!--ro, opt, bool, terminal management permission, desc:set terminal parameters such as backlight and Logo light-->true
</terminalMgr>
<shopSmart>
    <!--ro, opt, bool, permission for shop smart-->true
</shopSmart>
<materialReplace>
    <!--ro, opt, bool, permission to replace materials-->true
</materialReplace>
<scheduleRelease>
    <!--ro, opt, bool, permission to release schedules-->true
</scheduleRelease>
<scheduleReview>
    <!--ro, opt, bool, schedule checking-->true
</scheduleReview>
<addTerminal>
    <!--ro, opt, bool, terminal adding-->true
</addTerminal>
<terminalDispatch>
    <!--ro, opt, bool, permission to relocate terminals, desc:to relocate the terminals under this organization to lower-level organizations-->true
</terminalDispatch>
<addUser>
    <!--ro, opt, bool, the permission for editing users, desc:add, delete, and edit-->true
</addUser>
<addOrg>
    <!--ro, opt, bool, the permission for editing organizations, desc:add, delete, and edit-->true
</addOrg>
<logMgr>
    <!--ro, opt, bool, Log management-->true
</logMgr>
<dataUpdate>
    <!--ro, opt, bool, data update, desc:calling data-->true
</dataUpdate>
<insertMessage>
    <!--ro, opt, bool, the permission for cutting texts-->true
</insertMessage>
<notReview>
    <!--ro, opt, bool, free-reviewing permission-->true
</notReview>
<MgrTerminal>
    <!--ro, opt, object, the managed terminals or terminal group-->
    <TerminalNoList size="10">
        <!--ro, opt, object, terminal No. List, attr:size{req, int}-->

```

```

    <terminalNo>
    <!--ro, opt, int, terminal No., range:[1,10]-->1
  </terminalNo>
</TerminalNoList>
<TerminalGroupNoList size="10">
  <!--ro, opt, object, terminal group List, attr:size{req, int}-->
  <terminalGroupNo>
    <!--ro, opt, int, terminal group No., range:[1,10]-->1
  </terminalGroupNo>
</TerminalGroupNoList>
</MgrTerminal>
<subSysOrZoneArm>
  <!--ro, opt, bool, permission to arm partitions (areas)/zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
  <!--ro, opt, bool, permission to disarm partitions (areas)/zones-->true
</subSysOrZoneDisarm>
<subSysOrZoneClearArm>
  <!--ro, opt, bool, permission to clear alarms triggered in partitions (areas)/zones-->true
</subSysOrZoneClearArm>
<zoneBypass>
  <!--ro, opt, bool, permission to bypass zones-->true
</zoneBypass>
<zoneBypassRecover>
  <!--ro, opt, bool, permission to recover bypass of zones-->true
</zoneBypassRecover>
<operateOutput>
  <!--ro, opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
  <!--ro, opt, object, List of the partitions (areas) that support Linkage-->
  <subSystem>
    <!--ro, opt, int, the partition (area) that supports Linkage, range:[1,64]-->1
  </subSystem>
</supportLinkageSubSystemList>
<subSystemList>
  <!--ro, opt, object, partition (area) List, desc:users can Link partitions (areas) when adding or configuring operators-->
  <subSystem>
    <!--ro, opt, int, partition (area), range:[1,64]-->1
  </subSystem>
</subSystemList>
<factoryReset>
  <!--ro, opt, bool, restore to default parameters-->true
</factoryReset>
<arm>
  <!--ro, opt, bool, arm-->true
</arm>
<disarm>
  <!--ro, opt, bool, disarm-->true
</disarm>
<accessControl>
  <!--ro, opt, bool, access control-->true
</accessControl>
<restoreTamper>
  <!--ro, opt, bool, restore the tamper-proof setting, desc:refer to /ISAPI/Security/UserPermission/capabilities-->true
</restoreTamper>
<restoreConfirmedAlarm>
  <!--ro, opt, bool, restore the acknowledging alarm setting, desc:refer to /ISAPI/Security/UserPermission/capabilities-->true
</restoreConfirmedAlarm>
<LPListAudit>
  <!--ro, opt, bool, permission for blocklist and allowList operations,
  desc:users with this permission can import/export/edit/delete/search for blocklist and allowList. The related URIs are as follows:
  URI for searching for the license plate blocklist and allowList: /ISAPI/Traffic/channels/<ID>/searchLPListAudit;
  URI for importing/exporting the license plate blocklist and allowList data: /ISAPI/Traffic/channels/<ID>/LicensePlateAuditData;
  URI for editing the license plate List: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/record?format=json;
  URI for deleting the license plate blocklist and allowList: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/delete?format=json-->true
  </LPListAudit>
</applicationWarehouse>
  <!--ro, opt, bool, APP Library management permission, desc:this node is not returned if it is not supported. With management permission, users can
  perform operations such as importing, deleting and starting HEOP APP-->true
</applicationWarehouse>
<subSystemManagement>
  <!--ro, opt, bool, permission to manage partitions (areas), desc:all permissions for partition (area) management-->true
</subSystemManagement>
<userOfModifyKeypadPasswordList>
  <!--ro, opt, array, List of users allowed to change keypad password, subType:object-->
  <userOfModifyKeypadPassword>
    <!--ro, opt, object, users allowed to change keypad password-->
  </subUserType>
  <!--ro, opt, enum, type of sub-users, subType:string, desc:"cCloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
  "cCloudOperator", "LocalOperator"-->cCloudAdmin
  </subUserType>
</userOfModifyKeypadPassword>
</userOfModifyKeypadPasswordList>
<userOfModifyUserPermissionList>
  <!--ro, opt, array, List of users allowed to edit user permissions, subType:object-->
  <userOfModifyUserPermission>
    <!--ro, opt, object, users allowed to edit user permissions-->
  </subUserType>
  <!--ro, opt, enum, type of sub-users, subType:string, desc:"cCloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
  "cCloudOperator", "LocalOperator"-->cCloudAdmin
  </subUserType>
</userOfModifyUserPermission>
</userOfModifyUserPermissionList>

```

```

<userOfModifyPircamPermissionList>
  <!--ro, opt, array, List of users who can edit the permission to receive pircam camera video or picture event, subType:object-->
  <userOfModifyPircamPermission>
    <!--ro, opt, object, user who can edit the permission to receive pircam camera video or picture event-->
    <subUserType>
      <!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
      "cloudOperator", "LocalOperator"-->cloudAdmin
    </subUserType>
  </userOfModifyPircamPermission>
</userOfModifyPircamPermissionList>
<userOfModifyIPCPermissionList>
  <!--ro, opt, array, List of users who can edit the permission to receive network camera video or picture event, subType:object-->
  <userOfModifyIPCPermission>
    <!--ro, opt, object, user who can edit the permission to receive network camera video or picture event-->
    <subUserType>
      <!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
      "cloudOperator", "LocalOperator"-->cloudAdmin
    </subUserType>
  </userOfModifyIPCPermission>
</userOfModifyIPCPermissionList>
<userOfModifyUploadPermissionList>
  <!--ro, opt, array, List of users who can edit the uploading permission, subType:object-->
  <userOfModifyUploadPermission>
    <!--ro, opt, object, user who can edit the uploading permission-->
    <subUserType>
      <!--ro, opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
      "cloudOperator", "LocalOperator"-->cloudAdmin
    </subUserType>
  </userOfModifyUploadPermission>
</userOfModifyUploadPermissionList>
<falseAlarmMaterialLibMgr>
  <!--ro, opt, bool, permission for false alarm Library management-->true
</falseAlarmMaterialLibMgr>
<TIPConfig>
  <!--ro, opt, bool, whether the user has the permission of TIP configuration-->true
</TIPConfig>
<networkCfg>
  <!--ro, opt, bool, whether the user has the permission of network configuration, desc:including wireless and wired network-->true
</networkCfg>
<ISUPAccess>
  <!--ro, opt, bool, whether the user has the permission of ISUP access configuration, desc:it refers to the ISUP configuration permission for a device
  to access. Field: ISUP Access Management-->true
</ISUPAccess>
<timeMgr>
  <!--ro, opt, bool, whether the user has the permission of time management, desc:it refers to the permission of managing system time and time zone.
  Field: Time Management-->true
</timeMgr>
<HDDMgr>
  <!--ro, opt, bool, whether the user has the permission of HDD management, desc:field: HDD Management-->true
</HDDMgr>
<statusUploadPermission>
  <!--ro, opt, object, status event permission-->
  <panelPowerStatus>
    <!--ro, opt, bool, whether the user has the permission to receive security control panel's power status event, desc:the security control panel's
    power status refers to battery status of AC power, storage battery power, etc. The corresponding CID event codes are: [1301,3301,1302,3302,1311,3311,1311]--
    >true
  </panelPowerStatus>
  <panelCommunicateStatus>
    <!--ro, opt, bool, whether the user has the permission to receive security control panel's network status event, desc:the security control panel's
    network status refers to mobile network, SIM card, WIFI, wired network, etc. The corresponding CID event codes are:
    [1351,3351,1352,3352,1350,3350,1350,3350,1350,3350,1344,3344,1350,3350,1350,3350,1949,3949,1350,1948]-->true
  </panelCommunicateStatus>
  <zoneStatus>
    <!--ro, opt, bool, whether the user has the permission to receive zone status event, desc:the zones refer to all zones linked to the user, and the
    zone status refers to wireless detector's power, temperature, online status, etc., and peripheral module's connection status, sensor error status, etc. The
    corresponding CID event codes are: [1306,3306,1570,3570,1144,3144,1380,3380,1158,3158,1159,3159]-->true
  </zoneStatus>
  <externalDeviceStatus>
    <!--ro, opt, bool, whether the user has the permission to receive peripheral status event, desc:the peripherals refer to all peripherals linked to
    the user, such as repeater, wireless repeater, expansion module AC power, wireless peripheral, wireless sounder, keypad, and card reader; the peripheral
    status refers to the status of these devices. The corresponding CID event codes are:
    [1312,3312,1319,3319,1312,3312,1384,3384,1301,3301,1311,3311,1301,3301,1381,3381,1384,3384,1381,3381,1381,3381,1381,3381,1384,3384,1384,3384,1501,3501,1501,
    3501,1306,3306,1306,3306,1306,3306,1306,3306]-->true
  </externalDeviceStatus>
</statusUploadPermission>
<eventUploadPermission>
  <!--ro, opt, object, event uploading permission, desc:the uploading protocol of security control panel event is CID (event identifier: cidEvent). For
  definition values of CID, refer to the CID nodes on PMP platform-->
  <panicAlarm>
    <!--ro, opt, bool, whether the user has the permission to receive emergency alarm event, desc:emergency alarm event is triggered by card, keyfob, or
    keypad, and can be triggered in zones. The corresponding CID event codes are: [1120,3120]-->true
  </panicAlarm>
  <medicalAlarm>
    <!--ro, opt, bool, whether the user has the permission to receive medical alarm event, desc:medical alarm event is triggered via card, keyfob, or
    keypad by a person. The corresponding CID event codes are: [1100,3100]-->true
  </medicalAlarm>
  <fireAlarm>
    <!--ro, opt, bool, whether the user has the permission to receive fire alarm event, desc:fire alarm event is uploaded when a fire occurs. The
    corresponding CID event codes are: [1111,3111,1114,3114]-->true
  </fireAlarm>
  <smartAlarm>
    <!--ro, opt, bool, whether the user can have the permission to receive intelligent alarm event, desc:intelligent alarm events include intrusion
    detection alarm, line crossing alarm, fire source alarm, temperature alarm, motion detection alarm, and region entrance/exiting alarm. The corresponding CID

```

```

event codes are: [1131,3131,1131,3131,1112,3112,1158,3158,1159,3159,1158,3158,1159,3159,1131,3131,1134]-->true
</smartAlarm>
<panelTamper>
  <!--ro, opt, bool, whether the user has the permission to receive security control panel tampering event, desc:the security control panel tampering
event refers to the alarm event triggered when the security control panel is being tampered. The corresponding CID event codes are: [1137,3137]-->true
</panelTamper>
<externalDeviceTamper>
  <!--ro, opt, bool, whether the user has the permission to receive peripheral tampering event, desc:the peripherals include extension module,
wireless repeater, wireless peripheral, and the peripheral tampering event refers to the alarm triggered when the linked peripheral is being tampered. The
corresponding CID event codes are: [1144,3144,1144,3144,1144,3144,1144,3144]-->true
</externalDeviceTamper>
<zoneAlarm>
  <!--ro, opt, bool, whether the user has the permission to receive zone alarm event and zone tampering event, desc:the zone alarm events include
instant alarm, time out alarm, burglary alarm, 24-hour alarm, delay alarm, device moving alarm, detector blocking alarm, zone detector tampering alarm,
water leak alarm. The corresponding CID event codes are: [1120,3120,1100,3100,1111,3111,1114,3114,1162,3162]-->true
</zoneAlarm>
<panelOperation>
  <!--ro, opt, bool, whether the user has the permission to receive security control panel's operation event, desc:the security control panel's
operation events refer to the events triggered when some operations are performed on the security control panel, including disarming, away/stay/auto/key
zone arming/disarming, clearing alarm, pacing, forward/backward programming, control panel reset, entering incorrect password. The corresponding CID event
codes are: [1401,3401,3441,1403,3403,1409,3409,1121,1406,1452,1455,1454,3250,1865,1607,3607,1627,1628,1305,1461]-->true
</panelOperation>
</eventUploadPermission>
<FDLibPermissionList>
  <!--ro, opt, object, permission list of face picture Library control-->
<FDLibPermission>
  <!--ro, opt, object, permission of face picture Library control-->
  <FDID>
    <!--ro, req, string, face picture Library ID, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
  </FDID>
  <FDName>
    <!--ro, opt, string, face picture Library name, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
  </FDName>
  <FDLibControl>
    <!--ro, req, bool, whether to enable the permission of face picture Library control, desc:permission of face picture Library control includes
adding/deleting/editing/getting the information in the Library-->true
  </FDLibControl>
</FDLibPermission>
</FDLibPermissionList>
<picSearch>
  <!--ro, opt, bool, whether to enable the permission of picture search, desc:true-enabled, false-disabled. See details in /ISAPI/ContentMgmt/search--
>true
</picSearch>
<pircamVideoAndPicture>
  <!--ro, opt, bool, whether to enable the permission to receive pircam camera video or picture event, desc:when an alarm is triggered, if this
permission is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or
picture. There is no corresponding CID event code-->true
</pircamVideoAndPicture>
<IPCVideoAndPicture>
  <!--ro, opt, bool, whether to enable the permission to receive network camera video or picture event, desc:when an alarm is triggered, if this
permission is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or
picture. There is no corresponding CID event code-->true
</IPCVideoAndPicture>
</remotePermission>
<passwordValidity>
  <!--ro, opt, int, password validity period, range:[0,365], desc:unit: day; value range: [0, 365]; only the administrator can edit-->1
</passwordValidity>
<subUserType>
  <!--ro, opt, enum, type of sub-users, subType:string, desc:"cCloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
"cCloudOperator", "LocalOperator"-->cCloudAdmin
</subUserType>
<internalParametersAdjustment>
  <!--ro, opt, bool-->true
</internalParametersAdjustment>
</UserPermission>

```

12.1.8.3 Set permissions for a single user

Request URL

POT /ISAPI/Security/UserPermission/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<UserPermission xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user permissions, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, range:[1,32]-->test
  </id>

```

```

<userID>
  <!--req, string, user ID, range:[1,32]-->test
</userID>
<userType>
  <!--req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission), "operator"
(operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)-->admin
</userType>
<localPermission>
  <!--opt, object, local permission node-->
  <backup>
    <!--opt, bool, whether to enable the backup permission (of all channels), desc:true (enable), false (disable)-->true
  </backup>
  <record>
    <!--opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
  </record>
  <playBack>
    <!--opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
  </playBack>
  <preview>
    <!--opt, bool, whether to enable the Live view permission (of all channels), desc:true (enable), false (disable)-->true
  </preview>
  <privacyMaskClose>
    <!--opt, bool-->false
  </privacyMaskClose>
  <videoChannelPermissionList>
    <!--opt, object, permission configuration List for video channel-->
    <videoChannelPermission>
      <!--opt, object, permission configuration for a single video channel-->
      <id>
        <!--req, string, video channel ID, range:[1,32]-->test
      </id>
      <playBack>
        <!--opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
      </playBack>
      <preview>
        <!--opt, bool, whether to enable the Live view permission of the current channel, desc:true (enable), false (disable)-->true
      </preview>
      <record>
        <!--opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
      </record>
      <backup>
        <!--opt, bool, whether to enable the backup permission of the current channel, desc:true (enable), false (disable)-->true
      </backup>
      <playBackDoubleVerification>
        <!--opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)-->true
      </playBackDoubleVerification>
      <backupDoubleVerification>
        <!--opt, bool, whether to enable double verification permission for backup of the current channel, desc:true (enable), false (disable)-->true
      </backupDoubleVerification>
      <privacyMaskClose>
        <!--opt, bool-->false
      </privacyMaskClose>
    </videoChannelPermission>
  </videoChannelPermissionList>
  <ptzControl>
    <!--req, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
  </ptzControl>
  <ptzChannelPermissionList>
    <!--opt, object, PTZ control permission List of the video channels-->
    <ptzChannelPermission>
      <!--req, object, PTZ control permission of a specified video channel-->
      <id>
        <!--req, string, video channel ID, range:[1,32]-->test
      </id>
      <ptzControl>
        <!--opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
      </ptzControl>
    </ptzChannelPermission>
  </ptzChannelPermissionList>
  <logOrStateCheck>
    <!--opt, bool, whether to enable permission for checking logs and device status, desc:true (enable), false (disable)-->true
  </logOrStateCheck>
  <parameterConfig>
    <!--opt, bool, whether to enable the permission for remote parameters setting, desc:true (enable), false (disable)-->true
  </parameterConfig>
  <restartOrShutdown>
    <!--opt, bool, whether to enable the permission to reboot or power off, desc:true (enable), false (disable)-->true
  </restartOrShutdown>
  <upgrade>
    <!--opt, bool, whether to enable the upgrade permission, desc:true (enable), false (disable)-->true
  </upgrade>
  <upgradeNotShow>
    <!--opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure
compatibility, true (support), false or this node is not returned (not support)-->true
  </upgradeNotShow>
  <manageChannel>
    <!--opt, bool, whether to enable the channel management permission, desc:true (enable), false (disable)-->true
  </manageChannel>
  <AIModelManagement>
    <!--opt, bool, whether to enable permission for operation on the AI model packet, desc:true (enable), false (disable)-->true
  </AIModelManagement>
  <AITaskManagement>
    <!--opt, bool, whether to enable permission for management and operation on AI tasks, desc:true (enable), false (disable)-->true
  </AITaskManagement>

```



```

<subSysOrZoneArm>
  <!--opt, bool, permission to arm partitions (areas) / zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
  <!--opt, bool, permission to disarm partitions (areas) / zones-->true
</subSysOrZoneDisarm>
<operateOutput>
  <!--opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
  <!--opt, array, list of the partitions (areas) that support Linkage, subType:object, desc:N/A-->
  <subSystem>
    <!--opt, int, the partition (area) that supports Linkage, range:[1,64]-->1
  </subSystem>
</supportLinkageSubSystemList>
<subSystemList>
  <!--opt, array, partition (area) list, subType:object, desc:partitions (areas) can be Linked when adding or configuring operators N/A-->
  <subSystem>
    <!--opt, int, partition (area), range:[1,64]-->1
  </subSystem>
</subSystemList>
<localFileMgr>
  <!--opt, bool, whether the user has the permission of Local folder management, desc:it refers to the permission of adding, deleting, editing, and
getting local files-->true
</localFileMgr>
<LEDConfig>
  <!--opt, bool, whether the user has the permission to edit LED configuration-->true
</LEDConfig>
<logMgr>
  <!--opt, bool, whether the user has the Log management permission-->true
</logMgr>
<TIPConfig>
  <!--opt, bool, whether the user has the permission of TIP configuration-->true
</TIPConfig>
<securityFunctionCfg>
  <!--opt, bool, whether the user has the permission of security check function configuration, desc:it is used for permission configuration for security
check devices-->true
</securityFunctionCfg>
<networkCfg>
  <!--opt, bool, whether the user has the permission of network configuration, desc:including wireless and wired network-->true
</networkCfg>
<ISUPAccess>
  <!--opt, bool, whether the user has the permission of ISUP access configuration, desc:it refers to the ISUP configuration permission for a device to
access. Field: ISUP Access Management-->true
</ISUPAccess>
<timeMgr>
  <!--opt, bool, whether the user has the permission of time management, desc:it refers to the permission of managing system time and time zone. Field:
Time Management-->true
</timeMgr>
<HDDMgr>
  <!--opt, bool, whether the user has the permission of HDD management, desc:field: HDD Management-->true
</HDDMgr>
<securitySystemMaintenance>
  <!--opt, bool, permission for security inspection system maintenance, desc:the permission includes testing detector, ray source, and operation
keyboard, and diagnosing system, maintaining records, and performing other tests on, e.g., main control board, I/O board, and ray source light-->true
</securitySystemMaintenance>
<securityCheckingPicExport>
  <!--opt, bool, permission to export baggage security check pictures-->true
</securityCheckingPicExport>
<securityCheckingPicDelete>
  <!--opt, bool, permission to delete baggage security check pictures-->true
</securityCheckingPicDelete>
<operationScheduleList>
  <!--opt, array, operation schedule list, subType:object, dep:or,
{$.UserPermissionList.UserPermission.operationScheduleEnabled,eq,true}-->
  <operationSchedule>
    <!--opt, object, operation schedule-->
    <operationScheduleEnabled>
      <!--opt, bool, whether to enable operation schedule-->true
    </operationScheduleEnabled>
    <operationDetailItem>
      <!--opt, enum, operation detailed item, subType:string, desc:"all", "arm" (arming), "disarm" (disarming), "operateOutput" (operating relays)-->arm
    </operationDetailItem>
    <timeRange>
      <!--opt, object, time range-->
      <beginTime>
        <!--opt, time, start time-->00:00:00
      </beginTime>
      <endTime>
        <!--opt, time, end time-->00:00:00
      </endTime>
    </timeRange>
  </operationSchedule>
</operationScheduleList>
</localPermission>
<remotePermission>
  <!--opt, object, remote permission node-->
  <record>
    <!--opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
  </record>
  <playBack>
    <!--opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
  </playBack>

```



```

<preview>
  <!--opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
</preview>
<privacyMaskClose>
  <!--opt, bool-->false
</privacyMaskClose>
<videoChannelPermissionList>
  <!--opt, object, List of configuring permission for video channel-->
  <videoChannelPermission>
    <!--opt, object, configure the permission for a single video channel-->
    <id>
      <!--req, string, video channel ID, range:[1,32]-->test
    </id>
    <preview>
      <!--opt, bool, whether to enable the live view permission of the current channel, desc:true (enable), false (disable)-->true
    </preview>
    <playBack>
      <!--opt, bool, whether to enable the playback permission of the current channel, desc:true (enable), false (disable)-->true
    </playBack>
    <record>
      <!--opt, bool, whether to enable the recording permission of the current channel, desc:true (enable), false (disable)-->true
    </record>
    <playBackDoubleVerification>
      <!--opt, bool, whether to enable double verification permission for playback of the current channel, desc:true (enable), false (disable)-->true
    </playBackDoubleVerification>
    <privacyMaskClose>
      <!--opt, bool-->false
    </privacyMaskClose>
  </videoChannelPermission>
</videoChannelPermissionList>
<ptzControl>
  <!--opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
</ptzControl>
<ptzChannelPermissionList>
  <!--opt, object, PTZ control permission list of the video channels-->
  <ptzChannelPermission>
    <!--opt, object, PTZ control permission of a specified video channel-->
    <id>
      <!--req, string, video channel ID, range:[1,32], desc:/ISAPI/PTZCtrl/channels/ID-->test
    </id>
    <ptzControl>
      <!--opt, bool, whether to enable the PTZ control permission of the current channel, desc:true (enable), false (disable)-->true
    </ptzControl>
  </ptzChannelPermission>
</ptzChannelPermissionList>
<logOrStateCheck>
  <!--opt, bool, whether to enable permission for checking logs and device status-->true
</logOrStateCheck>
<parameterConfig>
  <!--opt, bool, whether to enable the permission for remote parameters setting-->true
</parameterConfig>
<restartOrShutdown>
  <!--opt, bool, whether to enable the permission to reboot or power off-->true
</restartOrShutdown>
<upgrade>
  <!--opt, bool, whether to enable the upgrade permission-->true
</upgrade>
<upgradeNotShow>
  <!--opt, bool, the upgrade is not displayed, desc:whether to support upgrade permission both for the admin user and installer user to ensure compatibility, true (support), false or this node is not returned (not support)-->true
</upgradeNotShow>
<voiceTalk>
  <!--opt, bool, whether to enable the two-way audio permission (of all channels)-->true
</voiceTalk>
<transparentChannel>
  <!--opt, bool, whether to enable the transparent channel, desc:namely, the serial port control-->true
</transparentChannel>
<contorlLocalOut>
  <!--opt, bool, whether to enable the local output control permission-->true
</contorlLocalOut>
<manageChannel>
  <!--opt, bool, whether to enable the channel management permission-->true
</manageChannel>
<alarmOutOrUpload>
  <!--opt, bool, whether to enable permission for alarm uploading and output-->true
</alarmOutOrUpload>
<IRAID>
  <!--opt, object, permission for iRAID quick configuration-->
  <rapidConfiguration>
    <!--opt, bool, whether to enable permission for iRAID quick configuration-->true
  </rapidConfiguration>
  <reset>
    <!--opt, bool, whether to enable permission for restoring iRAID configuration-->true
  </reset>
</IRAID>
<LEDConfig>
  <!--opt, bool, whether it allows the user to edit the LED configuration-->true
</LEDConfig>
<AIModelManagement>
  <!--opt, bool, whether to enable permission for operation on the AI model packet-->true
</AIModelManagement>
<AITaskManagement>
  <!--opt, bool, whether to enable permission for management and operation on AI tasks-->true
</AITaskManagement>

```

```

</MaterialMgr>
<materialReview>
  <!--opt, bool, material reviewing permission, desc:information release-->true
</materialReview>
<programReview>
  <!--opt, bool, program reviewing permission, desc:information release-->true
</programReview>
<uploadMaterial>
  <!--opt, bool, material upload, desc:information release-->true
</uploadMaterial>
<addProgram>
  <!--opt, bool, program editing permission, desc:add, delete, and edit-->true
</addProgram>
<addSchedule>
  <!--opt, bool, schedule editing permission, desc:add, delete, and edit-->true
</addSchedule>
<programPlay>
  <!--opt, bool, information playing permission-->true
</programPlay>
<terminalMgr>
  <!--opt, bool, terminal management permission, desc:set terminal parameters such as backlight and Logo light-->true
</terminalMgr>
<shopSmart>
  <!--opt, bool, permission for shop smart-->true
</shopSmart>
<materialReplace>
  <!--opt, bool, permission to replace materials-->true
</materialReplace>
<scheduleRelease>
  <!--opt, bool, permission for releasing schedule-->true
</scheduleRelease>
<scheduleReview>
  <!--opt, bool, schedule checking-->true
</scheduleReview>
<addTerminal>
  <!--opt, bool, terminal adding-->true
</addTerminal>
<terminalDispatch>
  <!--opt, bool, permission to relocate terminals, desc:to relocate the terminals under this organization to lower-level organizations-->true
</terminalDispatch>
<addUser>
  <!--opt, bool, the permission for editing users, desc:add, delete, and edit-->true
</addUser>
<addOrg>
  <!--opt, bool, the permission for editing organizations, desc:add, delete, and edit-->true
</addOrg>
<logMgr>
  <!--opt, bool, Log management-->true
</logMgr>
<dataUpdate>
  <!--opt, bool, data update, desc:calling data-->true
</dataUpdate>
<insertMessage>
  <!--opt, bool, the permission for cutting texts-->true
</insertMessage>
<notReview>
  <!--opt, bool, the permission for free-review-->true
</notReview>
<MgrTerminal>
  <!--opt, object, the managed terminals or terminal group-->
  <TerminalNoList size="10">
    <!--opt, object, terminal No. List, attr:size{req, int}-->
    <terminalNo>
      <!--opt, int, terminal No., range:[1,10]-->1
    </terminalNo>
  </TerminalNoList>
  <TerminalGroupNoList size="10">
    <!--opt, object, terminal group List, attr:size{req, int}-->
    <terminalGroupNo>
      <!--opt, int, terminal group No., range:[1,10]-->1
    </terminalGroupNo>
  </TerminalGroupNoList>
</MgrTerminal>
<subSysOrZoneArm>
  <!--opt, bool, permission to arm partitions (areas) / zones-->true
</subSysOrZoneArm>
<subSysOrZoneDisarm>
  <!--opt, bool, permission to disarm partitions (areas) / zones-->true
</subSysOrZoneDisarm>
<subSysOrZoneClearArm>
  <!--opt, bool, permission to clear alarms triggered in partitions (areas) / zones-->true
</subSysOrZoneClearArm>
<zoneBypass>
  <!--opt, bool, permission to bypass zones-->true
</zoneBypass>
<zoneBypassRecover>
  <!--opt, bool, permission to recover bypass of zones-->true
</zoneBypassRecover>
<operateOutput>
  <!--opt, bool, permission to operate relays-->true
</operateOutput>
<supportLinkageSubSystemList>
  <!--opt, object, List of the partitions (areas) that support linkage-->
  <subSystem>

```

```

    <!--opt, int, the partition (area) that supports Linkage, range:[1,64]-->1
  </subSystem>
</supportLinkageSubSystemList>
<subSystemList>
  <!--opt, object, partition (area) List, desc:partitions (areas) can be Linked when adding or configuring operators-->
  <subSystem>
    <!--opt, int, partition (area), range:[1,64]-->1
  </subSystem>
</subSystemList>
<factoryReset>
  <!--opt, bool, restore to default parameters-->true
</factoryReset>
<arm>
  <!--opt, bool, arm-->true
</arm>
<disarm>
  <!--opt, bool, disarm-->true
</disarm>
<accessControl>
  <!--opt, bool, access control-->true
</accessControl>
<restoreTamper>
  <!--opt, bool, restore the tamper-proof setting, desc:for usage details, refer to /ISAPI/Security/UserPermission/capabilities-->true
</restoreTamper>
<restoreConfirmedAlarm>
  <!--opt, bool, restore the acknowledging alarm setting, desc:for usage details, refer to /ISAPI/Security/UserPermission/capabilities-->true
</restoreConfirmedAlarm>
<LPListAudit>
  <!--opt, bool, permission for blocklist and allowList operations,
  desc:users with this permission can import/export/edit/delete/search for blocklist and allowList. The related URIs are as follows:
  URI for searching for the license plate blocklist and allowList: /ISAPI/Traffic/channels/<ID>/searchLPListAudit;
  URI for importing/exporting the license plate blocklist and allowList data: /ISAPI/Traffic/channels/<ID>/LicensePlateAuditData;
  URI for editing the license plate list: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/record?format=json;
  URI for deleting the license plate blocklist and allowList: /ISAPI/Traffic/channels/<channelID>/LicensePlateAuditData/delete?format=json-->true
  </LPListAudit>
<applicationWarehouse>
  <!--opt, bool, APP Library management permission, desc:with management permission, users can perform operations such as importing, deleting and
  starting HEOP APP-->true
</applicationWarehouse>
<subSystemManagement>
  <!--opt, bool, permission to manage partitions (areas), desc:all permissions for partition (area) management-->true
</subSystemManagement>
<userOfModifyKeypadPasswordList>
  <!--opt, array, List of users allowed to change keypad password, subType:object-->
  <userOfModifyKeypadPassword>
    <!--opt, object, users allowed to change keypad password-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyKeypadPassword>
</userOfModifyKeypadPasswordList>
<userOfModifyUserPermissionList>
  <!--opt, array, List of users allowed to edit user permissions, subType:object-->
  <userOfModifyUserPermission>
    <!--opt, object, users allowed to edit users permission-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyUserPermission>
</userOfModifyUserPermissionList>
<userOfModifyPircamPermissionList>
  <!--opt, array, List of users who can edit the permission to receive pircam camera video or picture event, subType:object-->
  <userOfModifyPircamPermission>
    <!--opt, object, user who can edit the permission to receive pircam camera video or picture event-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyPircamPermission>
</userOfModifyPircamPermissionList>
<userOfModifyIPCPermissionList>
  <!--opt, array, List of users who can edit the permission to receive network camera video or picture event, subType:object-->
  <userOfModifyIPCPermission>
    <!--opt, object, user who can edit the permission to receive network camera video or picture event-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyIPCPermission>
</userOfModifyIPCPermissionList>
<userOfModifyUploadPermissionList>
  <!--opt, array, List of users who can edit the uploading permission, subType:object-->
  <userOfModifyUploadPermission>
    <!--opt, object, user who can edit the uploading permission-->
  <subUserType>
    <!--opt, enum, type of sub-users, subType:string, desc:"cloudAdmin", "LocalAdmin", "installerAdmin", "installEmployee", "LocalInstaller",
    "cloudOperator", "LocalOperator"-->cloudAdmin
  </subUserType>
  </userOfModifyUploadPermission>
</userOfModifyUploadPermissionList>
<falseAlarmMaterialLibMgr>
  <!--opt, bool, permission for false alarm library management, desc:
  </falseAlarmMaterialLibMgr>

```

```

<!--opt, bool, permission for false alarm library management-->true
</falseAlarmMaterialLibMgr>
<TIPConfig>
  <!--opt, bool, whether the user has the permission of TIP configuration-->true
</TIPConfig>
<networkCfg>
  <!--opt, bool, whether the user has the permission of network configuration, desc:including wireless and wired network-->true
</networkCfg>
<ISUPAccess>
  <!--opt, bool, whether the user has the permission of ISUP access configuration, desc:it refers to the ISUP configuration permission for a device to
access. Field: ISUP Access Management-->true
</ISUPAccess>
<timeMgr>
  <!--opt, bool, whether the user has the permission of time management, desc:it refers to the permission of managing system time and time zone. Field:
Time Management-->true
</timeMgr>
<HDDMgr>
  <!--opt, bool, whether the user has the permission of HDD management, desc:field: HDD Management-->true
</HDDMgr>
<statusUploadPermission>
  <!--opt, object, status event permission-->
  <panelPowerStatus>
    <!--opt, bool, whether the user has the permission to receive security control panel's power status event, desc:the security control panel's power
status refers to battery status of AC power, storage battery power, etc. The corresponding CID event codes are: [1301,3301,1302,3302,1311,3311,1311]-->true
  </panelPowerStatus>
  <panelCommunicateStatus>
    <!--opt, bool, whether the user has the permission to receive security control panel's network status event, desc:the security control panel's
network status refers to mobile network, SIM card, WIFI, wired network, etc. The corresponding CID event codes are:
[1351,3351,1352,3352,1350,3350,1350,3350,1344,3344,1350,3350,1350,3350,1949,3949,1350,1948]-->true
  </panelCommunicateStatus>
  <zoneStatus>
    <!--opt, bool, whether the user has the permission to receive zone status event, desc:the zones refer to all zones linked to the user, and the zone
status refers to wireless detector's power, temperature, online status, etc., and peripheral module's connection status, sensor error status, etc. The
corresponding CID event codes are: [1306,3306,1570,3570,1144,3144,1380,3380,1158,3158,1159,3159]-->true
  </zoneStatus>
  <externalDeviceStatus>
    <!--opt, bool, whether the user has the permission to receive peripheral status event, desc:the peripherals refer to all peripherals linked to the
user, such as repeater, wireless repeater, expansion module AC power, wireless peripheral, wireless sounder, keypad, and card reader; the peripheral status
refers to the status of these devices. The corresponding CID event codes are:
[1312,3312,1319,3319,1312,3312,1384,3384,1301,3301,1311,3311,1301,3301,1381,3381,1384,3384,1381,3381,1381,3381,1384,3384,1384,3384,1501,3501,1501,
3501,1306,3306,1306,3306,1306,3306,1306,3306]-->true
  </externalDeviceStatus>
</statusUploadPermission>
<eventUploadPermission>
  <!--opt, object, event uploading permission, desc:the uploading protocol of security control panel event is CID (event identifier: cidEvent). For
definition values of CID, refer to the CID nodes on PMP platform-->
  <panicAlarm>
    <!--opt, bool, whether the user has the permission to receive emergency alarm event, desc:emergency alarm event is triggered by card, keyfob, or
keypad, and can be triggered in zones. The corresponding CID event codes are: [1120,3120]-->true
  </panicAlarm>
  <medicalAlarm>
    <!--opt, bool, whether the user has the permission to receive medical alarm event, desc:medical alarm event is triggered via card, keyfob, or keypad
by a person. The corresponding CID event codes are: [1100,3100]-->true
  </medicalAlarm>
  <fireAlarm>
    <!--opt, bool, whether the user has the permission to receive fire alarm event, desc:fire alarm event is uploaded when a fire occurs. The
corresponding CID event codes are: [1111,3111,1114,3114]-->true
  </fireAlarm>
  <smartAlarm>
    <!--opt, bool, whether the user has the permission to receive intelligent alarm event, desc:intelligent alarm events include intrusion detection
alarm, line crossing alarm, fire source alarm, temperature alarm, motion detection alarm, and region entrance/exiting alarm. The corresponding CID event
codes are: [1131,3131,1131,3131,1112,3112,1158,3158,1159,3159,1158,3158,1159,3159,1131,3131,1134]-->true
  </smartAlarm>
  <panelTamper>
    <!--opt, bool, whether the user has the permission to receive security control panel tampering event, desc:the security control panel tampering
event refers to the alarm event triggered when the security control panel is being tampered. The corresponding CID event codes are: [1137,3137]-->true
  </panelTamper>
  <externalDeviceTamper>
    <!--opt, bool, whether the user has the permission to receive peripheral tampering event, desc:the peripherals include extension module, wireless
repeater, wireless peripheral, and the peripheral tampering event refers to the alarm triggered when the linked peripheral is being tampered. The
corresponding CID event codes are: [1144,3144,1144,3144,1144,3144,1144,3144]-->true
  </externalDeviceTamper>
  <zoneAlarm>
    <!--opt, bool, whether the user has the permission to receive zone alarm event and zone tampering event, desc:the zone alarm events include instant
alarm, time out alarm, burglary alarm, 24-hour alarm, delay alarm, device moving alarm, detector blocking alarm, zone detector tampering alarm, water leak
alarm. The corresponding CID event codes are: [1120,3120,1100,3100,1111,3111,1114,3114,1162,3162]-->true
  </zoneAlarm>
  <panelOperation>
    <!--opt, bool, whether the user has the permission to receive security control panel's operation event, desc:the security control panel's operation
events refer to the events triggered when some operations are performed on the security control panel, including disarming, away/stay/auto/key zone
arming/disarming, clearing alarm, pacing, forward/backward programming, control panel reset, entering incorrect password. The corresponding CID event codes
are: [1401,3401,3441,1403,3403,1409,3409,1121,1406,1452,1455,1454,3250,1865,1607,3607,1627,1628,1305,1461]-->true
  </panelOperation>
</eventUploadPermission>
<FDLibPermissionList>
  <!--ro, opt, object, permission list of face picture library control-->
  <FDLibPermission>
    <!--ro, opt, object, permission of face picture library control-->
    <FDID>
      <!--ro, req, string, face picture library ID, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
    </FDID>
    <FDName>
      <!--ro, opt, string, face picture library name, range:[1,64], desc:/ISAPI/Intelligent/FDLib-->test
    </FDName>
  </FDLibPermission>

```

```

<FDLibControl>
  <!--ro, req, bool, whether to enable the permission of face picture library control, desc:permission of face picture library control includes
adding/deleting/editing/getting the information in the library-->true
</FDLibControl>
</FDLibPermission>
</FDLibPermissionList>
<picSearch>
  <!--ro, opt, bool, whether to enable the permission of picture search, desc:true-enabled, false-disabled. See details in /ISAPI/ContentMgmt/search--
>true
</picSearch>
<pircamVideoAndPicture>
  <!--opt, bool, whether to enable the permission to receive pircam camera video or picture event, desc:when an alarm is triggered, if this permission
is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or picture. There
is no corresponding CID event code-->true
</pircamVideoAndPicture>
<IPCVideoAndPicture>
  <!--opt, bool, whether to enable the permission to receive network camera video or picture event, desc:when an alarm is triggered, if this permission
is not enabled, the user can only receive message instead of video or picture; if this permission is enabled, the user can receive video or picture. There
is no corresponding CID event code-->true
</IPCVideoAndPicture>
<internalParametersAdjustment>
  <!--opt, bool-->true
</internalParametersAdjustment>
<subSystemMode>
  <!--opt, enum, subType:string-->assigned
</subSystemMode>
</remotePermission>
<passwordValidity>
  <!--opt, int, password validity period, range:[0,365], desc:unit: day; value range: [0, 365]; only the administrator can edit-->1
</passwordValidity>
</UserPermission>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode
4	Invalid Operation	privacyProtectionFunctionConflictsWithAiByNvrDisableAiByNvrOfTheChannel	0x4000A5I

12.1.8.4 Get the configuration capability of a specific user

Request URL

GET /ISAPI/Security/users/<indexID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	user (index) ID (it indicates the single ID in the array)

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user, attr:version{req, string, protocolVersion}-->
  <id min="1" max="200">
    <!--ro, req, int, user parameter No., range:[1,200], attr:min{req, int},max{req, int}-->1
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable user, attr:opt{req, string}-->true
  </enabled>
  <userName min="1" max="32">
    <!--ro, req, string, user name, range:[1,32], attr:min{req, int},max{req, int}-->test
  </userName>
  <userNickName>
    <!--ro, opt, string, user's nickname, range:[1,32], desc:userNickName is the user's nickname on the Hik-Connect client, and the user can modify it. The web client can get userNickName via the device-->test
  </userNickName>
  <phoneNum min="1" max="11">
    <!--ro, opt, string, phone number, range:[1,11], attr:min{req, int},max{req, int}, desc:it can replace userName for Login. The sensitive information should be encrypted-->test
  </phoneNum>
  <emailAddress min="1" max="32">
    <!--ro, opt, string, email address, range:[1,32], attr:min{req, int},max{req, int}, desc:it can replace userName for Login. The sensitive information should be encrypted-->test
  </emailAddress>
  <oldPassword min="1" max="16">
    <!--ro, opt, string, old password before the current Logging-in, range:[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-->null
  </oldPassword>
  <password min="1" max="16">
    <!--ro, req, string, current password, range:[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-->null
  </password>
  <duressPasswordMode opt="auto,custom">
    <!--ro, opt, string, range:[1,16], attr:opt{req, string}-->null
  </duressPasswordMode>
  <duressPassword min="1" max="16">
    <!--ro, opt, string, duress password, which is used for special password which is under duress, the sensitive information should be encrypted, range:[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-->null
  </duressPassword>
  <keypadPassword min="1" max="16">
    <!--ro, opt, string, keypad password, range:[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-->null
  </keypadPassword>
  <singleKeypadEnable opt="true,false">
    <!--ro, opt, bool, whether to enable the one-time keypad password, attr:opt{req, string}-->true
  </singleKeypadEnable>
  <invalidKeypadPassword opt="true,false">
    <!--ro, opt, bool, whether the current keypad password is invalid, attr:opt{req, string}, desc:it is valid after setting singleKeypadEnable-->true
  </invalidKeypadPassword>
  <userOperateType opt="1,2,3">
    <!--ro, opt, enum, user type, subType:int, attr:opt{req, string}, desc:operator accounts shared from the cloud can log in to the cloud, the Web Client, the Local device, operator accounts created on the cloud and LAN can only log in to Local device 1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
  </userOperateType>
  <bondIpAddressList>
    <!--ro, opt, array, List of socket IP address that bound to the user, subType:object-->
    <bondIpAddress>
      <!--ro, opt, object, socket IP address that bound to the user-->
      <id>
        <!--ro, req, int, bounded IP address No., range:[1,16]-->1
      </id>
      <ipAddress>
        <!--ro, opt, string, bounded IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, bounded IPv6 address, range:[1,32]-->test
      </ipv6Address>
    </bondIpAddress>
  </bondIpAddressList>
  <bondMacAddressList>
    <!--ro, opt, array, List of MAC address that bound to the user, subType:object-->
    <bondMacAddress>
      <!--ro, opt, object, MAC address that bound to the user-->
      <id>
        <!--ro, req, int, bounded MAC address No., range:[1,16]-->1
      </id>
      <macAddress>
        <!--ro, opt, string, bounded MAC address, range:[1,32]-->test
      </macAddress>
    </bondMacAddress>
  </bondMacAddressList>
  <userLevel opt="Administrator,Operator,Viewer,installer,manufacturer,single,installerAdmin,installEmployee,Employee,DepartmentManager">
    <!--ro, opt, enum, user type, subType:string, attr:opt{req, string}, desc:"Administrator", "Operator", "Viewer" (viewer, which is not applied on devices currently), "installer", "manufacturer", "single" (single user, which is not applied on devices currently), "installerAdmin" (installation admin), "installEmployee" (installation employee), "Employee" (for attendance), "DepartmentManager" (department manager, for attendance)-->Administrator
  </userLevel>
  <attribute>
    <!--ro, opt, object, user's special attributes-->
    <inherent>
      <!--ro, opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false / the node is invalid (the user can be deleted)-->true
    </inherent>
  </attribute>
  <belongUserId>

```

```

<!--ro, opt, int, upper-level user ID, range:[0,16], desc:0 (no upper-level user)-->1
</belongUserId>
<belongOrgNo>
<!--ro, opt, int, organization No., range:[1,16]-->1
</belongOrgNo>
<cardList>
<!--ro, opt, object, List of cards that belong to the user-->
<card>
<!--ro, opt, object, card-->
<id>
<!--ro, opt, int, No., range:[1,16]-->1
</id>
<name>
<!--ro, opt, string, name, range:[1,32]-->test
</name>
</card>
</cardList>
<remoteCtrlList>
<!--ro, opt, array, List of keyfobs that belong to the user, subType:object-->
<remoteCtrl>
<!--ro, opt, object, keyfob-->
<id>
<!--ro, opt, int, No., range:[1,16]-->1
</id>
<name>
<!--ro, opt, string, name, range:[1,32]-->test
</name>
</remoteCtrl>
</remoteCtrlList>
<userNo min="501" max="null">
<!--ro, opt, int, displayed No. corresponding to the user ID, attr:min{req, int},max{req, int},
desc:the No. is used on the App and the Web Client and reporting alarms to ARC. It is valid to all user types. The format of the No. is 500 + id adding
userNo for supporting distinguishing users from user names on some ARC
scenario: department. residents in one department shall share a security control panel and ARC service. Only one resident shall be the administrator. for
this scenario, after adding residents as users on the ARC, user ID of the server and App users shall be linked, for contacting the corresponding user when
problems occur. All users shall know their numbers on the ARC after creating their App accounts-->1
</userNo>
<adminType opt="Cloud,LAN">
<!--ro, opt, enum, admin type, subType:string, attr:opt{req, string}, desc:for hybrid security control panel, it is used for distinguishing whether the
administrator account is created on the cloud or in the Local area network. It is displayed both on the App and the Web Client. "Cloud" (user account
created on the cloud), "LAN" (user account created on LAN)-->Cloud
</adminType>
<installerType opt="Cloud,LAN">
<!--ro, opt, enum, installer type, subType:string, attr:opt{req, string}, desc:for hybrid security control panel, it is used for distinguishing whether
the installer account is created on the cloud or in the Local area network. It is displayed both on the App and the Web Client. "Cloud" (user account
created on the cloud), "LAN" (user account created on LAN)-->Cloud
</installerType>
<PasswordPeriodPrompt>
<!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
expired shall be prompted during logging-in. password changing is not required in using-->
<passwordPeriodType opt="permanent,byDate" def="permanent">
<!--ro, req, enum, notification type of the password validity period, subType:string, attr:opt{req, string},def{req, string}, desc:"permanent" (take
effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take
effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
</passwordPeriodType>
<passwordPeriodbyDate min="1" max="1200" def="90">
<!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
{$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}, attr:min{req, int},max{req, int},def{req, int}-->90
</passwordPeriodbyDate>
</PasswordPeriodPrompt>
<userActivationStatus opt="true,false">
<!--ro, opt, bool, user activation status, attr:opt{req, string}, desc:true (activated; default value), false (inactivated)-->true
</userActivationStatus>
<invalidDuressPassword opt="true,false">
<!--ro, opt, bool, whether current duress password is invalid, attr:opt{req, string}-->true
</invalidDuressPassword>
<userID min="1" max="64">
<!--ro, opt, string, user No., range:[1,64], attr:min{req, int},max{req, int}, desc:UUID (unique ID of the user)-->test
</userID>
<passwordType opt="simple,complex">
<!--ro, opt, enum, password type, subType:string, attr:opt{req, string}, desc:simple, "complex"-->simple
</passwordType>
<temporaryLocalOperator>
<!--ro, opt, object, information of temporary Local operator, desc:the temporary Local operator can only log in on Local keypad-->
<enabled opt="true,false">
<!--ro, opt, bool, whether to enable the permission of temporary Local operator, attr:opt{req, string}-->true
</enabled>
<scheduledOperationList size="1">
<!--ro, opt, array, validity period List of the permission of temporary Local operator, subType:object, attr:size{req, int}-->
<scheduledOperation>
<!--ro, opt, object, validity period of the permission of temporary Local operator-->
<beginTime>
<!--ro, req, datetime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
</beginTime>
<endTime>
<!--ro, req, datetime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
</endTime>
</scheduledOperation>
</scheduledOperationList>
<maxValidTimeSpan min="1" max="720">
<!--ro, opt, int, maximum valid span for one time, range:[1,720], step:1, unit:h, attr:min{req, int},max{req, int}-->1
</maxValidTimeSpan>
</temporaryLocalOperator>
<IssuerPIN on="true,false">

```



```

<!--ro, opt, bool, whether it has a bound DN, attr:opt{req, string}, desc:a bound DN refers to a bound server directory.The process of the client
sending a request to establish session with the LDAP server is called binding.When binding, the client needs to specify the access account so as to access
the directory information on the server.The user searches for contents according to the bound directory.This node will be used only when logging in to the
device (GB35114) via pin (UKey)-->true
</IssuerDN>
<isSupportModifyAdminUserName>
<!--ro, opt, bool, support modifying the user name of administrator, desc:if this node is not returned or its value is false, the function is not
supported. This node will be returned when getting the capability of configuring the administrator information parameters-->true
</isSupportModifyAdminUserName>
<userGroupID min="0" max="3" def="0">
<!--ro, opt, int, user group ID, range:[0,3], attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist or its value is 0, it
indicates that there are no groups. As for the security inspection, the upper-level software supports dividing users into several groups and delivering the
group IDs to them, which is managed by the device.-->0
</userGroupID>
<isSupportWeakPassword opt="true,false">
<!--ro, opt, bool, whether it supports configuring the weak password, attr:opt{req, string}, desc:according to the requirements of the security red
line, weak password is not allowed to configure in the new version-->true
</isSupportWeakPassword>
<isSupportKeyWord opt="true,false">
<!--ro, opt, bool, whether it supports the key word in the password, attr:opt{req, string}, desc:true (support the key word in the password), false (not
support the key word in the password)-->true
</isSupportKeyWord>
<facePicture>
<!--ro, opt, object-->
<filePathType opt="URL,multipart">
<!--ro, req, enum, subType:string, attr:opt{req, string}-->multipart
</filePathType>
<filePath min="0" max="262000">
<!--ro, opt, string, range:[0,262000], attr:min{req, int},max{req, int}-->facePicture
</filePath>
</facePicture>
</User>

```

12.1.8.5 Get information about a single user

Request URL

GET /ISAPI/Security/users/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, user parameter No.-->1
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable user-->true
  </enabled>
  <userName>
    <!--ro, req, string, user name, range:[1,32]-->test
  </userName>
  <phoneNum>
    <!--ro, opt, string, phone number, range:[1,11], desc:the sensitive information should be encrypted-->test
  </phoneNum>
  <emailAddress>
    <!--ro, opt, string, Email Address, range:[1,32], desc:the sensitive information should be encrypted-->test
  </emailAddress>
  <duressPassword>
    <!--ro, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
  </duressPassword>
  <keypadPassword>
    <!--ro, opt, string, Keypad Password, range:[1,16], desc:the sensitive information should be encrypted. entering password is required when the security
control panel is armed or disarmed via keypad-->test
  </keypadPassword>
  <singleKeypadEnable>
    <!--ro, opt, bool, whether to enable the keypad password for one time-->true
  </singleKeypadEnable>
  <invalidKeypadPassword>
    <!--ro, opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
  </invalidKeypadPassword>
  <userOperateType>
    <!--ro, opt, enum, user type, subType:int,
desc:1 (if userOperateType is 1 or none, password is valid. if method is POST, password is required. otherwise this node is optional)
2 (if userOperateType is 2, keypadPassword is valid. if method is POST, keypadPassword is required. otherwise this node is optional)
-->1
  </userOperateType>
</User>

```



```

4 (if userOperateType is 2, keypaarasswora is valid. if metnoa is POST, keypaarasswora is required. otherwise this node is optional)
3 (if userOperateType is 3, both passwordkeypad and Password is valid. if method is POST, password and keypadPassword is required. otherwise this node is optional) 1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
</userOperateType>
<bondIpAddressList>
  <!--ro, opt, object, List of socket IP address that bound to the user-->
  <bondIpAddress>
    <!--ro, opt, object, socket IP address that bound to the user-->
    <id>
      <!--ro, req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
    </id>
    <ipAddress>
      <!--ro, opt, string, bounded IPv4 address, range:[1,32]-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, bounded IPv6 address, range:[1,128]-->test
    </ipv6Address>
  </bondIpAddress>
</bondIpAddressList>
<bondMacAddressList>
  <!--ro, opt, object, List of mac address that bound to the user-->
  <bondMacAddress>
    <!--ro, opt, object, mac address that bound to the user-->
    <id>
      <!--ro, req, int, bounded mac address No., range:[1,16], desc:It starts from 1-->1
    </id>
    <macAddress>
      <!--ro, opt, string, bounded mac address, range:[1,32]-->test
    </macAddress>
  </bondMacAddress>
</bondMacAddressList>
<userLevel>
  <!--ro, opt, enum, user Level, subType:string, desc:"Administrator", "Operator", "Viewer" (viewer, which is not applied on devices currently),
"installer", "manufacturer", "single" (single user, which is not applied on devices currently), "installerAdmin" (installation admin), "installEmployee"
(installation employee), "Employee" (for attendance), "DepartmentManager" (department manager, for attendance)-->Administrator
</userLevel>
<attribute>
  <!--ro, opt, object, user's special attributes-->
  <inherent>
    <!--ro, opt, bool, whether it can be deleted, desc:true: the user cannot be deleted. false / the node is invalid: the user can be deleted-->true
  </inherent>
</attribute>
<cardList>
  <!--ro, opt, object, List of cards that belong to the user-->
  <card>
    <!--ro, opt, object, card-->
    <id>
      <!--ro, opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
  </card>
</cardList>
<remoteCtrlList>
  <!--ro, opt, object, List of keyfobs that belong to the user-->
  <remoteCtrl>
    <!--ro, opt, object, keyfob-->
    <id>
      <!--ro, opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
  </remoteCtrl>
</remoteCtrlList>
<userNo>
  <!--ro, opt, int, displayed No. corresponding to the user ID, desc:the No. is used on the App and the Web Client and reporting alarms to ARC. it is
valid to all user types. the format of the No. is 500 + id-->501
</userNo>
<adminType>
  <!--ro, opt, enum, admin type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the administrator account
is created on the cloud or in the Local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN"
(user account created on LAN)-->Cloud
</adminType>
<installerType>
  <!--ro, opt, enum, installer type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the installer account
is created on the cloud or in the Local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN"
(user account created on LAN)-->Cloud
</installerType>
<PasswordPeriodPrompt>
  <!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
expired shall be prompted during logging-in, but the expired password does not affect the usage-->
  <passwordPeriodType>
    <!--ro, req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take
effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password
expired shall be prompted in time, but password changing is not required in using-->permanent
  </passwordPeriodType>
  <passwordPeriodbyDate>
    <!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
{$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
  </passwordPeriodbyDate>
</PasswordPeriodPrompt>
<userActivationStatus>

```

```

    <!--ro, opt, bool, user activation status, desc:true (activated; default value), false (inactivated)-->true
</userActivationStatus>
<invalidDuressPassword>
    <!--ro, opt, bool, whether current duress password is invalid, desc:if the duress password is not configured, the value is true-->true
</invalidDuressPassword>
<passwordType>
    <!--ro, opt, enum, subType:string-->simple
</passwordType>
<temporaryLocalOperator>
    <!--ro, opt, object-->
    <enabled>
        <!--ro, req, bool-->true
    </enabled>
    <scheduledOperationList>
        <!--ro, opt, array, subType:object, dep:or,{$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->
        <scheduledOperation>
            <!--ro, opt, object-->
            <beginTime>
                <!--ro, req, datetime-->1970-01-01T00:00:00+08:00
            </beginTime>
            <endTime>
                <!--ro, req, datetime-->1970-01-01T00:00:00+08:00
            </endTime>
        </scheduledOperation>
    </scheduledOperationList>
    <validPeriodStatus>
        <!--ro, opt, enum, subType:string-->notReached
    </validPeriodStatus>
    <remainingValidTime>
        <!--ro, opt, int, range:[0,720], step:1, unit:h-->1
    </remainingValidTime>
</temporaryLocalOperator>
<IssuerDN>
    <!--ro, opt, bool-->true
</IssuerDN>
</User>

```

12.1.8.6 Set information for a single user

Request URL

PUT /ISAPI/Security/users/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>
<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
    <!--req, object, user, attr:version{req, string, protocolVersion}-->
    <id>
        <!--req, int, user parameter No., range:[1,16], desc:supported range: [1,16]-->1
    </id>
    <enabled>
        <!--opt, bool, whether to enable user-->true
    </enabled>
    <userName>
        <!--req, string, user name, range:[1,32]-->test
    </userName>
    <phoneNum>
        <!--wo, opt, string, phone number, range:[1,11], desc:the sensitive information should be encrypted-->test
    </phoneNum>
    <emailAddress>
        <!--wo, opt, string, email address, range:[1,32], desc:the sensitive information should be encrypted-->test
    </emailAddress>
    <password>
        <!--wo, opt, string, password, range:[1,16], desc:the sensitive information should be encrypted-->test
    </password>
    <duressPasswordMode>
        <!--wo, opt, enum, subType:object-->auto
    </duressPasswordMode>
    <duressPassword>
        <!--wo, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
    </duressPassword>
    <keypadPassword>
        <!--wo, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the security control panel is armed or disarmed via keypad-->test
    </keypadPassword>
    <invalidkeypadPassword>
        <!--opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
    </invalidkeypadPassword>
    <bondIpAddressList>

```

```

<!--opt, object, List of socket IP address that bound to the user-->
<bondIpAddress>
  <!--opt, object, socket IP address that bound to the user-->
  <id>
    <!--req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
  </id>
  <ipAddress>
    <!--opt, string, bounded IPv4 address, range:[1,32]-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, bounded IPv6 address, range:[1,32]-->test
  </ipv6Address>
</bondIpAddress>
</bondIpAddressList>
<bondMacAddressList>
  <!--opt, object, List of MAC address that bound to the user-->
  <bondMacAddress>
    <!--opt, object, MAC address that bound to the user-->
    <id>
      <!--req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
    </id>
    <macAddress>
      <!--opt, string, bounded MAC address, range:[1,32]-->test
    </macAddress>
  </bondMacAddress>
</bondMacAddressList>
<attribute>
  <!--opt, object, user's special attributes-->
  <inherent>
    <!--opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false / the node is invalid (the user can be deleted)-->true
  </inherent>
</attribute>
<loginPassword>
  <!--req, string, confirm password, range:[1,16], desc:confirming password is required when adding and editing user information. Encrypting the sensitive information is required when the confirm password is transmitted via HTTP-->King726014
</loginPassword>
<cardList>
  <!--opt, object, List of cards that belong to the user-->
  <card>
    <!--opt, object, card-->
    <id>
      <!--opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--opt, string, name, range:[1,32]-->test
    </name>
  </card>
</cardList>
<remoteCtrlList>
  <!--opt, object, List of keyfobs that belong to the user-->
  <remoteCtrl>
    <!--opt, object, keyfob-->
    <id>
      <!--opt, int, No., range:[1,16]-->1
    </id>
    <name>
      <!--opt, string, name, range:[1,32]-->test
    </name>
  </remoteCtrl>
</remoteCtrlList>
<PasswordPeriodPrompt>
  <!--opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password expired shall be prompted during logging-in, but the expired password does not affect the usage-->
  <passwordPeriodType>
    <!--req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
  </passwordPeriodType>
  <passwordPeriodbyDate>
    <!--opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
    {$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
  </passwordPeriodbyDate>
</PasswordPeriodPrompt>
<temporaryLocalOperator>
  <!--opt, object, information of temporary Local operator, desc:the temporary Local operator can only Log in on Local keypad-->
  <scheduledOperationList>
    <!--opt, array, validity period List of the permission of temporary Local operator, subType:object, dep:and,
    {$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->
    <scheduledOperation>
      <!--opt, object, validity period of the permission of temporary Local operator-->
      <beginTime>
        <!--req, datetime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
      </beginTime>
      <endTime>
        <!--req, datetime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
      </endTime>
    </scheduledOperation>
  </scheduledOperationList>
</temporaryLocalOperator>
<IssuerDN>
  <!--opt, bool, whether it has a bound DN, desc:a bound DN refers to a bound server directory. The process of the client sending a request to establish session with the LDAP server is called binding. When binding, the client needs to specify the access account so as to access the directory information on the server. The user searches for contents according to the bound directory. N/A-->true
  </IssuerDN>

```

```

</IssuerDN>
<userGroupID>
  <!--opt, int, user group ID, range:[0,3], desc:if this node does not exist or its value is 0, it indicates that there are no groups. As for the security
inspection, the upper-level software supports dividing users into several groups and delivering the group IDs to them, which is managed by the device.-->0
</userGroupID>
<facePicture>
  <!--opt, object-->
  <filePathType>
    <!--req, enum, subType:string-->multipart
  </filePathType>
  <filePath>
    <!--opt, string, range:[0,262000]-->test
  </filePath>
</facePicture>
</User>

```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
User	[Message content]	application/xml	--	--	--
facePicture	[Binary picture data]	image/jpeg	facePicture	facePicture.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```

--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value

```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, used for fast debugging-->badXmlFormat
  </description>
  <lockStatus>
    <!--ro, opt, enum, lock status, subType:string, desc:"Lock", "unlock". It is returned when verifying LoginPassword failed-->lock
  </lockStatus>
  <resLockTime>
    <!--ro, opt, int, range:[0,1800], unit:s, dep:or,{$.ResponseStatus.LockStatus,eq,Lock}-->1
  </resLockTime>
  <retryTimes>
    <!--ro, opt, int, remaining attempts, range:[0,6], desc:it is returned when verifying LoginPassword failed-->1
  </retryTimes>
  <lockType>
    <!--ro, opt, enum, subType:string, dep:or,{$.ResponseStatus.LockStatus,eq,Lock}-->lockIP
  </lockType>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	exDevNotExist	0x40008016	--

12.1.8.7 Get information about all users

Request URL

GET /ISAPI/Security/users

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UserList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, user list, subType:object, attr:version{req, string, protocolVersion}-->
  <User>
    <!--ro, opt, object, user-->
    <id>
      <!--ro, req, int, user parameter No., range:[1,200]-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable user-->true
    </enabled>
    <userName>
      <!--ro, req, string, user name, range:[1,32]-->test
    </userName>
    <userNickName>
      <!--ro, opt, string, user's nickname, range:[1,32], desc:userNickName is the user's nickname on the Hik-Connect client, and the user can modify it. The web client can get userNickName via the device-->test
    </userNickName>
    <phoneNum>
      <!--ro, opt, string, phone number, range:[1,13], desc:the sensitive information should be encrypted-->test
    </phoneNum>
    <emailAddress>
      <!--ro, opt, string, email address, range:[1,32], desc:the sensitive information should be encrypted-->test
    </emailAddress>
    <duressPassword>
      <!--ro, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
    </duressPassword>
    <keypadPassword>
```

```

    <!--ro, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the
security control panel is armed or disarmed via keypad-->test
  </keypadPassword>
  <singleKeypadEnable>
    <!--ro, opt, bool, whether to enable single keypad operation-->true
  </singleKeypadEnable>
  <invalidKeypadPassword>
    <!--ro, opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
  </invalidKeypadPassword>
  <userOperateType>
    <!--ro, opt, enum, user operation type, subType:int,
desc:1 (if userOperateType is 1 or none, password is valid. if method is POST, password is required. otherwise this node is optional)
2 (if userOperateType is 2, keypadPassword is valid. if method is POST, keypadPassword is required. otherwise this node is optional)
3 (if userOperateType is 3, both passwordKeypad and Password is valid. if method is POST, password and keypadPassword is required. otherwise this node is
optional) 1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
  </userOperateType>
  <bondIpAddressList>
    <!--ro, opt, object, List of socket IP address that bound to the user-->
    <bondIpAddress>
      <!--ro, opt, object, socket IP address that bound to the user-->
      <id>
        <!--ro, req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
      </id>
      <ipAddress>
        <!--ro, opt, string, bounded IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, bounded IPv6 address, range:[1,128]-->test
      </ipv6Address>
    </bondIpAddress>
  </bondIpAddressList>
  <bondMacAddressList>
    <!--ro, opt, object, List of MAC address that bound to the user-->
    <bondMacAddress>
      <!--ro, opt, object, MAC address that bound to the user-->
      <id>
        <!--ro, req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
      </id>
      <macAddress>
        <!--ro, opt, string, bounded MAC address, range:[1,32]-->test
      </macAddress>
    </bondMacAddress>
  </bondMacAddressList>
  <userLevel>
    <!--ro, opt, enum, user level, subType:string, desc:"Administrator", "Operator", "Viewer", "installer", "manufacturer", "single" (single user),
"installerAdmin" (installation admin), "installEmployee" (installation employee), "Employee", "DepartmentManager"-->Administrator
  </userLevel>
  <attribute>
    <!--ro, opt, object, user's special attributes-->
    <inherent>
      <!--ro, opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false / the node is invalid (the user can be deleted)-->true
    </inherent>
  </attribute>
  <cardList>
    <!--ro, opt, object, List of cards that belong to the user-->
    <card>
      <!--ro, opt, object, card-->
      <id>
        <!--ro, opt, int, No., range:[1,16]-->1
      </id>
      <name>
        <!--ro, opt, string, name, range:[1,32]-->test
      </name>
    </card>
  </cardList>
  <remoteCtrlList>
    <!--ro, opt, object, List of keyfobs that belong to the user-->
    <remoteCtrl>
      <!--ro, opt, object, keyfob-->
      <id>
        <!--ro, opt, int, No., range:[1,16]-->1
      </id>
      <name>
        <!--ro, opt, string, name, range:[1,32]-->test
      </name>
    </remoteCtrl>
  </remoteCtrlList>
  <userNo>
    <!--ro, opt, int, displayed No. corresponding to the user ID, desc:the No. is used on the App and the Web Client and reporting alarms to ARC. it is
valid to all user types. the format of the No. is 500 + id-->501
  </userNo>
  <adminType>
    <!--ro, opt, enum, admin type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the administrator account
is created on the cloud or in the local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN"
(user account created on LAN)-->Cloud
  </adminType>
  <installerType>
    <!--ro, opt, enum, installer type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the installer account
is created on the cloud or in the local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN"
(user account created on LAN)-->Cloud
  </installerType>
  <PasswordPeriodPrompt>
    <!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
expired shall be prompted during logging-in. but the expired password does not affect the usage-->
  </PasswordPeriodPrompt>

```

```

expected shall be prompted during logging in, but the expected password does not affect the usage.
    <passwordPeriodType>
        <!--ro, req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take
effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password
expired shall be prompted in time, but password changing is not required in using-->permanent
    </passwordPeriodType>
    <passwordPeriodbyDate>
        <!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
{$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate)-->90
    </passwordPeriodbyDate>
</PasswordPeriodPrompt>
<userActivationStatus>
    <!--ro, opt, bool, user activation status, desc:true (activated; default value), false (inactivated)-->true
</userActivationStatus>
<invalidDuressPassword>
    <!--ro, opt, bool, whether current duress password is invalid, desc:if the duress password is not configured, the value is true-->true
</invalidDuressPassword>
<userID>
    <!--ro, opt, string, user No., range:[1,64], desc:UUID (unique ID of the user)-->test
</userID>
<passwordType>
    <!--ro, opt, enum, password type, subType:string,
desc:1. Attendance check devices support setting simple password and complex password. The password is used for device activation and login.
2. The password should contain 6 digits. "simple", "complex"-->simple
</passwordType>
<temporaryLocalOperator>
    <!--ro, opt, object, information of temporary Local operator, desc:the temporary Local operator can only log in on local keypad-->
    <enabled>
        <!--ro, req, bool, whether to enable the permission of temporary Local operator-->true
    </enabled>
    <scheduledOperationList>
        <!--ro, opt, array, validity period list of the permission of temporary Local operator, subType:object, dep:or,
{$.UserList.User.temporaryLocalOperator.enabled,eq,true)-->
        <scheduledOperation>
            <!--ro, opt, object, validity period of the permission of temporary Local operator-->
            <beginTime>
                <!--ro, req, datetime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
            </beginTime>
            <endTime>
                <!--ro, req, datetime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
            </endTime>
        </scheduledOperation>
    </scheduledOperationList>
    <validPeriodStatus>
        <!--ro, opt, enum, validity period status, subType:string, desc:"notReached" (the validity period has not started), "available" (within the validity
period), "expired"-->notReached
    </validPeriodStatus>
    <remainingValidTime>
        <!--ro, opt, int, remaining validity period, range:[0,720], step:1, unit:h-->1
    </remainingValidTime>
</temporaryLocalOperator>
<userGroupID>
    <!--ro, opt, int, user group ID, range:[0,1], desc:if this node does not exist or its value is 0, it indicates that there are no groups. As for the
security inspection, the upper-level software supports dividing users into several groups and delivering the group IDs to them, which is managed by the
device.-->0
    </userGroupID>
</User>
</UserList>

```

12.1.8.8 Set information about all users

Request URL

PUT /ISAPI/Security/users

Query Parameter

None

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<UserList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user list, attr:version{req, string, protocolVersion}-->
  <User>
    <!--opt, object, user-->
    <id>
      <!--req, int, user parameter No., range:[1,64]-->1
    </id>
    <enabled>
      <!--opt, bool, whether to enable user-->true
    </enabled>
    <userName>
      <!--req, string, user name, range:[1,32]-->test
    </userName>
    <password>
      <!--wo, req, string, password, range:[1,16], desc:the sensitive information should be encrypted-->test
    </password>
    <keypadPassword>
      <!--wo, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the security control panel is armed or disarmed via keypad-->test
    </keypadPassword>
    <userOperateType>
      <!--opt, enum, user type, subType:int, desc:1 (network user), 2 (keypad user), 3 (network and keypad user). When userOperateType is 1 or none, the node password is valid and it is required when the method is POST. When userOperateType is 2, the node keypadPassword is valid and it is required when the method is POST. When userOperateType is 3, both nodes password and keypadPassword are valid and they are required when the method is POST.-->1
    </userOperateType>
    <bondIpAddressList>
      <!--opt, object, List of socket IP address that bound to the user-->
      <bondIpAddress>
        <!--opt, object, socket IP address that bound to the user-->
        <id>
          <!--req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <ipAddress>
          <!--opt, string, bounded IPv4 address, range:[1,32]-->test
        </ipAddress>
        <ipv6Address>
          <!--opt, string, bounded IPv6 address, range:[1,32]-->test
        </ipv6Address>
      </bondIpAddress>
    </bondIpAddressList>
    <bondMacAddressList>
      <!--opt, object, List of MAC address that bound to the user-->
      <bondMacAddress>
        <!--opt, object, MAC address that bound to the user-->
        <id>
          <!--req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <macAddress>
          <!--opt, string, bounded MAC address, range:[1,32]-->test
        </macAddress>
      </bondMacAddress>
    </bondMacAddressList>
    <userLevel>
      <!--opt, enum, user level, subType:string, desc:"Administrator", "Operator", "Viewer", "installer", "manufacturer", "single" (single user), "installerAdmin" (installation admin), "installEmployee" (installation employee)-->Administrator
    </userLevel>
    <attribute>
      <!--opt, object, user's special attributes-->
      <inherent>
        <!--opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false/the node is invalid (the user can be deleted)-->true
      </inherent>
    </attribute>
    <loginPassword>
      <!--req, string, confirm password, range:[1,16], desc:confirming password is required when adding and editing user information. Encrypting the sensitive information is required when the confirm password is transmitted via HTTP-->King726014
    </loginPassword>
    <PasswordPeriodPrompt>
      <!--opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password expired shall be prompted during logging-in. password changing is not required in using-->
      <passwordPeriodType>
        <!--req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
      </passwordPeriodType>
      <passwordPeriodbyDate>
        <!--opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and, {$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
      </passwordPeriodbyDate>
    </PasswordPeriodPrompt>
    <userID>
      <!--opt, string, user No., range:[1,64], desc:UUID (unique ID of the user)-->test
    </userID>
  </User>
</UserList>

```

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <lockStatus>
    <!--ro, opt, enum, lock status, subType:string, desc:"Lock", "unlock". It is returned when verifying LoginPassword failed-->lock
  </lockStatus>
  <retryTimes>
    <!--ro, opt, int, remaining attempts, range:[0,6], desc:it is returned when verifying LoginPassword failed-->1
  </retryTimes>
</ResponseStatus>
```

12.2 Network Configuration

12.2.1 HTTP Listening Management

12.2.1.1 Delete receiving server(s)

Request URL

DELETE /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

12.2.1.2 Delete a HTTP listening server

Request URL

DELETE /ISAPI/Event/notification/httpHosts/<hostID>

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	Add a single alarm host through POST /ISAPI/Event/notification/httpHosts?security=&iv=. It returns the alarm host ID for success.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

12.2.1.3 Check whether the HTTP listening server is working normally

Request URL

POST /ISAPI/Event/notification/httpHosts/<hostID>/test

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, event subscription parameters, see details in the message of XML_SubscribeEvent, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, listening host ID-->test
  </id>
  <url>
    <!--req, string, URL-->test
  </url>
  <protocolType>
    <!--req, enum, protocol type, subType:string, desc:"HTTP,HTTPS,EHome". When the value of this node is EHome,this message can be used to configure the information (such as IP address and port No.) for the AMS (Alarm Management Server) of ISUP 5.0 to Listen on. For communication via ISUP 5.0 (MQTT),the listening parameters can be configured and obtained; For other communication methods (such as communication via HTTP,HTTPS,EZVIZ Open Platform,etc.),the listening parameters can only be obtained and cannot be configured-->HTTPS
  </protocolType>
  <parameterFormatType>
    <!--req, enum, alarm/event information format, subType:string, desc:"XML,JSON"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:"ipaddress, hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IP address, dep:and,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
```

```

</ipAddress>
<ipv6Address>
  <!--opt, string, IPv6 address, dep:and,{$.HostNameNotification.addressingFormatType,eq,ipaddress}-->test
</ipv6Address>
<portNo>
  <!--opt, int, port-->1
</portNo>
<userName>
  <!--opt, string, user name-->test
</userName>
<password>
  <!--opt, string, password-->test
</password>
<httpAuthenticationMethod>
  <!--req, enum, HTTP authentication method, subType:string, desc:"MD5digest,none"-->MD5digest
</httpAuthenticationMethod>
<ANPR>
  <!--opt, object-->
  <detectUploadPicturesType>
    <!--opt, enum, subType:string-->all
  </detectUploadPicturesType>
  <videoUploadEnabled>
    <!--opt, bool-->false
  </videoUploadEnabled>
</ANPR>
<Extensions>
  <!--opt, object-->
  <intervalBetweenEvents>
    <!--opt, int-->1
  </intervalBetweenEvents>
</Extensions>
<uploadImagesDataType>
  <!--opt, enum, subType:string-->URL
</uploadImagesDataType>
<httpBroken>
  <!--opt, bool-->true
</httpBroken>
<SubscribeEvent>
  <!--opt, object-->
  <heartbeat>
    <!--opt, int, heartbeat interval time, unit:s, desc:heartbeat interval time-->1
  </heartbeat>
  <eventMode>
    <!--req, enum, "all, List", subType:string, desc:"all,List"-->all
  </eventMode>
  <EventList>
    <!--opt, array, event list, it is valid only when eventMode is "List", subType:object-->
    <Event>
      <!--opt, object-->
      <type>
        <!--req, string, type, desc:type-->test
      </type>
      <minorAlarm>
        <!--opt, string, alarm sub type, desc:alarm sub type-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, exception sub type, desc:exception sub type-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, operation sub type, desc:operation sub type-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "LocalURL" (device Local URL), "cloudStorageURL" (cloud storage URL)-->binary
      </pictureURLType>
      <channels>
        <!--opt, string, channel No.-->test
      </channels>
    </Event>
  </EventList>
</channels>
  <!--opt, string, "1,2,3,4..."-->test
</channels>
<pictureURLType>
  <!--opt, enum, alarm picture format, subType:string, desc:alarm picture format-->binary
</pictureURLType>
<ChangedUploadSub>
  <!--opt, object, subscribe to messages-->
  <interval>
    <!--opt, int, the lifecycle of arm GUID, unit:s, desc:the lifecycle of arm GUID-->1
  </interval>
  <StatusSub>
    <!--opt, object, status-->
    <all>
      <!--opt, bool, whether to subscribe to all-->true
    </all>
    <channel>
      <!--opt, bool, subscribe to channel status, desc:reporting is not required if all is true-->true
    </channel>
  </StatusSub>
</hd>

```

```

        <!--opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
    </hd>
    <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
    </capability>
    </StatusSub>
    </ChangedUploadSub>
    </SubscribeEvent>
    </HttpHostNotification>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<HttpHostTestResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
    <!--ro, req, object, attr:version{req, string, protocolVersion}-->
    <errorDescription>
        <!--ro, req, enum, error description, subType:string, desc:error description-->ok
    </errorDescription>
</HttpHostTestResult>

```

12.2.1.4 Set the parameters of a listening host

Request URL

PUT /ISAPI/Event/notification/httpHosts/<hostID>

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
    <!--req, object, listening host, attr:version{req, string, protocolVersion}-->
    <id>
        <!--req, string, listening host ID, range:[1,10]-->test
    </id>
    <url>
        <!--req, string, URL-->test
    </url>
    <protocolType>
        <!--req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
    </protocolType>
    <parameterFormatType>
        <!--req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
    </parameterFormatType>
    <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
    </addressingFormatType>
    <hostName>
        <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
    </hostName>
    <ipAddress>
        <!--opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IP address-->test
    </ipAddress>
    <ipv6Address>
        <!--opt, string, IPv6 Address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IPv6 Address-->test
    </ipv6Address>
    <portNo>
        <!--opt, int, port number-->1
    </portNo>
    <userName>
        <!--opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </userName>
    <password>
        <!--opt, string, password, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </password>
    <httpAuthenticationMethod>
        <!--req, enum, authentication method, subType:string, desc:"MD5digest"(MD5), "none", "base64"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
        <!--opt, object, ANPR-->
        <detectioUploadPicturesType>
            <!--opt, enum, uploaded pictures type, subType:string, desc:"all", "LicensePlatePicture", "detectionPicture"(detected picture)-->all
        </detectioUploadPicturesType>
    </ANPR>
    <Extensions>
        <!--opt, object, range-->
        <intervalBetweenEvents>

```

```

    <!--opt, int, event interval-->1
  </intervalBetweenEvents>
</Extensions>
<uploadImagesDataType>
  <!--opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
</uploadImagesDataType>
<httpBroken>
  <!--opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
</httpBroken>
<SubscribeEvent>
  <!--opt, object, picture uploading modes of all events which contain pictures-->
  <heartbeat>
    <!--opt, int, Heartbeat Interval Time-->30
  </heartbeat>
  <eventMode>
    <!--req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "List" (only Listed alarms need to be reported)-->all
  </eventMode>
  <EventList>
    <!--opt, array, event list, subType:object-->
    <Event>
      <!--opt, object, channel information Linked to event-->
      <type>
        <!--req, enum, event type, subType:string, desc:see details in event types: "ADAS"(advanced driving assistance system), "ADASAlarm"(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent", "CDsStatus", "DBD" (driving behavior detection) "GPSUpload", "HFDP"(frequently appeared person), "IO"(I/O Alarm), "IOTD", "LES", "LFPD"(Low frequency person detection), "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioException", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskError", "diskFull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fieldDetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "ilLaccess"(invalid access), "ipConflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "LineDetection"(Line crossing detection), "ListSyncException"(List synchronization exception), "loitering"(loitering detection), "LowHDDTemperature"(HDD Low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modeError", "nicBroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry" (staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"(people overstay detection), "rollOver", "running"(people running), "safetyHelmetDetection"(hard hat detection), "sceneChangedDetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelterAlarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storing full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading List alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoLoss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format of the specified event, subType:string, desc:"binary", "LocalURL" (Local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL)-->binary
      </pictureURLType>
      <channels>
        <!--opt, string, listen to the events on the specified channel No. List, desc:if all channels are being listened to, the node shall not be applied. if some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
      </channels>
    </Event>
  </EventList>
</channels>
  <!--opt, string, listen to the specified channel No. List, desc:if all channels are being listened to, the node shall not be applied. if some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
</channels>
  <pictureURLType>
    <!--opt, enum, alarm picture format, subType:string, desc:"binary", "LocalURL" (Local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is Local URL of the device-->binary
  </pictureURLType>
  <changedUploadSub>
    <!--opt, object, subscribe to messages-->
  </changedUploadSub>
</interval>
  <!--opt, int, the Lifecycle of armina GUID, desc:within the interval, if the client software does not reconnect to the device, a new GUID will be

```

```

generated by the device-->5
</interval>
<StatusSub>
  <!--opt, object, sub status-->
  <all>
    <!--opt, bool, whether to subscribe to all-->true
  </all>
  <channel>
    <!--opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
  </channel>
  <hd>
    <!--opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
  </hd>
  <capability>
    <!--opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the
value of <all> is true-->true
  </capability>
</StatusSub>
</ChangedUploadSub>
</SubscribeEvent>
<PackingSpaceRecognition>
  <!--opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event:
PackingSpaceRecognition-->
  <uploadPicturesType>
    <!--req, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes" (upload specified types of pictures), "notUpload" (not upload
pictures)-->all
  </uploadPicturesType>
  <PicturesTypes>
    <!--opt, array, specified list of uploaded picture type, subType:object, dep:and,
{$.HttpHostNotification.PackingSpaceRecognition.uploadPicturesType,eq,picturesTypes}-->
  <picturesType>
    <!--opt, enum, uploaded picture type, subType:string, desc:"backgroundImage" (captured background picture), "plateImage" (license plate thumbnail)--
>backgroundImage
  </picturesType>
</PicturesTypes>
</PackingSpaceRecognition>
<enabled>
  <!--opt, bool-->true
</enabled>
<network>
  <!--opt, enum, subType:int-->1
</network>
</HttpHostNotification>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:Version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK"(succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot"(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>

```

12.2.1.5 Get the parameters of a listening host

Request URL

GET /ISAPI/Event/notification/httpHosts/<hostID>

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, Listening host, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, Listening host ID, range:[1,10]-->test
  </id>
  <url>
    <!--ro, req, string, URL-->test
  </url>
  <protocolType>
    <!--ro, req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocolType>
  <parameterFormatType>
    <!--ro, req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IP address-->test
  </ipAddress>
  <ipV6Address>
    <!--ro, opt, string, IPv6 address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IPv6 address-->test
  </ipV6Address>
  <portNo>
    <!--ro, opt, int, port number-->1
  </portNo>
  <userName>
    <!--ro, opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
  </userName>
  <httpAuthenticationMethod>
    <!--ro, req, enum, authentication method, subType:string, desc:"MD5digest" (MD5), "none", "base64"-->MD5digest
  </httpAuthenticationMethod>
  <ANPR>
    <!--ro, opt, object, ANPR-->
    <uploadPicturesType>
      <!--ro, opt, enum, uploaded pictures type, subType:string, desc:"all", "LicensePlatePicture" (License plate picture), "detectionPicture" (detected picture)-->all
    </uploadPicturesType>
  </ANPR>
  <Extensions>
    <!--ro, opt, object, range-->
    <intervalBetweenEvents>
      <!--ro, opt, int, event interval-->1
    </intervalBetweenEvents>
  </Extensions>
  <uploadImagesDataType>
    <!--ro, opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
  </uploadImagesDataType>
  <httpBroken>
    <!--ro, opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
  </httpBroken>
  <SubscribeEvent>
    <!--ro, opt, object, picture uploading modes of all events which contain pictures-->
    <heartbeat>
      <!--ro, opt, int, heartbeat interval time-->30
    </heartbeat>
    <eventMode>
      <!--ro, req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "List" (only listed alarms need to be reported)-->all
    </eventMode>
    <EventList>
      <!--ro, opt, array, event list, subType:object-->
      <Event>
        <!--ro, opt, object, channel information linked to event-->
        <type>
          <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm" (advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (Logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch" (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioException", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hddBadBlock"(HDD bad sector detection event), "hddImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illAccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "Linedetection"(Line crossing detection), "ListSyncException"(List synchronization exception), "Loitering"(Loitering detection), "LowHDDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(mode disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "poeException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
```

"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention" (people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm" (angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelterAlarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storing full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading List alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoLoss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection

```

</type>
<minorAlarm>
  <!--ro, opt, string, alarm sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
</minorAlarm>
<minorException>
  <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
</minorException>
<minorOperation>
  <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
</minorOperation>
<minorEvent>
  <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
</minorEvent>
<pictureURLType>
  <!--ro, opt, enum, alarm picture format of the specified event, subType:string, desc:"binary", "LocalURL" (Local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL)-->binary
</pictureURLType>
<channels>
  <!--ro, opt, string, listen to the events on the specified channel No. List, desc:if all channels are being listened to, the node shall not be applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
</channels>
</Event>
</EventList>
<channels>
  <!--ro, opt, string, listen to the specified channel No. List, desc:if all channels are being listened to, the node shall not be applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
</channels>
<pictureURLType>
  <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "LocalURL" (Local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->binary
</pictureURLType>
<ChangedUploadSub>
  <!--ro, opt, object, subscribe to messages-->
<interval>
  <!--ro, opt, int, the lifecycle of arming GUID, desc:within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->5
</interval>
<StatusSub>
  <!--ro, opt, object, sub status-->
<all>
  <!--ro, opt, bool, whether to subscribe to all-->true
</all>
<channel>
  <!--ro, opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
</channel>
<hd>
  <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
</hd>
<capability>
  <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
</capability>
</StatusSub>
</ChangedUploadSub>
</SubscribeEvent>
<PackingSpaceRecognition>
  <!--ro, opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event: PackingSpaceRecognition-->
  <uploadPicturesType>
    <!--ro, req, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes"(upload specified types of pictures), "notUpload"(not upload pictures)-->all
  </uploadPicturesType>
  <PicturesTypes>
    <!--ro, opt, array, specified list of uploaded picture types, subType:object, dep:and, {$.HttpPostNotification.PackingSpaceRecognition.uploadPicturesType,eq,picturesTypes)-->
  <picturesType>
    <!--ro, opt, enum, uploaded picture type, subType:string, desc:"backgroundImage"(captured background picture), "plateImage"(license plate thumbnail)-->backgroundImage
  </picturesType>
</PicturesTypes>
</PackingSpaceRecognition>
<enabled>
  <!--ro, opt, bool-->true
</enabled>
<network>

```



```

    <!--ro, opt, enum, subType:int-->1
  </network>
</HttpHostNotification>

```

12.2.1.6 Get the capabilities of listening hosts parameters

Request URL

GET /ISAPI/Event/notification/httpHosts/capabilities?type= <type>

Query Parameter

Parameter Name	Parameter Type	Description
type	enum	custom refers to that the listening service is independent and unaffected by the upload center linkage method. default refers to that the listening service is controlled by the linkage method of the uploading center. When the type is empty or default, the listening service is the default.

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotificationCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capabilities of subscribing to event types and event channels, attr:version{req, string, protocolVersion}-->
  <hostNumber>
    <!--ro, req, int, the number of listening hosts-->0
  </hostNumber>
  <urlLen min="1" max="10">
    <!--ro, req, string, URL Length, attr:min{opt, int},max{req, int}-->test
  </urlLen>
  <url>
    <!--ro, opt, string, URL-->test
  </url>
  <protocolType opt="HTTP,HTTPS,EHome">
    <!--ro, req, enum, protocol type, subType:string, attr:opt{req, string}, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocolType>
  <parameterFormatType opt="XML,querystring,JSON">
    <!--ro, req, enum, alarm parameter format type, subType:string, attr:opt{req, string}, desc:"xml" (XML format), "querystring", "json" (JSON format)-->XML
  </parameterFormatType>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, address format type, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName min="1" max="64">
    <!--ro, opt, string, domain name, attr:min{req, int},max{req, int}, desc:it is valid when addressingFormatType is "hostname"-->test
  </hostName>
  <ipAddress opt="ipv4,ipv6">
    <!--ro, opt, string, IP address, attr:opt{req, string}, desc:it is valid when addressingFormatType is "ipaddress"-->test
  </ipAddress>
  <portNo min="0" max="10">
    <!--ro, opt, string, port number, range:[0,65535], attr:min{req, int},max{req, int}-->1
  </portNo>
  <userNameLen min="0" max="10">
    <!--ro, opt, string, user name length, attr:min{req, int},max{req, int}-->1
  </userNameLen>
  <passwordLen min="0" max="10">
    <!--ro, opt, string, password length, attr:min{req, int},max{req, int}-->1
  </passwordLen>
  <httpAuthenticationMethod opt="MD5digest,none,base64,basic">
    <!--ro, req, enum, HTTP authentication method, subType:string, attr:opt{req, string}, desc:"MD5digest"(MD5), "none", "base64"-->MD5digest
  </httpAuthenticationMethod>
  <ANPR>
    <!--ro, opt, object, ANPR-->
    <detectioUploadPicturesType opt="all,licensePlatePicture,detectionPicture,notUpload">
      <!--ro, opt, enum, uploaded pictures type, subType:string, attr:opt{req, string}, desc:"all", "LicensePlatePicture" (license plate picture), "detectionPicture" (detected picture), "notUpload" (no picture upload)-->all
    </detectioUploadPicturesType>
    <alarmHttpPushProtocol opt="baseline,custom">
      <!--ro, opt, string, alarm HTTP push protocol, attr:opt{req, string}-->baseline
    </alarmHttpPushProtocol>
    <videoUploadEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </videoUploadEnabled>
  </ANPR>
  <HttpHostUploadCtrl>
    <!--ro, opt, object, the capability of reporting alarms-->
  <postDataTypes>
    <!--ro, opt, string, checkpoint data dictionary-->test
  </postDataTypes>

```

```

</postDataTypes>
<epoliceDataTypes>
  <!--ro, opt, string, intersection violation data dictionary-->test
</epoliceDataTypes>
<eventDataTypes>
  <!--ro, opt, string, event data dictionary-->test
</eventDataTypes>
<evidenceDataTypes>
  <!--ro, opt, string, enforcement data dictionary-->test
</evidenceDataTypes>
<uploadInterval min="1" max="2000">
  <!--ro, req, string, the default upload interval is 50 (1~2000ms), attr:min{req, int},max{req, int}-->0
</uploadInterval>
<uploadTimeout min="1" max="6000">
  <!--ro, req, string, the default timeout duration is 5000 (1~6000ms), attr:min{req, int},max{req, int}-->0
</uploadTimeout>
<noPlateUploadEnabled opt="true,false">
  <!--ro, req, bool, whether to upload no license plate data, attr:opt{req, string}-->true
</noPlateUploadEnabled>
<historyEnabled opt="true,false">
  <!--ro, req, bool, whether to upload history data, attr:opt{req, string}-->true
</historyEnabled>
<uploadTimePlan>
  <!--ro, opt, object, upload event schedule-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <timeList>
    <!--ro, opt, object, time list-->
    <timeSpan>
      <!--ro, opt, object, time-->
      <timeId min="0" max="1">
        <!--ro, req, string, time No., attr:min{req, int},max{req, int}-->0
      </timeId>
      <startHour min="0" max="23">
        <!--ro, opt, string, start hour, attr:min{req, int},max{req, int}-->0
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, string, start minute, attr:min{req, int},max{req, int}-->0
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, string, end hour, attr:min{req, int},max{req, int}-->0
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, string, end minute, attr:min{req, int},max{req, int}-->0
      </endMinute>
    </timeSpan>
  </timeList>
</uploadTimePlan>
<roadMaintDataTypes>
  <!--ro, opt, string, road maintenance type-->pothole,crack,transverseCrack
</roadMaintDataTypes>
</HttpHostUploadCtrl>
<Extensions>
  <!--ro, opt, object, expand-->
  <intervalBetweenEvents min="0" max="10">
    <!--ro, opt, string, event interval, attr:min{req, int},max{req, int}-->0
  </intervalBetweenEvents>
</Extensions>
<uploadImagesDataType opt="URL,binary">
  <!--ro, opt, enum, uploaded picture type, subType:string, attr:opt{req, string}, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
</uploadImagesDataType>
<httpBroken opt="true,false" def="true">
  <!--ro, opt, bool, whether to enable the automatic network replenishment, attr:opt{req, string},def{req, string}-->true
</httpBroken>
<SubscribeEventCap>
  <!--ro, opt, object, subscribe to changing status of capability set-->
  <heartbeat min="0" max="180">
    <!--ro, opt, string, heartbeat interval time, attr:min{req, int},max{req, int}-->1
  </heartbeat>
  <channelMode opt="all,list">
    <!--ro, opt, enum, channel mode, subType:string, attr:opt{req, string}, desc:"all" (the device does not support subscribing to event channels separately, "list" (the device support subscribing to event channels separately). If both channelMode and eventMode return all, the device does not support subscribing to event types and event channels.-->all
  </channelMode>
  <eventMode opt="all,list">
    <!--ro, opt, enum, event mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels and events on the device), "list" (subscribe to all channels and events on the device)-->all
  </eventMode>
  <EventList>
    <!--ro, opt, array, event list, subType:object-->
    <Event>
      <!--ro, opt, object, event subscription information-->
      <type>
        <!--ro, req, enum, see details in event type list, subType:string-->AccessControllerEvent
      </type>
      <minorAlarm opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor alarm type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401
      </minorAlarm>
      <minorException opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor exception type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401
      </minorException>
    </Event>
  </EventList>

```

```

    </minorException>
    <minorOperation opt="0x400,0x401,0x402,0x403">
        <!--ro, opt, string, minor operation type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
    </minorOperation>
    <minorEvent opt="0x01,0x02,0x03,0x04">
        <!--ro, opt, string, minor event type, attr:opt{req, string}, desc:"IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401
    </minorEvent>
    <pictureURLType opt="binary,localURL,cloudStorageURL,EZVIZURL" def="cloudStorageURL">
        <!--ro, opt, enum, alarm picture format, subType:string, attr:opt{req, string},def{req, string}, desc:"binary", "LocalURL" (Local URL),
"cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL)-->cloudStorageURL
    </pictureURLType>
</Event>
</EventList>
<pictureURLType opt="binary,localURL,cloudStorageURL,EZVIZURL" def="cloudStorageURL">
    <!--ro, opt, enum, picture uploading modes of all events which contain pictures, subType:string, attr:opt{req, string},def{req, string},
desc:"binary", "LocalURL" (Local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL). If the node is applied, <pictureURLType> of <Event> will
be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is
binary, and for back-end devices is Local URL of the device-->cloudStorageURL
</pictureURLType>
<ChangedUploadSub>
    <!--ro, opt, object, upload changes-->
    <interval>
        <!--ro, opt, string, the Lifecycle of arming GUID, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not
reconnect to the device, a new GUID will be generated by the device-->1
    </interval>
    <StatusSub>
        <!--ro, opt, bool, subscribe to status-->true
    <all>
        <!--ro, opt, bool, whether to subscribe to all-->true
    </all>
    <channel>
        <!--ro, opt, bool, subscribe to channel status, desc:it is not required if the value of <all> is true-->true
    </channel>
    <hd>
        <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
    </hd>
    <capability>
        <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the
value of <all> is true-->true
    </capability>
</StatusSub>
</ChangedUploadSub>
</SubscribeEventCap>
<PackingSpaceRecognition>
    <!--ro, opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event:
PackingSpaceRecognition. if it is not supported by the device, all pictures will be uploaded by default-->
    <uploadPicturesType opt="all,picturesTypes,notUpload" def="all">
        <!--ro, req, enum, uploaded picture type, subType:string, attr:opt{req, string},def{req, string}, desc:"all", "picturesTypes" (upload specified types
of pictures), "notUpload" (not upload pictures)-->all
    </uploadPicturesType>
    <PicturesTypes size="2">
        <!--ro, opt, array, specified list of uploaded picture types, subType:object, dep:and,
{$.HttpHostNotificationList[*].HttpHostNotification.PackingSpaceRecognition.uploadPicturesType,eq,picturesTypes}, attr:size{req, int)-->
    <pictureType opt="backgroundImage,plateImage">
        <!--ro, opt, enum, uploaded picture type, subType:string, attr:opt{req, string}, desc:"backgroundImage" (captured background picture), "plateImage"
(License plate thumbnail)-->backgroundImage
    </pictureType>
</PicturesTypes>
</PackingSpaceRecognition>
<enabled>
    <!--ro, opt, bool, whether to enable-->true
</enabled>
<network opt="1,2,3">
    <!--ro, opt, enum, network, subType:int, attr:opt{req, string}, desc:1 (wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
</network>
<method opt="POST,PUT,GET" def="POST">
    <!--ro, opt, enum, request method, subType:string, attr:opt{req, string},def{req, string}, desc:"POST", "PUT", "GET"-->POST
</method>
<pictureAttachEnabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable attaching pictures, attr:opt{req, string},def{req, bool}, desc:true by default-->true
</pictureAttachEnabled>
<contentType opt="multipart" def="multipart">
    <!--ro, opt, enum, HTTP content format, subType:string, attr:opt{req, string},def{req, string}, desc:"multipart" (form format) by default. It is the
format for all uploaded events, not limited to pictures only-->multipart
</contentType>
<eventTemplateID min="1" max="255">
    <!--ro, opt, int, event message upload template ID, attr:min{req, int},max{req, int}, desc:related URL (GET
/ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-->1
</eventTemplateID>
<customTriggersEnabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable custom linkage, attr:opt{req, string},def{req, bool}, desc:true (listening service's linkage method is unaffected
by upload center's linkage method, that is, independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening
service)-->>false
</customTriggersEnabled>
<checkResponseEnabled opt="true,false" def="true">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool)-->true
</checkResponseEnabled>
</HttpHostNotificationCap>

```

12.2.1.7 Set IP address of receiving server(s)

Request URL

PUT /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, Listening host List, subType:object, attr:version{req, string, protocolVersion}-->
  <HttpHostNotification>
    <!--opt, object, Listening host-->
    <id>
      <!--req, string, ID-->test
    </id>
    <url>
      <!--req, string, URL-->test
    </url>
    <protocolType>
      <!--req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
    </protocolType>
    <parameterFormatType>
      <!--req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
    </parameterFormatType>
    <addressingFormatType>
      <!--req, enum, address types, subType:string, desc:"hostname" (host name), "ipaddress" (ip address)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:only one of IPv4 and IPv6 address exists when the address type is "ipaddress"-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:only one of IPv4 and IPv6 address exists when the address type is "ipaddress"-->test
    </ipv6Address>
    <portNo>
      <!--opt, int, port No.-->1
    </portNo>
    <userName>
      <!--opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </userName>
    <password>
      <!--opt, string, password, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </password>
    <httpAuthenticationMethod>
      <!--req, enum, authentication method, subType:string, desc:"MD5digest" (MD5), "none"-->MD5digest
    </httpAuthenticationMethod>
    <videoUploadEnabled>
      <!--opt, bool-->false
    </videoUploadEnabled>
    <ANPR>
      <!--opt, object, ANPR-->
      <detectioUploadPicturesType>
        <!--opt, enum, uploaded pictures type, subType:string, desc:uploaded pictures type-->all
      </detectioUploadPicturesType>
    </ANPR>
    <Extensions>
      <!--opt, object, range-->
      <intervalBetweenEvents>
        <!--opt, int, event interval-->1
      </intervalBetweenEvents>
    </Extensions>
    <uploadImagesDataType>
      <!--opt, enum, picture data type, subType:string, desc:picture data type-->URL
    </uploadImagesDataType>
    <httpBroken>
      <!--opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
    </httpBroken>
    <SubscribeEvent>
      <!--opt, object, picture uploading modes of all events which contain pictures-->
      <heartbeat>
        <!--opt, int, heartbeat interval-->30
      </heartbeat>
      <eventMode>
        <!--req, enum, event mode, subType:string, desc:event mode-->all
      </eventMode>
      <EventList>
        <!--opt, array, event List, subType:object-->
        <Event>
          <!--opt, object, channel information Linked to event-->
          <type>
            <!--req, enum, event type, subType:string, desc:event type-->mixedTargetDetection
          </type>
        </Event>
      </EventList>
    </SubscribeEvent>
  </HttpHostNotification>
</HttpHostNotificationList>
```



```

<!--opt, int, event message template id, desc:related URL (GET /ISAPI/Event/notification/eventTemplate?format=json) is to get template list-->1
</eventTemplateID>
<customTriggersEnabled>
  <!--opt, bool, whether to enable custom linkage, desc:true (listening service's Linkage method is unaffected by upload center's Linkage method, that
is, independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening service)-->>false
</customTriggersEnabled>
<checkResponseEnabled>
  <!--opt, bool-->true
</checkResponseEnabled>
</HttpHostNotification>
</HttpHostNotificationList>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:detailed information of custom error returned by device applications, used
for fast debugging-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:none-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>

```

12.2.1.8 Add a listening host

Request URL

POST /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, listening host, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, listening host ID-->test
  </id>
  <url>
    <!--req, string, URL-->test
  </url>
  <protocolType>
    <!--req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocolType>
  <parameterFormatType>
    <!--req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IP address-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 Address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IPv6 Address-->test
  </ipv6Address>
  <portNo>
    <!--opt, int, port number-->1
  </portNo>
</HttpHostNotification>

```

```

</user>
<userName>
  <!--opt, string, user name, dep:and,{$.HostNameNotification.httpAuthenticationMethod,eq,MD5digest}-->test
</userName>
<password>
  <!--opt, string, password, dep:and,{$.HostNameNotification.httpAuthenticationMethod,eq,MD5digest}-->test
</password>
<httpAuthenticationMethod>
  <!--req, enum, authentication method, subType:string, desc:"MD5digest"(MD5), "none", "base64"-->MD5digest
</httpAuthenticationMethod>
<ANPR>
  <!--opt, object, ANPR-->
  <uploadPicturesType>
    <!--opt, enum, uploaded pictures type, subType:string, desc:"all", "licensePlatePicture", "detectionPicture"(detected picture)-->all
  </uploadPicturesType>
</ANPR>
<Extensions>
  <!--opt, object, range-->
  <intervalBetweenEvents>
    <!--opt, int, event interval-->1
  </intervalBetweenEvents>
</Extensions>
<uploadImagesDataType>
  <!--opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
</uploadImagesDataType>
<httpBroken>
  <!--opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
</httpBroken>
<SubscribeEvent>
  <!--opt, object, picture uploading modes of all events which contain pictures-->
  <heartbeat>
    <!--opt, int, Heartbeat Interval Time, unit:s-->30
  </heartbeat>
  <eventMode>
    <!--req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "list" (only listed alarms need to be reported)-->all
  </eventMode>
  <eventList>
    <!--opt, array, event list, subType:object-->
  <event>
    <!--opt, object, channel information linked to event-->
    <type>
      <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
      (advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
      controller event), "CDStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
      person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFDP"(rarely appeared person detection), "PALMismatch"
      (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
      "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
      "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioException", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
      spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
      detection), "diskUnformat"(disk unformatted), "diskError", "diskFull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
      "faceCapture", "faceSnapModeling", "faceDetection", "failDown"(People Falling Down), "faultAlarm", "fieldDetection"(intrusion detection), "fireDetection",
      "fireEscapeDetection", "fLowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
      detection event), "hdImpact"(HDD impact detection event), "heatMap"(heat map alarm), "highHDDTemperature"(HDD high temperature detection event),
      "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "ilLAccess"(invalid access), "ipcTransferAbnormal", "ipConflict"(IP address
      conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "linedetection"(line crossing detection),
      "listSyncException"(list synchronization exception), "Loitering"(loitering detection), "LowHDDTemperature"(HDD low temperature detection event),
      "mixedTargetDetection"(multi-target-type detection), "modelError", "nicBroken"(network disconnected), "nodeOffline"(node disconnected),
      "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
      "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
      "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
      detection), "poeException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
      "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
      "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
      (people overstay detection), "roLover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
      (angular acceleration alarm), "severeHDDFailure"(HDD major fault detection), "shelterAlarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
      detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingException"(distance exception), "sysStorFull"(storing full alarm of cluster system),
      "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
      alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tolCodeInfo"(QR code information report), "tossing"(thrown object detection),
      "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcgResult", "versionAbnormal"(cluster version exception), "videoException",
      "videoLoss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent",
      "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison
      event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching
      consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison" (human body comparison),
      "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
    </type>
  <minorAlarm>
    <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
    "AccessControllerEvent"-->0x400,0x401,0x402,0x403
  </minorAlarm>
  <minorException>
    <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
    is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
  </minorException>
  <minorOperation>
    <!--opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
    is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
  </minorOperation>
  <minorEvent>
    <!--opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
    "AccessControllerEvent"-->0x01,0x02,0x03,0x04
  </minorEvent>
  <pictureURLType>
    <!--opt, enum, alarm picture format of the specified event, subType:string, desc:"binary", "LocalURL" (Local URL), "cloudStorageURL" (cloud
    storage URL), "EZVIZURL" (EZ URL)-->binary

```



```

</pictureURLType>
<channels>
  <!--opt, string, listen to the events on the specified channel No. List, desc:if all channels are being listened to, the node shall not be
  applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
</channels>
</Event>
</EventListener>
<channels>
  <!--opt, string, listen to the specified channel No. List, desc:if all channels are being listened to, the node shall not be applied. If some channels
  are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
</channels>
<pictureURLType>
  <!--opt, enum, alarm picture format, subType:string, desc:"binary", "LocalURL" (Local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ
  URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not
  applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices
  is local URL of the device-->binary
</pictureURLType>
<ChangedUploadSub>
  <!--opt, object, subscribe to messages-->
<interval>
  <!--opt, int, the lifecycle of arming GUID, desc:within the interval, if the client software does not reconnect to the device, a new GUID will be
  generated by the device-->5
</interval>
<StatusSub>
  <!--opt, object, sub status-->
<all>
  <!--opt, bool, whether to subscribe to all-->true
</all>
<channel>
  <!--opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
</channel>
<hd>
  <!--opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
</hd>
<capability>
  <!--opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the
  value of <all> is true-->true
</capability>
</StatusSub>
</ChangedUploadSub>
</SubscribeEvent>
<PackingSpaceRecognition>
  <!--opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event:
  PackingSpaceRecognition-->
  <uploadPicturesType>
    <!--req, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes" (upload specified types of pictures), "notUpload" (not upload
    pictures)-->all
  </uploadPicturesType>
  <PicturesTypes>
    <!--opt, array, specified list of uploaded picture type, subType:object, dep:and,
    {$.HttpHostNotification.PackingSpaceRecognition.uploadPicturesType,eq,picturesTypes}-->
    <picturesType>
      <!--opt, enum, uploaded picture type, subType:string, desc:"backgroundImage" (captured background picture), "plateImage" (license plate thumbnail)--
      >backgroundImage
    </picturesType>
  </PicturesTypes>
</PackingSpaceRecognition>
<enabled>
  <!--opt, bool-->true
</enabled>
<network>
  <!--opt, enum, subType:int-->1
</network>
</HttpHostNotification>

```

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK"(succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <id>
    <!--ro, req, string, ID, desc:if adding a listening host succeeded, a security control panel ID will be returned as the listening host ID. Related URL: /ISAPI/Event/notification/httpHosts/<ID>-->test
  </id>
</ResponseStatus>
```

12.2.1.9 Get parameters of all listening hosts

Request URL

GET /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, listening host list, subType:object, attr:version{req, string, protocolVersion}-->
  <HttpHostNotification>
    <!--ro, opt, object, Listening host-->
    <id>
      <!--ro, req, string, subscribe to ID, range:[1,10]-->test
    </id>
    <url>
      <!--ro, req, string, URL-->test
    </url>
    <protocolType>
      <!--ro, req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
    </protocolType>
    <parameterFormatType>
      <!--ro, req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
    </parameterFormatType>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}, desc:IP address-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 Address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}, desc:IPv6 Address-->test
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port number-->1
    </portNo>
    <userName>
      <!--ro, opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </userName>
    <httpAuthenticationMethod>
      <!--ro, req, enum, authentication method, subType:string, desc:"MD5digest"(MD5), "none", "base64"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
      <!--ro, opt, object, ANPR-->
      <detectionUploadPicturesType>
        <!--ro, opt, enum, uploaded pictures type, subType:string, desc:"all", "LicensePlatePicture", "detectionPicture"(detected picture)-->all
      </detectionUploadPicturesType>
      <videoUploadEnabled>
        <!--ro, opt, bool-->false
      </videoUploadEnabled>
```

```

</ANPR>
<Extensions>
  <!--ro, opt, object, range-->
  <intervalBetweenEvents>
    <!--ro, opt, int, event interval-->1
  </intervalBetweenEvents>
</Extensions>
<uploadImagesDataType>
  <!--ro, opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
</uploadImagesDataType>
<httpBroken>
  <!--ro, opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
</httpBroken>
<SubscribeEvent>
  <!--ro, opt, object, picture uploading modes of all events which contain pictures-->
  <heartbeat>
    <!--ro, opt, int, heartbeat interval time-->30
  </heartbeat>
  <eventMode>
    <!--ro, req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "list" (only listed alarms need to be reported)-->all
  </eventMode>
  <eventList>
    <!--ro, opt, array, event list, subType:object-->
    <Event>
      <!--ro, opt, object, channel information linked to event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system),
        "ADASAlarm"(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent"
        (access controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFDP"(frequently
        appeared person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (Logistics scanning event), "LFDP"(rarely appeared person detection),
        "PALMismatch" (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing
        up detection), "TMA"(thermometry alarm), "TPMA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving",
        "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioException", "behaviorResult"(abnormal event detection),
        "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis",
        "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status
        monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fieldDetection"
        (intrusion detection), "fireDetection", "fireEscapeDetection", "fLowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people
        gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDDTemperature"(HDD high
        temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "ilAccess"(invalid access),
        "ipcTransferAbnormal", "ipConflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection),
        "Linedetection"(Line crossing detection), "ListSyncException"(List synchronization exception), "Loitering"(Loitering detection), "LowHDDTemperature"(HDD Low
        temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node
        disconnected), "nonPoliceIntrusion", "overspeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection),
        "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting",
        "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"
        (playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection",
        "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection),
        "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance",
        "regionExiting", "retention"(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection),
        "scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelterAlarm"(video tampering alarm),
        "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"
        (storing full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature
        difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information
        report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"
        (cluster version exception), "videoException", "videoLoss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay
        detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control),
        "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent",
        "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"
        (human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--ro, opt, string, alarm sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
        is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, exception sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
        event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
        event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
        is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format of the specified event, subType:string, desc:alarm picture format of the specified event-->binary
      </pictureURLType>
      <channels>
        <!--ro, opt, string, listen to the events on the specified channel No. list, desc:if all channels are being listened to, the node shall not be
        applied. if some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
      </channels>
    </Event>
  </EventList>
</channels>
  <!--ro, opt, string, listen to the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. if some
  channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--ro, opt, enum, alarm picture format, subType:string, desc:the node indicates the picture types of all events which contain pictures to be
    uploaded. if the node is applied, the pictureURLType of the Event will not take effect. if the node is not applied, the pictures reported from the device by
    default mode: the default data type of uploaded pictures captured from front-end devices is binary. the default data type of uploaded pictures reported from
    storage devices is Local URL of the device.-->binary
  </pictureURLType>

```

```

Storage devices to detect one of the correct reading,
</pictureURLType>
<ChangedUploadSub>
  <!--ro, opt, object, subscribe to messages-->
  <interval>
    <!--ro, opt, int, the Lifecycle of arming GUID, desc:if GUID is not reconnected in the interval, a new GUID will be generated since the device
start a new arm period-->5
  </interval>
  <StatusSub>
    <!--ro, opt, object, sub status-->
    <all>
      <!--ro, opt, bool, subscribe to all?-->true
    </all>
    <channel>
      <!--ro, opt, bool, subscribe to channel status, desc:reporting is not required if all is true-->true
    </channel>
    <hd>
      <!--ro, opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
    </hd>
    <capability>
      <!--ro, opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
    </capability>
  </StatusSub>
</ChangedUploadSub>
</SubscribeEvent>
<PackingSpaceRecognition>
  <!--ro, opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event:
PackingSpaceRecognition-->
  <uploadPicturesType>
    <!--ro, opt, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes" (upload specified types of pictures), "notUpload" (not upload
pictures)-->all
  </uploadPicturesType>
  <PicturesTypes>
    <!--ro, opt, array, specified list of uploaded picture type, subType:object, dep:and,
{$.HttpHostNotificationList[*].HttpHostNotification.PackingSpaceRecognition.uploadPicturesType,eq,picturesTypes}-->
  <picturesType>
    <!--ro, opt, enum, uploaded picture type, subType:string, desc:"backgroundImage" (captured background picture), "plateImage" (license plate
thumbnail)-->backgroundImage
  </picturesType>
</PicturesTypes>
</PackingSpaceRecognition>
<enabled>
  <!--ro, opt, bool, whether to enable-->true
</enabled>
<network>
  <!--ro, opt, enum, network, subType:int, desc:1 (wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
</network>
<method>
  <!--ro, opt, enum, request method, subType:string, desc:"POST", "PUT", "GET"-->POST
</method>
<pictureAttachEnabled>
  <!--ro, opt, bool, whether to enable attaching pictures, desc:true by default-->true
</pictureAttachEnabled>
<contentType>
  <!--ro, opt, enum, HTTP content format, subType:string, desc:"multipart" (form format) by default. It is the format for all uploaded events, not
limited to pictures only-->multipart
</contentType>
<eventTemplateID>
  <!--ro, opt, int, event message template ID, desc:related URL (GET /ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-
-->1
</eventTemplateID>
<customTriggersEnabled>
  <!--ro, opt, bool, whether to enable custom Linkage, desc:true (listening service's Linkage method is unaffected by upload center's Linkage method,
that is, independent), false or node doesn't exist (use the previous Logic, that upload center sends data to listening service)-->>false
</customTriggersEnabled>
<checkResponseEnabled>
  <!--ro, opt, bool-->true
</checkResponseEnabled>
</HttpHostNotification>
</HttpHostNotificationList>

```

12.2.2 IP Address Management

12.2.2.1 Get capability of a specific network interface

Request URL

GET /ISAPI/System/Network/interfaces/<interfaceID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NetworkInterface xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <IPAddress>
    <!--ro, req, object, IP address-->
    <ipVersion opt="v4,v6,dual">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->v4
    </ipVersion>
    <addressingType opt="static,dynamic,api">
      <!--ro, req, enum, address type, subType:string, attr:opt{req, string}-->static
    </addressingType>
    <ipAddress>
      <!--ro, opt, string, IP address-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string, sub-net mask-->test
    </subnetMask>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address-->test
    </ipv6Address>
    <bitMask>
      <!--ro, opt, string-->test
    </bitMask>
    <DefaultGateway>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IP address-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </DefaultGateway>
    <PrimaryDNS>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </PrimaryDNS>
    <SecondaryDNS>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IP address-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </SecondaryDNS>
    <DNSEnable opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </DNSEnable>
    <IPv6Mode>
      <!--ro, opt, object-->
      <ipV6AddressingType opt="ra,manual,dhcp">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->ra
      </ipV6AddressingType>
      <ipV6AddressList>
        <!--ro, opt, array, subType:object-->
        <v6Address>
          <!--ro, opt, object, IPv6 address-->
          <id min="1" max="10">
            <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
          </id>
          <type opt="ra,manual,dhcp">
            <!--ro, opt, string, attr:opt{req, string}-->test
          </type>
          <address min="1" max="10">
            <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
          </address>
          <bitMask min="1" max="10">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
          </bitMask>
        </v6Address>
      </ipV6AddressList>
    </IPv6Mode>
    <DoubleNetRoute>
      <!--ro, opt, object-->
      <ipAddress min="1" max="10">
        <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
      </ipAddress>
      <subnetMask min="1" max="10">
        <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
      </subnetMask>
    </DoubleNetRoute>
  </IPAddress>

```

```

</subnetMask>
</DoubleNetRoute>
<LANConfigMode opt="manual,auto">
  <!--ro, opt, string, attr:opt{req, string}-->test
</LANConfigMode>
<DoubleNetRouteList size="3">
  <!--ro, opt, object, attr:size{req, int}-->
  <DoubleNetRoute>
    <!--ro, opt, object-->
    <ipAddress min="1" max="10">
      <!--ro, opt, string, IP address, attr:min{req, int},max{req, int}-->test
    </ipAddress>
    <subnetMask min="1" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </subnetMask>
  </DoubleNetRoute>
</DoubleNetRouteList>
</IPAddress>
<Wireless>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <wirelessNetworkMode opt="infrastructure,adhoc">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->infrastructure
  </wirelessNetworkMode>
  <channel opt="1,2,3,4,5,6,7,8,9,10,11,12,13,14,auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
  </channel>
  <ssid min="1" max="10">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ssid>
  <wmmEnabled>
    <!--ro, opt, bool-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--ro, opt, object-->
    <securityMode opt="disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-enterprise">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->disable
    </securityMode>
    <WEP>
      <!--ro, opt, object-->
      <authenticationType opt="open,sharedkey,auto">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->open
      </authenticationType>
      <defaultTransmitKeyIndex min="1" max="10">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength opt="64,128">
        <!--ro, opt, enum, subType:int, attr:opt{req, string}-->64
      </wepKeyLength>
      <EncryptionKeyList>
        <!--ro, opt, array, subType:object-->
        <encryptionKey>
          <!--ro, opt, string, encryption key-->test
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA>
      <!--ro, opt, object-->
      <algorithmType opt="TKIP,AES,TKIP/AES">
        <!--ro, req, enum, algorithm type, subType:string, attr:opt{req, string}-->TKIP
      </algorithmType>
      <sharedKey>
        <!--ro, req, string-->test
      </sharedKey>
      <wpaKeyLength min="8" max="64">
        <!--ro, req, int, range:[8,64], attr:min{req, int},max{req, int}-->8
      </wpaKeyLength>
    </WPA>
    <support64bitKey opt="WPA-personal,WPA2-personal">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->WPA-personal
    </support64bitKey>
  </WirelessSecurity>
</AccessPointList>
  <!--ro, opt, array, subType:object-->
  <InterfaceDisplay>
    <!--ro, opt, object-->
    <TableElementList>
      <!--ro, opt, array, subType:object-->
      <TableElement opt="SSID,workingMode,securityMode,channel,signalStrength,speed,connectionStatus">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->securityMode
      </TableElement>
    </TableElementList>
  </InterfaceDisplay>
</AccessPointList>
<isSupportConnectStatus>
  <!--ro, opt, bool-->true
</isSupportConnectStatus>
<workScene opt="computerRoom,monitorTerminal,centralTerminal">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->computerRoom
</workScene>
<protocol opt="802.11ac">

```

```

    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->802.11ac
</protocol>
<hideSsid>
    <!--ro, opt, bool-->true
</hideSsid>
<ChannelConfig>
    <!--ro, opt, object-->
    <transmitPower opt="9,12,15,18,21,24,27">
        <!--ro, opt, enum, subType:int, attr:opt{req, string}-->9
    </transmitPower>
    <ChannelCountryList>
        <!--ro, opt, array, subType:object-->
        <Country>
            <!--ro, opt, object-->
            <countryID>
                <!--ro, req, int-->1
            </countryID>
            <ChannelList>
                <!--ro, opt, object-->
                <Node>
                    <!--ro, opt, object-->
                    <width opt="auto,20,40,80">
                        <!--ro, req, enum, subType:string, attr:opt{req, string}-->auto
                    </width>
                    <channel>
                        <!--ro, req, int-->1
                    </channel>
                </Node>
            </ChannelList>
        </Country>
    </ChannelCountryList>
</ChannelConfig>
<rate opt="2.4GHz,5GHz,auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
</rate>
<dialUpGroupNo min="1" max="10">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</dialUpGroupNo>
<BaseNoisevalue min="1" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</BaseNoisevalue>
</Wireless>
<Discovery>
    <!--ro, opt, object-->
    <UPnP>
        <!--ro, req, object-->
        <enabled opt="true,false">
            <!--ro, req, bool, attr:opt{req, string}-->true
        </enabled>
    </UPnP>
    <Zeroconf>
        <!--ro, opt, object-->
        <enabled opt="true,false">
            <!--ro, req, bool, attr:opt{req, string}-->true
        </enabled>
    </Zeroconf>
</Discovery>
<Link>
    <!--ro, opt, object-->
    <MACAddress>
        <!--ro, req, string-->test
    </MACAddress>
    <autoNegotiation opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
    </autoNegotiation>
    <speed opt="10,100,1000">
        <!--ro, req, int, "10,100,1000,10000", attr:opt{req, string}-->1
    </speed>
    <duplex opt="half,full">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->half
    </duplex>
    <MTU min="1" max="10">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </MTU>
    <linkMode opt="optical,electrical">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->optical
    </linkMode>
</Link>
<defaultConnection opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
</defaultConnection>
<ActiveMulticast>
    <!--ro, opt, object-->
    <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <streamID opt="main">
        <!--ro, req, string, attr:opt{req, string}-->test
    </streamID>
    <ipV4Address min="1" max="10">
        <!--ro, opt, string, IPv4 address, attr:min{req, int},max{req, int}-->test
    </ipV4Address>
    <ipV6Address min="1" max="10">
        <!--ro, opt, string, IPv6 address, attr:min{req, int},max{req, int}-->test
    </ipV6Address>

```

```

<!--ro, opt, int, port No., attr:min{req, int},max{req, int}-->1
</port>
</ActiveMulticast>
<macAddress min="1" max="10">
  <!--ro, opt, string, MAC address, attr:min{req, int},max{req, int}-->test
</macAddress>
<EthernetPortList size="4">
  <!--ro, opt, array, network interface information, subType:object, attr:size{req, int}-->
  <EthernetPort>
    <!--ro, opt, object-->
    <id min="1" max="4">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </id>
    <MACAddress>
      <!--ro, req, string, MAC address-->test
    </MACAddress>
    <status opt="connected,disconnect">
      <!--ro, opt, enum, status, subType:string, attr:opt{req, string}-->connected
    </status>
    <speed opt="10,100,1000,10000">
      <!--ro, req, enum, speed, subType:int, attr:opt{req, string}, desc:"10,100,1000,10000"-->10
    </speed>
  </EthernetPort>
</EthernetPortList>
<Extensions>
  <!--ro, opt, object-->
  <NetworkCardType opt="normal,internal,external,main">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
  </NetworkCardType>
</Extensions>
<interfaceName min="1" max="10">
  <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
</interfaceName>
<LANConfigMode opt="manual,auto">
  <!--ro, opt, string, attr:opt{req, string}-->test
</LANConfigMode>
</NetworkInterface>

```

statusCode	statusString	subStatusCode	errorCode	des
4	Invalid Operation	theCurrentPoeModeDoesNotSupportNetworkParameterConfiguration	0x4000A214	--

12.2.2.2 Set the IP address

Request URL

PUT /ISAPI/System/Network/interfaces/<interfaceID>/ipAddress

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<IPAddress xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, IP address parameters, attr:version{req, string, protocolVersion}-->
  <ipVersion>
    <!--req, enum, IP address version, subType:string, desc:IP address version-->v4
  </ipVersion>
  <addressingType>
    <!--req, enum, address type, subType:string, desc:address type-->static
  </addressingType>
  <ipAddress>
    <!--opt, string, IPv4 address-->test
  </ipAddress>
  <subnetMask>
    <!--opt, string, IPv4 subnet mask-->test
  </subnetMask>
  <ipv6Address>
    <!--opt, string, IPv6 address-->test
  </ipv6Address>
  <bitMask>
    <!--opt, string, IPv6 subnet mask-->test
  </bitMask>
  <DefaultGateway>
    <!--opt, object, default gateway-->
    <ipAddress>
      <!--opt, string, IPv4 address-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address-->test
    </ipv6Address>
  </DefaultGateway>
  <PrimaryDNS>
    <!--opt, object-->
    <ipAddress>
      <!--opt, string, IPv4 address-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address-->test
    </ipv6Address>
  </PrimaryDNS>
  <SecondaryDNS>
    <!--opt, object, reserved DNS-->
    <ipAddress>
      <!--opt, string, IPv4 address-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address-->test
    </ipv6Address>
  </SecondaryDNS>
  <DNSEnable>
    <!--opt, bool-->true
  </DNSEnable>
  <Ipv6Mode>
    <!--opt, object-->
    <ipV6AddressingType>
      <!--opt, enum, subType:string-->ra
    </ipV6AddressingType>
    <ipV6AddressList>
      <!--opt, array, subType:object-->
      <v6Address>
        <!--opt, object, IPv6 address-->
        <id>
          <!--opt, string-->test
        </id>
        <type>
          <!--opt, enum, subType:string-->ra
        </type>
        <address>
          <!--opt, string, IPv6 address-->test
        </address>
        <bitMask>
          <!--opt, string-->test
        </bitMask>
      </v6Address>
    </ipV6AddressList>
  </Ipv6Mode>
  <DoubleNetRoute>
    <!--opt, object-->
    <ipAddress>
      <!--opt, string, IP address-->test
    </ipAddress>
    <subnetMask>
      <!--opt, string, subnet mask-->test
    </subnetMask>
  </DoubleNetRoute>
  <LANConfigMode>
    <!--opt, enum, subType:string-->manual
  </LANConfigMode>
</IPAddress>

```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
3	Device Error	deviceError	0x30000001	--
4	Invalid Operation	invalidOperation	0x40000006	--
6	Invalid Content	badIPv4Address	0x60000004	--
6	Invalid Content	badIPv6Address	0x60000005	--
6	Invalid Content	conflictIPv4Address	0x60000006	--
6	Invalid Content	conflictIPv6Address	0x60000007	--
6	Invalid Content	badNetMask	0x6000000E	--

12.2.2.3 Get the IP address parameters

Request URL

GET /ISAPI/System/Network/interfaces/<interfaceID>/ipAddress

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<IPAddress xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <ipVersion>
    <!--ro, req, enum, subType:string-->v4
  </ipVersion>
  <addressingType>
    <!--ro, req, enum, subType:string-->static
  </addressingType>
  <ipAddress>
    <!--ro, opt, string-->test
  </ipAddress>
  <subnetMask>
    <!--ro, opt, string-->test
  </subnetMask>
  <ipv6Address>
    <!--ro, opt, string-->test
  </ipv6Address>
  <bitMask>
    <!--ro, opt, string-->test
  </bitMask>
  <DefaultGateway>
    <!--ro, opt, object-->
    <ipAddress>
      <!--ro, opt, string-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string-->test
    </ipv6Address>
  </DefaultGateway>
  <PrimaryDNS>
    <!--ro, opt, object-->
    <ipAddress>
      <!--ro, opt, string-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string-->test
    </ipv6Address>
  </PrimaryDNS>
  <SecondaryDNS>
    <!--ro, opt, object-->
    <ipAddress>
      <!--ro, opt, string-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string-->test
    </ipv6Address>
  </SecondaryDNS>
  <DNSEnable>
    <!--ro, opt, bool-->true
  </DNSEnable>
  <Ipv6Mode>
    <!--ro, opt, object-->
    <ipV6AddressingType>
      <!--ro, opt, enum, subType:string-->ra
    </ipV6AddressingType>
    <ipV6AddressList>
      <!--ro, opt, array, subType:object-->
      <v6Address>
        <!--ro, opt, object-->
        <id>
          <!--ro, opt, string-->test
        </id>
        <type>
          <!--ro, opt, enum, subType:string-->ra
        </type>
        <address>
          <!--ro, opt, string-->test
        </address>
        <bitMask>
          <!--ro, opt, string-->test
        </bitMask>
      </v6Address>
    </ipV6AddressList>
  </Ipv6Mode>
  <DoubleNetRoute>
    <!--ro, opt, object-->
    <ipAddress>
      <!--ro, opt, string-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string-->test
    </subnetMask>
  </DoubleNetRoute>
  <LANConfigMode>
    <!--ro, opt, enum, subType:string-->manual
  </LANConfigMode>
</IPAddress>

```

12.2.2.4 Set the connection parameters of a specific network interface

Request URL

PUT /ISAPI/System/Network/interfaces/<interfaceID>/link

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Link xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NIC connection information, attr:version{req, string, protocolVersion}-->
  <MACAddress>
    <!--req, string, MAC address-->test
  </MACAddress>
  <autoNegotiation>
    <!--req, bool, whether to enable the auto-negotiation-->true
  </autoNegotiation>
  <speed>
    <!--opt, enum, link speed, subType:int, desc:Link speed-->10
  </speed>
  <duplex>
    <!--opt, enum, "half,full", subType:string, desc:"half" (half duplex), "full" (full duplex)-->half
  </duplex>
  <MTU>
    <!--req, int, MTU-->1
  </MTU>
  <linkMode>
    <!--opt, enum, connect mode, subType:string, desc:"optical", "electrical"-->optical
  </linkMode>
</Link>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

12.2.2.5 Get the connection parameters of a specific NIC

Request URL

GET /ISAPI/System/Network/interfaces/<interfaceID>/link

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Link xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NIC connection information, attr:version{req, string, protocolVersion}-->
  <MACAddress>
    <!--ro, req, string, MAC address-->test
  </MACAddress>
  <autoNegotiation>
    <!--ro, req, bool, whether to enable the auto-negotiation-->true
  </autoNegotiation>
  <speed>
    <!--ro, opt, enum, link speed, subType:int, desc:10 (10M), 100 (100M), 1000 (1000M)-->10
  </speed>
  <duplex>
    <!--ro, opt, enum, duplex mode, subType:string, desc:"half" (half duplex), "full" (full duplex)-->half
  </duplex>
  <MTU>
    <!--ro, req, int, MTU-->1
  </MTU>
  <linkMode>
    <!--ro, opt, enum, connection mode, subType:string, desc:"optical", "electrical"-->optical
  </linkMode>
</Link>
```

12.2.2.6 Set parameters of a specific network interface

Request URL

PUT /ISAPI/System/Network/interfaces/<interfaceID>?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--
serialNumber	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NetworkInterface xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <IPAddress>
    <!--req, object-->
    <ipVersion>
      <!--req, enum, IP address version, subType:string-->v4
    </ipVersion>
    <addressingType>
      <!--req, enum, address type, subType:string-->static
    </addressingType>
    <ipAddress>
      <!--opt, string, IPv4 address-->test
    </ipAddress>
    <subnetMask>
      <!--opt, string-->test
    </subnetMask>
    <ipV6Address>
      <!--opt, string, IPv6 address-->test
    </ipV6Address>
    <bitMask>
      <!--opt, string-->test
    </bitMask>
    <DefaultGateway>
      <!--opt, object, default gateway-->
      <ipAddress>
        <!--opt, string, IPv4 address-->test
      </ipAddress>
      <ipV6Address>
        <!--opt, string, IPv6 address-->test
      </ipV6Address>
    </DefaultGateway>
    <PrimaryDNS>
      <!--opt, object-->
      <ipAddress>
        <!--opt, string, IPv4 address-->test
      </ipAddress>
      <ipV6Address>
        <!--opt, string, IPv6 address-->test
      </ipV6Address>
    </PrimaryDNS>
  </NetworkInterface>
```

```

<SecondaryDNS>
  <!--opt, object-->
  <ipAddress>
    <!--opt, string, IPv4 address-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address-->test
  </ipv6Address>
</SecondaryDNS>
<DNSEnable>
  <!--opt, bool-->true
</DNSEnable>
<Ipv6Mode>
  <!--opt, object-->
  <ipV6AddressingType>
    <!--opt, enum, subType:string-->ra
  </ipV6AddressingType>
  <ipv6AddressList>
    <!--opt, array, subType:object-->
    <v6Address>
      <!--opt, object, IPv6 address-->
      <id>
        <!--opt, string-->test
      </id>
      <type>
        <!--opt, enum, subType:string-->ra
      </type>
      <address>
        <!--opt, string, IPv6 address-->test
      </address>
      <bitMask>
        <!--opt, string-->test
      </bitMask>
    </v6Address>
  </ipv6AddressList>
</Ipv6Mode>
<DoubleNetRoute>
  <!--opt, object-->
  <ipAddress>
    <!--opt, string, IP address-->test
  </ipAddress>
  <subnetMask>
    <!--opt, string, sub-net mask-->test
  </subnetMask>
</DoubleNetRoute>
<LANConfigMode>
  <!--opt, enum, subType:string-->manual
</LANConfigMode>
<DoubleNetRouteList>
  <!--opt, array, subType:object-->
  <DoubleNetRoute>
    <!--opt, object-->
    <ipAddress>
      <!--opt, string, IP address-->test
    </ipAddress>
    <subnetMask>
      <!--opt, string-->test
    </subnetMask>
  </DoubleNetRoute>
</DoubleNetRouteList>
</IPAddress>
<Wireless>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <wirelessNetworkMode>
    <!--opt, enum, subType:string-->infrastructure
  </wirelessNetworkMode>
  <channel>
    <!--opt, string-->test
  </channel>
  <ssid>
    <!--opt, string-->test
  </ssid>
  <wmmEnabled>
    <!--opt, bool-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--opt, object-->
    <securityMode>
      <!--opt, enum, subType:string-->disable
    </securityMode>
    <WEP>
      <!--opt, object, dep:or,{$.NetworkInterface.Wireless.WireLessSecurity.securityMode,eq,WEP}-->
      <authenticationType>
        <!--req, enum, authentication type, subType:string-->open
      </authenticationType>
      <defaultTransmitKeyIndex>
        <!--req, int-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength>
        <!--opt, int, range:[64,128]-->1
      </wepKeyLength>
    </WEP>
  </WirelessSecurity>
</Wireless>

```

```

    </wepkeyLength>
    <EncryptionKeyList>
      <!--opt, array, subType:object-->
      <encryptionKey>
        <!--req, string-->test
      </encryptionKey>
    </EncryptionKeyList>
  </WEP>
  <WPA>
    <!--opt, object, dep:or,{$.NetworkInterface.Wireless.WirelessSecurity.securityMode,eq,WPA}-->
    <algorithmType>
      <!--req, enum, algorithm type, subType:string-->TKIP
    </algorithmType>
    <sharedKey>
      <!--req, string-->test
    </sharedKey>
    <wpaKeyLength>
      <!--req, int, range:[8,63]-->1
    </wpaKeyLength>
  </WPA>
</WirelessSecurity>
<workScene>
  <!--opt, enum, subType:string-->computerRoom
</workScene>
<protocol>
  <!--opt, enum, protocol type, subType:string-->802.11ac
</protocol>
<protocolRealTime>
  <!--opt, string-->test
</protocolRealTime>
<hideSsid>
  <!--opt, bool-->true
</hideSsid>
<ChannelConfig>
  <!--opt, object-->
  <width>
    <!--opt, enum, subType:string-->auto
  </width>
  <autoWidth>
    <!--ro, opt, string-->test
  </autoWidth>
  <channel>
    <!--opt, string-->test
  </channel>
  <autoChannel>
    <!--ro, opt, string-->test
  </autoChannel>
  <transmitPower>
    <!--opt, enum, subType:int-->9
  </transmitPower>
  <transmitPowerRealTime>
    <!--opt, int-->1
  </transmitPowerRealTime>
  <countryID>
    <!--opt, int-->1
  </countryID>
</ChannelConfig>
<rate>
  <!--opt, enum, subType:string-->2.4GHz
</rate>
<dialUpGroupNo>
  <!--opt, string-->test
</dialUpGroupNo>
<baseNoisevalue>
  <!--ro, opt, int, step:1-->1
</baseNoisevalue>
</Wireless>
<Discovery>
  <!--opt, object-->
  <UPnP>
    <!--req, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
  </UPnP>
  <Zeroconf>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
  </Zeroconf>
</Discovery>
<Link>
  <!--opt, object-->
  <MACAddress>
    <!--req, string-->test
  </MACAddress>
  <autoNegotiation>
    <!--req, bool-->true
  </autoNegotiation>
  <speed>
    <!--req, enum, "10,100,1000,10000", subType:int, desc:"10,100,1000,10000"-->10
  </speed>
  <duplex>

```

```

    <!--req, enum, subType:string-->half
</duplex>
<MTU>
    <!--req, int-->1
</MTU>
<linkMode>
    <!--opt, enum, subType:string-->optical
</linkMode>
</Link>
<defaultConnection>
    <!--opt, bool, whether it is the default network connection-->true
</defaultConnection>
<ActiveMulticast>
    <!--opt, object-->
    <enabled>
        <!--req, bool-->true
    </enabled>
    <streamID>
        <!--req, enum, subType:string-->main
    </streamID>
    <ipV4Address>
        <!--opt, string-->test
    </ipV4Address>
    <ipV6Address>
        <!--opt, string-->test
    </ipV6Address>
    <port>
        <!--opt, int-->1
    </port>
</ActiveMulticast>
<macAddress>
    <!--opt, string-->test
</macAddress>
<EthernetPortList>
    <!--opt, array, ethernet port information list, subType:object-->
    <EthernetPort>
        <!--opt, object, ethernet port information-->
        <id>
            <!--req, int, range:[1,4]-->1
        </id>
        <MACAddress>
            <!--req, string-->test
        </MACAddress>
        <status>
            <!--opt, enum, connection status, subType:string, desc:disconnect"-->connected
        </status>
        <speed>
            <!--req, enum, "10,100,1000,10000", subType:string, desc:"10,100,1000,10000"-->10
        </speed>
    </EthernetPort>
</EthernetPortList>
<Extensions>
    <!--opt, object-->
    <NetworkCardType>
        <!--opt, enum, subType:string-->normal
    </NetworkCardType>
</Extensions>
<interfaceName>
    <!--opt, string-->test
</interfaceName>
</NetworkInterface>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
    <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
    <requestURL>
        <!--ro, req, string, request URL-->null
    </requestURL>
    <statusCode>
        <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
        (Invalid XML Content), 7 (Reboot Required)-->0
    </statusCode>
    <statusString>
        <!--ro, req, enum, status description, subType:string, desc:"OK", "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML
        Content", "Reboot"-->OK
    </statusString>
    <subStatusCode>
        <!--ro, req, string, sub status code, desc:read-only,describe the error reason in detail-->OK
    </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	wifiIsOpenedUnableToConnectWiredNetwork	0x4000A56F	--
4	Invalid Operation	theNetworkCardHasNotEnabledTheIcmpv6Protocol	0x4000A5E4	--

12.2.2.7 Get the information of a specific network interface.

Request URL

GET /ISAPI/System/Network/interfaces/<interfaceID>?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--
serialNumber	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NetworkInterface xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, IP address parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, NIC index-->test
  </id>
  <IPAddress>
    <!--ro, req, object, IP address parameters-->
    <ipVersion>
      <!--ro, req, enum, IP address version, subType:string-->v4
    </ipVersion>
    <addressingType>
      <!--ro, req, enum, address type, subType:string-->static
    </addressingType>
    <ipAddress>
      <!--ro, opt, string, IPv4 address-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string, IPv4 subnet mask-->test
    </subnetMask>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address-->test
    </ipv6Address>
    <bitMask>
      <!--ro, opt, string, IPv6 subnet mask-->test
    </bitMask>
    <DefaultGateway>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IPv4 address-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </DefaultGateway>
    <PrimaryDNS>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IPv4 address-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </PrimaryDNS>
    <SecondaryDNS>
      <!--ro, opt, object-->
      <ipAddress>
        <!--ro, opt, string, IPv4 address-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address-->test
      </ipv6Address>
    </SecondaryDNS>
    <DNSEnable>
      <!--ro, opt, bool-->true
    </DNSEnable>
    <IPv6Mode>
      <!--ro, opt, object-->
    </IPv6Mode>
  </IPAddress>

```



```

<!--ro, opt, enum, subType:string-->ra
</ipV6AddressingType>
<ipV6AddressList>
  <!--ro, opt, array, subType:object-->
  <v6Address>
    <!--ro, opt, object, IPv6 address-->
    <id>
      <!--ro, opt, string-->test
    </id>
    <type>
      <!--ro, opt, enum, subType:string-->ra
    </type>
    <address>
      <!--ro, opt, string, IPv6 address-->test
    </address>
    <bitMask>
      <!--ro, opt, string, IPv6 subnet mask-->test
    </bitMask>
  </v6Address>
</ipV6AddressList>
</Ipv6Mode>
<DoubleNetRoute>
  <!--ro, opt, object-->
  <ipAddress>
    <!--ro, opt, string, IP address-->test
  </ipAddress>
  <subnetMask>
    <!--ro, opt, string, subnet mask-->test
  </subnetMask>
</DoubleNetRoute>
<LANConfigMode>
  <!--ro, opt, enum, subType:string-->manual
</LANConfigMode>
<DoubleNetRouteList>
  <!--ro, opt, array, subType:object-->
  <DoubleNetRoute>
    <!--ro, opt, object-->
    <ipAddress>
      <!--ro, opt, string, IP address-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string, subnet mask-->test
    </subnetMask>
  </DoubleNetRoute>
</DoubleNetRouteList>
</IPAddress>
<Wireless>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <wirelessNetworkMode>
    <!--ro, opt, enum, subType:string-->infrastructure
  </wirelessNetworkMode>
  <channel>
    <!--ro, opt, string-->test
  </channel>
  <ssid>
    <!--ro, opt, string, SSID (Service Set Identifier)-->test
  </ssid>
  <wmmEnabled>
    <!--ro, opt, bool-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--ro, opt, object-->
    <securityMode>
      <!--ro, opt, enum, subType:string-->disable
    </securityMode>
    <WEP>
      <!--ro, opt, object, dep:or,{$.NetworkInterface.Wireless.WirelessSecurity.securityMode,eq,WEP}-->
      <authenticationType>
        <!--ro, req, enum, authentication type, subType:string-->open
      </authenticationType>
      <defaultTransmitKeyIndex>
        <!--ro, req, int-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength>
        <!--ro, opt, int, range:[64,128]-->1
      </wepKeyLength>
      <EncryptionKeyList>
        <!--ro, opt, array, subType:object-->
        <encryptionKey>
          <!--ro, req, string, access key-->test
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA>
      <!--ro, opt, object, dep:or,{$.NetworkInterface.Wireless.WirelessSecurity.securityMode,eq,WPA}-->
      <algorithmType>
        <!--ro, req, enum, algorithm type, subType:string-->TKIP
      </algorithmType>
      <sharedKey>

```

```

        <!--ro, req, string-->test
    </sharedKey>
    <wpaKeyLength>
        <!--ro, req, int, range:[8,63]-->1
    </wpaKeyLength>
</WPA>
</WirelessSecurity>
<workScene>
    <!--ro, opt, enum, subType:string-->computerRoom
</workScene>
<protocol>
    <!--ro, opt, enum, protocol type, subType:string-->802.11ac
</protocol>
<protocolRealTime>
    <!--ro, opt, string, real-time mode-->test
</protocolRealTime>
<hideSsid>
    <!--ro, opt, bool-->true
</hideSsid>
<ChannelConfig>
    <!--ro, opt, object-->
    <width>
        <!--ro, opt, enum, subType:string-->auto
    </width>
    <autoWidth>
        <!--ro, opt, string-->test
    </autoWidth>
    <channel>
        <!--ro, opt, string-->test
    </channel>
    <autoChannel>
        <!--ro, opt, string-->test
    </autoChannel>
    <transmitPower>
        <!--ro, opt, enum, subType:int-->9
    </transmitPower>
    <transmitPowerRealTime>
        <!--ro, opt, int-->1
    </transmitPowerRealTime>
    <countryID>
        <!--ro, opt, int-->1
    </countryID>
</ChannelConfig>
<rate>
    <!--ro, opt, enum, subType:string-->2.4GHz
</rate>
<dialUpGroupNo>
    <!--ro, opt, string-->test
</dialUpGroupNo>
<baseNoisevalue>
    <!--ro, opt, int, step:1-->1
</baseNoisevalue>
</Wireless>
<Discovery>
    <!--ro, opt, object-->
    <UPnP>
        <!--ro, req, object-->
        <enabled>
            <!--ro, req, bool-->true
        </enabled>
    </UPnP>
    <Zeroconf>
        <!--ro, opt, object-->
        <enabled>
            <!--ro, req, bool-->true
        </enabled>
    </Zeroconf>
</Discovery>
<Link>
    <!--ro, opt, object-->
    <MACAddress>
        <!--ro, req, string, MAC address-->test
    </MACAddress>
    <autoNegotiation>
        <!--ro, req, bool-->true
    </autoNegotiation>
    <speed>
        <!--ro, req, enum, "10,100,1000,10000", subType:int, desc:"10,100,1000,10000"-->10
    </speed>
    <duplex>
        <!--ro, req, enum, subType:string-->half
    </duplex>
    <MTU>
        <!--ro, req, int, MTU-->1
    </MTU>
    <linkMode>
        <!--ro, opt, enum, connect mode, subType:string-->optical
    </linkMode>
</Link>
<defaultConnection>
    <!--ro, opt, bool, wether to enable default network connection-->true
</defaultConnection>
<ActiveMulticast>
    <!--ro, opt, object-->

```

```

<!--ro, opt, object-->
<enabled>
  <!--ro, req, bool, whether to enable or not-->true
</enabled>
<streamID>
  <!--ro, req, enum, subType:string-->main
</streamID>
<ipV4Address>
  <!--ro, opt, string, IPv4 address-->test
</ipV4Address>
<ipV6Address>
  <!--ro, opt, string, IPv6 address-->test
</ipV6Address>
<port>
  <!--ro, opt, int, port No.-->1
</port>
</ActiveMulticast>
<macAddress>
  <!--ro, opt, string, MAC Address-->test
</macAddress>
<EthernetPortList>
  <!--ro, opt, array, network interface information list, subType:object-->
  <EthernetPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int, range:[1,4]-->1
    </id>
    <MACAddress>
      <!--ro, req, string, MAC Address-->test
    </MACAddress>
    <status>
      <!--ro, opt, enum, connection status, subType:string, desc:disconnect"-->connected
    </status>
    <speed>
      <!--ro, req, enum, speed, subType:string, desc:"10,100,1000,10000"-->10
    </speed>
  </EthernetPort>
</EthernetPortList>
<Extensions>
  <!--ro, opt, object, extended information-->
  <NetworkCardType>
    <!--ro, opt, enum, subType:string-->normal
  </NetworkCardType>
</Extensions>
<interfaceName>
  <!--ro, opt, string-->test
</interfaceName>
</NetworkInterface>

```

12.2.3 ISUP Access Management

12.2.3.1 Get the configuration capability of accessing servers via ISUP (EHome)

Request URL

GET /ISAPI/System/Network/Ehome/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<Ehome xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration capability of accessing servers via ISUP, attr:version{req, string, protocolVersion}, desc:0-make no sense, 1-
  automatic, 2-wired network preferred, 3-wired network, 4-3G/4G/GPRS-->
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:whether to enable-->>false
  </enabled>
  <id min="1" max="1">
    <!--ro, opt, string, ISUP center ID, attr:min{req, int},max{req, int}-->test
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, string, address type, attr:opt{req, string}-->ipaddress
  </addressingFormatType>
  <GPRSAddressingFormatType opt="ipaddress">
    <!--ro, opt, string, GPRS address type. If this node is not returned, the address type will be determined by addressingFormatType which indicates that
    the address types configured by different networks do not need to be distinguished, attr:opt{req, string}-->ipaddress
  </GPRSAddressingFormatType>
  <hostName min="1" max="128">
    <!--ro, opt, string, host name, attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="1" max="16">
    <!--ro, opt, string, IPv4 address, attr:min{req, int},max{req, int}-->192.168.1.112
  </ipAddress>
  <ipv6Address min="1" max="128">
    <!--ro, opt, string, IPv6 address, attr:min{req, int},max{req, int}-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="1" max="65535">
    <!--ro, opt, int, port No., attr:min{req, int},max{req, int}-->74
  </portNo>
  <deviceID min="1" max="64">
    <!--ro, req, string, attr:min{req, int},max{req, int}-->test1234
  </deviceID>
  <registerStatus opt="true,false">
    <!--ro, opt, bool, registration status, attr:opt{req, string}-->>true
  </registerStatus>
  <ServerAddressList>
    <!--ro, opt, array, List of servers, subType:object-->
    <ServerAddress>
      <!--ro, opt, object, server information-->
      <id>
        <!--ro, req, enum, server type, subType:string, desc:ISUP center ID-->alarm
      </id>
      <ipAddress>
        <!--ro, req, string, server IP address-->192.168.1.113
      </ipAddress>
      <portNo>
        <!--ro, req, int, server port No.-->76
      </portNo>
    </ServerAddress>
  </ServerAddressList>
  <key min="1" max="64">
    <!--ro, opt, string, secret key, which is used for registration via ISUP5.0 and OTAP, dep:and,{$.Ehome.protocolVersion,eq,OTAP},
    {$.Ehome.protocolVersion,eq,v5.0}, attr:min{req, int},max{req, int}-->test
  </key>
  <version min="1" max="64">
    <!--ro, opt, string, version, attr:min{req, int},max{req, int}-->test
  </version>
  <voiceDeviceType opt="bluetooth,client,local">
    <!--ro, opt, string, two-way audio device type: "bluetooth"-bluetooth device, "client", "local", attr:opt{req, string}-->test
  </voiceDeviceType>
  <protocolVersion opt="v2.0,v2.6,v4.0,v5.0,OTAP">
    <!--ro, opt, string, protocol version, attr:opt{req, string}-->v5.0
  </protocolVersion>
  <network opt="0,1,2,3,4,5,6">
    <!--ro, opt, int, network type, attr:opt{req, string}-->1
  </network>
  <pictureServerID min="1" max="64">
    <!--ro, opt, int, Linked picture storage server ID (related URI: /ISAPI/System/PictureServer), attr:min{req, int},max{req, int}-->1
  </pictureServerID>
  <ISUPID min="1" max="64">
    <!--ro, opt, int, ISUP multi-platform ID, attr:min{req, int},max{req, int}-->1
  </ISUPID>
  <isSupportTest>
    <!--ro, opt, bool, whether it supports server test via ISUP (EHome)-->>true
  </isSupportTest>
  <periodicTestEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable periodic test, attr:opt{req, string}, desc:when it is enabled, the host will send a Test event to the server every
    configured interval for checking whether the connection is normal (similar to heartbeat detection)-->>true
  </periodicTestEnabled>
  <periodicTestTime min="10" max="86400">
    <!--ro, opt, int, periodic test interval,unit: second. This node is valid when <periodicTestEnabled> is true, attr:min{req, int},max{req, int},
    desc:periodic test interval,unit: second. This node is valid when <periodicTestEnabled> is true-->50
  </periodicTestTime>
  <isSupportMulticenterTest>
    <!--ro, opt, bool-->true
  </isSupportMulticenterTest>
</Ehome>

```

12.2.3.2 Set the EHome server access parameters

Request URL

PUT /ISAPI/System/Network/Ehome?centerID= <centerID>

Query Parameter

Parameter Name	Parameter Type	Description
centerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Ehome xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--opt, bool-->false
  </enabled>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, this node is valid only when <addressingFormatType> is "hostname", range:[1,128]-->time.windows.com
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,16], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], desc:port No.-->74
  </portNo>
  <deviceID>
    <!--req, string, port No., range:[1,64]-->test1234
  </deviceID>
  <registerStatus>
    <!--ro, opt, bool, device ID-->true
  </registerStatus>
  <ServerAddressList>
    <!--ro, opt, array, list of servers, subType:object-->
    <ServerAddress>
      <!--ro, opt, object, server information-->
      <id>
        <!--ro, req, enum, server type, subType:string, desc:server type-->alarm
      </id>
      <ipAddress>
        <!--ro, req, string, server IP address, range:[1,16]-->192.168.1.113
      </ipAddress>
      <portNo>
        <!--ro, req, int, server port No., range:[1,65535]-->76
      </portNo>
    </ServerAddress>
  </ServerAddressList>
  <key>
    <!--ro, opt, string, secret key, which is used for registration via ISUP5.0 and OTAP, range:[1,64], dep:and,{$.Ehome.protocolVersion,eq,OTAP},
    {$.Ehome.protocolVersion,eq,v5.0}-->test
  </key>
  <version>
    <!--ro, opt, string, version number, range:[1,64]-->2.3.0.4
  </version>
  <voiceDeviceType>
    <!--opt, enum, two-way audio device type, subType:string, desc:"bluetooth" (bluetooth device), "client", "Local"-->bluetooth
  </voiceDeviceType>
  <protocolVersion>
    <!--opt, enum, protocol version No., subType:string, desc:if this node is set to "v2.6", "v4.0" or "v5.0", the device will firstly use this protocol to
    register. If this node is not returned, the protocol version will be determined by <version-->v5.0
  </protocolVersion>
  <network>
    <!--opt, enum, 0-make no sense,1-automatic,2-wired network preferred,3-wired network,4-3G/4G/GPRS, subType:string, desc:0-make no sense,1-automatic,2-
    wired network preferred,3-wired network,4-3G/4G/GPRS-->1
  </network>
</Ehome>
```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, read-only, describe the error reason in detail, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	操作错误	ISUPAndEZVIZExclusive	0x4000215A	--

12.2.3.3 Get the ISUP server access parameters

Request URL

GET /ISAPI/System/Network/Ehome?centerID=<centerID>

Query Parameter

Parameter Name	Parameter Type	Description
centerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Ehome xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, opt, bool, desc:by default, it is disabled-->>false
  </enabled>
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"ipaddress"-IP address, "hostname"-host name-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name,this node is valid only when <addressingFormatType> is "hostname", range:[1,128]-->time.windows.com
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,16], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], desc:port No.-->74
  </portNo>
  <deviceID>
    <!--ro, req, string, ISUP device serial No., range:[1,64]-->test1234
  </deviceID>
  <registerStatus>
    <!--ro, opt, bool, ro,registration status-->>true
  </registerStatus>
  <ServerAddressList>
    <!--ro, opt, array, server information List, subType:object-->
    <ServerAddress>
      <!--ro, opt, object, server information-->
      <id>
        <!--ro, req, enum, server type, subType:string, desc:ID-->alarm
      </id>
      <ipAddress>
        <!--ro, req, string, server IP address, range:[1,16]-->192.168.1.113
      </ipAddress>
      <portNo>
        <!--ro, req, int, server port No., range:[1,65535]-->76
      </portNo>
    </ServerAddress>
  </ServerAddressList>
  <key>
    <!--ro, opt, string, encryption key of ISUP version 5.0,which is required for ISUP version 50, range:[1,64], dep:and,{$.Ehome.protocolVersion,eq,OTAP},
    {$.Ehome.protocolVersion,eq,v5.0}-->test
  </key>
  <version>
    <!--ro, opt, string, version No., range:[1,64]-->2.3.0.4
  </version>
  <voiceDeviceType>
    <!--ro, opt, enum, two-way audio device type, subType:string, desc:"bluetooth"-bluetooth device,"client","local"-->bluetooth
  </voiceDeviceType>
  <protocolVersion>
    <!--ro, opt, enum, ISUP version No., subType:string, desc:If this node is not returned,the protocol version will be determined by <version>: v2.0, v2.6,
    v4.0, v5.0, OTAP;-->v5.0
  </protocolVersion>
  <network>
    <!--ro, opt, enum, Network, subType:string, desc:0-make no sense,1-automatic,2-wired network preferred,3-wired network,4-3G/4G/GPRS-->1
  </network>
</Ehome>
```

12.2.4 Picture Storage Protocol Access Management

12.2.4.1 Get the configuration capability of the picture storage server

Request URL

GET /ISAPI/System/PictureServer/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "PictureServerInformation": {
    /*ro, opt, object, picture server information*/
    "pictureServerType": {
      /*ro, opt, object, picture server type*/
      "@opt": "tomcat,VRB,cLoudStorage,KMS,null",
```

```

/*ro, opt, string, optional value*/
"#text": "cloudStorage"
/*ro, req, string, picture server type*/
},
"addressingFormatType": {
/*ro, opt, object, address type of the picture storage server*/
"@opt": "ipaddress,hostname",
/*ro, opt, string, optional value*/
"#text": "ipaddress"
/*ro, req, string, address type of the picture storage server*/
},
"hostName": {
/*ro, opt, object, domain name of the picture storage server*/
"@min": 0,
/*ro, opt, int, minimum value*/
"@max": 64,
/*ro, opt, int, maximum value*/
"#text": "pc3-netsdk"
/*ro, opt, string, domain name of the picture storage server*/
},
"ipv4Address": {
/*ro, opt, object, IPv4 server address of the picture storage server*/
"@min": 0,
/*ro, opt, int, minimum value*/
"@max": 64,
/*ro, opt, int, maximum value*/
"#text": "10.17.133.46"
/*ro, opt, string, IPv4 server address of the picture storage server*/
},
"ipv6Address": {
/*ro, opt, object, IPv6 server address of the picture storage server*/
"@min": 0,
/*ro, opt, int, minimum value*/
"@max": 128,
/*ro, opt, int, maximum value*/
"#text": "2001:A304:6101:1::E0:F726:4E58"
/*ro, opt, string, IPv6 server address of the picture storage server*/
},
"portNo": {
/*ro, opt, object, picture storage server port*/
"@min": 1024,
/*ro, opt, int, minimum value*/
"@max": 65535,
/*ro, opt, int, maximum value*/
"#text": 8000
/*ro, req, int, picture storage server port*/
},
"underlyProtocol": {
/*ro, opt, object, Communication Protocol*/
"@opt": "http,https"
/*ro, opt, string, Communication Protocol*/
},
"cloudStorage": {
/*ro, opt, object, configuration of cloud storage server parameters*/
"cloudManageHttpPort": {
/*ro, opt, object, HTTP port No. for central management of the cloud storage server*/
"@min": 1024,
/*ro, opt, int, minimum value*/
"@max": 65535,
/*ro, opt, int, maximum value*/
"#text": 8009
/*ro, opt, int, HTTP port No. for central management of the cloud storage server*/
},
"cloudTransDataPort": {
/*ro, opt, object, data transmission port No. of the cloud storage server*/
"@min": 1024,
/*ro, opt, int, minimum value*/
"@max": 65535,
/*ro, opt, int, maximum value*/
"#text": 9001
/*ro, opt, int, data transmission port No. of the cloud storage server*/
},
"cloudCmdPort": {
/*ro, opt, object, signaling port No. of the cloud storage server*/
"@min": 1024,
/*ro, opt, int, minimum value*/
"@max": 65535,
/*ro, opt, int, maximum value*/
"#text": 8008
/*ro, opt, int, signaling port No. of the cloud storage server*/
},
"cloudHeartBeatPort": {
/*ro, opt, object, heartbeat port No. of the cloud storage server*/
"@min": 1024,
/*ro, opt, int, minimum value*/
"@max": 65535,
/*ro, opt, int, maximum value*/
"#text": 9999
/*ro, opt, int, heartbeat port No. of the cloud storage server*/
},
"cloudStorageHttpPort": {
/*ro, opt, object, HTTP port No. of the cloud storage server*/
"@min": 1024,
/*ro, opt, int, minimum value*/

```



```

/*ro, opt, int, maximum value*/
"@max": 65535,
/*ro, opt, int, maximum value*/
"#text": 8083
/*ro, opt, int, HTTP port No. of the cloud storage server*/
},
"cloudUsername": {
/*ro, opt, object, user name of the cloud storage server*/
"@min": 0,
/*ro, opt, int, minimum value*/
"@max": 32,
/*ro, opt, int, maximum value*/
"#text": "username"
/*ro, opt, string, user name of the cloud storage server*/
},
"cloudPassword": {
/*ro, opt, object, user name of the cloud storage server*/
"@min": 0,
/*ro, opt, int, minimum value*/
"@max": 32,
/*ro, opt, int, maximum value*/
"#text": "12345"
/*ro, opt, string, user name of the cloud storage server*/
},
"cloudPoolId": {
/*ro, opt, object, cloud storage pool ID (int format)*/
"@min": 1,
/*ro, opt, int, minimum value*/
"@max": 4294967295,
/*ro, opt, float, maximum value*/
"#text": 1
/*ro, opt, int, cloud storage pool ID*/
},
"cloudProtocolVersion": {
/*ro, opt, object, protocol version of the cloud storage server*/
"@min": 0,
/*ro, opt, int, minimum value*/
"@max": 32,
/*ro, opt, int, maximum value*/
"#text": "V2.0"
/*ro, req, string, protocol version of the cloud storage server*/
},
"cloudAccessKey": {
/*ro, opt, object, access key of the cloud storage server*/
"@min": 0,
/*ro, opt, int, minimum value*/
"@max": 64
/*ro, opt, int, maximum value*/
},
"cloudSecretKey": {
/*ro, opt, object, secret key of the cloud storage server*/
"@min": 0,
/*ro, opt, int, minimum value*/
"@max": 64
/*ro, opt, int, maximum value*/
},
"resourceID": {
/*ro, opt, object, cloud storage pool ID (string format), dep:and,{$.PictureServerInformation.cloudStorage.cloudProtocolVersion,eq,V2.0},
desc:both resourceID and cloudPoolId can be used to configure the cloud storage pool ID. But different from cloudPoolId, the value of resourceID can be in
string format. When both resourceID and cloudPoolId are configured, cloudPoolId has priority over resourceID. If this node is not supported, you can
configure the pool ID via cloudPoolId.*/
"@min": 1,
/*ro, opt, int, the minimum length of the value*/
"@max": 256
/*ro, opt, int, the maximum length of the value*/
}
},
"pictureServerID": {
/*ro, opt, object, picture storage server ID*/
"@min": 1,
/*ro, opt, int, minimum value*/
"@max": 2,
/*ro, opt, int, maximum value*/
"#text": 1
/*ro, opt, int, picture storage server ID: /ISAPI/System/PictureServer*/
}
}
}

```

12.2.4.2 Set the picture storage server parameters

Request URL

PUT /ISAPI/System/PictureServer?format=json

Query Parameter

None

Request Message

```

{
  "PictureServerInformation": {
    /*req, object, picture server information*/
    "pictureServerType": "cloudStorage",
    /*req, enum, picture server type, subType:string, desc:"tomact", "VRB", "cloudStorage", "KMS". When the value is null, it indicates that the servers
are all closed*/
    "addressingFormatType": "ipaddress",
    /*req, enum, address type of the picture storage server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)*/
    "hostName": "pc3-netsdk",
    /*opt, string, domain name of the picture storage server*/
    "ipv4Address": "10.17.133.46",
    /*opt, string, IPv4 server address of the picture storage server*/
    "ipv6Address": "2001:A304:6101:1::E0:F726:4E58",
    /*opt, string, IPv6 server address of the picture storage server*/
    "portNo": 1024,
    /*req, int, picture storage server port, range:[1024,65535]*/
    "underlyingProtocol": "http",
    /*opt, enum, communication protocol, subType:string, desc:"http", "https"*/
    "cloudStorage": {
      /*opt, object, configuration of cloud storage server parameters, dep:and,{$.PictureServerInformation.pictureServerType,eq,cloudStorage}*/
      "cloudManageHttpPort": 6120,
      /*opt, int, HTTP port No. for central management of the cloud storage server, which can be used for downloading the URLs of pictures and
intelligent structuralization data, range:[1024,65535]*/
      "cloudTransDataPort": 9001,
      /*opt, int, data transmission port No. of the cloud storage server, range:[1024,65535]*/
      "cloudCmdPort": 6011,
      /*opt, int, signaling port No. of the cloud storage server, range:[1024,65535]*/
      "cloudHeartBeatPort": 9999,
      /*opt, int, heartbeat port No. of the cloud storage server, range:[1024,65535]*/
      "cloudStorageHttpPort": 8083,
      /*opt, int, HTTP port No. of the cloud storage server, range:[1024,65535]*/
      "cloudUsername": "username",
      /*opt, string, user name of the cloud storage server, range:[0,32]*/
      "cloudPassword": "12345",
      /*opt, string, user name of the cloud storage server*/
      "cloudPoolId": 1,
      /*opt, int, cloud storage pool ID (int format), range:[1,4294967295], desc:by default, it is 1 if there is no configured device*/
      "cloudProtocolVersion": "V2.0",
      /*req, string, protocol version of the cloud storage server, range:[0,32]*/
      "cloudAccessKey": "test",
      /*opt, string, access key of the cloud storage server, range:[0,64], dep:and,
{$.PictureServerInformation.cloudStorage.cloudProtocolVersion,eq,V2.0}*/
      "cloudSecretKey": "test",
      /*opt, string, secret key of the cloud storage server, range:[0,64], dep:and,
{$.PictureServerInformation.cloudStorage.cloudProtocolVersion,eq,V2.0}*/
      "resourceID": "test"
      /*opt, string, source pool ID (string format), range:[1,256], dep:and,{$.PictureServerInformation.cloudStorage.cloudProtocolVersion,eq,V2.0},
desc:both resourceID and cloudPoolId can be used to configure the cloud storage pool ID. But different from cloudPoolId, the value of resourceID can be in
string format. When both resourceID and cloudPoolId are configured, cloudPoolId has priority over resourceID. If this node is not supported, you can
configure the pool ID via cloudPoolId.*/
    }
  }
}

```

Response Message

```

{
  "requestURL": "test",
  /*ro, opt, string, URI*/
  "statusCode": "test",
  /*ro, opt, string, status code*/
  "statusString": "test",
  /*ro, opt, string, status description*/
  "subStatusCode": "test",
  /*ro, opt, string, sub status code*/
  "errorCode": 1,
  /*ro, req, int, error code*/
  "errorMsg": "ok"
  /*ro, req, string, error message*/
}

```

12.2.4.3 Get the parameters of the picture storage server

Request URL

GET /ISAPI/System/PictureServer?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "PictureServerInformation": {
    /*ro, req, object, picture server information*/
    "pictureServerType": "cloudStorage",
    /*ro, req, enum, picture server type, subType:string, desc:"tomact", "VRB", "cloudStorage", "KMS". When the value is null, it indicates that the
servers are all closed*/
    "addressingFormatType": "ipaddress",
    /*ro, req, enum, address type of the picture storage server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)*/
    "hostName": "pc3-netsdk",
    /*ro, opt, string, domain name of the picture storage server*/
    "ipv4Address": "10.17.133.46",
    /*ro, opt, string, IPv4 server address of the picture storage server*/
    "ipv6Address": "2001:A304:6101:1::E0:F726:4E58",
    /*ro, opt, string, IPv6 server address of the picture storage server*/
    "portNo": 1024,
    /*ro, req, int, picture storage server port, range:[1024,65535]*/
    "underlyingProtocol": "http",
    /*ro, opt, enum, communication protocol, subType:string, desc:"http", "https"*/
    "cloudStorage": {
      /*ro, opt, object, configuration of cloud storage server parameters, dep:and,{$.PictureServerInformation.pictureServerType,eq,cloudStorage}*/
      "cloudManageHttpPort": 6120,
      /*ro, opt, int, HTTP port No. for central management of the cloud storage server, which can be used for downloading the URLs of pictures and
intelligent structuralization data, range:[1024,65535]*/
      "cloudTransDataPort": 9001,
      /*ro, opt, int, data transmission port No. of the cloud storage server, range:[1024,65535]*/
      "cloudCmdPort": 6011,
      /*ro, opt, int, signaling port No. of the cloud storage server, range:[1024,65535]*/
      "cloudHeartBeatPort": 9999,
      /*ro, opt, int, heartbeat port No. of the cloud storage server, range:[1024,65535]*/
      "cloudStorageHttpPort": 8083,
      /*ro, opt, int, HTTP port No. of the cloud storage server, range:[1024,65535]*/
      "cloudUsername": "username",
      /*ro, opt, string, user name of the cloud storage server, range:[0,32]*/
      "cloudPassword": "12345",
      /*ro, opt, string, user name of the cloud storage server*/
      "cloudPoolId": 1,
      /*ro, opt, int, cloud storage pool ID (int format), range:[1,4294967295], desc:by default, it is 1 if there is no configured device*/
      "cloudProtocolVersion": "V2.0",
      /*ro, req, string, protocol version of the cloud storage server, range:[0,32]*/
      "cloudAccessKey": "test",
      /*ro, opt, string, access key of the cloud storage server, range:[0,64], dep:and,
{$.PictureServerInformation.cloudStorage.cloudProtocolVersion,eq,V2.0}*/
      "cloudSecretKey": "test",
      /*ro, opt, string, secret key of the cloud storage server, range:[0,64], dep:and,
{$.PictureServerInformation.cloudStorage.cloudProtocolVersion,eq,V2.0}*/
      "resourceID": "test"
      /*ro, opt, string, cloud storage pool ID (string format), range:[1,256], dep:and,
{$.PictureServerInformation.cloudStorage.cloudProtocolVersion,eq,V2.0}, desc:both resourceID and cloudPoolId can be used to configure the cloud storage
pool ID. But different from cloudPoolId, the value of resourceID can be in string format. When both resourceID and cloudPoolId are configured, cloudPoolId
has priority over resourceID. If this node is not supported, you can configure the pool ID via cloudPoolId.*/
    }
  }
}
```

12.2.5 WiFi Network Management

12.2.5.1 Get a specific Wi-Fi hotspot

Request URL

GET /ISAPI/System/Network/interfaces/<interfaceID>/wireless/accessPointList/<pointsID>

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--
pointsID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<accessPoint xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <networkMode>
    <!--ro, opt, enum, network mode, subType:string, desc:"infrastructure,adhoc"-->infrastructure
  </networkMode>
  <channel>
    <!--ro, opt, string, channel No.-->test
  </channel>
  <ssid>
    <!--ro, req, string-->test
  </ssid>
  <speed>
    <!--ro, opt, int, speed, unit:Mbps-->0
  </speed>
  <signalStrength>
    <!--ro, opt, int, signal strength-->0
  </signalStrength>
  <securityMode>
    <!--ro, req, enum, "disable,WEP,WPA-personal,WPA2-personal,WPA-RADIUS,WPA-enterprise,WPA2-enterprise", subType:string, desc:"disable, WEP, WPA-personal, WPA2-personal, WPA-RADIUS, WPA-enterprise, WPA2-enterprise"-->disable
  </securityMode>
  <connected>
    <!--ro, opt, bool-->true
  </connected>
  <connecting>
    <!--ro, opt, bool, whether the device is connecting to the Wi-Fi, desc:whether the device is connecting to the Wi-Fi-->true
  </connecting>
  <ipAddress>
    <!--ro, opt, string, IPv4 Address-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->test
  </ipv6Address>
  <subnetMask>
    <!--ro, opt, string, subnet mask-->test
  </subnetMask>
  <defaultGateway>
    <!--ro, opt, string, default gateway-->test
  </defaultGateway>
</accessPoint>

```

12.2.5.2 Get the wireless connection status of a network port

Request URL

GET /ISAPI/System/Network/interfaces/<interfaceID>/wireless/connectStatus

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WirelessConnectStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, wireless connection status, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable Wi-Fi-->true
  </enabled>
  <status>
    <!--ro, opt, enum, Wi-Fi connection status, subType:string, desc:"fail"-connecting failed, "success"-connected, "connecting"-connecting, "nomodule"-no module-->fail
  </status>
  <ssid>
    <!--ro, opt, string, SSID-->test
  </ssid>
  <signalStrength>
    <!--ro, opt, int, signal strength, range:[0,100]-->1
  </signalStrength>
  <ipAddress>
    <!--ro, opt, string, IP address-->test
  </ipAddress>
</WirelessConnectStatus>
```

12.2.5.3 Set Wi-Fi parameters of a specific network interface

Request URL

PUT /ISAPI/System/Network/interfaces/<interfaceID>/wireless

Query Parameter

Parameter Name	Parameter Type	Description
interfaceID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Wireless xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <wirelessNetworkMode>
    <!--opt, enum, subType:string-->infrastructure
  </wirelessNetworkMode>
  <channel>
    <!--opt, string-->test
  </channel>
  <ssid>
    <!--opt, string-->test
  </ssid>
  <wmmEnabled>
    <!--opt, bool-->true
  </wmmEnabled>
  <WirelessSecurity>
    <!--opt, object-->
    <securityMode>
      <!--opt, enum, subType:string-->disable
    </securityMode>
    <WEP>
      <!--opt, object-->
      <authenticationType>
        <!--opt, enum, subType:string-->open
      </authenticationType>
      <defaultTransmitKeyIndex>
        <!--req, int-->1
      </defaultTransmitKeyIndex>
      <wepKeyLength>
        <!--opt, int, range:[64,128]-->1
      </wepKeyLength>
      <EncryptionKeyList>
        <!--opt, array, subType:object-->
        <encryptionKey>
          <!--opt, string-->test
        </encryptionKey>
      </EncryptionKeyList>
    </WEP>
    <WPA>
      <!--opt, object-->
      <algorithmType>
        <!--opt, enum, subType:string-->TKIP
      </algorithmType>
      <sharedKey>
        <!--req, string-->test
      </sharedKey>
    </WPA>
  </WirelessSecurity>
</Wireless>
```

```

</wpaKeyLength>
  <!--req, int, range:[8,63]-->1
</wpaKeyLength>
</WPA>
</WirelessSecurity>
<workScene>
  <!--opt, enum, subType:string-->computerRoom
</workScene>
<protocol>
  <!--opt, enum, subType:string-->802.11ac
</protocol>
<protocolRealTime>
  <!--opt, string-->test
</protocolRealTime>
<hideSsid>
  <!--opt, bool-->true
</hideSsid>
<ChannelConfig>
  <!--opt, object-->
  <width>
    <!--opt, enum, subType:string-->auto
  </width>
  <autoWidth>
    <!--opt, string-->test
  </autoWidth>
  <channel>
    <!--opt, string-->test
  </channel>
  <autoChannel>
    <!--opt, string-->test
  </autoChannel>
  <transmitPower>
    <!--opt, enum, subType:int-->9
  </transmitPower>
  <transmitPowerRealTime>
    <!--opt, int-->1
  </transmitPowerRealTime>
  <countryID>
    <!--opt, int-->1
  </countryID>
</ChannelConfig>
<rate>
  <!--opt, enum, subType:string-->2.4GHz
</rate>
<dialUpGroupNo>
  <!--opt, string-->test
</dialUpGroupNo>
<baseNoisevalue>
  <!--opt, int-->1
</baseNoisevalue>
<ignoreAddedNetwork>
  <!--opt, bool-->true
</ignoreAddedNetwork>
<ssidType>
  <!--opt, enum, subType:string-->wirelessSsid
</ssidType>
<extendTime>
  <!--opt, int-->0
</extendTime>
<remainingTime>
  <!--opt, int-->0
</remainingTime>
<EIRPEnabled>
  <!--opt, bool-->true
</EIRPEnabled>
<antennaGain>
  <!--opt, int, range:[0,20], unit:dBm-->1
</antennaGain>
<id>
  <!--opt, int-->0
</id>
<operateType>
  <!--opt, enum, subType:string-->add
</operateType>
<bssid>
  <!--opt, string-->test
</bssid>
</Wireless>

```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, read-only,status code: 0,1-OK,2-Device Busy,3-Device Error,4-Invalid Operation,5-Invalid XML Format,6-Invalid XML Content,7-Reboot
    Required,9-Additional Error, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description: OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML Content,Reboot,Additional
    Error, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, read-only,describe the error reason in detail-->OK
  </subStatusCode>
</ResponseStatus>
```

12.3 Information Security

12.3.1 Security Upgrading of Network Communication Protocols

12.3.1.1 Get the configuration capability of the security mode level of the private protocol

Request URL

GET /ISAPI/Security/CommuMode/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "CommuMode": {
    /*ro, req, object, security mode of the private protocol*/
    "mode": {
      /*ro, req, object, security mode level of the private protocol*/
      "@opt": "compatibilityMode,safeMode"
      /*ro, req, string, options, desc:"compatibilityMode" (compatibility mode), "safeMode" (security mode)*/
    }
  }
}
```

12.3.1.2 Get the parameters of the security mode level of the private protocol

Request URL

GET /ISAPI/Security/CommuMode?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "CommuMode": {
    /*ro, req, object, security mode of the private protocol*/
    "mode": "safeMode"
    /*ro, req, enum, security mode level of the private protocol, subType:string, desc:"compatibilityMode" (compatibility mode), "safeMode" (security
    mode)*/
  }
}
```

12.3.1.3 Set the parameters of the security mode level of the private protocol

Request URL

PUT /ISAPI/Security/CommuMode?format=json

Query Parameter

None

Request Message

```
{
  "CommuMode": {
    /*req, object*/
    "mode": "safeMode"
    /*req, enum, security mode level of the private protocol, subType:string, desc:"compatibilityMode" (compatibility mode), "safeMode" (security mode)*/
  }
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

12.4 Video (General)

12.4.1 Image Adjustment

12.4.1.1 Set image parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version(req, string, protocolVersion)-->
  <id>
    <!--req, int-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <videoInputID>
    <!--req, int-->1
  </videoInputID>
  <Focusconfiguration>
    <!--opt, object-->
    <focusStyle>
      <!--req, enum, subType:string-->AUTO
    </focusStyle>
    <focusLimited>
      <!--opt, enum, subType:int-->10
    </focusLimited>
    <focusPosition>
      <!--opt, int, range:[1,10]-->1
    </focusPosition>
    <focusSpeed>
      <!--opt, int, range:[1,3]-->1
    </focusSpeed>
  </Focusconfiguration>
  <focusConfiguration>
```



```

<focussensitivity>
  <!--opt, int, range:[0,2]-->1
</focussensitivity>
<temperatureChangeAdaptEnabled>
  <!--opt, bool-->true
</temperatureChangeAdaptEnabled>
<relativeFocusPos>
  <!--opt, int, range:[0,4294967295]-->1
</relativeFocusPos>
<lensType>
  <!--opt, enum, subType:string-->1140
</lensType>
<highTemperaturePriority>
  <!--opt, bool-->true
</highTemperaturePriority>
<focusRange>
  <!--opt, enum, subType:string-->max
</focusRange>
<foucusDistanceMode>
  <!--opt, enum, subType:string-->compatible
</foucusDistanceMode>
</Focusconfiguration>
<LensInitialization>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
</LensInitialization>
<ImageFlip>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
<ImageFlipStyle>
  <!--opt, enum, subType:string-->LEFTRIGHT
</ImageFlipStyle>
<flipAngle>
  <!--opt, enum, subType:int-->90
</flipAngle>
</ImageFlip>
<ImageFreeze>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
</ImageFreeze>
<proportionalpan>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
</proportionalpan>
<WDR>
  <!--opt, object-->
  <mode>
    <!--req, enum, subType:string-->open
  </mode>
  <WDRLevel>
    <!--opt, int, range:[1,100]-->1
  </WDRLevel>
  <WDRContrastLevel>
    <!--opt, int, range:[1,100]-->1
  </WDRContrastLevel>
  <WDRLevel1>
    <!--opt, int, range:[1,100]-->1
  </WDRLevel1>
</WDR>
<BLC>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <BLCMode>
    <!--opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--opt, int, range:[0,10]-->0
  </BLCLevel>
  <BLCRegionList>
    <!--opt, array, subType:object-->
    <BLCRegion>
      <!--opt, object-->
      <id>
        <!--req, int, area No., range:[1,10]-->1
      </id>
      <RegionCoordinatesList>
        <!--req, array, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </BLCRegion>
    </BLCRegionList>
  </BLC>

```

```

        </positionY>
    </RegionCoordinates>
</RegionCoordinatesList>
</BLCRegion>
</BLCRegionList>
</BLC>
<NoiseReduce>
    <!--opt, object-->
    <mode>
        <!--req, enum, subType:string-->close
    </mode>
    <GeneralMode>
        <!--opt, object-->
        <generalLevel>
            <!--req, int, range:[0,100]-->1
        </generalLevel>
    </GeneralMode>
    <AdvancedMode>
        <!--opt, object-->
        <FrameNoiseReduceLevel>
            <!--req, int, range:[0,100]-->1
        </FrameNoiseReduceLevel>
        <InterFrameNoiseReduceLevel>
            <!--req, int, range:[0,100]-->1
        </InterFrameNoiseReduceLevel>
        <secondarySpatialNoiseReduceLevel>
            <!--opt, int, range:[0,100]-->1
        </secondarySpatialNoiseReduceLevel>
        <secondaryTemporalNoiseReduceLevel>
            <!--opt, int, range:[0,100]-->1
        </secondaryTemporalNoiseReduceLevel>
    </AdvancedMode>
</NoiseReduce>
<ImageEnhancement>
    <!--opt, object-->
    <enabled>
        <!--req, bool, whether to enable the function-->true
    </enabled>
    <ImageEnhancementLevel>
        <!--opt, int, range:[0,10]-->0
    </ImageEnhancementLevel>
</ImageEnhancement>
<DSS>
    <!--opt, object, DSS-->
    <enabled>
        <!--req, bool, whether to enable the function-->true
    </enabled>
    <DSSLLevel>
        <!--opt, enum, subType:string-->*1.25
    </DSSLLevel>
    <mode>
        <!--opt, enum, subType:string-->frame
    </mode>
    <slowShutterLevel>
        <!--opt, int, range:[1,6]-->1
    </slowShutterLevel>
</DSS>
<WhiteBalance>
    <!--opt, object, white balance-->
    <WhiteBalanceStyle>
        <!--opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
    <WhiteBalanceRed>
        <!--opt, int, range:[0,100]-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue>
        <!--opt, int, range:[0,100]-->1
    </WhiteBalanceBlue>
    <WhiteBalanceLevel>
        <!--opt, int, range:[0,100]-->1
    </WhiteBalanceLevel>
</WhiteBalance>
<Exposure>
    <!--opt, object-->
    <ExposureType>
        <!--req, enum, subType:string-->auto
    </ExposureType>
    <autoIrisLevel>
        <!--opt, int, range:[0,100]-->1
    </autoIrisLevel>
    <OverexposeSuppress>
        <!--opt, object-->
        <enabled>
            <!--req, bool, whether to enable the function-->true
        </enabled>
        <Type>
            <!--opt, enum, subType:string-->AUTO
        </Type>
        <DistanceLevel>
            <!--opt, int, focal length, range:[1,100]-->1
        </DistanceLevel>
        <shortIRDistanceLevel>
            <!--opt, int, range:[1,100]-->1

```

```

</shortIRDistanceLevel>
<longIRDistanceLevel>
  <!--opt, int, range:[1,100]-->1
</longIRDistanceLevel>
<supplementLightIntensity1>
  <!--opt, int, range:[1,100]-->1
</supplementLightIntensity1>
<supplementLightIntensity2>
  <!--opt, int, range:[1,100]-->1
</supplementLightIntensity2>
<supplementLightIntensity3>
  <!--opt, int, range:[1,100]-->1
</supplementLightIntensity3>
<supplementLightIntensity4>
  <!--opt, int, range:[1,100]-->1
</supplementLightIntensity4>
</OverexposeSuppress>
<pIris>
  <!--opt, object, aperture-->
  <pIrisType>
    <!--opt, enum, subType:string-->MANUAL
  </pIrisType>
  <IrisLevel>
    <!--opt, int, range:[1,100]-->1
  </IrisLevel>
</pIris>
<faceExposure>
  <!--opt, object-->
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <sensitivity>
    <!--opt, int, sensitivity, range:[0,100]-->0
  </sensitivity>
</faceExposure>
<PIrisGeneral>
  <!--opt, object-->
  <irisLevel>
    <!--opt, int, range:[1,100]-->1
  </irisLevel>
</PIrisGeneral>
</Exposure>
<Sharpness>
  <!--opt, object-->
  <SharpnessMode>
    <!--opt, enum, subType:string-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--req, int, range:[0,100]-->1
  </SharpnessLevel>
</Sharpness>
<gammaCorrection>
  <!--opt, object-->
  <gammaCorrectionEnabled>
    <!--opt, bool-->true
  </gammaCorrectionEnabled>
  <gammaCorrectionLevel>
    <!--opt, int, range:[0,100]-->1
  </gammaCorrectionLevel>
</gammaCorrection>
<Shutter>
  <!--opt, object-->
  <ShutterLevel>
    <!--opt, enum, subType:string-->1/1
  </ShutterLevel>
  <maxShutterLevelLimit>
    <!--opt, enum, subType:string-->1/1
  </maxShutterLevelLimit>
  <minShutterLevelLimit>
    <!--opt, enum, subType:string-->1/1
  </minShutterLevelLimit>
</Shutter>
<powerLineFrequency>
  <!--opt, object-->
  <powerLineFrequencyMode>
    <!--opt, enum, subType:string-->50hz
  </powerLineFrequencyMode>
</powerLineFrequency>
<Gain>
  <!--opt, object, gain-->
  <GainLevel>
    <!--opt, string, range:[0,10]-->test
  </GainLevel>
  <GainLimit>
    <!--opt, enum, gain, subType:string-->0
  </GainLimit>
</Gain>
<Color>
  <!--opt, object-->
  <brightnessLevel>
    <!--opt, int, brightness, range:[0,100]-->0
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast, range:[0,100]-->0
  </contrastLevel>

```

```

</contrastLevel>
<saturationLevel>
  <!--opt, int, saturation, range:[0,100]-->0
</saturationLevel>
<hueLevel>
  <!--opt, int, hue, range:[0,100]-->0
</hueLevel>
<grayScale>
  <!--opt, object-->
  <grayScaleMode>
    <!--opt, enum, subType:string-->indoor
  </grayScaleMode>
</grayScale>
<nightMode>
  <!--opt, bool-->true
</nightMode>
<sharpnessLevel>
  <!--opt, int, range:[0,100]-->0
</sharpnessLevel>
</Color>
<IrcutFilter>
  <!--opt, object-->
  <IrcutFilterType>
    <!--opt, enum, subType:string-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--opt, enum, subType:string-->low
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--opt, int, range:[0,10]-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--opt, enum, subType:string-->low
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--opt, int, range:[0,10]-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--opt, object-->
    <scheduleType>
      <!--req, enum, subType:string-->day
    </scheduleType>
    <TimeRange>
      <!--req, object, time range-->
      <beginTime>
        <!--req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--opt, object-->
    <eventType>
      <!--req, enum, event type, subType:string-->IO
    </eventType>
    <IrcutFilterAction>
      <!--req, enum, subType:string-->day
    </IrcutFilterAction>
  </EventTrigger>
</IrcutFilter>
<Scene>
  <!--opt, object-->
  <mode>
    <!--req, enum, subType:string-->indoor
  </mode>
</Scene>
<PTZ>
  <!--opt, object, PTZ-->
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
</PTZ>
<EPTZ>
  <!--opt, object, EPTZ-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <mode>
    <!--opt, enum, subType:string-->cruise
  </mode>
</EPTZ>
<EIS>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <EISLevel>
    <!--opt, int, range:[0,10]-->0
  </EISLevel>
</EIS>
<HLC>

```

```

<!--opt, object-->
<enabled>
  <!--req, bool, whether to enable the function-->true
</enabled>
<HLCLevel>
  <!--opt, int, range:[0,10]-->0
</HLCLevel>
</HLC>
<ZoomLimit>
  <!--opt, object-->
  <ZoomLimitRatio>
    <!--opt, enum, subType:int-->4
  </ZoomLimitRatio>
</ZoomLimit>
<corridor>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
</corridor>
<Dehaze>
  <!--opt, object-->
  <DehazeMode>
    <!--opt, enum, subType:string-->open
  </DehazeMode>
  <DehazeLevel>
    <!--opt, int, range:[0,10]-->0
  </DehazeLevel>
</Dehaze>
<ImageMode>
  <!--opt, enum, subType:string-->standard
</ImageMode>
<enableImageLossDetection>
  <!--opt, bool-->true
</enableImageLossDetection>
<CaptureMode>
  <!--opt, object, capture mode-->
  <mode>
    <!--req, enum, subType:string-->close
  </mode>
</CaptureMode>
<IrLight>
  <!--opt, object-->
  <mode>
    <!--req, enum, subType:string-->auto
  </mode>
  <brightnessLevel>
    <!--opt, int, range:[0,100]-->1
  </brightnessLevel>
  <sensitivityLevel>
    <!--opt, int, sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <brightnessLimit>
    <!--opt, int, range:[0,100]-->1
  </brightnessLimit>
  <triggerMode>
    <!--opt, enum, subType:string-->camera
  </triggerMode>
</IrLight>
<LensDistortionCorrection>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <level>
    <!--opt, enum, subType:string-->level1
  </level>
  <accurateLevel>
    <!--opt, int, range:[0,100]-->0
  </accurateLevel>
  <zoomedInDistantViewLevel>
    <!--opt, int, range:[0,100]-->0
  </zoomedInDistantViewLevel>
  <horizontalFOV>
    <!--opt, int, range:[0,100]-->0
  </horizontalFOV>
  <verticalFOV>
    <!--opt, int, range:[0,100]-->0
  </verticalFOV>
</LensDistortionCorrection>
<SupplementLight>
  <!--opt, object-->
  <mode>
    <!--opt, enum, subType:string, desc:"schedule", "off", "on", "auto", "manual"-->auto
  </mode>
  <Schedule>
    <!--opt, object-->
    <TimeRange>
      <!--req, object-->
      <beginTime>
        <!--req, time, start time-->00:08:00+08:00
      </beginTime>
      <endTime>
        <!--req, time, end time-->00:18:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
</SupplementLight>

```

```

</endTime>
</TimeRange>
</Schedule>
<brightnessLimit>
<!--opt, int, brightness limit of white supplement light, range:[0,100]-->80
</brightnessLimit>
<supplementLightMode>
<!--opt, enum, mode of supplement light, subType:string-->mixed
</supplementLightMode>
<irLightBrightness>
<!--opt, int, brightness of IR supplement light, range:[0,100], desc:this node is valid when supplementLightMode is set to "mixed" and
mixedLightBrightnessRegulatMode is set to "manual"-->80
</irLightBrightness>
<mixedLightBrightnessRegulatMode>
<!--opt, enum, brightness adjustment mode of mixed supplement light, subType:string, desc:"manual" (in this mode, you need to set the parameters
irLightBrightness and brightnessLimit), "auto" (brightness-related parameters are not required). This node is valid when supplementLightMode is set to
"mixed"-->auto
</mixedLightBrightnessRegulatMode>
<highIrLightBrightness>
<!--opt, int, brightness of far IR supplement light, range:[0,100], desc:it is shared by different supplement light modes, including "mixed",
"whiteLight", "colorVwWhiteLight", "irLight", and "dualLight"-->80
</highIrLightBrightness>
<highWhiteLightBrightness>
<!--opt, int, brightness of far white supplement light, range:[0,100], desc:it is shared by different supplement light modes, including "mixed",
"whiteLight", "colorVwWhiteLight", "irLight", and "dualLight"-->80
</highWhiteLightBrightness>
<lowIrLightBrightness>
<!--opt, int, brightness of near IR supplement light, range:[0,100], desc:it is shared by different supplement light modes, including "mixed",
"whiteLight", "colorVwWhiteLight", "irLight", and "dualLight"-->80
</lowIrLightBrightness>
<lowWhiteLightBrightness>
<!--opt, int, brightness of near white supplement light, range:[0,100], desc:it is shared by different supplement light modes, including "mixed",
"whiteLight", "colorVwWhiteLight", "irLight", and "dualLight"-->80
</lowWhiteLightBrightness>
<whiteLightBrightness>
<!--opt, int, light brightness (regardless of far or near light), range:[0,100], desc:it is shared by different supplement light modes, including
"mixed", "whiteLight", "colorVwWhiteLight", "irLight", and "dualLight", when the adjustment mode of the supplement light is "auto"-->80
</whiteLightBrightness>
<WhiteLightModeCfg>
<!--opt, object-->
<highWhiteLightBrightness>
<!--opt, int, brightness of far white supplement light, range:[0,100]-->80
</highWhiteLightBrightness>
<lowWhiteLightBrightness>
<!--opt, int, brightness of near white supplement light, range:[0,100]-->80
</lowWhiteLightBrightness>
<whiteLightBrightness>
<!--opt, int, white light brightness (regardless of far or near light), range:[0,100]-->80
</whiteLightBrightness>
</WhiteLightModeCfg>
<isAutoModeBrightnessCfg>
<!--opt, bool-->true
</isAutoModeBrightnessCfg>
<irLightbrightnessLimit>
<!--opt, int, brightness limit of IR supplement light, range:[0,100], desc:this node is valid when the value of Mode is "auto" and supplementLightMode
is "mixed"-->80
</irLightbrightnessLimit>
<whiteLightbrightnessLimit>
<!--opt, int, brightness limit of white supplement light, range:[0,100], desc:this node is valid when the value of Mode is "auto" and
supplementLightMode is "mixed"-->80
</whiteLightbrightnessLimit>
<isNotSupportMixedWhiteLight>
<!--opt, bool-->true
</isNotSupportMixedWhiteLight>
<EventIntelligenceModeCfg>
<!--opt, object-->
<brightnessRegulatMode>
<!--opt, enum, subType:string-->auto
</brightnessRegulatMode>
<whiteLightBrightness>
<!--opt, int, white light brightness (regardless of far or near light), range:[0,100]-->80
</whiteLightBrightness>
<irLightBrightness>
<!--opt, int, IR light brightness (regardless of far or near light), range:[0,100]-->80
</irLightBrightness>
<BrightnessCtrlList>
<!--opt, array, subType:object-->
<BrightnessCtrl>
<!--opt, object-->
<lightType>
<!--opt, enum, subType:string-->highWhite
</lightType>
<brightness>
<!--opt, int, supplement light brightness, range:[0,100]-->80
</brightness>
</BrightnessCtrl>
</BrightnessCtrlList>
</EventIntelligenceModeCfg>
<MultiAzimuthMixedModeCfg>
<!--opt, object-->
<brightnessRegulatMode>
<!--opt, enum, subType:string-->auto
</brightnessRegulatMode>

```

```

<LightCtrlList>
  <!--opt, array, subType:object-->
  <LightCtrl>
    <!--opt, object-->
    <azimuthType>
      <!--opt, enum, subType:string-->left
    </azimuthType>
    <lightType>
      <!--opt, enum, subType:string-->highWhite
    </lightType>
    <brightness>
      <!--opt, int, supplement light brightness, range:[0,100]-->80
    </brightness>
  </LightCtrl>
</LightCtrlList>
</MultiAzimuthMixedModeCfg>
<MultiAzimuthWhiteLightCfg>
  <!--opt, object-->
  <brightnessRegulatMode>
    <!--opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <azimuthType>
        <!--opt, enum, subType:string-->left
      </azimuthType>
      <lightType>
        <!--opt, enum, subType:string-->highWhite
      </lightType>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</MultiAzimuthWhiteLightCfg>
<MultiAzimuthIrLightCfg>
  <!--opt, object-->
  <brightnessRegulatMode>
    <!--opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <azimuthType>
        <!--opt, enum, subType:string-->left
      </azimuthType>
      <lightType>
        <!--opt, enum, subType:string-->highIrLight
      </lightType>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</MultiAzimuthIrLightCfg>
<PanoramicWhiteLightCfg>
  <!--opt, object-->
  <brightnessRegulatMode>
    <!--opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <lightID>
        <!--opt, int, range:[1,4]-->1
      </lightID>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</PanoramicWhiteLightCfg>
<PanoramicIrLightCfg>
  <!--opt, object-->
  <brightnessRegulatMode>
    <!--opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <lightID>
        <!--opt, int, range:[1,4]-->1
      </lightID>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</PanoramicIrLightCfg>

```

```

</PanoramaCtrlLight>
<whiteLightEnhanceEnabled>
  <!--opt, bool-->true
</whiteLightEnhanceEnabled>
<LightManualCtrlList>
  <!--opt, array, subType:object, dep:and,{$.ImageChannel.SupplementLight.mode,eq,manual}-->
  <LightCtrl>
    <!--opt, object-->
    <lightType>
      <!--opt, enum, subType:string-->highIrLight
    </lightType>
    <brightness>
      <!--opt, int, supplement Light brightness, range:[0,100]-->80
    </brightness>
  </LightCtrl>
</LightManualCtrlList>
</SupplementLight>
<OpticalDehaze>
  <!--opt, object-->
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
</OpticalDehaze>
<ManualRanging>
  <!--opt, object-->
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
</ManualRanging>
<OIS>
  <!--opt, object-->
  <mode>
    <!--req, enum, subType:string-->close
  </mode>
  <OISLevel>
    <!--opt, enum, subType:string-->high
  </OISLevel>
  <OISSensitivity>
    <!--opt, string-->test
  </OISSensitivity>
</OIS>
<isSupportLaserSpotManual>
  <!--opt, bool-->true
</isSupportLaserSpotManual>
<isSupportLaserSpotAdjustment>
  <!--opt, bool-->true
</isSupportLaserSpotAdjustment>
<DigitalZoom>
  <!--opt, object-->
  <ZoomRatio>
    <!--opt, enum, subType:string-->1x
  </ZoomRatio>
</DigitalZoom>
<Palettes>
  <!--opt, object-->
  <mode>
    <!--opt, enum, subType:string-->whiteHot
  </mode>
  <ColorateTarget>
    <!--opt, object-->
    <ColorateTargetModelList>
      <!--opt, array, subType:object-->
      <ColorateTargetMode>
        <!--opt, object-->
        <id>
          <!--req, int, desc:it starts from 1-->0
        </id>
        <mode>
          <!--req, enum, subType:string-->colorateHotAreae
        </mode>
        <enabled>
          <!--req, bool, whether to enable the function-->true
        </enabled>
        <TemperatureLimit>
          <!--req, object-->
          <minTemperature>
            <!--opt, float, the minimum temperature, range:[-20,150]-->0.000
          </minTemperature>
          <maxTemperature>
            <!--opt, float, the maximum temperature, range:[-20,150]-->0.000
          </maxTemperature>
        </TemperatureLimit>
        <Color>
          <!--req, object-->
          <R>
            <!--req, int, range:[0,255]-->0
          </R>
          <G>
            <!--req, int, G, range:[0,255]-->0
          </G>
          <B>
            <!--req, int, range:[0,255]-->0
          </B>
        </Color>
      </ColorateTargetMode>
    </ColorateTargetModelList>
  </ColorateTarget>
</Palettes>

```



```

        </ColorateTargetMode>
    </ColorateTargetModelList>
</ColorateTarget>
</Palettes>
<ExposureSync>
    <!--opt, object-->
    <enabled>
        <!--req, bool, whether to enable the function-->true
    </enabled>
</ExposureSync>
<BrightnessSuddenChangeSuppression>
    <!--opt, object-->
    <enabled>
        <!--req, bool, whether to enable the function-->true
    </enabled>
</BrightnessSuddenChangeSuppression>
<AGCCap>
    <!--opt, object-->
    <AGCMode>
        <!--opt, enum, subType:string-->self_Adaptive
    </AGCMode>
</AGCCap>
<RegionalImageEnhancementCap>
    <!--opt, object-->
    <mode>
        <!--req, enum, subType:string-->off
    </mode>
<CustomAreaParam>
    <!--opt, object-->
    <RegionCoordinatesList>
        <!--opt, array, area coordinate List, subType:object, range:[3,10]-->
    </RegionCoordinatesList>
    <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
    </RegionCoordinates>
    <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->0
    </positionX>
    <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
    </positionY>
</RegionCoordinates>
</RegionCoordinatesList>
</CustomAreaParam>
<DefaultAreaParamList>
    <!--opt, array, subType:object-->
    <DefaultAreaParam>
        <!--opt, object-->
    </DefaultAreaParam>
    <mode>
        <!--req, enum, subType:string-->up_60%
    </mode>
    <RegionCoordinatesList>
        <!--opt, array, area coordinate List, subType:object, range:[3,10]-->
    </RegionCoordinatesList>
    <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
    </RegionCoordinates>
    <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->0
    </positionX>
    <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
    </positionY>
</RegionCoordinates>
</RegionCoordinatesList>
</DefaultAreaParam>
</DefaultAreaParamList>
</RegionalImageEnhancementCap>
<faceExposure>
    <!--opt, object-->
    <faceExposureEnabled>
        <!--req, bool, whether to enable face picture exposure-->true
    </faceExposureEnabled>
    <faceExposureSensitivity>
        <!--req, int, range:[0,100]-->0
    </faceExposureSensitivity>
</faceExposure>
<lightHal>
    <!--opt, object-->
    <enabled>
        <!--req, bool, whether to enable the function-->true
    </enabled>
    <halLevel>
        <!--req, int, range:[0,100]-->0
    </halLevel>
    <halRange>
        <!--req, int, range:[0,2064]-->0
    </halRange>
</lightHal>
<TargetEnhancement>
    <!--opt, object-->
    <enabled>
        <!--req, bool, whether to enable the function-->true
    </enabled>
</TargetEnhancement>
<LaserLight>
    <!--opt, object-->

```

```

<mode>
  <!--opt, enum, subType:string-->auto
</mode>
<compensationRegionMode>
  <!--opt, enum, subType:string-->center
</compensationRegionMode>
<compensationLevel>
  <!--opt, enum, subType:string-->low
</compensationLevel>
<LaserColor>
  <!--opt, object-->
  <R>
    <!--req, int, range:[0,100]-->0
  </R>
  <G>
    <!--req, int, range:[0,100]-->0
  </G>
  <B>
    <!--req, int, range:[0,100]-->0
  </B>
</LaserColor>
</LaserLight>
<ExternalSupplementLight>
  <!--opt, object-->
  <mode>
    <!--opt, enum, subType:string-->close
  </mode>
  <BrightnessList>
    <!--opt, array, subType:object-->
    <Brightness>
      <!--opt, object-->
      <id>
        <!--opt, int-->0
      </id>
      <irLightBrightness>
        <!--opt, int, range:[0,100]-->0
      </irLightBrightness>
      <whiteLightBrightness>
        <!--opt, int, range:[0,100]-->0
      </whiteLightBrightness>
    </Brightness>
  </BrightnessList>
</ExternalSupplementLight>
<TempRange>
  <!--opt, object-->
  <mode>
    <!--req, enum, subType:string-->automatic
  </mode>
  <temperatureUpperLimit>
    <!--opt, float, range:[-20,550], unit:℃-->0.000
  </temperatureUpperLimit>
  <temperatureLowerLimit>
    <!--opt, float, range:[-20,550], unit:℃-->0.000
  </temperatureLowerLimit>
</TempRange>
<Beauty>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <whiteningStrength>
    <!--opt, int, range:[0,10]-->1
  </whiteningStrength>
  <skinSmoothingStrength>
    <!--opt, int, range:[0,10]-->1
  </skinSmoothingStrength>
</Beauty>
<MExposure>
  <!--opt, object-->
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <exposureLevel>
    <!--opt, string, range:[0,1000000]-->test
  </exposureLevel>
  <gainLevel>
    <!--opt, int, range:[0,100]-->0
  </gainLevel>
</MExposure>
<SensorRGBCtrl>
  <!--opt, object-->
  <sensorID>
    <!--opt, int, range:[0,1]-->0
  </sensorID>
</SensorRGBCtrl>
<YPbPrCalibration>
  <!--opt, object-->
  <mode>
    <!--opt, int, range:[0,2]-->1
  </mode>
</YPbPrCalibration>
<intrusionTriggeredLightingCfg>
  <!--opt, object-->
  <enabled>

```

```

<!--req, bool-->true
</enabled>
</intrusionTriggeredLightingCfg>
<shotProCfg>
  <!--opt, object-->
  <shotProMode>
    <!--req, enum, subType:string-->general
  </shotProMode>
</shotProCfg>
</ImageChannel>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	HLCMutexWDRorDehaze	0x40002008	--
4	Invalid Operation	WDRMutexDDSorDehaze	0x4000200A	--
4	Invalid Operation	slowShutterConflictsWithWdr	0x4000A64B	--
4	Invalid Operation	wdrConflictsWithDefog	0x4000A64C	--
4	Invalid Operation	defogConflictsWithWdr	0x4000A64D	--
4	Invalid Operation	wdrConflictsWithSlowShutter	0x4000A64E	--
4	Invalid Operation	theImageEffectsIsLockedUnlockAndTryAgain	0x4000A6C3	--

12.4.1.2 Get image configuration capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, temperature range capability, see XML_Cap_tempRange for details, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <videoInnutID>

```

```

<!--ro, req, int-->1
</videoInputID>
<sensorType>
  <!--ro, opt, int, sensor type-->0
</sensorType>
<FocusConfiguration>
  <!--ro, opt, object-->
  <focusStyle opt="AUTO,MANUAL,SEMIAUTOMATIC,distanceCalculate">
    <!--ro, req, string, attr:opt{req, string}-->test
  </focusStyle>
  <focusLimited opt="10,50,100,150,300,600,1000,2000,5000,20000,50000,65535">
    <!--ro, opt, int, attr:opt{req, string}-->1
  </focusLimited>
  <focusPosition min="1" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </focusPosition>
  <focusSpeed min="1" max="10">
    <!--ro, opt, int, range:[1,3], attr:min{req, int},max{req, int}-->1
  </focusSpeed>
  <focusSensitivity min="1" max="10">
    <!--ro, opt, int, range:[0,2], attr:min{req, int},max{req, int}-->1
  </focusSensitivity>
  <temperatureChangeAdaptEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </temperatureChangeAdaptEnabled>
  <relativeFocusPos min="1" max="10">
    <!--ro, opt, int, range:[0,4294967295], attr:min{req, int},max{req, int}-->1
  </relativeFocusPos>
  <lensType opt="1140,1050">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->1140
  </lensType>
  <highTemperaturePriority opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </highTemperaturePriority>
  <focusStatus opt="success,failure,working">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->success
  </focusStatus>
  <motorMovementType opt="manual,auto,armingLocation">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->manual
  </motorMovementType>
  <focusRange opt="min,medium,max">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->max
  </focusRange>
  <focusDistanceMode opt="compatible">
    <!--ro, opt, string, attr:opt{req, string}-->compatible
  </focusDistanceMode>
  <focusDistance min="0.2" max="999.9">
    <!--ro, opt, float, range:[0.2,999.9], step:0.1, unit:m, attr:min{req, float},max{req, float}-->0.2
  </focusDistance>
  <focusStrategy opt="DToF,general" def="general">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->DToF
  </focusStrategy>
</FocusConfiguration>
<LensInitialization>
  <!--ro, opt, object, Lens Initialization times-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</LensInitialization>
<ImageFlip>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <ImageFlipStyle opt="LEFTRIGHT,UPDOWN,CENTER,AUTO">
    <!--ro, opt, string, type, attr:opt{req, string}-->LEFTRIGHT
  </ImageFlipStyle>
  <flipAngle opt="90,180,270">
    <!--ro, opt, string, attr:opt{req, string}-->90
  </flipAngle>
</ImageFlip>
<ImageFreeze>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</ImageFreeze>
<proportionalpan>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</proportionalpan>
<WDR>
  <!--ro, opt, object-->
  <mode opt="open,close,auto,SDR,time">
    <!--ro, req, enum, WDR mode: "open,close,auto", subType:string, attr:opt{req, string}, desc: "WDR mode: "open,close,auto"-->open
  </mode>
  <WDRLevel min="1" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </WDRLevel>
  <WDRContrastLevel min="1" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </WDRContrastLevel>

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</WDRContrastLevel>
<WDRLevel1 min="1" max="100">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</WDRLevel1>
<timeMode>
  <!--ro, opt, object, dep:and,{$.WDR.mode,eq,time}-->
  <timeRangeList size="1">
    <!--ro, req, object, attr:size{req, int}-->
    <timeRange>
      <!--ro, req, object-->
      <startTime>
        <!--ro, req, time-->08:00:00
      </startTime>
      <endTime>
        <!--ro, req, time-->18:00:00
      </endTime>
    </timeRange>
  </timeRangeList>
</timeMode>
<WDRStreakInhibition opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->>false
</WDRStreakInhibition>
<WDRCloseToOpenFilterLevel min="1" max="9" def="5">
  <!--ro, opt, int, range:[1,9], attr:min{req, int},max{req, int},def{req, int}-->5
</WDRCloseToOpenFilterLevel>
<WDROpenToCloseFilterLevel min="1" max="9" def="5">
  <!--ro, opt, int, range:[1,9], attr:min{req, int},max{req, int},def{req, int}-->5
</WDROpenToCloseFilterLevel>
</WDR>
<BLC>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->>true
  </enabled>
  <BLCMode opt="UP,DOWN,LEFT,RIGHT,CENTER,MULTI-AREA,Region,AUTO">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->UP
  </BLCMode>
  <BLCLevel min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </BLCLevel>
  <BLCRegionList>
    <!--ro, opt, array, subType:object-->
    <BLCRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int, area No.-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, req, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
  <switchMode opt="time">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->time
  </switchMode>
  <timeMode>
    <!--ro, opt, object, dep:and,{$.BLC.switchMode,eq,time}-->
    <timeRangeList size="1">
      <!--ro, req, object, attr:size{req, int}-->
      <timeRange>
        <!--ro, req, object-->
        <startTime>
          <!--ro, req, time-->08:00:00
        </startTime>
        <endTime>
          <!--ro, req, time-->18:00:00
        </endTime>
      </timeRange>
    </timeRangeList>
  </timeMode>
</BLC>
<NoiseReduce>
  <!--ro, opt, object-->
  <mode opt="close,general,advanced">
    <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, mode-->
    <generalLevel min="0" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object-->
    <FrameNoiseReductionLevel min="0" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </FrameNoiseReductionLevel>
  </AdvancedMode>

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<!--ro, req, int, attr:min{req, int},max{req, int}-->1
</FrameNoiseReduceLevel>
<InterFrameNoiseReduceLevel min="0" max="10">
  <!--ro, req, int, attr:min{req, int},max{req, int}-->1
</InterFrameNoiseReduceLevel>
<secondarySpatialNoiseReduceLevel min="0" max="100">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</secondarySpatialNoiseReduceLevel>
<secondaryTemporalNoiseReduceLevel min="0" max="100">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</secondaryTemporalNoiseReduceLevel>
</AdvancedMode>
</NoiseReduce>
<ImageEnhancement>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <ImageEnhancementLevel min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </ImageEnhancementLevel>
</ImageEnhancement>
<DSS>
  <!--ro, opt, object, DSS-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <DSSLevel opt="*1.25,*1.5,*2,*3*4,*6,*8,*12,*16,*24,*32,*48,*64,*96,*128,*256,*512,auto">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </DSSLevel>
  <mode opt="frame,level">
    <!--ro, opt, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->frame
  </mode>
  <slowShutterLevel min="1" max="6">
    <!--ro, opt, int, range:[1,6], attr:min{req, int},max{req, int}-->1
  </slowShutterLevel>
</DSS>
<WhiteBalance>
  <!--ro, opt, object, white balance-->
  <WhiteBalanceStyle opt="manual, auto1, auto2, locked, outdoor, indoor, daylightlamp, sodiumlight, autotrace, onece, autooutdoor, autosodiumlight, mercury, incandescentlight, warmlight, naturalight, customColorTemperature, highTemperature">
    <!--ro, req, string, attr:opt{req, string}-->test
  </WhiteBalanceStyle>
  <WhiteBalanceRed min="-127" max="127">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue min="-127" max="127">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </WhiteBalanceBlue>
  <WhiteBalanceLevel min="0" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </WhiteBalanceLevel>
  <customColorTemperatureLevel min="1800" max="10000" def="3000">
    <!--ro, opt, int, range:[1800,10000], attr:min{req, int},max{req, int},def{opt, int}-->3000
  </customColorTemperatureLevel>
  <WhiteBalanceRedFineTuning min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.WhiteBalanceStyle,eq>manual}, attr:min{req, int},max{req, int}-->1
  </WhiteBalanceRedFineTuning>
  <WhiteBalanceBlueFineTuning min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.WhiteBalanceStyle,eq>manual}, attr:min{req, int},max{req, int}-->1
  </WhiteBalanceBlueFineTuning>
  <autoMode opt="colorTemperature,video,videoAndcolorTemperature" def="video">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->video
  </autoMode>
  <manualMode opt="colorFineTuning,colorTemperatureAdjust" def="colorFineTuning">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->colorFineTuning
  </manualMode>
</WhiteBalance>
<Exposure>
  <!--ro, opt, object-->
  <ExposureType opt="auto,IrisFirst,ShutterFirst,gainFirst>manual,pIris,T5280-PQ1,T5289-PQ1,T1140-PQ1,T2712-PQ1,HV1250P-MPIR,pc,pIris-General,HZ1140P-8MPIRA,HZ1135P-8MPSPGA,MV1550P-12MPIR,MZ1545P-12MP,HZ0618P-4MPSPG,MZ1050P-12MP,SF5014M-12MP,SF3018M-10MP,MZ1140M,MZ2514M-10MP,MZ7018M-8MP,MZ2516M-10MPIR,indoor50HZ,indoor60HZ,lowGhosting,customize">
    <!--ro, req, string, attr:opt{req, string}-->test
  </ExposureType>
  <autoIrisLevel min="0" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
  <Type opt="AUTO,MANUAL">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->AUTO
  </Type>
  <DistanceLevel min="1" max="100">
    <!--ro, opt, int, Focal Length, attr:min{req, int},max{req, int}-->1
  </DistanceLevel>
  <shortIRDistanceLevel min="1" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </shortIRDistanceLevel>
  <longIRDistanceLevel min="1" max="100">

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    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</longIRDistanceLevel>
<supplementLightIntensity1 min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
</supplementLightIntensity1>
<supplementLightIntensity2 min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
</supplementLightIntensity2>
<supplementLightIntensity3 min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
</supplementLightIntensity3>
<supplementLightIntensity4 min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
</supplementLightIntensity4>
<hightLightDistanceLevel min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{opt, int},max{opt, int}-->50
</hightLightDistanceLevel>
<lowLightDistanceLevel min="1" max="100">
    <!--ro, opt, int, range:[1,100], attr:min{opt, int},max{opt, int}-->50
</lowLightDistanceLevel>
</OverexposeSuppress>
<pIris>
    <!--ro, opt, object, aperture-->
    <pIrisType opt="AUTO,MANUAL">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->MANUAL
    </pIrisType>
    <IrisLevel min="1" max="100">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </IrisLevel>
</pIris>
<faceExposure>
    <!--ro, opt, object-->
    <enabled opt="true,false">
        <!--ro, opt, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <sensitivity>
        <!--ro, opt, int, sensitivity-->0
    </sensitivity>
</faceExposure>
<PIrisGeneral>
    <!--ro, opt, object-->
    <irisLevel min="1" max="100" def="50">
        <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int},def{req, int}-->1
    </irisLevel>
</PIrisGeneral>
<exposureModeDisplayCap opt="auto,IrisFirst,ShutterFirst,gainFirst>manual">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
</exposureModeDisplayCap>
<combinationParams min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->50
</combinationParams>
<lowGhostingExposure>
    <!--ro, opt, object-->
    <lowGhostingShutterLevel opt="1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/250">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->1/100
    </lowGhostingShutterLevel>
</lowGhostingExposure>
<exposureCompensateLevel min="-100" max="100" def="0">
    <!--ro, opt, int, range:[-100,100], attr:min{req, int},max{req, int},def{req, int}-->0
</exposureCompensateLevel>
<exposureControl opt="center,region,point">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->center
</exposureControl>
</Exposure>
<Sharpness>
    <!--ro, opt, object-->
    <SharpnessMode opt="manual,auto">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->manual
    </SharpnessMode>
    <SharpnessLevel min="0" max="255">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </SharpnessLevel>
</Sharpness>
<gammaCorrection>
    <!--ro, opt, object-->
    <gammaCorrectionEnabled opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
    </gammaCorrectionEnabled>
    <gammaCorrectionLevel min="0" max="10">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </gammaCorrectionLevel>
</gammaCorrection>
<Shutter>
    <!--ro, opt, object-->
    <ShutterLevel
opt="1/1,1/2,1/3,1/6,1/12,1/25,1/50,1/60,1/75,1/100,1/120,1/125,1/150,1/175,1/15,1/20,1/30,1/60,1/180,1/200,1/250,1/500,1/1000,1/2000,1/4000,1/8000,1/5000,1/10000,1/225,1/300,1/400,1/425,1/600,1/750,1/1250,1/1750,1/2500,1/3500,1/6000,1/30000,1/100000">
        <!--ro, opt, string, attr:opt{req, string}-->test
    </ShutterLevel>
    <maxShutterLevelLimit
opt="1/1,1/2,1/3,1/6,1/12,1/25,1/50,1/60,1/75,1/100,1/120,1/125,1/150,1/175,1/15,1/20,1/30,1/60,1/180,1/200,1/250,1/500,1/1000,1/2000,1/4000,1/8000,1/5000,1/10000,1/225,1/300,1/400,1/425,1/600,1/750,1/1250,1/1750,1/2500,1/3500,1/6000,1/30000,1/100000">
        <!--ro, opt, string, attr:opt{req, string}-->test
    </maxShutterLevelLimit>

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</maxShutterLevelLimit>
<minShutterLevelLimit
opt="1/1,1/2,1/3,1/6,1/12,1/25,1/50,1/60,1/75,1/100,1/120,1/125,1/150,1/175,1/15,1/20,1/30,1/60,1/180,1/200,1/250,1/500,1/1000,1/2000,1/4000,1/8000,1/5000,1/10000,1/225,1/300,1/400,1/425,1/600,1/750,1/1250,1/1750,1/2500,1/3500,1/6000,1/30000,1/100000">
  <!--ro, opt, string, attr:opt{req, string}-->test
</minShutterLevelLimit>
</Shutter>
<powerLineFrequency>
  <!--ro, opt, object-->
  <powerLineFrequencyMode opt="50hz,60hz">
    <!--ro, opt, string, mode, attr:opt{req, string}-->test
  </powerLineFrequencyMode>
</powerLineFrequency>
<Gain>
  <!--ro, opt, object, gain-->
  <GainLevel min="0" max="10">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </GainLevel>
  <GainLimit opt="0,400,900,1300,1800,2200,2600,3100,3500,4000,4400,4800,5300,5700,6200,6600">
    <!--ro, opt, string, gain, attr:opt{req, string}-->test
  </GainLimit>
  <minExposureGainLevel min="0" max="10">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </minExposureGainLevel>
  <maxExposureGainLevel min="0" max="10">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </maxExposureGainLevel>
</Gain>
<Color>
  <!--ro, opt, object, image adjustment capability-->
  <brightnessLevel min="0" max="255">
    <!--ro, opt, int, brightness, attr:min{req, int},max{req, int}-->0
  </brightnessLevel>
  <contrastLevel min="0" max="255">
    <!--ro, opt, int, contrast, attr:min{req, int},max{req, int}-->0
  </contrastLevel>
  <saturationLevel min="0" max="255">
    <!--ro, opt, int, saturation, attr:min{req, int},max{req, int}-->0
  </saturationLevel>
  <hueLevel min="0" max="100">
    <!--ro, opt, int, hue, attr:min{req, int},max{req, int}-->0
  </hueLevel>
  <grayScale>
    <!--ro, opt, object, gray scale-->
    <grayScaleMode opt="indoor,outdoor">
      <!--ro, opt, string, gray scale mode, attr:opt{req, string}-->test
    </grayScaleMode>
  </grayScale>
  <nightMode opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </nightMode>
  <sharpnessLevel min="0" max="255" def="0">
    <!--ro, opt, int, attr:min{req, int},max{req, int},def{req, int}-->0
  </sharpnessLevel>
  <hueMode opt="warm,cold">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->warm
  </hueMode>
  <saturationMode opt="gorgeous,common">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->common
  </saturationMode>
</Color>
<DayNightGate>
  <!--ro, opt, object-->
  <brightnessThreld min="0" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </brightnessThreld>
  <nightModeCtrl>
    <!--ro, opt, object-->
    <nightModeCtrlMode opt="manual,time,auto">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->manual
    </nightModeCtrlMode>
    <manualMode>
      <!--ro, opt, object-->
      <manualModeVal opt="day,night">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
      </manualModeVal>
    </manualMode>
    <timeMode>
      <!--ro, opt, object-->
      <scheduleType opt="day,range">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
      </scheduleType>
      <switchList>
        <!--ro, opt, array, subType:object-->
        <switch>
          <!--ro, opt, object-->
          <nightModeTimeId min="0" max="1">
            <!--ro, opt, int, range:[0,1], attr:min{req, int},max{req, int}-->1
          </nightModeTimeId>
          <nightModeStartHour min="0" max="23">
            <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
          </nightModeStartHour>
          <nightModeStartMinute min="0" max="59">
            <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
          </nightModeStartMinute>
        </switch>
      </switchList>
    </timeMode>
  </nightModeCtrl>
</DayNightGate>

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</nightModeStartMinute>
<nightModeEndHour min="0" max="23">
  <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
</nightModeEndHour>
<nightModeEndMinute min="0" max="59">
  <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
</nightModeEndMinute>
</switch>
</switchList>
<timeScheduleList size="2">
  <!--ro, opt, object, attr:size{req, int}-->
  <timeSchedule>
    <!--ro, opt, object-->
    <triggerData>
      <!--ro, opt, object-->
      <startTime>
        <!--ro, opt, object-->
        <monthOfYear min="1" max="12">
          <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->1
        </monthOfYear>
        </startTime>
        <endTime>
          <!--ro, opt, object-->
          <monthOfYear min="1" max="12">
            <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->12
          </monthOfYear>
          </endTime>
        </triggerData>
        <timeSwitchList size="2">
          <!--ro, opt, object, attr:size{req, int}-->
          <timeSwitch>
            <!--ro, opt, object-->
            <timeID min="0" max="1">
              <!--ro, req, int, range:[0,1], attr:min{req, int},max{req, int}-->1
            </timeID>
            <startHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </startHour>
            <startMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </startMinute>
            <endHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </endHour>
            <endMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </endMinute>
          </timeSwitch>
        </timeSwitchList>
      </timeSchedule>
    </timeScheduleList>
  </timeMode>
  <autoMode>
    <!--ro, opt, object-->
    <autoSwitchThresholdVal min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
    </autoSwitchThresholdVal>
    <autoNightSwitchThresholdVal min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
    </autoNightSwitchThresholdVal>
    <autoDayToNightDelayTime min="0" max="120">
      <!--ro, opt, int, range:[0,120], attr:min{req, int},max{req, int}-->1
    </autoDayToNightDelayTime>
    <autoNightToDayDelayTime min="0" max="120">
      <!--ro, opt, int, range:[0,120], attr:min{req, int},max{req, int}-->1
    </autoNightToDayDelayTime>
    <ICRChange opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->>false
    </ICRChange>
    <lightChange opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->>false
    </lightChange>
    <WDRChange opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->>false
    </WDRChange>
  </autoMode>
</nightModeCtrl>
</DayNightGate>
<IrcutFilter>
  <!--ro, opt, object-->
  <IrcutFilterType opt="auto,day,night,schedule,eventTrigger,videoAuto,IrcutFilterType,darkFighterX,darkFighterXAuto,darkFighterXSchedule,colorVuAuto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel min="0" max="7">
    <!--ro, opt, int, range:[0,7], attr:min{req, int},max{req, int}-->4
  </dayToNightFilterLevel>
  <dayToNightFilterTime min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel min="0" max="7">
    <!--ro, opt, int, range:[0,7], attr:min{req, int},max{req, int}-->4
  </nightToDayFilterLevel>
  <nightToDayFilterTime min="0" max="10">

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    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </nightToDayFilterTime>
</Schedule>
<!--ro, opt, object-->
<scheduleType opt="day,night">
  <!--ro, req, enum, subType:string, attr:opt{req, string}-->day
</scheduleType>
<TimeRange>
  <!--ro, req, object, time period-->
  <beginTime>
    <!--ro, req, time, start time-->00:00:00+08:00
  </beginTime>
  <endTime>
    <!--ro, req, time, end time-->00:00:00+08:00
  </endTime>
</TimeRange>
</Schedule>
<EventTrigger>
  <!--ro, opt, object-->
  <eventType opt="IO,VMD">
    <!--ro, req, enum, event type, subType:string, attr:opt{req, string}, desc:event type-->IO
  </eventType>
  <IrcutFilterAction opt="day,night">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->day
  </IrcutFilterAction>
</EventTrigger>
</IrcutFilter>
<Scene>
  <!--ro, opt, object-->
  <mode opt="indoor,outdoor,standard,dimLight,riverSeaProtection,perimeterProtection,forestFirePrevention">
    <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->indoor
  </mode>
</Scene>
<PTZ>
  <!--ro, opt, object, PTZ-->
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</PTZ>
<EPTZ>
  <!--ro, opt, object, EPTZ-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <mode opt="cruise,autoTrack">
    <!--ro, opt, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->cruise
  </mode>
</EPTZ>
<EIS>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <EISLevel min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </EISLevel>
</EIS>
<HLC>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <HLCMode opt="close,open,auto,timing">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->close
  </HLCMode>
  <HLCLevel min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </HLCLevel>
  <timePlanList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <startTime>
      <!--ro, req, time-->08:00:00
    </startTime>
    <endTime>
      <!--ro, req, time-->10:00:00
    </endTime>
  </timePlanList>
</HLC>
<ZoomLimit>
  <!--ro, opt, object-->
  <ZoomLimitRatio opt="4,8,16,32,64">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->4
  </ZoomLimitRatio>
</ZoomLimit>
<corridor>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</corridor>
<Dehaze>
  <!--ro, opt, object-->
  <DehazeMode opt="open,close,auto,advancedMode">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->open
  </DehazeMode>

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</DehazeMode>
<DehazeLevel min="0" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
</DehazeLevel>
<dehazeIntensity min="0" max="100">
  <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.Dehaze.DehazeMode,eq,advancedMode}, attr:min{req, int},max{req, int}-->0
</dehazeIntensity>
<smoothIntensity min="0" max="100">
  <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.Dehaze.DehazeMode,eq,advancedMode}, attr:min{req, int},max{req, int}-->0
</smoothIntensity>
</Dehaze>
<ImageMode opt="standard,indoor,outdoor,dimLight,bright,soft,vivid">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->standard
</ImageMode>
<enableImageLossDetection opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</enableImageLossDetection>
<CaptureMode>
  <!--ro, opt, object, capture mode-->
  <mode opt="close,1920*1080@30fps">
    <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->close
  </mode>
</CaptureMode>
<IrLight>
  <!--ro, opt, object-->
  <mode opt="auto>manual">
    <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->auto
  </mode>
  <brightnessLevel min="0" max="10">
    <!--ro, opt, int, brightness, attr:min{req, int},max{req, int}-->1
  </brightnessLevel>
  <sensitivityLevel min="0" max="100">
    <!--ro, opt, int, sensitivity, attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <brightnessLimit min="0" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </brightnessLimit>
  <triggerMode opt="camera,photosensitive">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->camera
  </triggerMode>
</IrLight>
<LensDistortionCorrection>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <level opt="level1,level2,level3,custom">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->level1
  </level>
  <accurateLevel min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </accurateLevel>
  <zoomedInDistantViewLevel min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </zoomedInDistantViewLevel>
  <horizontalFOV min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </horizontalFOV>
  <verticalFOV min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </verticalFOV>
  <isSupportLevelTypeAccurate>
    <!--ro, opt, bool-->true
  </isSupportLevelTypeAccurate>
  <supportPrompt opt="levelPrompt">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->levelPrompt
  </supportPrompt>
  <isLDCModifyResolution>
    <!--ro, opt, bool-->true
  </isLDCModifyResolution>
</LensDistortionCorrection>
<SupplementLight>
  <!--ro, opt, object-->
  <mode opt="schedule,off,on,auto>manual">
    <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:"schedule", "off", "on", "auto", "manual"-->auto
  </mode>
  <Schedule>
    <!--ro, opt, object-->
    <TimeRange>
      <!--ro, req, object-->
      <beginTime>
        <!--ro, req, time, start time-->00:08:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->00:18:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <brightnessLimit min="0" max="100">
    <!--ro, opt, int, brightness limit of white supplement light, attr:min{req, int},max{req, int}, desc:brightness limit of white supplement light-->80
  </brightnessLimit>
  <supplementLightMode
opt="mixed,whiteLight,colorVulwhiteLight,irLight,dualLight,close,eventIntelligence,multiAzimuthMixed,multiAzimuthWhiteLight,multiAzimuthIrLight,panoramicWhit

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eLight,panoramicIrLight">
  <!--ro, opt, enum, mode of supplement light, subType:string, attr:opt{req, string}, desc:mode of supplement light-->mixed
</supplementLightMode>
<irLightBrightness min="0" max="100">
  <!--ro, opt, int, brightness of IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:this node is valid when supplementLightMode
is set to "mixed" and mixedLightBrightnessRegulatMode is set to "manual"-->80
</irLightBrightness>
<mixedLightBrightnessRegulatMode opt="manual,auto">
  <!--ro, opt, enum, brightness adjustment mode of mixed supplement light, subType:string, attr:opt{req, string}, desc:"manual" (in this mode, you need
to set the parameters irLightBrightness and brightnessLimit), "auto" (brightness-related parameters are not required). This node is valid when
supplementLightMode is set to "mixed"-->auto
</mixedLightBrightnessRegulatMode>
<highIrLightBrightness min="0" max="100">
  <!--ro, opt, int, brightness of far IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVulWhiteLight", "irLight", and "dualLight"-->80
</highIrLightBrightness>
<highWhiteLightBrightness min="0" max="100">
  <!--ro, opt, int, brightness of far white supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVulWhiteLight", "irLight", and "dualLight"-->80
</highWhiteLightBrightness>
<lowIrLightBrightness min="0" max="100">
  <!--ro, opt, int, brightness of near IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVulWhiteLight", "irLight", and "dualLight"-->80
</lowIrLightBrightness>
<lowWhiteLightBrightness min="0" max="100">
  <!--ro, opt, int, brightness of near white supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different
supplement light modes, including "mixed", "whiteLight", "colorVulWhiteLight", "irLight", and "dualLight"-->80
</lowWhiteLightBrightness>
<whiteLightBrightness min="0" max="100">
  <!--ro, opt, int, light brightness (regardless of far or near light), range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different
supplement light modes, including "mixed", "whiteLight", "colorVulWhiteLight", "irLight", and "dualLight", when the adjustment mode of the supplement light
is "auto"-->80
</whiteLightBrightness>
<whiteLightModeCfg>
  <!--ro, opt, object-->
  <highWhiteLightBrightness min="0" max="100">
    <!--ro, opt, int, brightness of far white supplement light, range:[0,100], attr:min{req, int},max{req, int}>80
  </highWhiteLightBrightness>
  <lowWhiteLightBrightness min="0" max="100">
    <!--ro, opt, int, brightness of near white supplement light, range:[0,100], attr:min{req, int},max{req, int}>80
  </lowWhiteLightBrightness>
  <whiteLightBrightness min="0" max="100">
    <!--ro, opt, int, white light brightness (regardless of far or near light), range:[0,100], attr:min{req, int},max{req, int}>80
  </whiteLightBrightness>
  <whiteLightbrightLimit min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SupplementLight.mixedLightBrightnessRegulatMode,eq,auto}, attr:min{req, int},max{req, int}-
-->80
  </whiteLightbrightLimit>
</whiteLightModeCfg>
<isAutoModeBrightnessCfg>
  <!--ro, opt, bool-->true
</isAutoModeBrightnessCfg>
<irLightbrightnessLimit min="0" max="100">
  <!--ro, opt, int, brightness limit of IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:this node is valid when the value of
Mode is "auto" and supplementLightMode is "mixed"-->80
</irLightbrightnessLimit>
<whiteLightbrightnessLimit min="0" max="100">
  <!--ro, opt, int, brightness limit of white supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:this node is valid when the value
of Mode is "auto" and supplementLightMode is "mixed"-->80
</whiteLightbrightnessLimit>
<isNotSupportMixedWhiteLight>
  <!--ro, opt, bool-->true
</isNotSupportMixedWhiteLight>
<EventIntelligenceModeCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode opt="manual,auto">
    <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:supplement light adjustment mode-->auto
  </brightnessRegulatMode>
  <whiteLightBrightness min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}>80
  </whiteLightBrightness>
  <irLightBrightness min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}>80
  </irLightBrightness>
  <BrightnessCtrlList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}>
    <BrightnessCtrl>
      <!--ro, opt, object-->
      <lightType opt="highWhite,lowWhite,highIrLight,lowIrLight">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}>highWhite
      </lightType>
      <brightness min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}>80
      </brightness>
    </BrightnessCtrl>
  </BrightnessCtrlList>
  <irLightbrightLimit min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}>80
  </irLightbrightLimit>
  <whiteLightbrightLimit min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}>80
  </whiteLightbrightLimit>
</EventIntelligenceModeCfg>
<MultiAxiomativeModeCfg>

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<!--ro, opt, object-->
<brightnessRegulatMode opt="manual,auto">
  <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:supplement light adjustment mode-->auto
</brightnessRegulatMode>
<LightCtrlList size="12">
  <!--ro, opt, array, subType:object, attr:size{req, int}-->
  <LightCtrl>
    <!--ro, opt, object-->
    <azimuthType opt="left,right">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->left
    </azimuthType>
    <lightType opt="highWhite,midWhite,lowWhite,highIrLight,midIrLight,lowIrLight">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->highWhite
    </lightType>
    <brightness min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
    </brightness>
  </LightCtrl>
</LightCtrlList>
<regulatSensitivityType opt="fast,medium,slow">
  <!--ro, opt, enum, subType:string, dep:and,{$.SupplementLight.MultiAzimuthMixedModeCfg.brightnessRegulatMode,eq,auto}, attr:opt{req, string}-->fast
</regulatSensitivityType>
</MultiAzimuthMixedModeCfg>
<MultiAzimuthWhiteLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode opt="manual,auto">
    <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:supplement light adjustment mode-->auto
  </brightnessRegulatMode>
  <LightCtrlList size="6">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <LightCtrl>
      <!--ro, opt, object-->
      <azimuthType opt="left,right">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->left
      </azimuthType>
      <lightType opt="highWhite,midWhite,lowWhite,white">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->highWhite
      </lightType>
      <brightness min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
  <regulatSensitivityType opt="fast,medium,slow">
    <!--ro, opt, enum, subType:string, dep:and,{$.SupplementLight.MultiAzimuthWhiteLightCfg.brightnessRegulatMode,eq,auto}, attr:opt{req, string}-->fast
  </regulatSensitivityType>
</MultiAzimuthWhiteLightCfg>
<MultiAzimuthIrLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode opt="manual,auto">
    <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:supplement light adjustment mode-->auto
  </brightnessRegulatMode>
  <LightCtrlList size="6">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <LightCtrl>
      <!--ro, opt, object-->
      <azimuthType opt="left,right">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->left
      </azimuthType>
      <lightType opt="highIrLight,midIrLight,lowIrLight,irLight">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->highIrLight
      </lightType>
      <brightness min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</MultiAzimuthIrLightCfg>
<PanoramicWhiteLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode opt="manual,auto">
    <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:supplement light adjustment mode-->auto
  </brightnessRegulatMode>
  <LightCtrlList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <LightCtrl>
      <!--ro, opt, object-->
      <lightID min="1" max="4">
        <!--ro, opt, int, range:[1,4], attr:min{req, int},max{req, int}-->1
      </lightID>
      <brightness min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</PanoramicWhiteLightCfg>
<PanoramicIrLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode opt="manual,auto">
    <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:supplement light adjustment mode-->auto
  </brightnessRegulatMode>
  <LightCtrlList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->

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<LightCtrl>
  <!--ro, opt, object-->
  <lightID min="1" max="4">
    <!--ro, opt, int, range:[1,4], attr:min{req, int},max{req, int}-->1
  </lightID>
  <brightness min="0" max="100">
    <!--ro, opt, int, supplement light brightness, range:[0,100], attr:min{req, int},max{req, int}-->80
  </brightness>
</LightCtrl>
</LightCtrlList>
</PanoramicIrLightCfg>
<whiteLightEnhanceEnabled opt="true,false" def="false">
  <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->true
</whiteLightEnhanceEnabled>
<LightManualCtrlList size="12">
  <!--ro, opt, array, subType:object, dep:and,{$.SupplementLight.mode,eq>manual}, attr:size{req, int}-->
  <LightCtrl>
    <!--ro, opt, object-->
    <lightType opt="highIrLight,midIrLight,lowIrLight,irLight,highWhite,midWhite,lowWhite,white">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->highWhite
    </lightType>
    <brightness min="0" max="100">
      <!--ro, opt, int, supplement light brightness, range:[0,100], attr:min{req, int},max{req, int}-->80
    </brightness>
  </LightCtrl>
</LightManualCtrlList>
<IrLightModeCfg>
  <!--ro, opt, object-->
  <irLightbrightLimit min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
  </irLightbrightLimit>
</IrLightModeCfg>
<mixedModeCfg>
  <!--ro, opt, object-->
  <whiteLightbrightLimit min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SupplementLight.mixedLightBrightnessRegulatMode,eq:auto}, attr:min{req, int},max{req, int}-->80
  </whiteLightbrightLimit>
  <irLightbrightLimit min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
  </irLightbrightLimit>
</mixedModeCfg>
<colorVulWhiteLightModeCfg>
  <!--ro, opt, object-->
  <whiteLightbrightLimit min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SupplementLight.mixedLightBrightnessRegulatMode,eq:auto}, attr:min{req, int},max{req, int}-->100
  </whiteLightbrightLimit>
</colorVulWhiteLightModeCfg>
</SupplementLight>
<OpticalDehaze>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</OpticalDehaze>
<ManualRanging>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</ManualRanging>
<OIS>
  <!--ro, opt, object-->
  <mode opt="close,ordinary,experts" default="ordinary">
    <!--ro, req, enum, mode, subType:string, attr:opt{req, string},default{opt, string}, desc:wDR mode: "open,close,auto"-->close
  </mode>
  <OISLevel opt="high,middle,low" default="middle">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},default{opt, string}-->high
  </OISLevel>
  <OISSensitivity opt="high,middle,low" default="middle">
    <!--ro, opt, string, attr:opt{req, string},default{opt, string}-->test
  </OISSensitivity>
</OIS>
<isSupportLaserSpotManual>
  <!--ro, opt, bool-->true
</isSupportLaserSpotManual>
<isSupportLaserSpotAdjustment>
  <!--ro, opt, bool-->true
</isSupportLaserSpotAdjustment>
<DigitalZoom>
  <!--ro, opt, object-->
  <ZoomRatio opt="1x,2x,3x,4x,8x,16x,32x">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->1x
  </ZoomRatio>
</DigitalZoom>
<Palettes>
  <!--ro, opt, object-->
  <mode
opt="WhiteHot,BlackHot,Fusion1,Rainbow,Fusion2,Ironbow1,Ironbow2,Sepia,Color1,Color2,IceFire,Rain,RedHot,GreenHot,DeepBlue,Color3,Birdwatch,blueRed,winter,s
ummer,greenYellow,blueYellow,customize">
    <!--ro, opt, enum, mode, subType:string, attr:opt{req, string}, desc:mode-->WhiteHot
  </mode>

```

```

<ColorateTarget>
  <!--ro, opt, object, mode="WhiteHot", "target coloration", desc:mode="WhiteHot"-->
  <enhanceMode opt="palette">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->palette
  </enhanceMode>
  <ColorateTargetModelList>
    <!--ro, opt, array, subType:object-->
    <ColorateTargetMode>
      <!--ro, opt, object, List-->
      <id>
        <!--ro, req, int, No., desc:it starts from 1-->0
      </id>
      <mode opt="colorateHotAreae, colorateIntervalArea, colorateColdArea">
        <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->colorateHotAreae
      </mode>
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->>true
      </enabled>
      <TemperatureLimit>
        <!--ro, req, object-->
        <minTemperature min="-20.0" max="150.0" default="0.0">
          <!--ro, opt, float, the minimum temperature, attr:min{req, float},max{req, float},default{opt, float}-->0.000
        </minTemperature>
        <maxTemperature min="-20.0" max="150.0" default="0.0">
          <!--ro, opt, float, the maximum temperature, attr:min{req, float},max{req, float},default{opt, float}-->0.000
        </maxTemperature>
      </TemperatureLimit>
      <Color>
        <!--ro, req, object, image adjustment capability-->
        <R>
          <!--ro, req, int-->0
        </R>
        <G>
          <!--ro, req, int, G-->0
        </G>
        <B>
          <!--ro, req, int-->0
        </B>
      </Color>
    </ColorateTargetMode>
  </ColorateTargetModelList>
</ColorateTarget>
<customizeColorList size="10">
  <!--ro, opt, object, dep:and,{$.Palettes.mode,eq,customize}, attr:size{req, int}-->
  <customizeColorItem>
    <!--ro, req, object-->
    <customizeColorID min="1" max="10" def="1">
      <!--ro, req, int, range:[1,10], attr:min{req, int},max{req, int},def{req, int}-->1
    </customizeColorID>
    <maxTempCustomizeColor>
      <!--ro, req, object-->
      <R>
        <!--ro, req, int-->1
      </R>
      <G>
        <!--ro, req, int-->1
      </G>
      <B>
        <!--ro, req, int-->1
      </B>
    </maxTempCustomizeColor>
    <minTempCustomizeColor>
      <!--ro, req, object-->
      <R>
        <!--ro, req, int-->1
      </R>
      <G>
        <!--ro, req, int-->1
      </G>
      <B>
        <!--ro, req, int-->1
      </B>
    </minTempCustomizeColor>
  </customizeColorItem>
</customizeColorList>
</Palettes>
<LensExposureSync>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->>true
  </enabled>
</LensExposureSync>
<GapList>
  <!--ro, opt, object-->
  <Gap>
    <!--ro, opt, object-->
    <gapID>
      <!--ro, req, int-->0
    </gapID>
    <RectList size="16">
      <!--ro, opt, object, attr:size{req, int}-->
      <Rect>
        <!--ro, opt, object, area-->
        <rectID>
          <!--ro, req, int, region ID-->0
        </rectID>
      </Rect>
    </RectList>
  </Gap>
</GapList>

```

```

        </rectID>
        <height>
            <!--ro, req, int, height-->0
        </height>
        <width>
            <!--ro, req, int, width-->0
        </width>
        <x>
            <!--ro, req, int-->0
        </x>
        <y>
            <!--ro, req, int-->0
        </y>
    </Rect>
</RectList>
</Gap>
</GapList>
</LensExposureSync>
<BrightnessSuddenChangeSuppressionCap>
    <!--ro, opt, object-->
    <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
</BrightnessSuddenChangeSuppressionCap>
<AGCCap>
    <!--ro, opt, object-->
    <AGCMode opt="self_Adaptive,histogram,linear">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->self_Adaptive
    </AGCMode>
</AGCCap>
<RegionalImageEnhancementCap>
    <!--ro, opt, object-->
    <mode opt="off,customArea,up_60%,down_60%,left_60%,right_60%,center_50%,center_75%">
        <!--ro, req, string, mode, attr:opt{req, string}-->test
    </mode>
    <CustomAreaParam>
        <!--ro, opt, object-->
        <RegionCoordinatesList size="10">
            <!--ro, opt, array, area coordinate List, subType:object, range:[3,10], attr:size{req, int}-->
            <RegionCoordinates>
                <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
                <positionX>
                    <!--ro, req, int, X-coordinate, range:[0,1000]-->0
                </positionX>
                <positionY>
                    <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
                </positionY>
            </RegionCoordinates>
        </RegionCoordinatesList>
    </CustomAreaParam>
    <DefaultAreaParamList>
        <!--ro, opt, array, subType:object-->
        <DefaultAreaParam>
            <!--ro, opt, object-->
            <mode opt="up_60%,down_60%,left_60%,right_60%,center_50%,center_75%">
                <!--ro, req, string, mode, attr:opt{req, string}-->test
            </mode>
            <RegionCoordinatesList size="10">
                <!--ro, opt, array, area coordinate List, subType:object, range:[3,10], attr:size{req, int}-->
                <RegionCoordinates>
                    <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
                    <positionX>
                        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
                    </positionX>
                    <positionY>
                        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
                    </positionY>
                </RegionCoordinates>
            </RegionCoordinatesList>
        </DefaultAreaParam>
    </DefaultAreaParamList>
</RegionalImageEnhancementCap>
<faceExposure>
    <!--ro, opt, object-->
    <faceExposureEnabled opt="true,false">
        <!--ro, req, bool, whether to enable face picture exposure, attr:opt{req, string}-->true
    </faceExposureEnabled>
    <faceExposureSensitivity min="0" max="100">
        <!--ro, req, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </faceExposureSensitivity>
</faceExposure>
<lightHal>
    <!--ro, opt, object-->
    <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <halLevel min="0" max="100">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->0
    </halLevel>
    <halRange min="0" max="2064">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->0
    </halRange>
</lightHal>

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<TargetEnhancementCap>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</TargetEnhancementCap>
<LaserLight>
  <!--ro, opt, object-->
  <mode opt="auto,manual">
    <!--ro, opt, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->auto
  </mode>
  <compensationRegionMode opt="center,adaptive,full_Screen">
    <!--ro, opt, string, attr:opt{req, string}-->center
  </compensationRegionMode>
  <compensationLevel opt="low,medium,high">
    <!--ro, opt, string, attr:opt{req, string}-->medium
  </compensationLevel>
  <LaserColor>
    <!--ro, opt, object-->
    <R min="0" max="100" def="0">
      <!--ro, req, int, attr:min{req, int},max{req, int},def{req, int}-->0
    </R>
    <G min="0" max="100" def="0">
      <!--ro, req, int, attr:min{req, int},max{req, int},def{req, int}-->0
    </G>
    <B min="0" max="100" def="0">
      <!--ro, req, int, attr:min{req, int},max{req, int},def{req, int}-->0
    </B>
  </LaserColor>
</LaserLight>
<ExternalSupplementLightCap>
  <!--ro, opt, object-->
  <mode opt="close,oneMixedLight" def="close">
    <!--ro, opt, enum, mode, subType:string, attr:opt{req, string},def{req, string}, desc:WDR mode: "open,close,auto"-->close
  </mode>
  <BrightnessList>
    <!--ro, opt, array, subType:object-->
    <Brightness>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, int-->0
      </id>
      <irLightBrightness min="0" max="100" def="100">
        <!--ro, opt, int, brightness limit of IR supplement Light, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->0
      </irLightBrightness>
      <whiteLightBrightness min="0" max="100" def="1">
        <!--ro, opt, int, brightness limit of white supplement Light, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->0
      </whiteLightBrightness>
    </Brightness>
  </BrightnessList>
</ExternalSupplementLightCap>
<TempRange>
  <!--ro, opt, object-->
  <mode opt="automatic,manual,ROI,upperLimit,lowerLimit">
    <!--ro, req, enum, WDR mode: "open,close,auto", subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->automatic
  </mode>
  <temperatureUpperLimit min="-20" max="550" def="150">
    <!--ro, opt, float, range:[-20,550], unit:℃, attr:min{req, float},max{req, float},def{req, float}-->0.000
  </temperatureUpperLimit>
  <temperatureLowerLimit min="-20" max="550" def="0">
    <!--ro, opt, float, range:[-20,550], unit:℃, attr:min{req, float},max{req, float},def{req, float}-->0.000
  </temperatureLowerLimit>
  <ROIRegionCoordinatesList size="10">
    <!--ro, opt, array, subType:object, range:[3,10], attr:size{req, int}-->
    <RegionCoordinates>
      <!--ro, opt, object-->
      <positionX>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </positionX>
      <positionY>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </positionY>
    </RegionCoordinates>
  </ROIRegionCoordinatesList>
</TempRange>
<BeautyCap>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <whiteningStrength min="0" max="10">
    <!--ro, opt, int, range:[0,10], attr:min{req, int},max{req, int}-->1
  </whiteningStrength>
  <skinSmoothingStrength min="0" max="10">
    <!--ro, opt, int, range:[0,10], attr:min{req, int},max{req, int}-->1
  </skinSmoothingStrength>
</BeautyCap>
<ManualRangingCursor>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <mode opt="once,continuous">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->continuous
  </mode>

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</mode>
</ManualRangingCursor>
<MEposure>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, opt, bool, whether to enable the function-->true
  </enabled>
  <exposureLevel min="0" max="1000000">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </exposureLevel>
  <gainLevel min="0" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </gainLevel>
</MEposure>
<SensorRGBCtrl>
  <!--ro, opt, object-->
  <sensorID opt="0,1">
    <!--ro, opt, int, attr:opt{req, string}-->0
  </sensorID>
</SensorRGBCtrl>
<YPbPrCalibration>
  <!--ro, opt, object-->
  <mode opt="0,1,2">
    <!--ro, opt, int, mode, attr:opt{req, string}-->1
  </mode>
</YPbPrCalibration>
<isSupportDOFAdjust>
  <!--ro, opt, bool-->true
</isSupportDOFAdjust>
<isSupportFocalLengthType>
  <!--ro, opt, bool-->true
</isSupportFocalLengthType>
<isSupportLDCI>
  <!--ro, opt, bool-->true
</isSupportLDCI>
<intrusionTriggeredLightingCfg>
  <!--ro, opt, object-->
  <enabled opt="true,false" def="false">
    <!--ro, req, bool, attr:opt{req, string},def{req, bool}-->>false
  </enabled>
</intrusionTriggeredLightingCfg>
<shotProCfg>
  <!--ro, opt, object-->
  <shotProMode opt="close,general,enhancement" def="general">
    <!--ro, req, string, attr:opt{req, string},def{req, string}-->general
  </shotProMode>
  <isNeedReboot>
    <!--ro, req, string-->true
  </isNeedReboot>
</shotProCfg>
<FFC>
  <!--ro, opt, object-->
  <mode opt="schedule,temperature,close,auto">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->schedule
  </mode>
  <compensationTime opt="10,20,30,40,50,60,120,180,240">
    <!--ro, opt, enum, subType:int, dep:and,{$.FFC.mode,eq,schedule}, attr:opt{req, string}-->10
  </compensationTime>
  <customCompensationTime min="10" max="120">
    <!--ro, opt, int, range:[10,120], unit:min, attr:min{req, int},max{req, int}-->10
  </customCompensationTime>
</FFC>
<DDE>
  <!--ro, opt, object-->
  <mode opt="general,advanced,close">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, dep:and,{$.DDE.mode,eq,general}-->
    <enhanceLevel min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->5
    </enhanceLevel>
  </GeneralMode>
</DDE>
<carAutoExp>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <mode opt="general,advanced">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->general
  </mode>
  <timeRangeList size="2">
    <!--ro, req, array, subType:object, attr:size{req, int}-->
    <timeRange>
      <!--ro, req, object-->
      <beginTime>
        <!--ro, req, time-->00:00:00
      </beginTime>
      <endTime>
        <!--ro, req, time-->00:00:00
      </endTime>
    </timeRange>
  </timeRangeList>

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</timeRangeList>
<expoShutter min="0" max="1000000">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->40000
</expoShutter>
<expoGainLevel min="1" max="100">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->100
</expoGainLevel>
<expoSpeedLevel min="1" max="100">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->10
</expoSpeedLevel>
</carAutoExp>
<isSupportImageCorrection>
  <!--ro, opt, bool-->true
</isSupportImageCorrection>
<isSupportMotionModeParams>
  <!--ro, opt, bool-->true
</isSupportMotionModeParams>
<isSupportAntiBandingParams>
  <!--ro, opt, bool-->true
</isSupportAntiBandingParams>
<isSupportAntiHeatwaveParams>
  <!--ro, opt, bool-->true
</isSupportAntiHeatwaveParams>
<isSupportWhiteLightEnhancementParams>
  <!--ro, opt, bool-->true
</isSupportWhiteLightEnhancementParams>
<BLWebDisplay>
  <!--ro, opt, object-->
  <mode opt="close,BLC,WDR,autoWDR,SDR,HLC,autoHLC">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->BLC
  </mode>
</BLWebDisplay>
<isSupportRestoreImageWhiteBalanceParam>
  <!--ro, opt, bool-->true
</isSupportRestoreImageWhiteBalanceParam>
<isSupportRestoreImageAdjustmentParam>
  <!--ro, opt, bool-->true
</isSupportRestoreImageAdjustmentParam>
<isSupportRestoreImageExposureParam>
  <!--ro, opt, bool-->true
</isSupportRestoreImageExposureParam>
<isSupportRestoreImageEnhanceParam>
  <!--ro, opt, bool-->true
</isSupportRestoreImageEnhanceParam>
<isSupportRestoreImageAdjustParam>
  <!--ro, opt, bool-->true
</isSupportRestoreImageAdjustParam>
<isSupportMovingTargetShutterParams>
  <!--ro, opt, bool-->true
</isSupportMovingTargetShutterParams>
<isSupportControlSteplessDigitalZoom>
  <!--ro, opt, bool-->true
</isSupportControlSteplessDigitalZoom>
<isSupportGetSteplessDigitalZoom>
  <!--ro, opt, bool-->true
</isSupportGetSteplessDigitalZoom>
<isSupportSmartNoiseReduce>
  <!--ro, opt, bool-->true
</isSupportSmartNoiseReduce>
<isSupportAFDFocusAssist>
  <!--ro, opt, bool-->true
</isSupportAFDFocusAssist>
<smartNoiseReduce>
  <!--ro, opt, object, dep:and,{$.ImageChannel.isSupportSmartNoiseReduce,eq,true}-->
  <level min="1" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </level>
  <noiseReduceGain>
    <!--ro, opt, object-->
    <mode opt="auto,manual">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
    </mode>
    <autoGainThreshold min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->50
    </autoGainThreshold>
    <manualGainLevel opt="1,2,3">
      <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
    </manualGainLevel>
  </noiseReduceGain>
</smartNoiseReduce>
<isSupportSearchThermometryTemplateMap>
  <!--ro, opt, bool-->true
</isSupportSearchThermometryTemplateMap>
<isSupportGetMovingTargetShutterRecommParams>
  <!--ro, opt, bool-->true
</isSupportGetMovingTargetShutterRecommParams>
<isSupportModifyMovementCustomParams>
  <!--ro, opt, bool-->true
</isSupportModifyMovementCustomParams>
<isSupportSearchMovementCustomParams>
  <!--ro, opt, bool-->true
</isSupportSearchMovementCustomParams>
<isSupportManualSwitchingICR>
  <!--ro, opt, bool-->true
</isSupportManualSwitchingICR>

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```

<!--ro, opt, bool-->true
</isSupportManualSwitchingICR>
<isSupportZoomFocalLengthType>
  <!--ro, opt, bool-->true
</isSupportZoomFocalLengthType>
<NoiseReduceExt>
  <!--ro, opt, object-->
  <mode opt="close,general,trailingNoiseBalance,advanced">
    <!--ro, req, enum, subType:string, attr:opt(req, string)-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object-->
    <generalLevel min="0" max="255">
      <!--ro, opt, int, attr:min(req, int),max(req, int)-->50
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object-->
    <spatialLevel min="0" max="100">
      <!--ro, opt, int, attr:min(req, int),max(req, int)-->50
    </spatialLevel>
    <temporalLevel min="0" max="100">
      <!--ro, opt, int, attr:min(req, int),max(req, int)-->50
    </temporalLevel>
  </AdvancedMode>
  <trailingNoiseBalance>
    <!--ro, opt, object-->
    <level min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min(req, int),max(req, int)-->1
    </level>
  </trailingNoiseBalance>
</NoiseReduceExt>
<isSupportSmartSuperResolution>
  <!--ro, opt, bool-->true
</isSupportSmartSuperResolution>
<isSupportHighlightInhibitionParams>
  <!--ro, opt, bool-->true
</isSupportHighlightInhibitionParams>
<isSupportImagePerspectiveParams>
  <!--ro, opt, bool-->true
</isSupportImagePerspectiveParams>
<isSupportZoomLensMotionParams>
  <!--ro, opt, bool-->true
</isSupportZoomLensMotionParams>
<isSupportFocusLensMotionParams>
  <!--ro, opt, bool-->true
</isSupportFocusLensMotionParams>
<isSupportExportImageISPbinFile>
  <!--ro, opt, bool-->true
</isSupportExportImageISPbinFile>
<isSupportAlgStreamRecommParams>
  <!--ro, opt, bool-->true
</isSupportAlgStreamRecommParams>
<isSupportGetHighFrameRateStatus>
  <!--ro, opt, bool-->true
</isSupportGetHighFrameRateStatus>
<isSupportImageStitchingAdjustParams>
  <!--ro, opt, bool-->true
</isSupportImageStitchingAdjustParams>
<isSupportHotHighlightInhibitionParams>
  <!--ro, opt, bool-->true
</isSupportHotHighlightInhibitionParams>
<isSupportJPEGParam>
  <!--ro, opt, bool-->true
</isSupportJPEGParam>
<isSupportDarkEnhance>
  <!--ro, opt, bool-->true
</isSupportDarkEnhance>
<isSupportBrightEnhance>
  <!--ro, opt, bool-->true
</isSupportBrightEnhance>
<isSupportIcr>
  <!--ro, opt, bool-->true
</isSupportIcr>
<isSupportMultishut>
  <!--ro, opt, bool-->true
</isSupportMultishut>
<isSupportSectionCtrl>
  <!--ro, opt, bool-->true
</isSupportSectionCtrl>
<isSupportGrayRange>
  <!--ro, opt, bool-->true
</isSupportGrayRange>
<isSupportCapImage>
  <!--ro, opt, bool-->true
</isSupportCapImage>
<isSupportPlateBright>
  <!--ro, opt, bool-->true
</isSupportPlateBright>
<NoiseReduce2D>
  <!--ro, opt, object-->
  <noiseReduce2DEnable>
    <!--ro, opt, bool-->true
  </noiseReduce2DEnable>

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```
<noiseReduce2DLevel min="0" max="255">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
</noiseReduce2DLevel>
</NoiseReduce2D>
<BrightEnhance>
  <!--ro, opt, object-->
  <brightEnhanceLevel min="0" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
  </brightEnhanceLevel>
</BrightEnhance>
<isSupportBackFrontLightingPlanCfg>
  <!--ro, opt, bool-->true
</isSupportBackFrontLightingPlanCfg>
<isSupportKeepFocusParams>
  <!--ro, opt, bool-->true
</isSupportKeepFocusParams>
<isSupportRawStreamExposureSyncParams>
  <!--ro, opt, bool-->true
</isSupportRawStreamExposureSyncParams>
<isSupportZoomLaserLightCalibrationParams>
  <!--ro, opt, bool-->true
</isSupportZoomLaserLightCalibrationParams>
<isSupportRegionalExposureList>
  <!--ro, opt, bool-->true
</isSupportRegionalExposureList>
<isSupportExportImageKMatrixFile>
  <!--ro, opt, bool-->true
</isSupportExportImageKMatrixFile>
<isSupportExportPaletteFile>
  <!--ro, opt, bool-->true
</isSupportExportPaletteFile>
<isSupportSteplessDigitalZoomStep>
  <!--ro, opt, bool-->true
</isSupportSteplessDigitalZoomStep>
<isSupportLensInitializationStatus>
  <!--ro, opt, bool-->true
</isSupportLensInitializationStatus>
<isSupportScreenAntiFlickerParams>
  <!--ro, opt, bool-->true
</isSupportScreenAntiFlickerParams>
<isSupportMovementParams>
  <!--ro, opt, bool-->true
</isSupportMovementParams>
<isSupportCorridorRevolveTest>
  <!--ro, opt, bool-->true
</isSupportCorridorRevolveTest>
<isSupportSpecialPalettesParams>
  <!--ro, opt, bool-->true
</isSupportSpecialPalettesParams>
<isSupportModifyLaserFocusParams>
  <!--ro, opt, bool-->true
</isSupportModifyLaserFocusParams>
<calibrationFile>
  <!--ro, opt, object-->
  <mode opt="factory,user,auto">
    <!--ro, opt, string, attr:opt{req, string}-->factory
  </mode>
  <operate opt="import,export,search">
    <!--ro, opt, string, attr:opt{req, string}-->import
  </operate>
</calibrationFile>
<SplitJointPara>
  <!--ro, opt, object-->
  <AvsParam>
    <!--ro, opt, object-->
    <horizontalFOV min="1000" max="1800000">
      <!--ro, opt, int, range:[1000,180000], attr:min{req, int},max{req, int}-->1000
    </horizontalFOV>
    <verticalFOV min="1000" max="1800000">
      <!--ro, opt, int, range:[1000,180000], attr:min{req, int},max{req, int}-->1000
    </verticalFOV>
    <roll min="1000" max="1800000">
      <!--ro, opt, int, range:[1000,180000], attr:min{req, int},max{req, int}-->1000
    </roll>
    <pitch min="1000" max="1800000">
      <!--ro, opt, int, range:[1000,180000], attr:min{req, int},max{req, int}-->1000
    </pitch>
    <centerX min="1000" max="1800000">
      <!--ro, opt, int, range:[1000,180000], attr:min{req, int},max{req, int}-->1000
    </centerX>
    <centerY min="1000" max="1800000">
      <!--ro, opt, int, range:[1000,180000], attr:min{req, int},max{req, int}-->1000
    </centerY>
  </AvsParam>
  <AvsLutAttr>
    <!--ro, opt, object-->
    <caliMode opt="0,1,2">
      <!--ro, opt, int, attr:opt{req, string}-->1
    </caliMode>
    <avsDistance min="0.0" max="1000.0">
      <!--ro, opt, float, range:[0.0,1000.0], attr:min{req, float},max{req, float}-->1.0
    </avsDistance>
  </AvsLutAttr>
  <quickMode opt="1,2,3">
    <!--ro, opt, int, attr:opt{req, string}-->1
  </quickMode>
</SplitJointPara>
```

```

<!--ro, opt, int, attr:opt{req, string}-->1
</quickMode>
</SplitJointPara>
<sceneNum>
<!--ro, opt, int, range:[1,16]-->1
</sceneNum>
</calibrationFile>
<isSupportImportFocusRawData>
<!--ro, opt, bool-->true
</isSupportImportFocusRawData>
<isSupportSearchMappingZoomFocus>
<!--ro, opt, bool-->true
</isSupportSearchMappingZoomFocus>
<isSupportMovementRawDataAsyncUpload>
<!--ro, opt, bool-->true
</isSupportMovementRawDataAsyncUpload>
<isSupportSearchAsyncUploadStatus>
<!--ro, opt, bool-->true
</isSupportSearchAsyncUploadStatus>
<isSupportSensorAdjustment>
<!--ro, opt, bool-->true
</isSupportSensorAdjustment>
<isSupportSearchRawDataList>
<!--ro, opt, bool-->true
</isSupportSearchRawDataList>
<isSupportRawDataParams>
<!--ro, opt, bool-->true
</isSupportRawDataParams>
</ImageChannel>

```

12.4.1.3 Set the image adjustment parameters in auto mode of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/color

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--opt, object, gray scale-->
    <grayScaleMode>
      <!--opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, bool, whether to enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the saturation cannot be adjusted-->
  </nightMode>
  <redBrightnessLevel>
    <!--opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--opt, int, range:[0,100]-->0
  </sharpnessLevel>
</Color>

```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

12.4.1.4 Get the image adjustment parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/color?parameterType= <parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it will be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--ro, opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--ro, opt, object-->
    <grayScaleMode>
      <!--ro, opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--ro, opt, bool, whether to enable night mode-->true
  </nightMode>
  <redBrightnessLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--ro, opt, int, sharpness, range:[0,100]-->0
  </sharpnessLevel>
  <hueMode>
    <!--ro, opt, enum, subType:string-->warm
  </hueMode>
</Color>
```

12.4.1.5 Get the image processing parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it will be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <FocusConfiguration>
    <!--ro, opt, object-->
    <focusStyle>
      <!--ro, req, enum, subType:string-->AUTO
    </focusStyle>
    <focusLimited>
      <!--ro, opt, enum, subType:int-->10
    </focusLimited>
    <focusPosition>
      <!--ro, opt, int, range:[1,10]-->1
    </focusPosition>
    <focusSpeed>
      <!--ro, opt, int, range:[1,3]-->1
    </focusSpeed>
    <focusSensitivity>
      <!--ro, opt, int, range:[0,2]-->1
    </focusSensitivity>
    <temperatureChangeAdaptEnabled>
      <!--ro, opt, bool-->true
    </temperatureChangeAdaptEnabled>
    <relativeFocusPos>
      <!--ro, opt, int, range:[0,4294967295]-->1
    </relativeFocusPos>
    <lensType>
      <!--ro, opt, enum, subType:string-->1140
    </lensType>
    <highTemperaturePriority>
      <!--ro, opt, bool-->true
    </highTemperaturePriority>
    <focusRange>
      <!--ro, opt, enum, subType:string-->max
    </focusRange>
    <focusDistanceMode>
      <!--ro, opt, enum, subType:string-->compatible
    </focusDistanceMode>
    <focusDistance>
      <!--ro, opt, float, range:[0.2,999.9], step:0.1, unit:m, dep:and,{$.ImageChannel.Focusconfiguration.focusStyle,eq,distanceCalculate}-->0.2
    </focusDistance>
    <focusStrategy>
      <!--ro, opt, enum, subType:string-->DToF
    </focusStrategy>
  </FocusConfiguration>
  <LensInitialization>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </LensInitialization>
  <ImageFlip>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </ImageFlip>
</ImageChannel>
```



```

</enabled>
<ImageFlipStyle>
  <!--ro, opt, enum, subType:string-->LEFTRIGHT
</ImageFlipStyle>
<flipAngle>
  <!--ro, opt, enum, subType:int-->90
</flipAngle>
</ImageFlip>
<ImageFreeze>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</ImageFreeze>
<proportionalpan>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</proportionalpan>
<WDR>
  <!--ro, opt, object-->
  <mode>
    <!--ro, req, enum, subType:string-->open
  </mode>
  <WDRLevel>
    <!--ro, opt, int, range:[1,100]-->1
  </WDRLevel>
  <WDRContrastLevel>
    <!--ro, opt, int, range:[1,100]-->1
  </WDRContrastLevel>
  <WDRLevel1>
    <!--ro, opt, int, range:[1,100]-->1
  </WDRLevel1>
  <WDRStreakInhibition>
    <!--ro, opt, bool-->>false
  </WDRStreakInhibition>
  <WDRCloseToOpenFilterLevel>
    <!--ro, opt, int, range:[1,9]-->5
  </WDRCloseToOpenFilterLevel>
  <WDROpenToCloseFilterLevel>
    <!--ro, opt, int, range:[1,9]-->5
  </WDROpenToCloseFilterLevel>
</WDR>
<BLC>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <BLCMode>
    <!--ro, opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </BLCLevel>
  <BLCRegionList>
    <!--ro, opt, array, subType:object-->
    <BLCRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int, range:[1,10]-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
<NoiseReduce>
  <!--ro, opt, object-->
  <mode>
    <!--ro, req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object-->
    <generalLevel>
      <!--ro, req, int, range:[0,100]-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object-->
    <FrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->1
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>

```

```

<!--ro, req, int, range:[0,100]-->1
</InterFrameNoiseReduceLevel>
<secondarySpatialNoiseReduceLevel>
<!--ro, opt, int, range:[0,100]-->1
</secondarySpatialNoiseReduceLevel>
<secondaryTemporalNoiseReduceLevel>
<!--ro, opt, int, range:[0,100]-->1
</secondaryTemporalNoiseReduceLevel>
</AdvancedMode>
</NoiseReduce>
<ImageEnhancement>
<!--ro, opt, object-->
<enabled>
<!--ro, req, bool-->true
</enabled>
<ImageEnhancementLevel>
<!--ro, opt, int, range:[0,10]-->0
</ImageEnhancementLevel>
</ImageEnhancement>
<DSS>
<!--ro, opt, object-->
<enabled>
<!--ro, req, bool-->true
</enabled>
<DSSLevel>
<!--ro, opt, enum, subType:string-->*1.25
</DSSLevel>
<mode>
<!--ro, opt, enum, subType:string-->frame
</mode>
<slowShutterLevel>
<!--ro, opt, int, range:[1,6]-->1
</slowShutterLevel>
</DSS>
<WhiteBalance>
<!--ro, opt, object-->
<WhiteBalanceStyle>
<!--ro, opt, enum, subType:string-->auto
</WhiteBalanceStyle>
<WhiteBalanceRed>
<!--ro, opt, int, range:[-127,127]-->1
</WhiteBalanceRed>
<WhiteBalanceBlue>
<!--ro, opt, int, range:[-127,127]-->1
</WhiteBalanceBlue>
<WhiteBalanceLevel>
<!--ro, opt, int, range:[0,100]-->1
</WhiteBalanceLevel>
<customColorTemperatureLevel>
<!--ro, opt, int, range:[1800,10000]-->1
</customColorTemperatureLevel>
<WhiteBalanceRedFineTuning>
<!--ro, opt, int, range:[0,100]-->1
</WhiteBalanceRedFineTuning>
<WhiteBalanceBlueFineTuning>
<!--ro, opt, int, range:[0,100]-->1
</WhiteBalanceBlueFineTuning>
<autoMode>
<!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,auto}-->video
</autoMode>
<manualMode>
<!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq>manual}-->colorFineTuning
</manualMode>
</WhiteBalance>
<Exposure>
<!--ro, opt, object-->
<ExposureType>
<!--ro, req, enum, subType:string-->auto
</ExposureType>
<autoIrisLevel>
<!--ro, opt, int, range:[0,100]-->1
</autoIrisLevel>
<OverexposeSuppress>
<!--ro, opt, object-->
<enabled>
<!--ro, req, bool-->true
</enabled>
<Type>
<!--ro, opt, enum, subType:string-->AUTO
</Type>
<DistanceLevel>
<!--ro, opt, int, range:[1,100]-->1
</DistanceLevel>
<shortIRDistanceLevel>
<!--ro, opt, int, range:[1,100]-->1
</shortIRDistanceLevel>
<longIRDistanceLevel>
<!--ro, opt, int, range:[1,100]-->1
</longIRDistanceLevel>
<supplementLightIntensity1>
<!--ro, opt, int, range:[1,100]-->1
</supplementLightIntensity1>
<supplementLightIntensity2>

```

```

    <!--ro, opt, int, range:[1,100]-->1
  </supplementLightIntensity2>
  <supplementLightIntensity3>
    <!--ro, opt, int, range:[1,100]-->1
  </supplementLightIntensity3>
  <supplementLightIntensity4>
    <!--ro, opt, int, range:[1,100]-->1
  </supplementLightIntensity4>
</OverexposeSuppress>
<pIris>
  <!--ro, opt, object-->
  <pIrisType>
    <!--ro, opt, enum, subType:string-->MANUAL
  </pIrisType>
  <IrisLevel>
    <!--ro, opt, int, range:[1,100]-->1
  </IrisLevel>
</pIris>
<faceExposure>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <sensitivity>
    <!--ro, opt, int, range:[0,100]-->0
  </sensitivity>
</faceExposure>
<PIrisGeneral>
  <!--ro, opt, object-->
  <irisLevel>
    <!--ro, opt, int, range:[1,100]-->1
  </irisLevel>
</PIrisGeneral>
<lowGhostingExposure>
  <!--ro, opt, object-->
  <lowGhostingShutterLevel>
    <!--ro, opt, enum, subType:string-->1/100
  </lowGhostingShutterLevel>
</lowGhostingExposure>
<exposureCompensateLevel>
  <!--ro, opt, int, range:[-100,100]-->0
</exposureCompensateLevel>
<exposureControl>
  <!--ro, opt, enum, subType:string-->center
</exposureControl>
</Exposure>
<Sharpness>
  <!--ro, opt, object-->
  <SharpnessMode>
    <!--ro, opt, enum, subType:string-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--ro, req, int, range:[0,100]-->1
  </SharpnessLevel>
</Sharpness>
<gammaCorrection>
  <!--ro, opt, object-->
  <gammaCorrectionEnabled>
    <!--ro, opt, bool-->true
  </gammaCorrectionEnabled>
  <gammaCorrectionLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </gammaCorrectionLevel>
</gammaCorrection>
<Shutter>
  <!--ro, opt, object-->
  <ShutterLevel>
    <!--ro, opt, enum, subType:string-->1/1
  </ShutterLevel>
  <maxShutterLevelLimit>
    <!--ro, opt, enum, subType:string-->1/1
  </maxShutterLevelLimit>
  <minShutterLevelLimit>
    <!--ro, opt, enum, subType:string-->1/1
  </minShutterLevelLimit>
</Shutter>
<powerLineFrequency>
  <!--ro, opt, object-->
  <powerLineFrequencyMode>
    <!--ro, opt, enum, subType:string-->50hz
  </powerLineFrequencyMode>
</powerLineFrequency>
<Gain>
  <!--ro, opt, object-->
  <GainLevel>
    <!--ro, opt, string, range:[0,10]-->test
  </GainLevel>
  <GainLimit>
    <!--ro, opt, enum, subType:string-->0
  </GainLimit>
  <minExposureGainLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </minExposureGainLevel>
  <maxExposureGainLevel>
    <!--ro, opt, int, range:[0,10]-->10
  </maxExposureGainLevel>

```

```

<!--ro, opt, int, range:[0,10]-->0
</maxExposureGainLevel>
</Gain>
<Color>
  <!--ro, opt, object-->
  <brightnessLevel>
    <!--ro, opt, int, range:[0,255]-->0
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, range:[0,255]-->0
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, range:[0,255]-->0
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </hueLevel>
  <grayScale>
    <!--ro, opt, object-->
    <grayScaleMode>
      <!--ro, opt, enum, subType:string-->indoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--ro, opt, bool-->true
  </nightMode>
  <sharpnessLevel>
    <!--ro, opt, int, range:[0,255]-->0
  </sharpnessLevel>
  <hueMode>
    <!--ro, opt, enum, subType:string-->warm
  </hueMode>
  <saturationMode>
    <!--ro, opt, enum, subType:string-->common
  </saturationMode>
</Color>
<DayNightGate>
  <!--ro, opt, object-->
  <brightnessThreld>
    <!--ro, opt, int-->1
  </brightnessThreld>
  <nightModeCtrl>
    <!--ro, opt, object-->
    <nightModeCtrlMode>
      <!--ro, opt, enum, subType:string-->manual
    </nightModeCtrlMode>
    <manualMode>
      <!--ro, opt, object-->
      <manualModeVal>
        <!--ro, opt, enum, subType:string-->day
      </manualModeVal>
    </manualMode>
  </nightModeCtrl>
  <timeMode>
    <!--ro, opt, object-->
    <scheduleType>
      <!--ro, opt, enum, subType:string-->day
    </scheduleType>
    <switchList>
      <!--ro, opt, array, subType:object-->
      <switch>
        <!--ro, opt, object-->
        <nightModeTimeId>
          <!--ro, opt, int-->1
        </nightModeTimeId>
        <nightModeStartHour>
          <!--ro, opt, int, range:[0,23]-->1
        </nightModeStartHour>
        <nightModeStartMinute>
          <!--ro, opt, int, range:[0,59]-->1
        </nightModeStartMinute>
        <nightModeEndHour>
          <!--ro, opt, int, range:[0,23]-->1
        </nightModeEndHour>
        <nightModeEndMinute>
          <!--ro, opt, int, range:[0,59]-->1
        </nightModeEndMinute>
      </switch>
    </switchList>
  </timeMode>
  <timeScheduleList>
    <!--ro, opt, array, subType:object, range:[0,2]-->
    <timeSchedule>
      <!--ro, opt, object-->
      <triggerData>
        <!--ro, opt, object-->
        <startTime>
          <!--ro, opt, object-->
          <monthOfYear>
            <!--ro, opt, int, range:[1,12]-->0
          </monthOfYear>
        </startTime>
        <endTime>
          <!--ro, opt, object-->
          <monthOfYear>

```

```

        <!--ro, opt, int, range:[1,12]-->0
    </monthOfYear>
</endTime>
</triggerData>
<timeSwitchList>
    <!--ro, opt, array, subType:object, range:[0,1]-->
    <timeSwitch>
        <!--ro, opt, object-->
        <startHour>
            <!--ro, opt, int, range:[0,23]-->1
        </startHour>
        <startMinute>
            <!--ro, opt, int, range:[0,59]-->1
        </startMinute>
        <endHour>
            <!--ro, opt, int, range:[0,23]-->1
        </endHour>
        <endMinute>
            <!--ro, opt, int, range:[0,59]-->1
        </endMinute>
    </timeSwitch>
</timeSwitchList>
</timeSchedule>
</timeScheduleList>
</timeMode>
<autoMode>
    <!--ro, opt, object-->
    <autoSwitchThresholdVal>
        <!--ro, opt, int, range:[1,100]-->1
    </autoSwitchThresholdVal>
</autoMode>
</nightModeCtrl>
</DayNightGate>
<IrcutFilter>
    <!--ro, opt, object-->
    <IrcutFilterType>
        <!--ro, opt, enum, subType:string-->auto
    </IrcutFilterType>
    <dayToNightFilterLevel>
        <!--ro, opt, int, range:[0,7]-->4
    </dayToNightFilterLevel>
    <dayToNightFilterTime>
        <!--ro, opt, int-->1
    </dayToNightFilterTime>
    <nightToDayFilterLevel>
        <!--ro, opt, int, range:[0,7]-->4
    </nightToDayFilterLevel>
    <nightToDayFilterTime>
        <!--ro, opt, int, range:[0,10]-->1
    </nightToDayFilterTime>
</Schedule>
    <!--ro, opt, object-->
    <scheduleType>
        <!--ro, req, enum, subType:string-->day
    </scheduleType>
    <TimeRange>
        <!--ro, req, object-->
        <beginTime>
            <!--ro, req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
            <!--ro, req, time-->00:00:00+08:00
        </endTime>
    </TimeRange>
</Schedule>
<EventTrigger>
    <!--ro, opt, object-->
    <eventType>
        <!--ro, req, enum, subType:string-->IO
    </eventType>
    <IrcutFilterAction>
        <!--ro, req, enum, subType:string-->day
    </IrcutFilterAction>
</EventTrigger>
</IrcutFilter>
<Scene>
    <!--ro, opt, object-->
    <mode>
        <!--ro, req, enum, subType:string-->indoor
    </mode>
</Scene>
<PTZ>
    <!--ro, opt, object-->
    <enabled>
        <!--ro, opt, bool-->true
    </enabled>
</PTZ>
<EPTZ>
    <!--ro, opt, object-->
    <enabled>
        <!--ro, req, bool-->true
    </enabled>
    <mode>
        <!--ro, req, enum, subType:string-->indoor
    </mode>

```

```

    <!--ro, opt, enum, subType:string-->cruise
  </mode>
</EPTZ>
<EIS>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->>true
  </enabled>
  <EISLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </EISLevel>
</EIS>
<HLC>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->>true
  </enabled>
  <HLCLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </HLCLevel>
</HLC>
<ZoomLimit>
  <!--ro, opt, object-->
  <ZoomLimitRatio>
    <!--ro, opt, enum, subType:int-->4
  </ZoomLimitRatio>
</ZoomLimit>
<corridor>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->>true
  </enabled>
</corridor>
<Dehaze>
  <!--ro, opt, object-->
  <DehazeMode>
    <!--ro, opt, enum, subType:string-->open
  </DehazeMode>
  <DehazeLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </DehazeLevel>
  <dehazeIntensity>
    <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.Dehaze.DehazeMode,eq,advancedMode}-->0
  </dehazeIntensity>
  <smoothIntensity>
    <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.Dehaze.DehazeMode,eq,advancedMode}-->0
  </smoothIntensity>
</Dehaze>
<ImageMode>
  <!--ro, opt, enum, subType:string-->standard
</ImageMode>
<enableImageLossDetection>
  <!--ro, opt, bool-->true
</enableImageLossDetection>
<CaptureMode>
  <!--ro, opt, object-->
  <mode>
    <!--ro, req, enum, subType:string-->close
  </mode>
</CaptureMode>
<IrLight>
  <!--ro, opt, object-->
  <mode>
    <!--ro, req, enum, subType:string-->auto
  </mode>
  <brightnessLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </brightnessLevel>
  <sensitivityLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </sensitivityLevel>
  <brightnessLimit>
    <!--ro, opt, int, range:[0,100]-->1
  </brightnessLimit>
  <triggerMode>
    <!--ro, opt, enum, subType:string-->camera
  </triggerMode>
</IrLight>
<LensDistortionCorrection>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <level>
    <!--ro, opt, enum, subType:string-->level1
  </level>
  <accurateLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </accurateLevel>
  <zoomedInDistantViewLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </zoomedInDistantViewLevel>
  <horizontalFOV>
    <!--ro, opt, int, range:[0,100]-->0

```

```

</horizontalFOV>
<verticalFOV>
  <!--ro, opt, int, range:[0,100]-->0
</verticalFOV>
</LensDistortionCorrection>
<SupplementLight>
  <!--ro, opt, object-->
  <mode>
    <!--ro, opt, enum, subType:string-->auto
  </mode>
  <Schedule>
    <!--ro, opt, object-->
    <TimeRange>
      <!--ro, req, object-->
      <beginTime>
        <!--ro, req, time-->00:08:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, time-->00:18:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <brightnessLimit>
    <!--ro, opt, int, range:[0,100]-->80
  </brightnessLimit>
  <supplementLightMode>
    <!--ro, opt, enum, subType:string-->mixed
  </supplementLightMode>
  <irLightBrightness>
    <!--ro, opt, int, range:[0,100]-->80
  </irLightBrightness>
  <mixedLightBrightnessRegulatMode>
    <!--ro, opt, enum, subType:string-->auto
  </mixedLightBrightnessRegulatMode>
  <highIrLightBrightness>
    <!--ro, opt, int, range:[0,100]-->80
  </highIrLightBrightness>
  <highWhiteLightBrightness>
    <!--ro, opt, int, range:[0,100]-->80
  </highWhiteLightBrightness>
  <lowIrLightBrightness>
    <!--ro, opt, int, range:[0,100]-->80
  </lowIrLightBrightness>
  <lowWhiteLightBrightness>
    <!--ro, opt, int, range:[0,100]-->80
  </lowWhiteLightBrightness>
  <whiteLightBrightness>
    <!--ro, opt, int, range:[0,100]-->80
  </whiteLightBrightness>
  <WhiteLightModeCfg>
    <!--ro, opt, object-->
    <highWhiteLightBrightness>
      <!--ro, opt, int, range:[0,100]-->80
    </highWhiteLightBrightness>
    <lowWhiteLightBrightness>
      <!--ro, opt, int, range:[0,100]-->80
    </lowWhiteLightBrightness>
    <whiteLightBrightness>
      <!--ro, opt, int, range:[0,100]-->80
    </whiteLightBrightness>
    <whiteLightbrightnessLimit>
      <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SuppLementLight.mixedLightBrightnessRegulatMode,eq,auto}-->80
    </whiteLightbrightnessLimit>
    <highWhiteLightBrightnessLimit>
      <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SuppLementLight.mode,eq,auto}-->80
    </highWhiteLightBrightnessLimit>
    <middleWhiteLightBrightnessLimit>
      <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SuppLementLight.mode,eq,auto}-->80
    </middleWhiteLightBrightnessLimit>
    <lowWhiteLightBrightnessLimit>
      <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SuppLementLight.mode,eq,auto}-->80
    </lowWhiteLightBrightnessLimit>
  </WhiteLightModeCfg>
  <isAutoModeBrightnessCfg>
    <!--ro, opt, bool-->true
  </isAutoModeBrightnessCfg>
  <irLightbrightnessLimit>
    <!--ro, opt, int, range:[0,100]-->80
  </irLightbrightnessLimit>
  <whiteLightbrightnessLimit>
    <!--ro, opt, int, range:[0,100]-->80
  </whiteLightbrightnessLimit>
  <isNotSupportMixedWhiteLight>
    <!--ro, opt, bool-->true
  </isNotSupportMixedWhiteLight>
  <EventIntelligenceModeCfg>
    <!--ro, opt, object-->
    <brightnessRegulatMode>
      <!--ro, opt, enum, subType:string-->auto
    </brightnessRegulatMode>
    <whiteLightBrightness>
      <!--ro, opt, int, range:[0,100]-->80
    </whiteLightBrightness>

```

```

<irLightBrightness>
  <!--ro, opt, int, range:[0,100]-->80
</irLightBrightness>
<BrightnessCtrlList>
  <!--ro, opt, array, subType:object-->
  <BrightnessCtrl>
    <!--ro, opt, object-->
    <lightType>
      <!--ro, opt, enum, subType:string-->highWhite
    </lightType>
    <brightness>
      <!--ro, opt, int, range:[0,100]-->80
    </brightness>
  </BrightnessCtrl>
</BrightnessCtrlList>
<irLightbrightLimit>
  <!--ro, opt, int, range:[0,100]-->80
</irLightbrightLimit>
<whiteLightbrightLimit>
  <!--ro, opt, int, range:[0,100]-->80
</whiteLightbrightLimit>
</EventIntelligenceModeCfg>
<MultiAzimuthMixedModeCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode>
    <!--ro, opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--ro, opt, array, subType:object-->
    <LightCtrl>
      <!--ro, opt, object-->
      <azimuthType>
        <!--ro, opt, enum, subType:string-->left
      </azimuthType>
      <lightType>
        <!--ro, opt, enum, subType:string-->highWhite
      </lightType>
      <brightness>
        <!--ro, opt, int, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</MultiAzimuthMixedModeCfg>
<MultiAzimuthWhiteLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode>
    <!--ro, opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--ro, opt, array, subType:object-->
    <LightCtrl>
      <!--ro, opt, object-->
      <azimuthType>
        <!--ro, opt, enum, subType:string-->left
      </azimuthType>
      <lightType>
        <!--ro, opt, enum, subType:string-->highWhite
      </lightType>
      <brightness>
        <!--ro, opt, int, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</MultiAzimuthWhiteLightCfg>
<MultiAzimuthIrLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode>
    <!--ro, opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--ro, opt, array, subType:object-->
    <LightCtrl>
      <!--ro, opt, object-->
      <azimuthType>
        <!--ro, opt, enum, subType:string-->left
      </azimuthType>
      <lightType>
        <!--ro, opt, enum, subType:string-->highIrLight
      </lightType>
      <brightness>
        <!--ro, opt, int, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</MultiAzimuthIrLightCfg>
<PanoramicWhiteLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode>
    <!--ro, opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--ro, opt, array, subType:object-->
    <LightCtrl>
      <!--ro, opt, object-->

```



```

<!--ro, opt, int, range:[1,4]-->1
</lightID>
<brightness>
<!--ro, opt, int, range:[0,100]-->80
</brightness>
</LightCtrl>
</LightCtrlList>
</PanoramicWhiteLightCfg>
<PanoramicIrLightCfg>
<!--ro, opt, object-->
<brightnessRegulatMode>
<!--ro, opt, enum, subType:string-->auto
</brightnessRegulatMode>
<LightCtrlList>
<!--ro, opt, array, subType:object-->
<LightCtrl>
<!--ro, opt, object-->
<lightID>
<!--ro, opt, int, range:[1,4]-->1
</lightID>
<brightness>
<!--ro, opt, int, range:[0,100]-->80
</brightness>
</LightCtrl>
</LightCtrlList>
</PanoramicIrLightCfg>
<whiteLightEnhanceEnabled>
<!--ro, opt, bool-->true
</whiteLightEnhanceEnabled>
<LightManualCtrlList>
<!--ro, opt, array, subType:object, dep:and,{$.ImageChannel.SupplementLight.mode,eq>manual}-->
<LightCtrl>
<!--ro, opt, object-->
<lightType>
<!--ro, opt, enum, subType:string-->highIrLight
</lightType>
<brightness>
<!--ro, opt, int, range:[0,100]-->80
</brightness>
</LightCtrl>
</LightManualCtrlList>
<IrLightModeCfg>
<!--ro, opt, object-->
<irLightbrightLimit>
<!--ro, opt, int, range:[0,100]-->80
</irLightbrightLimit>
</IrLightModeCfg>
<mixedModeCfg>
<!--ro, opt, object-->
<whiteLightbrightLimit>
<!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SupplementLight.mixedLightBrightnessRegulatMode,eq:auto}-->80
</whiteLightbrightLimit>
<irLightbrightLimit>
<!--ro, opt, int, range:[0,100]-->80
</irLightbrightLimit>
</mixedModeCfg>
<colorVwWhiteLightModeCfg>
<!--ro, opt, object-->
<whiteLightbrightLimit>
<!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SupplementLight.mixedLightBrightnessRegulatMode,eq:auto}-->100
</whiteLightbrightLimit>
</colorVwWhiteLightModeCfg>
</SupplementLight>
<OpticalDehaze>
<!--ro, opt, object-->
<enabled>
<!--ro, opt, bool-->true
</enabled>
</OpticalDehaze>
<ManualRanging>
<!--ro, opt, object-->
<enabled>
<!--ro, opt, bool-->true
</enabled>
</ManualRanging>
<OIS>
<!--ro, opt, object-->
<mode>
<!--ro, req, enum, subType:string-->close
</mode>
<OISLevel>
<!--ro, opt, enum, subType:string-->high
</OISLevel>
<OISSensitivity>
<!--ro, opt, string-->test
</OISSensitivity>
</OIS>
<isSupportLaserSpotManual>
<!--ro, opt, bool-->true
</isSupportLaserSpotManual>
<isSupportLaserSpotAdjustment>
<!--ro, opt, bool-->true

```

```

</isSupportLaserSpotAdjustment>
<DigitalZoom>
  <!--ro, opt, object-->
  <ZoomRatio>
    <!--ro, opt, enum, subType:string-->1x
  </ZoomRatio>
</DigitalZoom>
<Palettes>
  <!--ro, opt, object-->
  <mode>
    <!--ro, opt, enum, subType:string-->WhiteHot
  </mode>
  <ColorateTarget>
    <!--ro, opt, object-->
    <enhanceMode>
      <!--ro, opt, enum, subType:string-->palette
    </enhanceMode>
    <ColorateTargetModeList>
      <!--ro, opt, array, subType:object-->
      <ColorateTargetMode>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->0
        </id>
        <mode>
          <!--ro, req, enum, subType:string-->colorateHotAreae
        </mode>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <TemperatureLimit>
          <!--ro, req, object-->
          <minTemperature>
            <!--ro, opt, float, range:[-20,150]-->0.000
          </minTemperature>
          <maxTemperature>
            <!--ro, opt, float, range:[-20,150]-->0.000
          </maxTemperature>
        </TemperatureLimit>
        <Color>
          <!--ro, req, object-->
          <R>
            <!--ro, req, int, range:[0,255]-->0
          </R>
          <G>
            <!--ro, req, int, range:[0,255]-->0
          </G>
          <B>
            <!--ro, req, int, range:[0,255]-->0
          </B>
        </Color>
        <paletteMode>
          <!--ro, opt, enum, subType:string, dep:and,{$.ImageChannel.Palettes.ColorateTarget.mode,eq,palette}-->WhiteHot
        </paletteMode>
        <ColorateTargetMode>
        </ColorateTargetMode>
      </ColorateTargetModeList>
    </ColorateTarget>
  </Palettes>
<ExposureSync>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</ExposureSync>
<BrightnessSuddenChangeSuppression>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</BrightnessSuddenChangeSuppression>
<AGCCap>
  <!--ro, opt, object-->
  <AGCMode>
    <!--ro, opt, enum, subType:string-->self_Adaptive
  </AGCMode>
</AGCCap>
<RegionalImageEnhancementCap>
  <!--ro, opt, object-->
  <mode>
    <!--ro, req, enum, subType:string-->off
  </mode>
  <CustomAreaParam>
    <!--ro, opt, object-->
    <RegionCoordinatesList>
      <!--ro, opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object-->
        <positionX>
          <!--ro, req, int, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </CustomAreaParam>

```

```

</RegionCoordinatesList>
</CustomAreaParam>
<DefaultAreaParamList>
  <!--ro, opt, array, subType:object-->
  <DefaultAreaParam>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->up_60%
    </mode>
    <RegionCoordinatesList>
      <!--ro, opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object-->
        <positionX>
          <!--ro, req, int, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </DefaultAreaParam>
</DefaultAreaParamList>
</RegionalImageEnhancementCap>
<faceExposure>
  <!--ro, opt, object-->
  <faceExposureEnabled>
    <!--ro, req, bool-->true
  </faceExposureEnabled>
  <faceExposureSensitivity>
    <!--ro, req, int, range:[0,100]-->0
  </faceExposureSensitivity>
</faceExposure>
<lightHal>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <halLevel>
    <!--ro, req, int, range:[0,100]-->0
  </halLevel>
  <halRange>
    <!--ro, req, int, range:[0,2064]-->0
  </halRange>
</lightHal>
<TargetEnhancement>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</TargetEnhancement>
<LaserLight>
  <!--ro, opt, object-->
  <mode>
    <!--ro, opt, enum, subType:string-->auto
  </mode>
  <compensationRegionMode>
    <!--ro, opt, enum, subType:string-->center
  </compensationRegionMode>
  <compensationLevel>
    <!--ro, opt, enum, subType:string-->low
  </compensationLevel>
  <LaserColor>
    <!--ro, opt, object-->
    <R>
      <!--ro, req, int, range:[0,100]-->0
    </R>
    <G>
      <!--ro, req, int, range:[0,100]-->0
    </G>
    <B>
      <!--ro, req, int, range:[0,100]-->0
    </B>
  </LaserColor>
</LaserLight>
<ExternalSupplementLight>
  <!--ro, opt, object-->
  <mode>
    <!--ro, opt, enum, subType:string-->close
  </mode>
  <BrightnessList>
    <!--ro, opt, array, subType:object-->
    <Brightness>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, int-->0
      </id>
      <irLightBrightness>
        <!--ro, opt, int, range:[0,100]-->0
      </irLightBrightness>
      <whiteLightBrightness>
        <!--ro, opt, int, range:[0,100]-->0
      </whiteLightBrightness>
    </Brightness>
  </BrightnessList>

```

```

    </Brightness>
  </BrightnessList>
</ExternalSupplementLight>
<TempRange>
  <!--ro, opt, object-->
  <mode>
    <!--ro, req, enum, subType:string-->automatic
  </mode>
  <temperatureUpperLimit>
    <!--ro, opt, float, range:[-20,550], unit:'C'-->0.000
  </temperatureUpperLimit>
  <temperatureLowerLimit>
    <!--ro, opt, float, range:[-20,550], unit:'C'-->0.000
  </temperatureLowerLimit>
</TempRange>
<Beauty>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <whiteningStrength>
    <!--ro, opt, int, range:[0,10]-->1
  </whiteningStrength>
  <skinSmoothingStrength>
    <!--ro, opt, int, range:[0,10]-->1
  </skinSmoothingStrength>
</Beauty>
<MExposure>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <exposureLevel>
    <!--ro, opt, string, range:[0,1000000]-->test
  </exposureLevel>
  <gainLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </gainLevel>
</MExposure>
<SensorRGBCtrl>
  <!--ro, opt, object-->
  <sensorID>
    <!--ro, opt, int, range:[0,1]-->0
  </sensorID>
</SensorRGBCtrl>
<YPbPrCalibration>
  <!--ro, opt, object-->
  <mode>
    <!--ro, opt, int, range:[0,2]-->1
  </mode>
</YPbPrCalibration>
<intrusionTriggeredLightingCfg>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</intrusionTriggeredLightingCfg>
<shotProCfg>
  <!--ro, opt, object-->
  <shotProMode>
    <!--ro, req, enum, subType:string-->general
  </shotProMode>
</shotProCfg>
<carAutoExp>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <mode>
    <!--ro, req, enum, subType:string-->general
  </mode>
  <timeRangeList>
    <!--ro, req, array, subType:object-->
    <timeRange>
      <!--ro, req, object-->
      <beginTime>
        <!--ro, req, time-->00:00:00
      </beginTime>
      <endTime>
        <!--ro, req, time-->00:00:00
      </endTime>
    </timeRange>
  </timeRangeList>
  <expoShutter>
    <!--ro, opt, int, range:[0,100000]-->40000
  </expoShutter>
  <expoGainLevel>
    <!--ro, opt, int, range:[1,100]-->100
  </expoGainLevel>
  <expoSpeedLevel>
    <!--ro, opt, int, range:[1,100]-->10
  </expoSpeedLevel>
</carAutoExp>

```

```

<noiseReduce2D>
  <!--ro, opt, object-->
  <noiseReduce2DEnable>
    <!--ro, req, bool-->true
  </noiseReduce2DEnable>
  <noiseReduce2DLevel>
    <!--ro, opt, int, range:[0,255]-->1
  </noiseReduce2DLevel>
</NoiseReduce2D>
<BrightEnhance>
  <!--ro, opt, object-->
  <brightEnhanceEnabled>
    <!--ro, req, bool-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--ro, req, int, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
<NoiseReduceExt>
  <!--ro, opt, object-->
  <mode>
    <!--ro, req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object-->
    <generalLevel>
      <!--ro, opt, int, range:[0,255]-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object-->
    <spatialLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </spatialLevel>
    <temporalLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </temporalLevel>
  </AdvancedMode>
  <trailingNoiseBalance>
    <!--ro, opt, object-->
    <level>
      <!--ro, opt, int, range:[0,100]-->1
    </level>
  </trailingNoiseBalance>
</NoiseReduceExt>
<PlateBright>
  <!--ro, opt, object-->
  <plateBrightEnabled>
    <!--ro, req, bool-->true
  </plateBrightEnabled>
  <plateBrightSensitivity>
    <!--ro, opt, int, range:[1,100]-->1
  </plateBrightSensitivity>
  <correctFactorEnabled>
    <!--ro, opt, bool-->true
  </correctFactorEnabled>
  <correctFactor>
    <!--ro, opt, int, range:[0,100]-->1
  </correctFactor>
</PlateBright>
<ImageMultishut>
  <!--ro, opt, object-->
  <doubleShutEnable>
    <!--ro, req, bool-->true
  </doubleShutEnable>
  <codecType>
    <!--ro, req, enum, subType:string-->all_ia
  </codecType>
  <threeShutEnable>
    <!--ro, opt, bool-->true
  </threeShutEnable>
  <doubleVideoEnable>
    <!--ro, opt, bool-->true
  </doubleVideoEnable>
</ImageMultishut>
<JPEGParam>
  <!--ro, opt, object-->
  <JPEGSize>
    <!--ro, opt, int, range:[64,8196], unit:KB-->64
  </JPEGSize>
  <PlateJPEGSize>
    <!--ro, opt, int, range:[1,255], unit:KB-->25
  </PlateJPEGSize>
  <MergeJPEGSize>
    <!--ro, opt, int, range:[64,8196], unit:KB-->64
  </MergeJPEGSize>
  <EXIFInformationEnabled>
    <!--ro, opt, bool-->true
  </EXIFInformationEnabled>
</JPEGParam>
<DarkEnhance>
  <!--ro, opt, object-->
  <darkEnhanceEnabled>
    <!--ro, req, bool-->true
  </darkEnhanceEnabled>

```

```

</darkEnhanceEnabled>
<darkEnhanceLevel>
  <!--ro, opt, int, range:[1,100]-->1
</darkEnhanceLevel>
</DarkEnhance>
<SectionCtrl>
  <!--ro, opt, object-->
  <sectionCtrlEnabled>
    <!--ro, req, bool-->true
  </sectionCtrlEnabled>
</SectionCtrl>
<GrayRange>
  <!--ro, opt, object-->
  <grayValueType>
    <!--ro, req, enum, subType:string-->highContrast
  </grayValueType>
</GrayRange>
<FFC>
  <!--ro, opt, object-->
  <mode>
    <!--ro, req, enum, subType:string-->schedule
  </mode>
  <compensationTime>
    <!--ro, opt, enum, subType:int, dep:and,{$.FFC.mode,eq,schedule}-->10
  </compensationTime>
  <customCompensationTime>
    <!--ro, opt, int, range:[10,120], unit:min-->10
  </customCompensationTime>
</FFC>
</ImageChannel>

```

12.4.2 Light Supplement Strategy Management

12.4.2.1 Set supplement light parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/supplementLight

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>
<SupplementLight xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--opt, enum, adjustment mode of supplement light, subType:string, desc:"schedule", "off", "on", "auto", "manual"-->auto
  </mode>
  <Schedule>
    <!--opt, object-->
    <TimeRange>
      <!--req, object-->
      <beginTime>
        <!--req, time, start time-->00:08:00+08:00
      </beginTime>
      <endTime>
        <!--req, time, end time-->00:18:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <brightnessLimit>
    <!--opt, int, brightness limit of white supplement light, desc:the brightness can be adjust when the node mode is set to "on"-->80
  </brightnessLimit>
  <supplementLightMode>
    <!--opt, enum, mode of supplement light, subType:string, desc:illumination mode: "mixed"-hybrid,"whiteLight"-white light,"close"-disabled-->mixed
  </supplementLightMode>
  <irLightBrightness>
    <!--opt, int, brightness of IR supplement light, range:[0,100], desc:this node is valid when supplementLightMode is set to "mixed" and mixedLightBrightnessRegulatMode is set to "manual"-->80
  </irLightBrightness>
  <mixedLightBrightnessRegulatMode>
    <!--opt, enum, brightness adjustment mode of mixed supplement light, subType:string, desc:"manual" (in this mode, you need to set the parameters irLightBrightness and brightnessLimit), "auto" (brightness-related parameters are not required). This node is valid when supplementLightMode is set to "mixed"-->auto
  </mixedLightBrightnessRegulatMode>
  <highIrLightBrightness>
    <!--opt, int, brightness of far IR supplement light, range:[0,100], desc:it is shared by different supplement light modes, including "mixed", "whiteLight", "colorVWhiteLight", "irLight", and "dualLight"-->80
  </highIrLightBrightness>
  <highWhiteLightBrightness>

```

```

    <!--opt, int, brightness of far white supplement light, range:[0,100], desc:it is shared by different supplement light modes, including "mixed",
    "whiteLight", "colorVwWhiteLight", "irLight", and "dualLight"-->80
  </highWhiteLightBrightness>
  <lowIrLightBrightness>
    <!--opt, int, brightness of near IR supplement light, range:[0,100], desc:it is shared by different supplement light modes, including "mixed",
    "whiteLight", "colorVwWhiteLight", "irLight", and "dualLight"-->80
  </lowIrLightBrightness>
  <lowWhiteLightBrightness>
    <!--opt, int, brightness of near white supplement light, range:[0,100], desc:it is shared by different supplement light modes, including "mixed",
    "whiteLight", "colorVwWhiteLight", "irLight", and "dualLight"-->80
  </lowWhiteLightBrightness>
  <whiteLightBrightness>
    <!--opt, int, light brightness (regardless of far or near light), range:[0,100], desc:it is shared by different supplement light modes, including
    "mixed", "whiteLight", "colorVwWhiteLight", "irLight", and "dualLight", when the adjustment mode of the supplement light is "auto"-->80
  </whiteLightBrightness>
  <whiteLightModeCfg>
    <!--opt, object-->
    <highWhiteLightBrightness>
      <!--opt, int, brightness of far white supplement light, range:[0,100]-->80
    </highWhiteLightBrightness>
    <lowWhiteLightBrightness>
      <!--opt, int, brightness of near white supplement light, range:[0,100]-->80
    </lowWhiteLightBrightness>
    <whiteLightBrightness>
      <!--opt, int, white light brightness (regardless of far or near light), range:[0,100]-->80
    </whiteLightBrightness>
  </whiteLightModeCfg>
  <irLightbrightnessLimit>
    <!--opt, int, brightness limit of IR supplement light, range:[0,100], desc:this node is valid when the value of Mode is "auto" and supplementLightMode
    is "mixed".-->80
  </irLightbrightnessLimit>
  <whiteLightbrightnessLimit>
    <!--opt, int, brightness limit of white supplement light, range:[0,100], desc:this node is valid when the value of Mode is "auto" and
    supplementLightMode is "mixed".-->80
  </whiteLightbrightnessLimit>
  <EventIntelligenceModeCfg>
    <!--opt, object-->
    <brightnessRegulatMode>
      <!--opt, enum, subType:string-->auto
    </brightnessRegulatMode>
    <whiteLightBrightness>
      <!--opt, int, white light brightness (regardless of far or near light), range:[0,100]-->80
    </whiteLightBrightness>
    <irLightBrightness>
      <!--opt, int, IR light brightness (regardless of far or near light), range:[0,100]-->80
    </irLightBrightness>
    <BrightnessCtrlList>
      <!--opt, array, subType:object-->
    <BrightnessCtrl>
      <!--opt, object-->
      <lightType>
        <!--opt, enum, subType:string-->highWhite
      </lightType>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </BrightnessCtrl>
  </BrightnessCtrlList>
  <EventIntelligenceModeCfg>
  <MultiAzimuthMixedModeCfg>
    <!--opt, object-->
    <brightnessRegulatMode>
      <!--opt, enum, subType:string-->auto
    </brightnessRegulatMode>
    <LightCtrlList>
      <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <azimuthType>
        <!--opt, enum, subType:string-->left
      </azimuthType>
      <lightType>
        <!--opt, enum, subType:string-->highWhite
      </lightType>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
  <regulatSensitivityType>
    <!--opt, enum, subType:string, dep:and,{$.SupplementLight.MultiAzimuthMixedModeCfg.brightnessRegulatMode,eq,auto}-->fast
  </regulatSensitivityType>
  <MultiAzimuthMixedModeCfg>
  <MultiAzimuthWhiteLightCfg>
    <!--opt, object-->
    <brightnessRegulatMode>
      <!--opt, enum, subType:string-->auto
    </brightnessRegulatMode>
    <LightCtrlList>
      <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <azimuthType>

```

```

    <!--opt, enum, subType:string-->left
  </azimuthType>
  <lightType>
    <!--opt, enum, subType:string-->highWhite
  </lightType>
  <brightness>
    <!--opt, int, supplement light brightness, range:[0,100]-->80
  </brightness>
</LightCtrl>
</LightCtrlList>
<regulatSensitivityType>
  <!--opt, enum, subType:string, dep:and,{$.SupplementLight.MultiAzimuthWhiteLightCfg.brightnessRegulatMode,eq,auto}-->fast
</regulatSensitivityType>
</MultiAzimuthWhiteLightCfg>
<MultiAzimuthIrLightCfg>
  <!--opt, object-->
  <brightnessRegulatMode>
    <!--opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <azimuthType>
        <!--opt, enum, subType:string-->left
      </azimuthType>
      <lightType>
        <!--opt, enum, subType:string-->highIrLight
      </lightType>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</MultiAzimuthIrLightCfg>
<PanoramicWhiteLightCfg>
  <!--opt, object-->
  <brightnessRegulatMode>
    <!--opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <lightID>
        <!--opt, int, range:[1,4]-->1
      </lightID>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</PanoramicWhiteLightCfg>
<PanoramicIrLightCfg>
  <!--opt, object-->
  <brightnessRegulatMode>
    <!--opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--opt, array, subType:object-->
    <LightCtrl>
      <!--opt, object-->
      <lightID>
        <!--opt, int, range:[1,4]-->1
      </lightID>
      <brightness>
        <!--opt, int, supplement light brightness, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</PanoramicIrLightCfg>
<whiteLightEnhanceEnabled>
  <!--opt, bool-->true
</whiteLightEnhanceEnabled>
<LightManualCtrlList>
  <!--opt, array, subType:object, dep:and,{$.SupplementLight.mode,eq>manual}-->
  <LightCtrl>
    <!--opt, object-->
    <lightType>
      <!--opt, enum, subType:string-->highIrLight
    </lightType>
    <brightness>
      <!--opt, int, supplement light brightness, range:[0,100]-->80
    </brightness>
  </LightCtrl>
</LightManualCtrlList>
</SupplementLight>

```

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	descriptio
4	Invalid Operation	failedToSwitchToIntelligentFillModeTheFlashingAlarmsIsEnabled	0x4000A3B3	--
4	Invalid Operation	ptIsNotLocatedAtThePositionOfTheFillLight	0x4000A661	--

12.4.2.2 Get supplement light parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/supplementLight

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SupplementLight xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, opt, enum, adjustment mode of supplement light, subType:string, desc:adjustment mode of supplement light,off,on,auto-->auto
  </mode>
  <Schedule>
    <!--ro, opt, object-->
    <TimeRange>
      <!--ro, req, object-->
      <beginTime>
        <!--ro, req, time, ISO8601 time-->00:08:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, time, ISO8601 time-->00:18:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <brightnessLimit>
    <!--ro, opt, int, brightness limit of supplement light, desc:brightness can be adjusted when mode is set to "on"-->80
  </brightnessLimit>
  <supplementLightMode>
    <!--ro, opt, enum, illumination mode, subType:string, desc:"mixed" (hybrid), "whiteLight" (white light), "close" (disabled)-->mixed
  </supplementLightMode>
  <irLightBrightness>
    <!--ro, opt, int, brightness of IR supplement light, range:[0,100], desc:this node is valid only when supplementLightMode is set to "mixed" and mixedLightBrightnessRegulatMode is set to "manual"-->80
  </irLightBrightness>
  <mixedLightBrightnessRegulatMode>
    <!--ro, opt, enum, brightness adjustment mode of hybrid supplement light, subType:string, desc:this node is valid only when supplementLightMode is set to "mixed"-->auto
  </mixedLightBrightnessRegulatMode>
  <highIrLightBrightness>
```

```

<!--ro, opt, int, brightness of far IR Light, range:[0,100], desc:brightness of far IR Light,which is between 0 and 100; this node cannot be configured
together with node <irLightBrightness>-->80
</highIrLightBrightness>
<highWhiteLightBrightness>
<!--ro, opt, int, brightness of far white Light, range:[0,100], desc:brightness of far white Light,which is between 0 and 100; this node cannot be
configured together with node <brightnessLimit>-->80
</highWhiteLightBrightness>
<lowIrLightBrightness>
<!--ro, opt, int, brightness of near IR Light, range:[0,100], desc:brightness of near IR Light,which is between 0 and 100; this node cannot be
configured together with node <irLightBrightness>-->80
</lowIrLightBrightness>
<lowWhiteLightBrightness>
<!--ro, opt, int, brightness of near white Light, range:[0,100], desc:brightness of near white Light,which is between 0 and 100; this node cannot be
configured together with node <brightnessLimit>-->80
</lowWhiteLightBrightness>
<whiteLightBrightness>
<!--ro, opt, int, white Light brightness, range:[0,100], desc:white Light brightness,which is between 0 and 100-->80
</whiteLightBrightness>
<whiteLightModeCfg>
<!--ro, opt, object-->
<highWhiteLightBrightness>
<!--ro, opt, int, brightness of far white Light, range:[0,100]-->80
</highWhiteLightBrightness>
<lowWhiteLightBrightness>
<!--ro, opt, int, brightness of near white Light, range:[0,100]-->80
</lowWhiteLightBrightness>
<whiteLightBrightness>
<!--ro, opt, int, white Light brightness, range:[0,100]-->80
</whiteLightBrightness>
</WhiteLightModeCfg>
<irLightbrightnessLimit>
<!--ro, opt, int, IR Light brightness limit, range:[0,100], desc:this node is valid only when mode is set to "auto" and supplementLightMode is set to
"mixed"-->80
</irLightbrightnessLimit>
<whiteLightbrightnessLimit>
<!--ro, opt, int, white Light brightness limit, range:[0,100], desc:this node is valid only when mode is set to "auto" and supplementLightMode is set to
"mixed"-->80
</whiteLightbrightnessLimit>
<EventIntelligenceModeCfg>
<!--ro, opt, object-->
<brightnessRegulatMode>
<!--ro, opt, enum, subType:string-->auto
</brightnessRegulatMode>
<whiteLightBrightness>
<!--ro, opt, int, white Light brightness, range:[0,100]-->80
</whiteLightBrightness>
<irLightBrightness>
<!--ro, opt, int, IR Light brightness, range:[0,100]-->80
</irLightBrightness>
<BrightnessCtrlList>
<!--ro, opt, array, subType:object-->
<BrightnessCtrl>
<!--ro, opt, object-->
<lightType>
<!--ro, opt, enum, subType:string-->highWhite
</lightType>
<brightness>
<!--ro, opt, int, range:[0,100]-->80
</brightness>
</BrightnessCtrl>
</BrightnessCtrlList>
</EventIntelligenceModeCfg>
<MultiAzimuthMixedModeCfg>
<!--ro, opt, object-->
<brightnessRegulatMode>
<!--ro, opt, enum, subType:string-->auto
</brightnessRegulatMode>
<LightCtrlList>
<!--ro, opt, array, subType:object-->
<LightCtrl>
<!--ro, opt, object-->
<azimuthType>
<!--ro, opt, enum, subType:string-->left
</azimuthType>
<lightType>
<!--ro, opt, enum, subType:string-->highWhite
</lightType>
<brightness>
<!--ro, opt, int, range:[0,100]-->80
</brightness>
</LightCtrl>
</LightCtrlList>
<regulatSensitivityType>
<!--ro, opt, enum, subType:string, dep:and,{$.SupplementLight.MultiAzimuthMixedModeCfg.brightnessRegulatMode,eq,auto}-->fast
</regulatSensitivityType>
</MultiAzimuthMixedModeCfg>
<MultiAzimuthWhiteLightCfg>
<!--ro, opt, object-->
<brightnessRegulatMode>
<!--ro, opt, enum, subType:string-->auto
</brightnessRegulatMode>
<LightCtrlList>
<!--ro, opt, array, subType:object-->
<LightCtrl>
<!--ro, opt, object-->
<azimuthType>
<!--ro, opt, enum, subType:string-->left
</azimuthType>
<lightType>
<!--ro, opt, enum, subType:string-->highWhite
</lightType>
<brightness>
<!--ro, opt, int, range:[0,100]-->80
</brightness>
</LightCtrl>
</LightCtrlList>
<regulatSensitivityType>
<!--ro, opt, enum, subType:string, dep:and,{$.SupplementLight.MultiAzimuthMixedModeCfg.brightnessRegulatMode,eq,auto}-->fast
</regulatSensitivityType>
</MultiAzimuthWhiteLightCfg>

```

```

<LightCtrl>
  <!--ro, opt, object-->
  <azimuthType>
    <!--ro, opt, enum, subType:string-->left
  </azimuthType>
  <lightType>
    <!--ro, opt, enum, subType:string-->highWhite
  </lightType>
  <brightness>
    <!--ro, opt, int, range:[0,100]-->80
  </brightness>
</LightCtrl>
</LightCtrlList>
<regulatSensitivityType>
  <!--ro, opt, enum, subType:string, dep:and,{$.SupplementLight.MultiAzimuthWhiteLightCfg.brightnessRegulatMode,eq,auto}-->fast
</regulatSensitivityType>
</MultiAzimuthWhiteLightCfg>
<MultiAzimuthIrLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode>
    <!--ro, opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--ro, opt, array, subType:object-->
    <LightCtrl>
      <!--ro, opt, object-->
      <azimuthType>
        <!--ro, opt, enum, subType:string-->left
      </azimuthType>
      <lightType>
        <!--ro, opt, enum, subType:string-->highIrLight
      </lightType>
      <brightness>
        <!--ro, opt, int, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</MultiAzimuthIrLightCfg>
<PanoramicWhiteLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode>
    <!--ro, opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--ro, opt, array, subType:object-->
    <LightCtrl>
      <!--ro, opt, object-->
      <lightID>
        <!--ro, opt, int, range:[1,4]-->1
      </lightID>
      <brightness>
        <!--ro, opt, int, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</PanoramicWhiteLightCfg>
<PanoramicIrLightCfg>
  <!--ro, opt, object-->
  <brightnessRegulatMode>
    <!--ro, opt, enum, subType:string-->auto
  </brightnessRegulatMode>
  <LightCtrlList>
    <!--ro, opt, array, subType:object-->
    <LightCtrl>
      <!--ro, opt, object-->
      <lightID>
        <!--ro, opt, int, range:[1,4]-->1
      </lightID>
      <brightness>
        <!--ro, opt, int, range:[0,100]-->80
      </brightness>
    </LightCtrl>
  </LightCtrlList>
</PanoramicIrLightCfg>
<whiteLightEnhanceEnabled>
  <!--ro, opt, bool-->true
</whiteLightEnhanceEnabled>
<LightManualCtrlList>
  <!--ro, opt, array, subType:object, dep:and,{$.SupplementLight.mode,eq>manual}-->
  <LightCtrl>
    <!--ro, opt, object-->
    <lightType>
      <!--ro, opt, enum, subType:string-->highIrLight
    </lightType>
    <brightness>
      <!--ro, opt, int, range:[0,100]-->80
    </brightness>
  </LightCtrl>
</LightManualCtrlList>
</SupplementLight>

```

12.4.3 Video Encoding

12.4.3.1 Set the encoding parameters of multiple channels

Request URL

PUT /ISAPI/Streaming/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<StreamingChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, channel List, subType:object, attr:version{opt, string, protocolVersion}-->
  <StreamingChannel>
    <!--opt, object-->
    <id>
      <!--req, string, channel ID-->test
    </id>
    <channelName>
      <!--req, string, channel name-->test
    </channelName>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <Transport>
      <!--req, object-->
      <rtspPortNo>
        <!--opt, int, range:[0,65535]-->554
      </rtspPortNo>
      <maxPacketSize>
        <!--opt, int-->1
      </maxPacketSize>
      <audioPacketLength>
        <!--opt, int-->1
      </audioPacketLength>
      <audioInboundPacketLength>
        <!--opt, int-->1
      </audioInboundPacketLength>
      <audioInboundPortNo>
        <!--opt, int-->1
      </audioInboundPortNo>
      <videoSourcePortNo>
        <!--opt, int-->1
      </videoSourcePortNo>
      <audioSourcePortNo>
        <!--opt, int-->1
      </audioSourcePortNo>
      <ControlProtocolList>
        <!--req, array, subType:object-->
        <ControlProtocol>
          <!--opt, object-->
          <streamingTransport>
            <!--req, enum, subType:string, desc:"HTTP", "RTSP", "SHTTP", "SHTTP"-->RTSP
          </streamingTransport>
        </ControlProtocol>
      </ControlProtocolList>
      <Unicast>
        <!--opt, object-->
        <enabled>
          <!--req, bool, whether to enable the function-->true
        </enabled>
        <interfaceID>
          <!--opt, string-->test
        </interfaceID>
        <rtpTransportType>
          <!--opt, enum, subType:string-->RTP/UDP
        </rtpTransportType>
      </Unicast>
      <Multicast>
        <!--opt, object, multicast settings-->
        <enabled>
          <!--req, bool, whether to enable the function-->true
        </enabled>
        <userTriggerThreshold>
          <!--opt, int-->1
        </userTriggerThreshold>
        <destIPAddress>
          <!--opt, string-->test
        </destIPAddress>
        <videoDestPortNo>
          <!--opt, int-->1
        </videoDestPortNo>
        <audioDestPortNo>
          <!--opt, int-->1
        </audioDestPortNo>
      </Multicast>
    </Transport>
  </StreamingChannel>
</StreamingChannelList>
```

```

<destIPv6Address>
  <!--opt, string-->test
</destIPv6Address>
<ttl>
  <!--opt, int-->1
</ttl>
<activeMulticastEnabled>
  <!--opt, bool-->true
</activeMulticastEnabled>
<packagingFormat>
  <!--opt, enum, subType:string-->RTP
</packagingFormat>
<FecInfo>
  <!--opt, object-->
  <fecRatio>
    <!--req, int, range:[0,100]-->1
  </fecRatio>
  <fecDestPortNo>
    <!--opt, int-->1
  </fecDestPortNo>
</FecInfo>
</Multicast>
<Security>
  <!--opt, object, security configuration-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <certificateType>
    <!--req, enum, RTSP authentication method, subType:string, desc:"digest", "digest/baisc"-->digest
  </certificateType>
  <SecurityAlgorithm>
    <!--opt, object, security algorithm-->
    <algorithmType>
      <!--opt, enum, algorithm type, subType:string, desc:"MD5", "SHA256", "MD5/SHA256"-->MD5
    </algorithmType>
  </SecurityAlgorithm>
</Security>
<SRTPMulticast>
  <!--opt, object-->
  <SRTPVideoDestPortNo>
    <!--opt, int-->1
  </SRTPVideoDestPortNo>
  <SRTPAudioDestPortNo>
    <!--opt, int-->1
  </SRTPAudioDestPortNo>
</SRTPMulticast>
</Transport>
<Video>
  <!--opt, object, video configuration-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->test
  </dynVideoInputChannelID>
  <videoCodecType>
    <!--req, enum, subType:string-->H.264
  </videoCodecType>
  <videoScanType>
    <!--opt, enum, subType:string-->progressive
  </videoScanType>
  <videoResolutionWidth>
    <!--req, int, resolution (width)-->1
  </videoResolutionWidth>
  <videoResolutionHeight>
    <!--req, int, resolution (height)-->1
  </videoResolutionHeight>
  <videoResolutionName>
    <!--opt, enum, resolution name, subType:string-->3MP
  </videoResolutionName>
  <videoDiffResolutionList>
    <!--opt, array, resolution list, subType:object-->
    <videoDiffResolution>
      <!--opt, object, resolution-->
      <videoCodecType>
        <!--req, enum, subType:string-->H.264
      </videoCodecType>
      <videoResolutionWidth>
        <!--req, int, resolution (width)-->1
      </videoResolutionWidth>
      <videoResolutionHeight>
        <!--req, int, resolution (height)-->1
      </videoResolutionHeight>
    </videoDiffResolution>
  </videoDiffResolutionList>
  <videoPositionX>
    <!--opt, int-->1
  </videoPositionX>
  <videoPositionY>
    <!--opt, int-->1
  </videoPositionY>

```

```

</videoPositionY>
<videoQualityControlType>
  <!--opt, enum, subType:string-->CBR
</videoQualityControlType>
<constantBitRate>
  <!--opt, int-->1
</constantBitRate>
<fixedQuality>
  <!--opt, enum, image quality, subType:int, desc:1 (Lowest), 20 (Lower), 40 (Low), 60 (medium), 80 (higher), 100 (highest)-->1
</fixedQuality>
<vbrUpperCap>
  <!--opt, int-->1
</vbrUpperCap>
<vbrLowerCap>
  <!--opt, int-->1
</vbrLowerCap>
<maxFrameRate>
  <!--req, int-->1
</maxFrameRate>
<keyFrameInterval>
  <!--opt, int, unit:ms-->1
</keyFrameInterval>
<rotationDegree>
  <!--opt, int-->1
</rotationDegree>
<mirrorEnabled>
  <!--opt, bool-->true
</mirrorEnabled>
<snapshotImageType>
  <!--opt, enum, subType:string, desc:"JPEG", "GIF", "PNG"-->JPEG
</snapshotImageType>
<Mpeg4Profile>
  <!--opt, enum, subType:string, desc:"SP", "ASP"-->SP
</Mpeg4Profile>
<H264Profile>
  <!--opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</H264Profile>
<SVACProfile>
  <!--opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</SVACProfile>
<GovLength>
  <!--opt, int-->1
</GovLength>
<SVC>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <SVCMode>
    <!--opt, enum, subType:string, desc:"manual", "auto"-->manual
  </SVCMode>
  <protectLevel>
    <!--opt, enum, subType:string-->all
  </protectLevel>
</SVC>
<smoothing>
  <!--opt, int, range:[1,100]-->1
</smoothing>
<H265Profile>
  <!--opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</H265Profile>
<SmartCodec>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
</SmartCodec>
<vbrAverageCap>
  <!--opt, int-->1
</vbrAverageCap>
<IntelligentInfoDisplayMethod>
  <!--opt, enum, subType:string-->player
</IntelligentInfoDisplayMethod>
<LBREnabled>
  <!--opt, bool-->true
</LBREnabled>
<minimumResolutionSupportedBySmartCode>
  <!--opt, string-->640*512
</minimumResolutionSupportedBySmartCode>
<videoInputType>
  <!--opt, enum, subType:string-->SDI
</videoInputType>
<BPFrameInterval>
  <!--opt, enum, subType:string-->auto
</BPFrameInterval>
<DepthMapEnable>
  <!--opt, bool-->false
</DepthMapEnable>
</Video>
<Audio>
  <!--opt, object, audio settings-->
  <enabled>
    <!--req, bool-->true
  </enabled>

```

```

<audioInputChannelID>
  <!--req, string-->test
</audioInputChannelID>
<audioCompressionType>
  <!--req, enum, subType:string-->UNKNOWN
</audioCompressionType>
<audioInboundCompressionType>
  <!--opt, enum, subType:string-->UNKNOWN
</audioInboundCompressionType>
<audioBitRate>
  <!--opt, int-->1
</audioBitRate>
<audioSamplingRate>
  <!--opt, float, audio sampling rate-->0.0
</audioSamplingRate>
<audioResolution>
  <!--opt, int, audio stream bitrate-->1
</audioResolution>
<VoiceChanger>
  <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <level>
      <!--req, int, range: [-12,12]-->1
    </level>
  </VoiceChanger>
<audioInputType>
  <!--opt, enum, subType:string-->AudioIn
</audioInputType>
</Audio>
<enableCABAC>
  <!--opt, bool-->true
</enableCABAC>
<subStreamRecStatus>
  <!--ro, opt, bool-->true
</subStreamRecStatus>
<isSupportRefreshFrame>
  <!--opt, bool-->true
</isSupportRefreshFrame>
<isSupportBareDataOverlay>
  <!--opt, bool-->true
</isSupportBareDataOverlay>
<isSupportRTCPCfg>
  <!--opt, bool-->true
</isSupportRTCPCfg>
<customStreamEnable>
  <!--opt, bool-->true
</customStreamEnable>
</StreamingChannel>
</StreamingChannelList>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

12.4.3.2 Get the encoding parameters of multiple channels

Request URL

GET /ISAPI/Streaming/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<StreamingChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, channel list, subType:object, attr:version{opt, string, protocolVersion}-->
  <StreamingChannel>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, channel ID-->test
    </id>
    <channelName>
      <!--ro, req, string, channel name-->test
    </channelName>
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <Transport>
      <!--ro, req, object-->
      <rtspPortNo>
        <!--ro, opt, int, range:[0,65535]-->554
      </rtspPortNo>
      <maxPacketSize>
        <!--ro, opt, int-->1
      </maxPacketSize>
      <audioPacketLength>
        <!--ro, opt, int-->1
      </audioPacketLength>
      <audioInboundPacketLength>
        <!--ro, opt, int-->1
      </audioInboundPacketLength>
      <audioInboundPortNo>
        <!--ro, opt, int-->1
      </audioInboundPortNo>
      <videoSourcePortNo>
        <!--ro, opt, int-->1
      </videoSourcePortNo>
      <audioSourcePortNo>
        <!--ro, opt, int-->1
      </audioSourcePortNo>
      <ControlProtocolList>
        <!--ro, req, array, subType:object-->
        <ControlProtocol>
          <!--ro, opt, object-->
          <streamingTransport>
            <!--ro, req, enum, subType:string, desc:"HTTP", "RTSP", "SHTTP"-->RTSP
          </streamingTransport>
        </ControlProtocol>
      </ControlProtocolList>
      <Unicast>
        <!--ro, opt, object-->
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <interfaceID>
          <!--ro, opt, string-->test
        </interfaceID>
        <rtpTransportType>
          <!--ro, opt, enum, subType:string-->RTP/UDP
        </rtpTransportType>
      </Unicast>
      <Multicast>
        <!--ro, opt, object, Multicast Settings-->
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <userTriggerThreshold>
          <!--ro, opt, int-->1
        </userTriggerThreshold>
        <destIPAddress>
          <!--ro, opt, string-->test
        </destIPAddress>
        <videoDestPortNo>
          <!--ro, opt, int-->1
        </videoDestPortNo>
        <audioDestPortNo>
          <!--ro, opt, int-->1
        </audioDestPortNo>
        <destIPv6Address>
          <!--ro, opt, string-->test
        </destIPv6Address>
        <ttl>
          <!--ro, opt, int-->1
        </ttl>
        <activeMulticastEnabled>
          <!--ro, opt, bool-->true
        </activeMulticastEnabled>
        <packagingFormat>
          <!--ro, opt, enum, subType:string, desc:"RTP", "PS"-->RTP
        </packagingFormat>
      </Multicast>
    </Transport>
  </StreamingChannel>
</StreamingChannelList>
```



```

<FecInfo>
  <!--ro, opt, object-->
  <fecRatio>
    <!--ro, req, int, range:[0,100]-->1
  </fecRatio>
  <fecDestPortNo>
    <!--ro, opt, int-->1
  </fecDestPortNo>
</FecInfo>
</Multicast>
<Security>
  <!--ro, opt, object, security configuration-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <certificateType>
    <!--ro, req, enum, RTSP authentication method, subType:string, desc:"digest", "digest/baisc"-->digest
  </certificateType>
  <SecurityAlgorithm>
    <!--ro, opt, object, security algorithm-->
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:string, desc:"MD5", "SHA256", "MD5/SHA256"-->MD5
    </algorithmType>
  </SecurityAlgorithm>
</Security>
<SRTPMulticast>
  <!--ro, opt, object-->
  <SRTPVideoDestPortNo>
    <!--ro, opt, int-->1
  </SRTPVideoDestPortNo>
  <SRTPAudioDestPortNo>
    <!--ro, opt, int-->1
  </SRTPAudioDestPortNo>
</SRTPMulticast>
</Transport>
<Video>
  <!--ro, opt, object, video configuration-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->test
  </dynVideoInputChannelID>
  <videoCodecType>
    <!--ro, req, enum, video encoding type, subType:string-->H.264
  </videoCodecType>
  <videoScanType>
    <!--ro, opt, enum, subType:string-->progressive
  </videoScanType>
  <videoResolutionWidth>
    <!--ro, req, int, resolution width-->1
  </videoResolutionWidth>
  <videoResolutionHeight>
    <!--ro, req, int, resolution height-->1
  </videoResolutionHeight>
  <videoResolutionName>
    <!--ro, opt, enum, resolution name, subType:string-->3MP
  </videoResolutionName>
  <videoDiffResolutionList>
    <!--ro, opt, array, resolution list, subType:object-->
    <videoDiffResolution>
      <!--ro, opt, object, resolution-->
      <videoCodecType>
        <!--ro, req, enum, video encoding type, subType:string-->H.264
      </videoCodecType>
      <videoResolutionWidth>
        <!--ro, req, int, resolution width-->1
      </videoResolutionWidth>
      <videoResolutionHeight>
        <!--ro, req, int, resolution height-->1
      </videoResolutionHeight>
    </videoDiffResolution>
  </videoDiffResolutionList>
  <videoPositionX>
    <!--ro, opt, int-->1
  </videoPositionX>
  <videoPositionY>
    <!--ro, opt, int-->1
  </videoPositionY>
  <videoQualityControlType>
    <!--ro, opt, enum, subType:string-->CBR
  </videoQualityControlType>
  <constantBitRate>
    <!--ro, opt, int-->1
  </constantBitRate>
  <fixedQuality>
    <!--ro, opt, enum, Image Quality, subType:int-->1
  </fixedQuality>
  <vbrUpperCap>
    <!--ro, opt, int-->1

```

```

</vbrUpperCap>
<vbrLowerCap>
  <!--ro, opt, int-->1
</vbrLowerCap>
<maxFrameRate>
  <!--ro, req, int-->1
</maxFrameRate>
<keyFrameInterval>
  <!--ro, opt, int, I frame interval, unit:ms-->1
</keyFrameInterval>
<rotationDegree>
  <!--ro, opt, int-->1
</rotationDegree>
<mirrorEnabled>
  <!--ro, opt, bool-->true
</mirrorEnabled>
<snapshotImageType>
  <!--ro, opt, enum, subType:string, desc:"JPEG", "GIF", "PNG"-->JPEG
</snapshotImageType>
<Mpeg4Profile>
  <!--ro, opt, enum, subType:string, desc:"SP", "ASP"-->SP
</Mpeg4Profile>
<H264Profile>
  <!--ro, opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</H264Profile>
<SVACProfile>
  <!--ro, opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</SVACProfile>
<GovLength>
  <!--ro, opt, int, I frame interval-->1
</GovLength>
<SVC>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <SVCMode>
    <!--ro, opt, enum, subType:string, desc:"manual", "auto"-->manual
  </SVCMode>
  <protectLevel>
    <!--ro, opt, enum, subType:string-->all
  </protectLevel>
</SVC>
<smoothing>
  <!--ro, opt, int, range:[1,100]-->1
</smoothing>
<H265Profile>
  <!--ro, opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</H265Profile>
<SmartCodec>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
</SmartCodec>
<vbrAverageCap>
  <!--ro, opt, int-->1
</vbrAverageCap>
<IntelligentInfoDisplayMethod>
  <!--ro, opt, enum, subType:string-->player
</IntelligentInfoDisplayMethod>
<LBREnabled>
  <!--ro, opt, bool-->true
</LBREnabled>
<minimumResolutionSupportedBySmartCode>
  <!--ro, opt, string-->640*512
</minimumResolutionSupportedBySmartCode>
<videoInputType>
  <!--ro, opt, enum, subType:string-->SDI
</videoInputType>
<BPFrameInterval>
  <!--ro, opt, enum, subType:string-->auto
</BPFrameInterval>
<DepthMapEnable>
  <!--ro, opt, bool-->false
</DepthMapEnable>
</Video>
<Audio>
  <!--ro, opt, object, audio settings-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <audioInputChannelID>
    <!--ro, req, string-->test
  </audioInputChannelID>
  <audioCompressionType>
    <!--ro, req, enum, audio encoding type, subType:string-->UNKNOWN
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--ro, opt, enum, audio encoding type, subType:string-->UNKNOWN
  </audioInboundCompressionType>
  <audioBitRate>
    <!--ro, opt, int-->1
  </audioBitRate>

```

```

</audioResolution>
<audioSamplingRate>
  <!--ro, opt, float, Audio Sampling Rate-->0.0
</audioSamplingRate>
<audioResolution>
  <!--ro, opt, int, Audio Stream Bitrate-->1
</audioResolution>
<VoiceChanger>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <level>
    <!--ro, req, int, range: [-12,12]-->1
  </level>
</VoiceChanger>
<audioInputType>
  <!--ro, opt, enum, subType:string-->AudioIn
</audioInputType>
</Audio>
<enableCABAC>
  <!--ro, opt, bool-->true
</enableCABAC>
<subStreamRecStatus>
  <!--ro, opt, bool-->true
</subStreamRecStatus>
<isSupportRefreshFrame>
  <!--ro, opt, bool-->true
</isSupportRefreshFrame>
<isSupportBareDataOverlay>
  <!--ro, opt, bool-->true
</isSupportBareDataOverlay>
<isSupportRTCPCfg>
  <!--ro, opt, bool-->true
</isSupportRTCPCfg>
<customStreamEnable>
  <!--ro, opt, bool-->true
</customStreamEnable>
</StreamingChannel>
</StreamingChannelList>

```

12.4.3.3 Get the encoding parameters of a specified channel

Request URL

GET /ISAPI/Streaming/channels/<trackStreamID>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	It is equal to channel x 100 + streamType. For streamType, 1 (main stream), 2 (sub stream), 3 (third stream). For example, if trackStreamID is 101, it indicates the main stream of channel 1; if trackStreamID is 302, it indicates the sub stream of channel 3.

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<StreamingChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, encoding parameters, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel ID-->test
  </id>
  <channelName>
    <!--ro, req, string, channel name-->test
  </channelName>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <Transport>
    <!--ro, req, object, transmission configurations-->
    <maxPacketSize>
      <!--ro, opt, int, the maximum packet size-->1
    </maxPacketSize>
    <audioPacketLength>
      <!--ro, opt, int, audio packet length-->1
    </audioPacketLength>
    <audioInboundPacketLength>
      <!--ro, opt, int, audio input packet length-->1
    </audioInboundPacketLength>
  </Transport>
</StreamingChannel>

```

```

</audioInboundPacketLength>
<audioInboundPortNo>
  <!--ro, opt, int, audio input port No.-->1
</audioInboundPortNo>
<videoSourcePortNo>
  <!--ro, opt, int, video source port No.-->1
</videoSourcePortNo>
<audioSourcePortNo>
  <!--ro, opt, int, audio source port No.-->1
</audioSourcePortNo>
<ControlProtocolList>
  <!--ro, req, array, control protocol List, subType:object-->
  <ControlProtocol>
    <!--ro, opt, object, control protocol type-->
    <streamingTransport>
      <!--ro, req, enum, stream transmission protocol, subType:string, desc:"HTTP", "RTSP", "SHTTP"-->RTSP
    </streamingTransport>
  </ControlProtocol>
</ControlProtocolList>
<Unicast>
  <!--ro, opt, object, unicast configurations-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <interfaceID>
    <!--ro, opt, string, NIC ID-->test
  </interfaceID>
  <rtpTransportType>
    <!--ro, opt, enum, RTP transmission type, subType:string, desc:"RTP/UDP" (RTP over UDP), "RTP/TCP" (RTP over TCP)-->RTP/UDP
  </rtpTransportType>
</Unicast>
<Multicast>
  <!--ro, opt, object, multicast configurations-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <userTriggerThreshold>
    <!--ro, opt, int-->1
  </userTriggerThreshold>
  <destIPAddress>
    <!--ro, opt, string, destination IPv4 address-->test
  </destIPAddress>
  <videoDestPortNo>
    <!--ro, opt, int, video destination port No.-->1
  </videoDestPortNo>
  <audioDestPortNo>
    <!--ro, opt, int, audio destination port No.-->1
  </audioDestPortNo>
  <destIPv6Address>
    <!--ro, opt, string, destination IPv6 address-->test
  </destIPv6Address>
  <ttl>
    <!--ro, opt, int, Lifetime of a domain name parsing record in the DNS server-->1
  </ttl>
  <activeMulticastEnabled>
    <!--ro, opt, bool, whether to enable the active multicast-->true
  </activeMulticastEnabled>
  <packagingFormat>
    <!--ro, opt, enum, encapsulation format, subType:string, desc:"RTP", "PS"-->RTP
  </packagingFormat>
  <FecInfo>
    <!--ro, opt, object, Forward Error Correction (FEC) information-->
    <fecRatio>
      <!--ro, req, int, ratio of bandwidth consumed by FEC data, range:[0,100]-->1
    </fecRatio>
    <fecDestPortNo>
      <!--ro, opt, int, FEC destination port No.-->1
    </fecDestPortNo>
  </FecInfo>
</Multicast>
<Security>
  <!--ro, opt, object, security configuration-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <certificateType>
    <!--ro, req, enum, RTSP authentication method, subType:string, desc:"digest", "digest/baisc" (digest/basic)-->digest
  </certificateType>
  <SecurityAlgorithm>
    <!--ro, opt, object, security algorithm-->
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:string, desc:"MD5", "SHA256", "MD5/SHA256"-->MD5
    </algorithmType>
  </SecurityAlgorithm>
</Security>
<SRTPMulticast>
  <!--ro, opt, object, SRTP multicast configurations-->
  <SRTPVideoDestPortNo>
    <!--ro, opt, int, SRTP audio destination port No.-->1
  </SRTPVideoDestPortNo>
  <SRTPAudioDestPortNo>
    <!--ro, opt, int, SRTP video destination port No.-->1
  </SRTPAudioDestPortNo>
</SRTPMulticast>

```

```

</Transport>
<Video>
  <!--ro, opt, object, video configurations-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <videoInputChannelID>
    <!--ro, opt, string, analog channel ID-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string, digital channel ID-->test
  </dynVideoInputChannelID>
  <videoCodecType>
    <!--ro, req, enum, video encoding type, subType:string, desc:"UNKNOWN", "MPEG4", "MJPEG", "3GP", "H.264", "HK.264", "MPNG", "SVAC", "H.265",
    "H.264BP", "H.264HP", "H.264SVC", "MPEG2"-->H.264
  </videoCodecType>
  <videoScanType>
    <!--ro, opt, enum, scanning type of video, subType:string, desc:"progressive", "interlaced"-->progressive
  </videoScanType>
  <videoResolutionWidth>
    <!--ro, req, int, resolution (width)-->1
  </videoResolutionWidth>
  <videoResolutionHeight>
    <!--ro, req, int, resolution (height)-->1
  </videoResolutionHeight>
  <videoResolutionName>
    <!--ro, opt, enum, resolution, subType:string, desc:"3MP", "5MP", "none"-->3MP
  </videoResolutionName>
  <videoDiffResolutionList>
    <!--ro, opt, array, resolution list, subType:object-->
    <videoDiffResolution>
      <!--ro, opt, object, resolution-->
      <videoCodecType>
        <!--ro, req, enum, video encoding type, subType:string, desc:"MPEG4", "MJPEG", "3GP", "H.264", "HK.264", "MPNG", "SVAC", "H.265"-->H.264
      </videoCodecType>
      <videoResolutionWidth>
        <!--ro, req, int, resolution (width)-->1
      </videoResolutionWidth>
      <videoResolutionHeight>
        <!--ro, req, int, resolution (height)-->1
      </videoResolutionHeight>
    </videoDiffResolution>
  </videoDiffResolutionList>
  <videoPositionX>
    <!--ro, opt, int, video x-coordinate-->1
  </videoPositionX>
  <videoPositionY>
    <!--ro, opt, int, video y-coordinate-->1
  </videoPositionY>
  <videoQualityControlType>
    <!--ro, opt, enum, video bitrate type, subType:string, desc:"CBR" (constant bit rate), "VBR" (variable bit rate)-->CBR
  </videoQualityControlType>
  <constantBitRate>
    <!--ro, opt, int, constant bit rate-->1
  </constantBitRate>
  <fixedQuality>
    <!--ro, opt, enum, image quality, subType:int, desc:1 (lowest), 20 (Lower), 40 (Low), 60 (medium), 80 (higher), 100 (highest)-->1
  </fixedQuality>
  <vbrUpperCap>
    <!--ro, opt, int, bitrate upper limit-->1
  </vbrUpperCap>
  <vbrLowerCap>
    <!--ro, opt, int, bitrate lower limit-->1
  </vbrLowerCap>
  <maxFrameRate>
    <!--ro, req, int, video frame rate, desc:frame rate x 100, for example, for 22 frames, it is 2200; for 1/4 frame, it is 25-->1
  </maxFrameRate>
  <keyFrameInterval>
    <!--ro, opt, int, I frame interval, unit:ms-->1
  </keyFrameInterval>
  <rotationDegree>
    <!--ro, opt, int, rotation degree-->1
  </rotationDegree>
  <mirrorEnabled>
    <!--ro, opt, bool, whether to enable the mirror-->true
  </mirrorEnabled>
  <snapshotImageType>
    <!--ro, opt, enum, format of captured picture, subType:string, desc:"JPEG", "GIF", "PNG"-->JPEG
  </snapshotImageType>
  <Mpeg4Profile>
    <!--ro, opt, enum, MPEG-4 coding profiles, subType:string, desc:"SP", "ASP"-->SP
  </Mpeg4Profile>
  <H264Profile>
    <!--ro, opt, enum, H.264 coding profiles, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
  </H264Profile>
  <SVACProfile>
    <!--ro, opt, enum, SVAC encoding profiles, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
  </SVACProfile>
  <GovLength>
    <!--ro, opt, int, I frame interval-->1
  </GovLength>
  <SVC>
    <!--ro, opt, object, scalable video coding function-->

```

```

<enabled>
  <!--ro, req, bool, whether to enable-->true
</enabled>
<SVCMode>
  <!--ro, opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
</SVCMode>
<protectLevel>
  <!--ro, opt, enum, protection level, subType:string, desc:"all" (all parts), "nuclear" (core part)-->all
</protectLevel>
</SVC>
<smoothing>
  <!--ro, opt, int, stream smoothness level, range:[1,100], desc:1 (clear), 100 (smooth)-->1
</smoothing>
<H265Profile>
  <!--ro, opt, enum, H.264 coding profiles, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</H265Profile>
<SmartCodec>
  <!--ro, opt, object, smart codec-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
</SmartCodec>
<vbrAverageCap>
  <!--ro, opt, int, average bitrate, desc:it is valid when the SmartCodec is enabled-->1
</vbrAverageCap>
<IntelligentInfoDisplayMethod>
  <!--ro, opt, enum, intelligent information display mode, subType:string, desc:"player" (display), "non-player" (not display)-->player
</IntelligentInfoDisplayMethod>
<LBREnabled>
  <!--ro, opt, bool, whether to enable Low bit rate-->true
</LBREnabled>
<minimumResolutionSupportedBySmartCode>
  <!--ro, opt, string, supported minimum resolution of smart codec (including smart264 and smart265)-->640*512
</minimumResolutionSupportedBySmartCode>
<videoInputType>
  <!--ro, opt, enum, video input type, subType:string, desc:"SDI", "HDMI", "VGA", "AUTO"-->SDI
</videoInputType>
<BPFrameInterval>
  <!--ro, opt, enum, P-frame interval, subType:string, desc:"BBP", "BP", "P", "auto" (same as the source)-->auto
</BPFrameInterval>
<DepthMapEnable>
  <!--ro, opt, bool, whether to enable depth image, desc:if it is enabled, the video parameters of sub-stream for the second channel are not
  configurable, and the default size of output depth image is 960*540-->>false
</DepthMapEnable>
</Video>
<Audio>
  <!--ro, opt, object, audio settings-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <audioInputChannelID>
    <!--ro, req, string, digital channel ID-->test
  </audioInputChannelID>
  <audioCompressionType>
    <!--ro, req, enum, audio encoding type, subType:string, desc:"UNKNOWN", "G.711aLaw", "G.711uLaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
    "AC3", "AAC", "ADPCM", "MP2L2", "G722", "G.719", "AAC-LC", "AAC-LD", "Opus", "G.728", "G.722", "G.722.1.C"-->UNKNOWN
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"UNKNOWN", "G.711aLaw", "G.711uLaw", "G.726", "G.729", "G.729a", "G.729b",
    "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "G722", "G.719", "AAC-LC", "AAC-LD", "Opus", "G.728", "G.722", "G.722.1.C"-->UNKNOWN
  </audioInboundCompressionType>
  <audioBitRate>
    <!--ro, opt, int, audio encoding bit rate-->1
  </audioBitRate>
  <audioSamplingRate>
    <!--ro, opt, float, audio sampling rate-->0.0
  </audioSamplingRate>
  <audioResolution>
    <!--ro, opt, int, audio bit rate-->1
  </audioResolution>
  <VoiceChanger>
    <!--ro, opt, object, audio change-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <level>
      <!--ro, req, int, Level, range:[-12,12]-->1
    </level>
  </VoiceChanger>
  <audioInputType>
    <!--ro, opt, enum, audio input type, subType:string, desc:"AudioIn" (both VGA and HDMI are supported), "HDMI"-->AudioIn
  </audioInputType>
</Audio>
<enableCABAC>
  <!--ro, opt, bool, whether to enable the stream compression performance improvement-->true
</enableCABAC>
<subStreamRecStatus>
  <!--ro, opt, bool, sub stream recording status-->true
</subStreamRecStatus>
<isSupportRefreshFrame>
  <!--ro, opt, bool, whether frame refreshing is supported, desc:when smart264 is enabled-->true
</isSupportRefreshFrame>
<isSupportBareDataOverlay>
  <!--ro, opt, bool, whether bare data overlay is supported, desc:when smart264 is enabled-->true
</isSupportBareDataOverlay>

```

```

    <!--ro, opt, bool, whether raw data overlay is supported-->true
  </isSupportBareDataOverlay>
  <isSupportRTCPCfg>
    <!--ro, opt, bool, whether RTCP configuration is supported-->true
  </isSupportRTCPCfg>
  <customStreamEnable>
    <!--ro, opt, bool, whether the custom stream is enabled-->true
  </customStreamEnable>
  <lowLatencyEnabled>
    <!--ro, opt, bool, whether the stream Low Latency is enabled-->true
  </lowLatencyEnabled>
  <rebootRequired>
    <!--ro, opt, bool, whether restart is needed to take effect, desc:the configurations will take effect after restarting the device-->>false
  </rebootRequired>
</StreamingChannel>

```

12.4.3.4 Set the encoding parameters of a specific channel

Request URL

PUT /ISAPI/Streaming/channels/<trackStreamID>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	--

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>
<StreamingChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, channel ID-->test
  </id>
  <channelName>
    <!--req, string, channel name-->test
  </channelName>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <Transport>
    <!--req, object-->
    <maxPacketSize>
      <!--opt, int-->1
    </maxPacketSize>
    <audioPacketLength>
      <!--opt, int-->1
    </audioPacketLength>
    <audioInboundPacketLength>
      <!--opt, int-->1
    </audioInboundPacketLength>
    <audioInboundPortNo>
      <!--opt, int-->1
    </audioInboundPortNo>
    <videoSourcePortNo>
      <!--opt, int-->1
    </videoSourcePortNo>
    <audioSourcePortNo>
      <!--opt, int-->1
    </audioSourcePortNo>
    <ControlProtocolList>
      <!--req, array, subType:object-->
      <ControlProtocol>
        <!--opt, object-->
        <streamingTransport>
          <!--req, enum, subType:string, desc:"HTTP", "RTSP", "SHTTP"-->RTSP
        </streamingTransport>
      </ControlProtocol>
    </ControlProtocolList>
    <Unicast>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <interfaceID>
        <!--opt, string-->test
      </interfaceID>
      <rtpTransportType>
        <!--opt, enum, subType:string-->RTP/UDP
      </rtpTransportType>
    </Unicast>
    <Multicast>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
    </Multicast>
  </Transport>

```

```

<!--req, bool-->true
</enabled>
<userTriggerThreshold>
  <!--opt, int-->1
</userTriggerThreshold>
<destIPAddress>
  <!--opt, string-->test
</destIPAddress>
<videoDestPortNo>
  <!--opt, int-->1
</videoDestPortNo>
<audioDestPortNo>
  <!--opt, int-->1
</audioDestPortNo>
<destIPv6Address>
  <!--opt, string-->test
</destIPv6Address>
<ttl>
  <!--opt, int-->1
</ttl>
<activeMulticastEnabled>
  <!--opt, bool-->true
</activeMulticastEnabled>
<packagingFormat>
  <!--opt, enum, subType:string, desc:"RTP", "PS"-->RTP
</packagingFormat>
<FecInfo>
  <!--opt, object-->
  <fecRatio>
    <!--req, int, range:[0,100]-->1
  </fecRatio>
  <fecDestPortNo>
    <!--opt, int-->1
  </fecDestPortNo>
</FecInfo>
</Multicast>
<Security>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <certificateType>
    <!--req, enum, RTSP authentication method, subType:string, desc:"digest", "digest/basic"-->digest
  </certificateType>
  <SecurityAlgorithm>
    <!--opt, object, security algorithm-->
    <algorithmType>
      <!--opt, enum, algorithm type, subType:string, desc:"MD5", "SHA256", "MD5/SHA256"-->MD5
    </algorithmType>
  </SecurityAlgorithm>
</Security>
<SRTPMulticast>
  <!--opt, object-->
  <SRTPVideoDestPortNo>
    <!--opt, int-->1
  </SRTPVideoDestPortNo>
  <SRTPAudioDestPortNo>
    <!--opt, int-->1
  </SRTPAudioDestPortNo>
</SRTPMulticast>
</Transport>
<Video>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->test
  </dynVideoInputChannelID>
  <videoCodecType>
    <!--req, enum, video encoding type, subType:string-->H.264
  </videoCodecType>
  <videoScanType>
    <!--opt, enum, subType:string-->progressive
  </videoScanType>
  <videoResolutionWidth>
    <!--req, int, resolution width-->1
  </videoResolutionWidth>
  <videoResolutionHeight>
    <!--req, int, resolution height-->1
  </videoResolutionHeight>
  <videoResolutionName>
    <!--opt, enum, resolution name, subType:string-->3MP
  </videoResolutionName>
  <videoDiffResolutionList>
    <!--opt, array, subType:object-->
    <videoDiffResolution>
      <!--opt, object, resolution-->
      <videoCodecType>
        <!--req, enum, subType:string-->H.264
      </videoCodecType>

```



```

<videoResolutionWidth>
  <!--req, int, resolution width-->1
</videoResolutionWidth>
<videoResolutionHeight>
  <!--req, int, resolution height-->1
</videoResolutionHeight>
<videoDiffResolution>
</videoDiffResolution>
<videoDiffResolutionList>
<videoPositionX>
  <!--opt, int-->1
</videoPositionX>
<videoPositionY>
  <!--opt, int-->1
</videoPositionY>
<videoQualityControlType>
  <!--opt, enum, subType:string-->CBR
</videoQualityControlType>
<constantBitRate>
  <!--opt, int, constant bit rate-->1
</constantBitRate>
<fixedQuality>
  <!--opt, enum, subType:int-->1
</fixedQuality>
<vbrUpperCap>
  <!--opt, int-->1
</vbrUpperCap>
<vbrLowerCap>
  <!--opt, int-->1
</vbrLowerCap>
<maxFrameRate>
  <!--req, int-->1
</maxFrameRate>
<keyFrameInterval>
  <!--opt, int, unit:ms-->1
</keyFrameInterval>
<rotationDegree>
  <!--opt, int-->1
</rotationDegree>
<mirrorEnabled>
  <!--opt, bool-->true
</mirrorEnabled>
<snapshotImageType>
  <!--opt, enum, subType:string, desc:"JPEG", "GIF", "PNG"-->JPEG
</snapshotImageType>
<Mpeg4Profile>
  <!--opt, enum, subType:string, desc:"SP", "ASP"-->SP
</Mpeg4Profile>
<H264Profile>
  <!--opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</H264Profile>
<SVACProfile>
  <!--opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</SVACProfile>
<GovLength>
  <!--opt, int-->1
</GovLength>
<SVC>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <SVCMode>
    <!--opt, enum, subType:string, desc:"manual", "auto"-->manual
  </SVCMode>
  <protectLevel>
    <!--opt, enum, subType:string-->all
  </protectLevel>
</SVC>
<smoothing>
  <!--opt, int, range:[1,100]-->1
</smoothing>
<H265Profile>
  <!--opt, enum, subType:string, desc:"Baseline", "Main", "High", "Extended"-->Baseline
</H265Profile>
<SmartCodec>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
</SmartCodec>
<vbrAverageCap>
  <!--opt, int-->1
</vbrAverageCap>
<IntelligentInfoDisplayMethod>
  <!--opt, enum, subType:string-->player
</IntelligentInfoDisplayMethod>
<LBREnabled>
  <!--opt, bool-->true
</LBREnabled>
<minimumResolutionSupportedBySmartCode>
  <!--opt, string-->640*512
</minimumResolutionSupportedBySmartCode>
<videoInputType>
  <!--opt, enum, subType:string, desc:"AVI", "MP4", "H264", "H265"-->AVI
</videoInputType>

```

```

    <!--opt, enum, subType:string-->SUI
  </videoInputType>
  <BPFrameInterval>
    <!--opt, enum, subType:string-->auto
  </BPFrameInterval>
  <DepthMapEnable>
    <!--opt, bool-->false
  </DepthMapEnable>
</Video>
<Audio>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <audioInputChannelID>
    <!--req, string-->test
  </audioInputChannelID>
  <audioCompressionType>
    <!--req, enum, audio encoding type, subType:string-->UNKNOWN
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, enum, subType:string-->UNKNOWN
  </audioInboundCompressionType>
  <audioBitRate>
    <!--opt, int-->1
  </audioBitRate>
  <audioSamplingRate>
    <!--opt, float-->0.0
  </audioSamplingRate>
  <audioResolution>
    <!--opt, int-->1
  </audioResolution>
  <VoiceChanger>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <level>
      <!--req, int, range: [-12,12]-->1
    </level>
  </VoiceChanger>
  <audioInputType>
    <!--opt, enum, subType:string-->AudioIn
  </audioInputType>
</Audio>
<enableCABAC>
  <!--opt, bool-->true
</enableCABAC>
<subStreamRecStatus>
  <!--opt, bool-->true
</subStreamRecStatus>
<isSupportRefreshFrame>
  <!--opt, bool-->true
</isSupportRefreshFrame>
<isSupportBareDataOverlay>
  <!--opt, bool-->true
</isSupportBareDataOverlay>
<isSupportRTCPCfg>
  <!--opt, bool-->true
</isSupportRTCPCfg>
<customStreamEnable>
  <!--opt, bool-->true
</customStreamEnable>
</StreamingChannel>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
    (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
    "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>

```

statusCode	statusString	subStatusCode
1	OK	theAudioAndVideoStreamsHaveBeenReEncodedPleaseGetTheStreamsAgain
3	Device Error	highlightCompensationConflictsWithHighFrameRatePleaseDisableHighlightCompensation
3	Device Error	slowShutterConflictsWithHighFrameRatePleaseDisableSlowShutter
4	Invalid Operation	videoModeConflictsWithSmokeDetectionPleaseDisableSmokeDetectionAndTryAgain
4	Invalid Operation	switchingFailedNtscAndTheSmartApplicationCannotBeEnabledAtTheSameTime

12.4.3.5 Get the encoding capability of a specific channel

Request URL

GET /ISAPI/Streaming/channels/<trackStreamID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<StreamingChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, encoding parameter capability, attr:version{opt, string, protocolVersion}-->
  <id opt="1,2">
    <!--ro, req, string, channel ID, attr:opt{opt, string}-->1
  </id>
  <channelName min="0" max="0">
    <!--ro, req, string, channel name, attr:min{opt, string},max{opt, string}-->name1
  </channelName>
  <enabled opt="true,false" def="true">
    <!--ro, req, bool, whether to enable, attr:opt{opt, string},def{opt, string}-->true
  </enabled>
  <Transport>
    <!--ro, req, object-->
    <maxPacketSize opt="1000">
      <!--ro, opt, int, attr:opt{req, string}-->0
    </maxPacketSize>
    <audioPacketLength>
      <!--ro, opt, int-->0
    </audioPacketLength>
    <audioInboundPacketLength>
      <!--ro, opt, int-->0
    </audioInboundPacketLength>
    <audioInboundPortNo>
      <!--ro, opt, int-->0
    </audioInboundPortNo>
    <videoSourcePortNo>
      <!--ro, opt, int-->0
    </videoSourcePortNo>
    <audioSourcePortNo>
      <!--ro, opt, int-->0
    </audioSourcePortNo>
    <ControlProtocolList>
      <!--ro, req, array, subType:object-->
      <ControlProtocol>
        <!--ro, opt, object, list-->
        <streamingTransport opt="HTTP,RTSP,SHHTTP">
          <!--ro, req, enum, "HTTP,RTSP,SHHTTP,SRTP", subType:string, attr:opt{req, string}, desc:HTTP; RTSP; SHHTTP;-->HTTP
        </streamingTransport>
      </ControlProtocol>
    </ControlProtocolList>
    <Unicast>
      <!--ro, opt, object, "RTP/UDP,RTP/TCP"-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
      </enabled>
    </interfaceID>
    <!--ro, opt, string-->test
  </Transport>
</StreamingChannel>
```

```

</interfaceID>
<rtpTransportType opt="RTP/UDP,RTP/TCP">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->RTP/UDP
</rtpTransportType>
</Unicast>
<Multicast>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <userTriggerThreshold>
    <!--ro, opt, int-->0
  </userTriggerThreshold>
  <destIPAddress>
    <!--ro, opt, string-->test
  </destIPAddress>
  <videoDestPortNo>
    <!--ro, opt, int-->0
  </videoDestPortNo>
  <audioDestPortNo>
    <!--ro, opt, int-->0
  </audioDestPortNo>
  <destIPv6Address>
    <!--ro, opt, string-->test
  </destIPv6Address>
  <ttl>
    <!--ro, opt, int-->0
  </ttl>
  <activeMulticastEnabled>
    <!--ro, opt, bool-->true
  </activeMulticastEnabled>
  <packagingFormat opt="RTP,PS">
    <!--ro, opt, enum, container format, subType:string, attr:opt{req, string}, desc:RTP; PS;-->RTP
  </packagingFormat>
  <FecInfo>
    <!--ro, opt, object-->
    <fecRatio min="0" max="100">
      <!--ro, req, int, extra bandwidth occupation ratio of forward error correction (FEC) data, range:[0,100], attr:min{req, int},max{req, int},
desc:the default value is 0-->0
    </fecRatio>
    <fecDestPortNo>
      <!--ro, opt, int, port No. of FEC multicast, desc:the default port No. is specified by device-->0
    </fecDestPortNo>
  </FecInfo>
</Multicast>
<Security>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <certificateType opt="digest,digest/baisc" def="digest">
    <!--ro, req, enum, subType:string, attr:opt{req, string},def{req, string}-->digest
  </certificateType>
  <SecurityAlgorithm>
    <!--ro, opt, object, encryption algorithm, desc:this node is valid when certificateType is "digest" or "digest/basic"-->
    <algorithmType opt="MD5,SHA256,MD5/SHA256">
      <!--ro, opt, enum, algorithm type, subType:string, attr:opt{req, string}, desc:algorithm type: "MD5","SHA256","MD5/SHA256"-->MD5
    </algorithmType>
  </SecurityAlgorithm>
</Security>
<SRTPMulticast>
  <!--ro, opt, object-->
  <SRTPVideoDestPortNo min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </SRTPVideoDestPortNo>
  <SRTPAudioDestPortNo min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </SRTPAudioDestPortNo>
</SRTPMulticast>
</Transport>
<Video>
  <!--ro, opt, object-->
  <enabled opt="true">
    <!--ro, req, bool, attr:opt{opt, string}-->true
  </enabled>
  <videoInputChannelID opt="6">
    <!--ro, opt, string, attr:opt{opt, string}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID opt="H.264,H.265">
    <!--ro, opt, string, attr:opt{opt, string}-->1
  </dynVideoInputChannelID>
  <videoCodecType opt="H.264,H.265,MPEG2">
    <!--ro, req, enum, subType:string, attr:opt{opt, string}-->H.264
  </videoCodecType>
  <videoScanType opt="progressive,interlaced">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}, desc:progressive;interlaced;-->progressive
  </videoScanType>
  <videoResolutionWidth opt="352,704,1280,640,1920,3840,1536,3072,1440,720">
    <!--ro, req, enum, resolution width, subType:int, attr:opt{opt, string}, desc:352; 704; 1280; 640; 1920; 3840;-->1920
  </videoResolutionWidth>
  <videoResolutionHeight opt="288,576,720,512,1080,2160,432,864,540">
    <!--ro, req, enum, resolution height, subType:int, attr:opt{opt, string}, desc:288; 576; 720; 512; 1080; 536;-->1080
  </videoResolutionHeight>

```

```

<videoDiffResolutionList>
  <!--ro, opt, array, List, if this node exists, it indicates that the resolution depends on the node <videoCodecType>, and <videoResolutionWidth> and
  <videoResolutionHeight> are invalid. For forward compatibility, the maximum set of <videoCodecType>, <videoResolutionWidth>, and <videoResolutionHeight>,
  subType:object-->
  <videoDiffResolution>
    <!--ro, opt, object, resolution-->
    <videoCodecType opt="UNKNOWN,MPEG4,MJPEG,3GP,H.264,HK.264,MPNG,SVAC,H.265,H.264BP,H.264HP,H.264SVC">
      <!--ro, req, enum, video encoding type, subType:string, attr:opt{opt, string}, desc:UNKNOWN; MPEG4; MJPEG; 3GP; H.264; HK.264; MPNG; SVAC; H.265;
      H.264BP; H.264HP; H.264SVC;-->H.264SVC
    </videoCodecType>
    <videoResolutionWidth opt="352,704,1280,640,1920">
      <!--ro, req, enum, resolution width, subType:int, attr:opt{opt, string}, desc:352; 704; 1280; 640; 1920;-->1920
    </videoResolutionWidth>
    <videoResolutionHeight opt="288,576,720,512,1080,536">
      <!--ro, req, enum, resolution height, subType:int, attr:opt{opt, string}, desc:288; 576; 720; 512; 1080; 536;-->1080
    </videoResolutionHeight>
  </videoDiffResolution>
</videoDiffResolutionList>
<videoPositionX>
  <!--ro, opt, int-->1920
</videoPositionX>
<videoPositionY>
  <!--ro, opt, int-->1080
</videoPositionY>
<videoQualityControlType opt="CBR,VBR,MVBR,SVBR">
  <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->CBR
</videoQualityControlType>
<constantBitRate min="32" max="16384">
  <!--ro, opt, int, unit:kpbs, attr:min{opt, string},max{opt, string}-->6144
</constantBitRate>
<fixedQuality opt="1,20,30,45,60,75,90,100">
  <!--ro, opt, enum, subType:int, attr:opt{opt, string}, desc:1:Lowest;20:Lower;40:Low;60:medium;80:higher;100:highest;-->1
</fixedQuality>
<vbrUpperCap min="32" max="16384">
  <!--ro, opt, int, range:[32,16384], attr:min{opt, string},max{opt, string}-->16384
</vbrUpperCap>
<vbrLowerCap opt="32">
  <!--ro, opt, enum, subType:int, attr:opt{opt, string}, desc:32;-->32
</vbrLowerCap>
<maxFrameRate opt="0,5000,2500,2200,2000,1800,1600,1500,1200,1000,800,600,400,200,100,50,25,12,6">
  <!--ro, req, enum, subType:int, attr:opt{opt, string}, desc:0; 5000; 2500; 2200; 2000; 1800; 1600; 1500; 1200; 1000; 800; 600; 400; 200; 100; 50; 25;
  12; 6;-->5000
</maxFrameRate>
<keyFrameInterval min="40" max="16000">
  <!--ro, opt, int, range:[40,16000], unit:ms, attr:min{opt, string},max{opt, string}-->40
</keyFrameInterval>
<rotationDegree min="0" max="360">
  <!--ro, opt, int, range:[0,360], attr:min{opt, string},max{opt, string}-->360
</rotationDegree>
<mirrorEnabled>
  <!--ro, opt, bool-->true
</mirrorEnabled>
<snapShotImageType opt="JPEG,GIF,PNG">
  <!--ro, opt, enum, subType:string, attr:opt{opt, string}, desc:JPEG; GIF; PNG;-->JPEG
</snapShotImageType>
<Mpeg4Profile>
  <!--ro, opt, enum, subType:string, desc:SP; ASP;-->ASP
</Mpeg4Profile>
<H264Profile opt="Baseline,Main,High,Extended">
  <!--ro, opt, enum, subType:string, attr:opt{opt, string}, desc:Baseline; Main; High; Extended;-->Extended
</H264Profile>
<SVACProfile opt="Baseline,Main,High,Extended">
  <!--ro, opt, enum, subType:string, attr:opt{opt, string}, desc:Baseline; Main; High; Extended;-->Baseline
</SVACProfile>
<GovLength min="1" max="400">
  <!--ro, opt, int, range:[1,400], attr:min{opt, string},max{opt, string}-->1
</GovLength>
<SVC>
  <!--ro, opt, object-->
  <enabled opt="ture,false">
    <!--ro, req, bool, whether to enable, attr:opt{opt, string}-->true
  </enabled>
  <SVCMode opt="manual,auto">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}, desc>manual; auto;-->auto
  </SVCMode>
  <protectLevel opt="all,nuclear">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->nuclear
  </protectLevel>
</SVC>
<smoothing min="1" max="100">
  <!--ro, opt, int, range:[1,100], attr:min{opt, string},max{opt, string}-->1
</smoothing>
<SmartCodec>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
</SmartCodec>
<vbrAverageCap>
  <!--ro, opt, int-->6144
</vbrAverageCap>
<IntelligentInfoDisplayMethod opt="player,non-player">
  <!--ro, opt, enum, method of displaying intelligent information, subType:string, attr:opt{opt, string}, desc:"player","non-player"-->non-player
</IntelligentInfoDisplayMethod>

```

```

</streamingChannelConfiguration>
<LBRCap>
  <!--ro, opt, object-->
  <LBREnabled opt="true,false">
    <!--ro, req, bool, attr:opt{opt, string}-->true
  </LBREnabled>
  <supportVideoCodecType opt="H.264,H.265">
    <!--ro, req, enum, subType:string, attr:opt{opt, string}, desc:H.264; H.265;-->H.264
  </supportVideoCodecType>
  <supportPrompt opt="prompt1">
    <!--ro, req, enum, subType:string, attr:opt{opt, string}-->prompt1
  </supportPrompt>
</LBRCap>
<minimumResolutionSupportedBySmartCode>
  <!--ro, opt, string, the minimum resolution supported by smart coding (smart264 and smart265),e.g., "640*512", desc:if this node is returned, it
  indicates the smart code of current device (H7,H5) is Limited by the minimum resolution, and the upper layer can get the supported minimum resolution for
  smart coding of current device; for the resolution Lower than the minimum resolution does not support smart coding. If this node is not returned, it
  indicates the smart coding of current device (H3) is not Limited by minimum resolution-->512*512
  <minimumResolutionSupportedBySmartCode>
  <H265Profile opt="Baseline,Main,High,Extended">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}, desc:Baseline; Main; High; Extended;-->Extended
  </H265Profile>
  <videoInputType opt="SDI,HDMI,HDMI,AUTO" def="AUTO">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string},def{opt, string}-->AUTO
  </videoInputType>
  <supportVideoPrompt opt="videoPrompt1">
    <!--ro, req, enum, subType:string, attr:opt{opt, string}-->videoPrompt1
  </supportVideoPrompt>
  <BPFrameInterval opt="BBP,BP,P,auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
  </BPFrameInterval>
  <DepthMapEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->>false
  </DepthMapEnable>
</Video>
<Audio>
  <!--ro, opt, object, audio-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <audioInputChannelID opt="1">
    <!--ro, req, string, attr:opt{req, string}-->test
  </audioInputChannelID>
  <audioCompressionType opt="UNKNOWN,G.711aLaw,G.711uLaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,G722,G.719,AAC-LC,AAC-
  LD,Opus,G.728,G.722.1,G.722.1.C">
    <!--ro, req, enum, subType:string, attr:opt{req, string}, desc:UNKNOWN; G.711aLaw; G.711uLaw; G.726; G.729; G.729a; G.729b; PCM; MP3; AC3; AAC; ADPCM;
  MP2L2; G722; G.719; AAC-LC; AAC-LD; Opus; G.728; G.722.1; G.722.1.C; -->UNKNOWN
  </audioCompressionType>
  <audioInboundCompressionType opt="UNKNOWN,G.711aLaw,G.711uLaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,G722,G.719,AAC-LC,AAC-
  LD,Opus,G.728,G.722.1,G.722.1.C">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}, desc:UNKNOWN; G.711aLaw; G.711uLaw; G.726; G.729; G.729a; G.729b; PCM; MP3; AC3; AAC; ADPCM;
  MP2L2; G722; G.719; AAC-LC; AAC-LD; Opus; G.728; G.722.1; G.722.1.C; -->UNKNOWN
  </audioInboundCompressionType>
  <audioBitRate>
    <!--ro, opt, int, desc:unit: kbps-->0
  </audioBitRate>
  <audioSamplingRate>
    <!--ro, opt, float, unit: kHz-->0.000
  </audioSamplingRate>
  <audioResolution>
    <!--ro, opt, int, unit: bit-->0
  </audioResolution>
  <audioInputType opt="AudioIn,HDMI" def="AudioIn">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->AudioIn
  </audioInputType>
</Audio>
<enableCABAC>
  <!--ro, opt, bool, whether it supports compressing stream to improve performance-->true
</enableCABAC>
<subStreamRecStatus>
  <!--ro, opt, bool-->true
</subStreamRecStatus>
<isSupportRefreshFrame>
  <!--ro, opt, bool, whether it supports refreshing frames when Smart264 is enabled-->true
</isSupportRefreshFrame>
<isSupportBareDataOverlay>
  <!--ro, opt, bool-->true
</isSupportBareDataOverlay>
<isSupportRTCPCfg>
  <!--ro, opt, bool-->true
</isSupportRTCPCfg>
<customStreamEnable>
  <!--ro, opt, bool, whether the stream is custom stream-->true
</customStreamEnable>
<isSupportCDBurningEncryption>
  <!--ro, opt, bool-->true
</isSupportCDBurningEncryption>
<isSupportSmartCodeWithoutReStart>
  <!--ro, opt, bool-->true
</isSupportSmartCodeWithoutReStart>
<isSupportDynamicCapWithCondition>
  <!--ro, opt, bool, whether to support dynamic capability with conditions-->true
</isSupportDynamicCapWithCondition>
</StreamingChannel>

```

12.4.4 Video Input

12.4.4.1 Get the video input configuration parameters of the device

Request URL

GET /ISAPI/System/Video/inputs

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoInput xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video input, attr:version{req, string, protocolVersion}-->
  <VideoInputChannelList>
    <!--ro, opt, array, video input channel list, subType:object-->
    <VideoInputChannel>
      <!--ro, opt, object, video input channel-->
      <id>
        <!--ro, req, string, ID-->test
      </id>
      <inputPort>
        <!--ro, req, string, channel No.-->test
      </inputPort>
      <videoInputEnabled>
        <!--ro, opt, bool, whether to enable the function-->true
      </videoInputEnabled>
      <name>
        <!--ro, opt, string, channel name-->test
      </name>
      <videoFormat>
        <!--ro, opt, enum, "PAL", "NTSC", subType:string, desc:"PAL", "NTSC"-->PAL
      </videoFormat>
      <portType>
        <!--ro, opt, enum, port type, subType:string, desc:"SDI", "VGA", "HDMI", "VpPr"-->SDI
      </portType>
      <resDesc>
        <!--ro, opt, enum, resource description, subType:string-->NO VIDEO
      </resDesc>
      <channelType>
        <!--ro, opt, enum, channel type, subType:string, desc:channel type-->trial
      </channelType>
      <status>
        <!--ro, opt, enum, channel status, subType:string, desc:"onLine", "offline"-->online
      </status>
      <recordStatus>
        <!--ro, opt, enum, recording status, subType:string, desc:recording status-->recording
      </recordStatus>
      <channelDescription>
        <!--ro, opt, string, channel description, desc:channel description-->panorama
      </channelDescription>
    </VideoInputChannel>
  </VideoInputChannelList>
</VideoInput>
```

12.4.4.2 Get the configuration parameters of all video input channels of the device

Request URL

GET /ISAPI/System/Video/inputs/channels

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<VideoInputChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, video input channel, subType:object, attr:version{req, string, protocolVersion}-->
  <VideoInputChannel>
    <!--ro, opt, object, video input channel-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <inputPort>
      <!--ro, req, string, channel No.-->test
    </inputPort>
    <videoInputEnabled>
      <!--ro, opt, bool, whether to enable the function-->true
    </videoInputEnabled>
    <name>
      <!--ro, opt, string, channel name-->test
    </name>
    <videoFormat>
      <!--ro, opt, enum, "PAL", "NTSC", subType:string, desc:"PAL", "NTSC"-->PAL
    </videoFormat>
    <portType>
      <!--ro, opt, enum, port type, subType:string, desc:"SDI", "VGA", "HDMI", "YPbPr"-->SDI
    </portType>
    <resDesc>
      <!--ro, opt, enum, resouce description, subType:string-->NO VIDEO
    </resDesc>
    <channelType>
      <!--ro, opt, enum, channel type, subType:string, desc:channel type-->trial
    </channelType>
    <status>
      <!--ro, opt, enum, channel status, subType:string, desc:"onLine", "offLine"-->online
    </status>
    <recordStatus>
      <!--ro, opt, enum, recording status, subType:string, desc:recording status-->recording
    </recordStatus>
    <channelDescription>
      <!--ro, opt, string, channel description, range:[1,64], desc:channel description-->panorama
    </channelDescription>
  </VideoInputChannel>
</VideoInputChannelList>

```

12.4.4.3 Set the parameters of a specific video input channel of the device

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message


```
<?xml version="1.0" encoding="UTF-8"?>

<VideoInputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video input channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->test
  </id>
  <inputPort>
    <!--req, string, channel No.-->test
  </inputPort>
  <videoInputEnabled>
    <!--opt, bool, whether to enable the function-->true
  </videoInputEnabled>
  <name>
    <!--opt, string, channel name-->test
  </name>
  <videoFormat>
    <!--opt, enum, "PAL,NTSC", subType:string, desc:"PAL", "NTSC"-->PAL
  </videoFormat>
  <portType>
    <!--opt, enum, port type, subType:string, desc:"SDI", "VGA", "HDMI", "YPbPr"-->SDI
  </portType>
  <resDesc>
    <!--ro, opt, enum, subType:string-->NO VIDEO
  </resDesc>
  <channelType>
    <!--opt, enum, channel type, subType:string, desc:channel type-->trial
  </channelType>
  <status>
    <!--opt, enum, channel status, subType:string, desc:"onLine", "offLine"-->online
  </status>
  <recordStatus>
    <!--opt, enum, recording status, subType:string, desc:recording status-->recording
  </recordStatus>
</VideoInputChannel>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	enableOneSilentChannelCloseOneAnalogChannel	0x4000A23C	--

12.4.4.4 Get the capability of a video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoInputChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video input channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <inputPort>
    <!--ro, req, string, channel No.-->test
  </inputPort>
  <videoInputEnabled>
    <!--ro, opt, bool, whether to enable the function-->true
  </videoInputEnabled>
  <name>
    <!--ro, opt, string, channel name-->test
  </name>
  <videoFormat>
    <!--ro, opt, enum, "PAL,NTSC", subType:string, desc:"PAL", "NTSC"-->PAL
  </videoFormat>
  <portType>
    <!--ro, opt, enum, "SDI,VGA,HDMI,YpBPr", subType:string, desc:"SDI", "VGA", "HDMI", "YpBPr"-->SDI
  </portType>
  <resDesc>
    <!--ro, opt, enum, subType:string-->NO VIDEO
  </resDesc>
  <channelType>
    <!--ro, opt, enum, channel type, subType:string-->trial
  </channelType>
  <status>
    <!--ro, opt, enum, channel status, subType:string, desc:"onLine", "offLine"-->online
  </status>
  <recordStatus>
    <!--ro, opt, enum, recording status, subType:string-->recording
  </recordStatus>
  <channelDescription>
    <!--ro, opt, string, range:[1,64]-->panorama
  </channelDescription>
</VideoInputChannel>
```

12.5 Video Management

12.5.1 Storage Strategy

12.5.1.1 Get the parameters of current storage mode

Request URL

GET /ISAPI/ContentMgmt/Storage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<storage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, storage information, attr:version{req, string, protocolVersion}-->
  <hddList size="10">
    <!--ro, opt, object, HDD information List, attr:size{req, int}-->
    <hdd>
      <!--ro, opt, object, HDD information-->
      <id>
        <!--ro, req, string, HDD No.-->test
      </id>
      <hddName>
        <!--ro, req, string, HDD name-->test
      </hddName>
      <hddPath>
        <!--ro, opt, string, HDD path-->test
      </hddPath>
      <hddType>
        <!--ro, req, string, HDD type, desc:"IDE", "SATA", "eSATA", "NFS", "iSCSI", "Virtual Disk", "NTFS"-->test
      </hddType>
      <status>
        <!--ro, req, enum, HDD status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection failed), "repairing" (repairing), "formatting" (HDD is formatting), "notexist" (HDD does not exist), "unRecordHostFormatted" (unformatted in education sharing custom), "maintaining" (HDD is in maintaining), "suspending" (synchronizing), "syncError" (synchronization exception), "unloaded", "unSupported" (HDD is not supported)-->
      </status>
    </hdd>
  </hddList>
</storage>
```

system), manufacturing (now is in manufacturing), synchronizing (synchronizing), syncerror (synchronization exception), unclouded , unsupported (now is not supported by the device), "waiting" (waiting for formatting)-->ok

```
</status>
<capacity>
  <!--ro, req, float, capacity, unit:MB-->50.000
</capacity>
<freeSpace>
  <!--ro, req, float, free space, unit:MB-->20.000
</freeSpace>
<property>
  <!--ro, req, string, HDD property, desc:"RW" (read and write), "RO" (read-only), "Redund" (redundant)-->test
</property>
<group>
  <!--ro, opt, string, the No. of the disk group to which the HDD belongs-->test
</group>
<DataModelList>
  <!--ro, opt, array, data mode List, subType:object-->
  <DataMode>
    <!--ro, opt, object, data mode-->
    <type>
      <!--ro, req, string, application mode type, desc:"recordStorage" (video storage), "pictureCloudStorage" (picture to be saved in cloud storage), "fileStorage" (file storage)-->test
    </type>
    <occupancyRate>
      <!--ro, req, int, disk usage, range:[0,100], desc:the disk space occupied by the application mode-->0
    </occupancyRate>
  </DataMode>
</DataModelList>
<formatType>
  <!--ro, opt, enum, disk file type, subType:string, desc:"FAT32", "EXT4"-->FAT32
</formatType>
<ipAddress>
  <!--ro, opt, string, IPV4, desc:IPV4 address-->172.6.64.7
</ipAddress>
<ipv6Address>
  <!--ro, opt, string, IPV6, desc:IPV6 address-->1080:0:0:8:800:200C:417A
</ipv6Address>
<encryptionStatus>
  <!--ro, opt, enum, encryption status, subType:string, desc:"unencrypted", "encrypted", "verifyFailed" (verification failed)-->unencrypted
</encryptionStatus>
<logStorageEnable>
  <!--ro, opt, bool, whether the current disk is the Log storage disk-->true
</logStorageEnable>
<installationTime>
  <!--ro, opt, time, storage disk installation time-->12:05:00+08:00
</installationTime>
<hddModel>
  <!--ro, opt, string, HDD model-->test
</hddModel>
<hddSerialNumber>
  <!--ro, opt, string, HDD serial No.-->test
</hddSerialNumber>
<HDDDataType>
  <!--ro, opt, enum, disk data type, subType:string, desc:"trainingData" (data training disk). The DeepinMind will take the first disk as the data training disk to store data set, test set, and other data used for algorithm training. The client displays this disk only on the HDD management page of algorithm training, and the disk is managed separate from other disks.-->trainingData
</HDDDataType>
<hddMode>
  <!--ro, opt, enum, subType:string-->record
</hddMode>
</hdd>
</hddList>
<nasList size="10">
  <!--ro, opt, object, NAS information List, attr:size{req, int}-->
  <nas>
    <!--ro, opt, object, NAS information-->
    <id>
      <!--ro, req, string, index No.-->test
    </id>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host server name, dep:and,{$.nasList.nas.addressingFormatType,eq,hostname}, desc:it is valid when the value of addressingFormatType is "hostname"-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPV4 address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of addressingFormatType is "ipaddress"-->192.168.230.1
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPV6 address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of addressingFormatType is "ipaddress"-->CDD:910A:2222:5498:8475:1111:3900:2020
    </ipv6Address>
    <portNo>
      <!--ro, req, int, port No.-->1
    </portNo>
    <userName>
      <!--ro, opt, string, user name-->test
    </userName>
    <nasType>
      <!--ro, req, enum, NAS disk type, subType:string, desc:; "NFS", "iSCSI"-->NFS
    </nasType>
    <path>
```

```

    <!--ro, req, string, NAS disk path-->test
  </path>
  <status>
    <!--ro, req, enum, current status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection
failed), "repairing" (repairing), "formatting" (in formatting), "notexist" (HDD does not exist), "waiting" (waiting for formatting)-->ok
  </status>
  <capacity>
    <!--ro, req, float, capacity, desc:unit: MB-->0.000
  </capacity>
  <freeSpace>
    <!--ro, req, float, free space, desc:unit: MB-->0.000
  </freeSpace>
  <property>
    <!--ro, req, enum, property, subType:string, desc:"RW" (read and write), "RO" (read-only), "redund" (redundant)-->RW
  </property>
  <group>
    <!--ro, opt, string, the No. of the disk group to which the NAS device belongs-->test
  </group>
</nas>
</nasList>
<workMode opt="group,quota,extract,auto,age">
  <!--ro, opt, enum, working mode, subType:string, attr:opt{req, string}, desc:"group","quota","extract", "auto"-->group
</workMode>
<ssdList size="10">
  <!--ro, opt, object, SSD information List, attr:size{req, int}-->
  <ssd>
    <!--ro, opt, object, SSD information-->
    <id>
      <!--ro, req, string, index No.-->test
    </id>
    <ssdName>
      <!--ro, req, string, SSD name-->test
    </ssdName>
    <version>
      <!--ro, opt, string, SSD firmware version-->test
    </version>
  </ssd>
</ssdList>
<posInfoStoringEnabled>
  <!--ro, opt, bool, whether to enable POS information storage-->true
</posInfoStoringEnabled>
</storage>

```

12.5.1.2 Set the current storage mode parameters

Request URL

PUT /ISAPI/ContentMgmt/Storage

Query Parameter

None

Request Message

```

<?xml version="1.0" encoding="UTF-8"?>

<storage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, storage information, attr:version{req, string, protocolVersion}-->
  <hddList size="10">
    <!--opt, object, HDD information List, attr:size{req, int}-->
    <hdd>
      <!--opt, object, HDD information-->
      <id>
        <!--req, string, HDD No.-->test
      </id>
      <hddName>
        <!--req, string, HDD name-->test
      </hddName>
      <hddPath>
        <!--opt, string, HDD path-->test
      </hddPath>
      <hddType>
        <!--req, string, HDD type, desc:"IDE", "SATA", "eSATA", "NFS", "iSCSI", "Virtual Disk", "NTFS"-->test
      </hddType>
      <status>
        <!--req, enum, HDD status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection failed),
"repairing" (repairing), "formatting" (HDD is formatting), "notexist" (HDD does not exist), "unRecordHostFormatted" (unformatted in education sharing system),
"maintaining" (HDD is in maintaining), "synching" (synchronizing), "syncError" (synchronization exception), "unloaded", "waiting" (waiting for formatting),
"initializing"-->ok
      </status>
      <capacity>
        <!--req, float, capacity, unit:MB-->50.000
      </capacity>
      <freeSpace>
        <!--req, float, free space, unit:MB-->20.000
      </freeSpace>
      <property>
        <!--req, string, HDD property, desc:"RW" (read and write), "RO" (read-only), "Redund" (redundant)-->test
      </property>
    </hdd>
  </hddList>

```

```

</property>
<group>
  <!--opt, string, the No. of the disk group to which the HDD belongs-->test
</group>
<DataModelList>
  <!--opt, array, data mode List, subType:object-->
  <DataMode>
    <!--opt, object, data mode-->
    <type>
      <!--req, string, application mode type, desc:"recordStorage" (video storage), "pictureCloudStorage" (picture to be saved in cloud storage),
"fileStorage" (file storage), only one type can be selected-->test
    </type>
    <occupancyRate>
      <!--req, int, disk usage, range:[0,100], desc:the disk space occupied by the application mode-->0
    </occupancyRate>
  </DataMode>
</DataModelList>
<formatType>
  <!--opt, enum, disk file type, subType:string, desc:"FAT32", "EXT4"-->FAT32
</formatType>
<ipAddress>
  <!--opt, string, IPv4, desc:IPv4 address-->172.6.64.7
</ipAddress>
<ipv6Address>
  <!--opt, string, IPv6, desc:IPv6 address-->1080:0:0:0:8:800:200C:417A
</ipv6Address>
<encryptionStatus>
  <!--opt, enum, encryption status, subType:string, desc:"unencrypted", "encrypted", "verifyFailed" (verification failed)-->unencrypted
</encryptionStatus>
<logStorageEnable>
  <!--opt, bool, whether the current disk is a Log storage disk-->true
</logStorageEnable>
<installationTime>
  <!--opt, time, memory card installation time-->12:05:00+08:00
</installationTime>
<hddModel>
  <!--opt, string, HDD model-->test
</hddModel>
<hddSerialNumber>
  <!--opt, string, HDD serial No.-->test
</hddSerialNumber>
<hddMode>
  <!--ro, opt, enum, subType:string-->record
</hddMode>
</hdd>
</hddList>
<nasList size="10">
  <!--opt, object, NAS information List, attr:size(req, int)-->
  <nas>
    <!--opt, object, NAS information-->
    <id>
      <!--req, string, index No.-->test
    </id>
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress" (IP address), "hostname" (host server name)-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host server name, dep:and,{$.nasList.nas.addressingFormatType,eq,hostname}, desc:it is valid when the value of addressingFormatType
is "hostname"-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of addressingFormatType is
"ipaddress"-->192.168.230.1
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of addressingFormatType is
"ipaddress"-->CDD:910A:2222:5498:8475:1111:3900:2020
    </ipv6Address>
    <portNo>
      <!--req, int, port No.-->1
    </portNo>
    <userName>
      <!--opt, string, user name-->test
    </userName>
    <password>
      <!--opt, string, password-->test
    </password>
    <nasType>
      <!--req, enum, NAS disk type, subType:string, desc:"NFS", "iSCSI"-->NFS
    </nasType>
    <path>
      <!--req, string, NAS disk path-->test
    </path>
    <status>
      <!--req, enum, current status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection
failed), "repairing" (repairing), "formatting" (in formatting), "notexist" (HDD does not exist), "waiting" (waiting for formatting), "initializing"-->ok
    </status>
    <capacity>
      <!--req, float, capacity, desc:unit: MB-->0.000
    </capacity>
    <freeSpace>
      <!--req, float, free space, desc:unit: MB-->0.000
    </freeSpace>
  </nas>
</nasList>

```

```

    <property>
      <!--req, enum, property, subType:string, desc;; "RW" (read and write), "RO" (read-only), "redund" (redundant)-->RW
    </property>
  </group>
</group>
</nas>
</nasList>
<workMode opt="group,quota,extract,auto,age">
  <!--opt, enum, working mode, subType:string, attr:opt{req, string}, desc;; "group","quota","extract", "auto"-->group
</workMode>
<ssdList size="10">
  <!--opt, object, SSD information List, attr:size{req, int}-->
  <ssd>
    <!--opt, object, SSD information-->
    <id>
      <!--req, string, index No.-->test
    </id>
    <ssdName>
      <!--req, string, SSD name-->test
    </ssdName>
    <version>
      <!--opt, string, SSD firmware version-->test
    </version>
  </ssd>
</ssdList>
<posInfoStoringEnabled>
  <!--opt, bool, whether to enable POS information storage-->true
</posInfoStoringEnabled>
</storage>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>

```

12.5.1.3 Get the configuration capability of current storage working mode

Request URL

GET /ISAPI/ContentMgmt/Storage/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<storage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, storage information, attr:version{req, string, protocolVersion}-->
  <hddList size="10">
    <!--ro, opt, object, HDD information List, attr:size{req, int}-->
    <hdd>
      <!--ro, opt, object, HDD information-->
      <id>
        <!--ro, req, string, HDD No.-->test
      </id>
      <hddName>
        <!--ro, req, string, HDD name-->test
      </hddName>
    </hdd>
  </hddList>

```

```

</hddName>
<hddPath>
  <!--ro, opt, string, HDD path-->test
</hddPath>
<hddType>
  <!--ro, req, enum, HDD type, subType:string, desc:"IDE", "SATA", "eSATA", "NFS", "iSCSI", "Virtual Disk"-->IDE
</hddType>
<status>
  <!--ro, req, enum, HDD status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection failed), "repairing" (repairing), "formatting" (HDD is formatting), "notexist" (HDD does not exist), "unRecordHostFormatted" (unformatted in education sharing system), "maintaining" (HDD is in maintaining), "synching" (synchronizing), "syncError" (synchronization exception), "unloaded", "sizeMismatch" (HDD capacity mismatch), "unsupported" (HDD is not supported by the device)-->ok
</status>
<capacity>
  <!--ro, req, float, capacity, unit:MB-->50.000
</capacity>
<freeSpace>
  <!--ro, req, float, free space, unit:MB-->20.000
</freeSpace>
<property>
  <!--ro, req, enum, HDD property, subType:string, desc:"RW" (read and write), "RO" (read-only), "Redund" (redundant)-->RW
</property>
<group>
  <!--ro, opt, string, the No. of the disk group to which the HDD belongs-->test
</group>
<DataModeList>
  <!--ro, opt, array, data mode List, subType:object-->
  <DataMode>
    <!--ro, opt, object, data mode-->
    <type>
      <!--ro, req, enum, application mode type, subType:string, desc:"recordStorage" (video storage), "pictureCloudStorage" (picture to be saved in cloud storage), "fileStorage" (file storage)-->recordStorage
    </type>
    <occupancyRate>
      <!--ro, req, int, disk usage, range:[0,100], desc:the disk space occupied by the application mode-->0
    </occupancyRate>
  </DataMode>
</DataModeList>
<formatType>
  <!--ro, opt, enum, disk file type, subType:string, desc:"FAT32", "EXT4"-->FAT32
</formatType>
<ipAddress>
  <!--ro, opt, string, IPv4, desc:IPv4 address-->172.6.64.7
</ipAddress>
<ipv6Address>
  <!--ro, opt, string, IPv6, desc:IPv6 address-->1080:0:0:0:8:800:200C:417A
</ipv6Address>
<encryptionStatus>
  <!--ro, opt, enum, encryption status, subType:string, desc:"unencrypted", "encrypted", "verifyFailed" (verification failed)-->unencrypted
</encryptionStatus>
<logStorageEnable>
  <!--ro, opt, bool, whether the current disk is a Log storage disk-->true
</logStorageEnable>
<installationTime>
  <!--ro, opt, time, storage disk installation time-->12:05:00+08:00
</installationTime>
<hddModel>
  <!--ro, opt, string, HDD model-->test
</hddModel>
<hddSerialNumber>
  <!--ro, opt, string, HDD serial No.-->test
</hddSerialNumber>
<HDDDataType opt="trainingData">
  <!--ro, opt, enum, disk data type, subType:string, attr:opt{req, string}, desc:"trainingData" (data training disk). The DeepinMind will take the first disk as the data training disk to store data set, test set, and other data used for algorithm training. The client displays this disk only on the HDD management page of algorithm training, and the disk is managed separate from other disks.-->trainingData
</HDDDataType>
</hdd>
</hddList>
<nasList size="10">
  <!--ro, opt, object, NAS information List, attr:size{req, int}-->
  <nas>
    <!--ro, opt, object, NAS information-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, dep:and,{$.nasList.nas.addressingFormatType,eq,hostname}, desc:it is valid when the value of addressingFormatType is "hostname"-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of addressingFormatType is "ipaddress"-->192.168.230.1
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 Address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of addressingFormatType is "ipaddress"-->CDD:910A:2222:5498:8475:1111:3900:2020
    </ipv6Address>
    <portNo>
      <!--ro, req, int, port No.-->1
    </portNo>
  </nas>

```

```

<!--ro, opt, string, user name-->test
</userName>
<password>
<!--ro, opt, string, password-->test
</password>
<nasType>
<!--ro, req, enum, NAS type, subType:string, desc:"NFS", "iSCSI"-->NFS
</nasType>
<path>
<!--ro, req, string, NAS path-->test
</path>
<status>
<!--ro, req, enum, current status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection
failed), "repairing" (repairing), "formatting" (in formatting), "notexist" (HDD does not exist)-->ok
</status>
<capacity>
<!--ro, req, float, capacity, desc:unit: MB-->0.000
</capacity>
<freeSpace>
<!--ro, req, float, free space, desc:unit: MB-->0.000
</freeSpace>
<property>
<!--ro, req, enum, property, subType:string, desc:"RW" (read and write), "RO" (read-only), "redund" (redundant), "RDD"-->RW
</property>
<group>
<!--ro, opt, string, the No. of the disk group to which the NAS device belongs-->test
</group>
</nas>
</nasList>
<workMode opt="group,quota,extract">
<!--ro, opt, enum, working mode, subType:string, attr:opt{req, string}, desc:"group", "quota", "extract", "auto"-->group
</workMode>
<supportDataReconstruction opt="all,record">
<!--ro, opt, enum, the supported data type for reconstruction, subType:string, attr:opt{req, string}, desc:"all", "record" (recorded video)-->all
</supportDataReconstruction>
<isSupportSSDManagement>
<!--ro, opt, bool, whether the device supports SSD management, desc:related URL: /ISAPI/ContentMgmt/storage/ssd/<ID>/capabilities?format=json-->true
</isSupportSSDManagement>
<ssdList size="10">
<!--ro, opt, object, SSD information List, attr:size{req, int}-->
<ssd>
<!--ro, opt, object, SSD information-->
<id>
<!--ro, req, string, ID-->test
</id>
<ssdName>
<!--ro, req, string, SSD name-->test
</ssdName>
<version>
<!--ro, opt, string, SSD firmware version-->test
</version>
</ssd>
</ssdList>
<minHddCapacity>
<!--ro, opt, string, minimum HDD capacity supported by the device (with unit), desc:e.g., "8GB"-->test
</minHddCapacity>
<maxHddCapacity>
<!--ro, opt, string, minimum HDD capacity supported by the device (with unit), desc:e.g., "16TB"-->test
</maxHddCapacity>
<isSupportCapacity>
<!--ro, opt, bool, whether the device supports searching for total storage capacity, desc:related URL: /ISAPI/ContentMgmt/Storage/capacity?format=json--
>true
</isSupportCapacity>
<isSupportpoolConfig>
<!--ro, opt, bool, whether the device supports configuring resource pool-->true
</isSupportpoolConfig>
<isSupportRAIDDisk>
<!--ro, opt, bool, whether the device supports configuring information in all disks, desc:related URL:
/ISAPI/ContentMgmt/Storage/raid/RAIDDisk/capabilities?format=json-->true
</isSupportRAIDDisk>
<posInfoStoringEnabled opt="true,false">
<!--ro, opt, bool, whether the device supports POS information storage, attr:opt{req, string}-->true
</posInfoStoringEnabled>
<isSupportHddFormatVerify>
<!--ro, opt, bool, whether the device supports double verification of HDD formatting, desc:related URL:
/ISAPI/ContentMgmt/Storage/hdd/harddiskID/format; /ISAPI/ContentMgmt/Storage/hdd/specifyHddFormat?format=json-->true
</isSupportHddFormatVerify>
<isSupportNasFormatVerify>
<!--ro, opt, bool, whether the device supports double verification of NAS formatting, desc:related URL: /ISAPI/ContentMgmt/Storage/nas/<nasID>/format?
ipAddress=<ipAddress>-->true
</isSupportNasFormatVerify>
</storage>

```

12.6 Algorithm Scheduling

12.6.1 Algorithm Library

12.6.1.1 Get the capability of getting the algorithm library information

Request URL

GET /ISAPI/System/algorithmsInfo/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

richard.granado95@gmail.com
SYSTEMSTIC

```

{
  "AlgorithmsInfoCap": {
    /*ro, opt, object*/
    "type": {
      /*ro, opt, object, algorithm library type*/
      "@opt": ["face", "dsp", "intelligent", "calibration", "living", "iris"]
      /*ro, req, array, subType:string*/
    },
    "maxSize": 1,
    /*ro, req, int, supported number of algorithm libraries*/
    "name": {
      /*ro, opt, object, algorithm library name, desc:it is used to distinguish different algorithms*/
      "@min": 1,
      /*ro, opt, int, the minimum value*/
      "@max": 32
      /*ro, req, int, the maximum value*/
    },
    "versionInfo": {
      /*ro, req, object, algorithm library version information*/
      "major": {
        /*ro, opt, object, major version of the algorithm library*/
        "@min": 1,
        /*ro, opt, int, the minimum value*/
        "@max": 32
        /*ro, req, int, the maximum value*/
      },
      "minor": {
        /*ro, opt, object, minor version of the algorithm library*/
        "@min": 1,
        /*ro, opt, int, the minimum value*/
        "@max": 32
        /*ro, req, int, the maximum value*/
      },
      "revision": {
        /*ro, opt, object, revision version of the algorithm library*/
        "@min": 1,
        /*ro, opt, int, the minimum value*/
        "@max": 32
        /*ro, req, int, the maximum value*/
      }
    },
    "platform": {
      /*ro, opt, object, platform where the algorithm library runs*/
      "@min": 1,
      /*ro, opt, int, the minimum value*/
      "@max": 32
      /*ro, req, int, the maximum value*/
    },
    "accuracy": {
      /*ro, opt, object, computational accuracy of the algorithm library*/
      "@min": 1,
      /*ro, opt, int, the minimum value*/
      "@max": 32
      /*ro, opt, int, the maximum value*/
    },
    "encryption": {
      /*ro, opt, object, encryption method*/
      "@min": 1,
      /*ro, opt, int, the minimum value*/
      "@max": 32
      /*ro, req, int, the maximum value*/
    },
    "buildTime": {
      /*ro, opt, object, compilation date*/
      "@min": 1,
      /*ro, opt, int, the minimum value*/
      "@max": 32
      /*ro, req, int, the maximum value*/
    },
    "versionProperties": {
      /*ro, opt, object, version attribute*/
      "@min": 1,
      /*ro, opt, int, the minimum value*/
      "@max": 32
      /*ro, req, int, the maximum value*/
    },
    "modelDataLength": {
      /*ro, opt, object*/
      "@opt": [256, 272]
      /*ro, opt, array, subType:int*/
    }
  }
}

```

12.6.1.2 Get the algorithm library information

Request URL

GET /ISAPI/System/algorithmsInfo?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "AlgorithmsInfo": [
    /*ro, req, array, algorithm library information, subType:object*/
    {
      "type": "face",
      /*ro, req, enum, algorithm library type, subType:string, desc:"face" (face algorithm), "dsp" (DSP algorithm), "intelligent" (intelligent algorithm), "calibration" (calibration algorithm), "living" (face anti-spoofing algorithm)*/
      "versionInfo": {
        /*ro, req, object, algorithm library version*/
        "major": 1,
        /*ro, req, int, major version of the algorithm library*/
        "minor": 2,
        /*ro, req, int, minor version of the algorithm library*/
        "revision": 3
        /*ro, req, int, revision version of the algorithm library*/
      },
      "platform": "test",
      /*ro, req, string, platform where the algorithm library runs*/
      "accuracy": "test",
      /*ro, opt, string, accuracy of the algorithm library*/
      "encryption": "test",
      /*ro, req, string, encryption method*/
      "buildTime": "buildyyyymmdd",
      /*ro, req, string, compilation date, desc:buildyyyymmdd*/
      "versionProperties": "alpha",
      /*ro, req, enum, version attribute, subType:string, desc:"alpha" (preview version), "beta" (test version), "release" (release version),
      "custom"*/
      "modelDataLength": 272
      /*ro, opt, enum, subType:int, dep:and,{$.AlgorithmsInfo[*].type,eq,iris}*/
    }
  ]
}
```

12.7 Audio (General)

12.7.1 Audio Input

12.7.1.1 Set audio input parameters of a specific channel

Request URL

PUT /ISAPI/System/Audio/AudioIn/channels/<audioID>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioIn xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, single channel audio input parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index-->test
  </id>
  <MixAudioIn>
    <!--opt, object, audio mixing parameter-->
    <enabled>
      <!--req, bool, whether to enable the function-->>true
    </enabled>
    <audioInputType>
      <!--opt, enum, audio input type, subType:string, desc:"macIn" (microphone input), "LineIn" (Line in)-->macIn
    </audioInputType>
    <highPassFilter>
      <!--opt, int, high-pass filtering, range:[0,127], unit:Hz-->1
    </highPassFilter>
    <noiseMargin>
      <!--opt, int, noise threshold, range:[0,127], desc:unit: dB-->1
    </noiseMargin>
  </MixAudioIn>
</AudioIn>
```

```

<!--opt, object, parameters for controlling automatic frequency limitation-->
<FBCEnable>
  <!--req, bool, howling suppression-->true
</FBCEnable>
<mode>
  <!--req, enum, mode, subType:string, desc:"fast", "general", "slow"-->fast
</mode>
<filterQValue>
  <!--req, string, Q value of self-adoptive filters, desc:unit: Oct-->1
</filterQValue>
<staticFilterNum>
  <!--req, int, the number of static filters, range:[0,12]-->1
</staticFilterNum>
</AutoLimitWave>
<DSPPParameter>
  <!--opt, object, parameters of digital signal processing chip-->
  <FBCEnable>
    <!--req, bool, howling suppression-->true
  </FBCEnable>
  <volume>
    <!--req, int, input volume, range:[0,255]-->1
  </volume>
</DSPPParameter>
</MixAudioIn>
<AudioInVolumelist>
  <!--opt, object, audio input List-->
  <AudioInVlome>
    <!--opt, object, audio input-->
    <type>
      <!--req, enum, type, subType:string, desc:"audioInput", "audioOutput", "audioEncode"-->audioOutput
    </type>
    <volume>
      <!--req, int, volume-->1
    </volume>
  </AudioInVlome>
</AudioInVolumelist>
<automaticGainControl>
  <!--opt, int, auto gain configuration, desc:for each channel from MIC1 to MIC16-->1
</automaticGainControl>
<energyThreshold>
  <!--opt, int, energy threshold, desc:for each channel from MIC1 to MIC16-->1
</energyThreshold>
<PhantomPower>
  <!--opt, object, phantom power-->
  <enabled>
    <!--req, bool, whether to enable phantom power-->true
  </enabled>
</PhantomPower>
<audioInCodingFormat>
  <!--opt, enum, audio input encoding format, subType:string, desc:"G.711U", "G.711A", "OPUS", "AAC", "PCM". The default input encoding format varies with devices. For example, the default encoding format of the speaker in the current intercom product is G.711U-->G.711U
</audioInCodingFormat>
<audioInModeEnabled>
  <!--opt, bool, whether to allow selecting the audio input mode, desc:when enabled, the device collects audio according to the input mode (audioInMode) selected by the user; when disabled, users cannot configure the input mode, and the device automatically detects the input mode for audio collection by default, it is disabled-->true
</audioInModeEnabled>
<audioInMode>
  <!--opt, enum, audio input type, subType:string, dep:and,{$.AudioIn.audioInModeEnable,eq,true}, desc:the audio stream from microphone (left and right) input is being processed by DSP before being transmitted to the speaker and the sound is optimized; line in directly transmits the audio stream to the speaker without optimizing the sound; "LeftMic" (Left microphone), "rightMic" (right microphone), "bluetooth" (Bluetooth), "LineIn" (Line in), "earphone" (headphone input); "6.5mm" (6.5 mm input), "RCALeft" (RCA connector, Left channel input), "RCARight" (RCA connector, right channel input), "embeddedAudio" (embedded audio)-->leftMic
</audioInMode>
<audioInSensitivity>
  <!--opt, enum, audio input sensitivity, subType:string, dep:and,{$.AudioIn.audioInMode,eq,LineIn}, desc:this field represents audio sensitivity Level; it is valid when the audio input mode is Line in; the default value is high; "high" (high), "mid" (middle), "Low" (Low)-->high
</audioInSensitivity>
<AGCEnabled>
  <!--opt, bool, whether to enable audio auto gain, desc:when enabled, the gain of the amplifier circuit automatically adjusts according to the signal strength-->true
</AGCEnabled>
<noiseReductionEnabled>
  <!--opt, bool, whether to enable audio noise reduction, desc:when enabled, it can greatly reduce various types of noise in audio-->true
</noiseReductionEnabled>
<AECEnable>
  <!--opt, bool, echo cancellation-->true
</AECEnable>
<audioSamplingRate>
  <!--opt, int, audio sampling rate, desc:unit: kHz-->0
</audioSamplingRate>
<audioBitRate>
  <!--opt, int, audio bit rate, desc:0 (32Kbps), 1(64Kbps), 2 (128Kbps), 3 (256Kbps), 4 (512Kbps)-->0
</audioBitRate>
<noiseReductionLevel>
  <!--opt, int, noise reduction Level, range:[0,6], desc:0 indicates the noise reduction is disabled-->0
</noiseReductionLevel>
<AGCGain>
  <!--opt, int, range:[0,100], step:1, unit:%-->50
</AGCGain>
<directionalMikingEnabled>
  <!--opt, bool-->true

```

```
</directionalMikingEnabled>
</AudioIn>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MerrCode>
    <!--ro, opt, string-->0x00000000
  </MerrCode>
  <MerrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MerrDevSelfEx>
</ResponseStatus>
```

12.7.1.2 Get the audio input parameters of a channel

Request URL

GET /ISAPI/System/Audio/AudioIn/channels/<audioID>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioIn xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, single channel audio input parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index-->test
  </id>
  <MixAudioIn>
    <!--ro, opt, object, audio mixing parameter-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <audioInputType>
      <!--ro, opt, enum, audio input type, subType:string, desc:"macIn" (microphone input), "LineIn" (Line in)-->macIn
    </audioInputType>
    <highPassFilter>
      <!--ro, opt, int, high-pass filtering, range:[0,127], unit:Hz-->1
    </highPassFilter>
    <noiseMargin>
      <!--ro, opt, int, noise threshold, range:[0,127], desc:unit: dB-->1
    </noiseMargin>
    <AutoLimitWave>
      <!--ro, opt, object, parameters for controlling automatic frequency limitation-->
      <FBCEnable>
        <!--ro, req, bool, howling suppression-->true
      </FBCEnable>
      <mode>
        <!--ro, req, enum, mode, subType:string, desc:"fast", "general", "slow"-->fast
      </mode>
      <filterQValue>
        <!--ro, req, string, Q value of self-adaptive filters, desc:unit: Oct-->1
      </filterQValue>
    </AutoLimitWave>
  </MixAudioIn>
</AudioIn>
```

```

<staticFilterNum>
  <!--ro, req, int, the number of static filters, range:[0,12]-->1
</staticFilterNum>
</AutoLimitWave>
<DSPPParameter>
  <!--ro, opt, object, parameters of digital signal processing chip-->
  <FBCEnable>
    <!--ro, req, bool, howling suppression-->true
  </FBCEnable>
  <volume>
    <!--ro, req, int, input volume, range:[0,255]-->1
  </volume>
</DSPPParameter>
</MixAudioIn>
<AudioInVolumelist>
  <!--ro, opt, object, audio input list-->
  <AudioInVlome>
    <!--ro, opt, object, audio input-->
    <type>
      <!--ro, req, enum, type, subType:string, desc:"audioInput", "audioOutput", "audioEncode"-->audioOutput
    </type>
    <volume>
      <!--ro, req, int, volume-->1
    </volume>
  </AudioInVlome>
</AudioInVolumelist>
<automaticGainControl>
  <!--ro, opt, int, auto gain configuration, desc:for each channel from MIC1 to MIC16-->1
</automaticGainControl>
<energyThreshold>
  <!--ro, opt, int, energy threshold, desc:for each channel from MIC1 to MIC16-->1
</energyThreshold>
<PhantomPower>
  <!--ro, opt, object, phantom power-->
  <enabled>
    <!--ro, req, bool, whether to enable phantom power-->true
  </enabled>
</PhantomPower>
<audioInCodingFormat>
  <!--ro, opt, enum, audio input encoding format, subType:string, desc:"G.711U", "G.711A", "OPUS", "AAC", "PCM". The default input encoding format varies with devices. For example, the default encoding format of the speaker in the current intercom product is G.711U-->G.711U
</audioInCodingFormat>
<audioInModeEnabled>
  <!--ro, opt, bool, whether to allow selecting the audio input mode, desc:when enabled, the device collects audio according to the input mode (audioInMode) selected by the user; when disabled, users cannot configure the input mode, and the device automatically detects the input mode for audio collection by default, it is disabled-->true
</audioInModeEnabled>
<audioInMode>
  <!--ro, opt, enum, audio input mode, subType:string, dep:and,{$.AudioIn.audioInModeEnable,eq,true}, desc:the audio stream from microphone (left and right) input is being processed by DSP before being transmitted to the speaker and the sound is optimized; line in directly transmits the audio stream to the speaker without optimizing the sound; "LeftMic" (Left microphone), "RightMic" (right microphone), "bluetooth" (Bluetooth), "LineIn" (Line in), "earphone" (headphone input); "6.5mm" (6.5 mm input), "RCALeft" (RCA connector, Left channel input), "RCARight" (RCA connector, right channel input), "embeddedAudio" (embedded audio)-->leftMic
</audioInMode>
<audioInSensitivity>
  <!--ro, opt, enum, audio input sensitivity, subType:string, dep:and,{$.AudioIn.audioInMode,eq,LineIn}, desc:this field represents audio sensitivity level; it is valid when the audio input mode is line in; the default value is high; "high" (high), "mid" (middle), "Low" (Low)-->high
</audioInSensitivity>
<AGCEnable>
  <!--ro, opt, bool, whether to enable audio auto gain, desc:when enabled, the gain of the amplifier circuit automatically adjusts according to the signal strength-->true
</AGCEnable>
<noiseReductionEnabled>
  <!--ro, opt, bool, whether to enable audio noise reduction, desc:when enabled, it can greatly reduce various types of noise in audio-->true
</noiseReductionEnabled>
<AECEnable>
  <!--ro, opt, bool, echo cancellation-->true
</AECEnable>
<audioSamplingRate>
  <!--ro, opt, int, audio sample rate, desc:unit: kHz-->0
</audioSamplingRate>
<audioBitRate>
  <!--ro, opt, int, audio bit rate, desc:0 (32Kbps), 1(64Kbps), 2 (128Kbps), 3 (256Kbps), 4 (512Kbps)-->0
</audioBitRate>
<noiseReductionLevel>
  <!--ro, opt, int, noise reduction level, range:[0,6], desc:0 indicates the noise reduction is disabled-->0
</noiseReductionLevel>
<AGCGain>
  <!--ro, opt, int, range:[0,100], step:1, unit:%-->50
</AGCGain>
<directionalMikingEnabled>
  <!--ro, opt, bool-->true
</directionalMikingEnabled>
</AudioIn>

```

12.7.1.3 Get the audio input capability of a specified channel

Request URL

GET /ISAPI/System/Audio/AudioIn/channels/<audioID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	audio channel

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioInCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability set of single channel audio input, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index-->test
  </id>
  <MixAudioIn>
    <!--ro, opt, object, audio mixing parameter-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <audioInputType opt="micIn, lineIn">
      <!--ro, opt, enum, audio input type, subType:string, attr:opt{req, string}, desc:"micIn" (microphone input), "LineIn" (line in)-->micIn
    </audioInputType>
    <highPassFilter
      opt="0, 8, 16, 24, 31, 39, 47, 55, 63, 71, 79, 87, 94, 102, 110, 118, 126, 134, 142, 150, 157, 165, 173, 181, 189, 197, 205, 213, 220, 228, 236, 244, 252, 260, 268, 276, 283, 291, 299, 307, 315, 32
      3, 331, 339, 346, 354, 362, 370, 378, 386, 394, 402, 409, 417, 425, 433, 441, 449, 457, 465, 472, 480, 488, 496, 504, 512, 520, 528, 535, 543, 551, 559, 567, 575, 583, 591, 598, 606, 614, 622, 63
      0, 638, 646, 654, 661, 669, 677, 685, 693, 701, 709, 717, 724, 732, 740, 748, 756, 764, 772, 780, 787, 795, 803, 811, 819, 827, 835, 843, 850, 858, 866, 874, 882, 890, 898, 906, 913, 921, 929, 93
      7, 945, 953, 961, 969, 976, 984, 992, 1000" def="30">
      <!--ro, opt, int, high-pass filtering, range:[0,127], unit:Hz, attr:opt{req, string},def{req, string}-->1
    </highPassFilter>
    <noiseMargin opt="1000,660,657,653,650,647,643,640,637,633,-
      630,627,623,620,617,613,610,607,603,600,597,593,590,587,583,580,577,573,570,567,563,560,557,553,550,547,543,540,537,533,530,527,523,520,517,513,510,507,503,
      500,497,493,490,487,483,480,477,473,470,467,463,460,457,453,450,447,443,440,437,433,430,427,423,420,417,413,410,407,403,400,397,393,390,387,383,380,377,373,
      370,367,363,360,357,353,350,347,343,340,337,333,330,327,323,320,317,313,310,307,303,300,297,293,290,287,283,280,277,273,270,267,263,260,257,253,250,247,243,
      240" def="30">
      <!--ro, opt, int, noise threshold, range:[0,127], attr:opt{req, string},def{req, string}, desc:unit: dB-->1
    </noiseMargin>
    <AutoLimitWave>
      <!--ro, opt, object, parameters for controlling automatic frequency limitation-->
      <FBCEnable opt="true,false">
        <!--ro, req, bool, howling suppression, attr:opt{req, string}-->true
      </FBCEnable>
      <mode opt="fast,general,slow" def="general">
        <!--ro, req, enum, mode, subType:string, attr:opt{req, string},def{req, string}, desc:"fast", "general", "slow"-->fast
      </mode>
      <filterQValue opt="40,10" def="40">
        <!--ro, req, int, Q value of self-adaptive filters, attr:opt{req, string},def{req, string}, desc:unit: Oct-->1
      </filterQValue>
      <staticFilterNum min="0" max="12" def="0">
        <!--ro, req, int, the number of static filters, range:[0,12], attr:min{req, int},max{req, int},def{req, int}-->1
      </staticFilterNum>
    </AutoLimitWave>
    <DSPPParameter>
      <!--ro, opt, object, parameters of digital signal processing chip-->
      <FBCEnable>
        <!--ro, req, bool, howling suppression-->true
      </FBCEnable>
      <volume min="1" max="10">
        <!--ro, req, int, input volume, range:[0,255], attr:min{req, int},max{req, int}-->1
      </volume>
    </DSPPParameter>
  </MixAudioIn>
  <AudioInVolumelist>
    <!--ro, opt, object, audio input list-->
    <AudioInVlome>
      <!--ro, opt, object, audio input-->
      <type opt="audioInput,audioOutput,audioEncode">
        <!--ro, req, enum, type, subType:string, attr:opt{req, string}, desc:"audioInput", "audioOutput", "audioEncode"-->audioOutput
      </type>
      <volume min="0" max="127" def="50">
        <!--ro, req, int, volume, attr:min{req, int},max{req, int},def{req, int}-->1
      </volume>
    </AudioInVlome>
  </AudioInVolumelist>
  <automaticGainControl min="0" max="12" default="12">
    <!--ro, opt, int, auto gain configuration, attr:min{req, int},max{req, int},default{opt, string}, desc:for each channel from MIC1 to MIC16-->1
  </automaticGainControl>
  <energyThreshold min="-90" max="0" default="-55">
    <!--ro, opt, int, energy threshold, attr:min{req, int},max{req, int},default{opt, string}, desc:for each channel from MIC1 to MIC16-->1
  </energyThreshold>
  <PhantomPower>
    <!--ro, opt, object, phantom power-->
    <enabled opt="true,false" def="false">
      <!--ro, req, bool, whether to enable phantom power, attr:opt{req, string},def{opt, string}-->true
    </enabled>
  </PhantomPower>
</AudioInCap>
```

```

</PhantomPower>
<audioInCodingFormat opt="G.711U,G.711A,OPUS,PCM,AAC,MP3" def="G.711U">
  <!--ro, opt, enum, audio input encoding format, subType:string, dep:and,{$.AudioInCap.audioInModeEnable,eq,true}, attr:opt{req, string},def{req, string}, desc:"G.711U", "G.711A", "OPUS", "AAC", "PCM".
  The default input encoding format varies with devices. For example, the default encoding format of the speaker in the current intercom product is G.711U-->G.711U
</audioInCodingFormat>
<audioInModeEnabled>
  <!--ro, opt, bool, whether to allow selecting the audio input mode, desc:when enabled, the device collects audio according to the input mode
  (audioInMode) selected by the user; when disabled, users cannot configure the input mode, and the device automatically detects the input mode for audio
  collection by default, it is disabled-->true
</audioInModeEnabled>
<audioInMode opt="leftMic,rightMic,bluetooth,lineIn,earphone,6.5mm,RCAleft,RCAright,embeddedAudio,externalInput,handle">
  <!--ro, opt, enum, audio input sensitivity, subType:string, dep:and,{$.AudioInCap.audioInModeEnable,eq,true}, attr:opt{req, string}, desc:the audio stream from
  microphone (left and right) input is being processed by DSP before being transmitted to the speaker and the sound is optimized; line in directly transmits
  the audio stream to the speaker without optimizing the sound; "leftMic" (left microphone), "rightMic" (right microphone), "bluetooth" (Bluetooth), "lineIn"
  (line in), "earphone" (headphone input); "6.5mm" (6.5 mm input), "RCAleft" (RCA connector, left channel input), "RCAright" (RCA connector, right channel
  input), "embeddedAudio" (embedded audio)-->leftMic
</audioInMode>
<audioInSensitivity opt="high,mid,low">
  <!--ro, opt, enum, audio input sensitivity, subType:string, dep:and,{$.AudioInCap.audioInMode,eq,lineIn}, attr:opt{req, string}, desc:this field
  represents audio sensitivity level; it is valid when the audio input mode is line in; the default value is high; "high" (high), "mid" (middle), "low" (low)-
  >high
</audioInSensitivity>
<AGCEnabled opt="true,false" def="true">
  <!--ro, opt, bool, whether to enable audio auto gain, attr:opt{req, string},def{req, string}, desc:when enabled, the gain of the amplifier circuit
  automatically adjusts according to the signal strength-->true
</AGCEnabled>
<noiseReductionEnabled opt="true,false">
  <!--ro, opt, bool, whether to enable audio noise reduction, attr:opt{req, string}, desc:when enabled, it can greatly reduce various types of noise in
  audio-->true
</noiseReductionEnabled>
<AECEnable opt="true,false">
  <!--ro, opt, bool, echo cancellation, attr:opt{req, string}-->true
</AECEnable>
<audioSamplingRate opt="8,16,32" def="16">
  <!--ro, opt, int, audio sample rate, attr:opt{req, string},def{req, string}, desc:unit: kHz-->0
</audioSamplingRate>
<audioBitRate opt="1,2,3,4,5" def="0">
  <!--ro, opt, int, audio bit rate, attr:opt{req, string},def{req, string}, desc:0 (32Kbps), 1(64Kbps), 2 (128Kbps), 3 (256Kbps), 4 (512Kbps)-->0
</audioBitRate>
<isSupportAudioSceneSettings>
  <!--ro, opt, bool, whether the device supports configuring audio scene, desc:related URI:
  /ISAPI/System/Audio/AudioIn/channels/<audioID>/sceneSettings/capabilities?format=json-->true
</isSupportAudioSceneSettings>
<isSupportSoundSourcePosition>
  <!--ro, opt, bool, whether the device supports getting sound source location information by audio separation, desc:related URI:
  /ISAPI/System/Audio/AudioIn/channels/<audioID>/soundSourcePosition/capabilities?format=json-->true
</isSupportSoundSourcePosition>
<noiseReductionLevel min="0" max="6">
  <!--ro, opt, int, noise reduction level, range:[0,6], attr:min{req, int},max{req, int}, desc:0 indicates the noise reduction is disabled-->0
</noiseReductionLevel>
<AGCGain min="0" max="100" def="50">
  <!--ro, opt, int, range:[0,100], step:1, unit:%, attr:min{req, int},max{req, int},def{req, int}-->50
</AGCGain>
<directionalMikingEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</directionalMikingEnabled>
</AudioInCap>

```

12.7.1.4 Get the audio capabilities

Request URL

GET /ISAPI/System/Audio/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<AudioCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, audio capability, attr:version{req, string, protocolVersion}-->
  <audioInputNums>
    <!--ro, req, int, number of audio inputs-->1
  </audioInputNums>
  <audioOutputNums>
    <!--ro, req, int, number of audio outputs-->1
  </audioOutputNums>
  <mixAudioInSet>
    <!--ro, opt, bool, whether the device supports configuring parameters of mixing audio input, desc:related API: NET_DVR_MIX_AUDIOIN_CFG-->true
  </mixAudioInSet>
  <mixAudioOutSet>

```



```

    <!--ro, opt, bool, whether the device supports configuring parameters of mixing audio output, desc:related API: NET_DVR_MIX_AUDIOOUT_CFG-->true
</mixAudioOutSet>
<isSupportAudioMixing>
    <!--ro, opt, bool, whether the device supports audio mixing, desc:related URL: /ISAPI/System/Audio/Mixing/capabilities-->true
</isSupportAudioMixing>
<isSupportRemoveFrequencyPoint>
    <!--ro, opt, bool, whether the device supports clearing the howling frequency point-->true
</isSupportRemoveFrequencyPoint>
<isSupportAudioInConfig>
    <!--ro, opt, bool, whether the device supports configuring parameters of all audio inputs, desc:related URL: /ISAPI/System/Audio/AudioIn/capabilities-->true
</isSupportAudioInConfig>
<muteDuringPanTilt opt="MicIn">
    <!--ro, opt, string, whether the device supports muting during motion, attr:opt{req, string}, desc:only MicIn is supported now, corresponding filed
muteDuringPanTilt, related URL: /ISAPI/System/TwoWayAudio/channels/ID-->test
</muteDuringPanTilt>
<isSupportAudioOutConfig>
    <!--ro, opt, bool, whether the device supports configuring all audio outputs, desc:related URL: /ISAPI/System/Audio/AudioOut/capabilities?format=json-->true
</isSupportAudioOutConfig>
<isSupportSAIP>
    <!--ro, opt, bool, whether the device supports connecting to the independent audio track channel-->true
</isSupportSAIP>
<SAIPChannelsNum>
    <!--ro, opt, int, number of devices that supports connecting to the independent audio track channel, dep:and,{$.AudioCap.isSupportSAIP,eq,true}-->1
</SAIPChannelsNum>
<isSupportDataStorage>
    <!--ro, opt, bool, whether the device supports uploading the audio file, desc:related URL: /ISAPI/System/Audio/dataStorage?format=json-->true
</isSupportDataStorage>
<isSupportDataRelease>
    <!--ro, opt, bool, whether the device supports audio file releasing, desc:related URL: /ISAPI/System/Audio/dataRelease?format=json-->true
</isSupportDataRelease>
<isSupportAudioPlan>
    <!--ro, opt, bool, whether the device supports audio plan, desc:related URL: ISAPI/System/Audio/plan?format=json-->true
</isSupportAudioPlan>
<isSupportAudioTask>
    <!--ro, opt, bool, whether the device supports audio plan task, desc:related URL: /ISAPI/System/Audio/p/plan/<PlanID>/task?format=json-->true
</isSupportAudioTask>
<isSupportaudioCollection>
    <!--ro, opt, bool, whether the device supports audio collection, desc:related URL: /ISAPI/System/Audio/audioCollection/capabilities?format=json;
maintenance function, it is used for collecting audio to analyze audio problems-->true
</isSupportaudioCollection>
<isSupportWirelessPickUp>
    <!--ro, opt, bool, whether the device supports accessing wireless pick-up, desc:related URL: /ISAPI/System/Audio/wirelessPickUp/capabilities?
format=json-->true
</isSupportWirelessPickUp>
<audioOutputBroadcastNums opt="1,2,3,4">
    <!--ro, opt, int, number of audio broadcast outputs, attr:opt{opt, string}, desc:at present, this node is only supported by broadcast audio devices when
opt="1,2,3,4" is returned and the output depends on the actual node; when not returned, the output starts from 1 by default according to business type, the
audio devices classify the output ports into broadcast output port and audio output port considering the start No. of every output starts from 1, it should
return the specific number of supported broadcast outputs if this node is returned, it indicates that the device supports the audio output volume
configuration; otherwise, the device does not support. Corresponding URI: /ISAPI/System/Audio/AudioOut/broadcast/capabilities?format=json-->0
</audioOutputBroadcastNums>
<isSupportAudioOutMixing>
    <!--ro, opt, bool, whether the device supports outputting audio mixing, desc:related URL: /ISAPI/System/Audio/AudioOut/mixing/capabilities?format=json-->true
</isSupportAudioOutMixing>
<isSupportRTSPAudio>
    <!--ro, opt, bool, whether the device supports listening to the audio transmitted via RTSP in real time, desc:related URL:
/ISAPI/Streaming/audio/<audioID>, the audioID refers to the audio channel ID-->true
</isSupportRTSPAudio>
<AudioBoard>
    <!--ro, opt, object, whether the device supports audio board related configurations-->
<isSupportNetworkParamCfg>
    <!--ro, opt, bool, whether the device supports configuring network parameters of audio boards, desc:related URL:
/ISAPI/System/Audio/AudioBoard/NetworkParamCfg/capabilities?format=json-->true
</isSupportNetworkParamCfg>
</AudioBoard>
<isSupportRTSPAudioPlaybackByFile>
    <!--ro, opt, bool, whether the device supports playing back audio transmitted via RTSP according to files, desc:the RTSP streaming URL is playbackURI
returned by /ISAPI/ContentMgmt/recordAudio/search?format=json-->true
</isSupportRTSPAudioPlaybackByFile>
<isSupportAudioInStatusDetectionParams>
    <!--ro, opt, bool, whether the device supports status detection parameters of an audio input port, desc:GET or PUT
/ISAPI/System/Audio/AudioIn/<AudioInID>/AudioInStatusDetectionParams?format=json-->true
</isSupportAudioInStatusDetectionParams>
<isSupportGetAudioInStatus>
    <!--ro, opt, bool, whether the device supports getting the audio input port status, desc:GET /ISAPI/System/Audio/AudioIn/<AudioInID>/GetAudioInStatus?
format=json-->true
</isSupportGetAudioInStatus>
<isSupportAudioInStatusTimeUpload>
    <!--ro, opt, bool, whether the device supports scheduled uploading linkage parameters of audio input port status, desc:GET or PUT
/ISAPI/System/Audio/AudioIn/<AudioInID>/AudioInStatusTimeUpload?format=json-->true
</isSupportAudioInStatusTimeUpload>
<isSupportDeviceAudioCapture>
    <!--ro, opt, bool, whether the device supports audio collection, desc:GET/PUT /ISAPI/System/Audio/DeviceAudioCapture?format=json-->true
</isSupportDeviceAudioCapture>
<audioInputNumsEx>
    <!--ro, opt, int-->1
</audioInputNumsEx>
<isSupportEQParams>
    <!--ro, opt, bool-->true
</isSupportEQParams>
<isSupportEndAudioCollectionFunc>

```

```

<!--ro, opt, bool-->true
</isSupportSendAudioCalibrationCmd>
<isSupportOverloadSPLAlarmParams>
<!--ro, opt, bool-->true
</isSupportOverloadSPLAlarmParams>
<isSupportVoiceprintCmptAlarmParams>
<!--ro, opt, bool-->true
</isSupportVoiceprintCmptAlarmParams>
<isSupportAudioCodingPriorityParams>
<!--ro, opt, bool-->true
</isSupportAudioCodingPriorityParams>
<isSupportSceneMode>
<!--ro, opt, bool-->true
</isSupportSceneMode>
<isSupportSearchAudioMonitorPoint>
<!--ro, opt, bool-->true
</isSupportSearchAudioMonitorPoint>
<isSupportAudioMonitorPoint>
<!--ro, opt, bool-->true
</isSupportAudioMonitorPoint>
<isSupportModifyBatchAudioInCfg>
<!--ro, opt, bool-->true
</isSupportModifyBatchAudioInCfg>
<isSupportGetAudioInCfgList>
<!--ro, opt, bool-->true
</isSupportGetAudioInCfgList>
<isSupportSearchSoundPressureStatisticalData>
<!--ro, opt, bool-->true
</isSupportSearchSoundPressureStatisticalData>
</AudioCap>

```

12.7.1.5 Get parameters of all audio channels of the device

Request URL

GET /ISAPI/System/Audio/channels

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<AudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, List of audio channels, subType:object, attr:version{req, string, protocolVersion}-->
  <AudioChannel>
    <!--ro, opt, object, audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable audio-->true
    </enabled>
    <type opt="audioIn,audioOut">
      <!--ro, opt, enum, audio channel management, subType:string, attr:opt{req, string}, desc:"audioIn" (input audio), "audioOut" (output audio)-->audioIn
    </type>
    <name>
      <!--ro, opt, string, audio channel name-->test
    </name>
  </AudioChannel>
</AudioChannelList>

```

12.7.1.6 Get the configuration of a specific audio channel of the device

Request URL

GET /ISAPI/System/Audio/channels/<audioID>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, audio channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel index-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable audio-->true
  </enabled>
  <type opt="audioIn,audioOut">
    <!--ro, opt, enum, audio channel type, subType:string, attr:opt{req, string}, desc:"audioIn" (audio input), "audioOut" (audio output)-->audioIn
  </type>
  <name>
    <!--ro, opt, string, audio channel name-->test
  </name>
</AudioChannel>
```

12.7.2 Audio Output

12.7.2.1 Set audio output parameters of a specific channel

Request URL

PUT /ISAPI/System/Audio/AudioOut/channels/<audioID>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioOut xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->test
  </id>
  <MixAudioOut>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <modulatorEnable>
      <!--req, bool-->true
    </modulatorEnable>
    <postFilter>
      <!--req, bool-->true
    </postFilter>
    <limitPressure>
      <!--req, bool-->true
    </limitPressure>
    <modulatorValue>
      <!--req, int, the default value is 10, range:[0,127], desc:the default value is 10-->1
    </modulatorValue>
    <triggerTime>
      <!--opt, int, unit: ms, range:[1,872], unit:ms, desc:the default value is 4-->1
    </triggerTime>
    <freeTime>
      <!--opt, int, unit: ms, range:[26,46062], unit:ms, desc:the default value is 42-->26
    </freeTime>
    <compressThreshold>
      <!--req, int-->1
    </compressThreshold>
    <compressMode>
      <!--req, enum, subType:string-->soft
    </compressMode>
    <compressRate>
      <!--req, int, range:[0,127]-->1
    </compressRate>
    <recoveryGain>
      <!--req, int-->1
    </recoveryGain>
    <outputGain>
      <!--req, int-->1
    </outputGain>
    <outputMute>
```

```

    <!--opt, bool-->true
  </outputMute>
  <soundQualityHandling>
    <!--opt, bool-->true
  </soundQualityHandling>
  <outputVolume>
    <!--opt, int, range:[0,100]-->1
  </outputVolume>
</MixAudioOut>
<AudioOutVolumelist>
  <!--opt, object-->
  <AudioOutVlome>
    <!--opt, object-->
    <type>
      <!--req, enum, type, subType:string, desc:type-->audioOutput
    </type>
    <volume>
      <!--req, int, system volume, desc:system volume-->1
    </volume>
    <id>
      <!--opt, string, dep:and,{$.AudioOutCap.AudioOutVolumelist.AudioOutVlome.type,eq,specific}-->test
    </id>
    <name>
      <!--opt, string, dep:and,{$.AudioOutCap.AudioOutVolumelist.AudioOutVlome.type,eq,specific}-->test
    </name>
    <talkVolume>
      <!--opt, int, range:[0,100]-->0
    </talkVolume>
    <trebleVolume>
      <!--opt, int, range:[0,100]-->0
    </trebleVolume>
    <bassVolume>
      <!--opt, int, range:[0,100]-->0
    </bassVolume>
    <balanceVolume>
      <!--opt, int, range:[0,100]-->0
    </balanceVolume>
  </AudioOutVlome>
</AudioOutVolumelist>
<audioOutputInterfaceType>
  <!--opt, enum, subType:string-->RCA
</audioOutputInterfaceType>
<audioOutputInterfaceMode>
  <!--opt, enum, subType:string-->auto
</audioOutputInterfaceMode>
<audioBitRate>
  <!--opt, int, range:[32,196], unit:kpbs-->32
</audioBitRate>
</AudioOut>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>

```

12.7.2.2 Get the audio output parameters of a specified channel

Request URL

GET /ISAPI/System/Audio/AudioOut/channels/<audioID>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioOut xmlns="http://www.isapi.org/ver20/XMLSchema" version="20">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <MixAudioOut>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <modulatorEnable>
      <!--ro, req, bool-->true
    </modulatorEnable>
    <postFilter>
      <!--ro, req, bool-->true
    </postFilter>
    <limitPressure>
      <!--ro, req, bool-->true
    </limitPressure>
    <modulatorValue>
      <!--ro, req, int, phase modulation value, range:[0,127], desc:the default value is 10-->1
    </modulatorValue>
    <triggerTime>
      <!--ro, opt, int, unit: ms, range:[1,872], unit:ms, desc:the default value is 4-->1
    </triggerTime>
    <freeTime>
      <!--ro, opt, int, unit: ms, range:[26,46062], unit:ms, desc:the default value is 42-->26
    </freeTime>
    <compressThreshold>
      <!--ro, req, int-->1
    </compressThreshold>
    <compressMode>
      <!--ro, req, enum, compression mode, subType:string, desc:compression mode-->soft
    </compressMode>
    <compressRate>
      <!--ro, req, int, compression ratio, range:[0,127]-->1
    </compressRate>
    <recoveryGain>
      <!--ro, req, int-->1
    </recoveryGain>
    <outputGain>
      <!--ro, req, int, output gain-->1
    </outputGain>
    <outputMute>
      <!--ro, opt, bool, output mute-->true
    </outputMute>
    <soundQualityHandling>
      <!--ro, opt, bool, sound quality processing-->true
    </soundQualityHandling>
    <outputVolume>
      <!--ro, opt, int, output volume, range:[0,100]-->1
    </outputVolume>
  </MixAudioOut>
  <AudioOutVolumelist>
    <!--ro, opt, object-->
    <AudioOutVlome>
      <!--ro, opt, object-->
      <type>
        <!--ro, req, enum, type, subType:string, desc:type-->audioOutput
      </type>
      <volume>
        <!--ro, req, int, system volume, desc:system volume-->1
      </volume>
      <id>
        <!--ro, opt, string, dep:and,{$.AudioOutCap.AudioOutVolumelist.AudioOutVlome.type,eq,specific}-->test
      </id>
      <name>
        <!--ro, opt, string, dep:and,{$.AudioOutCap.AudioOutVolumelist.AudioOutVlome.type,eq,specific}-->test
      </name>
      <talkVolume>
        <!--ro, opt, int, range:[0,100]-->0
      </talkVolume>
      <mediaVolume>
        <!--ro, opt, int, range:[0,100]-->0
      </mediaVolume>
      <trebleVolume>
        <!--ro, opt, int, range:[0,100]-->0
      </trebleVolume>
      <bassVolume>
        <!--ro, opt, int, range:[0,100]-->0
      </bassVolume>
      <balanceVolume>
        <!--ro, opt, int, range:[0,100]-->0
      </balanceVolume>
      <percentOfMainAudioVolume>
        <!--ro, opt, int, range:[0,100]-->0
      </percentOfMainAudioVolume>
    </AudioOutVlome>
  </AudioOutVolumelist>
</AudioOut>
```

```

    </percentOfMainAudioVolume>
  </AudioOutVolume>
</AudioOutVolumeList>
<audioOutputInterfaceType>
  <!--ro, opt, enum, subType:string-->RCA
</audioOutputInterfaceType>
<audioOutputInterfaceMode>
  <!--ro, opt, enum, subType:string-->auto
</audioOutputInterfaceMode>
<audioBitRate>
  <!--ro, opt, int, range:[32,196], unit:kpbs-->32
</audioBitRate>
<audioPlaybackMode>
  <!--ro, opt, enum, subType:string-->smooth
</audioPlaybackMode>
<EQAdjust>
  <!--ro, opt, object-->
  <lowFrequency>
    <!--ro, opt, object-->
    <lowFrequencyValue>
      <!--ro, req, int, range:[20,20000], unit:Hz-->29
    </lowFrequencyValue>
    <lowFrequencyGain>
      <!--ro, req, int, range:[-30,15], unit:dB-->1
    </lowFrequencyGain>
  </lowFrequency>
  <highFrequency>
    <!--ro, opt, object-->
    <highFrequencyValue>
      <!--ro, req, int, range:[20,20000], unit:Hz-->0
    </highFrequencyValue>
    <highFrequencyGain>
      <!--ro, req, int, range:[-30,15], unit:dB-->1
    </highFrequencyGain>
  </highFrequency>
</EQAdjust>
</AudioOut>

```

12.7.2.3 Get the audio output capability of a specified channel

Request URL

GET /ISAPI/System/Audio/AudioOut/channels/<audioID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<AudioOutCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="20">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <MixAudioOut>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <modulatorEnbale opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </modulatorEnbale>
    <postFilter opt="false,true" default="true">
      <!--ro, req, bool, attr:opt{req, string},default{opt, string}-->true
    </postFilter>
    <limitPressure opt="false,true" default="true">
      <!--ro, req, bool, attr:opt{req, string},default{opt, string}-->true
    </limitPressure>
    <modulatorValue min="0" max="127" default="10">
      <!--ro, req, int, phase modulation value, range:[0,127], attr:min{req, int},max{req, int},default{opt, string}, desc:the default value is 10-->1
    </modulatorValue>
    <triggerTime>
      <!--ro, opt, int, unit: ms, range:[1,872], unit:ms, desc:the default value is 4-->1
    </triggerTime>
    <freeTime>
      <!--ro, opt, int, unit: ms, range:[26,46062], unit:ms, desc:the default value is 42-->26
    </freeTime>
    <compressThreshold opt="1 2 4 8" default="1">

```

```

<compressThreshold opt="1,2,4,8" default="2">
  <!--ro, req, int, attr:opt{req, string},default{opt, string}-->1
</compressThreshold>
<compressMode opt="soft,hard" default="soft">
  <!--ro, req, enum, compression mode, subType:string, attr:opt{req, string},default{opt, string}, desc:compression mode-->soft
</compressMode>
<compressRate min="0" max="127">
  <!--ro, req, int, compression ratio, attr:min{req, int},max{req, int}-->1
</compressRate>
<recoveryGain opt="1,2,4,8" default="2">
  <!--ro, req, int, attr:opt{req, string},default{opt, string}-->1
</recoveryGain>
<outputGain def="1">
  <!--ro, req, int, output gain, attr:def{opt, string}-->1
</outputGain>
<outputMute opt="true,false" def="true">
  <!--ro, opt, bool, output mute, attr:opt{req, string},def{req, string}-->true
</outputMute>
<soundQualityHandling opt="true,false">
  <!--ro, opt, bool, sound quality processing, attr:opt{req, string}-->true
</soundQualityHandling>
<outputVolume>
  <!--ro, opt, int, output volume, range:[0,100]-->1
</outputVolume>
</MixAudioOut>
<AudioOutVolumelist>
  <!--ro, opt, object-->
  <AudioOutVlome>
    <!--ro, opt, object-->
    <type opt="audioOutput,audioEncode,specific">
      <!--ro, req, enum, type, subType:string, attr:opt{req, string}, desc:type-->audioOutput
    </type>
    <volume min="0" max="127" def="50">
      <!--ro, req, int, system volume, attr:min{opt, int},max{opt, int},def{opt, int}, desc:system volume-->1
    </volume>
    <id>
      <!--ro, opt, string, dep:and,{$.AudioOutCap.AudioOutVolumelist.AudioOutVlome.type,eq,specific}-->test
    </id>
    <name>
      <!--ro, opt, string, dep:and,{$.AudioOutCap.AudioOutVolumelist.AudioOutVlome.type,eq,specific}-->test
    </name>
    <talkVolume min="0" max="15" def="5">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int},def{opt, int}-->0
    </talkVolume>
    <mediaVolume min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </mediaVolume>
    <trebleVolume min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </trebleVolume>
    <bassVolume min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </bassVolume>
    <balanceVolume min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </balanceVolume>
    <percentOfMainAudioVolume min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </percentOfMainAudioVolume>
  </AudioOutVlome>
</AudioOutVolumelist>
<audioOutputInterfaceType opt="RCA,3.5MM,usbPpeaker,localMegaphone,telephoneMegaphone">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->RCA
</audioOutputInterfaceType>
<audioOutputInterfaceMode opt="auto">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
</audioOutputInterfaceMode>
<audioBitRate min="32" max="196">
  <!--ro, opt, int, range:[32,196], step:1, unit:kbps, attr:min{req, int},max{req, int}-->32
</audioBitRate>
<audioPlaybackMode opt="smooth,balance,realtime,auto">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->smooth
</audioPlaybackMode>
<EQAdjust>
  <!--ro, opt, object-->
  <lowFrequency>
    <!--ro, opt, object-->
    <lowFrequencyValue min="20" max="20000" def="20">
      <!--ro, opt, int, range:[20,20000], unit:Hz, attr:min{req, int},max{req, int},def{req, int}-->20
    </lowFrequencyValue>
    <lowFrequencyGain min="-30" max="15" def="0">
      <!--ro, opt, int, range:[-30,15], unit:dB, attr:min{req, int},max{req, int},def{req, int}-->1
    </lowFrequencyGain>
  </lowFrequency>
  <highFrequency>
    <!--ro, opt, object-->
    <highFrequencyValue min="20" max="20000" def="20">
      <!--ro, opt, int, range:[20,20000], unit:Hz, attr:min{req, int},max{req, int},def{req, int}-->20
    </highFrequencyValue>
    <highFrequencyGain min="-30" max="15" def="0">
      <!--ro, opt, int, range:[-30,15], unit:dB, attr:min{req, int},max{req, int},def{req, int}-->1
    </highFrequencyGain>
  </highFrequency>
</EQAdjust>

```

```

<audioOutputInterfaceTypeEnabled opt="true,false" def="false">
  <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->false
</audioOutputInterfaceTypeEnabled>
</AudioOutCap>

```

12.8 Access Control (General)

12.8.1 Access Control Event Management

12.8.1.1 Get the capability of searching for access control events

Request URL

GET /ISAPI/AccessControl/AcsEvent/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "AcsEvent": {
    /*ro, req, object, access control events*/
    "AcsEventCond": {
      /*ro, opt, object, search conditions*/
      "searchID": {
        /*ro, req, object, search ID, it is used to check whether the current search requester is the same as the previous one. If they are the same,
        the search record will be stored in the device to speed up the next search*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
      },
      "searchResultPosition": {
        /*ro, req, object, the start position of the search result in the result list*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
      },
      "maxResults": {
        /*ro, req, object, the maximum number of search results that can be obtained by calling this URL*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
      },
      "major": {
        /*ro, opt, object, major alarm type (the type value should be transformed to the decimal number)*/
        "@opt": "0,1,2,3,5"
        /*ro, req, string, major type*/
      },
      "minorAlarm": {
        /*ro, opt, object, minor alarm type (the type value should be transformed to the decimal number)*/
        "@opt": "1024,1025,1026,1027..."
        /*ro, req, string, minor alarm type*/
      },
      "minorException": {
        /*ro, opt, object, minor exception type (the type value should be transformed to the decimal number)*/
        "@opt": "39,58,59,1024..."
        /*ro, req, string, minor exception type*/
      },
      "minorOperation": {
        /*ro, opt, object, minor operation type (the type value should be transformed to the decimal number)*/
        "@opt": "80,90,112,113..."
        /*ro, req, string, minor operation type*/
      },
      "minorEvent": {
        /*ro, opt, object, minor event type (the type value should be transformed to the decimal number)*/
        "@opt": "1,2,3,4..."
        /*ro, req, string, minor event type*/
      },
      "startTime": {
        /*ro, opt, object, start time*/
        "@min": 0,
        /*ro, req, int, the minimum value, range:[0,32]*/
        "@max": 32
        /*ro, req, int, the maximum value, range:[0,32]*/
      },
      "endTime": {
        /*ro, opt, object, end time*/

```



```

        "@min": 0,
        /*ro, req, int, the minimum value, range:[0,32]*/
        "@max": 32
        /*ro, req, int, the maximum value, range:[0,32]*/
    },
    "cardNo": {
        /*ro, opt, object, card No.*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    },
    "name": {
        /*ro, opt, object, name of the card holder*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    },
    "videoChannel": {
        /*ro, opt, object, video channel No. (this node is newly added to DeepinMind devices for attendance)*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    },
    "picEnable": "true,false",
    /*ro, opt, string, whether to include pictures*/
    "beginSerialNo": {
        /*ro, opt, object, start serial No.*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    },
    "endSerialNo": {
        /*ro, opt, object, end serial No.*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    },
    "employeeNoString": {
        /*ro, opt, object, employee No.*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    },
    "isAbnormalTemperature": {
        /*ro, opt, object, whether the skin-surface temperature is abnormal*/
        "@opt": [true, false]
        /*ro, req, array, options, subType:bool, desc:true (yes), false (no)*/
    },
    "temperatureSearchCond": {
        /*ro, opt, object, temperature search condition*/
        "@opt": ["all", "normal", "abnormal"]
        /*ro, req, array, options, subType:string*/
    },
    "isAttendanceInfo": {
        /*ro, opt, object, whether it contains attendance records*/
        "@opt": [true, false]
        /*ro, req, array, options, subType:bool, desc:true (yes), false (no)*/
    },
    "hasRecordInfo": {
        /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, req, array, subType:bool*/
    }
},
"InfoList": {
    /*ro, opt, object, information list*/
    "maxSize": 10,
    /*ro, opt, int, the maximum value*/
    "time": {
        /*ro, opt, object, time (UTC time)*/
        "@min": 0,
        /*ro, req, int, the minimum value, range:[0,32]*/
        "@max": 32
        /*ro, req, int, the maximum value, range:[0,32]*/
    },
    "netUser": {
        /*ro, opt, object, user name*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    },
    "remoteHostAddr": {
        /*ro, opt, object, remote host address*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    },
    "remoteHostPort": {
        /*ro, opt, object, remote host port*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 1
        /*ro, req, int, the maximum value*/
    }
}

```

```

/*ro, req, int, the maximum value*/
},
"cardNo": {
/*ro, opt, object, card No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"cardType": {
/*ro, opt, object, card type*/
  "@opt": "1,2,3,4,5,6,7,8"
/*ro, req, string, 1 (normal card), 2 (disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6 (super card), 7 (visitor
card), 8 (dismiss card)*/
},
"whitelistNo": {
/*ro, opt, object, allowlist No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"reportChannel": {
/*ro, opt, object, channel type for uploading alarms/events*/
  "@opt": "1,2,3"
/*ro, req, string, "1" (for uploading arming information), "2" (for uploading by central group 1), "3" (for uploading by central group 2)*/
},
"cardReaderKind": {
/*ro, opt, object, card reader type*/
  "@opt": "1,2,3,4"
/*ro, req, string, "1" (IC card reader), "2" (ID card reader), "3" (QR code scanner), "4" (fingerprint module)*/
},
"cardReaderNo": {
/*ro, opt, object, card reader No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"doorNo": {
/*ro, opt, object, door (floor) No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"verifyNo": {
/*ro, opt, object, multiple authentication No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"alarmInNo": {
/*ro, opt, object, alarm input No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"alarmOutNo": {
/*ro, opt, object, alarm output No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"caseSensorNo": {
/*ro, opt, object, event trigger No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"RS485No": {
/*ro, opt, object, RS-485 channel No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"multiCardGroupNo": {
/*ro, opt, object, group No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
},
"accessChannel": {
/*ro, opt, object, turnstile No.*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 1
/*ro, req, int, the maximum value*/
}

```

```

},
"deviceNo": {
/*ro, opt, object, device No.*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"distractControlNo": {
/*ro, opt, object, distributed access controller No.*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"employeeNo": {
/*ro, opt, object, employee No. (person ID)*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"localControllerID": {
/*ro, opt, object, distributed access controller No.*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"InternetAccess": {
/*ro, opt, object, network interface No.*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"type": {
/*ro, opt, object, zone type*/
"@opt": "0,1,2,3,4,5,6,7,8,9,10,11,255"
/*ro, req, string, 0 (instant alarm zone), 1 (24-hour zone), 2 (delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm zone), 6
(perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone), 9 (24-hour check zone), 10 (emergency door open zone), 11 (emergency door closed
zone), 255 (none)*/
},
"MACAddr": {
/*ro, opt, object, MAC Address*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"swipeCardType": {
/*ro, opt, object, card swiping type*/
"@opt": "0,1"
/*ro, req, string, "0" (invalid), "1" (QR code)*/
},
"serialNo": {
/*ro, opt, object, event serial No.*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"channelControllerID": {
/*ro, opt, object, lane controller No.*/
"@opt": "0,1"
/*ro, req, string, "1" (main lane controller), "2" (sub lane controller)*/
},
"channelControllerLampID": {
/*ro, opt, object, light board ID of lane controller*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"channelControllerIRAdaptorID": {
/*ro, opt, object, IR adapter No. of lane controller*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"channelControllerIREmitterID": {
/*ro, opt, object, active infrared intrusion detector No. of lane controller*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"userType": {
/*ro, opt, object, person types*/
"@opt": "normal,visitor,blacklist,administrators"
/*ro, req, string, "normal" (normal person (household)), "visitor" (visitor), "blacklist" (person in blacklist), "administrators"
(administrator)*/

```

```

    },
    "currentVerifyMode": {
      /*ro, opt, object, current authentication mode of the card reader*/
      "@opt":
        "cardAndPw, card, cardOrPw, fp, fpAndPw, fpOrCard, fpAndCard, fpAndCardAndPw, faceOrFpOrCardOrPw, faceAndFp, faceAndPw, faceAndCard, face, employeeNoAndPw, fpOrPw, employeeNoAndFp, employeeNoAndFpAndPw, faceAndFpAndCard, faceAndPwAndFp, employeeNoAndFace, faceOrfaceAndCard, fpOrface, cardOrfaceOrPw, iris, faceOrFpOrCardOrPwOrIris, faceOrCardOrPwOrIris"

      /*ro, req, string, "cardAndPw"-card+password, "card", "cardOrPw"-card or password, "fp"-fingerprint, "fpAndPw"-fingerprint+password, "fpOrCard"-fingerprint or card, "fpAndCard"-fingerprint+card, "fpAndCardAndPw"-fingerprint+card+password, "faceOrFpOrCardOrPw"-face or fingerprint or card or password, "faceAndFp"-face+fingerprint, "faceAndPw"-face+password, "faceAndCard"-face+card, "face", "employeeNoAndPw"-employee No.+password, "fpOrPw"-fingerprint or password, "employeeNoAndFp"-employee No.+fingerprint, "employeeNoAndFpAndPw"-employee No.+fingerprint+password, "faceAndFpAndCard"-face+fingerprint+card, "faceAndPwAndFp"-face+password+fingerprint, "employeeNoAndFace"-employee No.+face, "faceOrfaceAndCard"-face or face+card, "fpOrface"-fingerprint or face, "cardOrfaceOrPw"-card or face or password, "cardOrFpOrPw"-card or fingerprint or password*/
    },
    "QRCodeInfo": {
      /*ro, opt, object, QR code information*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 1
      /*ro, req, int, the maximum value*/
    },
    "thermometryUnit": {
      /*ro, opt, object, temperature unit*/
      "@opt": ["celsius", "fahrenheit", "kelvin"]
      /*ro, req, array, "celsius" (Celsius (default)), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin), subType:string*/
    },
    "currTemperature": {
      /*ro, opt, object, skin-surface temperature*/
      "@min": 1,
      /*ro, req, int, skin-surface temperature, which is accurate to one decimal place*/
      "@max": 1
      /*ro, req, int, skin-surface temperature, which is accurate to one decimal place*/
    },
    "isAbnormalTemperature": {
      /*ro, opt, object, whether the skin-surface temperature is abnormal*/
      "@opt": [true, false]
      /*ro, req, array, whether the skin-surface temperature is abnormal (true=yes), subType:bool*/
    },
    "RegionCoordinates": {
      /*ro, opt, object, coordinates of the skin-surface temperature*/
      "positionX": {
        /*ro, opt, object, X-coordinate*/
        "@min": 0,
        /*ro, req, int, the minimum value, normalized X-coordinate which is between 0 and 1000*/
        "@max": 1000
        /*ro, req, int, the maximum value, normalized X-coordinate which is between 0 and 1000*/
      },
      "positionY": {
        /*ro, opt, object, Y-coordinate*/
        "@min": 0,
        /*ro, req, int, the minimum value, normalized Y-coordinate which is between 0 and 1000*/
        "@max": 1000
        /*ro, req, int, the maximum value, normalized Y-coordinate which is between 0 and 1000*/
      }
    },
    "mask": {
      /*ro, opt, object, whether the person is wearing mask*/
      "@opt": ["unknown", "yes", "no"]
      /*ro, req, array, "unknown", "yes", "no", subType:string*/
    },
    "pictureURL": {
      /*ro, opt, object, URL of the captured picture*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 1
      /*ro, req, int, the maximum value*/
    },
    "filename": {
      /*ro, opt, object, file name, desc:if multiple pictures are returned, the file name of each picture should be unique*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 1
      /*ro, req, int, the maximum value*/
    },
    "attendanceStatus": {
      /*ro, opt, object, attendance status, desc:"undefined", "checkIn" (check-in), "checkOut" (check-out), "breakOut" (start of break), "breakIn" (end of break), "overtimeIn" (start of overtime), "overTimeOut" (end of overtime)*/
      "@opt": "undefined,checkIn,checkOut,breakOut,breakIn,overtimeIn,overTimeOut"
      /*ro, req, string, options*/
    },
    "label": {
      /*ro, opt, object, self-defined attendance name*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 1
      /*ro, req, int, the maximum value*/
    },
    "statusValue": {
      /*ro, opt, object, status value*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 1
      /*ro, req, int, the maximum value*/
    },
  },

```

```

    },
    "helmet": {
      /*ro, opt, object, whether the person is wearing hard hat, desc:"unknown", "yes", "no"*/
      "@opt": "unknown,yes,no"
      /*ro, req, string, options*/
    },
    "visibleLightPicUrl": {
      /*ro, opt, object, URL of the visible light picture*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 1
      /*ro, req, int, the maximum value*/
    },
    "thermalPicUrl": {
      /*ro, opt, object, URL of the thermal imaging picture*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 1
      /*ro, req, int, the maximum value*/
    },
    "HealthInfo": {
      /*ro, opt, object, health information*/
      "healthCode": {
        /*ro, opt, object, health code status*/
        "@opt": [0, 1, 2, 3, 4, 5, 6, 7]
        /*ro, req, array, options, subType:int, desc:0 (no request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red QR code),
5 (no such person), 6 (other error, e.g., searching failed due to API exception), 7 (searching for the health code timed out)*/
      },
      "NADCode": {
        /*ro, opt, object, nucleic acid test result, desc:0 (no result), 1 (negative, which means normal), 2 (positive, which means diagnosed), 3
(the result has expired)*/
        "@opt": [0, 1, 2, 3, 4]
        /*ro, req, array, options, subType:int*/
      },
      "travelCode": {
        /*ro, opt, object, trip code, desc:0 (no trip in the past 14 days), 1 (has left the current area left in the past 14 days), 2 (has been to
the high-risk area in the past 14 days), 3 (other)*/
        "@opt": [0, 1, 2, 3, 4]
        /*ro, req, array, options, subType:int*/
      },
      "travelInfo": {
        /*ro, opt, object*/
        "@min": 0,
        /*ro, req, int, step:1*/
        "@max": 15
        /*ro, req, int, step:1*/
      },
      "vaccineStatus": {
        /*ro, opt, object, whether the person is vaccinated, desc:0 (not vaccinated), 1 (vaccinated)*/
        "@opt": [0, 1, 2, 3, 4]
        /*ro, req, array, options, subType:int*/
      },
      "vaccineNum": {
        /*ro, opt, object*/
        "@min": 0,
        /*ro, req, int, step:1*/
        "@max": 3
        /*ro, req, int, step:1*/
      }
    },
    "meetingID": {
      /*ro, opt, object, meeting ID*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 32
      /*ro, req, int, the maximum value*/
    },
    "PersonInfoExtends": {
      /*ro, opt, object, extended nodes for the additional person information*/
      "maxSize": 3,
      /*ro, req, int, range of members in the array*/
      "id": {
        /*ro, opt, object, extended ID of the additional person information*/
        "@min": 1,
        /*ro, req, int, the minimum value*/
        "@max": 32
        /*ro, req, int, the maximum value*/
      },
      "value": {
        /*ro, opt, object, extended content of the additional person information*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 100
        /*ro, req, int, the maximum value*/
      }
    },
    "name": {
      /*ro, opt, object, name*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 128
      /*ro, req, int, the maximum value*/
    },
    "FaceRect": {

```

```
/*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
    "height": {
        /*ro, req, object, height*/
        "@min": 0.000,
        /*ro, req, float, the minimum value*/
        "@max": 1.000
        /*ro, req, float, the maximum value*/
    },
    "width": {
        /*ro, req, object, width*/
        "@min": 0.000,
        /*ro, req, float, the minimum value*/
        "@max": 1.000
        /*ro, req, float, the maximum value*/
    },
    "x": {
        /*ro, req, object, X-coordinate of the upper-left corner of the frame*/
        "@min": 0.000,
        /*ro, req, float, the minimum value*/
        "@max": 1.000
        /*ro, req, float, the maximum value*/
    },
    "y": {
        /*ro, req, object, Y-coordinate of the upper-left corner of the frame*/
        "@min": 0.000,
        /*ro, req, float, the minimum value*/
        "@max": 1.000
        /*ro, req, float, the maximum value*/
    }
},
"RecordInfo": {
    /*ro, opt, object*/
    "startTime": "1970-01-01T00:00:00+08:00",
    /*ro, opt, datetime*/
    "endTime": "1970-01-01T00:00:00+08:00",
    /*ro, opt, datetime*/
    "playbackURL": {
        /*ro, opt, object*/
        "@min": 0,
        /*ro, req, int, the minimum length*/
        "@max": 256
        /*ro, req, int, the maximum length*/
    }
},
"currentAuthenticationTimes": {
    /*ro, req, object*/
    "@min": 0,
    /*ro, req, int*/
    "@max": 255
    /*ro, req, int*/
},
"allowAuthenticationTimes": {
    /*ro, req, object*/
    "@min": 0,
    /*ro, req, int*/
    "@max": 255
    /*ro, req, int*/
}
}
}
```

Request Message

```

{
  "AcsEventCond": {
    /*req, object, access control events*/
    "searchID": "test",
    /*req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the
search record will be stored in the device to speed up the next search*/
    "searchResultPosition": 0,
    /*req, int, the start position of the search result in the result list; desc:in a single search, if you cannot get all the records in the result
list, you can mark the end position and get the following records after the marked position in the next search. If the maximum number of totalMatches
supported by the device is M and the number of totalMatches stored in the device now is N (N<=M), the valid range of this node is 0 to N-1*/
    "maxResults": 30,
    /*req, int, the maximum number of search results, which is defined by the device capability, will be returned if the value of maxResults reaches the
limit, desc:if maxResults exceeds the range returned by the device capability, the device will return the maximum number of search results according to the
device capability and will not return error message*/
    "major": 1,
    /*req, int, major type, desc:the type value should be transformed to the decimal number; see Access Control Event Types for details*/
    "minor": 1024,
    /*req, int, minor type, desc:the type value should be transformed to the decimal number; see Access Control Event Types for details*/
    "startTime": "1970-01-01T00:00:00+08:00",
    /*opt, datetime, start time (UTC time)*/
    "endTime": "1970-01-01T00:00:00+08:00",
    /*opt, datetime, end time (UTC time)*/
    "cardNo": "test",
    /*opt, string, card No.*/
    "name": "test",
    /*opt, string, name of the card holder*/
    "videoChannel": 1,
    /*opt, int, video channel No., range:[1,86400], desc:this node is newly added to DeepinMind devices for attendance*/
    "picEnable": true,
    /*opt, bool, whether to upload the picture along with the event information, desc:false (no), true (yes, default value); (1. all matched events will
be uploaded without pictures; 2. all matched events will be uploaded with pictures if there are any; 3. if this node is not configured, the default value is
true)*/
    "beginSerialNo": 1,
    /*opt, int, start serial No.*/
    "endSerialNo": 1,
    /*opt, int, end serial No.*/
    "employeeNoString": "test",
    /*opt, string, employee No. (person ID)*/
    "timeReverseOrder": true,
    /*opt, bool, whether to return events in descending order of time (later events will be returned first), desc:true (yes), false or this node is not
returned (no)*/
    "isAbnormalTemperature": true,
    /*opt, bool, whether the skin-surface temperature is abnormal*/
    "temperatureSearchCond": "all",
    /*opt, enum, temperature search condition, subType:string, desc:when this node and isAbnormalTemperature both exist, isAbnormalTemperature is
invalid; "all" (event with temperature), "normal" (event with normal temperature), "abnormal" (event with abnormal temperature)*/
    "isAttendanceInfo": true,
    /*opt, bool, whether it contains attendance records, desc:this node is newly added to HEOP protocol; if this node is true, main type, minor type,
employee No., name, and time will be returned*/
    "hasRecordInfo": true
  }
}

```

Response Message

```

{
  "AcsEvent": {
    /*ro, req, object, access control events*/
    "searchID": "test",
    /*ro, req, string, search ID, it is used to check whether the current search requester is the same as the previous one. If they are the same, the
search record will be stored in the device to speed up the next search*/
    "responseStatusStrg": "OK",
    /*ro, req, string, searching status description*/
    "numOfMatches": 1,
    /*ro, req, int, number of results returned this time*/
    "totalMatches": 1,
    /*ro, req, int, total number of matched results*/
    "InfoList": [
      /*ro, opt, array, information list, subType:object*/
      {
        "major": 1,
        /*ro, req, int, major alarm type*/
        "minor": 1,
        /*ro, req, int, minor alarm type*/
        "time": "2016-12-12T17:30:08+08:00",
        /*ro, req, string, time (UTC time)*/
        "netUser": "test",
        /*ro, opt, string, user name*/
        "remoteHostAddr": "test",
        /*ro, opt, string, remote host address*/
        "videoChannel": 1,
        /*ro, opt, int, video channel No., range:[1,86400], desc:this node is newly added to DeepinMind devices for attendance*/
        "cardNo": "test",
        /*ro, opt, string, card No.*/
        "cardType": 1,
        /*ro, opt, enum, card type, subType:int, desc:1 (normal card), 2 (disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6
(super card), 7 (visitor card), 8 (dismiss card)*/
        "whiteListNo": 1,

```

```

/*ro, opt, int, allowlist No.*/
"reportChannel": 1,
/*ro, opt, int, channel type for uploading alarm/event*/
"cardReaderKind": 1,
/*ro, opt, int, card reader type: 1 (IC card reader)*/
"cardReaderNo": 1,
/*ro, opt, int, card reader No.*/
"doorNo": 1,
/*ro, opt, int, door or floor No.*/
"verifyNo": 1,
/*ro, opt, int, multi-factor authentication No.*/
"alarmInNo": 1,
/*ro, opt, int, alarm input No.*/
"alarmOutNo": 1,
/*ro, opt, int, alarm output No.*/
"caseSensorNo": 1,
/*ro, opt, int, event trigger No.*/
"RS485No": 1,
/*ro, opt, int, RS-485 channel No.*/
"multiCardGroupNo": 1,
/*ro, opt, int, group No.*/
"accessChannel": 1,
/*ro, opt, int, RS-485 channel No.*/
"deviceNo": 1,
/*ro, opt, int, device No.*/
"distractControlNo": 1,
/*ro, opt, int, distributed controller No.*/
"employeeNoString": "test",
/*ro, opt, string, employee No. (person ID)*/
"localControllerID": 1,
/*ro, opt, int, distributed controller No.*/
"InternetAccess": 1,
/*ro, opt, int, network inter-face No.*/
"type": 1,
/*ro, opt, int, zone type, desc:0 (instant alarm zone), 1 (24-hour zone), 2 (delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm
zone), 6 (perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone), 9 (24-hour shock zone), 10 (emergency door open zone), 11 (emergency door
closed zone), 255 (none)*/
"MACAddr": "test",
/*ro, opt, string, MAC address*/
"swipeCardType": 1,
/*ro, opt, enum, card swiping type, subType:int, desc:0 (invalid), 1 (QR code)*/
"serialNo": 1,
/*ro, opt, int, event serial No.*/
"channelControllerID": 1,
/*ro, opt, int, lane controller ID*/
"channelControllerLampID": 1,
/*ro, opt, int, light board ID of lane controller, range:[1,255]*/
"channelControllerIRAdaptorID": 1,
/*ro, opt, int, IR adapter ID of lane controller, range:[1,255]*/
"channelControllerIREmitterID": 1,
/*ro, opt, int, active infrared intrusion detector No. of lane controller, range:[1,255]*/
"userType": "normal",
/*ro, opt, string, person type*/
"currentVerifyMode": "cardAndPw",
/*ro, opt, enum, current authentication mode of the card reader, subType:string, desc:"cardAndPw" (card + password); "card", "cardOrPw"
(card or password), "fp" (fingerprint), "fpAndPw" (fingerprint + password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint + card),
"fpAndCardAndPw" (fingerprint + card + password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face + fingerprint),
"faceAndPw" (face + password), "faceAndCard" (face + card), "face", "employeeNoAndPw" (employee No. +password), "fpOrPw" (fingerprint or password),
"employeeNoAndFp" (employee No. + fingerprint), "employeeNoAndFpAndPw" (employee No. + fingerprint + password), "faceAndFpAndCard" (face + fingerprint +
card), "faceAndPwAndFp" (face + password + fingerprint), "employeeNoAndFace" (employee No. + face), "faceOrfaceAndCard" (face or face + card), "fpOrface"
(fingerprint or face), "cardOrfaceOrPw" (card or face or password), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris),
"faceOrCardOrPwOrIris" (face or card or password or iris), "sleep", "invalid"*/
"QRCodeInfo": "test",
/*ro, opt, string, QR code information*/
"thermometryUnit": "celsius",
/*ro, opt, enum, temperature unit, subType:string, desc:"celsius" (Celsius, default value), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin)*/
"currTemperature": 36.5,
/*ro, opt, float, skin-surface temperature, which is accurate to one decimal place*/
"isAbnormalTemperature": true,
/*ro, opt, bool, whether the skin-surface temperature is abnormal (true=yes)*/
"RegionCoordinates": {
/*ro, opt, object, coordinates of the skin-surface temperature*/
"positionX": 254,
/*ro, opt, int, normalized X-coordinate which is between 0 and 1000*/
"positionY": 133,
/*ro, opt, int, normalized Y-coordinate which is between 0 and 1000*/
},
"mask": "unknown",
/*ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown"*/
"pictureURL": "test",
/*ro, opt, string, picture URL*/
"filename": "picture1",
/*ro, opt, string, file name, desc:if multiple pictures are returned at a time, filename of each picture should be unique*/
"attendanceStatus": "undefined",
/*ro, opt, enum, attendance status, subType:string, desc:"undefined", "checkIn" (check-in), "checkOut" (check-out), "breakOut" (start of
break), "breakIn" (end of break), "overtimeIn" (start of overtime), "overTimeOut" (end of overtime)*/
"label": "test",
/*ro, opt, string, custom attendance name*/
"statusValue": 1,
/*ro, opt, int, status value*/
"helmet": "unknown",
/*ro, opt, enum, whether the person wears a hard hat, subType:string, desc:"unknown", "yes", "no"*/
"visibleLightPicUrl": "test",
/*ro, opt, string, visible light picture URL*/

```



```

/*ro, opt, string, visible light picture url*/
"thermalPicUrl": "test",
/*ro, opt, string, URL of the thermal imaging picture*/
"appType": "attendance",
/*ro, opt, enum, application type, subType:string, desc:"attendance" (Time & Attendance module), "signIn" (Check-In module, which is only
used for FocSign products)*/
"HealthInfo": {
/*ro, opt, object, health information*/
"healthCode": 1,
/*ro, opt, enum, health code status, subType:int, desc:0 (no request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red
QR code), 5 (no such person), 6 (other error, e.g., searching failed due to API exception), 7 (searching for the health code timed out)*/
"NADCode": 1,
/*ro, opt, enum, nucleic acid test result, subType:int, desc:0 (no result), 1 (negative, which means normal), 2 (positive, which means
diagnosed), 3 (the result has expired)*/
"travelCode": 1,
/*ro, opt, enum, trip code, subType:int, desc:0 (no trip in the past 14 days), 1 (has left the current area in the past 14 days), 2 (has
been to the high-risk area in the past 14 days), 3 (other)*/
"travelInfo": "test",
/*ro, opt, string*/
"vaccineStatus": 1,
/*ro, opt, enum, whether the person is vaccinated, subType:int, desc:0 (not vaccinated), 1 (vaccinated)*/
"vaccineNum": 1
/*ro, opt, int, step:1*/
},
"meetingID": "test",
/*ro, opt, string, meeting ID*/
"PersonInfoExtends": [
/*ro, opt, array, additional person information, subType:object, desc:this node displays additional person information on the device*/
{
"id": 1,
/*ro, opt, int, extended ID of the additional person information, range:[1,32], desc:related URL:
/ISAPI/AccessControl/personInfoExtendName?format=json; this node is used for displaying the name of value; if ID does not exists, it starts from 1*/
"value": "test"
/*ro, opt, string, extended content of the additional person information*/
}
],
"name": "test",
/*ro, opt, string, name, desc:person name*/
"FaceRect": {
/*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
"height": 1.000,
/*ro, req, float, height, range:[0.000,1.000]*/
"width": 1.000,
/*ro, req, float, width, range:[0.000,1.000]*/
"x": 0.000,
/*ro, req, float, X-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
"y": 0.000
/*ro, req, float, Y-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
},
"RecordInfo": {
/*ro, opt, object*/
"startTime": "1970-01-01T00:00:00+08:00",
/*ro, opt, datetime, recording start time*/
"endTime": "1970-01-01T00:00:00+08:00",
/*ro, opt, datetime, recording end time*/
"playbackURL": "rtsp://10.65.130.168:554/ISAPI/Streaming/tracks/201/?starttime=20190213T091134Z&endtime=20190213T092116Z"
/*ro, opt, string, range:[0,256]*/
},
"currentAuthenticationTimes": 1,
/*ro, opt, int, range:[0,255], step:1*/
"allowAuthenticationTimes": 1
/*ro, opt, int, range:[0,255], step:1*/
}
]
}
}

```

12.8.1.3 Get the capability of getting total number of access control events by specific conditions

Request URL

GET /ISAPI/AccessControl/AcsEventTotalNum/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "AcsEvent": {
    /*ro, opt, object*/
    "AcsEventTotalNumCond": {
      /*ro, opt, object, search conditions*/
      "major": {

```

```

/*ro, req, object, major alarm type*/
"@opt": "0,1,2,3,5"
/*ro, opt, string, major alarm type*/
},
"minorAlarm": {
/*ro, req, object, minor alarm type*/
"@opt": "1024,1025,1026,1027"
/*ro, opt, string, minor alarm type*/
},
"minorException": {
/*ro, req, object, minor exception type*/
"@opt": "39,58,59,1024"
/*ro, opt, string, minor exception type*/
},
"minorOperation": {
/*ro, req, object, minor operation type*/
"@opt": "80,90,112,113"
/*ro, opt, string, minor operation type*/
},
"minorEvent": {
/*ro, opt, object, minor event type*/
"@opt": "1,2,3,4"
/*ro, opt, string, minor event type*/
},
"startTime": {
/*ro, opt, object, start time*/
"@min": 1,
/*ro, opt, int, start time (UTC time)*/
"@max": 1
/*ro, opt, int, end time (UTC time)*/
},
"endTime": {
/*ro, opt, object, end time*/
"@min": 1,
/*ro, opt, int, start time (UTC time)*/
"@max": 1
/*ro, opt, int, end time (UTC time)*/
},
"cardNo": {
/*ro, opt, object, card No.*/
"@min": 1,
/*ro, opt, int, card No.*/
"@max": 32
/*ro, opt, int, card No.*/
},
"name": {
/*ro, opt, object, name of the card holder*/
"@min": 1,
/*ro, opt, int, name of the card holder*/
"@max": 32
/*ro, opt, int, name of the card holder*/
},
"picEnable": "true,false",
/*ro, opt, string*/
"beginSerialNo": {
/*ro, opt, object, start serial No.*/
"@min": 1,
/*ro, opt, int, start serial No.*/
"@max": 1
/*ro, opt, int, start serial No.*/
},
"endSerialNo": {
/*ro, opt, object, end serial No.*/
"@min": 1,
/*ro, opt, int, end serial No.*/
"@max": 1
/*ro, opt, int, end serial No.*/
},
"employeeNoString": {
/*ro, opt, object, employee No. (person ID)*/
"@min": 1,
/*ro, opt, int, employee No. (person ID)*/
"@max": 32
/*ro, opt, int, employee No. (person ID)*/
}
},
"totalNum": {
/*ro, req, object*/
"@min": 1,
/*ro, opt, int*/
"@max": 1
/*ro, opt, int*/
}
}
}

```

12.8.1.4 Get the total number of access control events by specific conditions

Request URL

POST /ISAPI/AccessControl/AcsEventTotalNum?format=json

Query Parameter

None

Request Message

```
{
  "AcsEventTotalNumCond": {
    /*req, object*/
    "major": 1,
    /*req, int, major alarm type, desc:(the type value should be transformed to the decimal number), refer to Access Control Event Types for details*/
    "minor": 1024,
    /*req, int, sub type, step:1, desc:(the type value should be transformed to the decimal number),refer to Access Control Event Types for details*/
    "startTime": "1970-01-01+08:00",
    /*opt, date, start time (UTC time)*/
    "endTime": "1970-01-01+08:00",
    /*opt, date, end time (UTC time)*/
    "cardNo": "test",
    /*opt, string, card No.*/
    "name": "test",
    /*opt, string, name of the card holder*/
    "picEnable": true,
    /*opt, bool, whether to upload the picture along with the event information, desc:whether to contain pictures: "true"-yes,"false"-no*/
    "beginSerialNo": 1,
    /*opt, int, start serial No.*/
    "endSerialNo": 100,
    /*opt, int, end serial No.*/
    "employeeNoString": "test"
    /*opt, string, employee No. (person ID), range:[1,32]*/
  }
}
```

Response Message

```
{
  "AcsEventTotalNum": {
    /*ro, req, object*/
    "totalNum": 1,
    /*ro, req, int, total number of events that match the search conditions*/
    "existedEventNum": 1
    /*ro, opt, int*/
  }
}
```

12.8.1.5 Getting arming information

Request URL

GET /ISAPI/AccessControl/DeployInfo

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeployInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming information, attr:version{req, string, protocolVersion}-->
  <DeployList size="5">
    <!--ro, opt, array, arming list, subType:object, attr:size{req, int}-->
    <Content>
      <!--ro, opt, object, subscribe to messages-->
      <deployNo>
        <!--ro, req, int, arming No.-->1
      </deployNo>
      <deployType>
        <!--ro, req, enum, arming type, subType:int-->1
      </deployType>
      <protocolType>
        <!--ro, opt, enum, protocol type, subType:string, dep:or,{$.DeployInfo.DeployList[*].Content.deployType,eq,2},
        {$.DeployInfo.DeployList[*].Content.deployType,eq,3}, desc:"HTTP", "HTTPS"-->HTTP
      </protocolType>
      <ipAddr>
        <!--ro, req, string, IP address-->test
      </ipAddr>
      <port>
        <!--ro, opt, int, port No., range:[1,65535]-->1
      </port>
      <eventType>
        <!--ro, opt, enum, subType:string-->AccessController
      </eventType>
    </Content>
  </DeployList>
</DeployInfo>
```

12.8.1.6 Getting arming information capability

Request URL

GET /ISAPI/AccessControl/DeployInfo/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeployInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming information, attr:version{req, string, protocolVersion}-->
  <DeployList size="5">
    <!--ro, opt, array, arming list, subType:object, attr:size{req, int}-->
    <Content>
      <!--ro, opt, object-->
      <deployNo min="1" max="10">
        <!--ro, req, int, arming No., attr:min{req, int},max{req, int}-->1
      </deployNo>
      <deployType opt="0,1,2,3">
        <!--ro, req, int, arming type, attr:opt{req, string}-->1
      </deployType>
      <protocolType opt="HTTP,HTTPS">
        <!--ro, opt, enum, protocol type, subType:string, attr:opt{req, string}, desc:"HTTP", "HTTPS"-->HTTP
      </protocolType>
      <ipAddr min="1" max="10">
        <!--ro, req, string, IP address, attr:min{req, int},max{req, int}-->test
      </ipAddr>
      <port min="0" max="10">
        <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}-->1
      </port>
      <eventType opt="AccessController,Consumer,AccessControllerAndConsumer">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->AccessController
      </eventType>
    </Content>
  </DeployList>
</DeployInfo>
```

12.8.1.7 Access control event

EventType:AccessControllerEvent

```
{
  "ipAddress": "172.6.64.7",
  /*...*/
}
```

```

/*ro, req, string, IPv4 address of the device that triggers the alarm*/
"ipv6Address": "1080:0:0:8:800:200C:417A",
/*ro, opt, string, IPv6 address of the device that triggers the alarm*/
"portNo": 80,
/*ro, opt, int, communication port No. of the device that triggers the alarm*/
"protocol": "HTTP",
/*ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetsDK, the channel No. is the
video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
"macAddress": "01:17:24:45:D9:F4",
/*ro, opt, string, MAC address*/
"channelID": 1,
/*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetsDK, the channel No. is the video
channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI
protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
"dateTime": "2004-05-03T17:30:08+08:00",
/*ro, req, datetime, alarm trigger time*/
"activePostCount": 1,
/*ro, req, int, times that the same alarm has been uploaded, desc:times that the same alarm has been uploaded*/
"eventType": "AccessControllerEvent",
/*ro, req, string, event type, desc:"AccessControllerEvent" (access control event)*/
"eventState": "active",
/*ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid event or event starts), "inactive" (invalid event or the event
ends). For the heartbeat, the node value indicates the heartbeat data, and it is uploaded every 10 seconds*/
"eventDescription": "AccessControllerEvent",
/*ro, req, string, event description, desc:"AccessControllerEvent" (access control event)*/
"deviceId": "test0123",
/*ro, opt, string, device ID (PUID), desc:this node must be returned when ISAPI event information is transmitted via ISUP*/
"AccessControllerEvent": {
/*ro, req, object, access control event information*/
"deviceName": "test",
/*ro, opt, string, device name*/
"majorEventType": 1,
/*ro, req, int, major alarm type, desc:the type value should be transformed to the decimal number; see Access Control Alarm Types for details*/
"subEventType": 1,
/*ro, req, int, minor alarm type, desc:the type value should be transformed to the decimal number; see Access Control Alarm Types for details*/
"inductiveEventType": "authenticated",
/*ro, opt, enum, inductive event type, subType:string, desc:this node is used by storage devices; for access control devices, this node is invalid;
"authenticated", "authenticationFailed", "openingDoor", "closingDoor", "doorException", "remoteOperation", "timeSynchronization", "deviceException",
"deviceRecovered", "alarmTriggered", "alarmRecovered" (arming restoring event), "callCenter"*/
"netUser": "test",
/*ro, opt, string, user name for network operations*/
"remoteHostAddr": "test",
/*ro, opt, string, remote host address*/
"cardNo": "test",
/*ro, opt, string, card No.*/
"cardType": 1,
/*ro, opt, enum, card type, subType:int, desc:1 (normal card), 2 (disability card), 3 (blocklist card), 4 (patrol card), 5 (duress card), 6 (super
card), 7 (visitor card), 8 (dismiss card)*/
"name": "test",
/*ro, opt, string, person name*/
"sex": "male",
/*ro, opt, enum, subType:string, desc:"male", "female"*/
"whitelistNo": 1,
/*ro, opt, int, allowlist No.*/
"reportChannel": 1,
/*ro, opt, enum, channel type for uploading alarms/events, subType:int, desc:1 (uploading in arming mode), 2 (uploading by central group 1), 3
(uploading by central group 2)*/
"cardReaderKind": 1,
/*ro, opt, enum, card reader type, subType:int, desc:1 (IC card reader), 2 (ID card reader), 3 (QR code scanner), 4 (fingerprint module)*/
"cardReaderNo": 1,
/*ro, opt, int, card reader No., step:1, desc:card reader No.*/
"doorNo": 1,
/*ro, opt, int, door (floor) No.*/
"verifyNo": 1,
/*ro, opt, int, multiple authentication No.*/
"alarmInNo": 1,
/*ro, opt, int, alarm input No.*/
"alarmOutNo": 1,
/*ro, opt, int, alarm output No.*/
"caseSensorNo": 1,
/*ro, opt, int, event trigger No.*/
"RS485No": 1,
/*ro, opt, int, RS-485 channel No.*/
"multiCardGroupNo": 1,
/*ro, opt, int, group No.*/
"accessChannel": 1,
/*ro, opt, int, turnstile No.*/
"deviceNo": 1,
/*ro, opt, int, device No.*/
"distractControlNo": 1,
/*ro, opt, int, distributed access controller No.*/
"employeeNo": 1,
/*ro, opt, int, employee No. (person ID)*/
"employeeNoString": "test",
/*ro, opt, string, employee No. (person ID), desc:if the node employeeNo exists or the value of employeeNoString can be converted to that of
employeeNo, this node is required. For the upper-layer platform or client software, the node employeeNoString will be parsed in priority; if
employeeNoString is not configured, the node employeeNo will be parsed*/
"employeeName": "test",
/*ro, opt, string, person name, desc:this node is only used for FocSign products*/
"localControllerID": 1,
/*ro, opt, int, distributed access controller No., desc:0 (access controller), 1 to 64 (distributed access controller No. 1 to distributed access
controller No. 64)*/
"InternetAccess": 1,

```

```

/*ro, opt, enum, network interface No., subType:int, desc:1 (upstream network interface No. 1), 2 (upstream network interface No. 2), 3 (downstream
network interface No. 1)*/
"type": 1,
/*ro, opt, enum, zone type, subType:int, desc:0 (instant zone), 1 (24-hour zone), 2 (delayed zone), 3 (internal zone), 4 (key zone), 5 (fire alarm
zone), 6 (perimeter zone), 7 (24-hour silent zone), 8 (24-hour auxiliary zone), 9 (24-hour shock zone), 10 (emergency door open zone), 11 (emergency door
closed zone), 255 (none)*/
"MACAddr": "test",
/*ro, opt, string, MAC address*/
"swipeCardType": 1,
/*ro, opt, enum, card swiping types, subType:int, desc:0 (invalid), 1 (QR code)*/
"serialNo": 1,
/*ro, opt, int, event serial No., range:[1,100000], desc:it starts at 1 and each record increases by 1. It will be overwritten repeatedly when
reaching the maximum value supported by the device*/
"channelControllerID": 1,
/*ro, opt, enum, lane controller ID, subType:int, desc:1 (main lane controller), 2 (sub-lane controller)*/
"channelControllerLampID": 1,
/*ro, opt, int, light board ID of lane controller, range:[1,255]*/
"channelControllerIRAdaptorID": 1,
/*ro, opt, int, IR adaptor ID of the lane controller, range:[1,255]*/
"channelControllerIREmitterID": 1,
/*ro, opt, int, active infrared intrusion detector No. of the lane controller, range:[1,255]*/
"userType": "normal",
/*ro, opt, enum, person type, subType:string, desc:"normal" (normal person (resident)), "visitor" (visitor), "blacklist" (person in the blacklist),
"administrators" (administrators)*/
"currentVerifyMode": "cardAndPw",
/*ro, opt, enum, current authentication mode of the card reader, subType:string, desc:"cardAndPw" (card+password), "card" (card), "cardOrPw" (card
or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw"
(fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password),
"faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee
No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp"
(face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face),
"cardOrfaceOrPw" (card or face or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face, fingerprint, card, password, or iris), "faceOrCardOrPwOrIris"
(face, card, password, or iris)*/
"currentEvent": true,
/*ro, opt, bool, whether it is a real-time event*/
"QRCodeInfo": "test",
/*ro, opt, string, QR code information*/
"thermometryResult": "success",
/*ro, opt, enum, temperature screening result, subType:string, desc:"success", "fail"*/
"thermometryUnit": "celsius",
/*ro, opt, enum, temperature unit, subType:string, desc:"celsius" (Celsius, default value), "fahrenheit" (Fahrenheit), "kelvin" (Kelvin)*/
"currTemperature": 36.1,
/*ro, opt, float, skin-surface temperature, which is accurate to one decimal place*/
"isAbnormalTemperature": true,
/*ro, opt, bool, whether the skin-surface temperature is abnormal*/
"RegionCoordinates": {
/*ro, opt, object, coordinates of the skin-surface temperature*/
"positionX": 0,
/*ro, opt, int, normalized X-coordinate which is between 0 and 1000, range:[0,1000]*/
"positionY": 0
/*ro, opt, int, normalized Y-coordinate which is between 0 and 1000, range:[0,1000]*/
},
"remoteCheck": true,
/*ro, opt, bool, whether remote verification is required: true=yes, false=no (default)*/
"mask": "unknown",
/*ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown", "yes", "no"*/
"frontSerialNo": 1,
/*ro, opt, int, serial No. of the previous event, desc:if this node does not exist, the platform will check whether the event loss occurred
according to the node serialNo. If both the serialNo and frontSerialNo are returned, the platform will check whether the event loss occurred according to
both nodes. It is mainly used to solve the problem that the serialNo is inconsistent after subscribing events or alarms*/
"attendanceStatus": "checkIn",
/*ro, opt, enum, attendance status, subType:string, desc:"checkIn" (check-in), "checkOut" (check-out), "breakOut" (start of break), "breakIn" (end
of break), "overTimeIn" (start of overtime), "overTimeOut" (end of overtime)*/
"label": "test",
/*ro, opt, string, self-defined attendance name*/
"statusValue": 1,
/*ro, opt, int, status value*/
"pictureURL": "test",
/*ro, opt, string, URL of the captured picture, range:[0,256]*/
"visibleLightURL": "test",
/*ro, opt, string, visible light picture URL of the thermal imaging camera, range:[0,256]*/
"thermalURL": "test",
/*ro, opt, string, URL of the thermal picture, range:[0,256]*/
"faceBasemapURL": "test",
/*ro, opt, string, range:[0,256]*/
"picturesNumber": 1,
/*ro, opt, int, number of captured pictures*/
"unlockType": "password",
/*ro, opt, enum, unlocking type, subType:string, desc:this node is returned when the minor type is MINOR_UNLOCK_RECORD; "password" (unlock by
password), "hijacking" (unlock by duress), "card" (unlock by card), "householder" (unlock by householder), "centerplatform" (unlock by management center),
"bluetooth" (unlock by bluetooth), "qrcode" (unlocked via QR code), "face" (unlock by recognizing face), "fingerprint" (unlock by fingerprint)*/
"classroomId": "test",
/*ro, opt, string, class ID*/
"classroomName": "test",
/*ro, opt, string, class name*/
"analysisModule": "signageApp",
/*ro, opt, enum, analysis module, subType:string, desc:this node is not returned, and the value is report via signage App; "signageApp" (signage
App), "faceSDK" (face picture SDK)*/
"customInfo": "test",
/*ro, opt, string, custom information*/
"helmet": "unknown",
/*ro, opt, enum, whether the person is wearing hard hat, subType:string, desc:"unknown", "yes", "no"*/
"purePwVerifyEnable": true,

```

```

/*ro, opt, bool, whether the device supports opening the door only by password,
desc:opening the door only by password:
the password in authentication method is person password; checking the repetition of person password is not supported by the device, it should be
performed by the upper-layer platform; adding, deleting, editing, and searching for person password locally is not supported by the device*/
"appType": "attendance",
/*ro, opt, enum, application type (for FocSign products), subType:string, desc:"attendance" (Time & Attendance module), "signIn" (Check-In module)*/
"HealthInfo": {
/*ro, opt, object, health information*/
"healthCode": 1,
/*ro, opt, enum, health code status, subType:int, desc:0 (no request), 1 (no health code), 2 (green QR code), 3 (yellow QR code), 4 (red QR
code), 5 (no such person), 6 (other error, e.g., searching failed due to API exception), 7 (searching for the health code timed out)*/
"NADCode": 1,
/*ro, opt, enum, nucleic acid test result, subType:int, desc:0 (no result), 1 (negative, which means normal), 2 (positive, which means
diagnosed), 3 (the result has expired)*/
"NADMsg": "test",
/*ro, opt, string, range:[0,64]*/
"NADTime": 1,
/*ro, opt, enum, subType:int*/
"travelCode": 1,
/*ro, opt, enum, trip code, subType:int, desc:0 (no trip in the past 14 days), 1 (has left the current area left in the past 14 days), 2 (has
been to the high-risk area in the past 14 days), 3 (other)*/
"travelInfo": "test",
/*ro, opt, string, trip information, desc:the empty string indicates that searching trip failed*/
"vaccineStatus": 1,
/*ro, opt, enum, whether the person is vaccinated, subType:int, desc:0 (not vaccinated), 1 (vaccinated)*/
"vaccineNum": 1,
/*ro, opt, int, step:1*/
"vaccineMsg": "test",
/*ro, opt, string, range:[0,64]*/
"ANTCode": 1,
/*ro, opt, enum, subType:int*/
"ANTMsg": "test"
/*ro, opt, string, range:[0,64]*/
},
"PhysicalInfo": {
/*ro, opt, object, BMI information, desc:this node is obtained after authentication by BMI scales which is connected to MinMoe terminals*/
"weight": 7000,
/*ro, opt, int, weight, unit:kg*/
"height": 18000
/*ro, opt, int, height, unit:cm*/
},
"meetingID": "test",
/*ro, opt, string, meeting ID, range:[1,32]*/
"PersonInfoExtends": [
/*ro, opt, array, additional person information, subType:object, desc:this node displays additional person information on the device*/
{
"id": 1,
/*ro, opt, int, extended ID of the additional person information, range:[1,32], desc:related URL: /ISAPI/AccessControl/personInfoExtendName?
format=json; this node is used for displaying the name of value; if ID does not exists, it starts from 1*/
"value": "test"
/*ro, opt, string, extended content of the additional person information*/
}
],
"customPrompt": "test",
/*ro, opt, string, custom prompt message, range:[1,128], desc:this node is displayed when the authentication result is authenticated, authentication
failed, or stranger*/
"FaceRect": {
/*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
"height": 1.000,
/*ro, req, float, height, range:[0.000,1.000]*/
"width": 1.000,
/*ro, req, float, width, range:[0.000,1.000]*/
"x": 0.000,
/*ro, req, float, X-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
"y": 0.000
/*ro, req, float, Y-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
},
"faceSimilarity": 90,
/*ro, opt, int, Similarity, range:[0,100]*/
"faceRecognitionDistance": 0.1,
/*ro, opt, float, unit:m*/
"eyesDistance": 20,
/*ro, opt, int, range:[0,100], step:1*/
"faceRecognitionFailedReason": "attackBlacklist",
/*ro, opt, enum, subType:string*/
"currentAuthenticationTimes": 1,
/*ro, opt, int, range:[0,255], step:1*/
"allowAuthenticationTimes": 1,
/*ro, opt, int, range:[0,255], step:1*/
"LocalAttendanceData": {
/*ro, opt, object*/
"attendanceResult": [
/*ro, opt, array, subType:object*/
{
"date": "1970-01-01",
/*ro, opt, date*/
"week": 1,
/*ro, opt, enum, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
"personalAttendanceStatus": "normal"
/*ro, opt, enum, subType:string*/
}
]
},
"hasRecord": true

```

```

/*ro, opt, bool*/
},
"URLCertificationType": "digest"
/*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (no authentication, it is used for the cloud protocol), "digest" (digest authentication, it is used for local picture URL returned by NVR or DVR)*/
}

```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
AccessControllerEvent	[Message content]	application/json	--	--	--
Picture	[Binary picture data]	image/jpeg	pictureImage	Picture.jpg	--
VisibleLight	[Binary picture data]	image/jpeg	visibleLight_image	VisibleLight.jpg	--
Thermal	[Binary picture data]	image/jpeg	thermal_image	Thermal.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```

--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value

```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

12.8.2 Access Control System Management

12.8.2.1 Get the functional capability of access control

Request URL

GET /ISAPI/AccessControl/capabilities

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>
<AccessControl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the functional capability of access control, attr:version{req, string, protocolVersion}-->
  <isSupportWiegandCfg>
    <!--ro, opt, bool, whether it supports Wiegand configuration, desc:true (support), this node is not returned (not support)-->true
  </isSupportWiegandCfg>
  <isSupportModuleStatus>
    <!--ro, opt, bool, whether it supports getting the status of secure door control unit, desc:true (support), this node is not returned (not support)-->true
  </isSupportModuleStatus>
  <isSupportSNAPConfig>
    <!--ro, opt, bool, whether it supports getting capture Linkage parameters, desc:true (support), this node is not returned (not support)-->true
  </isSupportSNAPConfig>

```



```

<LocalController>
  <!--ro, opt, object-->
  <isSupportLocalControllerManage>
    <!--ro, opt, bool, whether it supports distributed access controller management, desc:true (support), this node is not returned (not support)-->true
  </isSupportLocalControllerManage>
  <isSupportLocalControllerControl>
    <!--ro, opt, bool, whether it supports distributed access controller control, desc:true (support), this node is not returned (not support)-->true
  </isSupportLocalControllerControl>
</LocalController>
<isSupportUSBManage>
  <!--ro, opt, bool, whether it supports USB management of access control device, desc:true (support), this node is not returned (not support)-->true
</isSupportUSBManage>
<isSupportLadderControlRelay>
  <!--ro, opt, bool, whether it supports setting elevator controller relay, desc:true (support), this node is not returned (not support)-->true
</isSupportLadderControlRelay>
<isSupportIdentityTerminal>
  <!--ro, opt, bool, whether it supports face recognition terminal configuration, desc:true (support), this node is not returned (not support)-->true
</isSupportIdentityTerminal>
<isSupportDepartmentParam>
  <!--ro, opt, bool, whether it supports setting department parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportDepartmentParam>
<isSupportSchedulePlan>
  <!--ro, opt, bool, whether it supports setting shift schedule, desc:true (support), this node is not returned (not support)-->true
</isSupportSchedulePlan>
<isSupportAttendanceRule>
  <!--ro, opt, bool, Attendance Rule Configuration, desc:true (support), this node is not returned (not support)-->true
</isSupportAttendanceRule>
<isSupportOrdinaryClass>
  <!--ro, opt, bool, whether it supports setting normal shift parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportOrdinaryClass>
<isSupportWorkingClass>
  <!--ro, opt, bool, whether it supports setting man-hour shift parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportWorkingClass>
<isSupportAttendanceHolidayGroup>
  <!--ro, opt, bool, whether it supports setting holiday group for time and attendance, desc:true (support), this node is not returned (not support)-->true
</isSupportAttendanceHolidayGroup>
<isSupportAttendanceHolidayPlan>
  <!--ro, opt, bool, whether it supports setting holiday schedule for time and attendance, desc:true (support), this node is not returned (not support)-->true
</isSupportAttendanceHolidayPlan>
<isSupportWiegandRuleCfg>
  <!--ro, opt, bool, whether it supports setting Wiegand rule, desc:true (support), this node is not returned (not support)-->true
</isSupportWiegandRuleCfg>
<isSupportM1CardEncryptCfg>
  <!--ro, opt, bool, configuration capability of the M1 card encryption verification, desc:true (support), this node is not returned (not support)-->true
</isSupportM1CardEncryptCfg>
<isSupportDeployInfo>
  <!--ro, opt, bool, whether it supports getting arming information, desc:whether it supports getting arming information-->true
</isSupportDeployInfo>
<isSupportSubmarineBack>
  <!--ro, opt, bool, whether it supports specifying anti-passing back server, desc:true (support), this node is not returned (not support)-->true
</isSupportSubmarineBack>
<isSupportSubmarineBackHostInfo>
  <!--ro, opt, bool, whether it supports setting access controllers with anti-passing back enabled, desc:true (support), this node is not returned (not support)-->true
</isSupportSubmarineBackHostInfo>
<isSupportStartReaderInfo>
  <!--ro, opt, bool, whether it supports setting first card reader, desc:true (support), this node is not returned (not support)-->true
</isSupportStartReaderInfo>
<isSupportSubmarineBackReader>
  <!--ro, opt, bool, whether it supports setting anti-passing back card reader, desc:true (support), this node is not returned (not support)-->true
</isSupportSubmarineBackReader>
<isSupportServerDevice>
  <!--ro, opt, bool, whether it supports setting anti-passing back server information, desc:true (support), this node is not returned (not support)-->true
</isSupportServerDevice>
<isSupportReaderAcrossHost>
  <!--ro, opt, bool, whether it supports enabling cross-controller anti-passing back function of card reader, desc:true (support), this node is not returned (not support)-->true
</isSupportReaderAcrossHost>
<isSupportClearCardRecord>
  <!--ro, opt, bool, whether it supports clearing card swiping records in anti-passing back server, desc:true (support), this node is not returned (not support)-->true
</isSupportClearCardRecord>
<isSupportSubmarineBackMode>
  <!--ro, opt, bool, whether it supports setting anti-passing back mode, desc:true (support), this node is not returned (not support)-->true
</isSupportSubmarineBackMode>
<isSupportClearSubmarineBack>
  <!--ro, opt, bool, whether it supports clearing cross-controller anti-passing back information, desc:true (support), this node is not returned (not support)-->true
</isSupportClearSubmarineBack>
<isSupportSupplementLight>
  <!--ro, opt, bool-->true
</isSupportSupplementLight>
<isSupportFaceCompareCond>
  <!--ro, opt, bool, whether it supports configuring restriction condition parameters of face picture comparison, desc:whether it supports configuring restriction condition parameters of face picture comparison-->true
</isSupportFaceCompareCond>
<isSupportRightControllerAudio>
  <!--ro, opt, bool-->true
</isSupportRightControllerAudio>
<isSupportChannelControllerCfg>
  <!--ro, opt, bool-->true
</isSupportChannelControllerCfg>

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<isSupportGateDialAndInfo>
  <!--ro, opt, bool-->true
</isSupportGateDialAndInfo>
<isSupportGateStatus>
  <!--ro, opt, bool-->true
</isSupportGateStatus>
<isSupportGateIRStatus>
  <!--ro, opt, bool-->true
</isSupportGateIRStatus>
<isSupportGateRelatedPartsStatus>
  <!--ro, opt, bool-->true
</isSupportGateRelatedPartsStatus>
<isSupportChannelControllerAlarmLinkage>
  <!--ro, opt, bool-->true
</isSupportChannelControllerAlarmLinkage>
<isSupportChannelControllerAlarmOut>
  <!--ro, opt, bool-->true
</isSupportChannelControllerAlarmOut>
<isSupportChannelControllerAlarmOutControl>
  <!--ro, opt, bool-->true
</isSupportChannelControllerAlarmOutControl>
<isSupportChannelControllerTypeCfg>
  <!--ro, opt, bool-->true
</isSupportChannelControllerTypeCfg>
<isSupportRemoteCtrllerModeCfg>
  <!--ro, opt, bool-->true
</isSupportRemoteCtrllerModeCfg>
<isSupportRemoteControlDoor>
  <!--ro, opt, bool, whether it supports remote door,elevator,and Lock contro, desc:true (support), this node is not returned (not support)-->true
</isSupportRemoteControlDoor>
<isSupportUserInfo>
  <!--ro, opt, bool, whether it supports person management based on person, desc:true (support), this node is not returned (not support)-->true
</isSupportUserInfo>
<EmployeeNoInfo>
  <!--ro, opt, object, employee No. (person ID) information, desc:this node is valid only when the isSupportUserInfo is "true"-->
  <employeeNo min="1" max="32">
    <!--ro, opt, int, employee No. (person ID), attr:min{req, int},max{req, int}-->1
  </employeeNo>
  <characterType opt="any,number">
    <!--ro, opt, enum, employee No. (person) ID type, subType:string, attr:opt{req, string}, desc:"any"-any characters (default), "number"-digits (from 0
to 9), only one value can be returned-->any
  </characterType>
  <isSupportCompress>
    <!--ro, opt, bool, whether it supports compressing employee No. (person ID) for storage, desc:true (support), this node is not returned (not support)-->true
  </isSupportCompress>
</EmployeeNoInfo>
<isSupportCardInfo>
  <!--ro, opt, bool, whether it supports card management based on person, desc:true (support), this node is not returned (not support)-->true
</isSupportCardInfo>
<isSupportFDLib>
  <!--ro, opt, bool, whether it supports face picture Library management-->true
</isSupportFDLib>
<isSupportUserInfoDetailDelete>
  <!--ro, opt, bool, whether it supports deleting person information and permission, desc:true (support), this node is not returned (not support)-->true
</isSupportUserInfoDetailDelete>
<isSupportAuthCodeInfo>
  <!--ro, opt, bool, whether it supports authentication password management, desc:true (support), this node is not returned (not support)-->true
</isSupportAuthCodeInfo>
<isSupportFingerPrintCfg>
  <!--ro, opt, bool, whether it supports configuring fingerprint parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportFingerPrintCfg>
<isSupportFingerPrintDelete>
  <!--ro, opt, bool, whether it supports deleting fingerprints, desc:true (support), this node is not returned (not support)-->true
</isSupportFingerPrintDelete>
<isSupportCaptureFingerPrint>
  <!--ro, opt, bool, whether it supports collecting fingerprint information, desc:true (support), this node is not returned (not support)-->true
</isSupportCaptureFingerPrint>
<isSupportDoorStatusWeekPlanCfg>
  <!--ro, opt, bool, whether it supports configuring door control week schedule, desc:true (support), this node is not returned (not support)-->true
</isSupportDoorStatusWeekPlanCfg>
<isSupportVerifyWeekPlanCfg>
  <!--ro, opt, bool, whether it supports configuring week schedule of the card reader authentication mode, desc:true (support), this node is not returned
(not support)-->true
</isSupportVerifyWeekPlanCfg>
<isSupportCardRightWeekPlanCfg>
  <!--ro, opt, bool, whether it supports configuring week schedule of the access permission control, desc:true (support), this node is not returned (not
support)-->true
</isSupportCardRightWeekPlanCfg>
<isSupportDoorStatusHolidayPlanCfg>
  <!--ro, opt, bool, whether it supports configuring door control holiday schedule, desc:true (support), this node is not returned (not support)-->true
</isSupportDoorStatusHolidayPlanCfg>
<isSupportVerifyHolidayPlanCfg>
  <!--ro, opt, bool, whether it supports configuring holiday schedule of the card reader authentication mode, desc:true (support), this node is not
returned (not support)-->true
</isSupportVerifyHolidayPlanCfg>
<isSupportCardRightHolidayPlanCfg>
  <!--ro, opt, bool, whether it supports configuring holiday schedule of the access permission control, desc:true (support), this node is not returned
(not support)-->true
</isSupportCardRightHolidayPlanCfg>
<isSupportDoorStatusHolidayGroupCfg>
  <!--ro, opt, bool, whether it supports configuring holiday group of the door control schedule, desc:true (support), this node is not returned (not
support)-->true

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</isSupportDoorStatusHolidayGroupCfg>
<isSupportVerifyHolidayGroupCfg>
    <!--ro, opt, bool, whether it supports configuring holiday group of the control schedule of the card reader authentication mode, desc:true (support),
this node is not returned (not support)-->true
</isSupportVerifyHolidayGroupCfg>
<isSupportUserRightHolidayGroupCfg>
    <!--ro, opt, bool, whether it supports configuring holiday group of the access permission control schedule, desc:true (support), this node is not
returned (not support)-->true
</isSupportUserRightHolidayGroupCfg>
<isSupportDoorStatusPlanTemplate>
    <!--ro, opt, bool, whether it supports configuring door control schedule template, desc:true (support), this node is not returned (not support)-->true
</isSupportDoorStatusPlanTemplate>
<isSupportVerifyPlanTemplate>
    <!--ro, opt, bool, whether it supports configuring schedule template of the card reader authentication mode, desc:true (support), this node is not
returned (not support)-->true
</isSupportVerifyPlanTemplate>
<isSupportUserRightPlanTemplate>
    <!--ro, opt, bool, whether it supports configuring schedule template of the access permission control, desc:true (support), this node is not returned
(not support)-->true
</isSupportUserRightPlanTemplate>
<isSupportDoorStatusPlan>
    <!--ro, opt, bool, whether it supports configuring door control schedule, desc:true (support), this node is not returned (not support)-->true
</isSupportDoorStatusPlan>
<isSupportCardReaderPlan>
    <!--ro, opt, bool, whether it supports configuring control schedule of the card reader authentication mode, desc:true (support), this node is not
returned (not support)-->true
</isSupportCardReaderPlan>
<isSupportClearPlansCfg>
    <!--ro, opt, bool, whether it supports clearing the access control schedule parameters, desc:true (support), this node is not returned (not support)--
>true
</isSupportClearPlansCfg>
<isSupportRemoteControlBuzzer>
    <!--ro, opt, bool, whether it supports remotely controlling the buzzer of the card reader, desc:true (support), this node is not returned (not support)--
>true
</isSupportRemoteControlBuzzer>
<isSupportEventCardNoList>
    <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportEventCardNoList>
<isSupportEventCardLinkageCfg>
    <!--ro, opt, bool, whether it supports configuring event and card linkage parameters, desc:true (support), this node is not returned (not support)--
>true
</isSupportEventCardLinkageCfg>
<isSupportClearEventCardLinkageCfg>
    <!--ro, opt, bool, whether it supports clearing event and card linkage parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportClearEventCardLinkageCfg>
<isSupportAcsEvent>
    <!--ro, opt, bool, whether it supports searching for access control events, desc:true (support), this node is not returned (not support)-->true
</isSupportAcsEvent>
<isSupportAcsEventTotalNum>
    <!--ro, opt, bool, whether it supports getting total number of access control events by specific conditions, desc:true (support), this node is not
returned (not support)-->true
</isSupportAcsEventTotalNum>
<isSupportEventOptimizationCfg>
    <!--ro, opt, bool, whether it supports configuring event optimization, desc:true (support), this node is not returned (not support)-->true
</isSupportEventOptimizationCfg>
<isSupportAcsWorkStatus>
    <!--ro, opt, bool, whether it supports getting working status of the access control device, desc:true (support), this node is not returned (not
support)-->true
</isSupportAcsWorkStatus>
<isSupportDoorCfg>
    <!--ro, opt, bool, whether it supports configuring door parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportDoorCfg>
<isSupportCardReaderCfg>
    <!--ro, opt, bool, whether it supports configuring card reader parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportCardReaderCfg>
<isSupportAcsCfg>
    <!--ro, opt, bool, whether it supports configuring parameters of access control device, desc:true (support), this node is not returned (not support)--
>true
</isSupportAcsCfg>
<isSupportRemoteCheck>
    <!--ro, opt, bool, whether it supports verifying access control events remotely, desc:true (support), this node is not returned (not support)-->true
</isSupportRemoteCheck>
<isSupportMaskDetection>
    <!--ro, opt, bool, whether it supports mask detection, desc:true (support), this node is not returned (not support)-->true
</isSupportMaskDetection>
<isSupportGroupCfg>
    <!--ro, opt, bool, whether it supports configuring group parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportGroupCfg>
<isSupportClearGroupCfg>
    <!--ro, opt, bool, whether it supports clearing group parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportClearGroupCfg>
<isSupportMultiCardCfg>
    <!--ro, opt, bool, whether it supports configuring multiple authentication mode, desc:true (support), this node is not returned (not support)-->true
</isSupportMultiCardCfg>
<isSupportMultiDoorInterLockCfg>
    <!--ro, opt, bool, whether it supports configuring multi-door interlocking parameters, desc:true (support), this node is not returned (not support)--
>true
</isSupportMultiDoorInterLockCfg>
<isSupportAntiSneakCfg>
    <!--ro, opt, bool, whether it supports configuring anti-passing back parameters in the device, desc:true (support), this node is not returned (not
support)-->true
</isSupportAntiSneakCfg>
<isSupportCardReaderAntiSneakCfg>
    <!--ro, opt, bool, whether it supports configuring anti-passing back parameters for the card reader in the device, desc:true (support), this node is not

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    <!--ro, opt, bool, whether it supports configuring anti-passing back parameters for the card reader, desc:true (support), this node is not
returned (not support)-->true
</isSupportCardReaderAntiSneakCfg>
<isSupportClearAntiSneakCfg>
    <!--ro, opt, bool, whether it supports clearing anti-passing back parameters, desc:true (support), this node is not returned (not support)-->true
</isSupportClearAntiSneakCfg>
<isSupportClearAntiSneak>
    <!--ro, opt, bool, whether it supports clearing anti-passing back records in the device, desc:true (support), this node is not returned (not support)--
>true
</isSupportClearAntiSneak>
<isSupportSmsRelativeParam>
    <!--ro, opt, bool, whether it supports configuring message function, desc:true (support), this node is not returned (not support)-->true
</isSupportSmsRelativeParam>
<isSupportPhoneDoorRightCfg>
    <!--ro, opt, bool, whether it supports configuring the door permission linked to the mobile phone number: "true"-yes, this node is not returned-no,
desc:true (support), this node is not returned (not support)-->true
</isSupportPhoneDoorRightCfg>
<isSupportOSDPStatus>
    <!--ro, opt, bool, whether it supports searching for OSDP card reader status, desc:true (support), this node is not returned (not support)-->true
</isSupportOSDPStatus>
<isSupportOSDPModify>
    <!--ro, opt, bool, whether it supports editing OSDP card reader ID, desc:true (support), this node is not returned (not support)-->true
</isSupportOSDPModify>
<isSupportRemoteControlPWChcek>
    <!--ro, opt, bool, whether it supports verifying the password for remote door control, desc:true (support), this node is not returned (not support)--
>true
</isSupportRemoteControlPWChcek>
<isSupportRemoteControlPWCfg>
    <!--ro, opt, bool, whether it supports configuring the password for remote door contro, desc:true (support), this node is not returned (not support)--
>true
</isSupportRemoteControlPWCfg>
<isSupportLogModeCfg>
    <!--ro, opt, bool, whether it supports configuring log mode, desc:true (support), this node is not returned (not support)-->true
</isSupportLogModeCfg>
<isSupportQRCodeEncryption>
    <!--ro, opt, bool-->true
</isSupportQRCodeEncryption>
<isSupportQRCodeInfo>
    <!--ro, opt, bool-->true
</isSupportQRCodeInfo>
<isSupportBluetoothEncryptionInfo>
    <!--ro, opt, bool-->true
</isSupportBluetoothEncryptionInfo>
<isSupportBluetoothEncryptionVersion>
    <!--ro, opt, bool-->true
</isSupportBluetoothEncryptionVersion>
<FactoryReset>
    <!--ro, opt, object-->
<isSupportFactoryReset>
    <!--ro, opt, bool, whether it supports restoring to default settings by condition, desc:whether it supports restoring to default settings by
condition-->true
</isSupportFactoryReset>
    <mode opt="full,basic,part">
    <!--ro, opt, enum, conditions for restoring to default settings, subType:string, attr:opt{req, string}, desc:conditions for restoring to default
settings-->full
    </mode>
</FactoryReset>
<isSupportCaptureFace>
    <!--ro, opt, bool, whether it supports collecting face pictures, desc:true (support), this node is not returned (not support)-->true
</isSupportCaptureFace>
<isSupportCaptureInfraredFace>
    <!--ro, opt, bool, whether it supports collecting infrared face pictures, desc:true (support), this node is not returned (not support)-->true
</isSupportCaptureInfraredFace>
<isSupportFaceRecognizeMode>
    <!--ro, opt, bool, whether it supports configuring facial recognition mode, desc:true (support), this node is not returned (not support)-->true
</isSupportFaceRecognizeMode>
<isSupportCaptureCardInfo>
    <!--ro, opt, bool, whether it supports collecting card information, desc:true (support), this node is not returned (not support)-->true
</isSupportCaptureCardInfo>
<isSupportCaptureIDInfo>
    <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportCaptureIDInfo>
<isSupportCaptureRule>
    <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportCaptureRule>
<isSupportCapturePresetParam>
    <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportCapturePresetParam>
<isSupportOfflineCapture>
    <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportOfflineCapture>
<isSupportCardOperations>
    <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportCardOperations>
<isSupportNFCCfg>
    <!--ro, opt, bool, configuration capability of enabling NFC (Near-Field Communication) function, desc:true (support), this node is not returned (not
support)-->true
</isSupportNFCCfg>
<isSupportRFCardCfg>
    <!--ro, opt, bool, whether it supports enabling or disabling RF card recognition, desc:true (support), this node is not returned (not support)-->true
</isSupportRFCardCfg>
<isSupportTTSText>
    <!--ro, opt, bool, whether it supports configuring the text of the audio prompt, desc:true (support), this node is not returned (not support)-->true
</isSupportTTSText>

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<isSupportIDBlackListCfg>
  <!--ro, opt, bool, whether it supports applying ID card blockList, desc:true (support), this node is not returned (not support)-->true
</isSupportIDBlackListCfg>
<isSupportUserInfoAndRight>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportUserInfoAndRight>
<isSupportMaintenanceDataExport>
  <!--ro, opt, bool, whether it supports exporting maintenance data, desc:whether it supports exporting maintenance data-->true
</isSupportMaintenanceDataExport>
<isSupportLockTypeCfg>
  <!--ro, opt, bool, whether it supports configuring door Lock status when the device is powered off, desc:whether it supports configuring door Lock
status when the device is powered off-->true
</isSupportLockTypeCfg>
<isSupportSafetyHelmetDetection>
  <!--ro, opt, bool, whether it supports configuring hard hat detection, desc:true (support), this node is not returned (not support)-->true
</isSupportSafetyHelmetDetection>
<isSupportKeyCfgAttendance>
  <!--ro, opt, bool, whether it supports configuring parameters of attendance check by pressing the key, desc:true (support), this node is not returned
(not support)-->true
</isSupportKeyCfgAttendance>
<isSupportAsyncImportDatas>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportAsyncImportDatas>
<isSupportAsyncImportPic>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportAsyncImportPic>
<isSupportIDBlackListClear>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportIDBlackListClear>
<isSupportIDBlackListParamCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportIDBlackListParamCfg>
<isSupportIDBlackListTemplate>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportIDBlackListTemplate>
<isSupportCertificateCompareCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportCertificateCompareCfg>
<isSupportVisitorRightListCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportVisitorRightListCfg>
<isSupportVisitorRegisterInfoCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportVisitorRegisterInfoCfg>
<isSupportVisitorReceiptInfoCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportVisitorReceiptInfoCfg>
<isSupportVisitorParamCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportVisitorParamCfg>
<isSupportVisitorAdvanceParamCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportVisitorAdvanceParamCfg>
<isSupportVisitorRegisterAudio>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportVisitorRegisterAudio>
<isSupportFingerDataEncryption>
  <!--ro, opt, bool-->true
</isSupportFingerDataEncryption>
<isSupportAttendanceWeekPlan>
  <!--ro, opt, bool, whether it supports configuring parameters of the week attendance schedule, desc:true (support), this node is not returned (not
support)-->true
</isSupportAttendanceWeekPlan>
<isSupportClearAttendancePlan>
  <!--ro, opt, bool, whether it supports clearing the week attendance schedule, desc:true (support), this node is not returned (not support)-->true
</isSupportClearAttendancePlan>
<isSupportAttendanceStatusPlan>
  <!--ro, opt, bool-->true
</isSupportAttendanceStatusPlan>
<isSupportAttendanceMode>
  <!--ro, opt, bool, whether it supports configuring the attendance mode-->true
</isSupportAttendanceMode>
<isSupportAttendancePlanTemplate>
  <!--ro, opt, bool, whether it supports configuring the attendance schedule template, desc:true (support), this node is not returned (not support)-->true
</isSupportAttendancePlanTemplate>
<isSupportAttendancePlanTemplateList>
  <!--ro, opt, bool, whether it supports getting the List of attendance schedule templates, desc:true (support), this node is not returned (not support)--
>true
</isSupportAttendancePlanTemplateList>
<isSupportClearPictureCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportClearPictureCfg>
<isSupportEventStorageCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportEventStorageCfg>
<isSupportCardVerificationRule>
  <!--ro, opt, bool, whether it supports configuring card No. authentication mode, desc:true (support), this node is not returned (not support)-->true
</isSupportCardVerificationRule>
<isSupportFaceTemperatureEvent>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportFaceTemperatureEvent>
<isSupportQRCodeEvent>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportQRCodeEvent>

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</isSupportUserCodeEvent>
<isSupportIDCardInfoEvent>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportIDCardInfoEvent>
<isSupportTTSTextHolidayPlan>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportTTSTextHolidayPlan>
<isSupportPersonInfoExtendName>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportPersonInfoExtendName>
<isSupportCardDisplayCfg>
  <!--ro, opt, bool, true (support), this node is not returned (not support), desc:true (support), this node is not returned (not support)-->true
</isSupportCardDisplayCfg>
<isSupportBluetooth>
  <!--ro, opt, bool, whether it supports bluetooth configuration, desc:whether it supports bluetooth configuration-->true
</isSupportBluetooth>
<isSupportFaceCompareParamUpload>
  <!--ro, opt, bool-->true
</isSupportFaceCompareParamUpload>
<isSupportTemperatureMeasureCfg>
  <!--ro, opt, bool, whether it supports configuring temperature measurement parameters, desc:true (support), this node is not returned (not support)-->true
>true
</isSupportTemperatureMeasureCfg>
<isSupportTemperatureMeasureAreaCfg>
  <!--ro, opt, bool, whether it supports configuring parameters of the temperature measurement area, desc:true (support), this node is not returned (not support)-->true
</isSupportTemperatureMeasureAreaCfg>
<isSupportTemperatureMeasureAreaCalibrationCfg>
  <!--ro, opt, bool, whether it supports configuring calibration parameters of the temperature measurement area, desc:true (support), this node is not returned (not support)-->true
</isSupportTemperatureMeasureAreaCalibrationCfg>
<isSupportBlackObjectCfg>
  <!--ro, opt, bool, whether it supports configuring black body parameters, desc:true (support), this node is not returned (not support)-->true
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<isSupportHealthCodeCfg>
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<isSupportLocalPreviewPortraitDisplayParams>
  <!--ro, opt, bool-->true
</isSupportLocalPreviewPortraitDisplayParams>
<isSupportPersonnelChannelIRWorkMode>
  <!--ro, opt, bool-->true
</isSupportPersonnelChannelIRWorkMode>
<isSupportDoorOpenButtonCfg>
```

```

    <!--ro, opt, bool-->true
</isSupportDoorOpenButtonCfg>
<isSupportDoorLockCfg>
    <!--ro, opt, bool-->true
</isSupportDoorLockCfg>
<isSupportDoorbellCfg>
    <!--ro, opt, bool-->true
</isSupportDoorbellCfg>
<isSupportSearchDoorLinkageDeviceList>
    <!--ro, opt, bool-->true
</isSupportSearchDoorLinkageDeviceList>
<isSupportDoorLinkageDevice>
    <!--ro, opt, bool-->true
</isSupportDoorLinkageDevice>
<isSupportDoorLinkageDeviceParams>
    <!--ro, opt, bool-->true
</isSupportDoorLinkageDeviceParams>
<isSupportGetReaderOpenDoorPw>
    <!--ro, opt, bool-->true
</isSupportGetReaderOpenDoorPw>
<isSupportOneKeyUnbindDoorLinkageDevice>
    <!--ro, opt, bool-->true
</isSupportOneKeyUnbindDoorLinkageDevice>
<isSupportSearchAvailableLinkageDoor>
    <!--ro, opt, bool-->true
</isSupportSearchAvailableLinkageDoor>
<isSupportOpenGatePromptAngleCfg>
    <!--ro, opt, bool-->true
</isSupportOpenGatePromptAngleCfg>
<isSupportGetDoorbellEquipmentDiagram>
    <!--ro, opt, bool-->true
</isSupportGetDoorbellEquipmentDiagram>
<isSupportGetDoorLockEquipmentDiagram>
    <!--ro, opt, bool-->true
</isSupportGetDoorLockEquipmentDiagram>
<isSupportGetDoorOpenButtonEquipmentDiagram>
    <!--ro, opt, bool-->true
</isSupportGetDoorOpenButtonEquipmentDiagram>
<isSupportGetReaderOpenDoorDiagram>
    <!--ro, opt, bool-->true
</isSupportGetReaderOpenDoorDiagram>
<isSupportGetReaderEquipmentDiagram>
    <!--ro, opt, bool-->true
</isSupportGetReaderEquipmentDiagram>
<isSupportGetDoorLinkageDeviceNum>
    <!--ro, opt, bool-->true
</isSupportGetDoorLinkageDeviceNum>
<isSupportSearchFolderCustomAudio>
    <!--ro, opt, bool-->true
</isSupportSearchFolderCustomAudio>
<isSupportDeleteFolderCustomAudio>
    <!--ro, opt, bool-->true
</isSupportDeleteFolderCustomAudio>
<isSupportAddFolderCustomAudio>
    <!--ro, opt, bool-->true
</isSupportAddFolderCustomAudio>
<isSupportGetCustomAudioFolder>
    <!--ro, opt, bool-->true
</isSupportGetCustomAudioFolder>
<isSupportModifyCustomAudioFolder>
    <!--ro, opt, bool-->true
</isSupportModifyCustomAudioFolder>
<isSupportDeleteCustomAudioFolder>
    <!--ro, opt, bool-->true
</isSupportDeleteCustomAudioFolder>
<isSupportAddCustomAudioFolder>
    <!--ro, opt, bool-->true
</isSupportAddCustomAudioFolder>
<isSupportSendUSBKeyCertAndToken>
    <!--ro, opt, bool-->true
</isSupportSendUSBKeyCertAndToken>
<isSupportIDCardReadParams>
    <!--ro, opt, bool-->true
</isSupportIDCardReadParams>
<isSupportPreRemoteCheckEventAcknowledgment>
    <!--ro, opt, bool-->true
</isSupportPreRemoteCheckEventAcknowledgment>
</AccessControl>

```

12.8.3 Credential Recognition Module Management

12.8.3.1 Set the card reader parameters

Request URL

PUT /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	--

Request Message

```
{
  "CardReaderCfg": {
    /*req, object, card reader information*/
    "enable": true,
    /*req, bool, whether to enable, desc:true=yes, false=no*/
    "okLedPolarity": "cathode",
    /*opt, enum, OK LED polarity, subType:string, desc:"cathode", "anode"*/
    "errorLedPolarity": "cathode",
    /*opt, enum, Error LED polarity, subType:string, desc:"cathode", "anode"*/
    "buzzerPolarity": "cathode",
    /*opt, enum, buzzer polarity, subType:string, desc:"cathode", "anode"*/
    "swipeInterval": 1,
    /*opt, int, time interval of repeated authentication, unit:s, desc:it is valid for authentication modes such as fingerprint, card, face, etc.*/
    "pressTimeout": 1,
    /*opt, int, timeout to reset entry on keypad, unit:s*/
    "enableFailAlarm": true,
    /*opt, bool, whether to enable excessive failed authentication attempt alarm*/
    "maxReadCardFailNum": 1,
    /*opt, int, maximum number of failed authentication attempts*/
    "enableTamperCheck": true,
    /*opt, bool, whether to enable tampering detection*/
    "offlineCheckTime": 1,
    /*opt, int, time to detect after the card reader is offline, unit:s*/
    "fingerprintCheckLevel": 1,
    /*opt, enum, fingerprint recognition level, subType:int, desc:1-1/10 false acceptance rate (FAR), 2-1/100 false acceptance rate (FAR), 3-1/1000 false acceptance rate (FAR), 4-1/10000 false acceptance rate (FAR), 5-1/100000 false acceptance rate (FAR), 6-1/1000000 false acceptance rate (FAR), 7-1/10000000 false acceptance rate (FAR), 8-1/100000000 false acceptance rate (FAR), 9-3/100 false acceptance rate (FAR), 10-3/1000 false acceptance rate (FAR), 11-3/10000 false acceptance rate (FAR), 12-3/100000 false acceptance rate (FAR), 13-3/1000000 false acceptance rate (FAR), 14-3/10000000 false acceptance rate (FAR), 15-3/100000000 false acceptance rate (FAR), 16-Automatic Normal, 17-Automatic Secure, 18-Automatic More Secure (currently not support)*/
    "useLocalController": true,
    /*opt, bool, whether it is connected to the distributed controller*/
    "localControllerID": 1,
    /*opt, int, distributed controller No., range:[0,64], dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-unregistered. This field is valid only when useLocalController is "true"*/
    "localControllerReaderID": 1,
    /*opt, int, card reader ID of the distributed controller, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-unregistered. This field is valid only when useLocalController is "true"*/
    "cardReaderChannel": 1,
    /*opt, enum, communication channel No. of the card reader, subType:int, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-Wiegand or offline, 1-RS-485A, 2-RS-485B. This field is valid only when useLocalController is "true"*/
    "fingerprintImageQuality": 1,
    /*opt, enum, fingerprint image quality, subType:int, desc:1-low quality (V1), 2-medium quality (V1), 3-high quality (V1), 4-highest quality (V1), 5-low quality (V2), 6-medium quality (V2), 7-high quality (V2), 8-highest quality (V2)*/
    "fingerprintContrastTimeout": 1,
    /*opt, enum, fingerprint comparison timeout, subType:int, desc:it is between 1 and 20, unit: second, 255-infinite*/
    "fingerprintRecognizeInterval": 1,
    /*opt, enum, fingerprint scanning interval, subType:int, desc:it is between 1 and 10, unit: second, 255-no delay*/
    "fingerprintMatchFastMode": 1,
    /*opt, enum, fingerprint matching quick mode, subType:int, desc:1-quick mode 1, 2-quick mode 2, 3-quick mode 3, 4-quick mode 4, 5-quick mode 5, 255-automatic*/
    "fingerprintModuleSensitive": 1,
    /*opt, enum, fingerprint module sensitivity, subType:int, desc:fingerprint module sensitivity,which is between 1 and 8*/
    "fingerprintModuleLightCondition": "outdoor",
    /*opt, enum, fingerprint module light condition, subType:string, desc:"outdoor", "indoor"*/
    "faceMatchThresholdN": 1,
    /*opt, int, threshold of face picture 1:N comparison, range:[0,100], desc:threshold of face picture 1:N comparison,which is between 0 and 100*/
    "faceQuality": 1,
    /*opt, int, face picture quality, range:[0,100]*/
    "faceRecognizeTimeout": 1,
    /*opt, enum, face recognition timeout, subType:int, desc:it is between 1 and 20, unit: second, 255-infinite*/
    "faceRecognizeInterval": 1,
    /*opt, enum, face recognition interval, subType:int, desc:it is between 1 and 10, unit: second, 255-no delay*/
    "cardReaderFunction": ["fingerprint", "face", "fingerVein", "iris", "card"],
    /*opt, enumarray, card reader type, subType:string, desc:"fingerprint"-fingerprint, "face", "fingerVein"-finger vein, "iris". For example, ["fingerprint","face"] indicates that the card reader supports both fingerprint and face*/
    "cardReaderDescription": "Wiegand\u000485Offline",
    /*opt, string, card reader description, desc:if the card reader is the Wiegand card reader or if offline, this field will be set to "Wiegand" or "485Offline"*/
    "faceImageSensitometry": 1,
    /*opt, int, face picture exposure, range:[0,65535]*/
    "livingBodyDetect": true,
    /*opt, bool, whether to enable human detection*/
    "faceMatchThreshold1": 1,
    /*opt, int, threshold of face picture 1:1 comparison, range:[0,100], desc:threshold of face picture 1:1 comparison,which is between 0 and 100*/
    "buzzerTime": 1,
    /*opt, int, buzzing duration, range:[0,59999], unit:s, desc:0-long buzzing*/
    "faceMatch1SecurityLevel": 1,
    /*opt, enum, security level of face 1:1 recognition, subType:int, desc:1-normal, 2-high, 3-higher*/
    "faceMatchNSecurityLevel": 1,
    /*opt, enum, security level of face 1:N recognition, subType:int, desc:1-normal, 2-high, 3-higher*/
    "envirMode": "indoor",
    /*opt, enum, environment mode of face recognition, subType:string, desc:"indoor", "other"*/
    "liveDetLevelSet": "low",
  }
}
```

```

/*opt, enum, threshold level of liveness detection, subType:string, desc:"low", "middle"-medium, "high"*/
"liveDetThreshold": 1,
/*opt, int, range:[0,100]*/
"liveDetAntiAttackCntLimit": 1,
/*opt, int, number of anti-attacks of liveness detection,, range:[1,255], desc:this value should be configured as the same one on both client and device*/
"enableLiveDetAntiAttack": true,
/*opt, bool, whether to enable anti-attack for liveness detection*/
"liveDetAntiAttackLockedTime": 1,
/*opt, int, range:[0,300], unit:s*/
"supportDelFPByID": true,
/*opt, bool, whether the card reader supports deleting fingerprint by fingerprint ID, desc:true=yes, false-no*/
"fingerPrintCapacity": 1,
/*opt, int, fingerprint capacity*/
"fingerPrintNum": 1,
/*opt, int, number of added fingerprints*/
"defaultVerifyMode": "cardAndPw",
/*opt, enum, default authentication mode of the fingerprint and card reader (factory defaults):, subType:string, desc:factory defaults; "cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password), "faceAndFp" (face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw" (fingerprint or password), "employeeNoAndFp" (employee No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard" (face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrfaceAndCard" (face or face+card), "fpOrface" (fingerprint or face), "cardOrfaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint), "cardOrFpOrPw" (card or fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris), "faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or password), "employeeNoAndFaceAndPw" (employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint or password), "cardOrFpOrFaceOrIris" (card or fingerprint or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password), "cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+ris), "fpAndIri" (fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw" (face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
"faceRecognizeEnable": 1,
/*opt, enum, whether to enable facial recognition, subType:int, desc:1-enable, 2-disable, 3-attendance checked in/out by recognition of multiple faces*/
"FPAlgorithmVersion": "test",
/*opt, string, fingerprint algorithm library version, range:[1,32]*/
"cardReaderVersion": "test",
/*opt, string, card reader version, range:[1,32]*/
"enableReverseCardNo": true,
/*opt, bool, whether to enable reversing the card No.*/
"independSwipeIntervals": 0,
/*opt, int, time interval of person authentication, unit: second. This time interval is calculated for each person separately and is different from swipeInterval, desc:time interval of person authentication,unit: second. This time interval is calculated for each person separately and is different from swipeInterval*/
"maskFaceMatchThresholdN": 1,
/*opt, int, 1:N face picture (face with mask and normal background) comparison threshold, range:[0,100], desc:1:N face picture (face with mask and normal background) comparison threshold,value range: [0,100]*/
"maskFaceMatchThreshold1": 1,
/*opt, int, 1:1 face picture (face with mask and normal background) comparison threshold, range:[0,100], desc:1:1 face picture (face with mask and normal background) comparison threshold,value range: [0,100]*/
"faceMotionDetLevel": "low",
/*opt, enum, face motion detection level, subType:string, desc:"high", "medium", "low"*/
"showMode": "normal",
/*opt, enum, display mode, subType:string, desc:this node is not valid; simple mode indicates that the device displays authentication results exclude employee No., name, etc.; the device applies normal mode by default; advertisement mode indicates that the device displays both advertisement and authentication results; meeting mode indicates that the device displays check-in page of the conference; custom mode indicates that the device displays layout of the interface customized by users; "concise"-simple mode, "normal"-normal mode (default), "advertising"-advertisement mode, "meeting"-meeting mode, "selfDefine"-custom mode*/
"enableScreenOff": true,
/*opt, bool, whether to enable auto locking the screen*/
"screenOffTimeout": 1
/*opt, int, time, step:1, unit:s*/
}
}

```

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "OK",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
}

```

12.8.3.2 Get the card reader configuration parameters

Request URL

GET /ISAPI/AccessControl/CardReaderCfg/<cardReaderID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	--

Request Message

None

Response Message

```
{
  "CardReaderCfg": {
    /*ro, req, object, card reader information*/
    "enable": true,
    /*ro, req, bool, whether to enable, desc:true (enable), false (disable)*/
    "cardReaderName": "test",
    /*ro, opt, string, range:[0,64]*/
    "okLedPolarity": "cathode",
    /*ro, opt, enum, OK LED polarity, subType:string, desc:"cathode", "anode"*/
    "errorLedPolarity": "cathode",
    /*ro, opt, enum, Error LED polarity, subType:string, desc:"cathode", "anode"*/
    "buzzerPolarity": "cathode",
    /*ro, opt, enum, buzzer polarity, subType:string, desc:"cathode", "anode"*/
    "swipeInterval": 1,
    /*ro, opt, int, time interval of repeated authentication, unit:s, desc:which is valid for authentication modes such as fingerprint, card, face,
    etc.*/
    "pressTimeout": 1,
    /*ro, opt, int, timeout to reset entry on keypad, unit:s*/
    "enableFailAlarm": true,
    /*ro, opt, bool, whether to enable excessive failed authentication attempts alarm*/
    "maxReadCardFailNum": 1,
    /*ro, opt, int, maximum number of failed authentication attempts*/
    "enableTamperCheck": true,
    /*ro, opt, bool, whether to enable tampering detection*/
    "offlineCheckTime": 1,
    /*ro, opt, int, time to detect after the card reader is offline, unit:s*/
    "fingerprintCheckLevel": 1,
    /*ro, opt, enum, fingerprint recognition level, subType:int, desc:1-1/10 false acceptance rate (FAR), 2-1/100 false acceptance rate (FAR), 3-1/1000
    false acceptance rate (FAR), 4-1/10000 false acceptance rate (FAR), 5-1/100000 false acceptance rate (FAR), 6-1/1000000 false acceptance rate (FAR), 7-
    1/10000000 false acceptance rate (FAR), 8-1/100000000 false acceptance rate (FAR), 9-3/100 false acceptance rate (FAR), 10-3/1000 false acceptance rate
    (FAR), 11-3/10000 false acceptance rate (FAR), 12-3/100000 false acceptance rate (FAR), 13-3/1000000 false acceptance rate (FAR), 14-3/10000000 false
    acceptance rate (FAR), 15-3/100000000 false acceptance rate (FAR), 16-Automatic Normal, 17-Automatic Secure, 18-Automatic More Secure (currently not
    support)*/
    "useLocalController": true,
    /*ro, opt, bool, whether it is connected to the distributed controller*/
    "localControllerID": 1,
    /*ro, opt, int, distributed controller No., range:[0,64], dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:which is between 1 and 64, 0-
    unregistered. This field is valid only when useLocalController is "true"*/
    "localControllerReaderID": 1,
    /*ro, opt, int, card reader ID of the distributed controller, 0-unregistered, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:this field
    is valid only when useLocalController is "true"*/
    "cardReaderChannel": 1,
    /*ro, opt, enum, communication channel No. of the card reader, subType:int, dep:and,{$.CardReaderCfg.localControllerID,eq,true}, desc:0-Wiegand or
    offline, 1-RS-485A, 2-RS-485B. This field is valid only when useLocalController is "true"*/
    "fingerprintImageQuality": 1,
    /*ro, opt, enum, fingerprint image quality, subType:int, desc:1-low quality (V1), 2-medium quality (V1), 3-high quality (V1), 4-highest quality
    (V1), 5-low quality (V2), 6-medium quality (V2), 7-high quality (V2), 8-highest quality (V2)*/
    "fingerprintContrastTimeOut": 1,
    /*ro, opt, enum, fingerprint comparison timeout, subType:int, desc:fingerprint comparison timeout,which is between 1 and 20,unit: second,255-
    infinite*/
    "fingerprintRecognizeInterval": 1,
    /*ro, opt, enum, fingerprint scanning interval, subType:int, desc:fingerprint scanning interval,which is between 1 and 10,unit: second,255-no
    delay*/
    "fingerprintMatchFastMode": 1,
    /*ro, opt, enum, fingerprint matching quick mode, subType:int, desc:1-quick mode 1, 2-quick mode 2, 3-quick mode 3, 4-quick mode 4, 5-quick mode 5,
    255-automatic*/
    "fingerprintModuleSensitive": 1,
    /*ro, opt, enum, fingerprint module sensitivity, subType:int, desc:fingerprint module sensitivity,which is between 1 and 8*/
    "fingerprintModuleLightCondition": "outdoor",
    /*ro, opt, enum, fingerprint module light condition, subType:string, desc:"outdoor", "indoor"*/
    "faceMatchThresholdN": 1,
    /*ro, opt, int, threshold of face picture 1:N comparison,which is between 0 and 100, range:[0,100]*/
    "faceQuality": 1,
    /*ro, opt, int, face picture quality, range:[0,100]*/
    "faceRecognizeTimeOut": 1,
    /*ro, opt, enum, face recognition timeout, subType:int, desc:face recognition timeout,which is between 1 and 20,unit: second,255-infinite*/
    "faceRecognizeInterval": 1,
    /*ro, opt, enum, face recognition interval, subType:int, desc:face recognition interval,which is between 1 and 10,unit: second,255-no delay*/
    "cardReaderFunction": ["fingerprint", "face", "card", "voiceprint", "PPAndPV"],
    /*ro, opt, enumarray, card reader type, subType:string, desc:"fingerprint", "face", "fingerVein". For example, ["fingerprint", "face"] indicates
    that the card reader supports both fingerprint and face*/
    "cardReaderDescription": "WiegandOrOffline",
    /*ro, opt, string, card reader description, desc:if the card reader is the Wiegand card reader or if offline, this field will be set to "Wiegand" or
    "485Offline"*/
    "faceImageSensitometry": 1,
    /*ro, opt, int, face picture exposure, range:[0,65535]*/
    "faceMatchThresholdI": 1,
    /*ro, opt, int, threshold of face picture 1:1 comparison, range:[0,100]*/
    "buzzerTime": 1,
    /*ro, opt, int, buzzing duration, range:[0,5000], unit:s, desc:buzzing duration which is between 0 and 5000 unit: second,0-long buzzing*/
  }
}
```

```

/*ro, opt, int, buzzing duration, range:[0,9999], units: desc:buzzing duration,which is between 0 and 9999,unit: second,0-long buzzing*/
"faceMatch1SecurityLevel": 1,
/*ro, opt, enum, security level of face 1:1 recognition, subType:int, desc:1 (normal), 2 (high), 3 (higher)*/
"faceMatchNSecurityLevel": 1,
/*ro, opt, enum, security level of face 1:N recognition: 1-normal,2-high,3-higher, subType:int, desc:1 (normal), 2 (high), 3 (higher)*/
"envirMode": "other",
/*ro, opt, enum, environment mode of face recognition, subType:string, desc:"indoor", "other"*/
"livingBodyDetect": true,
/*ro, opt, bool, whether to enable human detection*/
"liveDetLevelSet": "low",
/*ro, opt, enum, threshold level of liveness detection, subType:string, desc:"low", "middle", "high"*/
"liveDetAntiAttackCntLimit": 1,
/*ro, opt, int, number of anti-attacks of liveness detection, range:[1,255], desc:this value should be configured as the same one on both client and
device*/
"enableLiveDetAntiAttack": true,
/*ro, opt, bool, whether to enable anti-attack for liveness detection*/
"supportDelFPByID": true,
/*ro, opt, bool, whether the card reader supports deleting fingerprint by fingerprint ID, desc:"true"-yes, "false"-no*/
"fingerprintCapacity": 1,
/*ro, opt, int, fingerprint capacity*/
"fingerprintNum": 1,
/*ro, opt, int, number of added fingerprints*/
"defaultVerifyMode": "cardAndPw",
/*ro, opt, enum, default authentication mode of the fingerprint and card reader (factory defaults), subType:string, desc:factory defaults;
"cardAndPw" (card+password), "card" (card), "cardOrPw" (card or password), "fp" (fingerprint), "fpAndPw" (fingerprint+password), "fpOrCard" (fingerprint or
card), "fpAndCard" (fingerprint+card), "fpAndCardAndPw" (fingerprint+card+password), "faceOrFpOrCardOrPw" (face or fingerprint or card or password),
"faceAndFp" (face+fingerprint), "faceAndPw" (face+password), "faceAndCard" (face+card), "face" (face), "employeeNoAndPw" (employee No.+password), "fpOrPw"
(fingerprint or password), "employeeNoAndFp" (employee No.+fingerprint), "employeeNoAndFpAndPw" (employee No.+fingerprint+password), "faceAndFpAndCard"
(face+fingerprint+card), "faceAndPwAndFp" (face+password+fingerprint), "employeeNoAndFace" (employee No.+face), "faceOrFaceAndCard" (face or face+card),
"fpOrFace" (fingerprint or face), "cardOrFaceOrPw" (card or face or password), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or fingerprint),
"cardOrFpOrPw" (card or fingerprint or password), "iris" (iris), "faceOrFpOrCardOrPwOrIris" (face or fingerprint or card or password or iris),
"faceOrCardOrPwOrIris" (face or card or password or iris), "cardOrFace" (card or face), "cardOrFaceOrFp" (card or face or password), "faceOrPw" (face or
password), "employeeNoAndFaceAndPw" (employee No.+face+password), "cardOrFaceOrFaceAndCard" (card or face or face+card), "faceOrFpOrPw" (face or fingerprint
or password), "cardOrFpOrFaceOrIris" (card or fingerprint or face or iris), "fpOrFaceOrIrisOrPw" (fingerprint or face or iris or password),
"cardOrFpOrIrisOrPw" (card or fingerprint or iris or password), "cardOrIrisOrPw" (card or iris or password), "cardAndIris" (card+iris), "fpAndIri"
(fingerprint+iris), "faceAndIris" (face+iris), "irisAndPw" (iris+password), "cardAndIrisAndPw" (card+iris+password), "faceAndIrisAndPw"
(face+iris+password), "cardAndFaceAndIris" (card+face+iris)*/
"faceRecognizeEnable": 1,
/*ro, opt, enum, whether to enable facial recognition, subType:int, desc:1 (enable), 2 (disable), 3 (attendance checked in/out by recognition of
multiple faces)*/
"FPAlgorithmVersion": "test",
/*ro, opt, string, fingerprint algorithm library version, range:[1,32]*/
"cardReaderVersion": "test",
/*ro, opt, string, card reader version, range:[1,32]*/
"enableReverseCardNo": true,
/*ro, opt, bool, whether to enable reversing the card No.*/
"independSwipeIntervals": 0,
/*ro, opt, int, time interval of person authentication, desc:unit: second. This time interval is calculated for each person separately and is
different from swipeInterval*/
"maskFaceMatchThresholdN": 1,
/*ro, opt, int, 1:N face picture (face with mask and normal background) comparison threshold, range:[0,100]*/
"maskFaceMatchThreshold1": 1,
/*ro, opt, int, 1:1 face picture (face with mask and normal background) comparison threshold, range:[0,100]*/
"faceMotionDetLevel": "low",
/*ro, opt, enum, face motion detection level, subType:string, desc:"high", "medium", "low"*/
"showMode": "normal",
/*ro, opt, enum, display mode, subType:string, desc:this node is not valid; simple mode indicates that the device displays authentication results
exclude employee No., name, etc.; the device applies normal mode by default; advertisement mode indicates that the device displays both advertisement and
authentication results; meeting mode indicates that the device displays check-in page of the conference; custom mode indicates that the device displays
layout of the interface customized by users; "concise"-simple mode, "normal"-normal mode (default), "advertising"-advertisement mode, "meeting"-meeting
mode, "selfDefine"-custom mode*/
"enableScreenOff": true,
/*ro, opt, bool, whether to enable auto locking the screen*/
"screenOffTimeout": 1,
/*ro, opt, int, time, step:1, unit:s*/
"QRCodeEnabled": false,
/*ro, opt, bool*/
"swipeCardEnabled": true,
/*ro, opt, bool*/
"pressEnabled": true,
/*ro, opt, bool*/
"buzzerEnabled": true,
/*ro, opt, bool*/
"elevatorCarID": 1,
/*ro, opt, int, range:[1,8]*/
}
}

```

12.8.3.3 Get the configuration capability of the card reader

Request URL

GET /ISAPI/AccessControl/CardReaderCfg/capabilities?format=json&cardReaderID=<cardReaderID>

Query Parameter

Parameter Name	Parameter Type	Description
cardReaderID	string	--

Request Message

None

Response Message

```
{
  "CardReaderCfg": {
    /*ro, req, object, Set Card Reader Parameters*/
    "cardReaderNo": {
      /*ro, opt, object, card reader No., desc:card reader No.*/
      "@min": 1,
      /*ro, opt, int, the minimum value*/
      "@max": 512,
      /*ro, opt, int, the maximum value*/
      "@opt": [1, 4]
      /*ro, opt, array, value range, subType:int, desc:turnstiles only support No.1 and No.4; returning the node "@min" and "@max" are not required*/
    },
    "hostCardReaderNo": {
      /*ro, opt, object*/
      "@min": 1,
      /*ro, opt, int*/
      "@max": 8
      /*ro, opt, int*/
    },
    "localExtensionCardReaderInfoList": [
      /*ro, opt, array, subType:object*/
      {
        "devType": "2doors",
        /*ro, req, enum, subType:string*/
        "devNum": 2
        /*ro, req, int, range:[1,2]*/
      }
    ],
    "enable": "true,false",
    /*ro, opt, string, whether to enable, desc:"true"-yes,"false"-no*/
    "cardReaderName": {
      /*ro, opt, object*/
      "@min": 0,
      /*ro, req, int*/
      "@max": 64
      /*ro, req, int*/
    },
    "okLedPolarity": {
      /*ro, opt, object, OK LED polarity*/
      "@opt": "cathode,anode"
      /*ro, req, string, options, desc:"cathode", "anode"*/
    },
    "errorLedPolarity": {
      /*ro, opt, object, error LED polarity*/
      "@opt": "cathode,anode"
      /*ro, req, string, options, desc:"cathode", "anode"*/
    },
    "buzzerPolarity": {
      /*ro, opt, object, buzzer polarity, desc:buzzer polarity: "cathode","anode"*/
      "@opt": "cathode,anode"
      /*ro, req, string, options, desc:"cathode", "anode"*/
    },
    "swipeInterval": {
      /*ro, opt, object, time interval of repeated authentication, desc:it is valid for authentication modes such as fingerprint, card, face, etc., unit:
second*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 255
      /*ro, req, int, the maximum value*/
    },
    "pressTimeout": {
      /*ro, opt, object, timeout to reset entry on keypad, desc:unit: second*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 255
      /*ro, req, int, the maximum value*/
    },
    "enableFailAlarm": "true,false",
    /*ro, opt, string, whether to enable excessive failed authentication attempts alarm*/
    "maxReadCardFailNum": {
      /*ro, opt, object, maximum number of failed authentication attempts*/
      "@min": 1,
      /*ro, req, int, the minimum value*/
      "@max": 255
      /*ro, req, int, the maximum value*/
    },
    "enableTamperCheck": "true,false",
    /*ro, opt, string, whether to enable tampering detection*/
    "offlineCheckTime": {
      /*ro, opt, object, time to detect after the card reader is offline, desc:unit: second*/

```

```

/*ro, opt, object, time to detect after the card reader is offline, desc:unit: second*/
  "@min": 1,
  /*ro, req, int, the minimum value*/
  "@max": 255
  /*ro, req, int, the maximum value*/
},
"fingerPrintCheckLevel": {
/*ro, opt, object, fingerprint recognition level*/
  "@opt": "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18"
  /*ro, req, string, options*/
},
"useLocalController": "true,false",
/*ro, opt, string, whether it is connected to the distributed controller*/
"localControllerID": {
/*ro, opt, object, distributed controller No., desc:it is between 1 and 64, 0 (unregistered). This field is valid only when useLocalController is
"true"*/
  "@min": 0,
  /*ro, req, int, the minimum value*/
  "@max": 64
  /*ro, req, int, the maximum value*/
},
"localControllerReaderID": {
/*ro, opt, object, card reader ID of the distributed controller, desc:0 (unregistered). This field is valid only when useLocalController is "true"*/
  "@min": 0,
  /*ro, req, int, the minimum value*/
  "@max": 4
  /*ro, req, int, the maximum value*/
},
"cardReaderChannel": {
/*ro, opt, object, communication channel No. of the card reader, desc:this field is valid only when useLocalController is "true"*/
  "@opt": "0,1,2"
  /*ro, req, string, options*/
},
"fingerPrintImageQuality": {
/*ro, opt, object, fingerprint image quality, desc:fingerprint image quality: 1 (low quality (V1)), 2 (medium quality (V1)), 3 (high quality (V1)),
4 (highest quality (V1)), 5 (low quality (V2)), 6 (medium quality (V2)), 7 (high quality (V2)), 8 (highest quality (V2))*/
  "@opt": "1,2,3,4,5,6,7,8"
  /*ro, req, string, options*/
},
"fingerPrintContrastTimeOut": {
/*ro, opt, object, fingerprint comparison timeout, desc:it is between 1 and 20, unit: second, 0 (infinite)*/
  "@min": 0,
  /*ro, req, int, the minimum value*/
  "@max": 20
  /*ro, req, int, the maximum value*/
},
"fingerPrintRecognizeInterval": {
/*ro, opt, object, fingerprint scanning interval, desc:it is between 1 and 10, unit: second, 0 (no delay)*/
  "@min": 0,
  /*ro, req, int, the minimum value*/
  "@max": 10
  /*ro, req, int, the maximum value*/
},
"fingerPrintMatchFastMode": {
/*ro, opt, object, fingerprint matching quick mode, desc:1 (quick mode 1), 2 (quick mode 2), 3 (quick mode 3), 4 (quick mode 4), 5 (quick mode 5), 0
(automatic)*/
  "@min": 0,
  /*ro, req, int, the minimum value*/
  "@max": 5
  /*ro, req, int, the maximum value*/
},
"fingerPrintModuleSensitive": {
/*ro, opt, object, fingerprint module sensitivity, desc:it is between 1 and 8*/
  "@min": 1,
  /*ro, req, int, the minimum value*/
  "@max": 8
  /*ro, req, int, the maximum value*/
},
"fingerPrintModuleLightCondition": {
/*ro, opt, object, fingerprint module light condition*/
  "@opt": "outdoor,indoor"
  /*ro, req, string, options, desc:"outdoor", "indoor"*/
},
"faceMatchThresholdN": {
/*ro, opt, object, threshold of face picture 1:N comparison, desc:it is between 0 and 100*/
  "@min": 0,
  /*ro, req, int, the minimum value*/
  "@max": 100
  /*ro, req, int, the maximum value*/
},
"faceQuality": {
/*ro, opt, object, face picture quality, desc:it is between 0 and 100*/
  "@min": 0,
  /*ro, req, int, the minimum value*/
  "@max": 100
  /*ro, req, int, the maximum value*/
},
"faceRecognizeTimeOut": {
/*ro, opt, object, face recognition timeout, desc:it is between 1 and 20, unit: second, 0 (infinite)*/
  "@min": 0,
  /*ro, req, int, the minimum value*/
  "@max": 20
  /*ro, req, int, the maximum value*/
},

```

```

"faceRecognizeInterval": {
/*ro, opt, object, face recognition interval, desc:it is between 1 and 10, unit: second, 0 (no delay)*/
  "@min": 0,
/*ro, req, int, the minimum value*/
  "@max": 10
/*ro, req, int, the maximum value*/
},
"cardReaderFunction": {
/*ro, opt, object, card reader type*/
  "@opt": "fingerPrint,face,fingerVein,iris,card,voiceprint,PPAndPV"
/*ro, req, string, options, desc:"fingerPrint" (fingerprint), "face", "fingerVein" (finger vein), "iris"*/
},
"cardReaderDescription": {
/*ro, opt, object, card reader description, desc:if the card reader is the Wiegand card reader or if offline, this field will be set to "Wiegand" or
"4850Offline"*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 16
/*ro, req, int, the maximum value*/
},
"isSupportOfflineSendPermission": true,
/*ro, opt, bool, whether to enable applying permissions offline on the card reader, desc:AIOT (Artificial Intelligence & Internet of Things)
supports apply person's permissions when the card reader is offline*/
"faceImageSensitometry": {
/*ro, opt, object, face picture exposure, desc:it is between 0 and 65535*/
  "@min": 0,
/*ro, req, int, the minimum value*/
  "@max": 65535
/*ro, req, int, the maximum value*/
},
"faceMatchThreshold1": {
/*ro, opt, object, threshold of face picture 1:1 comparison, desc:it is between 0 and 100*/
  "@min": 0,
/*ro, req, int, the minimum value*/
  "@max": 100
/*ro, req, int, the maximum value*/
},
"buzzerTime": {
/*ro, opt, object, buzzing duration, desc:it is between 0 and 5999, unit: second, 0 (long buzzing)*/
  "@min": 0,
/*ro, req, int, the minimum value*/
  "@max": 5999
/*ro, req, int, the maximum value*/
},
"faceMatch1SecurityLevel": {
/*ro, opt, object, security level of face 1:1 recognition*/
  "@opt": "1,2,3"
/*ro, req, string, options, desc:1 (normal), 2 (high), 3 (higher)*/
},
"faceMatchNSecurityLevel": {
/*ro, opt, object, security level of face 1:N recognition*/
  "@opt": "1,2,3"
/*ro, req, string, options, desc:1 (normal), 2 (high), 3 (higher)*/
},
"envirMode": {
/*ro, opt, object, environment mode of face recognition*/
  "@opt": "indoor,other"
/*ro, req, string, options, desc:"indoor", "other"*/
},
"livingBodyDetect": "true,false",
/*ro, opt, string, whether to enable human detection*/
"liveDetLevelSet": {
/*ro, opt, object, threshold level of liveness detection*/
  "@opt": "low,middle,high,general,enhancive,professional"
/*ro, req, string, options*/
},
"liveDetAntiAttackCntLimit": {
/*ro, opt, object, number of anti-attacks of liveness detection, desc:it is between 1 and 255. This value should be configured as the same one on
both client and device*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 255
/*ro, req, int, the maximum value*/
},
"enableLiveDetAntiAttack": "true,false",
/*ro, opt, string, whether to enable anti-attack for liveness detection, desc:whether to enable anti-attack for liveness detection*/
"supportDelFPByID": "true,false",
/*ro, opt, string, whether the card reader supports deleting the fingerprint by fingerprint ID, desc:"true" (yes), "false" (no)*/
"fingerPrintCapacity": {
/*ro, opt, object, maximum number of fingerprints that can be added*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 100
/*ro, req, int, the maximum value*/
},
"fingerPrintNum": {
/*ro, opt, object, number of added fingerprints*/
  "@min": 1,
/*ro, req, int, the minimum value*/
  "@max": 100
/*ro, req, int, the maximum value*/
},
"defaultVerifyMode": {

```

```

/*ro, opt, object, default authentication mode of the card reader, desc:default authentication mode of the card reader (factory defaults)*/
"@opt":
"cardAndPw,card,cardOrPw,fp,fpAndPw,fpOrCard,fpAndCard,fpAndCardAndPw,faceOrFpOrCardOrPw,faceAndFp,faceAndPw,faceAndCard,face,employeeNoAndPw,fpOrPw,employeeNoAndFp,employeeNoAndFpAndPw,faceAndFpAndCard,faceAndPwAndFp,employeeNoAndFace,faceOrFaceAndCard,fpOrFace,cardOrFaceOrPw,cardOrFace,cardOrFaceOrFp,faceOrPw,employeeNoAndFaceAndPw,cardOrFaceOrFaceAndCard,iris,faceOrFpOrCardOrPwOrIris,faceOrCardOrPwOrIris,cardOrFpOrFaceOrIris,fpOrFaceOrIrisOrPw,cardOrFpOrIrisOrPw,cardOrIrisOrPw,cardAndIris,fpAndIris,faceAndIris,irisAndPw,cardAndIrisAndPw,faceAndIrisAndPw,cardAndFaceAndIris,Pw,cardOrFpOrPw,faceOrFpOrPw,cardAndVp,faceAndVp,fpAndVp,pwAndVp,remoteControl,remoteControlAndFace,remoteControlOrFace,remoteControlOrCard,remoteControlOrFp,remoteControlOrPw,remoteControlOrCardOrFp,remoteControlOrCardOrFace,remoteControlOrCardOrPw,remoteControlOrFpOrFace,remoteControlOrFpOrPw,remoteControlOrFaceOrPw,remoteControlOrCardOrFpOrFace,remoteControlOrCardOrFpOrPw,remoteControlOrFpOrFaceOrPw,remoteControlOrCardOrFaceOrPw,remoteControlOrCardOrFpOrFaceOrPw,faceOrPPPv,cardOrPPPv,pwOrPPPv,faceOrCardOrPPPv,faceOrCardOrPwOrPPPv,faceAndPPPv,cardAndPPPv,pwAndPPPv,faceAndCardAndPPPv,faceAndCardAndPwAndPPPv,PPPv,faceAndCardAndPw"

/*ro, req, string, options*/
},
"FPAlgorithmVersion": {
/*ro, opt, object, fingerprint algorithm library version*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"cardReaderVersion": {
/*ro, opt, object, card reader version*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"faceRecognizeEnable": {
/*ro, opt, object, whether to enable facial recognition*/
"@opt": "1,2,3"
/*ro, req, string, options, desc:1 (enable), 2 (disable), 3 (attendance checked in/out by recognition of multiple faces)*/
},
"enableReverseCardNo": "true,false",
/*ro, opt, string, whether to enable reversing the card No.*/
"independSwipeIntervals": {
/*ro, opt, object, time interval of person authentication, desc:unit: second. This time interval is calculated for each person separately and is different from swipeInterval*/
"@min": 1,
/*ro, req, int, the minimum value*/
"@max": 1
/*ro, req, int, the maximum value*/
},
"maskFaceMatchThresholdN": {
/*ro, opt, object, 1:N face picture (face with mask and normal background) comparison threshold, desc:value range: [0,100]*/
"@min": 0,
/*ro, req, int, the minimum value*/
"@max": 100
/*ro, req, int, the maximum value*/
},
"maskFaceMatchThreshold1": {
/*ro, opt, object, 1:1 face picture (face with mask and normal background) comparison threshold, desc:value range: [0,100]*/
"@min": 0,
/*ro, req, int, the minimum value*/
"@max": 100
/*ro, req, int, the maximum value*/
},
"faceMotionDetLevel": {
/*ro, opt, object, face motion detection level*/
"@opt": ["low", "medium", "height"]
/*ro, req, array, options, subType:string*/
},
"showMode": {
/*ro, opt, object, display mode*/
"@opt": ["concise", "normal", "advertising", "meeting", "selfDefine"]
/*ro, req, array, options, subType:string*/
},
"enableScreenOff": {
/*ro, opt, object, whether to enable auto locking the screen*/
"@opt": [true, false]
/*ro, req, array, options, subType:bool*/
},
"screenOffTimeout": {
/*ro, opt, object, time*/
"@min": 0,
/*ro, req, int, the minimum value*/
"@max": 10
/*ro, req, int, the maximum value*/
},
"QRCodeEnabled": {
/*ro, opt, object*/
"@opt": [true, false]
/*ro, req, array, subType:bool*/
},
"method": {
/*ro, opt, object*/
"@opt": ["batchPut", "search"]
/*ro, req, array, subType:string*/
},
"cardReaderCfgType": {
/*ro, opt, object*/
"@opt": ["all", "byCardReader", "byRegion"]
/*ro, req, array, subType:string*/
},
"searchCardReaderParams": {
/*ro, opt, object*/

```

```

        "searchID": {
            /*ro, req, object*/
            "@min": 1,
            /*ro, req, int*/
            "@max": 32
            /*ro, req, int*/
        },
        "searchResultPosition": {
            /*ro, req, object*/
            "@min": 1,
            /*ro, req, int*/
            "@max": 256
            /*ro, req, int*/
        },
        "maxResults": {
            /*ro, req, object*/
            "@min": 1,
            /*ro, req, int*/
            "@max": 100
            /*ro, req, int*/
        },
        "readerIDList": {
            /*ro, opt, object*/
            "@size": 256,
            /*ro, req, int*/
            "@min": 1,
            /*ro, req, int*/
            "@max": 256
            /*ro, req, int*/
        }
    },
    "swipeCardEnabled": {
        /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, req, array, subType:bool*/
    },
    "pressEnabled": {
        /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, req, array, subType:bool*/
    },
    "buzzerEnabled": {
        /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, req, array, subType:bool*/
    },
    "elevatorCarID": {
        /*ro, opt, object*/
        "@min": 1,
        /*ro, req, int*/
        "@max": 8
        /*ro, req, int*/
    }
}
}
}

```

12.8.3.4 Get the configuration capability of enabling RF (Radio Frequency) card recognition

Request URL

GET /ISAPI/AccessControl/Configuration/RFCardCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "RFCardCfgCap": {
    /*ro, req, object*/
    "cardType": {
      /*ro, opt, object, card type, desc:"EMCard"-EM card, "M1Card"-M1 card, "CPUCard"-CPU card, "IDCard"-ID card, "DesfireCard"-DESFire card,
      "FelicaCard"-FeliCa card*/
      "@opt": ["EMCard", "M1Card", "CPUCard", "IDCard", "FelicaCard"]
      /*ro, req, array, options, subType:string*/
    },
    "enabled": {
      /*ro, opt, object, whether to enable RF card recognition*/
      "@opt": [true, false]
      /*ro, req, array, options, subType:bool*/
    }
  }
}

```

12.8.3.5 Get the parameters of enabling RF (Radio Frequency) card recognition

Request URL

GET /ISAPI/AccessControl/Configuration/RFCardCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "RFCardCfg": [
    /*ro, req, array, subType:object*/
    {
      "cardType": "EMCard",
      /*ro, req, enum, card type, subType:string, desc:"EMCard"(EM card), "M1Card"(M1 card), "CPUCard"(CPU card), "IDCard"(ID card), "DesfireCard"
      (DESFire card), "FelicaCard"(FeliCa card)*/
      "enabled": true
      /*ro, req, bool, whether to enable RF card recognition, desc:true=yes, false=no*/
    }
  ]
}

```

12.8.3.6 Set the parameters of enabling RF (Radio Frequency) card recognition

Request URL

PUT /ISAPI/AccessControl/Configuration/RFCardCfg?format=json

Query Parameter

None

Request Message

```

{
  "RFCardCfg": [
    /*req, array, the parameters of enabling RF (Radio Frequency) card recognition, subType:object*/
    {
      "cardType": "EMCard",
      /*req, enum, card type, subType:string, desc:"EMCard"(EM card), "M1Card"(M1 card), "CPUCard"(CPU card), "IDCard"(ID card), "DesfireCard"(DESFire
      card), "FelicaCard"(FeliCa card)*/
      "enabled": true
      /*req, bool, whether to enable RF card recognition, desc:"true"(yes), "false"(no)*/
    }
  ]
}

```

Response Message


```
{
  "requestURL": "test",
  /*ro, opt, string, URI*/
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "test",
  /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "test",
  /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

12.8.3.7 Set the condition parameters of face picture comparison

Request URL

PUT /ISAPI/AccessControl/FaceCompareCond

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceCompareCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, facial recognition parameters, attr:version{opt, string, protocolVersion}-->
  <faceWidthLowerLimit>
    <!--opt, int, face width threshold, range:[0,100], desc:when the detected face width is larger than this threshold, the following conditions will be ignored and the face picture comparison will be executed it has the highest priority-->50
  </faceWidthLowerLimit>
  <pitch>
    <!--opt, int, face raising or bowing angle, range:[0,90], desc:the smaller the better-->0
  </pitch>
  <yaw>
    <!--opt, int, face siding left or right angle, range:[0,90], desc:the smaller the better-->0
  </yaw>
  <width>
    <!--opt, int, face width, range:[0,100]-->50
  </width>
  <height>
    <!--opt, int, face height, range:[0,100]-->50
  </height>
  <leftBorder>
    <!--opt, int, Left border of face, range:[0,100]-->50
  </leftBorder>
  <rightBorder>
    <!--opt, int, right border of face, range:[0,100]-->50
  </rightBorder>
  <upBorder>
    <!--opt, int, top border of face, range:[0,100]-->50
  </upBorder>
  <bottomBorder>
    <!--opt, int, bottom border of face, range:[0,100]-->50
  </bottomBorder>
  <interorbitalDistance>
    <!--opt, int, pupillary distance, range:[0,100]-->50
  </interorbitalDistance>
  <faceScore>
    <!--opt, int, face score (face picture quality), range:[0,100], desc:the valid face score must be larger than this value-->50
  </faceScore>
  <maxDistance>
    <!--opt, enum, maximum recognition distance, subType:string, desc:"0.5" (0.5 m), "1" (1 m), "1.5" (1.5 m), "2" (2m), "auto" (automatic)-->0.5
  </maxDistance>
  <similarity>
    <!--opt, float, face picture comparison similarity-->50
  </similarity>
  <antiFake>
    <!--opt, int, face anti-spoofing parameters-->50
  </antiFake>
  <identifyType>
    <!--opt, enum, face recognition type, subType:string, desc:highest (the highest similarity, default), single (one picture whose similarity exceeds the threshold), multiple (multiple pictures whose similarity exceeds the threshold). If the value of this node is highest, it indicates all face pictures in the whole face picture library will be compared and the one with the highest similarity will be returned. If the value of this node is single, it indicates that the first picture whose similarity exceeds the threshold will be returned. If the value of this node is multiple, the first 30 pictures whose similarity exceeds the threshold will be returned-->highest
  </identifyType>
  <chooseType>
    <!--opt, enum, face recognition area, subType:string, desc:if "middle" is set, the device will recognize the middle face on the picture; if "biggest" is set, the device will recognize the biggest face on the picture; if "all" is set, the device will recognize all faces on the picture; "middle", "biggest", "all"; the default value is all-->all
  </chooseType>
  <enabled>
    <!--opt, enum, whether to enable face recognition, subType:string, desc:singleFace (recognizing a single face, default), close (disable facial
```

```

<!--opt, enum, whether to enable face recognition, subType:string, desc:singleFace (recognizing a single face, default), close (disable face
recognition), multiFace (recognizing multiple faces). This node can be used to enable facial recognition for the device and it will take effect in all
readers-->singleFace
</enabled>
<faceScoreEnabled>
  <!--opt, bool, whether to enable face scoring, desc:whether to enable face scoring-->true
</faceScoreEnabled>
<ageFaceMatchEnabled>
  <!--opt, bool, whether to enable age groups matching-->true
</ageFaceMatchEnabled>
<ageFaceMatchList>
  <!--opt, array, list of different age groups, subType:object, range:[0,5], dep:or,{$.FaceCompareCond.ageFaceMatchEnabled,eq,true}-->
  <ageFaceMatch>
    <!--opt, object, age groups matching-->
    <ageLevel>
      <!--opt, enum, age groups, subType:int, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are 18~40
years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
    </ageLevel>
    <ageFaceMatchThreshold1>
      <!--opt, int, matching threshold when authenticating via age groups (1:1), range:[0,100]-->1
    </ageFaceMatchThreshold1>
    <ageFaceMatchThresholdN>
      <!--opt, int, matching threshold when authenticating via age groups (1:N), range:[0,100]-->1
    </ageFaceMatchThresholdN>
    </ageFaceMatch>
  </ageFaceMatchList>
  <faceScoreThreshold1>
    <!--opt, int, matching threshold when face scoring via 1:N matching mode, range:[0,100], desc:it can rate the captured face pictures and is available
for person and ID comparison and authentication without ID card-->1
  </faceScoreThreshold1>
</FaceCompareCond>

```

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>

```

12.8.3.8 Get the facial recognition parameters.

Request URL

GET /ISAPI/AccessControl/FaceCompareCond

Query Parameter

None

Request Message

None

Response Message

```

<?xml version="1.0" encoding="UTF-8"?>

<FaceCompareCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, facial recognition parameters, attr:version{opt, string, protocolVersion}-->
  <faceWidthLowerLimit>
    <!--ro, opt, int, face width threshold, range:[0,100], desc:when the detected face width is larger than this threshold, the following conditions will be
ignored and the face picture comparison will be executed-->50
  </faceWidthLowerLimit>
  <pitch>
    <!--ro, opt, int, face pitching angle, range:[0,90], desc:the smaller the better-->0
  </pitch>
  <yaw>
    <!--ro, opt, int, face yawing angle, range:[0,90], desc:the smaller the better-->0
  </yaw>

```

```

<!--ro, opt, int, face yawing angle, range:[0,360], desc:the smaller, the better.-->0
</yaw>
<width>
<!--ro, opt, int, face width, range:[0,100]-->50
</width>
<height>
<!--ro, opt, int, face height, range:[0,100]-->50
</height>
<leftBorder>
<!--ro, opt, int, Left border of the face, range:[0,100]-->50
</leftBorder>
<rightBorder>
<!--ro, opt, int, right border of the face, range:[0,100]-->50
</rightBorder>
<upBorder>
<!--ro, opt, int, upper border of the face, range:[0,100]-->50
</upBorder>
<bottomBorder>
<!--ro, opt, int, Lower border of the face, range:[0,100]-->50
</bottomBorder>
<interorbitalDistance>
<!--ro, opt, int, pupillary distance, range:[0,100]-->50
</interorbitalDistance>
<faceScore>
<!--ro, opt, int, face score, range:[0,100], desc:the face to be detected is valid when its score is higher than the value of this node-->50
</faceScore>
<maxDistance>
<!--ro, opt, enum, maximum recognition distance, subType:string, desc:0.5 (0.5m), 1 (1m), 1.5 (1.5m), 2 (2m), auto (automatic)-->0.5
</maxDistance>
<similarity>
<!--ro, opt, float, face picture comparison similarity-->50
</similarity>
<antiFake>
<!--ro, opt, int, face anti-spoofing parameters-->50
</antiFake>
<identifyType>
<!--ro, opt, enum, facial recognition type, subType:string, desc:highest (the highest similarity, default), single (one picture whose similarity exceeds the threshold), multiple (multiple pictures whose similarity exceeds the threshold). If the value of this node is highest, it indicates all face pictures in the whole face picture library will be compared and the one with the highest similarity will be returned. If the value of this node is single, it indicates that the first picture whose similarity exceeds the threshold will be returned. If the value of this node is multiple, the first 30 pictures whose similarity exceeds the threshold will be returned-->highest
</identifyType>
<chooseType>
<!--ro, opt, enum, face selection type, subType:string, desc:middle (the face in the middle of the picture), biggest (the biggest face), all (all faces, default). If the value of this node is middle, only the face in the middle of a picture will be recognized. If the value of this node is biggest, the biggest face in a picture will be recognized. If the value of this node is all, all faces in a picture will be recognized-->all
</chooseType>
<enabled>
<!--ro, opt, enum, whether to enable facial recognition, subType:string, desc:singleFace (recognizing a single face, default), close (disable facial recognition), multiFace (recognizing multiple faces). This node can be used to enable facial recognition for the device and it will take effect in all readers-->singleFace
</enabled>
<faceScoreEnabled>
<!--ro, opt, bool, whether to enable face grading, desc:face grading is used to select the high-quality pictures during person and ID comparison or face picture collection. This field is used together with the faceScore-->true
</faceScoreEnabled>
<ageFaceMatchEnabled>
<!--ro, opt, bool, whether to enable different age groups matching-->true
</ageFaceMatchEnabled>
<ageFaceMatchList>
<!--ro, opt, array, list of different age groups, subType:object, range:[0,5], dep:or,{$.FaceCompareCond.ageFaceMatchEnabled,eq,true}-->
<ageFaceMatch>
<!--ro, opt, object, age groups matching-->
<ageLevel>
<!--ro, opt, enum, age groups, subType:int, desc:0 (child who are 0~6 years old), 1 (teenager who are 7~17years old), 2 (youth and prime who are 18~40 years old), 3 (middle age who are 41~65 years old), 4 (elderly who are more than 66 years old)-->1
</ageLevel>
<ageFaceMatchThreshold1>
<!--ro, opt, int, matching threshold when authenticating via age groups (1:1), range:[0,100]-->1
</ageFaceMatchThreshold1>
<ageFaceMatchThresholdN>
<!--ro, opt, int, matching threshold when authenticating via age groups (1:N), range:[0,100]-->1
</ageFaceMatchThresholdN>
</ageFaceMatch>
</ageFaceMatchList>
<faceScoreThreshold1>
<!--ro, opt, int, matching threshold when face scoring via 1:N matching mode, range:[0,100], desc:it can rate the captured face pictures and is available for person and ID comparison and authentication without ID card-->1
</faceScoreThreshold1>
</FaceCompareCond>

```

12.8.3.9 Get the capability of configuring facial recognition parameters.

Request URL

GET /ISAPI/AccessControl/FaceCompareCond/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceCompareCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capability of configuring facial recognition parameters, attr:version{req, string, protocolVersion}-->
  <faceWidthLowerLimit min="1" max="10">
    <!--ro, opt, int, face width threshold, range:[0,100], attr:min{req, int},max{req, int}, desc:When the width of the face to be detected exceeds the threshold, the following conditions will be ignored and the face picture comparison will be executed. That is, this parameter has the highest priority.-->1
    </faceWidthLowerLimit>
    <pitch min="1" max="10">
      <!--ro, opt, int, face pitching angle, range:[0,90], attr:min{req, int},max{req, int}, desc:The smaller, the better.-->1
    </pitch>
    <yaw min="1" max="10">
      <!--ro, opt, int, face yawing angle, range:[0,90], attr:min{req, int},max{req, int}, desc:The smaller, the better.-->1
    </yaw>
    <width min="1" max="10">
      <!--ro, opt, int, face width, range:[0,100], attr:min{req, int},max{req, int}-->1
    </width>
    <height min="1" max="10">
      <!--ro, opt, int, face height, range:[0,100], attr:min{req, int},max{req, int}-->1
    </height>
    <leftBorder min="1" max="10">
      <!--ro, opt, int, left border of the face, range:[0,100], attr:min{req, int},max{req, int}-->1
    </leftBorder>
    <rightBorder min="1" max="10">
      <!--ro, opt, int, right border of the face, range:[0,100], attr:min{req, int},max{req, int}-->1
    </rightBorder>
    <upBorder min="1" max="10">
      <!--ro, opt, int, upper border of the face, range:[0,100], attr:min{req, int},max{req, int}-->1
    </upBorder>
    <bottomBorder min="1" max="10">
      <!--ro, opt, int, lower border of the face, range:[0,100], attr:min{req, int},max{req, int}-->1
    </bottomBorder>
    <interorbitalDistance min="1" max="10">
      <!--ro, opt, int, pupillary distance, range:[0,100], attr:min{req, int},max{req, int}-->1
    </interorbitalDistance>
    <faceScore min="1" max="10">
      <!--ro, opt, int, face score, range:[0,100], attr:min{req, int},max{req, int}, desc:The face to be detected is valid when its score is higher than the value of this node.-->1
    </faceScore>
    <maxDistance opt="0.5,0.75,1,1.5,2,auto">
      <!--ro, opt, string, maximum recognition distance, attr:opt{req, string}, desc:This field takes precedence over interorbitalDistance, unit: m.-->test
    </maxDistance>
    <similarity min="0.0" max="1.0">
      <!--ro, opt, float, face picture comparison similarity, attr:min{req, int},max{req, int}-->0.000
    </similarity>
    <antiFake min="1" max="10">
      <!--ro, opt, int, face anti-spoofing parameters, attr:min{req, int},max{req, int}-->1
    </antiFake>
    <identifyType opt="highest,single,multi">
      <!--ro, opt, enum, facial recognition type, subType:string, attr:opt{req, string}, desc:highest (the highest similarity, default), single (one picture whose similarity exceeds the threshold), multiple (multiple pictures whose similarity exceeds the threshold). If the value of this node is highest, it indicates all face pictures in the whole face picture library will be compared and the one with the highest similarity will be returned. If the value of this node is single, it indicates that the first picture whose similarity exceeds the threshold will be returned. If the value of this node is multiple, the first 30 pictures whose similarity exceeds the threshold will be returned.-->highest
    </identifyType>
    <chooseType opt="middle,biggest,all">
      <!--ro, opt, enum, face selection type, subType:string, attr:opt{req, string}, desc:middle (the face in the middle of the picture), biggest (the biggest face), all (all faces, default). If the value of this node is middle, only the face in the middle of a picture will be recognized. If the value of this node is biggest, the biggest face in a picture will be recognized. If the value of this node is all, all faces in a picture will be recognized.-->middle
    </chooseType>
    <enabled opt="singleFace,close,multiFace">
      <!--ro, opt, enum, whether to enable facial recognition, subType:string, attr:opt{req, string}, desc:singleFace (recognizing a single face, default), close (disable facial recognition), multiFace (recognizing multiple faces). This node can be used to enable facial recognition for the device and it will take effect in all readers.-->singleFace
    </enabled>
    <faceScoreEnabled opt="true,false">
      <!--ro, opt, bool, whether to enable face grading, attr:opt{req, string}, desc:Face grading is used to select the high-quality pictures during person and ID comparison or face picture collection. This field is used together with the faceScore.-->true
    </faceScoreEnabled>
    <ageFaceMatchEnabled opt="true,false">
      <!--ro, opt, bool, whether to enable different age groups matching, attr:opt{req, string}-->true
    </ageFaceMatchEnabled>
    <ageFaceMatchList size="5">
      <!--ro, opt, array, List of different age groups, subType:object, range:[0,5], dep:or,{$.FaceCompareCond.ageFaceMatchEnabled,eq,true}, attr:size{req, int}-->
      <ageFaceMatch>
        <!--ro, opt, object, age groups matching-->
        <ageLevel opt="0,1,2,3,4">
          <!--ro, opt, enum, age groups, subType:int, attr:opt{req, string}, desc:0 (child who are 0-6 years old), 1 (teenager who are 7-17years old), 2 (youth and prime who are 18-40 years old), 3 (middle age who are 41-65 years old), 4 (elderly who are more than 66 years old)-->1
        </ageLevel>
        <ageFaceMatchThreshold1 min="0" max="100">
          <!--ro, opt, int, matching threshold when authenticating via age groups (1:1), range:[0,100], attr:min{req, int},max{req, int}-->1
        </ageFaceMatchThreshold1>
        <ageFaceMatchThresholdN min="0" max="100">
          <!--ro, opt, int, matching threshold when authenticating via age groups (1:N), range:[0,100], attr:min{req, int},max{req, int}-->1
        </ageFaceMatchThresholdN>
      </ageFaceMatch>
    </ageFaceMatchList>
  </FaceCompareCond>
```

```

<!--ro, opt, int, matching threshold when face scoring via 1:N matching mode, range:[0,100], attr:min{req, int},max{req, int}-->
</ageFaceMatchThreshold>
</ageFaceMatch>
</ageFaceMatchList>
<faceScoreThreshold1 min="0" max="100">
  <!--ro, opt, int, matching threshold when face scoring via 1:N matching mode, range:[0,100], attr:min{req, int},max{req, int}-->1
</faceScoreThreshold1>
</FaceCompareCond>

```

12.9 Credentials Collection

12.9.1 Card Online Collection

12.9.1.1 Get the capability of collecting card information

Request URL

GET /ISAPI/AccessControl/CaptureCardInfo/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "CardInfoCap": {
    /*ro, req, object*/
    "cardNo": {
      /*ro, opt, object, card No.*/
      "@min": 1,
      /*ro, req, int, the minimum length, range:[1,32]*/
      "@max": 32
      /*ro, req, int, the maximum length, range:[1,32]*/
    },
    "cardType": ["TypeA_M1", "TypeA_CPU", "TypeB", "ID_125K", "FelicaCard", "DesfireCard"],
    /*ro, opt, array, card type, subType:string, range:[1,6]*/
    "readerID": {
      /*ro, opt, object*/
      "@min": 1,
      /*ro, req, int*/
      "@max": 8
      /*ro, req, int*/
    }
  }
}

```

12.9.1.2 Collect card information by the card reading module of the device

Request URL

GET /ISAPI/AccessControl/CaptureCardInfo?format=json&readerID=<readerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--

Request Message

None

Response Message

```

{
  "CardInfo": {
    /*ro, req, object, card information*/
    "cardNo": "abcd1234",
    /*ro, req, string, card No.*/
    "cardType": "TypeA_M1",
    /*ro, opt, enum, card type, subType:string, desc:"TypeA_M1", "TypeA_CPU", "TypeB", "ID_125K", "FelicaCard" (FeliCa card), "DesfireCard" (DESFire card)*/
    "readerID": 1
    /*ro, opt, int, range:[1,8]*/
  }
}

```

12.9.2 Cards Management

12.9.2.1 Get card operation capability

Request URL

GET /ISAPI/AccessControl/CardOperations/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "CardOperationsCap": {
    /*ro, req, object, capability*/
    "SectionEncryption": {
      /*ro, opt, object, encrypt sector*/
      "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, enumeration, subType:string*/
      },
      "sectionNo": {
        /*ro, opt, object, section No.*/
        "@min": 0,
        /*ro, opt, int, minimum value*/
        "@max": 0
        /*ro, req, int, maximum value*/
      },
      "keyType": {
        /*ro, opt, object, verification key types: "private"-private key, "normal"-other valid keys*/
        "@opt": ["private", "normal"]
        /*ro, req, array, enumeration, subType:string*/
      },
      "password": {
        /*ro, opt, object, a hexadecimal verification key, this field is valid only when keyType is set to "normal"*/
        "@min": 0,
        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
      },
      "newKeyType": {
        /*ro, opt, object, new key types: "private"-private key, "normal"-other valid keys*/
        "@opt": ["private", "normal"]
        /*ro, req, array, enumeration, subType:string*/
      },
      "KeyA": {
        /*ro, opt, object, a hexadecimal key A password*/
        "@min": 0,
        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
      },
      "KeyB": {
        /*ro, opt, object, a hexadecimal key B password*/
        "@min": 0,
        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
      },
      "controlBits": {
        /*ro, opt, object, a hexadecimal control bit*/
        "@min": 0,
        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
      }
    }
  }
}

```

```

},
"Verification": {
/*ro, opt, object*/
"supportFunction": {
/*ro, opt, object, supported methods*/
"@opt": ["put", "get", "delete", "post"]
/*ro, req, array, enumeration, subType:string*/
},
"sectionNo": {
/*ro, opt, object, section No.*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"passwordType": {
/*ro, opt, object, password types: "KeyA" (default),"KeyB"*/
"@opt": ["KeyA", "KeyB"]
/*ro, req, array, enumeration, subType:string*/
},
"password": {
/*ro, opt, object, password*/
"@min": 0,
/*ro, req, int*/
"@max": 0
/*ro, req, int*/
}
},
"DataBlock": {
/*ro, opt, object*/
"supportFunction": {
/*ro, opt, object, supported methods*/
"@opt": ["put", "get", "delete", "post"]
/*ro, req, array, enumeration, subType:string*/
},
"addressOfBlock": {
/*ro, opt, object, block address*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"data": {
/*ro, opt, object, a hexBinary string,e.g., "f2345678abf2345678abf2345678abf2"*/
"@min": 0,
/*ro, req, int*/
"@max": 0
/*ro, req, int*/
}
},
"DataBlockCtrl": {
/*ro, opt, object*/
"supportFunction": {
/*ro, opt, object, supported methods*/
"@opt": ["put", "get", "delete", "post"]
/*ro, req, array, enumeration, subType:string*/
},
"addressOfBlock": {
/*ro, opt, object, block address*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"command": {
/*ro, opt, object, control command*/
"@opt": ["add", "minus", "copy", "paste"]
/*ro, req, array, enumeration, subType:string*/
},
"value": {
/*ro, opt, object, relative value to be changed, this field is valid only when the command is set to "add" or "minus"*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
}
},
"ControlBlock": {
/*ro, opt, object*/
"supportFunction": {
/*ro, opt, object, supported methods*/
"@opt": ["put", "get", "delete", "post"]
/*ro, req, array, enumeration, subType:string*/
},
"sectionNo": {
/*ro, opt, object, section No.*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"KeyA": {
/*ro, opt, object, a hexadecimal key A password*/
"@min": 0,

```

```

        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
    },
    "KeyB": {
        /*ro, opt, object, a hexadecimal key B password*/
        "@min": 0,
        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
    },
    "controlBits": {
        /*ro, opt, object, a hexadecimal control bit*/
        "@min": 0,
        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
    }
},
"CardProto": {
    /*ro, opt, object*/
    "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "protocol": {
        /*ro, opt, object, operation protocol types*/
        "@opt": ["TypeA", "TypeB", "TypeAB", "125K", "all"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "encryptionEnabled": {
        /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, opt, array, subType:bool*/
    }
},
"CardEncryption": {
    /*ro, opt, object*/
    "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "cardType": {
        /*ro, opt, object, card type*/
        "@opt": ["blank", "private", "encrypted"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "keyLen": {
        /*ro, opt, object, size of key for external authentication, this field is valid only when cardType is set to "encrypted"*/
        "@min": 0,
        /*ro, req, int, minimum value*/
        "@max": 0
        /*ro, req, int, maximum value*/
    },
    "key": {
        /*ro, opt, object, hexadecimal string, a 16-byte key content for external authentication*/
        "@min": 0,
        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
    }
},
"CardParam": {
    /*ro, opt, object*/
    "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "type": {
        /*ro, opt, object, card type*/
        "@opt": ["CPU1356", "PSAM1", "PSAM2", "PSAM3", "PSAM4"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "protocol": {
        /*ro, opt, object, operation protocol types*/
        "@opt": ["T0", "T1"]
        /*ro, req, array, enumeration, subType:string*/
    }
},
"CardResetResponse": {
    /*ro, opt, object*/
    "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "data": {
        /*ro, opt, object, a hexBinary string, e.g., "f2345678abf2345678abf2345678abf2"*/
        "@min": 0,
        /*ro, req, int*/

```



```

        "@max": 0
        /*ro, req, int*/
    }
},
"DataTrans": {
    /*ro, opt, object*/
    "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "content": {
        /*ro, opt, object, data to be passed through, which is encoded in Base64*/
        "@min": 0,
        /*ro, req, int*/
        "@max": 0
        /*ro, req, int*/
    }
},
"Issue": {
    /*ro, opt, object, capability of sending a request for card issuing and getting the current card issuing status and real-time card issuing results,
    related URIs: /ISAPI/AccessControl/CardOperations/localIssueRequest?format=json and /ISAPI/AccessControl/CardOperations/localIssueStatus?format=json*/
    "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, enumeration, subType:string*/
    },
    "LocalIssueRequest": {
        /*ro, req, object*/
        "operation": {
            /*ro, opt, object, operation type: "face"-issue card to be enrolled with face picture, "fingerprint"-issue card to be enrolled with
            fingerprint*/
            "@opt": ["face", "fingerprint"]
            /*ro, req, array, enumeration, subType:string*/
        },
        "FPIndex": {
            /*ro, opt, object, fingerprint storage index (card storage area). This field is valid when operation is "fingerprint"*/
            "@min": 0,
            /*ro, req, int, minimum value*/
            "@max": 0
            /*ro, req, int, maximum value*/
        },
        "facePic": {
            /*ro, opt, object, face picture type: "visible"-visible light picture, "infrared"-IR light picture. This field is valid when operation is
            "face"*/
            "@opt": ["visible", "infrared"]
            /*ro, req, array, enumeration, subType:string*/
        }
    },
    "LocalIssueRes": {
        /*ro, opt, object*/
        "status": {
            /*ro, opt, object, card issuing status: "ok"-succeeded, "failed"-card operation failed, "timeout"-timed out, "verifyFailure"-authentication
            failed, "noCard"-no card detected, "processing"-processing*/
            "@opt": ["ok", "failed", "processing", "timeout", "verifyFailure", "noCard"]
            /*ro, req, array, enumeration, subType:string*/
        },
        "cardNo": {
            /*ro, opt, object, issued card No.*/
            "@min": 0,
            /*ro, req, int*/
            "@max": 0
            /*ro, req, int*/
        },
        "cardErrorCode": {
            /*ro, opt, object*/
            "@opt": ["1"]
            /*ro, req, array, enumeration, subType:string*/
        }
    }
},
"localIssueCfg": {
    /*ro, opt, object, capability of configuring rule parameters for issuing smart cards, related URI:
    /ISAPI/AccessControl/CardOperations/localIssueCfg?format=json*/
    "validFP": {
        /*ro, opt, object, valid fingerprint ID. This field is valid for applying fingerprint to the card*/
        "@size": 2,
        /*ro, req, int*/
        "@min": 1,
        /*ro, req, int, minimum value*/
        "@max": 10
        /*ro, req, int, maximum value*/
    },
    "validFacePicture": {
        /*ro, opt, object, valid face picture type: "visible"-visible light picture, "infrared"-IR light picture. This field is valid for applying face
        picture to the card*/
        "@opt": ["visible", "infrared"]
        /*ro, req, array, enumeration, subType:string*/
    }
},
"ClearData": {
    /*ro, opt, object, capability of deleting data from the card, related URI: /ISAPI/AccessControl/CardOperations/clearData?format=json*/
    "supportFunction": {
        /*ro, opt, object, supported methods*/
    }
}

```

```

/*ro, opt, object, supported methods*/
"@opt": ["put", "get", "delete", "post"]
/*ro, req, array, enumeration, subType:string*/
},
"checkAll": {
/*ro, opt, object, whether to delete all data*/
"@opt": [true, false]
/*ro, req, array, enumeration, subType:bool*/
},
"checkFingerprint": {
/*ro, opt, object, whether to delete fingerprint data. This field is valid when checkAll is false or does not exist*/
"@opt": [true, false]
/*ro, req, array, enumeration, subType:bool*/
},
"fingerprints": {
/*ro, opt, object, list of addresses whether the fingerprints to be deleted are stored. This field is valid when checkFingerprint exists. If
this field does not exist, it indicates deleting all fingerprints*/
"@size": 2,
/*ro, req, int*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"checkFacePicture": {
/*ro, opt, object, whether to delete face data. This field is valid when checkAll is false or does not exist*/
"@opt": [true, false]
/*ro, req, array, enumeration, subType:bool*/
},
"checkCustom": {
/*ro, opt, object, whether to delete custom data. This field is valid when checkAll is false or does not exist*/
"@opt": [true, false]
/*ro, req, array, enumeration, subType:bool*/
},
"ClearDataRes": {
/*ro, opt, object*/
"status": {
/*ro, opt, object, card issuing status: "ok"-succeeded,"failed"-card operation failed,"timeout"-timed out,"verifyFailure"-authentication
failed,"noCard"-no card detected, "processing"-processing*/
"@opt": ["ok", "failed", "processing", "timeout", "verifyFailure", "noCard"]
/*ro, req, array, enumeration, subType:string*/
},
"cardErrorCode": {
/*ro, opt, object, error description*/
"@opt": ["1"]
/*ro, req, array, enumeration, subType:string*/
}
}
},
"CustomData": {
/*ro, opt, object, capability of setting custom card information, related URI: /ISAPI/AccessControl/CardOperations/customData?format=json*/
"supportFunction": {
/*ro, opt, object, supported methods*/
"@opt": ["put", "get", "delete", "post"]
/*ro, req, array, enumeration, subType:string*/
},
"address": {
/*ro, opt, object, start address for writing. By default the data will be written from the start address*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"length": {
/*ro, opt, object, length of source data to be written, it is 0 by default*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"data": {
/*ro, opt, object, a hexBinary string,e.g., "f2345678abf2345678abf2345678abf2"*/
"@min": 0,
/*ro, req, int*/
"@max": 0
/*ro, req, int*/
},
"CustomDataRes": {
/*ro, opt, object*/
"status": {
/*ro, opt, object, card issuing status: "ok"-succeeded,"failed"-card operation failed,"timeout"-timed out,"verifyFailure"-authentication
failed,"noCard"-no card detected, "processing"-processing*/
"@opt": ["ok", "failed", "processing", "timeout", "verifyFailure", "noCard"]
/*ro, req, array, enumeration, subType:string*/
},
"cardErrorCode": {
/*ro, opt, object, error description*/
"@opt": ["1"]
/*ro, req, array, enumeration, subType:string*/
}
}
},
"CustomDataSearchCond": {
/*ro, opt, object, search condition*/
"address": {

```

```

/*ro, opt, object, start address for writing. By default the data will be written from the start address*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"length": {
/*ro, opt, object, length of source data to be written, it is 0 by default*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
}
},
"CustomDataResult": {
/*ro, opt, object, result capability of searching for custom card information, related URI:
/ISAPI/AccessControl/CardOperations/customData/searchTask?format=json*/
"length": {
/*ro, opt, object, length of source data to be written, it is 0 by default*/
"@min": 0,
/*ro, req, int, minimum value*/
"@max": 0
/*ro, req, int, maximum value*/
},
"data": {
/*ro, opt, object, a hexBinary string,e.g., "f2345678abf2345678abf2345678abf2"*/
"@min": 0,
/*ro, req, int*/
"@max": 0
/*ro, req, int*/
},
"status": {
/*ro, opt, object, card issuing status: "ok"-succeeded, "failed"-card operation failed,"timeout"-timed out,"verifyFailure"-authentication
failed,"noCard"-no card detected,"processing"-processing*/
"@opt": ["ok", "failed", "processing", "timeout", "verifyFailure", "noCard"]
/*ro, req, array, enumeration, subType:string*/
},
"cardErrorCode": {
/*ro, opt, object, error description*/
"@opt": ["1"]
/*ro, req, array, enumeration, subType:string*/
}
}
},
"CardIssueStatus": {
/*ro, opt, object, capability of getting the smart card issuing status, related URI: /ISAPI/AccessControl/CardOperations/cardIssueStatus?
format=json*/
"status": {
/*ro, opt, object, card issuing status: "ok"-succeeded, "failed"-card operation failed,"timeout"-timed out,"verifyFailure"-authentication
failed,"noCard"-no card detected,"processing"-processing*/
"@opt": ["ok", "failed", "processing", "timeout", "verifyFailure", "noCard"]
/*ro, req, array, enumeration, subType:string*/
},
"cardNo": {
/*ro, opt, object, issued card No.*/
"@min": 0,
/*ro, req, int*/
"@max": 0
/*ro, req, int*/
},
"cardErrorCode": {
/*ro, opt, object, error description*/
"@opt": ["1"]
/*ro, req, array, enumeration, subType:string*/
},
"face": {
/*ro, opt, object, issuing status of the card containing the face picture: true-issued,false-not issued*/
"@opt": [true, false]
/*ro, req, array, enumeration, subType:bool*/
},
"fingerprint1": {
/*ro, opt, object, issuing status of the card containing fingerprint 1: true-issued,false-not issued*/
"@opt": [true, false]
/*ro, req, array, enumeration, subType:bool*/
},
"fingerprint2": {
/*ro, opt, object, issuing status of the card containing fingerprint 2: true-issued,false-not issued*/
"@opt": [true, false]
/*ro, req, array, enumeration, subType:bool*/
},
"customData": {
/*ro, opt, object, issuing status of the card containing custom information: true-issued,false-not issued*/
"@opt": [true, false]
/*ro, req, array, enumeration, subType:bool*/
}
}
}
}
}

```

12.9.2.2 Set card parameters (only available for CPU card)

Request URL

PUT /ISAPI/AccessControl/CardOperations/cardParam?format=json

Query Parameter

None

Request Message

```
{
  "CardParam": {
    /*req, object, card type*/
    "type": "CPU1356",
    /*req, enum, card type, subType:string, desc:"CPU1356", "PSAM1", "PSAM2", "PSAM3", "PSAM4"*/
    "protocol": "T0",
    /*req, enum, card protocol type, subType:string, desc:"T0", "T1"*/
    "encryptionEnabled": true
  }
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
}
```

12.9.2.3 Change the control block of a specific section of M1 card

Request URL

PUT /ISAPI/AccessControl/CardOperations/controlBlock?format=json

Query Parameter

None

Request Message

```
{
  "ControlBlock": {
    /*req, object*/
    "sectionNo": 1,
    /*req, int, section No., range:[0,15]*/
    "KeyA": "test",
    /*opt, string, key A password, desc:a hexadecimal string*/
    "KeyB": "test",
    /*opt, string, key B password, desc:a hexadecimal string*/
    "controlBits": "test"
  }
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
}
```

12.9.2.4 Get data block (only available for M1 card)

Request URL

GET /ISAPI/AccessControl/CardOperations/dataBlock/<address>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
address	string	--

Request Message

None

Response Message

```
{
  "DataBlock": {
    /*ro, req, object, data block*/
    "addressOfBlock": 1,
    /*ro, opt, int, block address, range:[0,63]*/
    "data": "test"
    /*ro, req, string, data*/
  }
}
```

12.9.2.5 Write data block (only available for M1 card)

Request URL

PUT /ISAPI/AccessControl/CardOperations/dataBlock/<address>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
address	string	--

Request Message

```
{
  "DataBlock": {
    /*req, object*/
    "addressOfBlock": 1,
    /*opt, int, block address, range:[0,63]*/
    "data": "test"
    /*req, string, a hexBinary character string*/
  }
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

12.9.2.6 Do operations (i.e., plus, minus, copy, and paste) on the data block

Request URL

PUT /ISAPI/AccessControl/CardOperations/dataBlock/control?format=json

Query Parameter

None

Request Message

```
{
  "DataBlockCtrl": {
    /*req, object*/
    "addressOfBlock": 1,
    /*req, int, block address, range:[0,63]*/
    "command": "add",
    /*req, enum, control commands, subType:string, desc:"add", "minus", "copy", "paste"*/
    "value": 1
  }
  /*opt, int, relative value to be changed*/
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
}
```

12.9.2.7 Transmit the data package (only available for CPU card)

Request URL

PUT /ISAPI/AccessControl/CardOperations/dataTrans?format=json

Query Parameter

None

Request Message

```
{
  "DataTrans": {
    /*req, object, data pass-through information*/
    "content": "test"
  }
  /*req, string, data to be passed through, desc:it is encoded by Base64*/
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred.*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred.*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:This field is required when the value of statusCode is not 1, and it corresponds to subStatusCode.*/
  "errorMsg": "ok"
  /*ro, opt, string, error details, desc:This field is required when the value of statusCode is not 1.*/
}
```

12.9.2.8 Set operation protocol type for the card

Request URL

PUT /ISAPI/AccessControl/CardOperations/protocol?format=json

Query Parameter

None

Request Message

```

{
  "CardProto": {
    /*req, object*/
    "protocol": "TypeA"
    /*req, enum, operation protocol type, subType:string, desc:"TypeA, TypeB, TypeAB, 125K, all"*/
  }
}

```

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}

```

12.9.2.9 Reset card parameters (only available for CPU card)

Request URL

GET /ISAPI/AccessControl/CardOperations/reset?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "CardResetResponse": {
    /*ro, req, object*/
    "data": "test"
    /*ro, req, string, resetting response information, desc:resetting response information*/
  }
}

```

12.9.2.10 Set the encryption parameters of a specific section (only available for M1 card)

Request URL

PUT /ISAPI/AccessControl/CardOperations/sectionEncryption?format=json

Query Parameter

None

Request Message

```

{
  "SectionEncryption": {
    /*req, object*/
    "sectionNo": 0,
    /*req, int, section No., range:[0,15]*/
    "keyType": "private",
    /*req, enum, key type, subType:string, desc:"private" (private key), "normal" (other valid keys)*/
    "password": "test",
    /*opt, string, a hexadecimal verification key, desc:this field is valid only when the keyType is "normal"*/
    "newKeyType": "private",
    /*req, enum, new key type, subType:string, desc:"private" (private key), "normal" (other valid keys)*/
    "KeyA": "test",
    /*opt, string, a hexadecimal password of key A, this field is valid only when the keyType is "normal", desc:this field is valid only when the keyType
is "normal"*/
    "KeyB": "test",
    /*opt, string, a hexadecimal password of key B, this field is valid only when the keyType is "normal", desc:this field is valid only when the keyType
is "normal"*/
    "controlBits": "test"
    /*opt, string, a hexadecimal control bit, desc:this field is valid only when the keyType is "normal"*/
  }
}

```

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error information, desc:this field is required when statusCode is not 1*/
}

```

12.9.2.11 Verify the password of the encrypted section (only available for M1 card).

Request URL

PUT /ISAPI/AccessControl/CardOperations/verification?format=json

Query Parameter

None

Request Message

```

{
  "Verification": {
    /*req, object*/
    "sectionNo": 1,
    /*req, int, section No., range:[0,15]*/
    "passwordType": "KeyA",
    /*opt, enum, password types, subType:string, desc:"KeyA", "KeyB"; the default value is KeyA*/
    "password": "test"
    /*opt, string, password, desc:password*/
  }
}

```

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}

```

12.9.3 Face Picture Collecting

12.9.3.1 Collect face picture information

Request URL

POST /ISAPI/AccessControl/CaptureFaceData

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceDataCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <captureInfrared>
    <!--opt, bool, whether to collect infrared face pictures simultaneously, desc:"true"-yes, "false"-no-->true
  </captureInfrared>
  <dataType>
    <!--opt, enum, data type of collected face pictures, subType:string, desc:"url" (default), "binary"-->url
  </dataType>
  <readerID>
    <!--opt, int, range:[1,8]-->1
  </readerID>
</CaptureFaceDataCond>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <CaptureFaceDataCond>
    <!--ro, opt, object-->
    <captureInfrared opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </captureInfrared>
    <dataType opt="url,binary">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->url
    </dataType>
  </CaptureFaceDataCond>
  <faceDataURL min="1" max="768">
    <!--ro, opt, string, face data URL, if this node does not exist, it indicates that there is no face data, attr:min{req, int},max{req, int}-->test
  </faceDataURL>
  <captureProgress min="1" max="100">
    <!--ro, opt, int, collection progress, range:[0,100], attr:min{req, int},max{req, int}-->1
  </captureProgress>
  <infraredFaceDataURL min="1" max="100">
    <!--ro, opt, string, infrared face data URL, if this node does not exist, it indicates that there is no infrared face data, attr:min{req, int},max{req,
int}-->test
  </infraredFaceDataURL>
  <modelData>
    <!--ro, opt, string-->test
  </modelData>
  <score>
    <!--ro, opt, int, face score (face picture quality), range:[0,100]-->0
  </score>
  <thermometryUnit>
    <!--ro, opt, string-->test
  </thermometryUnit>
  <currTemperature>
    <!--ro, opt, float-->0.0
  </currTemperature>
  <visibleLightURL>
    <!--ro, opt, string-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
</CaptureFaceData>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
CaptureFace	[Message content]	application/xml	--	--	--
FaceData	[Binary picture data]	image/jpeg		FaceDataac	--
InfraredFaceData	[Binary picture data]	image/jpeg		InfraredFaceData.jpg	--
faceMatting	[Binary picture data]	image/jpeg		faceMatting.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

statusCode	statusString	subStatusCode	errorCode	description
2	Device Busy	deviceBusy	0x20000004	--
3	Device Error	captureTimeout	0x30006000	--
4	Invalid Operation	fileUploadFailed	0x4000600A	--
4	Invalid Operation	faceLowQuality	0x40006010	--
4	Invalid Operation	facePictureAndIdPhotoMismatch	0x4000A1CD	--

12.9.3.2 Get the capability of collecting face picture information.

Request URL

GET /ISAPI/AccessControl/CaptureFaceData/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of collecting face picture information, attr:version{req, string, protocolVersion}-->
  <CaptureFaceDataCond>
    <!--ro, req, bool, whether to collect the infrared face data-->true
    <captureInfrared opt="true,false">
      <!--ro, opt, bool, whether to collect the infrared face data, attr:opt{req, string}-->true
    </captureInfrared>
    <dataType opt="url,binary">
      <!--ro, opt, enum, data type of collected face pictures, subType:string, attr:opt{req, string}, desc:url (URL, default),binary (binary)-->url
    </dataType>
  </CaptureFaceDataCond>
  <faceDataUrl min="1" max="768">
    <!--ro, opt, string, face data URL, range:[1,768], attr:min{req, int},max{req, int}-->1
  </faceDataUrl>
  <captureProgress min="1" max="10">
    <!--ro, req, int, collection progress, range:[1,10], attr:min{req, int},max{req, int}-->1
  </captureProgress>
  <infraredFaceDataUrl min="1" max="100">
    <!--ro, req, string, infrared picture URL, range:[1,100], attr:min{req, int},max{req, int}-->test
  </infraredFaceDataUrl>
  <modelData min="1" max="10">
    <!--ro, opt, string, face modeling data encoded by Base64, attr:min{req, int},max{req, int}-->test
  </modelData>
  <score min="0" max="100">
    <!--ro, opt, int, face score, range:[0,100], attr:min{req, int},max{req, int}-->80
  </score>
  <thermometryUnit opt="celsius,fahrenheit,kelvin">
    <!--ro, opt, string, temperature unit: celsius (Celsius, default), attr:opt{req, string}-->test
  </thermometryUnit>
  <currTemperature min="0" max="10">
    <!--ro, opt, float, face temperature which is accurate to one decimal place, attr:min{req, int},max{req, int}-->0.000
  </currTemperature>
  <visibleLightURL min="0" max="10">
    <!--ro, opt, string, URL of the visible light picture captured by the thermal camera, attr:min{req, int},max{req, int}-->test
  </visibleLightURL>
  <thermalURL min="0" max="10">
    <!--ro, opt, string, thermal picture URL, attr:min{req, int},max{req, int}-->test
  </thermalURL>
  <readerID min="1" max="8">
    <!--ro, opt, int, range:[1,8], attr:min{req, int},max{req, int}-->1
  </readerID>
</CaptureFaceData>
```

12.9.3.3 Get capability of getting face picture collection progress

Request URL

GET /ISAPI/AccessControl/CaptureFaceData/Progress/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <faceDataUrl min="0" max="10">
    <!--ro, opt, string, face data URL, range:[1,32], attr:min{req, int},max{req, int}, desc:face data URL,if this node does not exist,it indicates that there is no face data-->test
  </faceDataUrl>
  <captureProgress min="0" max="100">
    <!--ro, req, int, collection progress, range:[0,100], attr:min{req, int},max{req, int}, desc:collection progress,which is between 0 and 100,0-no face data collected,100-collected,the face data URL can be parsed only when the progress is 100-->1
  </captureProgress>
  <isCurRequestOver opt="true,false">
    <!--ro, opt, bool, whether the current collection request is completed, attr:opt{req, string}-->true
  </isCurRequestOver>
  <infraredFaceDataUrl min="0" max="10">
    <!--ro, opt, string, infrared face data URL, range:[1,32], attr:min{req, int},max{req, int}, desc:if this node does not exist, it indicates that there is no infrared face data-->test
  </infraredFaceDataUrl>
  <readerID min="1" max="8">
    <!--ro, opt, int, range:[1,8], attr:min{req, int},max{req, int}-->1
  </readerID>
  <requireReaderID>
    <!--ro, opt, bool-->true
  </requireReaderID>
</CaptureFaceData>
```

12.9.3.4 Get the progress of collecting face picture information

Request URL

GET /ISAPI/AccessControl/CaptureFaceData/Progress?readerID=<readerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	The intelligent host supports capturing faces by card reader.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFaceData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <faceDataUrl>
    <!--ro, opt, string, face data URL, range:[1,32]-->test
  </faceDataUrl>
  <captureProgress>
    <!--ro, req, int, collection progress, range:[0,100], desc:collection progress,which is between 0 and 100,0-no face data collected,100-collected,the face data URL can be parsed only when the progress is 100-->100
  </captureProgress>
  <isCurRequestOver>
    <!--ro, opt, bool, whether the current collection request is completed, desc:whether the current collection request is completed: "true"-yes, "false"-no-->true
  </isCurRequestOver>
  <infraredFaceDataUrl>
    <!--ro, opt, string, infrared face data URL, range:[1,32], desc:if this node does not exist, it indicates that there is no infrared face data-->test
  </infraredFaceDataUrl>
  <faceMattingURL>
    <!--ro, opt, string, range:[1,32]-->test
  </faceMattingURL>
</CaptureFaceData>
```

statusCode	statusString	subStatusCode	errorCode	description
4	Invalid Operation	fileUploadFailed	0x4000600A	--

12.9.4 Fingerprint Online Collection

12.9.4.1 Collect fingerprint information

Request URL

POST /ISAPI/AccessControl/CaptureFingerPrint

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFingerPrintCond xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, Collect fingerprint information conditions, attr:version{req, string, protocolVersion}-->
  <fingerNo>
    <!--req, int, finger No., range:[1,10]-->1
  </fingerNo>
</CaptureFingerPrintCond>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFingerPrint xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <fingerData>
    <!--ro, opt, string, fingerprint data, range:[1,768], desc:which, should be encoded by Base64-->test
  </fingerData>
  <fingerNo>
    <!--ro, req, int, finger No., range:[1,10]-->1
  </fingerNo>
  <fingerPrintQuality>
    <!--ro, req, int, fingerprint quality, range:[1,100]-->1
  </fingerPrintQuality>
</CaptureFingerPrint>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
CaptureFingerPrint	[Message content]	application/xml	--	--	--
fingerPrintPic	[Binary picture data]	image/jpeg		fingerPrintPic.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

statusCode	statusString	subStatusCode	errorCode	description
2	Device Busy	deviceBusy	0x20000004	--
3	Device Error	captureTimeout	0x30006000	--
3	Device Error	fingerPrintLowQuality	0x3000600B	--

12.9.4.2 Get the fingerprint collection capability

Request URL

GET /ISAPI/AccessControl/CaptureFingerPrint/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CaptureFingerPrint xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, collect fingerprint information, attr:version{req, string, protocolVersion}-->
  <CaptureFingerPrintCond>
    <!--ro, req, object, finger No.-->
    <fingerNo min="1" max="10">
      <!--ro, opt, int, fingerprint No., range:[1,10], attr:min{req, int},max{req, int}-->1
    </fingerNo>
  </CaptureFingerPrintCond>
  <fingerData min="1" max="768">
    <!--ro, opt, string, fingerprint data, range:[1,768], attr:min{req, int},max{req, int}-->test
  </fingerData>
  <fingerNo min="1" max="10">
    <!--ro, opt, int, fingerprint No., range:[1,10], attr:min{req, int},max{req, int}-->1
  </fingerNo>
  <fingerPrintQuality min="1" max="100">
    <!--ro, opt, int, fingerprint quality, range:[1,100], attr:min{req, int},max{req, int}-->1
  </fingerPrintQuality>
</CaptureFingerPrint>
```

12.9.5 Offline Collection

12.9.5.1 Get the offline collection capability

Request URL

GET /ISAPI/AccessControl/OfflineCapture/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "OfflineCaptureCap": {
    /*ro, req, object, capability*/
    "isSupportDownloadOfflineCaptureInfoTemplate": true,
    /*ro, opt, bool, whether it supports downloading the template of offline user list*/
    "isSupportUploadOfflineCaptureInfo": true,
    /*ro, opt, bool, whether it supports uploading files with offline user list*/
    "isSupportUserInfo": true,
    /*ro, opt, bool, whether it supports issuing the offline collection user information in JSON format*/
    "isSupportDownloadCaptureData": true,
    /*ro, opt, bool, whether it supports downloading collected data*/
    "isSupportDeleteAllData": true,
    /*ro, opt, bool, whether it supports deleting all collected data*/
    "isSupportDeleteTheData": true,
    /*ro, opt, bool, whether it supports deleting specific collected data*/
    "isSupportDeleteDataCollection": true,
    /*ro, opt, bool, whether it supports deleting collected data by conditions*/
    "isSupportOfflineCaptureDataEvent": true,
    /*ro, opt, bool, whether it supports offline collection event*/
    "isSupportOfflineCaptureDataEventConfirm": true,
    /*ro, opt, bool, whether it supports offline collection event acknowledgement, desc:/ISAPI/AccessControl/OfflineCapture/CaptureEventConfirm?
format=json*/
    "SearchTask": {
      /*ro, opt, object, Search capability*/
      "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, options, subType:string*/
      },
      "searchID": {
        /*ro, opt, object, search ID*/
        "@min": 0,
        /*ro, req, int, the minimum length*/
        "@max": 0
      }
    }
  }
}
```

```

/*ro, req, int, the maximum length*/
},
"maxResults": {
/*ro, opt, object, the maximum search results*/
"@min": 0,
/*ro, req, int, the minimum value*/
"@max": 0
/*ro, req, int, the maximum value*/
},
"captureNoList": {
/*ro, opt, object, collection No. list*/
"maxSize": 1,
/*ro, req, int, the maximum collection No.*/
"@min": 0,
/*ro, req, int, the minimum value*/
"@max": 0
/*ro, req, int, the maximum value*/
},
"employeeNoList": {
/*ro, opt, object, employee No. list*/
"maxSize": 1,
/*ro, req, int, the maximum employee No.*/
"@min": 0,
/*ro, req, int, the minimum length*/
"@max": 0
/*ro, req, int, the maximum length*/
},
"searchType": {
/*ro, opt, object, collection No.*/
"@opt": ["new", "modified"]
/*ro, req, array, options, subType:string*/
},
"DataCollections": {
/*ro, opt, object, list of matched records*/
"maxSize": 0,
/*ro, req, int, the maximum data number*/
"captureNo": {
/*ro, opt, object, collection No.*/
"@min": 0,
/*ro, req, int, the minimum value*/
"@max": 0
/*ro, req, int, the maximum value*/
},
"name": {
/*ro, opt, object, name*/
"@min": 0,
/*ro, req, int, the minimum length*/
"@max": 0
/*ro, req, int, the maximum length*/
},
"userType": {
/*ro, opt, object, person type, desc:"normal"-normal person (resident), "visitor"-visitor, "blacklist"-person in the blacklist. "patient",
"maintenance"-maintenance person, cleaner, etc.;*/
"@opt": ["normal", "visitor", "blacklist", "maintenance"]
/*ro, req, array, person type, subType:string*/
},
"employeeNo": {
/*ro, opt, object, employee No.*/
"@min": 0,
/*ro, req, int, the minimum length*/
"@max": 0
/*ro, req, int, the maximum length*/
},
"CardNoList": {
/*ro, opt, object, card No. list*/
"maxSize": 0,
/*ro, req, int, the maximum number of arrays*/
"cardNo": {
/*ro, opt, object, card No.*/
"@min": 0,
/*ro, req, int, the minimum length*/
"@max": 0
/*ro, req, int, the maximum length*/
},
"cardType": {
/*ro, opt, object, card type*/
"@opt": ["TypeA_M1", "TypeA_CPU", "TypeB", "ID_125K", "FelicaCard", "DesfireCard"]
/*ro, req, array, options, subType:string*/
}
},
"IDCardNo": {
/*ro, opt, object, ID card No.*/
"@min": 0,
/*ro, req, int, the minimum length*/
"@max": 0
/*ro, req, int, the maximum length*/
},
"Valid": {
/*ro, opt, object, parameters of the validity period*/
"enable": {
/*ro, req, object, whether to enable validity period*/
"@opt": [true, false]
/*ro, req, array, "true"- enable, "false"-disable, subType:bool*/
}
},

```

```

        "timeRangeBegin": "1970-01-01T00:00:00",
        /*ro, opt, string, start time that can be configured for beginTime, desc:if the device does not return this node, the default start time
that can be configured for beginTime is "1970-01-01T00:00:00"*/
        "timeRangeEnd": "2037-12-31T23:59:59",
        /*ro, opt, string, end time that can be configured for endTime, desc:if the device does not return this node, the default start time
that can be configured for beginTime is "1970-01-01T00:00:00"*/
        "timeType": {
            /*ro, opt, object, time type*/
            "@opt": ["local", "UTC"]
            /*ro, req, array, options, subType:string*/
        }
    },
    "FingerprintList": {
        /*ro, opt, object, fingerprint list*/
        "fingerprintID": {
            /*ro, opt, object, fingerprint No.*/
            "@min": 0,
            /*ro, req, int, the minimum value*/
            "@max": 0
            /*ro, req, int, the maximum value*/
        },
        "fingerprint": {
            /*ro, opt, object, fingerprint information*/
            "@min": 0,
            /*ro, req, int, the minimum length*/
            "@max": 0
            /*ro, req, int, the maximum length*/
        }
    },
    "faceURL": {
        /*ro, opt, object, Face picture URL*/
        "@min": 1,
        /*ro, req, int, the minimum length*/
        "@max": 256
        /*ro, req, int, the maximum length*/
    },
    "infraredFaceURL": {
        /*ro, opt, object*/
        "@min": 1,
        /*ro, req, int*/
        "@max": 256
        /*ro, req, int*/
    },
    "FaceFeature": {
        /*ro, opt, object, facial features*/
        "isSupportFaceRegion": true,
        /*ro, opt, bool, whether it supports configuring areas for facial feature detection*/
        "isSupportCommonPoint": true
        /*ro, opt, bool, whether it supports feature point coordinates*/
    },
    "isSupportRiskMark": true,
    /*ro, opt, bool, whether it supports risk data mark*/
    "dataType": {
        /*ro, opt, object, data type*/
        "@opt": ["new", "modified", "normal"]
        /*ro, req, array, options, subType:string*/
    },
    "IdentityInfo": {
        /*ro, opt, object, ID card information*/
        "chnName": {
            /*ro, opt, object, Chinese name*/
            "@min": 0,
            /*ro, req, int, the minimum length*/
            "@max": 0
            /*ro, req, int, the maximum length*/
        },
        "enName": {
            /*ro, opt, object, English name*/
            "@min": 0,
            /*ro, req, int, the minimum length*/
            "@max": 0
            /*ro, req, int, the maximum length*/
        },
        "sex": {
            /*ro, opt, object, gender*/
            "@opt": ["1", "2", "0"]
            /*ro, req, array, options, subType:string*/
        },
        "birth": {
            /*ro, opt, object, date of birth*/
            "@min": 0,
            /*ro, req, int, the minimum length*/
            "@max": 0
            /*ro, req, int, the maximum length*/
        },
        "addr": {
            /*ro, opt, object, address*/
            "@min": 0,
            /*ro, req, int, the minimum length*/
            "@max": 0
            /*ro, req, int, the maximum length*/
        },
        "IDCardNo": {

```



```

/*ro, opt, object, ID card No.*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"issuingAuthority": {
/*ro, opt, object, issuing authority*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"startDate": {
/*ro, opt, object, start date of validity period*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"endDate": {
/*ro, opt, object, end date of validity period*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"nation": {
/*ro, opt, object, N/A*/
  "@min": 1,
  /*ro, req, int, the minimum value, range:[1,56]*/
  "@max": 56
  /*ro, req, int, the maximum value, range:[1,56]*/
},
"passNo": {
/*ro, opt, object, entry-exit permit No.*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"issueNumber": {
/*ro, opt, object, issued times*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"certificateType": {
/*ro, opt, object, certificate type*/
  "@opt": ["1", "2", "3"]
  /*ro, req, array, options, subtype:string*/
},
"permanentResidenceCardNo": {
/*ro, opt, object, permanent resident visa No.*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"nationalityOrAreaCode": {
/*ro, opt, object, country/region code*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"version": {
/*ro, opt, object, certificate version No.*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"receivingAuthorityCode": {
/*ro, opt, object, acceptance authority code*/
  "@min": 0,
  /*ro, req, int, the minimum length*/
  "@max": 0
  /*ro, req, int, the maximum length*/
},
"FingerprintList": {
/*ro, opt, object, fingerprint information list*/
  "maxSize": 0,
  /*ro, req, int, the maximum number of arrays*/
  "fingerprint": {
/*ro, opt, object, fingerprint information*/
    "@min": 0,
    /*ro, req, int, the minimum length*/
    "@max": 0
    /*ro, req, int, the maximum length*/
  }
},
"nic": {

```

```

        /*ro, opt, object, certificate picture information*/
        "@min": 0,
        /*ro, req, int, the minimum length*/
        "@max": 0
        /*ro, req, int, the maximum length*/
    }
},
"CardIssueStatus": {
    /*ro, opt, object, issuing status list of cards containing face pictures and fingerprints*/
    "@size": 0,
    /*ro, req, int, capability of number of elements in the array*/
    "face": {
        /*ro, opt, object, issuing status of the card with face information*/
        "@opt": [true, false]
        /*ro, req, array, options, subType:bool*/
    },
    "fingerprint1": {
        /*ro, opt, object, issuing status of the card with fingerprint 1 information*/
        "@opt": [true, false]
        /*ro, req, array, options, subType:bool*/
    },
    "fingerprint2": {
        /*ro, opt, object, issuing status of the card with fingerprint 2 information*/
        "@opt": [true, false]
        /*ro, req, array, options, subType:bool*/
    }
}
},
"RuleInfo": {
    /*ro, opt, object, rule list*/
    "reqAdminRights": [true, false],
    /*ro, opt, array, whether the administrator permission is required, subType:bool*/
    "enableCardNoLenAuto": [true, false],
    /*ro, opt, array, whether to enable length self-adaption of the card serial No., subType:bool*/
    "maxSize": 0,
    /*ro, opt, int, the maximum number of rule list*/
    "supportFunction": {
        /*ro, opt, object, supported methods*/
        "@opt": ["put", "get", "delete", "post"]
        /*ro, req, array, options, subType:string*/
    },
    "dataType": {
        /*ro, opt, object, data type*/
        "@opt": ["name", "employeeNo", "IDCardNo", "IDCardSerialNo", "IDCardDetails", "card", "fingerprint", "face"]
        /*ro, req, array, options, subType:string*/
    },
    "enable": [true, false],
    /*ro, opt, array, whether to collect and display data, subType:bool*/
    "uniqueCheck": [true, false],
    /*ro, opt, array, whether to enable uniqueness verification, subType:bool*/
    "len": [
        /*ro, opt, array, the capability list will be returned according to the data type, subType:object*/
        {
            "dataType": "name",
            /*ro, req, enum, data type, subType:string, desc:data type*/
            "@min": 0,
            /*ro, req, int, the minimum length*/
            "@max": 0
            /*ro, req, int, the maximum length*/
        }
    ],
    "num": [
        /*ro, opt, array, the capability list will be returned according to the data type, subType:object*/
        {
            "dataType": "card",
            /*ro, req, enum, data type, subType:string, desc:data type*/
            "@min": 0,
            /*ro, req, int, the minimum value*/
            "@max": 0
            /*ro, req, int, the maximum value*/
        }
    ],
    "fingerprintIDs": {
        /*ro, opt, object, fingerprint No.*/
        "maxSize": 0,
        /*ro, req, int, the maximum value*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 0
        /*ro, req, int, the maximum value*/
    },
    "enableLocalIssueCard": {
        /*ro, opt, object, whether to enable issuing smart cards locally*/
        "@opt": [true, false]
        /*ro, req, array, options, subType:bool*/
    },
    "isLocalStorage": {
        /*ro, opt, object, whether to store face picture and fingerprint information in the device locally*/
        "@opt": [true, false]
        /*ro, req, array, options, subType:bool*/
    }
}
},

```

```
captureProgress": {  
    /*ro, opt, object, collection progress*/  
    "supportFunction": {  
        /*ro, opt, object, supported methods*/  
        "@opt": ["put", "get", "delete", "post"]  
        /*ro, req, array, options, subtype:string*/  
    },  
    "reqCaptureNum": {  
        /*ro, opt, object, total number of persons to be collected*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "completelyCaptureNum": {  
        /*ro, opt, object, number of completely collected persons*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "partiallyCaptureNum": {  
        /*ro, opt, object, number of partially collected persons*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "reqFaceNum": {  
        /*ro, opt, object, number of faces to be collected*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "faceNum": {  
        /*ro, opt, object, number of collected faces*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "reqFingerprintNum": {  
        /*ro, opt, object, number of fingerprints to be collected*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "fingerprintNum": {  
        /*ro, opt, object, number of collected fingerprints*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "reqCardNum": {  
        /*ro, opt, object, number of cards to be collected*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "cardNum": {  
        /*ro, opt, object, number of collected cards*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "reqIDCardNum": {  
        /*ro, opt, object, number of ID cards to be collected*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "IDCardNum": {  
        /*ro, opt, object, number of collected ID cards*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "reqIssueNum": {  
        /*ro, opt, object, number of persons to be issued with smart cards*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    },  
    "IssuedNum": {  
        /*ro, opt, object, number of persons that have been issued with smart cards*/  
        "@min": 0,  
        /*ro, req, int, the minimum value*/  
        "@max": 0  
        /*ro, req, int, the maximum value*/  
    }  
}
```

```

    string, options*/

    object, exporting progress*/
    0,
    , int, the minimum value*/
    0
    , int, the maximum value*/

    "details": {
        , failing details uploading*/
        : {
            object, detailed description*/
            0,
            , int, the minimum length*/
            0
            , int, the maximum length*/

        "actions": {
            , offline collection deleting (by conditions)*/
            "list": {
                object, employee No. list*/
                : 1,
                , int, the maximum number of employee No.*/
                0,
                , int, the minimum length*/
                0
                , int, the maximum length*/
            }
        }
    }
}

```

Request URL

Query Parameter

Request Message

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "OK",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok",
  /*ro, opt, string, error information, desc:this field is required when statusCode is not 1*/
  "MErrCode": "0x00000000",
  /*ro, opt, string*/
  "MErrDevSelfEx": "0x00000000"
  /*ro, opt, string*/
}

```

12.9.5.3 Download offline data

Request URL

POST /ISAPI/AccessControl/OfflineCapture/DataCollections/downloadTask?format=json

Query Parameter

None

Request Message

```

{
  "DataCollectionsCond": {
    /*req, object, export condition*/
    "id": "test",
    /*req, string, downloading ID*/
    "dataType": "binary"
    /*req, enum, file data type, subType:string, desc:file data type*/
  }
}

```

Response Message

```

{
  "DataCollections": {
    /*ro, req, object, offline data*/
    "dataType": "binary",
    /*ro, req, enum, file data type, subType:string, desc:file data type*/
    "fileUrl": "test"
    /*ro, opt, string, file URL, range:[1,256]*/
  }
}

```

12.9.5.4 Search for the collected data

Request URL

POST /ISAPI/AccessControl/OfflineCapture/DataCollections/searchTask?format=json

Query Parameter

None

Request Message

```

{
  "SearchTaskCond": {
    /*req, object, search condition*/
    "searchID": "test",
    /*req, string, search ID which is used to check whether the upper-layer clients are the same one, range:[1,32]*/
    "searchResultPosition": 0,
    /*req, int, result filter*/
    "maxResults": 1,
    /*req, int, result filter, range:[1,2000000000], step:1*/
    "captureNoList": [1, 2],
    /*opt, array, search condition, subType:int, desc:collection No. list. If the collection No. is not configured, it will search all data according to searchResultPosition*/
    "searchType": "new",
    /*opt, enum, search condition, subType:string, desc:"new"-search and only return newly added data, "modified"-search and only return edited data. By default all data will be searched*/
    "EmployeeNoList": [
      /*opt, array, subType:object*/
      {
        "employeeNo": "test"
        /*opt, string, employee ID, range:[1,32]*/
      }
    ]
  }
}

```

Response Message

```

{
  "SearchTaskResponse": {
    /*ro, req, object*/
    "searchID": "test",
    /*ro, req, string, search ID which is used to check whether the upper-layer clients are the same one, range:[1,32]*/
    "responseStatusStrg": "OK",
    /*ro, opt, enum, status search, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).*/
    "numOfMatches": 1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches": 1,
    /*ro, opt, int, total number of matched results*/
    "DataCollections": [
      /*ro, opt, array, searched matched data information, subType:object*/
      {
        "lastCaptureNo": 1,
        /*ro, req, int, last collection No., range:[1,2000000000]*/
        "captureNo": 1,
        /*ro, req, int, current collection No., range:[1,2000000000]*/
        "name": "test",
        /*ro, opt, string, name*/
        "userType": "normal",
        /*ro, opt, enum, subType:string*/
        "employeeNo": "test",
        /*ro, opt, string, employee ID*/
        "IDCardNo": "test",
        /*ro, opt, string, ID Card No.*/
        "Valid": {
          /*ro, opt, object*/
          "enable": true,
          /*ro, req, bool*/
          "beginTime": "2017-08-01T17:30:08+08:00",
          /*ro, req, datetime*/
          "endTime": "2017-08-01T17:30:08+08:00",
          /*ro, req, datetime*/
          "timeType": "local"
          /*ro, opt, enum, subType:string*/
        },
        "CardNoList": [
          /*ro, opt, array, card No. list, subType:object*/
          {
            "cardNo": "test",
            /*ro, opt, string, card No., range:[1,32]*/
            "cardType": "TypeA_M1"
            /*ro, opt, enum, card type, subType:string, desc:"TypeA_M1", "TypeA_CPU", "TypeB", "ID_125K", "FelicaCard", "DesfireCard"*/
          }
        ],
        "FingerprintList": [
          /*ro, opt, array, fingerprint data list, subType:object*/
          {
            "fingerprintID": 1,
            /*ro, opt, int, fingerprint No.*/
            "fingerprint": "test"
            /*ro, opt, string, fingerprint information, desc:base64 decoding*/
          }
        ],
        "faceURL": "test",
        /*ro, opt, string, face picture URL, range:[1,256]*/
        "FaceFeature": {
          /*ro, req, object, feature information of face picture matting*/
          "Region": {
            /*ro, req, object, area, desc:the origin is the upper-left corner of the screen*/
            "height": 0.120,

```

```

        /*ro, req, float, height, range:[0.000,1.000]*/
        "width": 0.120,
        /*ro, req, float, width, range:[0.000,1.000]*/
        "x": 0.120,
        /*ro, req, float, X-coordinate of the upper-left corner, range:[0.000,1.000]*/
        "y": 0.120
        /*ro, req, float, Y-coordinate of the upper-left corner, range:[0.000,1.000]*/
    },
    "LeftEyePoint": {
        /*ro, opt, object, coordinates of the left eye, desc:the origin is the upper-left corner of the screen*/
        "x": 0.120,
        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
        "y": 0.120
        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
    },
    "RightEyePoint": {
        /*ro, opt, object, coordinates of the right eye, desc:the origin is the upper-left corner of the screen*/
        "x": 0.120,
        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
        "y": 0.120
        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
    },
    "LeftMouthPoint": {
        /*ro, opt, object, coordinates of the left mouth corner, desc:the origin is the upper-left corner of the screen*/
        "x": 0.120,
        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
        "y": 0.120
        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
    },
    "RightMouthPoint": {
        /*ro, opt, object, coordinates of the right mouth corner, desc:the origin is the upper-left corner of the screen*/
        "x": 0.120,
        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
        "y": 0.120
        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
    },
    "NoseTipPoint": {
        /*ro, opt, object, coordinates of the nose, desc:the origin is the upper-left corner of the screen*/
        "x": 0.120,
        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
        "y": 0.120
        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
    }
},
"riskDataMark": true,
/*ro, opt, bool, whether to mark risk data: "true" mark the data as the risk data and person and ID comparison failed, "false" or this field
is not returned-the data is normal*/
"dataType": "new",
/*ro, opt, enum, data type and status:, subType:string, desc: "new"-newly added data, "modified"-edited data, "normal"-unchanged data*/
"IdentityInfo": {
    /*ro, req, object, ID card information*/
    "chnName": "test",
    /*ro, opt, string, Chinese name*/
    "enName": "test",
    /*ro, opt, string, English name*/
    "sex": "1",
    /*ro, opt, enum, gender, subType:string, desc: "male", "female"*/
    "birth": "1990-02-24",
    /*ro, opt, string, date of birth*/
    "addr": "test",
    /*ro, opt, string, address*/
    "IDCardNo": "test",
    /*ro, opt, string, ID Number*/
    "issuingAuthority": "test",
    /*ro, opt, string, issuing authority*/
    "startDate": "test",
    /*ro, opt, string, start date of the effective period*/
    "endDate": "test",
    /*ro, opt, string, end date of the effective period*/
    "nation": 1,
    /*ro, opt, int, N/A*/
    "passNo": "test",
    /*ro, opt, string, entry-exit permit No.*/
    "issueNumber": "test",
    /*ro, opt, string, issued times*/
    "certificateType": "1",
    /*ro, opt, enum, certificate type, subType:string*/
    "permanentResidenceCardNo": "test",
    /*ro, opt, string, permanent resident visa No.*/
    "nationalityOrAreaCode": "test",
    /*ro, opt, string, country/region code*/
    "version": "test",
    /*ro, opt, string, certificate version No.*/
    "receivingAuthorityCode": "test",
    /*ro, opt, string, acceptance authority code*/
    "FingerprintList": [
        /*ro, opt, array, fingerprint information list, subType:object*/
        {
            "fingerprint": "test"
            /*ro, opt, string, fingerprint information, desc:base64 decoding*/
        }
    ],
    "pic": "test"
    /*ro, opt, string, certificate picture information, desc:which should be encoded by Base64, encrypted and decrypted by a specific

```


Request URL

GET /ISAPI/AccessControl/OfflineCapture/dataOutput/progress?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "DataOutputProgress": {
    /*ro, req, object*/
    "progress": 1
    /*ro, req, int, exporting progress, range:[1,100]*/
  }
}
```

12.9.5.7 Export the offline collected data

Request URL

PUT /ISAPI/AccessControl/OfflineCapture/dataOutput?format=json

Query Parameter

None

Request Message

```
{
  "DataOutputCfg": {
    /*req, object*/
    "password": "test",
    /*req, string, password for exporting*/
    "type": "UsbDisk"
    /*opt, enum, exporting method, subType:string, desc:exporting method*/
  }
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "OK",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok",
  /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
  "MErrCode": "0x00000000",
  /*ro, opt, string*/
  "MErrDevSelfEx": "0x00000000"
  /*ro, opt, string*/
}
```

12.9.5.8 Get the progress of uploading the user list of offline collection

Request URL

GET /ISAPI/AccessControl/OfflineCapture/InfoFile/progress?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "InfoFileProgress": {
    /*ro, req, object*/
    "percent": 100
    /*ro, req, int, percentage of the uploading progress, range:[1,100]*/
  }
}

```

12.9.5.9 Upload the user list of offline collection

Request URL

POST /ISAPI/AccessControl/OfflineCapture/InfoFile?format=json

Query Parameter

None

Request Message

```

{
  "InfoFile": {
    /*req, object*/
    "dataType": "binary",
    /*req, enum, URL, subType:string, desc:URL*/
    "fileUrl": "test"
    /*opt, string, this node is valid when the value of dataType is "url"*/
  }
}

```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
设备未解析，如填写1	[Message content]	application/json	--	--	--
设备未实现	[Stream data]	application/octet-stream		设备未解析，文件真实名称（带后缀.xls）	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```

--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value

```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}

```

12.9.5.10 Download user list template of offline collection

Request URL

POST /ISAPI/AccessControl/OfflineCapture/InfoFileTemplateDownload?format=json

Query Parameter

None

Request Message

```

{
  "InfoFileTemplateCond": {
    /*req, object*/
    "dataType": "binary"
    /*req, enum, URL, subType:string, desc:URL*/
  }
}

```

Response Message

```

{
  "InfoFileTemplate": {
    /*ro, req, object, template file*/
    "dataType": "binary",
    /*ro, req, enum, file data type, subType:string, desc:file data type*/
    "fileUrl": "test"
    /*ro, opt, string, file URL, range:[1,256], desc:this field is valid when the value of dataType is "url"*/
  }
}

```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
设备未实现	[Message content]	application/json	--	--	--
dataCollections	[Binary picture data]	image/jpeg		infoFileTemplate.xls	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```

--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value

```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

12.9.5.11 Search for the offline collection progress

Request URL

GET /ISAPI/AccessControl/OfflineCapture/progress?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "CaptureProgress": {
    /*ro, req, object*/
    "reqCaptureNum": 1,
    /*ro, opt, int, total number of persons to be collected*/
    "completelyCaptureNum": 1,
    /*ro, opt, int, number of completely collected persons*/
    "partiallyCaptureNum": 1,
    /*ro, opt, int, number of partially collected persons*/
    "reqFaceNum": 1,
    /*ro, opt, int, number of faces to be collected*/
    "faceNum": 1,
    /*ro, opt, int, number of collected faces*/
    "reqFingerprintNum": 1,
    /*ro, opt, int, number of fingerprints to be collected*/
    "fingerprintNum": 1,
    /*ro, opt, int, number of collected fingerprints*/
    "reqCardNum": 1,
    /*ro, opt, int, number of cards to be collected*/
    "cardNum": 1,
    /*ro, opt, int, number of collected cards*/
    "reqIDCardNum": 1,
    /*ro, opt, int, number of ID cards to be collected*/
    "IDCardNum": 1,
    /*ro, opt, int, number of collected ID cards*/
    "reqIssueNum": 1,
    /*ro, opt, int, number of persons to be issued with smart cards*/
    "IssuedNum": 1,
    /*ro, opt, int, number of persons that have been issued with smart cards*/
  }
}
```

12.9.5.12 Set the parameters of offline collection rules

Request URL

PUT /ISAPI/AccessControl/OfflineCapture/ruleInfo?format=json

Query Parameter

None

Request Message

```

{
  "RuleInfo": {
    /*req, object, rule information*/
    "reqAdminRights": true,
    /*req, bool, whether the administrator permission is required, desc:whether the administrator permission is required*/
    "enableCardNoLenAuto": true,
    /*opt, bool, whether to enable length self-adaption of the card serial No., desc:the priority of this node is higher than len*/
    "RuleList": [
      /*opt, array, rule list, subType:object, desc:it contains rules for collecting different types of data*/
      {
        "dataType": "name",
        /*req, enum, data type, subType:string, desc:"name", "employeeNo" (employee No.), "IDCardNo" (ID card No.), "IDCardSerialNo" (ID card serial No.), "IDCardDetails" (ID card details), "card", "fingerprint" (fingerprint), "face"*/
        "enable": true,
        /*req, bool, whether to collect and display*/
        "uniqueCheck": true,
        /*opt, bool, whether to enable uniqueness verification; "false" (default), desc:this mode is valid when dataType is "name". For other data types, this node is the read-only optional parameter*/
        "len": 128,
        /*opt, int, data length, desc:this node is valid when dataType is "name", "employeeNo" or "card". The default data length of name is 128. For other data types, this node is the read-only optional parameter. If it is not supported, this node will not be returned*/
        "num": 2,
        /*opt, int, number of collected data, desc:this node is valid when dataType is "fingerprint" or "card"*/
        "fingerprintIDs": [1, 2]
        /*opt, array, ID list of fingerprints that need to be collected, subType:int, desc:this node is valid when dataType is "fingerprint"*/
      },
      {
        "enableLocalIssueCard": true,
        /*opt, bool, whether to enable issuing smart cards locally, desc:whether to enable issuing smart cards locally*/
        "isLocalStorage": false
        /*opt, bool, whether to store face picture and fingerprint information in the device locally, desc:whether to store face picture and fingerprint information in the device locally*/
      }
    ],
    "enableLocalIssueCard": true,
    /*opt, bool, whether to enable issuing smart cards locally, desc:whether to enable issuing smart cards locally*/
    "isLocalStorage": false
    /*opt, bool, whether to store face picture and fingerprint information in the device locally, desc:whether to store face picture and fingerprint information in the device locally*/
  }
}

```

Response Message

```

{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}

```

12.9.5.13 Get the parameters of offline collection rules

Request URL

GET /ISAPI/AccessControl/OfflineCapture/ruleInfo?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "RuleInfo": {
    /*ro, req, object, rule information*/
    "reqAdminRights": true,
    /*ro, req, bool, whether the administrator permission is required, desc:whether the administrator permission is required: "true"-yes, "false"-no*/
    "enableCardNoLenAuto": true,
    /*ro, opt, bool, whether to enable length self-adaption of the card serial No. The priority of this field is higher than len, desc:whether to enable
length self-adaption of the card serial No. The priority of this field is higher than len*/
    "RuleList": [
      /*ro, opt, array, rule list, subType:object, desc:this node contains rules for collecting different types of data*/
      {
        "dataType": "name",
        /*ro, req, enum, data type, subType:string, desc:"name", "employeeNo" (employee No.), "IDCardNo" (ID card No.), "IDCardSerialNo" (ID card
serial No.), "IDCardDetails" (ID card details), "card", "fingerprint" (fingerprint), "face"*/
        "enable": true,
        /*ro, req, bool, whether to collect and display*/
        "uniqueCheck": true,
        /*ro, opt, bool, whether to enable uniqueness verification, desc:this field is valid when dataType is "name". For other data types, this
field is the read-only optional parameter*/
        "len": 128,
        /*ro, opt, int, data length, desc:this field is valid when dataType is "name", "employeeNo" or "card". The default data length of name is
128. For other data types, this field is the read-only optional parameter. If it is not supported, this field will not be returned*/
        "num": 2,
        /*ro, opt, int, number of collected data, desc:this field is valid when dataType is "fingerprint" or "card"*/
        "fingerprintIDs": [1, 2]
        /*ro, opt, array, ID list of fingerprints that need to be collected, subType:int, desc:this field is valid when dataType is "fingerprint"*/
      }
    ],
    "enableLocalIssueCard": true,
    /*ro, opt, bool, whether to enable issuing smart cards locally, desc:whether to enable issuing smart cards locally*/
    "isLocalStorage": false
    /*ro, opt, bool, whether to store face picture and fingerprint information in the device locally, desc:whether to store face picture and fingerprint
information in the device locally*/
  }
}

```

12.9.5.14 Get the details of failing to upload the files with offline collection user list

Request URL

GET /ISAPI/AccessControl/OfflineCapture/uploadFailedDetails?format=json

Query Parameter

None

Request Message

None

Response Message

```

{
  "UploadFailedDetails": {
    /*ro, req, object*/
    "description": "test"
    /*ro, req, string, detailed descriptions of failed uploading*/
  }
}

```

12.10 Security Control Device (General)

12.10.1 Event Message Push Management

12.10.1.1 Get the parameters of the report uploading method

Request URL

GET /ISAPI/SecurityCP/ReportCenterCfg/<centerID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
centerID	string	--

Request Message

None

Response Message

```
{
  "ReportCenterCfg": {
    /*ro, req, object*/
    "enable": true,
    /*ro, opt, bool, whether to enable the function*/
    "ChanAlarmMode": [
      /*ro, opt, array, alarm channel of the center group, subType:object*/
      {
        "id": 1,
        /*ro, opt, enum, channel ID, subType:int, desc:channel ID: 1-main channel,2-backup channel 1,3-backup channel 2,4-backup channel 3*/
        "chanAlarmMode": "T1"
        /*ro, opt, enum, alarm channel mode, subType:string, desc:"T1" (T1 channel), "T2" (T2 channel), "N1" (N1 channel), "N2" (N2 channel), "G1" (G1 channel), "G2" (G2 channel), "N3" (N3 channel), "N4" (N4 channel)*/
      }
    ]
  }
}
```

12.10.1.2 Set the parameters of the report uploading method

Request URL

PUT /ISAPI/SecurityCP/ReportCenterCfg/<centerID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
centerID	string	--

Request Message

```
{
  "ReportCenterCfg": {
    /*req, object*/
    "enable": true,
    /*opt, bool, whether to enable uploading report*/
    "ChanAlarmMode": [
      /*opt, array, alarm channel of the center group, subType:object*/
      {
        "id": 1,
        /*opt, enum, channel ID, subType:int, desc:1 (main channel), 2 (backup channel 1), 3 (backup channel 2), 4 (backup channel 3)*/
        "chanAlarmMode": "T1"
        /*opt, enum, alarm channel mode, subType:string, desc:"T1" (T1 channel), "T2" (T2 channel), "N1" (N1 channel), "N2" (N2 channel), "G1" (G1 channel), "G2" (G2 channel), "N3" (N3 channel), "N4" (N4 channel)*/
      }
    ]
  }
}
```

Response Message

```
{
  "statusCode": 1,
  /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
  "statusString": "ok",
  /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "subStatusCode": "ok",
  /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
  "errorCode": 1,
  /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
  "errorMsg": "ok"
  /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

12.10.1.3 Get the configuration capability of the report uploading method

Request URL

GET /ISAPI/SecurityCP/ReportCenterCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
  "ReportCenterCfg": {
    /*ro, req, object*/
    "CenterID": {
      /*ro, opt, object, center group No.*/
      "@min": 1,
      /*ro, opt, int, the minimum value*/
      "@max": 1
      /*ro, opt, int, the maximum value*/
    },
    "enable": "true,false",
    /*ro, opt, string, whether to enable uploading report*/
    "ChanAlarmMode": {
      /*ro, opt, object, alarm channel of the center group*/
      "maxSize": 1,
      /*ro, opt, int, the maximum number of channels*/
      "id": {
        /*ro, opt, object, channel ID, desc:1 (main channel), 2 (backup channel 1), 3 (backup channel 2), 4 (backup channel 3)*/
        "@min": 1,
        /*ro, opt, int, the minimum value*/
        "@max": 2
        /*ro, opt, int, the maximum value*/
      },
      "chanAlarmMode": {
        /*ro, opt, object, alarm channel mode, desc:"T1" (T1 channel), "T2" (T2 channel), "N1" (N1 channel), "N2" (N2 channel), "G1" (G1 channel), "G2"
        (G2 channel), "N3" (N3 channel), "N4" (N4 channel)*/
        "@opt": "T1,T2,N1,N2,G1,G2,N3,N4"
        /*ro, opt, string*/
      }
    }
  }
}
```

13 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on <https://tpp.hikvision.com> and visit our Training Center: <https://tpp.hikvision.com/tpp/Training>. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.